Remarks on mixed topological-holomorphic strings

Closed Hamiltonian BF theory and reduced CE/PV central operations for a Dirac brane probe

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Abstract

We work in the formal-local mixed holomorphic-topological string field theory on

$$X = \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}, \qquad L = \mathbb{R}_{\mathrm{brane}} \times \{s = z_1 = z_2 = 0\},$$

completed at the mixed Dirac brane whose holomorphic normal disk is $(\operatorname{Spf} \mathbb{C}[[z_1, z_2]], dz_1 \wedge dz_2)$. The theorem package proved in this local model has three cores: the Dirac brane probe, the Hamiltonian BF closed sector, and the reduced Hamiltonian CE/PV central-operation theorem, together with Matlis principal parts and the Weyl/Moyal–Capelli quantum coefficient surface, finite radial/Weyl certificates, and a finite-window non-scalar Costello/QME obstruction criterion.

On a stack of N probes the holomorphic coordinates become $\phi_1, \phi_2 \in \mathfrak{gl}_N$. The open sector is the Dirac-normalized first-order system

$$S_\partial = \int_{\mathbb{R}} \operatorname{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big), \qquad Q\psi = [\phi_1, \phi_2].$$

The closed sector is Hamiltonian BF theory for the shifted cotangent Lie algebra

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1],$$

with the Hamiltonian bracket induced by $dz_1 \wedge dz_2$, the local BV pairing supplied by continuous evaluation $\mathfrak{h}^\vee_{\mathrm{cont}} \otimes \mathfrak{h} \to \mathbb{C}$, and the BF action

$$S_{\mathrm{BF}} = \int_{\mathbb{R}^2 \times \mathbb{C}^2} \big\langle \beta,\ (d_{\mathbb{R}^2} + \bar\partial)\alpha + \tfrac{1}{2}\{\alpha, \alpha\} \big\rangle\, dz_1 dz_2.$$

The classical master equation is the Jacobi identity for $\mathfrak{h}$ plus the coadjoint-module identity.

The native holomorphic object is not a curve vertex algebra. It is the two-complex-dimensional Hamiltonian holomorphic factorization algebra

$$\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}(B) = C^\bullet_{\mathrm{CE,cont}}\big(\Omega^{0,\bullet}_c(B) \widehat{\otimes} \mathfrak{g}\big)$$

on holomorphic polydisks $B \subset \mathbb{C}^2$, with mixed enhancement

$$\mathcal{F}^{\mathrm{mix}}_{\mathrm{Ham}}(U \times B) = C^\bullet_{\mathrm{CE,cont}}\big(\Omega^\bullet_c(U) \widehat{\otimes} \Omega^{0,\bullet}_c(B) \widehat{\otimes} \mathfrak{g}\big)$$

for $U \subset \mathbb{R}^2_{\mathrm{top}}$. This is the manuscript's native holomorphic E_2-type/factorization object. Its brane avatars are the Weyl/Moyal open-string algebra, matrix or supermatrix cyclic trace models, and support-local principal-part currents on brane intervals.

The inner CE/PV theorem is a reduced Hamiltonian central-operation statement. For a finite-dimensional Lie algebra $\mathfrak{l}$,

$$C^\bullet_{\mathrm{CE}}(\mathfrak{l} \ltimes \mathfrak{l}^\vee[1]) \simeq \mathrm{PV}(\mathbf{S}(\mathfrak{l})), \qquad c^I \mapsto \theta^I, \quad u_I \mapsto O_I.$$

For the formal disk this becomes the coordinate completed map

$$\Phi_{\mathrm{coord}} \colon C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}}).$$

It is a dg coordinate theorem. The bracketed P_0 form is asserted only after a bracket-admissible completion $B \subset \mathrm{PV}_{\mathrm{coord}}$ is chosen; the Maurer–Cartan kernel Θ_B is asserted only when the diagonal tensor converges in a kernel-admissible product. This is not the unreduced factorization-centre theorem.

The cotangent coefficients are Grothendieck–Matlis principal parts, not polynomial one-ψ descendants:

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \subset H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2,$$
$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \qquad a, b \geq 0,\ a + b > 0.$$

Their coadjoint action is

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with negative targets zero. The PBW one-ψ classes $\Psi_{a,b}$ share bidegree labels with $\rho_{a,b}$ but live in a different Hamiltonian module. The $\mathcal{D}_\hbar$ Fourier–Rees bridge

$$z_1^a z_2^b \mapsto \partial_1^a \partial_2^b e_0 \mapsto (-1)^{a+b} \hbar^{a+b} a! b! \rho_{a,b}$$

is a polarization comparison; the same-action obstruction forbids a flat polynomial realization of $\mathcal{P}$ with the original action.

The quantum target uses the Weyl/Moyal convention

$$f * g = m \exp\left(\frac{\hbar}{2}(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})\right)(f \otimes g), \qquad \{f, g\}_\hbar = \hbar^{-1}[f, g]_*.$$

The raw commutator has first and third coefficients 1 and 1/24. At finite N the normal-ordered quantum comoment carries the Capelli contact term

$$\widehat{\mu}_\epsilon = \mathrm{Tr}\big(\epsilon(YX - XY + \hbar N I)\big),$$

and the scalar QME obstruction in the ordinary scalar-reduced $\mathfrak{gl}_N$ branch is

$$\mathrm{Ob}_{\mathrm{sc}} = \hbar N\, \omega(f, g) \int_{\mathbb{R}} a(t) b(t) \gamma_1(t)\, dt, \qquad [\mathrm{Ob}_{\mathrm{sc}}] = \hbar N [\bar{c}].$$

The central-character and balanced-supertrace branches cancel this scalar class by changing the source or open model before the reduced class forms; any other cancellation must supply an explicit null-homotopy with leading class $-\hbar N [\bar{c}]$.

The proved quantum surface is componentwise. For an admissible weight and interval I,

$$\mathfrak{S}_{\hbar,w}(I) = \mathcal{K}^w_{\mathrm{BV}} \times \{0_{\mathrm{sc}}\} \times A^{\mathrm{pp},w}_{\partial,\hbar}(I) \times T_{\mathrm{Ham},\hbar}.$$

Its factors are weighted coefficient/BV-kernel convergence, balanced-supertrace scalar QME cancellation, reduced weighted Moyal principal-part current brackets, and the degree-zero Hamiltonian

trace comparison relative to the radial/Weyl normal-form gate. The finite certificates now close the edge families $(2, b)$, $(b, 2)$ through $b = 12$, the balanced diagonal through $(6, 6)$, and all non-edge bidegrees of total degree at most 12. The missing all-bidegree theorem is the uniform splitting of the free normal-diagram correction map $C_{a,b}$; the first balanced case outside the current certificate range is $(7, 7)$, which is open rather than obstructed.

The non-scalar Costello graph/QME surface is now a finite-window criterion. After constructing the scalar-contact lift $\widehat{\sigma}_{\mathrm{sc}}$, one must kill

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$$

with finite-window counterterms, a curved bulk-to-defect kernel, and compatible centrality homotopies. The closed rows α_{sc} and α_{II} vanish in the constructed finite-window habitat. The first exposed larger source-face row is θ_3:

$$\mathsf{E}_{\theta_3,M} = -K^{M}_{\Theta_3}(\zeta_{M,\nu_3}), \qquad \ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1.$$

It is not solved by naming a counterterm. It is solved only by a CE ancestor, a scalar-zero Costello local counterterm, or a complete companion-face table whose differential gives $d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$ with Roos-compatible finite-window projections.

Curve chiral and Zhu objects are reduced surfaces, not native objects. A $\mathbb{C} \times \mathbb{R}$ reduction must specify the projection, the retained z_2-mode or principal-part coefficient system, the image of the brane, the pushed-forward bracket, the BV pairing, the anomaly matching, and the Dirac BRST differential before any $Q^2_{\mathrm{BRST}} = 0$ or Zhu comparison is asserted. The normal Ω-background is likewise a theorem surface, not a consequence of the preceding constructions: one must use the brane-preserving normal torus on

$$N_L X = \mathbb{R}_s \oplus \mathbb{C}_{z_1} \oplus \mathbb{C}_{z_2}, \qquad T_\Omega = \mathbb{C}^*_{\epsilon_s} \times \mathbb{C}^*_{\epsilon_1} \times \mathbb{C}^*_{\epsilon_2}, \qquad t \text{ fixed},$$

construct

$$Q_\Omega = Q + \iota_{V_\Omega}, \qquad Q^2_\Omega = L_{V_\Omega},$$

invert the moving normal weights, and use a line-valued or $\hbar_\omega$-weighted Hamiltonian bracket off the self-dual locus $\epsilon_1 + \epsilon_2 = 0$. It must also state the residue/Euler normalization and build the resulting stratified factorization algebra on (X, L). A protected stratified trace or physical Ω-large-N trace requires, in addition, the brane-defect QME, trace-state normalization, and the same non-scalar obstruction vanishing.

No compact Calabi–Yau, global BCOV, CoHA, quintic, HKQ/CDGP/GV/OSV, Abel–Jacobi, MNOP/chiral-volume, Vol III, Igusa, Borcherds/BKM, or sister-volume consequence follows from this local theorem. Such a target requires a matched-conventions descent theorem and vanishing, with specified null-homotopies, of the corresponding period, anomaly, QME, convergence, hyperdescent, six-relation, and operadic obstruction classes.

Claim status and dependencies: coordinate CE/PV and admissible Koszul layers

Formal-local theorem data. The formal-local interface is attached to the shifted cotangent Hamiltonian Lie algebra

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$

Its visible interface is stratified by proof strength:

$$\begin{gathered}\text{finite cotangent CE/PV, proved}\\ \Downarrow\\ \text{coordinate formal-disk CE/PV, proved as dg algebra}\\ \Downarrow\\ \text{bracket-admissible } P_0 \text{ realizations, proved after } B\\ \Downarrow\\ \text{kernel-admissible MC kernels, proved after datum}\\ \Downarrow\\ \text{Tate bar-cobar, relative to the admissible envelope.}\end{gathered}$$

The reduced CE/PV central-operation generator map is fixed once and for all:

$$c^I \longmapsto \theta^I, \qquad u_I \longmapsto O_I.$$

The admissible interface datum is

$$\mathfrak{U}_{\mathfrak{h}} = (\Phi_{\mathrm{coord}}, \{\Phi^B_{P_0}\}_B, \{\Theta_B\}_B),$$

where B ranges over bracket-admissible subalgebras in the second component and kernel-admissible data in the third. This is not a categorical universal object unless a category of admissible completions and a universal property are supplied. The theory is extended by the primitive higher-ψ extraction, the $\mathcal{D}_\hbar$ Fourier–Rees bridge, the weighted finite-window regulator, the endpoint criteria, and the unreduced factorization-current lift datum.

Admissibility principle. A frontier extension is admissible only when a target theorem supplies the matched-conventions datum and proves vanishing, with null-homotopies, of the period, anomaly, QME, convergence, hyperdescent, six-relation, descent, and operadic obstruction vector.

External-comparison quarantine. CoHA, compact Calabi–Yau/quintic, OSV/GV, Abel–Jacobi, MNOP, Igusa/BKM, and global BCOV data are external only. Without a separate matched-conventions theorem with comparison maps and vanishing obstruction vector, they prove none of the local QME, Ext, cyclic, Weyl/Moyal, radial-parts, or kernel claims; their gaps are external source obligations, not core proof obligations.

Proof route. For the proof route read the local theorem dependencies, then the local theorem, then the obstruction complexes in Section 4. For theorem names read the status table below together with Theorem 2.3. For residual obligations read the admissible Tate, unreduced factorization-current, one-antifield, componentwise quantum coefficient, non-scalar QME, and descent rows. For cross-volume uses read only the firewall row: no CoHA, compact Calabi–Yau, quintic, OSV/GV, Abel–Jacobi, MNOP, Vol III, Igusa/BKM, or global BCOV claim supports or follows from the local QME, Ext, cyclic, Weyl/Moyal, radial-parts, or kernel theorem surface without the named matched-conventions descent theorem.

Admissibility stages. Stage A0 is the finite cotangent CE/PV identity. Stage A1 is the coordinate formal-disk CE/PV identity Φ_{coord}, proved as a completed dg commutative algebra statement. Stage A2 is a bracket-admissible P_0-realization $\Phi^B_{P_0} : \Phi^{-1}_{\mathrm{coord}}(B) \to B$. Here B is an explicitly topologized complete Hausdorff subalgebra with continuous inclusion in the coordinate product, continuous multiplication, continuous d^{coord}_{π}, and a continuous closed Schouten contraction bracket whose transport to $\Phi^{-1}_{\mathrm{coord}}(B)$ is again continuous. Boundary Hamiltonian uses also require B to contain the relevant O_f's and their forced differentials $d^{\mathrm{coord}}_{\pi} O_f$. Stage A3 is a kernel-admissible Maurer–Cartan kernel Θ_B, proved after a specified completed convolution dg Lie algebra $\mathcal{K}_B$ in which the diagonal sums converge; without that object there is no kernel theorem. Stage A4 is an admissible Tate or relative bar-cobar/Quillen envelope: strict Mittag–Leffler exactness and kernel existence are constructed in the finite-window habitat, while Quillen equivalence remains relative to the stated admissible model-envelope hypotheses. Stage A5 is the stable scalar-reduced Dirac trace comparison. Stage A6 splits. Stage A6a is the componentwise quantum coefficient surface of Theorem 3.270: weighted BV kernel convergence, balanced-supertrace scalar QME cancellation, reduced weighted Moyal principal-part current brackets, and the finite-certificate-relative radial/Weyl trace comparison with its dual-potential obstruction $\Omega^{\mathrm{rad}}_{a,b}$, decorated PBW Stokes gate $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$, and signed-row failure criterion. Stage A6b has a proved minimal full-equivariant finite-window branch: Theorem E.42 kills $R^{\mathrm{ns}}_{n,M}$, $\mathfrak{o}^{\mathrm{ns}}_{n,M}$, and $C_{n,M}$ for every $n \geq 1$ only in the named minimal marked habitat. The larger non-scalar Costello graph/QME construction problem of Problem 3.271 is the exact obstruction problem obtained before the scalar-projection lift tower $o^{(r)}_{\sigma,w}$ vanishes, the filtered projection $\widehat{\sigma}_{\mathrm{sc}}$ is constructed, the named obstruction class in $H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$ is killed by compatible counterterms, and a curved bulk-to-defect kernel is built. Stage A7 is any external matched-conventions descent problem. Its rows are quarantine gates, not prerequisites for the core local theorem; they are not consequences of the local stages.

Koszul comparison layers. Layer 0 is the operadic Lie/coComm Koszul input $\mathrm{Lie}^{!} \simeq \mathrm{coComm}[-1]$. Layer 1 is the Tate CE/bar-cobar identity $\Omega C^{\mathrm{CE}}_{*}(\mathfrak{g}) \simeq U(\mathfrak{g})$ only in an admissible lower-central Tate-pronilpotent envelope whose strict Mittag–Leffler exactness has been constructed. Layer 2 is the corresponding Tate Quillen equivalence, relative to the transferred model-envelope hypotheses. Layer 3 is the coordinate CE/PV reduced P_0 central-operation theorem with $c^I \mapsto \theta^I$ and $u_I \mapsto O_I$. Layer 4 is the PBW/Rees special-fibre shadow

$$U_\hbar(\mathfrak{g}) = T(\mathfrak{g})[[\hbar]]/\langle xy - (-1)^{|x||y|} yx - \hbar[x,y]\rangle, \qquad U_\hbar(\mathfrak{g})/(\hbar) \cong S(\mathfrak{h} \oplus \mathfrak{h}^{\vee}[1]).$$

Layer 5 is the Matlis/Grothendieck cotangent module

$$\mathfrak{h}^{\vee}_{\mathrm{cont}} \cong \mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\rho_{a,b}, \qquad z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}.$$

Layer 6 is the brane-line factorization-current comparison

$$\mathfrak{h}_I = \Omega^{\bullet}_c(I)\widehat{\otimes}\mathfrak{h}, \qquad u_{a(t)dt\otimes f} \longmapsto B_f(a).$$

Layer 7 is the stable large-N bulk-boundary horizon; in this ledger it is a trace-state construction surface, not evidence for the local Hamiltonian BF/Moyal core.

Status table.

Claim surface	Status	Boundary
Coordinate formal-disk CE/PV theorem	proved as dg coordinate algebra	Φ_{coord} is a finite-test/pro-window cochain theorem; it is not a strict global P_0 theorem

Bracket-admissible P_0 realization	proved after B	$\Phi^{B}_{P_0}$ is asserted only for complete $B \subset \mathrm{PV}_{\mathrm{coord}}$ stable under the coordinate differential d^{coord}_{π}, closed under multiplication, and equipped with a continuous closed contraction bracket whose transported source bracket is closed
Kernel-admissible MC kernel	proved after datum	Θ_B is a theorem inside a specified $(B, \mathcal{K}_B)$ where the diagonal sums converge; $\mathfrak{o}^{\mathrm{Cas}}$ proves that the strict product/direct-sum endpoint has no such kernel
Full $Z^{P_0}_{\mathrm{fact}}$ of the open BV algebra	centrality/QME obstruction-vector problem	construct product and P_0-centrality homotopies, stratified descent, and a Costello brane-defect QME kernel; the exact obstruction vector is $(\mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}, \mathrm{ob}^{P_0}_{\mathrm{cent}}, \mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}, \delta^{C/R}_{\mathrm{Weiss}})$
Full open BV identification	reduced-sector theorem plus unreduced kernel problem	only the scalar-reduced connected Hamiltonian sector is used; the unreduced identification requires the principal-part current lift datum, coefficient kernel, brane-defect propagator, and compatible centrality homotopies
Admissible formal conilpotent Koszul duality	relative theorem	strict ML exactness and compatible kernels are constructed; bar-cobar and module-category equivalences require the named finite-window model-envelope, conilpotence, PBW, transfer, smallness, and cyclic trace hypotheses; failure is the vector O_{AdmTate} with window, topology, conilpotence, MC, cobar, PBW, P_0, and cyclic components
Matlis cotangent module	proved	$\mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1)$ is a residue-annihilator, not a naive R-submodule quotient
Radial/Weyl trace comparison	finite certificates plus exact dual-potential obstruction	formal Moyal coefficients, reduced open-line Wick weights, Capelli triangular normalization, free normal ordering, stable trace-diagram injectivity, and the supplied finite certificates are proved; the all-bidegree theorem is $\Omega^{\mathrm{rad}}_{a,b} = 0$, equivalently decorated PBW Stokes $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$; failure is exactly a signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$ with $\lambda(E^{+}_{a,b}) - \phi(T_{a,b}) \neq 0$

Quantum coefficient surface	proved componentwise	weighted BV kernel convergence, balanced-supertrace scalar zero, reduced weighted Moyal principal-part current brackets, and the finite-certificate-relative degree-zero trace comparison; not a graph/QME construction
Minimal full-equivariant finite-window non-scalar QME branch	proved all orders in named habitat	in the minimal marked list generated by strict current-interface rows, α_{sc}, and its subset-deletion closure, with no added positive-order seed, CE-source, graph-counterterm, or nonzero scalar-zero row, $R^{\mathrm{ns}}_{n,\mathcal{M}} = \mathfrak{o}^{\mathrm{ns}}_{n,\mathcal{M}} = C_{n,\mathcal{M}} = 0$ for every $n \geq 1$; this is not a theorem about larger graph packages
Larger non-scalar Costello graph/QME construction	obstruction-class problem	first kill $o^{(r)}_{\sigma,w}$, then kill $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$ and the finite-window pair $\mathfrak{O}^{\mathrm{ns}}_n$; the curved bulk-to-defect kernel and homotopies are the exact construction target
QME cancellation mechanisms	exact scalar alternatives	unreduced central-character, opposite central-extension, and balanced-supertrace source/model replacements are proved; ordinary scalar-reduced $\mathfrak{gl}_N$ has first obstruction $\hbar N[\bar{c}]$, and a same-branch closed-exchange repair must supply leading class $-\hbar N[\bar{c}]$
External compact-CY/CoHA frontier material	quarantined comparison, no transfer theorem	no support for the local QME, Ext, cyclic, Weyl/Moyal, radial-parts, or kernel claims until $\mathrm{Ob}_{\mathfrak{C}_{\mathrm{tar}}}$ vanishes with null-homotopies and comparison morphisms; missing fixtures stay external

Local graph/QME branch statuses.

Branch	Status	Exact boundary
Balanced scalar-contact projection	proved on named habitats	$\widehat{\sigma}_{\mathrm{bal,sc}} = 0$ is a filtered chain map on balanced scalar-contact habitats and on balanced graph/counterterm subhabitats; outside these habitats the obstruction is the first nonzero $o^{(r)}_{\sigma,w}(I)$
Full-equivariant scalar shadow	proved on finite marked habitats	codimension-two closed finite-window marked Costello lists and full-equivariant Brauer marker tensors have zero cyclic-supertrace shadow for $V = \mathbb{C}^{N\|N}$; missing boundary entries, signs, marker transports, or closure equations are the obstruction
Finite closed graph list	finite-window row-closure datum	a finite seed set becomes a theorem-grade list exactly after all field-differential, BV-edge, collision, counterterm, and source faces close with compatible codimension-two signs and marker transports; until then the package has no defined non-scalar $H^1/\varprojlim^1$ class
Minimal full-equivariant finite-window array	proved all orders in named habitat	the order-0 current row has $R^{\mathrm{row}}_{0,\mathcal{M}} = S_{0,\mathcal{M}} = 0$; the full-equivariant one-cross scalar-contact row is zero as a local functional for $V = \mathbb{C}^{N\|N}$; in the minimal subset-deletion closure $R^{\mathrm{ns}}_{n,\mathcal{M}} = \mathfrak{o}^{\mathrm{ns}}_{n,\mathcal{M}} = C_{n,\mathcal{M}} = 0$ for all $n \geq 1$
Larger finite-window graph packages	row-data obstruction problem	after any added positive-order seed, CE-source face, graph-counterterm face, nonzero scalar-zero row, or nonminimal lower counterterm, the package must supply graph type, labels, marker tensor, incidence signs, row values, scalar gates, and truncation coefficients before the $H^1/\varprojlim^1$ non-scalar class is defined
Regular-density curvature certificate	proved under certificate	for a closed regular-density graph package with zero scalar projection and compatible regular lower counterterms, the residual lies in $\mathcal{X}^{\mathrm{reg}}$ and η^{reg}, κ^{reg}, $\beta^{\mathrm{cl,reg}}$, μ^{cl}, λ^{cl} vanish; absent certificate, $\eta^{\mathrm{reg}}_{n_0,\mathcal{M}}$ and then $\kappa^{\mathrm{reg}}_{n_0}$ decide the branch
Wavefront eta branch	obstruction theorem	non-smooth wavefront labels survive in $\mathcal{S}_c(I) = \mathcal{D}'_c(I)/\mathcal{D}^{\mathrm{reg}}_c(I)$; the obstruction is $\eta^{\mathrm{reg}}_{n_0,\mathcal{M}} = [\mathfrak{s}^{\mathrm{WF}}_{n_0,\mathcal{M}}]$, killed only by a degree-zero distributional primitive in the original quotient complex
Radial trace-diagram branch	finite certificates plus exact cokernel obstruction	the moment-ideal primitive problem is sharpened to $\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b}$; known finite certificates vanish in the recorded range, while the all-bidegree theorem is decorated PBW Stokes $D^{\square}_{a,b} = C^+_{a,b}\partial_2$, with failure a signed $B^*_{a,b}$-row

Failed-surface status ledger. The following frontier surfaces are not downgraded. Each is either proved theorem data already present, a construction target whose missing datum is named, or an exact obstruction-theorem target.

(i) *Native E_2 all-window Bochner–Martinelli transfer.* Status: construction target. The native $\mathbb{C}^2$ factorization object plus BMK current-limit data, finite Matlis moment maps, and the arity-two output projection $p_N[ix, iy]$ are proved data. Strict finite-current transfer is obstructed by $\mathrm{Ob}_N^{\text{BMK-curr}}$, with boundary-mode obstruction $z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1}$; the projected finite-window L_∞ escape repair has Jacobiator $(N+1)(N+2)\rho_{0,1}$. The positive all-window theorem presently available is the one-pair analytic pro-Matlis retract $P_\Pi^{\text{an}} = \tau P_{\text{an}}$ with $\pi_N P_\Pi^{\text{an}} = p_N$. The strict native all-window current theorem would have to construct cutoff kernels, a direct-sum moment map, a compact-current section of the pro-product tower, a full formal Hamiltonian action, collar estimates, ternary homotopy, and inverse-limit control; failure is the exact obstruction theorem for $\mathrm{Ob}_{\text{BM}}^{\Pi}$.

(ii) $\mathbb{C} \times \mathbb{R}$, *VOA, BRST, and Zhu gates.* Status: construction target. The required interface is

$$\mathfrak{I}_{\text{ch}} = (\mathcal{R}_{\mathbb{C}\times\mathbb{R}}, \mathcal{F}_{\text{red}}, V_{\text{red}}, Q_{\text{BRST}}, \mathbf{1}, T, Y, \text{wt}, \zeta_\hbar, H_{\text{anom}})$$

with a controlled $\mathbb{C} \times \mathbb{R}$ reduction retaining the z_2-mode or full principal-part coefficient system, together with bracket, BV-pairing, brane, product, factorization-to-vertex, Zhu/Moyal, and stable Hochschild/HKR/CE-PV comparison and anomaly comparison. Before that construction, any curve VOA or Zhu assertion is false transfer. Failure is the exact obstruction vector

$$\mathrm{Ob}_{\mathfrak{I}_{\text{ch}}} = (\mathrm{Ob}_{\text{red}}, \mathrm{Ob}_{\text{vert}}, \mathrm{Ob}_{\text{Zhu}}, \mathrm{Ob}_{\text{nat}}).$$

(iii) *Strict product/direct-sum endpoints and Casimir kernels.* Status: exact obstruction-theorem target for the strict endpoint and construction target for admissible replacements. The strict endpoint is governed by $\mathfrak{o}^{\text{Cas}}$; the stronger positive theorem must build a weighted coefficient habitat with convergent Casimir, continuous bracket, coevaluation, and bar-cobar compatibility.

(iv) *Equivariant CE/PV refined bracket and completion.* Status: construction target. The target is a T_Ω-graded completed habitat with diagonal kernel, $Q_\Omega = Q + \iota_{V_\Omega}$, $Q_\Omega^2 = L_{V_\Omega}$, coefficient ring, continuous bracket, and normalization. Failure is $\mathrm{ob}_{\Omega,\text{CE/PV}}$.

(v) *Open A_∞ Koszul and cyclic acceptance.* Status: proved cyclic/Koszul trace core plus construction target for the admissible open A_∞ comparison. The relative theorem requires the acceptance datum K_{I}–K_{II}: finite-window cyclic source towers, reduced cyclic chains, the corrected finite-window Loday–Quillen–Tsygan single-cycle comparison, an admissible filtered open A_∞-algebra, closed CE/PV target, twisting cochain, completed cobar morphism, exact cone/Roos obstruction complex, and cyclicity/BV-pairing conventions. The raw PBW stable trace source does not satisfy this theorem: the full cyclic normal-ordering contraction fails, the $\Psi_{a,b} \to \rho_{a,b}$ module bridge is obstructed, the strict Casimir endpoint is absent, and the full disk is not made lower-central pronilpotent by a naive split. Failure is the exact obstruction tuple

$$\mathcal{O}_{\text{open}} = (\mathcal{O}_{\text{cyc}}, \mathcal{O}_{\Psi-\rho}, \mathcal{O}_{\text{Cas}}, \mathcal{O}_{\text{lc}}, H^*(\mathrm{Cone}(\mathrm{gr}\,\kappa_\tau)), \varprojlim{}^1 H(\Omega C_{\text{op}}), \varprojlim{}^1 H(B_{\text{CE/PV}}), H^*(\mathrm{Cone}\,\lambda_{\text{LQT}})).$$

The LQT component is now theorem-grade only on the Koszul-window quotient $A_{\psi,K} = A_\psi / F^{>K} A_\psi$, the CE-length window $\leq L$, and on the stable single-cycle image $\mathrm{Prim}_{N,K,L}^{\text{1cyc}} = \mathrm{im}\,\Theta_{N,K,L}$, with $\lambda_{\text{LQT},K,L} = \Theta_{N,K,L}^{-1} P_{\text{1cyc}}$. The accepted projection is the stable block-sum Hopf Eulerian idempotent

$$P_{\text{1cyc}} = \log^*(\mathrm{id}) = \sum_{m=1}^{L} \frac{(-1)^{m-1}}{m} (\mathrm{id} - u\epsilon)^{*m},$$

in the range $N \geq \max(K, L+2)$, or $N \geq W + 2$ under one bound; in that specialization $A_{\psi,W}$, $\Theta_{N,W}$, and $\lambda_{\text{LQT},W}$ abbreviate $K, L \leq W$. The same-rank operation that deletes fixed-rank multicycles is false as a chain map because $d_{\text{CE}}(\Theta_1(a) \wedge \Theta_1(b))$ has a single-cycle commutator term while deletion before

d_{CE} gives zero. The finite-rank bridge uses the normalized signed average $\beta_{M,K,L}$ over injections into the L labeled diagonal blocks and the read-back map $\alpha_{M,K,L}$, with $\alpha_{M,K,L}\beta_{M,K,L} = \mathrm{id}$; its idempotent is

$$P^{\mathrm{sep}}_{M,K,L} = \beta_{M,K,L} P_{1\mathrm{cyc}} \alpha_{M,K,L}, \qquad M = L \max(K, L+2),$$

while the same-rank chain-homotopy bridge is blocked by

$$\Gamma_N = d_{\mathrm{same}} F_N - F_N d_{\oplus}.$$

The unqualified finite-window primitive statement is blocked by

$$O_{\mathrm{LQT},K,L} = (O_{\mathrm{win}}, O_{\mathrm{Prim}}, O_{\mathrm{stab}}, O_{\mathrm{scalar}}, O_{\mathrm{CC}}, O_{\mathrm{Roos}}),$$

where O_{scalar} is the failure of killing $\Theta_1(\psi)$ alone to define a chain quotient: $d_{\mathrm{CE}}\Theta_1(\psi) = \Theta_1(xy - yx)$.

(vi) *Full open BV factorization center and unreduced cotangent lift.* Status: construction target with exact obstruction vector. The missing theorem data are distributional principal-part currents, boundary representatives, product and P_0-centrality homotopies, coefficient kernel, brane-defect propagator, and QME solution. Failure is $(\mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}, \mathrm{ob}^{P_0}_{\mathrm{cent}}, \mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}, \delta^{\check{C}/R}_{\mathrm{Weiss}})$, with the last entry the Cech-cone/Roos gluing class of the stratified factorization row.

(vii) *PBW/Rees source action and polynomial-descendant target.* Status: exact obstruction-theorem target for same-action polynomial descendants, and construction target for the changed principal-part quantum target. A positive theorem must build the homotopical Rees action and naturality; failure is the categorical obstruction to a polynomial sub-bimodule with the same Hamiltonian action.

(viii) *Analytic Costello specialization and first/third graph normalization.* Status: construction target. The reduced Wick/Moyal coefficients 1 and 1/24 are proved algebraic data. The stronger theorem must construct the elliptic parent, reduction, propagator, measure, counterterms, anomaly cancellation, and graph-normalization rows; failure is the local anomaly/normalization obstruction theorem.

(ix) *Ordinary scalar-reduced $\mathfrak{gl}_N$ QME.* Status: exact obstruction-theorem target. The ordinary branch has first scalar class $\hbar N[\bar{c}]$. It is repaired only by the proved central-character or balanced-supertrace replacements, or by a closed-exchange term whose leading class is $-\hbar N[\bar{c}]$; $W_{\mathrm{ext}} = 0$ or exact leading class leaves the branch obstructed.

(x) *Non-scalar Costello/QME beyond the minimal marked branch.* Status: construction target with exact obstruction pair. The minimal full-equivariant finite-window branch is proved only in its named habitat. In that habitat the binary centrality row is the zero-row theorem $A^M_{\min} : 0 \to 0$, $r^M_{\min} = 0$, $h^M_{\min} = 0$, with zero Roos primitive class. A larger theorem must construct $\widehat{\sigma}_{\mathrm{sc}}$, closed row arrays, residual vectors, boundary matrices, truncation identities, and zero Roos class. For an enlarged binary Costello row the local equation is $A^M h^M = -r^M$ after the scalar gate, row basis, primitive columns, and transition matrices have been supplied. Failure is $(([\mathfrak{o}^{\mathrm{ns}}_M])_M, \lambda)$, with the closed-exchange refinement $(\beta^{\mathrm{cl}}, \mu^{\mathrm{cl}}, \lambda^{\mathrm{cl}})$.

(xi) *θ_3 primitive and companion-face table.* Status: exact obstruction-theorem target until one of the accepted constructive witnesses is supplied. The present one-row scalar-zero complex has $K^0 = 0$, $K^1 = \mathbb{C}\mathsf{E}_{\theta_3}$, $d = 0$, and $\ell_{\theta_3}(\mathsf{E}_{\theta_3}) = 1$. Thus there is no scalar-zero counterterm inside the supplied one-row complex. In the tested two-u, Hamiltonian-degree-≤ 3 CE source envelope, a left-cokernel q satisfies $q d_{\mathrm{CE}} = 0$ and $q(\zeta_{M,\nu_3}) = 1$, so no pure CE ancestor exists in that finite envelope. The eight-face companion attempt has nonzero coordinate row $(2, -2, 3, 2, -1, 1, -2, -3)$ and its diagonal transport to $N = 2$ leaves $2E_{\nu_{02}} - 2E_{\nu_{20}}$. The canonical lower source-coordinate candidate $D^2_{02,20} = K^2_{\Theta_3}(u_{a,H_{1,0}} u_{b,H_{0,1}})$ is source-exact because $d_{\mathrm{CE}}(u_{a,H_{1,0}} u_{b,H_{0,1}}) = 2\zeta_{\nu_{02}} - 2\zeta_{\nu_{20}}$, but it is not yet local-functional-exact. In the Bianchi-exposed lower basis $(E^2_{\nu_{02}}, E^2_{\nu_{20}}, \mathfrak{b}^2_{02,20})$,

$$A^2_D = (-2, 2, 1)^T, \qquad r_2 = (2, -2, 0)^T,$$

and the cokernel

$$\widetilde{\lambda}_{02,\mathfrak{b}} = \frac{1}{2}(E^2_{\nu_{02}})^* + (\mathfrak{b}^2_{02,20})^*$$

kills A_D^2 and detects r_2. A repair must therefore leave the obstructed envelope by adding a controlled degree-zero local counterterm primitive killing $\mathfrak{b}^2_{02,20}$, higher or marked CE-source rows, or a complete transport-compatible companion table with scalar-zero, signed residual zero, codimension-two closure, truncation cocycle, Bianchi transport, and zero Roos class.

(xii) *Larger finite-window graph packages and marked scalar shadow.* Status: construction target. Scalar-contact and marked-shadow vanishing are proved only on named habitats. Full packages require all boundary entries, incidence signs, marker transports, scalar gates, row bases, locality data, and truncation matrices; failure is the first undefined or nonzero finite-window cokernel row.

(xiii) *Regular-density and wavefront-current graph branches.* Status: theorem-grade on named habitats, with exact wavefront obstruction outside them. The positive branch is the regular-density graph calculus on $\mathcal{D}_c^{\mathrm{reg}}(I) = C_c^\infty(I)dt$, the graphwise tuple relation $\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M)$, and, for ordinary collision-conormal kernels, the two maximal sign-conic finite-scaling habitats $\mathcal{D}'_{c,\Lambda_\pm,\mathrm{fs}}(I)$. A theorem for arbitrary $\mathcal{D}'_c(I)$ currents is absent because products of pushed-forward currents with Hadamard graph kernels are not canonical unless Hörmander transversality, finite scaling degree, diagonal extension, counterterms, and pushforward wavefront bounds are supplied. Failure in the singular quotient is the class

$$\eta^{\mathrm{reg}}_{n_0,M} \in H^1(\mathfrak{Q}^\bullet_{w,M}(I)/X^{\mathrm{reg}}_{w,M}(I)),$$

and failure before cohomology is the microlocal no-canonical-graph-contraction obstruction for arbitrary $\mathcal{D}'_c(I)$-labels.

(xiv) *Radial all-N image and signed-row obstruction surface.* Status: construction target with exact cokernel obstruction. The finite certificates, edge strips, and free normal forms are proved data. The all-bidegree theorem is the vanishing of

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b},$$

equivalently decorated PBW Stokes $D^\square_{a,b} = C^+_{a,b}\partial_2$. Failure is a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ detecting $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. The $(7,7)$ row is open computation, not a proved nonzero class.

(xv) *Normal Ω finite-window QME habitat.* Status: construction target. Localization data are not QME data. The positive theorem must build the $R^{N,M}_\Omega$-package with normal contraction, residue/Euler normalization, equivariant propagator, scalar projection, kernel counterterms, zero $\varprojlim^1 H^0$, and centrality homotopies. Failure is the corresponding obstruction-class theorem.

(xvi) *Stratified factorization on (X, L).* Status: construction target. The missing theorem is $\mathcal{F}^{\mathrm{str}}_{\Omega,N}$ with branch choice, collar module, products, descent, centrality homotopies, and QME curvature. Failure is the exact obstruction vector $\mathrm{Ob}^{\mathrm{int}}_{\mathrm{str},\Omega,N}$, with the source-primary class $\mathrm{ob}^{\mathrm{src}}_{\mathrm{CFG}}$ kept as an external matched-conventions gate.

(xvii) *Protected trace and physical Ω large-N state.* Status: construction target. Algebraic stable trace generators are not a protected state. The theorem must construct $\omega_{N,\Omega}$, topology $\mathcal{T}_\Omega$, normalization $\nu_{\Omega,N}$, kernels, counterterms, Ward identities, and large-N cumulant estimates; failure is $\mathrm{Ob}_{\Omega,\mathrm{phys}}$.

(xviii) *Global Weiss/Ran descent and external comparison firewalls.* Status: construction target for descent and matched-conventions comparisons, external quarantine otherwise. The theorem must provide period, source comparison, cotangent boundary, P_0-center homotopies, descent data, comparison maps, and vanishing obstruction vector. Failure is the descent or matched-conventions obstruction theorem.

(xix) *Full Hamiltonian finite-dimensional truncation and Tate kernel.* Status: construction target with exact Tate/Casimir obstruction. A naive finite-dimensional truncation is not a Lie/BV kernel theorem. The positive theorem must build weighted finite-window coefficient kernels with coherent heat/gauge fixing, coevaluation, admissible weights, and endpoint control; failure is the finite-window/Tate Casimir obstruction.

Supremum controller classifications. The ledger is a theorem controller, not a weakening table. Every frontier claim below is admissible only in the strongest true form named in the right-hand clause. The controller statuses are: *proved*, *theorem-surface-with-missing-construction*, *exact obstruction*, *false transfer*, and *external comparison*.

(i) *Native holomorphic E_2 / $\mathbb{C}^2$ factorization.* Classification: *proved* for the CE/factorization object and the semidirect Hamiltonian–cotangent collision algebra; *proved BMK current-limit data, finite Matlis moments, and arity-two output projection*; and *exact obstruction* for the all-window strict Bochner–Martinelli transfer. The admissible object is

$$\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}(B) = C^{\bullet}_{\mathrm{CE,cont}}\big(\Omega^{0,\bullet}_{c}(B)\widehat{\otimes}(\mathfrak{h} \ltimes \mathcal{P}[1])\big),$$

with $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1)$. The finite-window output projection $p_N[ix, iy]$ identifies the arity-two operations with the Hamiltonian bracket and the Matlis coadjoint action; it is not a proved native finite-window homotopy. Strict support-local finite-current transfer is obstructed by $\mathrm{Ob}^{\text{BMK-curr}}_N$, and the projected finite-window L_∞ escape repair has nonzero Jacobiator $(N+1)(N+2)\rho_{0,1}$. The positive all-window theorem currently proved is the one-pair analytic pro-Matlis retract. The stronger strict native all-window current theorem is exactly the vanishing, or constructive replacement, of

$$\mathrm{Ob}^{\Pi}_{\mathrm{BM}} = (\mathrm{ob}_{\oplus}, \mathrm{ob}_{I_\Pi}, \mathrm{ob}_{\mathfrak{h},\Pi}, \mathrm{ob}_{\mathrm{collar},\Pi}, \mathrm{ob}_{3,\Pi}, \mathrm{ob}_{\mathrm{unif},\Pi}),$$

meaning uniform moment restriction to the direct-sum Matlis module, a compact-current section of the pro-product tower, a full formal Hamiltonian action on the chosen target, collar removal under extension-by-zero factorization, vanishing of the first transferred ternary obstruction, and uniform estimates in the m-adic window. The first, second, third, and sixth entries are nonzero in the native current/product topology; the collar and ternary entries are exact primitive construction targets. It is not a curve VOA theorem and not a Costello QME theorem.

(ii) *Controlled $\mathbb{C} \times \mathbb{R}$ reduction.* Classification: *theorem-surface-with-missing-construction*. A curve/topological shadow is admissible only after a reduction datum

$$\mathcal{R}_{\mathbb{C}\times\mathbb{R}} = (\pi, \kappa_{\mathrm{top}}, \kappa_{\mathrm{hol}}, B_{z_2}, \pi_!, L_{\mathrm{red}}, \langle -,- \rangle_{\mathrm{red}}, \{-,-\}_{\mathrm{red}}, H_{\mathrm{anom}})$$

is constructed. Here π fixes the surviving t, z_1 directions, κ_{top} and κ_{hol} are the topological and holomorphic kernels, B_{z_2} records the retained z_2-mode or principal-part coefficient system, $\pi_!$ is the compact-support or residue pushforward, and the remaining entries prove the pushed-forward bracket, BV pairing, brane image, and anomaly comparison. Without this datum, importing a $\mathbb{C} \times \mathbb{R}$ chiral/topological theorem is a *false transfer*.

(iii) *Reduced Dirac BRST curve algebra and Zhu bridge.* Classification: *theorem-surface-with-missing-construction* after the controlled reduction, and *false transfer* before it. The strongest admissible theorem must construct a reduced BRST complex with $Q^2_{\mathrm{BRST}} = 0$, a one-complex-dimensional factorization-to-vertex package

$$(V_L, \mathbf{1}_L, T_L, Y_L), \qquad Y_L(a,w)b = \sum_{n\in\mathbb{Z}} (a_{(n)}b)w^{-n-1},$$

finite singular orders, locality, and a continuous Zhu or zero-mode map

$$\zeta_{L,\hbar} \colon (\bar{A}_\hbar, *) \to \mathrm{Zhu}(V_L)$$

compatible with $[J_{f,(0)}, \Theta_\rho] = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}$. The existing Weyl/Moyal calculation supplies neither Y_L nor $\zeta_{L,\hbar}$. In this manuscript such a generic V_L datum is admissible only after it factors through $\mathcal{R}_{\mathbb{C}\times\mathbb{R}}$ and is identified with the retained-mode object V_{red}. Equivalently, an externally constructed chiral algebra enters only as V_{red} in $\mathfrak{I}_{\mathrm{ch}}$ after $\mathcal{R}_{\mathbb{C}\times\mathbb{R}}$ has retained the z_2-coefficient or principal-part system and after

$$\mathrm{Ob}_{\mathfrak{I}_{\mathrm{ch}}} = (\mathrm{Ob}_{\mathrm{red}}, \mathrm{Ob}_{\mathrm{vert}}, \mathrm{Ob}_{\mathrm{Zhu}}, \mathrm{Ob}_{\mathrm{nat}})$$

has been killed by the theorem or named as the obstruction-theorem target. Before that reduction datum it is *false transfer*, not native $\mathbb{C}^2$-factorization data.

(iv) *Normal Ω-background.* Classification: *theorem-surface-with-missing-construction*. The native deformation is normal scaling of

$$N_L X = \mathbb{R}_s \oplus \mathbb{C}_{z_1} \oplus \mathbb{C}_{z_2}, \qquad t \text{ fixed},$$

with

$$V_\Omega = \epsilon_s s\partial_s + \epsilon_1 z_1 \partial_{z_1} + \epsilon_2 z_2 \partial_{z_2}, \qquad Q_\Omega = Q + \iota_{V_\Omega}, \quad Q_\Omega^2 = L_{V_\Omega}.$$

The strongest admissible theorem constructs the mixed Cartan model and the localized deformation retract

$$\mathsf{C}_\Omega = (\pi_L, j_L, h_\Omega), \qquad Q_\Omega h_\Omega + h_\Omega Q_\Omega = \mathrm{id} - j_L \pi_L$$

over the finite-window localization

$$R_\Omega^{N,M} = \mathbb{C}[\epsilon_s, \epsilon_1, \epsilon_2, \hbar_\omega][\chi^{-1} \mid \chi \in \mathsf{X}_{N,M}],$$

where $\mathsf{X}_{N,M}$ contains the nonzero moving characters $n\epsilon_s + a\epsilon_1 + b\epsilon_2$ and $n\epsilon_s - a\epsilon_1 - b\epsilon_2$ that occur in the chosen window. The self-dual specialization is formed before further inversion. The exact residuals are $\mathrm{ob}_{\Omega,\mathrm{Cartan}}$ and $\mathrm{ob}_{\Omega,\mathrm{contr}}$.

(v) *Equivariant CE/PV and refined Hamiltonian bracket.* Classification: *proved coordinate formula plus global-completion obstruction theorem surface*. The self-dual locus $\epsilon_1 + \epsilon_2 = 0$ has scalar Hamiltonian bracket. Off that locus the Poisson tensor has weight $-(\epsilon_1 + \epsilon_2)$, so the scalar bracket exists only after adjoining the line parameter

$$[f, g]_\Omega = \hbar_\omega \{f, g\}, \qquad w(\hbar_\omega) = \epsilon_1 + \epsilon_2.$$

The stable coordinate theorem keeps $c_{a,b} \mapsto \theta_{a,b}$ and $u_{a,b} \mapsto O_{a,b}$, with

$$d_{\mathrm{CE},\Omega} c^K = -\frac{1}{2}\hbar_\omega C_{IJ}^K c^I c^J, \qquad d_{\mathrm{CE},\Omega} u_K = \hbar_\omega C_{KJ}^L u_L c^J,$$

and the matching PV tensor $\pi_\Omega = \frac{1}{2}\hbar_\omega C_{IJ}^K O_K \theta^I \theta^J$. The stronger global theorem is a weighted basic-Cartan prefactorization comparison over finite normal windows, with bracket-admissible topology, diagonal kernel $\Theta_{\Omega,w,B}$, resonance retention, normalization branch, and Weiss/Roos descent. Its obstruction vector is

$$\mathrm{Ob}^{\mathrm{glob}}_{\Omega,\mathrm{CE/PV}} = (\mathrm{ob}_{\Omega,\mathrm{Cartan}}, \mathrm{ob}_{\Omega,\mathrm{top}}, \mathrm{ob}_{\Omega,\mathrm{br}}, \mathrm{ob}_{\Omega,\mathrm{diag}}, \mathrm{ob}_{\Omega,\mathrm{res}}, \mathrm{ob}_{\Omega,\mathrm{norm}}, \mathrm{ob}_{\Omega,\mathrm{fact}}, \mathrm{ob}_{\Omega,\mathrm{Roos}}, \mathrm{ob}_{\Omega,\mathrm{QME}}).$$

(vi) *Omega quantum parameters.* Classification: *notation gate, not a QME theorem*. The ledger distinguishes

$$\hbar_{\mathrm{QME}}, \qquad \hbar_{\mathrm{W}}, \qquad \hbar_\omega.$$

The first is the Costello/BV graph loop parameter, the second is the Weyl/Moyal deformation parameter, and the third is the equivariant line parameter with $w(\hbar_\omega) = \epsilon_1 + \epsilon_2$. A Weyl branch may set $\hbar_{\mathrm{W}} = \hbar_\omega$; no branch sets $\hbar_{\mathrm{QME}}$ equal to either parameter without an explicit convention and a separate QME obstruction calculation.

(vii) *Stratified factorization on* (X, L). Classification: *theorem-surface-with-missing-construction*. The stronger theorem is a functor

$$\mathcal{F}^{\mathrm{str}}_{\Omega,N} : \mathrm{Disk}^{\mathrm{str}}_{X,L} \to \mathrm{Ch}_{R_\Omega^{N,M}[[\hbar_{\mathrm{QME}}]]}$$

with bulk values, brane values $A^q_{N,\Omega}$ over the Weyl parameter $\hbar_{\mathrm{W}}$ or $A^{\mathrm{pp},w}_{\partial,\Omega,\hbar_{\mathrm{W}}}$, a collar module $\mathcal{M}_{\Omega,N}$, products μ_{str}, and stratified Weiss descent. Its exact internal obstruction vector is

$$\mathrm{Ob}^{\mathrm{int}}_{\mathrm{str},\Omega,N} = (\delta_{\mathrm{pref}}, \delta_{\mathrm{assoc}}, \delta^{\check{C}/R}_{\mathrm{Weiss}}, \mathrm{ob}_{\mathrm{collar}}, \mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}, \mathrm{ob}^{P_0}_{\mathrm{cent}}, \mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}, \mathrm{ob}_{\Omega-\mathrm{basic}}, \mathrm{ob}_{\mathrm{tr-top}}).$$

where

$$\delta^{\check{C}/R}_{\mathrm{Weiss}} = \left(\{H^\bullet(\Delta^{N,M}_{V,\mathcal{U}})\}_{V,\mathcal{U},N,M}, \mathfrak{w}^{\check{C}/R}_{V,\Omega,N,M}\right).$$

The source-primary class $\mathrm{ob}^{\mathrm{src}}_{\mathrm{CFG}}$ is external to this internal vector. The Weiss component is not a holim slogan: for a collared stratified Weiss cover $\mathcal{U}$ it is the cone

$$\Delta^{N,M}_{V,\mathcal{U}} = \mathrm{Cone}\Big(\mathcal{F}^{\mathrm{str}}_{\Omega,N,M}(V) \to \mathrm{Tot}\,\check{C}^{\bullet}(\mathcal{U};\mathcal{F}^{\mathrm{str}}_{\Omega,N,M})\Big),$$

together with the Roos compatibility class $\mathfrak{w}^{\check{C}/R}_{V,\Omega,N,M}$ for refinements, windows, interval inclusions, and weight transports.

(viii) *Protected stratified trace.* Classification: *theorem-surface-with-missing-construction*. The algebraic stable trace theorem supplies candidate generators only. A protected trace requires a continuous state

$$\omega_{N,\Omega}\colon \int_{L\subset X} \mathcal{F}^{\mathrm{str}}_{\Omega,N} \to R^{N,M}_{\Omega}[[\hbar_{\mathrm{QME}}]]$$

together with normalization $\nu_{\Omega,N}$, topology $\mathcal{T}_{\Omega}$, brane-defect kernel and counterterms, product and P_0-centrality homotopies, and Ward identities for d_{QME}, Q_{Ω}, the moment map, and the brane curvature. The residue branch has $\nu^{\mathrm{res}}_{\Omega} = 1$. The Euler branch has

$$\nu^{\mathrm{Eu}}_{\Omega} = \sigma_5(\epsilon_5\epsilon_1\epsilon_2)^{-1}.$$

These are separate branches until a comparison theorem fixes the sign and factor.

(ix) *Costello graph/QME brane-defect package.* Classification: *exact obstruction*. The normal Ω-background and the componentwise quantum coefficient surface do not construct the Costello propagator, counterterms, or brane-defect QME kernel. The admissible theorem must produce $K_{\Omega,N}$ with

$$\mathrm{Curv}(K_{\Omega,N}) = d_{\mathrm{QME}}K_{\Omega,N} + \frac{1}{2}[K_{\Omega,N}, K_{\Omega,N}]_{\hbar_{\mathrm{QME}}} = 0,$$

after the scalar projection tower $o^{(r)}_{\sigma,w}$, the non-scalar class in $H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$, and the finite-window pairs $\mathfrak{O}^{\mathrm{ns}}_n$ are killed by compatible counterterms and centrality homotopies. The certified θ_3 row is the model case for this rule: in the supplied one-row scalar-zero complex

$$Q^0_{\theta_3,M} = 0, \qquad Q^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, \qquad d_M = 0,$$

the functional $\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$ proves a nonzero finite-row class. The admissible upgrade is not to soften the non-scalar QME theorem, but to add one of the missing constructive witnesses: a CE ancestor $\eta_{\theta_3,M}$ with $d_{\mathrm{CE}}\eta_{\theta_3,M} = \zeta_{M,\nu_3}$ and $d_M K^M_{\Theta_3}(\eta_{\theta_3,M}) = -\mathsf{E}_{\theta_3,M}$; a scalar-zero Costello local counterterm $C^{\mathrm{ct}}_{\theta_3,M}$ with $d_M C^{\mathrm{ct}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$; or a complete companion-face table changing the row residual to zero. The lower source-coordinate candidate $D^2_{02,20} = K^2_{\Theta_3}(\zeta^0_2)$ adds only source exactness until the Bianchi defect

$$\mathfrak{b}^2_{02,20} = d_{\mathrm{ns},2}K^2_{\Theta_3}(\zeta^0_2) - K^2_{\Theta_3}(d_{\mathrm{CE}}\zeta^0_2)$$

is killed. In the exposed lower basis the obstruction is $A^2_D = (-2,2,1)^T$, $r_2 = (2,-2,0)^T$, detected by $\widetilde{\lambda}_{02,\mathfrak{b}} = \frac{1}{2}(E^2_{\nu_{02}})^* + (\mathfrak{b}^2_{02,20})^*$. The current habitat contains no $B^2_{02,20}$ counterterm: the target $(0,0,-1)^T = -\mathfrak{b}^2_{02,20}$ is outside the image of A^2_D, detected by $\widetilde{\lambda}_{02,\mathfrak{b}}((0,0,-1)^T) = -1$. The exact next target is a scalar-zero local-functional column $A^2_B = (0,0,-1)^T$ with locality or wavefront admissibility and finite-window transport. In a tower the named transport cocycle is

$$\Delta^1_{M,N} = -\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N,$$

and a compatible family B^M kills it only when

$$\Delta^1_{M,N} = d_{\mathrm{ns},N}(\pi_{M,N}B^M - B^N).$$

These witnesses must therefore kill both the H^1 class of $\Delta^1_{M,N}$ and the secondary $\varprojlim^1 H^0$ primitive-compatibility class. A named C_{θ_3} without one of these differentials is not evidence.

(x) *Radial/Weyl finite certificates and decorated Stokes gate.* Classification: *proved finite certificates plus exact dual-potential obstruction theorem*. The radial comparison is no longer a bare external input in the ledger. The proved algebraic surface consists of the formal Weyl/Moyal coefficients, the reduced open-line Wick weights, the Capelli triangular normalization, the free indexed normal-ordering formula, stable trace-diagram injectivity, the all-N correction $K_{4,4}$, and the finite certificate range recorded in Section 4. For every bidegree put

$$R^{\mathrm{free}}_{a,b} = \mathcal{N}_{\mathrm{free}}(E_{a,b,N}) - \mathcal{N}_{\mathrm{free}}\left(\sum_W c^{\mathrm{cl}}_W \operatorname{Tr}(WM)\right),$$

and

$$\mathfrak{o}^{\mathrm{rad}}_{a,b} = [R^{\mathrm{free}}_{a,b}] \in \operatorname{coker}\big(C_{a,b}\colon \ker\partial \longrightarrow \mathsf{Diag}_{a,b}\big).$$

A kernel splitting of $C_{a,b}$ is one presentation of the same problem, but the controlling theorem surface is the sharper dual-potential formulation. With $B_{a,b}(W) = (\partial W, C^+_{a,b}(W))$ and $\eta_{a,b} = (T_{a,b}, E^+_{a,b})$, this is

$$\Omega^{\mathrm{rad}}_{a,b} = [\eta_{a,b}] \in \operatorname{coker} B_{a,b}.$$

Failure is exactly a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. The ordinary square-cell necklace cycle mechanism is proved; the current missing theorem is the decorated PBW Stokes identity for $D^{\square}_{a,b} = C^+_{a,b}\partial_2$. The first unclosed certificate targets $(6,8)$, $(8,6)$, and $(7,7)$ are exact-basis frontier computations in the present solver, not nonzero obstruction rows.

(xi) *Physical Ω-large-N trace-state theorem.* Classification: *theorem-surface-with-missing-construction*; external physical or factorization-homology analogies are *external comparison* until matched to this local pair. The strongest admissible theorem fixes the state and topology, defines connected cumulants by

$$\langle F_1, \dots, F_k\rangle^c_{N,\Omega} = [t_1 \cdots t_k] \log \omega_{N,\Omega}\left(\exp\Big(\sum_i t_i F_i\Big)\right),$$

and proves, in $\mathcal{T}_\Omega$,

$$\langle F_1, \dots, F_k\rangle^c_{N,\Omega} \sim \sum_{g\geq 0} N^{2-2g-k} F^{\Omega}_{g;k}(F_1, \dots, F_k)$$

with the chosen $\hbar$, N-scaling and Ward identities. Degreewise stable invariant theory is not this theorem.

Theorem data.

(i) *Stage A0: finite cotangent CE/PV.* For each finite-dimensional Lie algebra $\mathfrak{l}$,

$$\Phi_{\mathfrak{l}}\colon C^{\bullet}_{\mathrm{CE}}(\mathfrak{l} \ltimes \mathfrak{l}^\vee[1]) \xrightarrow{\sim} \mathrm{PV}(\mathbf{S}(\mathfrak{l}))$$

with $c \mapsto \theta$, $u \mapsto O$. The formal-disk coordinate comparison is reconstructed from admissible finite coordinate tests and pro-window coordinate cochains. It becomes a completed bracketed statement only in a mixed or weighted topology where the contraction bracket is continuous. The strict ambient product/direct-sum completion is obstructed by the infinite diagonal.

(ii) *Stage A1: coordinate formal disk CE/PV.* The completed formal-disk comparison is first a coordinate dg algebra isomorphism

$$\Phi_{\mathrm{coord}}\colon C^{\bullet}_{\mathrm{CE,coord}}(\mathfrak{g}) \longrightarrow \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}}).$$

It becomes a P_0 comparison only after choosing a bracket-admissible subalgebra B. If B is kernel-admissible, the diagonal kernel Θ_B is an additional theorem datum whose existence is exactly the convergence of the two diagonal sums in the chosen convolution habitat. Its coordinate transition obstruction terms are

$$\mathfrak{o}^{\mathrm{br}}_{M,N}, \qquad \mathfrak{o}^{\mathrm{coad}}_{M,N}.$$

The separate strict-kernel and ambient-product bracket obstruction is $\mathfrak{o}^{\mathrm{Cas}}_{M,N}$.

(iii) *Stage A2: bracket-admissible P_0 realization.* A bracket-admissible $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ is complete, stable under d_π^{coord}, closed under multiplication, and closed under a continuous Schouten contraction bracket. Its source $\Phi^{-1}_{\mathrm{coord}}(B)$ must be closed under the transported bracket. Under these hypotheses

$$\Phi^B_{P_0} : \Phi^{-1}_{\mathrm{coord}}(B) \xrightarrow{\sim} B$$

is the completed dg P_0-comparison.

(iv) *Stage A3: kernel-admissible Maurer–Cartan tensor.* A bracket-admissible B is kernel-admissible when the diagonal sums

$$\Theta_B = \sum_I H_I \otimes \theta^I + \sum_I \eta^I \otimes O_I$$

converge in the completed tensor product attached to B. Then Θ_B is the corresponding Maurer–Cartan tensor, checked coefficientwise from the Jacobi and coadjoint identities in that completed convolution dg Lie algebra:

$$d\Theta_B + \frac{1}{2}[\Theta_B, \Theta_B] = 0.$$

The strict product/direct-sum endpoint has no such kernel: $\mathfrak{o}^{\mathrm{Cas}}$ is the obstruction. The weighted and strict ML habitats supply replacement coefficient categories; they are regulators, not universal endpoints for all tail-sensitive cohomology.

(v) *Stage A4: admissible Tate bar-cobar/Quillen envelope.* For an admissible Tate replacement $\mathfrak{g}_{\mathrm{adm}}$,

$$B_{\mathrm{adm}} = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}), \qquad B^!_{\mathrm{adm}} \simeq \widehat{U}(\mathfrak{g}_{\mathrm{adm}}).$$

The strict finite-window Mittag–Leffler habitat constructs exact inverse limits, $\varprojlim^1 = 0$, paired-window tensor products, continuous duals, and compatible kernels. The bar-cobar/Quillen theorem is then relative to an admissible model envelope: conilpotent CE coalgebra, PBW completeness, completed symmetric associated graded, transfer and smallness, and an admissible transferred P_0-algebra model structure.

(vi) *Reduced CE/PV central-operation generator map.* The Hamiltonian CE ghost c^I becomes the Schouten vector θ^I. The shifted-cotangent CE coordinate u_I becomes the Hamiltonian boundary coordinate O_I. Boundary Hamiltonian observables come from u_I.

(vii) *Dirac open Koszul model.* The brane stack is the derived commuting-pair locus:

$$S_\partial = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]), \qquad Q\psi = [\phi_1, \phi_2].$$

The displayed action is in the Dirac-normalized variables obtained from the raw shifted-Chern–Simons coefficient fields; the Koszul complex is the derived zero fibre of the commuting-pair equation, not an assertion that the finite-N moment-map components form a regular sequence. Stable trace invariants are taken degreewise; for total word degree d, the stable range is $N \geq d$.

(viii) *Primitive higher-ψ extraction.* The primitive Koszul complex has nonconstant Hamiltonian H_0 classes and positive Hamiltonian one-ψ primitive H_1 classes $\Psi_{a,b}$ with $a + b > 0$. The scalar $\Psi_{0,0} = \mathrm{Tr}(\psi)$ is the removed odd scalar line, not a Matlis principal-part class. The higher primitive sectors satisfy $H_p = 0$ for $p \geq 2$ in the stable finite-bidegree cyclic word model. The one-ψ action is

$$\{\overline{\mathrm{Tr}(\phi_1^p \phi_2^q)}, \widetilde{\Psi}_{a,b}\} = (pb - qa)\widetilde{\Psi}_{a+p-1,b+q-1}.$$

(ix) *Matlis cotangent module.* The cotangent coefficient module is

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \subset H^2_{\mathfrak{m}}(R)\, dz_1 dz_2, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2.$$

Its coadjoint action is

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}.$$

(x) *Polynomial-descendant obstruction.* Polynomial one-ψ representatives have locally nilpotent linear Hamiltonian action, while

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}$$

is not locally nilpotent on principal parts. Any polynomial realization of $\mathcal{P}$ must therefore change the target, the action, or the category.

(xi) *$\mathcal{D}_\hbar$ Fourier–Rees bridge.* With

$$\mathcal{D}_\hbar = \mathbb{C}[[\hbar]]\langle z_i, \partial_i\rangle/([\partial_i, z_j] = \hbar\delta_{ij}),$$

the Fourier delta lattice gives

$$z_1^a z_2^b \mapsto \partial_1^a \partial_2^b e_0 \mapsto (-1)^{a+b}\hbar^{a+b} a!b!\rho_{a,b}.$$

Before inverting $\hbar$ the image is the Rees sublattice $\bigoplus \hbar^{a+b}\mathbb{C}[[\hbar]]\rho_{a,b}$. The bridge changes polarization; same-action Rees families from $\Psi_{a,b}$ to $\rho_{a,b}$ are obstructed.

(xii) *Quantum Hamiltonian and Moyal sector.* The Weyl convention is

$$f * g = m\exp\left(\frac{\hbar}{2}P\right)(f \otimes g), \qquad \{f, g\}_\hbar = \hbar^{-1}[f, g]_*.$$

The first and third raw commutator coefficients are 1 and 1/24. These formal coefficients, the reduced open-line Wick midpoint weights, the Capelli triangular normalization, and the free trace-diagram normal-ordering formula are proved. The all-N ordinary Weyl-trace comparison is controlled by the quantum shear congruence of Proposition F.14. In the matrix Weyl algebra this is first the explicit moment-ideal primitive problem of Lemma F.15: construct $A^j{}_i(a, b, N)$ with $E_{a,b,N} = \sum_{i,j} A^j{}_i(a, b, N)\widetilde{\mu}^i{}_j$. The sharper free-diagram form is the dual-potential obstruction theorem

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b},$$

equivalently the decorated PBW Stokes identity $D^\square_{a,b} = C^+_{a,b}\partial_2$; failure is a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. Finite certificates prove vanishing in the recorded windows and exhibit the nontrivial correction $K_{4,4}$. The descendant sector is governed by the principal-part Moyal target and the no-$\hbar$-adic same-action intertwiner.

(xiii) *Componentwise quantum coefficient surface.* The proved quantum surface is

$$\mathfrak{S}_{\hbar,w}(I) = \mathcal{K}^w_{\mathrm{BV}} \times \{0_{\mathrm{sc}}\} \times A^{\mathrm{pp},w}_{\partial,\hbar}(I) \times T_{\mathrm{Ham},\hbar}.$$

The first factor records weighted coefficient-kernel convergence and graphwise weighted boundedness. The second records the balanced $\mathfrak{gl}(N|N)$ supertrace scalar QME cancellation. The third is the reduced weighted Moyal principal-part current target, with coadjoint action $\mathrm{ad}^\vee_{f,\hbar}$ rather than the polynomial one-ψ PBW action. The fourth is the degree-zero Hamiltonian trace comparison, relative to the finite-certificate radial/Weyl gate: the obstruction is $\Omega^{\mathrm{rad}}_{a,b}$, equivalently decorated PBW Stokes $D^\square_{a,b} = C^+_{a,b}\partial_2$, with failure certified by a signed $B^*_{a,b}$-row. This is a genuine theorem surface, not the non-scalar Costello graph/QME construction.

The next theorem obligation first asks for the scalar-projection lift: for a complete d_{QME}-stable scalar-contact filtration, the successive classes

$$o^{(r)}_{\sigma,w}(I) \in H^1\Big(\mathrm{Hom}\big(\mathrm{gr}_F\,\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I), C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]]\big)_{-r}\Big)$$

must vanish. In the ordinary scalar-reduced $\mathfrak{gl}_N$ branch the first class evaluates on z_1, z_2 as $\hbar N\gamma_1$, so no filtered scalar projection exists without an external leading class $-\hbar N[\bar{c}]$ or a source/model replacement. The balanced supertrace branch has proved zero filtered projections on the balanced scalar-contact habitat,

on balanced graph/counterterm subhabitats, and on codimension-two closed full-equivariant marked finite-window habitats. For any larger graph/counterterm habitat not shown to lie in one of these classes, the scalar-contact extension remains the obstruction tower $o^{(r)}_{\sigma,w}(I)$. Only after the scalar lift is built does the next obstruction become the vanishing of

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$$

inside the filtered complex $\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$ of Problem 3.271. Concretely, one must construct finite-window local counterterms whose obstruction pair

$$\mathfrak{O}^{\mathrm{ns}}_n = \big(([\mathfrak{o}^{\mathrm{ns}}_{n,M}])_M, \lambda_n\big)$$

vanishes at every $\hbar$-order, build the curved bulk-to-defect kernel, and prove that the non-scalar class vanishes with compatible centrality homotopies. The minimal full-equivariant finite-window branch is already closed by Theorem E.42: its positive-order row sums, non-scalar classes, and counterterms are all zero in the named minimal marked habitat. This closure does not supply the row data or curved kernel for any larger graph package.

(xiv) *Weighted finite-window regulator.* Admissible weights w, w' are related by diagonal finite-window transport $R^{w_0}_{w,w'}$. The regulator theorem identifies the Mittag-Leffler local-functional complexes and carries Ob_w to $\mathrm{Ob}_{w'}$. Vanishing of the mixed brane-defect QME class is independent of the admissible weight. Weight transport preserves the non-scalar obstruction pair $\mathfrak{O}^{\mathrm{ns}}_n$; it does not kill a nonzero class.

(xv) *Endpoint criteria.* An endpoint for $w_q(d) = q^d$ is admissible when conditions (E1)–(E5) hold. The obstruction tuple is

$$\mathfrak{o}_{\mathrm{BV,end}} = (\mathfrak{o}_{\mathrm{Cas}}, \mathfrak{o}_{\mathrm{def}}, \mathfrak{o}_{\mathrm{RG}}, \mathfrak{o}_{\mathrm{QME}}).$$

The strict product/direct-sum endpoint fails because no tensor projects to all finite-window Casimirs $C^{w_0}_{\mathfrak{h}}$.

(xvi) *Unreduced factorization-current lift.* Distributional principal-part currents give the target for cotangent boundary modes. The lift datum consists of boundary representatives, product centrality homotopies, P_0-bracket homotopies, the coefficient kernel, the brane-defect propagator, and the QME solution. The scalar component is

$$\mathrm{Ob}_{\mathrm{sc}} = \hbar N\, \omega(f, g) \int_{\mathbb{R}} a(t) b(t) \gamma_{\mathbf{1}}(t)\, dt,$$

with associated graded formal class $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}; \mathbb{C})[[\hbar]]$. In the original scalar-reduced ordinary $\mathfrak{gl}_N$ model, the first scalar-lift obstruction is this class, detected already on the linear Hamiltonian window. Closed-exchange data can repair that branch only if its leading scalar class is $-\hbar N[\bar{c}]$. The minimal algebraic defect-supported central-contact cone with generator e_M, $\Xi_M(e_M) = -\hbar N \bar{c}_M$, $W_M = e_M$, and $C_M = 0$ kills the finite-window scalar classes only as a changed algebraic source branch; it is not yet a Costello-local closed-exchange realization. The genuine ordinary branch is obstructed first by $o^{(1),\mathrm{sc}}_{\sigma,M} = \hbar N[\bar{c}_M]$, and then by β^{sc}_M, $\beta^{\mathrm{sc}}_{\mathrm{cl}}$, $\mu^{\mathrm{sc}}_{\mathrm{cl}}$, and λ^{sc}_C for any supplied closed-exchange tower. The central-character, opposite central-extension, and balanced-supertrace mechanisms instead replace the source or open model before the reduced scalar class forms. In the opposite extension $J_{\mathrm{op}} = -k^{\vee}$ satisfies $d_{\mathrm{CE}} J_{\mathrm{op}} = -\hbar N \bar{c}$; after forgetting K_{cl}, the class $\hbar N[\bar{c}]$ returns on $\bar{A}$. Outside the minimal full-equivariant finite-window branch of Theorem E.42, the non-scalar lift still requires the curved kernel, product centrality homotopies, P_0-bracket homotopies, current-compatible Costello locality, and vanishing of the residual class in $H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$.

(xvii) *Global and cross-volume descent.* A target theorem supplies a matched-conventions datum $\mathfrak{C}_{\mathrm{tar}}$. The admissibility vector is

$$\mathrm{Ob}_{\mathfrak{C}_{\mathrm{tar}}} = (\mathrm{per}_{\mathrm{tar}}, \mathrm{ob}_{\mathrm{WR}}, \mathrm{ob}_{\mathrm{anom}}, \mathrm{ob}_{\mathrm{QME}}, \mathrm{ob}_{\mathrm{conv}}, \mathrm{ob}_{\mathrm{hyp}}, \mathrm{ob}_{6r}, \mathrm{ob}_{\mathrm{MD}}).$$

It must vanish, with compatible null-homotopies and comparison morphisms, before any CoHA, compact BCOV, quintic, OSV/GV, Abel–Jacobi, MNOP, Vol III, Igusa/BKM, or global Calabi–Yau consequence is deduced from the formal-local theorem. The compact-CY fixture split is a source-routing construction only:

it separates CoHA vocabulary from HKQ, CDGP, GV, OSV, Abel–Jacobi, conifold, Stokes, and arithmetic fixtures. It is not theorem support for a compact-CY transfer and supplies no evidence for the local QME, Ext, cyclic, Weyl/Moyal, radial-parts, or kernel theorem surfaces.

Local dictionary

Local Calabi–Yau/unimodularity datum. In the core local theorem surface, "Calabi–Yau" means only the following formal orientation/unimodularity datum. It is a definition, not a compact-target hypothesis.

Datum	**Definition and force**
Ω_{loc}	The fixed holomorphic volume and symplectic form on the formal disk $$\widehat{\mathbb{C}^2}_0 = \operatorname{Spf} \mathbb{C}[[z_1, z_2]], \qquad \Omega_{\mathrm{loc}} = dz_1 \wedge dz_2.$$ It trivializes the local canonical line and fixes the residue orientation. No compact manifold, no global moduli space, and no threefold Hodge theory are part of the datum.
Poisson tensor and bracket	The inverse bivector is $$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}, \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$ Polynomial calculations use $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$; completed CE/PV calculations use $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$. Constants are removed because they generate the zero Hamiltonian vector field.
Hamiltonian vector field	The convention is $$\iota_{X_f} \Omega_{\mathrm{loc}} = -df, \qquad X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}.$$ Thus $X_f(g) = \{f, g\}$ and $[X_f, X_g] = X_{\{f,g\}}$ after projecting constants.
Unimodularity / divergence-free condition	For a holomorphic vector field $X = a_1 \partial_{z_1} + a_2 \partial_{z_2}$, $$\mathcal{L}_X \Omega_{\mathrm{loc}} = (\operatorname{div}_{\Omega_{\mathrm{loc}}} X)\Omega_{\mathrm{loc}}, \qquad \operatorname{div}_{\Omega_{\mathrm{loc}}} X = \partial_{z_1} a_1 + \partial_{z_2} a_2.$$ Every Hamiltonian vector field is divergence-free: $$\operatorname{div}_{\Omega_{\mathrm{loc}}} X_f = -\partial_{z_1}\partial_{z_2} f + \partial_{z_2}\partial_{z_1} f = 0.$$ Conversely, on the formal disk the holomorphic Poincare lemma identifies $\mathbf{Ker}(\partial_{\Omega_{\mathrm{loc}}} \colon \mathrm{PV}^1 \to \mathrm{PV}^0)$ with Hamiltonians modulo constants. This is the unimodularity assertion used in the paper.
Closed BV pairing	For the Hamiltonian BF fields $$\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\mathfrak{h}[1] \oplus \mathfrak{h}^\vee_{\mathrm{cont}}[2]\big),$$ the BV pairing is the compactly supported local pairing obtained by wedging the form factors, integrating against Ω_{loc}, and applying the continuous evaluation $\mathfrak{h}^\vee_{\mathrm{cont}} \otimes \mathfrak{h} \to \mathbb{C}$. It has degree -1. This abstract pairing is not a Costello coefficient coevaluation kernel unless a Tate/weighted category makes the Casimir converge.

Datum	Definition and force
Residue / cyclic trace density	The same orientation fixes the principal-part residue pairing $$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2, \qquad \mathrm{Res}_0(\rho_{a,b} z_1^m z_2^n) = \delta_{a,m}\delta_{b,n},$$ with $a + b > 0$ in the reduced cotangent module. Hamiltonian vector fields preserve $dz_1 dz_2$, so residue integration by parts gives the coadjoint action. On the open brane, the top Ext class is the residue orientation dual to $dz_1 \wedge dz_2$, and the cyclic trace is $$\mathrm{Tr}_{\mathrm{open}}(\eta \otimes u \otimes M) = \int_{\mathbb{R}} \eta\, [\varepsilon_1\varepsilon_2] u\ \mathrm{Tr}_N(M).$$
Difference from compact CY_3 BCOV	BCOV supplies source conventions: Dolbeault polyvectors, divergence from a volume form, and cotangent BV language. It is not a theorem input for the local $2_{\mathbb{R}} + 2_{\mathbb{C}}$ Hamiltonian surface model. The local datum is only $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$ with its Hamiltonian Lie algebra, continuous cotangent module, BV pairing, and residue/cyclic trace.

Admissible and inadmissible uses.

Use	Status
Admissible	Use Ω_{loc} to define $\partial_{\Omega_{\mathrm{loc}}}$, the Poisson bracket, Hamiltonian vector fields modulo constants, the local BF action, the degree -1 BV pairing, residue principal parts, Ext orientation, cyclic traces, and integration-by-parts/coadjoint formulas.
Admissible with proof	Use a global holomorphic symplectic surface only after reducing to a formal Darboux neighbourhood and recording the period obstruction to global Hamiltonian primitives. Use a Costello graph/QME theorem only after supplying the elliptic parent, gauge fixing, convergent coefficient kernel, counterterm scheme, and brane-defect wavefront rules.
Inadmissible	Do not import compact Calabi–Yau threefold hypotheses, compact Hodge theory, compact-CY chiral homology, unproved one-dimensional vertex-algebra structure, period integrals, holomorphic anomaly equations, Gromov–Witten or Gopakumar–Vafa expansions, MNOP/PT/DT comparisons, quintic computations, OSV/GV, Abel–Jacobi or normal-function claims, CoHA structures, Igusa products, Borcherds lifts, BKM algebras, or Vol III comparison theorems into the core local SFT.
Inadmissible	Do not infer a compact BCOV theorem, Costello–Li anomaly cancellation, ordinary $\mathfrak{gl}_N$ QME cancellation, radial-parts theorem, finite-N trace normalization, or non-scalar closed-open factorization-centre morphism from the word "Calabi–Yau". Each such claim requires its own matched-conventions theorem and its own obstruction calculation.

Local chiral/factorization taxonomy. In this manuscript the word "chiral", when it is used informally, means the constructed local Hamiltonian chiral/factorization object below, or one of its mixed, brane, or current avatars. The construction is present; the missing extra theorem, if desired, is only the extraction of a one-complex-dimensional vertex/OPE algebra from it. The word never means a compact Calabi–Yau chiral

homology object, a CoHA, a quintic/GV/OSV structure, or a Borcherds–Kac–Moody denominator unless a separate matched-conventions theorem supplies that target.

Object	Definition and exclusions
Formal Hamiltonian holomorphic factorization algebra on $\mathbb{C}^2$	The constructed local chiral/factorization algebra is $$\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}},$$ the formal-disk Hamiltonian BF object on $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$. For a holomorphic polydisk B its dg Lie input is $$\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g}, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1], \quad \mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1,$$ with differential $\bar\partial$ plus the Hamiltonian BF differential. The formal stalk of observables is the completed CE/PV object $$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \simeq \mathrm{PV}_{\mathrm{coord}}(\mathcal{A}_{\partial,\mathrm{coord}})$$ only in the coordinate/admissible sense recorded below. This is a two-complex-dimensional holomorphic factorization algebra in the manuscript's local sense of "chiral". It is not yet the one-complex-dimensional vertex-algebra specialization.
Mixed topological-current enhancement over $\mathbb{R}^2$	For a topological open $U \subset \mathbb{R}^2_{\mathrm{top}}$ and holomorphic polydisk $B \subset \mathbb{C}^2_{\mathrm{hol}}$, the mixed current input is $$\mathfrak{g}^{\mathrm{mix}}_{U,B} = \Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]).$$ Its observables are locally constant in U and holomorphic in B. The brane-line current algebras below are the support-local one-real-dimensional restrictions of this mixed topological current object. They are not compact-CY chiral homology and not a VOA obtained from a complex curve.
Weyl/Moyal open-string brane algebra and cyclic avatars	The degree-zero quantum brane target is the Weyl/Moyal deformation of the formal symplectic disk. Put $$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$ Then $$f * g = m \circ \exp\left(\frac{\hbar}{2}P\right)(f \otimes g),$$ $$\{f, g\}_\hbar = \hbar^{-1}(f * g - g * f),$$ with constants quotiented in the Hamiltonian Lie algebra $\bar{A}_\hbar$. At finite N the open avatar is the matrix Weyl algebra generated by $(\phi_1)^i{}_j$, $(\phi_2)^k{}_l$ with $$[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar\, \delta^i_l \delta^k_j,$$ followed by the quantum moment-map reduction and the connected trace projection. Cyclic and supercyclic bar-cobar avatars are trace or supertrace word complexes for matrix or supermatrix branes; they are not CoHA objects, compact-CY chiral homology, or BKM denominator data.

Object	Definition and exclusions
Defect/principal-part current algebra	For an interval $I \subset \mathbb{R}_{\mathrm{brane}}$, $$\mathfrak{h}_I = \Omega_c^\bullet(I)\widehat{\otimes}\mathfrak{h}, \qquad \mathcal{P}_I = \mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P}, \qquad \mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1],$$ where $$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) \subset H^2_{\mathfrak{m}}(\mathbb{C}[[z_1,z_2]])\, dz_1 dz_2.$$ The reduced target is $$A^{\mathrm{pp}}_\partial(I) = \widehat{\mathbf{S}}(\Omega_c^0(I)\widehat{\otimes}\bar{A})\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P}[1]),$$ with brackets $$\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab),$$ $$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB),$$ $$\{\Theta_\rho(B), \Theta_\sigma(C)\} = 0.$$ This is a support-local real-current P_0 algebra with Matlis principal parts in two holomorphic variables. It is not the singular OPE algebra of a one-dimensional vertex algebra.
One-dimensional vertex/OPE restriction boundary	The constructed local chiral/factorization algebra above does not require the following data. They are necessary only for the stronger assertion that it has been restricted to a one-complex-dimensional vertex/OPE algebra. Such a restriction requires extra datum $$\mathfrak{R}_L = (\iota_L, \mathcal{B}_\perp, \mathsf{R}_{L,\mathcal{B}_\perp}, V_L, \mathbf{1}_L, T_L, Y_L, \zeta_{L,\hbar}),$$ where $\iota_L\colon L = \mathrm{Spf}\,\mathbb{C}[[w]] \hookrightarrow \widehat{\mathbb{C}^2}_0$ is a chosen holomorphic line or defect germ, $\mathcal{B}_\perp$ is a transverse boundary condition or normal vacuum, and $\mathsf{R}_{L,\mathcal{B}_\perp}$ is a restriction functor producing a one-complex-dimensional holomorphic factorization algebra $$\mathcal{F}_{L,\mathcal{B}_\perp}(D) = \mathsf{R}_{L,\mathcal{B}_\perp}(\mathcal{F}_{\mathbb{C}^2})(D)$$ for disks $D \subset L$. The functor must be built from a specified tubular-neighbourhood, shriek/star restriction, or defect-coupling construction, and must prove independence of the auxiliary transverse radius, functoriality for disjoint disks, and translation covariance along the coordinate w. The tuple also includes the restricted state space V_L, vacuum, translation, state-field map, and Zhu/zero-mode map. In the $\mathbb{C}\times\mathbb{R}$ interface used below, such a generic V_L datum is admissible only when it factors through $\mathcal{R}_{\mathbb{C}\times\mathbb{R}}$ and is identified with V_{red} in $\mathfrak{I}_{\mathrm{ch}}$. The present formal $\mathbb{C}^2$ Hamiltonian data construct $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$ and its mixed current avatars, but they do not choose L, $\mathcal{B}_\perp$, or $\mathsf{R}_{L,\mathcal{B}_\perp}$.

Object	Definition and exclusions
OPE data for a restricted curve object	After such a restriction one must construct $$V_L = H^\bullet(\mathcal{F}_{L,\mathcal{B}_\perp}(\Delta_0)), \qquad Y_L(a,w)b = \sum_{n\in\mathbb{Z}} (a_{(n)}b)\, w^{-n-1}.$$ The factorization product for two disks centered at w and 0 must have a finite-pole expansion $$\mu_{w,0}(a \boxtimes b) \sim \sum_{n\geq 0} (a_{(n)}b)\, w^{-n-1} + \mu^{\mathrm{reg}}_{w,0}(a,b),$$ with $a_{(n)}b = 0$ for $n \gg 0$, a vacuum $\mathbf{1}_L$, translation $[T_L, Y_L(a,w)] = \partial_w Y_L(a,w)$, and locality $$(w_1 - w_2)^N [Y_L(a,w_1), Y_L(b,w_2)] = 0$$ for $N \gg 0$. These Laurent modes and finite singular orders are not part of the two-complex-dimensional holomorphic factorization algebra or of the real interval current algebra.
Compatibility with the Weyl/Moyal brane algebra	A one-dimensional vertex/OPE restriction must also specify how the degree-zero brane algebra acts on the restricted object. Equivalently it must supply a continuous Zhu/zero-mode map $$\zeta_{L,\hbar}\colon (\bar{A}_\hbar, *) \longrightarrow \mathrm{Zhu}(V_L)$$ or fields $J_f(w)$ satisfying $$J_f(w)J_g(0) \sim \frac{\kappa_L(f,g)}{w^2} + \frac{J_{\{f,g\}_\hbar}(0)}{w},$$ with the central level κ_L specified, and on the cotangent current sector $$[J_{f,(0)}, \Theta_\rho] = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}.$$ Equivalently, $\zeta_{L,\hbar}(f * g) = \zeta_{L,\hbar}(f)\zeta_{L,\hbar}(g)$. The existing Weyl/Moyal calculation supplies $f * g$ and the reduced interval-current brackets only; it does not supply $\zeta_{L,\hbar}$, the fields $J_f(w)$, the central level, or the OPE singular expansion.

Object	Definition and exclusions
Constructed chiral algebra interface tuple	An externally constructed curve chiral algebra may enter this manuscript only as the reduced vertex object V_{red} in $$\mathfrak{I}_{\mathrm{ch}} = (\mathcal{R}_{\mathbb{C}\times\mathbb{R}}, \mathcal{F}_{\mathrm{red}}, V_{\mathrm{red}}, Q_{\mathrm{BRST}}, \mathbf{1}, T, Y, \mathrm{wt}, \zeta_\hbar, H_{\mathrm{anom}}).$$ Here $\mathcal{R}_{\mathbb{C}\times\mathbb{R}}$ is a controlled pushforward to $\mathbb{R}_t \times \mathbb{C}_{z_1}$ retaining the z_2-mode or the full two-index principal-part coefficient system $$\mathfrak{h}_{\mathrm{red}} = \mathbb{C}[[z_1]][[z_2]]/\mathbb{C}\cdot 1, \qquad \mathcal{P}_{\mathrm{red}} \subset H^2_{(z_1,z_2)}(\mathbb{C}[[z_1,z_2]])\, dz_1 dz_2.$$ $\mathcal{F}_{\mathrm{red}}$ denotes the resulting pushed-forward factorization algebra on $\mathbb{R}_t \times \mathbb{C}_{z_1}$; until the reduction obstruction vanishes it is a theorem target, not native $\mathbb{C}^2$ data. The interface must prove the pushed-forward bracket, BV pairing, brane image, factorization-to-vertex package, BRST nilpotence, Zhu/zero-mode map, Capelli normalization, Moyal multiplicativity, the completed stable Hochschild/HKR/CE-PV comparison, and anomaly transport. Its exact obstruction vector is $$\mathrm{Ob}_{\mathfrak{I}_{\mathrm{ch}}} = (\mathrm{Ob}_{\mathrm{red}}, \mathrm{Ob}_{\mathrm{vert}}, \mathrm{Ob}_{\mathrm{Zhu}}, \mathrm{Ob}_{\mathrm{nat}}).$$ Before this datum is supplied, a curve VOA or Zhu assertion is false transfer. After it is supplied, V_{red} is still a reduced shadow of $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$, not the native $\mathbb{C}^2$ holomorphic factorization algebra.

Normal Ω-background theorem surface. This block is the dictionary for the brane-preserving equivariant deformation. It is a theorem surface: it records the objects and the identities that must be constructed in the mixed de Rham/Dolbeault coefficient category. It is not a Costello graph/QME theorem and does not produce physical trace-state asymptotics.

Object or convention	Definition and force
Ambient pair and fixed stratum	The local pair is $$X = \mathbb{R}^2_{t,s} \times \mathbb{C}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}.$$ The normal bundle of the brane line is $$N_L X = \mathbb{R}_s \oplus \mathbb{C}_{z_1} \oplus \mathbb{C}_{z_2}.$$ The Ω-background used here is normal scaling of this bundle with t fixed. A literal rotation in the (t,s)-plane is not native to this brane problem because it does not leave L as the fixed line.

Object or convention	Definition and force
Equivariant parameters	Write $$T_\Omega = \mathbb{C}^*_{\epsilon_s} \times \mathbb{C}^*_{\epsilon_1} \times \mathbb{C}^*_{\epsilon_2}.$$ The parameters $\epsilon_s, \epsilon_1, \epsilon_2$ are equivariant weights. They are not the Ext generators $\varepsilon_1, \varepsilon_2$ used in the open trace orientation. The action is $$(s, z_1, z_2) \mapsto (\lambda_s s, \lambda_1 z_1, \lambda_2 z_2), \qquad t \mapsto t.$$ Its infinitesimal vector field is $$V_\Omega = \epsilon_s s \partial_s + \epsilon_1 z_1 \partial_{z_1} + \epsilon_2 z_2 \partial_{z_2}.$$
Equivariant differential	The Cartan differential on the local mixed complex is $$Q_\Omega = Q + \iota_{V_\Omega}, \qquad Q_\Omega^2 = L_{V_\Omega}.$$ Thus Q_Ω is an honest differential on the T_Ω-invariant subcomplex. Away from invariants the correct object is the T_Ω-equivariant curved/Cartan package with curvature L_{V_Ω}, not an ordinary dg complex with $Q_\Omega^2 = 0$.
Normal-weight localization ring	For a finite (N, M)-window let $\mathsf{X}_{N,M}$ be the finite set of nonzero moving characters that occur in that window. It contains the normal Hamiltonian and residue-dual characters $$n\epsilon_s + a\epsilon_1 + b\epsilon_2, \qquad n\epsilon_s - a\epsilon_1 - b\epsilon_2$$ whenever the corresponding moving summands occur. The localized coefficient ring is $$R_\Omega^{N,M} = \mathbb{C}[\epsilon_s, \epsilon_1, \epsilon_2, \hbar_\omega][\chi^{-1} \mid \chi \in \mathsf{X}_{N,M}].$$ Equivalently, one inverts only the nonzero weights present in the finite normal window. This is the algebraic habitat in which moving normal modes may be contracted. There is no default inversion of $\epsilon_s\epsilon_1\epsilon_2$ unless the Euler-localization branch has been declared. The self-dual branch is a specialization with its own allowed inversions; base-change first, then invert only the surviving nonzero characters, so $\epsilon_1 + \epsilon_2$ is not inverted on the self-dual branch.
Normal contraction datum	The stronger theorem surface is the construction of a localized deformation retract $$\mathsf{C}_\Omega = (\pi_L, j_L, h_\Omega) \colon \mathcal{E}_\Omega(X) \rightleftarrows \mathcal{E}_\Omega(L)$$ for the mixed de Rham/Dolbeault coefficient complex, with $$\pi_L j_L = \mathrm{id}, \qquad Q_\Omega h_\Omega + h_\Omega Q_\Omega = \mathrm{id} - j_L \pi_L$$ on the localized invariant normal complex. On a homogeneous normal summand of nonzero weight λ, h_Ω is the Euler/Koszul homotopy divided by λ, with signs fixed by the de Rham degree in the s-direction and the Dolbeault degree in the z_i-directions. The present dictionary records this contraction datum as required data; it does not assert that the Costello propagator and counterterms have already been transported through it.

Object or convention	Definition and force
Hamiltonian and cotangent weights	For $H_{a,b} = z_1^a z_2^b$, $a + b > 0$, $$w(H_{a,b}) = a\epsilon_1 + b\epsilon_2.$$ For the principal-part residue basis, $$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2, \qquad w(\rho_{a,b}) = -a\epsilon_1 - b\epsilon_2.$$ The holomorphic symplectic character is $$\chi_\omega = \epsilon_1 + \epsilon_2, \qquad w(dz_1 \wedge dz_2) = \chi_\omega, \qquad w(P) = -\chi_\omega.$$ The Weyl/Moyal deformation parameter, when present, has $$w(\hbar_{\mathrm{W}}) = \chi_\omega.$$
Three $\hbar$-parameters	The Omega convention distinguishes three formal parameters: $$\hbar_{\mathrm{QME}}, \qquad \hbar_{\mathrm{W}}, \qquad \hbar_\omega.$$ Here $\hbar_{\mathrm{QME}}$ is the Costello/BV loop-counting parameter in graph and curvature equations; $\hbar_{\mathrm{W}}$ is the Weyl/Moyal deformation parameter in the open brane algebra; and $\hbar_\omega$ is the equivariant line parameter of weight χ_ω used to scalarize the Poisson bracket. A Weyl branch may identify $\hbar_{\mathrm{W}} = \hbar_\omega$. No branch identifies $\hbar_{\mathrm{QME}}$ with either parameter unless the theorem states that convention explicitly.
Self-dual bracket	On the self-dual locus $$\epsilon_1 + \epsilon_2 = 0$$ the Poisson tensor has weight 0. The Hamiltonian bracket $\{f, g\}$ is then a scalar T_Ω-equivariant Lie bracket on $\mathfrak{h}$ after quotienting constants.
Refined bracket off the self-dual locus	Off the self-dual locus the ordinary Poisson bracket has weight $$w(\{f, g\}) = w(f) + w(g) - (\epsilon_1 + \epsilon_2).$$ It is therefore a bracket valued in the inverse symplectic-character line. The scalar refined bracket is obtained only after adjoining the line parameter $\hbar_\omega$ of weight $w(\hbar_\omega) = \epsilon_1 + \epsilon_2$: $$[f, g]_\Omega = \hbar_\omega \{f, g\}.$$ In the Weyl/Moyal specialization one may identify $\hbar_\omega$ with the deformation parameter $\hbar_{\mathrm{W}}$. Without this line or parameter, the refined Hamiltonian algebra is line-valued, not the old scalar Hamiltonian Lie algebra.

Object or convention	Definition and force
Equivariant CE/PV generator rule	The equivariant CE/PV theorem surface keeps the same generator labels $$c_{a,b} \mapsto \theta_{a,b}, \qquad u_{a,b} \mapsto O_{a,b},$$ but all differentials and brackets are read in the T_Ω-graded completed coordinate habitat. In the refined convention the contraction bracket and the d_π-terms carry the factor $\hbar_\omega$. In the self-dual convention this factor has weight 0 and may be suppressed.
Equivariant quantum brane algebra	The degree-zero open quantum brane target is the equivariant Hamiltonian reduction $$A^q_{N,\Omega} = \mathrm{Weyl}_{\hbar_{\mathrm{W}}}(\mathrm{Mat}^2_N)//_{\hbar_{\mathrm{W}}} GL_N, \qquad YX - XY + \hbar_{\mathrm{W}} NI = 0,$$ with $$w(X) = \epsilon_1, \qquad w(Y) = \epsilon_2, \qquad w(\hbar_{\mathrm{W}}) = \epsilon_1 + \epsilon_2.$$ The moment-map equation is weight homogeneous. Its scalar $U(1)$ contact term is the same Capelli line as in the non-equivariant dictionary, now graded by $\epsilon_1 + \epsilon_2$.
Residue normalization	The residue convention uses the formal residue orientation $$\mathrm{Res}_{z_1=z_2=0}\left(z_1^{-1} z_2^{-1} dz_1 dz_2\right) = 1$$ and the open top Ext coefficient $[\varepsilon_1 \varepsilon_2]$. Its normalization constant is $$\nu^{\mathrm{res}}_\Omega = 1.$$ In this convention the brane trace is normalized by the residue/cyclic density and no smooth normal Euler denominator is silently inserted. Any $\epsilon_s \epsilon_1 \epsilon_2$ factor must be written explicitly as a comparison constant.
Euler-localization normalization	The smooth equivariant localization convention pushes from X to L with the inverse normal Euler class $$e_{T_\Omega}(N_L X)^{-1} = (\epsilon_s \epsilon_1 \epsilon_2)^{-1},$$ up to the orientation sign σ_s fixed by the chosen real s-density and the complex orientation $dz_1 \wedge dz_2$. Its normalization constant is $$\nu^{\mathrm{Eu}}_\Omega = \sigma_s (\epsilon_s \epsilon_1 \epsilon_2)^{-1}.$$ A theorem comparing this convention with the residue convention must state the sign and the exact factor. Until then the two normalizations are separate branches.

Object or convention	Definition and force
Stratified factorization datum	The brane-localized object is a stratified mixed HT factorization datum on the pair (X, L), $$\mathcal{F}_{\Omega}^{\mathrm{str}}(X, L) = \big(\mathcal{F}_{\Omega, X\setminus L}, \mathcal{F}_{\Omega, L}, \mathcal{M}_{\Omega, L}, j^*, i^!, \mu_{\mathrm{str}}\big),$$ where the open stratum is the T_Ω-equivariant mixed bulk factorization algebra, the closed stratum is the brane-line current object, and $\mathcal{M}_{\Omega,L}$ is the defect module. Protected observables become stratified factorization-homology traces only after the brane-defect QME and the trace-state normalization are supplied.
Open obstruction vector	The residual theorem obligations are recorded as $$\begin{aligned}\mathrm{Ob}_\Omega = \big(&\mathrm{ob}_{\Omega,\mathrm{Cartan}}, \mathrm{ob}_{\Omega,\mathrm{contr}},\\ &\mathrm{ob}_{\Omega,\mathrm{CE/PV}}, \mathrm{ob}_{\Omega,\mathrm{norm}},\\ &\mathrm{ob}_{\Omega,\mathrm{strat}}, \mathrm{ob}_{\Omega,\mathrm{QME}},\\ &\mathrm{ob}_{\Omega,N\to\infty}\big).\end{aligned}$$ These name, respectively, the mixed Cartan model, the localized normal contraction, the equivariant CE/PV bracket comparison, the residue-versus-Euler normalization theorem, the stratified factorization structure, the Costello brane-defect graph/QME package, and the physical large-N trace-state theorem.

Canonical maps and obstruction symbols.

Symbol	Meaning
$\Phi_{\mathfrak{l}}$	Finite-dimensional cotangent CE/PV map $$C^\bullet_{\mathrm{CE}}(\mathfrak{l} \ltimes \mathfrak{l}^\vee[1]) \to \mathrm{PV}(\mathbf{S}(\mathfrak{l})), \qquad c \mapsto \theta, \quad u \mapsto O.$$
Φ_{coord}	Formal-disk coordinate CE/PV map reconstructed from admissible finite coordinate tests and pro-window coordinate cochains for $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$ and $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$. It is a completed coordinate dg commutative algebra isomorphism before any P_0 bracket is asserted.
$\Phi^B_{P_0}$	Restriction of Φ_{coord} to a bracket-admissible subalgebra $B \subset \mathrm{PV}_{\mathrm{coord}}(\mathcal{A}_{\partial,\mathrm{coord}})$: $$\Phi^B_{P_0} : \Phi^{-1}_{\mathrm{coord}}(B) \xrightarrow{\sim} B.$$ This is the completed P_0 comparison only because B carries a specified complete Hausdorff topology, continuous inclusion in the coordinate product, continuous multiplication, continuous d^{coord}_π, and a continuous closed contraction bracket. Boundary Hamiltonians inside this P_0 comparison require the relevant O_f and $d^{\mathrm{coord}}_\pi O_f$ to lie in B.

Symbol	Meaning
Θ_B	Kernel-admissible Maurer–Cartan tensor in a specified completed convolution dg Lie algebra $\mathcal{K}_B$ $$\Theta_B = \sum_I H_I \otimes \theta^I + \sum_I \eta^I \otimes O_I$$ attached to B. It is asserted only when the displayed diagonal sums converge and the convolution differential and bracket are continuous in $\mathcal{K}_B$.
η^I	Shifted cotangent generator $\eta^I = (H_I)^\vee[1] \in \mathfrak{h}^\vee_{\mathrm{cont}}[1]$, for $I = (a, b)$ and $a + b > 0$. It is the source of the O_I coordinate in the CE/PV map.
$\mathfrak{U}_\mathfrak{h}$	Admissible formal-local interface datum, not a categorical universal object unless a category of admissible completions and a universal property are supplied, $$\mathfrak{U}_\mathfrak{h} = (\Phi_{\mathrm{coord}}, \{\Phi^B_{P_0}\}_B, \{\Theta_B\}_B),$$ where B ranges over bracket-admissible subalgebras in the middle term and kernel-admissible data in the last term.
B_{fin}, E_q	Model admissibility tests. B_{fin} is the finite-support coordinate subalgebra on a finite window. E_q, for $q > 1$, is the weighted Köthe test target with seminorms $$p_n(x) = \sup_{a,b} \lvert x_{a,b}\rvert q^{a+b}(1 + a + b)^n.$$ It is bracket-admissible only after the finite-word tensor convention and the corresponding convolution estimate are fixed; it is kernel-admissible only if the diagonal tensor converges.
$\mathfrak{o}^{\mathrm{br}}, \mathfrak{o}^{\mathrm{coad}}$	Transition obstruction terms for bracket and coadjoint compatibility in the pro/mixed coordinate CE/PV comparison.
$\mathfrak{o}^{\mathrm{Cas}}$	Strict product/direct-sum kernel obstruction for the Casimir, propagator, and Costello endpoint problem. It is not part of the coordinate CE/PV identity.
$\mathfrak{o}_{\mathrm{BV,end}}$	Endpoint obstruction tuple $(\mathfrak{o}_{\mathrm{Cas}}, \mathfrak{o}_{\mathrm{def}}, \mathfrak{o}_{\mathrm{RG}}, \mathfrak{o}_{\mathrm{QME}})$ controlling a Costello-compatible coefficient endpoint.
$\mathrm{Ob}_{\mathfrak{C}_{\mathrm{tar}}}$	Matched-conventions descent vector $(\mathrm{per}_{\mathrm{tar}}, \mathrm{ob}_{\mathrm{WR}}, \mathrm{ob}_{\mathrm{anom}}, \mathrm{ob}_{\mathrm{QME}}, \mathrm{ob}_{\mathrm{conv}}, \mathrm{ob}_{\mathrm{hyp}}, \mathrm{ob}_{6r}, \mathrm{ob}_{\mathrm{MD}})$, with target-specific Vol III, Igusa/BKM, or sister-volume components appended when such a target is claimed.

Non-scalar Costello/QME dictionary. The entries in this block are the local obstruction habitat for Problem 3.271. A scalar-symbol map is associated-graded data. A scalar-contact projection is a filtered chain map. Reduced principal-part brackets live after de Rham-current contraction. A non-scalar Costello graph realization is the unreduced QME equation below.

Symbol or datum	Meaning
$\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$	Filtered brane-defect local-functional complex $$\varprojlim_{M}\Big(\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}^{M}_{w}\vert_{I};A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)),d_M\Big),\qquad d_M=Q+\{I^{w}_{\mathrm{o},M},-\}_{\hbar,M}.$$ The inverse limit is taken over finite Hamiltonian windows with the $\hbar$-adic, scalar-contact, and finite-window filtrations. The differential has degree +1. This is the target habitat for the bulk-to-defect kernel and QME obstruction; it is not the reduced current algebra $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$.
$F^{\bullet}\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$	Complete decreasing closed scalar-contact filtration preserved by d_{QME}. In the ordinary scalar-reduced branch one may take $F^r=J^r_{\mathrm{sc}}$, where J_{sc} is the closed ideal of functionals with zero associated-graded scalar-contact symbol. In the balanced branch it is the closed filtration generated by identity-endomorphism contractions inside the balanced scalar-contact habitat. Its role is to define the associated-graded scalar symbol and the obstruction tower for lifting that symbol.
σ_{sc}	Associated-graded scalar-symbol extractor $$\sigma_{\mathrm{sc}}\colon\ \mathrm{gr}_F\,\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)\longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]].$$ It has cochain degree 0, but it is not by itself a projection of the filtered QME complex. On a degree-one scalar contact with two Hamiltonian inputs it records the Lie two-cocycle coefficient of γ_1. In the ordinary branch this leading symbol is $\hbar N[\bar{c}]$.
$\widehat{\sigma}_{\mathrm{sc}}$	Actual scalar-contact projection: a continuous filtered degree-0 cochain map $$\widehat{\sigma}_{\mathrm{sc}}\colon\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)\to C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]]$$ whose associated graded is σ_{sc}. It exists exactly when the scalar-lift obstruction classes $o^{(r)}_{\sigma,w}(I)$ vanish successively with compatible finite-window choices. The residual non-scalar complex is defined only after this map has been chosen. On a balanced scalar-contact habitat the proved projection is the zero chain map $\widehat{\sigma}_{\mathrm{bal,sc}}=0$.
$o^{(r)}_{\sigma,w}(I)$	Successive scalar-contact lift obstruction. For a filtered linear lift s of σ_{sc}, the chain defect $\delta_s=d_{\mathrm{CE}}s-sd_{\mathrm{QME}}$ lowers $F^{\bullet}$. Its first nonzero part gives $$o^{(r)}_{\sigma,w}(I)\in H^1\Big(\mathrm{Hom}\big(\mathrm{gr}_F\,\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I),C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]]\big)_{-r}\Big).$$ In the ordinary scalar-reduced $\mathfrak{gl}_N$ branch the first two-input evaluation is $\hbar N\,\omega(f,g)\gamma_1$, nonzero on $(f,g)=(z_1,z_2)$.
$\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}$	Current source for the unreduced non-scalar lift, $$\big(\Omega^{\bullet}_{c}(I)\widehat{\otimes}\mathfrak{h}_{w,\hbar}\big)\ltimes\big(\mathcal{D}'_{c}(I)\widehat{\otimes}\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}\big)[1].$$ The topology is the completed current topology with the weighted finite-window and $\hbar$-adic completions. On CE coordinates, $c_{B,\rho}$ has degree 1 and $u_{a(t)dt\otimes f}$ has degree 0. Before scalar reduction one may replace the Hamiltonian source by the central-character source.

Symbol or datum	Meaning
$\kappa_{\hbar,w,I}$	Continuous bulk-to-defect kernel, equivalently a degree-0 Maurer–Cartan candidate in the completed convolution Lie algebra, $$\kappa_{\hbar,w,I} : C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}) \longrightarrow \mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I),$$ with generator values $$u_{a(t)dt\otimes f} \mapsto B^{\mathrm{BV}}_f(a), \qquad c_{B,\rho} \mapsto \Theta^{\mathrm{BV}}_\rho(B).$$ Its reduced image must recover $B_f(a)$ and $\Theta_\rho(B)$, but the kernel itself lives before scalar reduction, before de Rham-current contraction, and before replacing unreduced open observables by freely adjoined current generators.
$\mathrm{Ob}_{w,\partial,\hbar}$	Curvature cocycle of the bulk-to-defect kernel, $$\mathrm{Ob}_{w,\partial,\hbar} = d_{\mathrm{QME}}\kappa_{\hbar,w,I} + \frac{1}{2}[\kappa_{\hbar,w,I}, \kappa_{\hbar,w,I}]_\hbar.$$ It has QME degree 1. The scalar branch first requires $\widehat{\sigma}_{\mathrm{sc}}(\mathrm{Ob}_{w,\partial,\hbar}) = 0$. Only then does its residual class lie in the non-scalar complex.
$\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}$	Residual non-scalar obstruction after the scalar-contact projection has been constructed and the scalar component has been killed: $$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1\big(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}}\big).$$ This class is not defined from the scalar-symbol map alone and is not killed by the reduced principal-part bracket formulas.
$\mathcal{Q}^{\bullet}_{w,M}(I)$	Finite-window residual complex $$\mathcal{Q}^{\bullet}_{w,M}(I) = \ker(\widehat{\sigma}_{\mathrm{sc},M}) \subset \Big(\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}\big(\mathcal{E}^M_w|_I; A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)\big), d_M\Big).$$ The bracket, interaction, and differential are the transported finite-window structures supplied by the acceptance gate, not naive truncations of the unweighted Hamiltonian bracket.
$C_{n,M}$, $C^{w_0}_{\Gamma,w}(\epsilon)$	Finite-window counterterms. At order $\hbar^n$, $C_{n,M} \in \mathcal{Q}^0_{w,M}(I)$ must satisfy $$d_M C_{n,M} = -\mathfrak{v}^{\mathrm{ns}}_{n,M}, \qquad \pi_{M,N} C_{n,M} = C_{n,N}.$$ Graph representatives $C^{w_0}_{\Gamma,w}(\epsilon)$ must also commute with window truncation and weight transport. They are degree-zero local functionals; they are not arbitrary finite changes of the reduced current target.
$\mathfrak{O}^{\mathrm{ns}}_n$	Order-n finite-window obstruction vector $$\mathfrak{O}^{\mathrm{ns}}_n = \big(([\mathfrak{v}^{\mathrm{ns}}_{n,M}])_M, \lambda_n\big),$$ where the first component lies in $\varprojlim_M H^1(\mathcal{Q}^{\bullet}_{w,M}(I))$, and $\lambda_n \in \varprojlim^1_M H^0(\mathcal{Q}^{\bullet}_{w,M}(I))$ is the primitive-compatibility class. Its vanishing is exactly the finite-window criterion for compatible order-n counterterms.

Symbol or datum	Meaning
$\mathsf{E}_{\theta_3,M}, C_{\theta_3,M}$	First certified larger-row non-scalar obstruction and its permitted primitive. In the supplied one-row package, $$Q^0_{\theta_3,M} = 0, \qquad Q^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, \qquad d_M = 0,$$ and the residual is $\mathfrak{o}^{\mathrm{ns}}_{\theta_3,M} = \mathsf{E}_{\theta_3,M}$. The covector $\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$ proves that this class is nonzero in the supplied subcomplex. A valid $C_{\theta_3,M}$ is not a formal name: it must be one of three explicit witnesses, $$K^M_{\Theta_3}(\eta_{\theta_3,M}), \qquad C^{\mathrm{ct}}_{\theta_3,M}, \qquad \text{or a companion-face primitive,}$$ with differential $d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$, scalar-contact value zero, and vanishing Roos compatibility class under finite-window projection.
I_N, η, χ_N	Moment-ideal and scalar primitive data. The quantum Hamiltonian reduction uses the left moment-map ideal $I_N = \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$ and then removes the scalar empty trace. The scalar primitive lives on the unreduced Hamiltonian source $A = \mathbb{C}[z_1, z_2]$: $$\eta(f) = -[f]_0, \qquad d_{\mathrm{CE}}\eta = \omega, \qquad \chi_N(f) = N[f]_0 = -N\eta(f).$$ Thus $\hbar N\omega + d_{\mathrm{CE}}(\hbar\chi_N) = 0$ before scalar reduction. This primitive does not descend to $\bar{A} = A/\mathbb{C}\cdot 1$ and is not a counterterm in the original scalar-reduced QME complex.
$E_{a,b,N}, A^j{}_i(a,b,N)$	Quantum shear moment-ideal primitive datum in the matrix Weyl algebra. Put $$E_{a,b,N} = \sum_{j=0}^{b} \mathrm{Tr}(Y^j X^a Y^{b-j}) - J_N\Big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\Big).$$ The all-N shear congruence required for the radial/Weyl image is equivalent to $$E_{a,b,N} \in I_N \quad \Longleftrightarrow \quad E_{a,b,N} = \sum_{i,j} A^j{}_i(a,b,N)\widehat{\mu}^i{}_j$$ for coefficients $A^j{}_i(a,b,N) \in \mathcal{W}_N$. This is a degree-zero left-ideal primitive problem in the quantum moment-map ideal, not the scalar CE primitive η, not a scalar-contact projection, and not a Costello QME counterterm.
Reduced principal-part brackets	Target-side brackets in $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$ after scalar reduction and de Rham-current contraction: $$\{B_f(a), B_g(b)\}_\hbar = B_{\{f,g\}_\hbar}(ab), \quad \{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$ $$\{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0.$$ They use multiplication of a distribution by a smooth test function, never a product of two distributions. They prove the reduced current interface and do not supply Costello wavefront estimates, unreduced centrality homotopies, or the non-scalar QME.

Symbol or datum	Meaning
Unreduced non-scalar QME realization	The full graph theorem is the construction of $\kappa_{\hbar,w,I}$ and compatible counterterms $C_{\hbar,w}$. Put $K_{\hbar,w,I} = \kappa_{\hbar,w,I} + C_{\hbar,w}$. The equation is $$\mathrm{Curv}(K_{\hbar,w,I}) = d_{\mathrm{QME}} K_{\hbar,w,I} + \frac{1}{2}[K_{\hbar,w,I}, K_{\hbar,w,I}]_\hbar = 0.$$ It requires the Costello finite-scale BV propagator and Laplacian, locality and counterterm estimates, scalar-contact chain projection, vanishing of $\mathfrak{O}_n^{\mathrm{ns}}$ for every order, and unreduced product and P_0-bracket homotopies against arbitrary open observables.

Common local rings and quotients.

Symbol	Meaning
$R = \mathbb{C}[[z_1, z_2]]$	Complete local ring of the formal symplectic disk.
$\mathfrak{m} = (z_1, z_2)$	Maximal ideal of functions vanishing at the brane point.
$\mathfrak{h} = R/\mathbb{C} \cdot 1$	Reduced Hamiltonian Lie algebra of the formal disk. Constants are removed because they generate the zero Hamiltonian vector field.
$\mathfrak{g}_{\mathrm{str}}$	Strict shifted cotangent Hamiltonian Lie algebra $$\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$ Its Hamiltonian side is a product completion and its cotangent side is continuous dual/direct-sum in residue coordinates. This is the formal-local CE/PV object, but not by itself a Costello coefficient kernel.
$\mathfrak{g}_w$	Weighted finite-window Tate replacement used as a regulator. It carries the same coordinate-window CE/PV formulas after passage through the admissible tower, but its weight topology is auxiliary data and is not the strict cotangent completion.
$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$	Polynomial Hamiltonian sector used in explicit trace and Moyal calculations.
$\bar{A}_{\mathrm{desc}}$	Abstract polynomial label space for PBW one-ψ descendants. It is not the Matlis principal-part module.
$[\bar{c}]$	Distinguished scalar $U(1)$ anomaly class in $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$, realized by the constant term of the Poisson bracket and by the Capelli shift.
N	Rank of the brane probe stack. Degreewise stable trace statements are formed at fixed total word degree and then for N in the stable range.
D	Taylor truncation degree when a finite quotient $\mathfrak{m}/\mathfrak{m}^{D+1}$ is used. It is not the brane rank N.

Large-N and quotient conventions.

Symbol or operation	Meaning
$\mathrm{Prim}_{\mathrm{st}}$	Degreewise stable primitive invariant sector. In total word degree d, the Procesi–Razmyslov stable range is $N \geq d$.
$\mathrm{Conn}_{N\to\infty}$	Connected single-trace cumulant extraction after finite-N normalization and passage to the stable range. It removes disconnected products.
Hamiltonian scalar quotient	The closed scalar $1 \in \mathbb{C}[z_1, z_2]$ and the open empty trace $\mathrm{Tr}_N(1) = N$ are omitted as primitive Hamiltonian generators. The unit of the symmetric algebra remains.

Separation rule. The next three blocks are disjoint. PBW special-fibre labels $H_{a,b}$, $O_{a,b}$, $\Psi_{a,b}$, CE/PV labels c^I, u_I, θ^I, O_I, and Matlis labels $\rho_{a,b}$ live in different modules unless a specific theorem supplies a comparison. The Fourier–Rees theorem matches associated-graded labels after Fourier normalization through the Fourier delta Rees lattice $\mathcal{D}_\hbar/(z_1, z_2)$. Its Čech realization carries the factor $(-1)^{a+b}\hbar^{a+b}a!b!$, so before $\hbar$ is inverted it is the Rees sublattice, and as a module statement it uses a Fourier-twisted polarization rather than the original Hamiltonian coadjoint action.

Source–target status table.

Source	Target	Status	Forbidden identification
coordinate-window tower / mixed-continuous replacement attached to $\mathfrak{g}_{\mathrm{str}}$	$\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$	coordinate dg identity; bracketed comparison only after choosing a bracket-admissible B	not the full unreduced factorization centre without additional centrality and descent data
bracket-admissible $B \subset \mathrm{PV}_{\mathrm{coord}}$	P_0 realization $\Phi^B_{P_0}$	proved after completeness, d^{coord}_π-stability, product stability, and continuous contraction bracket are part of B's data	not the strict product/direct-sum completion
kernel-admissible B	Maurer–Cartan kernel Θ_B	accepted only after diagonal convergence in the chosen completed tensor product is proved	not a strict Casimir kernel for the ambient product completion
$C^\bullet_{\mathrm{CE}}(\mathfrak{g})$ / $\mathrm{PV}(A)$	reduced P_0 central-operation model	local Poisson cochain model	not the ordinary centre and not the unreduced $Z^{P_0}_{\mathrm{fact}}$ without centrality homotopies
$U_\hbar(\mathfrak{g})/(\hbar)$ PBW special fibre	$\mathbf{S}(\mathfrak{h} \oplus \mathfrak{h}^\vee[1])$ and then selected $\mathbf{S}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1])$ labels	associated-graded label comparison	not a same-action identification of $\Psi_{a,b}$ with $\rho_{a,b}$

Source	Target	Status	Forbidden identification
admissible Tate / relative bar-cobar envelope	$\widehat{U}(\mathfrak{g}_{\mathrm{adm}})$	accepted only after conilpotent CE chains, exact continuous duality, PBW completeness, and strong convergence are supplied	not available for the raw full formal disk without admissibility data
c^I	θ^I	proved generator rule	not a boundary Hamiltonian coordinate
u_I	O_I and brane-line Hamiltonian coordinates $B_f(a)$	proved generator rule	not a Hamiltonian CE ghost
$C^\bullet_{\mathrm{CE}}(\mathfrak{g}_I)$ on intervals	reduced brane-line factorization target	cofactorization/ opposite-interval source; factorization only after the CE coproduct is paired with target multiplication	not a same-variance morphism of raw factorization algebras
$\Psi_{a,b}$	stable primitive one-ψ PBW special-fibre class	proved in the stable trace complex	not $\mathfrak{h}^\vee_{\mathrm{cont}}$, not $\mathcal{P}$, and not a principal-part current
$\rho_{a,b}$	$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$	proved by residue/Matlis comparison	not a polynomial descendant and not an R-submodule quotient
$\mathsf{F}_{\mathrm{Rees}}$	$\bigoplus \hbar^{a+b}\mathbb{C}[[\hbar]]\rho_{a,b}$	proved after Fourier polarization	not a same-action bridge $\Psi_{a,b} \cong \rho_{a,b}$
$\Omega^\bullet_c(I)\otimes\mathfrak{h}$	brane-line Hamiltonian currents	proved reduced current model	$C^\infty_c(I)$ denotes only the degree-zero summand
$\mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1]$	principal-part current sector $\Theta_\rho(B)$	proved reduced post-contraction	does not require a product of two distributions
formal Moyal algebra $\tilde{A}_\hbar$	degree-zero Weyl coefficient target	proved algebraically in the reduced setting with raw weights 1 and 1/24	not a finite-N radial-parts theorem by itself

Source	Target	Status	Forbidden identification
finite-N radial-parts target	stable connected trace realization of the Moyal coefficients	finite-certificate-relative theorem datum: the formal coefficients, Capelli normalization, free normal-diagram obstruction $\mathfrak{o}^{\mathrm{rad}}_{a,b}$, proved edge PBW strip, and the remaining uniform splitting or left-cokernel theorem	not supplied by formal Moyal coefficients or by the cited radial-parts sources alone; not a Costello analytic graph/QME realization
Costello brane-defect kernel	closed-open graph/QME specialization	open obstruction datum	not inferred from the formal Moyal or reduced open-line computation
scalar cocycle $[\bar{c}]$	$U(1)$ anomaly / Capelli contact line	proved distinguished class	not an exhaustive classification of unreduced QME cancellation
External global bulk-boundary target	compact BCOV / Calabi–Yau / Vol III / Igusa–BKM consequence	open matched-conventions descent problem	not a consequence of the local Koszul Layers 0–6

PBW special-fibre open trace labels.

Symbol	Meaning
$H_{a,b} = z_1^a z_2^b$	Polynomial Hamiltonian label in $\bar{A}$, with $a + b > 0$. In the PBW special fibre it labels the scalar-reduced trace observable $O_{a,b}$.
$O_{a,b}$	Open ghost-zero trace class $\overline{\mathrm{Tr}(\phi_1^a \phi_2^b)}$, with $a + b > 0$ in the reduced Hamiltonian label set.
$\Psi_{a,b}$	Primitive one-ψ stable open class, represented by the symmetrised trace with one ψ, a copies of ϕ_1, and b copies of ϕ_2, with $a + b > 0$ after removing the scalar descendant line. It lies in the PBW special-fibre open shadow and carries the corresponding polynomial action. In the reduced comparison it supplies the primitive indecomposable P_0-shadow label.
$\mathsf{F}_{\mathrm{Rees}}$	Fourier–Rees label bridge of Theorem A.10. It sends the polynomial label $z_1^a z_2^b$ to the Fourier-delta label $\partial_1^a \partial_2^b e_0$, whose Čech realization is $(-1)^{a+b}\hbar^{a+b} a! b! \rho_{a,b}$. After inverting $\hbar$ one may divide by this Rees factor and read the principal-part label $\rho_{a,b}$. It changes polarization and is not an $\mathfrak{h}$-module identification between $\Psi_{a,b}$ and $\rho_{a,b}$.

The shared indices on $H_{a,b}$, $\Psi_{a,b}$, and $\rho_{a,b}$ are labels, not an $\mathfrak{h}$-module identification. The span of the $\Psi_{a,b}$ is not $\mathfrak{h}^{\vee}_{\mathrm{cont}}$, not $\mathcal{P}$, and not a principal-part current module.

Closed CE/PV labels.

Symbol	Meaning
$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$	Shifted cotangent Lie algebra of Hamiltonian canonical transformations.
H_I	Hamiltonian generator in $\mathfrak{h}$.
η^I	Shifted cotangent generator $(H_I)^\vee[1]$, with $I = (a, b)$ and $a + b > 0$.
c^I	Chevalley–Eilenberg coordinate dual to a Hamiltonian generator $H_I \in \mathfrak{h}$. Under Φ_{coord}, $c^I \mapsto \theta^I$.
u_I	Chevalley–Eilenberg coordinate dual to the shifted cotangent generator. Under Φ_{coord}, $u_I \mapsto O_I$. Boundary Hamiltonian observables come from u_I, not from c^I.
θ^I	Polyvector coordinate corresponding to c^I under the coordinate CE/PV isomorphism.
O_I	Hamiltonian observable coordinate corresponding to u_I under the coordinate CE/PV isomorphism.
Φ_{coord}	Coordinate dg CE/PV map determined on generators by $c^I \mapsto \theta^I$ and $u_I \mapsto O_I$. Its P_0 restriction is $\Phi^B_{P_0}$ only after a bracket-admissible B is chosen.

Matlis principal-part labels.

Symbol	Meaning
$\mathcal{P}$	Reduced principal-part module $\mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \subset H^2_{\mathfrak{m}}(R)\, dz_1 dz_2$, identified with $\mathfrak{h}^\vee_{\mathrm{cont}}$.
$\rho_{a,b}$	Principal-part residue current $z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$, $a + b > 0$, with the Matlis coadjoint action.
$z_1^p z_2^q \cdot \rho_{a,b}$	Coadjoint action $-(pb - qa + p - q)\rho_{a-p+1,b-q+1}$. This is the Hamiltonian coadjoint principal-part action, not the Matlis R-module action and not the PBW action on $\Psi_{a,b}$.
$\mathcal{P}_I$	$\mathcal{P}_I = \mathcal{D}'_c(I) \widehat{\otimes} \mathcal{P}$, the compactly supported distributional principal-part currents on an interval I.

Thus $\Psi_{a,b}$ and $\rho_{a,b}$ live in different modules:

$$\Psi_{a,b} \in \text{PBW special-fibre open homology}, \qquad \rho_{a,b} \in \mathcal{P} \cong \mathfrak{h}^\vee_{\mathrm{cont}}.$$

No ordinary $\mathfrak{h}$-module isomorphism between these two spans is used or asserted.

Interval-current and boundary labels.

Symbol	Meaning
$\mathfrak{h}_I$	$\mathfrak{h}_I = \Omega_c^\bullet(I)\widehat{\otimes}\mathfrak{h}$, the compactly supported de Rham Hamiltonian currents on an interval $I \subset \mathbb{R}$. Degree-zero test functions appear only after taking the explicit measurement notation.
$\mathfrak{g}_I$	$\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$, the shifted cotangent current algebra on I, with principal-part currents in the shifted cotangent direction.
$B_f(a)$	Smeared boundary Hamiltonian observable $\int_I a(t)\overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt$. Its CE source is $u_{a(t)dt\otimes f}$.
$A_\partial^{\mathrm{Ham}}(I)$	Reduced Hamiltonian brane-line P_0 factorization algebra $\widehat{\mathbf{S}}(\mathfrak{h}_I)$.
$A_\partial^{\mathrm{pp}}(I)$	Reduced principal-part defect-current model after de Rham-current contraction $\widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{P}_I[1])$. An unreduced BV observable algebra maps to it only after the centrality homotopies, coefficient kernel, brane-defect propagator, descent data, and BV/QME solution are supplied.
$\Theta_\rho(B)$	Principal-part current observable with $B \in \mathcal{D}_c'(I)$ and $\rho \in \mathcal{P}$. It is the boundary notation for $B \otimes \rho[1]$ in the reduced model and pairs with smooth test functions by $B(a)\,\mathrm{Res}(\rho f)$.
aB	Multiplication of a compactly supported distribution B by a smooth test function $a \in \Omega_c^0(I)$. The reduced brackets use this operation and never require multiplying two distributions.

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1 Three principles and the admissibility discipline

After the symplectic primitive, scalar quotient, and residue normalization are fixed, the formal-local CE/PV theory and its admissible Koszul-duality enhancement are organized by three source principles. Each is physical in origin; each enters as algebraic theorem data. Together they specify the open brane complex, the closed Hamiltonian BF complex, the closed-open CE/PV map, the Tate coefficient envelope, the current lift datum, and the obstruction complexes controlling global use. A separate admissibility discipline governs repairs: a failed strong sentence is not replaced by a weak true slogan, but by the strongest true theorem obtained after adding the missing object, topology, kernel, homotopy, counterterm, finite-window row, or obstruction class.

1.1 *Dirac: a brane stack is a constrained probe*

Fix the formal holomorphic symplectic disk

$$\widehat{\mathbb{C}^2}_0 = \operatorname{Spf} \mathbb{C}[[z_1, z_2]], \qquad \omega = dz_1 \wedge dz_2.$$

A stack of N branes supported on the topological line probes this disk by replacing the normal coordinates by matrices

$$z_1 \rightsquigarrow \phi_1, \qquad z_2 \rightsquigarrow \phi_2, \qquad \phi_i \in \mathfrak{gl}_N.$$

The pulled-back symplectic form is

$$\Omega_N = \operatorname{Tr}(d\phi_1 \wedge d\phi_2),$$

with primitive $\operatorname{Tr}(\phi_1\, d\phi_2)$. Change of basis in the probe stack is gauge. Its moment map is

$$\mu(\epsilon) = -\operatorname{Tr}(\epsilon[\phi_1, \phi_2]), \qquad \epsilon \in \mathfrak{gl}_N.$$

Dirac reduction therefore imposes the first-class constraint

$$[\phi_1, \phi_2] = 0.$$

After the primitive $\operatorname{Tr}(\phi_1\, d\phi_2)$ and moment map convention have been fixed, the corresponding first-order world-line action is

$$S_\partial = \int_{\mathbb{R}} \operatorname{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

In BV form the constraint is the Koszul differential

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0,$$

where $\psi = A^*$. Thus the open algebra is not postulated as a trace algebra. It is the algebra of gauge-invariant functions on the derived commuting-pair locus. Lemma 3.6 and Proposition 3.7 give the formal proof.

1.2 *Witten convention: closed fields target central operations*

Witten's open string field theory supplies the cyclic open-field-theory source convention. A closed insertion approaching a brane gives an operation on compatible boundary observables. In a topological direction the algebraic target for such an operation is centrality up to derived Poisson homotopies. The formal-local target is therefore

$$Z^{P_0}_{\mathrm{red,Ham}}(\mathrm{Obs}_{\mathrm{open}}),$$

not $\mathrm{Obs}_{\mathrm{open}}$ itself. The reduced Hamiltonian sector supplies the cochain CE/PV theorem; the unreduced factorization-current theorem is formulated by adding the boundary representatives, centrality homotopies, coefficient kernel, and QME solution demanded by that target.

For the formal symplectic disk the infinitesimal closed symmetries are Hamiltonian vector fields. Constants generate the zero vector field, so the Hamiltonian Lie algebra is

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1, \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

In the local model the closed BV algebra of canonical transformations is the shifted cotangent theory

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$

The finite-window and coordinate-completed CE/PV theorem is the cochain identity

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \cong \mathrm{PV}_{\mathrm{coord}}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big),$$

with generator dictionary

$$c^I \mapsto \theta^I, \qquad u_I \mapsto O_I.$$

The $Z^{P_0}_{\mathrm{red,Ham}}$ interpretation is made only after imposing the finite-window or mixed/weighted continuity condition that makes the bracket continuous; it is not a statement about the full ambient product completion. The boundary Hamiltonian observable is the image of the shifted cotangent coordinate u_I. It is not the image of the Hamiltonian ghost coordinate c^I. This distinction is the type check behind the factorization-current theorem. At the unreduced BV level the same type check defines the theorem datum: boundary representatives for u_I, product centrality homotopies, P_0-bracket homotopies, coefficient kernel, and mixed brane-defect QME solution.

The polynomial category is eliminated by a local-nilpotence criterion. In the polynomial unreduced BV/Koszul category the linear Hamiltonians act by constant translations of ϕ_2 and ϕ_1, hence locally nilpotently on every finite polynomial observable and on its Q-cohomology. The principal-part cotangent module has the opposite behavior:

$$z_1^r \cdot \rho_{a,b} = (-1)^r (b+1)\cdots(b+r)\rho_{a,b+r} \neq 0, \qquad r \geq 1,$$

and similarly for z_2. Thus a same-action unreduced lift cannot be hidden inside polynomial one-ψ representatives. The boundary target is therefore the distributional principal-part current sector, with its centrality homotopies stated as theorem data.

The QME obstruction has an independent scalar component. With the Capelli/Weyl convention used below,

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\ \mathrm{Tr}(A) \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

After inserting the central boundary ghost γ_1, $|\gamma_1| = 1$, the scalar contact contributes

$$\mathrm{Ob}_{\mathrm{sc}} = \hbar N\, \omega(f, g) \int_{\mathbb{R}} a(t)b(t)\gamma_1(t)\, dt.$$

Its associated graded class is $\hbar N[\bar{c}]$. The reduced Hamiltonian source has no constant Hamiltonian line on which the primitive $\eta(f) = -[f]_0$ could live. A fixed scalar-reduced ordinary $\mathfrak{gl}_N$ lift therefore requires a closed-exchange contribution W_{ext} whose leading class satisfies

$$[w_{\mathrm{ext}}] = -\hbar N[\bar{c}] \qquad \text{in } H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}).$$

Alternatively, the unreduced central-character source absorbs the scalar symbol before scalar reduction, while a Casimir-balanced supertrace model makes the scalar representative zero. The obstruction complex controls any further mechanism by the same degree-one class.

1.3 Grothendieck: cotangent modes are residue currents

The continuous dual of functions on a formal disk is not another copy of polynomial functions. It is the local-cohomology module of residue currents. For

$$R = \mathbb{C}[[z_1, z_2]], \qquad \mathfrak{m} = (z_1, z_2),$$

the Matlis dualizing module is

$$H^2_{\mathfrak{m}}(R)\, dz_1 dz_2 = \bigoplus_{a,b\geq 0} \mathbb{C}\, z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2.$$

After removing the scalar Hamiltonian, the cotangent module is

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) = \bigoplus_{a+b>0} \mathbb{C}\, \rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2.$$

The residue pairing

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \mathrm{Res}_0\, z_1^{m-a-1} z_2^{n-b-1}\, dz_1 dz_2 = \delta_{a,m}\delta_{b,n}$$

identifies $\mathfrak{h}^\vee_{\mathrm{cont}}$ with $\mathcal{P}$. The coadjoint action is the principal-part formula

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}.$$

This is the cotangent sector used by the closed theory. The one-ψ open classes $\Psi_{a,b}$ carry the same labels but a different $\mathfrak{h}$-module structure. They form a PBW special-fibre shadow; they do not realize the principal-part module. The no-go theorem 3.234 records the formal obstruction.

1.4 Supremum discipline: repair by construction

Every frontier statement in this manuscript is read in its supremum form. Suppose a proposed theorem T fails in the currently displayed habitat. The admissible repair is not the largest harmless weakening of T. It is the smallest enlargement of the mathematical habitat in which the intended mechanism is a theorem, or else the exact obstruction proving that no such enlargement of the prescribed type can work. Thus a failed theorem surface is not a stable status. It must be reconstituted until it is either a proved theorem with all missing data supplied, or a proved obstruction theorem naming the cohomology class, cokernel functional, Roos class, finite-window matrix row, graph incidence table, or source fixture that prevents the original statement.

Concretely, a failed bracket statement demands the missing continuous bracket habitat, scalar quotient, line-valued parameter, or weight character; a failed kernel statement demands the missing convergence topology, diagonal tensor, Bochner–Martinelli homotopy, finite-window transport, or source-face

table; a failed QME statement demands the filtered local-functional complex, scalar-contact projection, bulk-to-defect kernel, counterterm primitive, centrality homotopies, and the non-scalar obstruction class. If any of these data are unavailable, the manuscript records the precise class or construction problem rather than demoting the theorem into analogy.

Thus the current non-scalar QME surface is not the vague assertion that "the QME is open". It is the finite-window criterion in $\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$: the scalar-contact lift must exist, the reduced class in

$$H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$$

must vanish, and each row must pass the matrix equation

$$A^M c^M = -r^M$$

with Roos-compatible primitives. The first exposed larger row is θ_3. In the supplied one-row complex

$$Q^0_{\theta_3,M} = 0, \qquad Q^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M},$$

the covector $\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$ is an exact obstruction. Supremum repair therefore asks for a CE ancestor, a scalar-zero Costello counterterm primitive, or a complete companion-face table whose differential or signed CE boundary kills $\mathsf{E}_{\theta_3,M}$. Merely naming C_{θ_3} is not mathematics.

The same discipline governs the radial/Weyl input. The finite normal-form certificates are not discarded because they stop before the all-bidegree theorem, and they are not inflated into that theorem either. They define the stronger true target: construct a functorial splitting of the free normal-diagram correction maps

$$C_{a,b}\colon\ \ker \partial \longrightarrow \mathsf{Diag}_{a,b},$$

or prove a left-cokernel functional detecting a genuine obstruction. Likewise the native chiral object is not weakened to a curve vertex algebra; it remains the $\mathbb{C}^2$ holomorphic factorization algebra, with curve/Zhu structures admitted only after the controlled $\mathbb{C} \times \mathbb{R}$ reduction retaining the z_2-mode or principal-part system.

1.5 *Deduction of the local theorem*

The three principles have disjoint jobs in the formal-local theorem.

(i) Dirac determines the open BV complex and the stable trace observables.

(ii) Witten's source convention motivates the closed-open target as a derived P_0-centre; the local CE/PV theorem supplies the proof in the reduced Hamiltonian sector.

(iii) Grothendieck determines the continuous cotangent coefficients and separates principal parts from polynomial descendants.

The resulting theorem decomposes into independent component theorems: the Dirac probe theorem; the stable trace and one-ψ calculation; the CE/PV central-operation theorem; the Matlis principal-part theorem; the brane-line factorization-current theorem; the reduced principal-part defect-current theorem; the degree-zero Moyal/Weyl quantum theorem; and the scalar $U(1)$ anomaly theorem. The frontier components are theorem data: the admissible and endpoint Tate envelope, the unreduced cotangent factorization-centre morphism, the one-antifield quantum lift, the Costello brane-defect QME realization, and matched-conventions global descent. Each is governed by the obstruction complex displayed in Section 4.

2 Introduction

2.1 *Local measurements, constraints, and brackets*

The object of study is a formal local constrained mechanical system. Let

$$X = \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}} = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}.$$

The coordinate t is the brane direction and s is the topological normal direction. The holomorphic normal data is the formal Darboux disk

$$(\widehat{\mathbb{C}^2}_0, \omega), \qquad \omega = dz_1 \wedge dz_2.$$

If this disk is obtained from an ambient holomorphic symplectic surface, the formal holomorphic Darboux theorem is the only input: the paper then forgets the ambient surface and works on the completed normal disk. The defect is a *point in* $\mathbb{C}^2$, *dragged along* $\mathbb{R}_t$; neither compact Calabi–Yau geometry nor a curve vertex algebra is part of the local datum.

The two holomorphic normal coordinates z_1, z_2 describe the position of the defect inside the symplectic disk. A stack of N identical probes at the defect represents each coordinate by an $N \times N$ matrix:

$$z_1 \rightsquigarrow \phi_1, \qquad z_2 \rightsquigarrow \phi_2, \qquad \phi_i \in \mathfrak{gl}_N.$$

The symplectic form pulls back to the matrix-valued two-form $\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2)$, and the symplectic potential along the world-line is

$$\theta_N = \mathrm{Tr}(\phi_1\, d\phi_2).$$

Acting on a probe stack by an internal change of basis is a gauge operation. The gauge generator is the $\mathfrak{gl}_N$-valued moment map

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big), \qquad \epsilon \in \mathfrak{gl}_N,$$

and demanding gauge-invariance of the world-line dynamics imposes the *first-class constraint*

$$[\phi_1, \phi_2] = 0.$$

The Lagrange multiplier for this constraint is a connection $A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N$. The boundary action is the standard first-order Dirac action attached to the fixed potential θ_N and enforcing the first-class constraint:

$$S_\partial = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big). \tag{1}$$

When this action is extracted from the shifted open Chern–Simons coefficient fields below, we first make the explicit linear Dirac-normalization of those coefficient fields. The sign convention is therefore part of the local model, not an additional theorem. This finite ghost-zero open model is the constrained mechanics of an N-probe stack on the formal symplectic normal disk, with the matrix moment-map equation as its first-class constraint. Lemma 3.6 records the precise constrained-mechanics statement and identifies A as the multiplier.

The BV reduction is algebraic: the constraint becomes a Koszul differential generated by a single odd matrix antifield $\psi = A^* \in \mathfrak{gl}_N[1]$ with

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0.$$

The reduced local polynomial algebra at a brane point is

$$\mathcal{R}_N = \mathbb{C}[\phi_1{}^i{}_j, \phi_2{}^i{}_j] \otimes \Lambda(\psi^i{}_j),$$

and $H^0(\mathcal{R}_N^{GL_N}, Q)$ is, by definition, the algebra of GL_N-invariant local observables on the Dirac-reduced phase space of commuting matrix pairs. This algebra is the open trace algebra of the defect. The BV lift used here is the algebraic ghost-zero lift of (1); this lift is recorded in Lemma 3.5 and Proposition 3.7.

2.2 *The Hamiltonian Lie algebra of the symplectic disk*

The corresponding closed system is dictated by the same canonical mechanics. A closed infinitesimal symmetry of the formal symplectic disk is a Hamiltonian vector field $X_f = \{f, -\}$. Constants act trivially: $X_c = 0$ for $c \in \mathbb{C}$. The closed Hamiltonian Lie algebra is therefore the formal power series modulo constants,

$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1, \qquad \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1,$$

with bracket

$$\{f, g\} = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g \pmod{\mathbb{C}}.$$

Polynomial calculations use $\bar{A}$; closed Chevalley–Eilenberg/Koszul calculations use the completed $\widehat{\mathfrak{h}}$. Continuous duality on $\widehat{\mathfrak{h}}$ produces $\mathfrak{h}^\vee = \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$, with the Taylor-monomial basis $\delta_{k,l}$ for $k + l > 0$. We use $\bar{A}_{\mathrm{desc}}$ for an abstract copy of $\bar{A}$ whose nonconstant basis vectors are represented on the brane by the first descendant cycles $\Psi_{k,l}$ (Construction 3.195).

Remark 2.1 *(Scalar Hamiltonian and center-of-mass term).* The quotient by constants is load-bearing. In the reduced Hamiltonian algebra of the formal disk,

$$\{z_1, z_2\} = 1 \equiv 0 \quad \text{in } \bar{A}.$$

On the unreduced brane phase space, however,

$$\{\mathrm{Tr}(\phi_1), \mathrm{Tr}(\phi_2)\} = N.$$

This is the center-of-mass symplectic pair of the stack. Every bracket statement below involving linear Hamiltonians is therefore a statement after projection to the Hamiltonian quotient by scalar trace classes. The same term reappears quantum mechanically as the Capelli shift in Remark 3.247.

BCOV [1] is the source-convention template for the closed B-model/Kodaira-Spencer field: Dolbeault polyvectors, the divergence operator, and a cotangent BV formalism on Calabi–Yau threefolds. That compact threefold theory is not used as a transfer theorem for the present formal-local $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ surface model. Here the local Hamiltonian sector is defined and later checked directly: after fixing the shifted BV cotangent convention, the closed target is the cotangent theory of canonical transformations of the formal holomorphic-symplectic disk, namely Hamiltonian BF theory for the Tate Lie algebra

$$\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}^\vee[1], \tag{2}$$

where the shift [1] records the BV degree of the cotangent field and the Lie bracket on $\mathfrak{g}$ is the semi-direct sum of the Hamiltonian bracket on $\widehat{\mathfrak{h}}$ with the coadjoint action on $\mathfrak{h}^\vee[1]$.

The closed local observables are the continuous Chevalley–Eilenberg cochains

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{grav}} = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big).$$

There are two universal statements here, and they are kept separate. First, the cotangent CE/PV identity is a cochain-level derived-centre statement for the formal disk itself. For finite cotangent Lie algebras the generator map is

$$c^I \longmapsto \theta^I, \qquad u_I \longmapsto O_I,$$

where c^I is dual to a Hamiltonian generator, u_I is dual to the shifted cotangent generator, θ^I is the Schouten vector-field coordinate, and O_I is the boundary Hamiltonian function. The completed formal disk is obtained by compatible finite-coordinate tests. Second, the promotion from cochains to an enveloping/Koszul-dual algebra is a bar-cobar statement. It is not asserted for the strict formal Hamiltonian algebra without extra hypotheses. Whenever the coefficient category supplies an admissible Tate, nilpotent, relative, or weighted replacement $\mathfrak{g}_{\mathrm{adm}}$, the Koszul dual is

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}})^{!} \simeq \widehat{U(\mathfrak{g}_{\mathrm{adm}})}.$$

For the strict object $\mathfrak{g}$, failure of those hypotheses is the universal obstruction datum recorded below: Casimir convergence in the chosen Tate pair, the low-degree linear translation term, the $H_2 \cong \mathfrak{sl}_2$ non-pronilpotent sector, the associated-graded cone of the filtered cobar map, and the possible Milnor $\varprojlim^1$ defect. The PBW Rees algebra $U_\hbar(\mathfrak{g})$ has completed special fibre

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}\big(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^\vee[1]\big). \tag{3}$$

This special fibre is only a PBW shadow. The open one-ψ descendants $\Psi_{a,b}$ live in this associated-graded label system. The deformation-level cotangent module is instead the continuous dual $P = \mathfrak{h}^\vee_{\mathrm{cont}}$, realized later by Matlis principal parts $\rho_{a,b}$. No theorem below identifies $\Psi_{a,b}$ with $\rho_{a,b}$ as $\mathfrak{h}$-modules or as quantum boundary observables. Equations (2)–(3) say what the closed sector is: *the shifted cotangent theory of canonical transformations of the disk*. Proposition 3.13 below derives $\widehat{\mathfrak{h}}$ from divergence-closed polyvectors on the formal disk.

2.3 Notation index for the local theorem

The local theorem uses four alphabets. They are kept separate throughout.

Symbol	Meaning
$R = \mathbb{C}[[z_1, z_2]]$	Complete local ring of the formal symplectic disk.
$\mathfrak{h} = R/\mathbb{C} \cdot 1$	Reduced Hamiltonian Lie algebra; constants have zero Hamiltonian vector field.
$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$	Polynomial Hamiltonian subalgebra used in trace and Moyal computations.
$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$	Shifted cotangent Lie algebra of Hamiltonian canonical transformations.
c^I, u_I	CE coordinates: c^I is dual to a Hamiltonian generator; u_I is dual to the shifted cotangent generator.
θ^I, O_I	PV coordinates: $c^I \mapsto \theta^I, u_I \mapsto O_I$ under the CE/PV map.
$O_{a,b}$	Open ghost-zero trace class $\overline{\mathrm{Tr}(\phi_1^a \phi_2^b)}$, with $a + b > 0$ in the reduced Hamiltonian label set.
$\Psi_{a,b}$	Primitive one-ψ PBW special-fibre class, with $a + b > 0$ after removing the scalar descendant line. It is not a Matlis principal part.
$\rho_{a,b}$	Principal-part residue current $z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$, realizing $\mathfrak{h}^\vee_{\mathrm{cont}}$.
$B_f(a)$	Smeared boundary Hamiltonian $\int a(t) \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt$; it is the boundary name for the image of the shifted-cotangent CE coordinate represented by $a(t)dt \otimes f$.

Remark 2.2 (Separation convention). The boundary Hamiltonian observable comes from the shifted-cotangent coordinate u, not from the Hamiltonian CE coordinate c. The polynomial one-ψ class $\Psi_{a,b}$ and the principal part $\rho_{a,b}$ have the same bidegree label but live in different modules. The CE/PV comparison identifies a reduced Hamiltonian P_0-central-operation model; it is not the unreduced P_0-factorization-centre theorem.

2.4 *Reduced Hamiltonian measurement and CE/PV central operations*

A closed Hamiltonian $f(z_1, z_2)$ is *measured* by the brane by substitution of the matrix-valued normal coordinates:

$$J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

At finite N one chooses a word ordering. In $H^0(\mathcal{R}_N^{GL_N}, Q)$ adjacent swaps are exact, $Q\,\mathrm{Tr}(\psi v u) = \mathrm{Tr}([\phi_1, \phi_2] v u)$, so the ordering ambiguity disappears modulo the constraint. In the stable connected limit $N \to \infty$, $J(f)$ is a well-defined Hamiltonian observable on the brane. The matrix Poisson bracket gives

$$\{J(f), J(g)\}_\partial = J(\{f, g\}_{\mathfrak{h}}), \tag{4}$$

after removal of the scalar Hamiltonian. Equation (4) is the ordinary canonical bracket of the probe system, the statement that the probe brane is a representation of the Hamiltonian Lie algebra of the disk. On monomials,

$$\{\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}\} = \overline{(kn - lm)\,\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1})}. \tag{5}$$

This computation exposes the obstruction that controls the paper. The full formal Hamiltonian algebra has no bracket-compatible finite Lie quotient system obtained by cutting Taylor degree: linear Hamiltonians lower degree, projected brackets fail Jacobi, and the diagonal cotangent Casimir does not

live in the strict product/direct-sum endpoint. Therefore the finite cotangent CE/PV theorem cannot be transported to the disk by pretending that finite coordinate windows are finite Lie algebras.

The survivor is different and sharper. Finite coordinate projections test coefficients only; they assemble to a completed coordinate dg algebra isomorphism

$$\Phi_{\mathrm{coord}}\colon C^{\bullet}_{\mathrm{CE,coord}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}}).$$

A P_0 comparison exists only after a bracket-admissible completion B is chosen, and a Maurer–Cartan kernel exists only after the diagonal tensor converges in the corresponding kernel-admissible tensor product. This is the reduced Hamiltonian central-operation model. It is not the unreduced factorization centre of the open factorization algebra.

The map $f \mapsto J(f)$ is the *moment-map shadow* of this reduced formal CE/PV central-operation map. Witten's centrality slogan enters here as a local algebraic criterion proved in the reduced Hamiltonian model, not as a transfer theorem from BCOV. Write $Z^{P_0}_{\mathrm{red,Ham}}(A;B) = B$ when $B \subset \mathrm{PV}(A)$ is the chosen bracket-admissible Poisson-CE model. The boundary observable $J(f) = \overline{\mathrm{Tr}\, f(\phi_1,\phi_2)}$ is the image, under that identification, of the coordinate dual to the shifted cotangent generator (*not* of the coordinate dual to f itself); the closed Hamiltonian ghost becomes the Hamiltonian vector-field operation on the brane.

THEOREM 2.3 *(Reduced Hamiltonian central-operation criterion, local P_0 form).* Let A_∂ be a locally constant classical P_0 factorization algebra on the brane line, in the reduced algebraic model, and let $F \in A_\partial(I)$ be the boundary image of a closed insertion supported in an interval I. The statement that this insertion may cross any open insertion without changing the local theory is equivalent, in the classical algebraic model, to an operation in the P_0-central-operation complex of A_∂. Explicitly, for every open observable $G \in A_\partial(J)$ there are homotopies natural for inclusions and disjoint factorization products whose degree-zero shadows are

$$F \cdot G \simeq (-1)^{|F||G|} G \cdot F, \qquad \{F,G\}_{P_0} \simeq 0$$

when the closed operation is a scalar central insertion. For Hamiltonian insertions the bracket component is not zero; it is the moment-map operation

$$\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab).$$

Thus the theorem-level datum proved here is the reduced Hamiltonian P_0-central-operation datum. A morphism to the unreduced derived P_0-factorization centre, including centrality homotopies against arbitrary open observables and principal-part cotangent currents, is Problem 3.60.

Proof. In a locally constant factorization algebra on the line, moving a closed insertion through an open insertion compares the two factorization products corresponding to the two orders on the line. A chain homotopy between those products is exactly the Hochschild centrality datum of the associated E_1-algebra stalk. The P_0 refinement records, in the same operation complex, compatibility with the classical Poisson bracket. Thus product-centrality and bracket-centrality are part of one algebraic datum.

For the Hamiltonian generators of the probe system this datum can be computed. The canonical equal-time bracket is

$$\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta^i_l \delta^k_j\, \delta(t-s).$$

After smearing against $a(t)b(s)$, the delta function becomes the product ab, and the matrix symplectic form contracts the Hamiltonian derivatives to $\{f, g\}$. Therefore $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$. Scalar closed insertions have zero Hamiltonian vector field, so their bracket component is null homotopic. This proves the local algebraic centrality principle used here and also shows exactly what is missing from the unreduced cotangent statement: centrality homotopies for arbitrary open observables and principal-part cotangent currents. □

2.5 *Scalar reduction and the U(1) center anomaly*

Quotienting by constants on the closed side and by the unit trace $\mathrm{Tr}(1) = N$ on the open side removes the constant-Hamiltonian central extension. Its extension class is the U(1) center anomaly: a distinguished nontrivial scalar-extension class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ obstructing the lift of the scalar-reduced Hamiltonian Poisson bracket on $\bar{A}$ to the unreduced polynomial Hamiltonian algebra $\mathfrak{h}_{\mathrm{poly}} = \mathbb{C}[z_1, z_2]$, and the corresponding obstruction to lifting the scalar-reduced trace bracket to the unreduced brane bracket with its Chan-Paton central line $\mathbb{C} \cdot I$. Theorem 2.5 makes this precise on the closed side; Theorem 2.6 on the open side; Theorem 2.8 identifies the same class with the Moyal/Capelli $\hbar N$ shift. Every subsequent theorem touching $\mathfrak{h}$ or $\mathcal{A}_\partial$ carries the flag "before/after killing the scalar" explicitly.

Notation. Write $\mathfrak{h}_{\mathrm{poly}} = \mathbb{C}[z_1, z_2]$ for the unreduced polynomial Hamiltonian Lie algebra, including the constant Hamiltonians; the bracket on $\mathfrak{h}_{\mathrm{poly}}$ is $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ and takes values again in $\mathfrak{h}_{\mathrm{poly}}$. Define

$$\omega : \mathfrak{h}_{\mathrm{poly}} \otimes \mathfrak{h}_{\mathrm{poly}} \longrightarrow \mathbb{C}, \qquad \omega(f, g) = [\,\{f, g\}\,]_0,$$

the constant Taylor coefficient of $\{f, g\}$. On monomials,

$$\omega(z_1^k z_2^l, z_1^m z_2^n) = \begin{cases} kn - lm, & (k+m, l+n) = (1,1), \\ 0, & \text{otherwise}, \end{cases}$$

hence ω is concentrated in the unique bidegree where the bracket multidegree $(k+m-1, l+n-1)$ is $(0,0)$; this is exactly the bidegree the manuscript discards as "scalar trace".

Lemma 2.4. ω is a Lie 2-cocycle on $\mathfrak{h}_{\mathrm{poly}}$. On $\mathfrak{h}_{\mathrm{poly}}$ itself it is a coboundary; on the scalar-reduced quotient $\bar{A} = \mathfrak{h}_{\mathrm{poly}}/\mathbb{C} \cdot 1$ the same formula represents a nontrivial class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$.

Proof. The Jacobi identity for $\{-,-\}$ on $\mathfrak{h}_{\mathrm{poly}}$ asserts

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0,$$

and projecting each summand to its constant Taylor coefficient gives the cocycle identity

$$\omega(f, \{g, h\}) + \omega(g, \{h, f\}) + \omega(h, \{f, g\}) = 0.$$

Antisymmetry $\omega(g, f) = -\omega(f, g)$ is inherited from antisymmetry of the bracket. Hence $\omega \in Z^2_{\mathrm{Lie}}(\mathfrak{h}_{\mathrm{poly}}, \mathbb{C})$.

On $\mathfrak{h}_{\mathrm{poly}}$ the cocycle is a coboundary. Define $\eta : \mathfrak{h}_{\mathrm{poly}} \to \mathbb{C}$ by $\eta(f) = -[f]_0$, the negative of the constant Taylor coefficient. Then for all $f, g \in \mathfrak{h}_{\mathrm{poly}}$, $\delta\eta(f, g) = -\eta(\{f, g\}) = [\{f, g\}]_0 = \omega(f, g)$, so $\omega = \delta\eta$.

After projection to $\bar{A} = \mathfrak{h}_{\mathrm{poly}}/\mathbb{C} \cdot 1$ the primitive η does not descend, since $\eta(1) = -1 \neq 0$. The cocycle ω does descend, because it already vanishes on the line $\mathbb{C} \cdot 1$. The class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ is nontrivial: any putative primitive $\bar{\eta} : \bar{A} \to \mathbb{C}$ would lift, by setting $\tilde{\eta}(1) = 0$, to a primitive of ω on $\mathfrak{h}_{\mathrm{poly}}$. The lifted $\tilde{\eta}$ would satisfy $\tilde{\eta}(\{z_1, z_2\}) = \tilde{\eta}(1) = 0$, but $\omega(z_1, z_2) = 1 \neq 0$, a contradiction. □

THEOREM 2.5 *(U(1) center anomaly: closed side).* The class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ of Lemma 2.4 is the scalar-axis obstruction class to lifting the Lie algebra structure on $\bar{A}$ to the Lie algebra structure on the unreduced polynomial Hamiltonians $\mathfrak{h}_{\mathrm{poly}}$ along the abelian extension

$$0 \longrightarrow \mathbb{C} \cdot 1 \longrightarrow \mathfrak{h}_{\mathrm{poly}} \longrightarrow \bar{A} \longrightarrow 0.$$

Equivalently, $\mathfrak{h}_{\mathrm{poly}}$ is isomorphic to the central extension $\bar{A}_{[\bar{c}]} = \bar{A} \oplus \mathbb{C}K$ with bracket

$$[\,(f, aK), (g, bK)\,] = (\{f, g\}_{\bar{A}},\ \bar{c}(f, g)\,K),$$

and $\mathbb{C} \cdot [\bar{c}] \subset H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ is the distinguished anomaly line attached to this constant-Hamiltonian extension.

Proof. Central extensions of $\bar{A}$ by $\mathbb{C}$ are classified by $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ via the standard Chevalley-Eilenberg argument. The exact sequence above is a central extension because the constant Hamiltonian $1 \in \mathfrak{h}_{\mathrm{poly}}$ has zero Hamiltonian vector field, so $\mathbb{C} \cdot 1$ is central in $\mathfrak{h}_{\mathrm{poly}}$. The class of this extension is the obstruction to a Lie-algebra splitting; computing the class on monomial generators,

$$[z_1^k z_2^l, z_1^m z_2^n]_{\mathfrak{h}_{\mathrm{poly}}} = \{z_1^k z_2^l, z_1^m z_2^n\} = (kn - lm) z_1^{k+m-1} z_2^{l+n-1},$$

and projecting to $\mathbb{C} \cdot 1$ keeps only the $(0, 0)$-Taylor coefficient ω of Lemma 2.4. Thus the extension class is $[\bar{c}]$, and Lemma 2.4 shows it is nontrivial. The same lemma shows that the scalar-extension line $\mathbb{C} \cdot [\bar{c}]$ is nonzero; no classification of all of $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ is asserted. Splitting this extension is exactly the problem of lifting back to $\mathfrak{h}_{\mathrm{poly}}$. □

THEOREM 2.6 *(U(1) center anomaly: open side).* At finite N the trace algebra Lie bracket $\{F, G\} = \mathrm{Tr}(\partial_{\phi_1} F\, \partial_{\phi_2} G - \partial_{\phi_2} F\, \partial_{\phi_1} G)$ on $\mathbb{C}[\mathfrak{gl}_N \oplus \mathfrak{gl}_N]^{GL_N}$ satisfies

$$\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$$

in the unreduced ring, and zero in the scalar-reduced quotient $H^0_{\mathrm{red}} = H^0/\mathbb{C} \cdot \mathrm{Tr}(1)$. The scalar-axis central extension class of the unreduced trace algebra is the image of $[\bar{c}]$ under the boundary evaluation map

$$\partial^{\mathrm{full}}_{\mathrm{bb},N} : \mathbb{C}[z_1, z_2] \longrightarrow \mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}, \qquad f \longmapsto \mathrm{Tr}\, f(\phi_1, \phi_2),$$

i.e. the same anomaly class is realised on the brane as the U(1) center of $\mathfrak{gl}_N$ with central element I.

Proof. The matrix Poisson bracket gives, on monomials of bidegree $(1, 0)$ and $(0, 1)$,

$$\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = \mathrm{Tr}\big((\partial_{\phi_1} \mathrm{Tr}\,\phi_1)(\partial_{\phi_2} \mathrm{Tr}\,\phi_2) - (\partial_{\phi_2} \mathrm{Tr}\,\phi_1)(\partial_{\phi_1} \mathrm{Tr}\,\phi_2)\big) = \mathrm{Tr}(I \cdot I) - 0 = N.$$

Hence the unreduced bracket realises the constant N as a scalar element. In $H^0_{\mathrm{red}} = H^0/\mathbb{C} \cdot \mathrm{Tr}(1)$ this scalar is killed, matching the projection on the closed side. The boundary evaluation map $\partial^{\mathrm{full}}_{\mathrm{bb},N}$ sends $1 \in \mathbb{C}[z_1, z_2]$ to $\mathrm{Tr}(I) = N$ and satisfies, for monomials with $(k + m, l + n) = (1, 1)$,

$$\partial^{\mathrm{full}}_{\mathrm{bb},N}(\{z_1^k z_2^l, z_1^m z_2^n\}) = (kn - lm) \cdot \mathrm{Tr}(I) = (kn - lm) \cdot N,$$

which equates the closed cocycle ω with the open trace bracket on the central direction. Hence the extension class on the brane is the image of $[\bar{c}]$. Since $\mathbb{C} \cdot I \subset \mathfrak{gl}_N$ is the U(1) centre with character Tr, the brane realisation of $[\bar{c}]$ is exactly the U(1) centre of $\mathfrak{gl}_N$. □

Remark 2.7. Every theorem that uses $\mathfrak{h}$, $\bar{A}$, or $\mathcal{A}_\partial = H^0_{\mathrm{red}}$ lives *after* killing the scalar. The unreduced statement requires keeping the central line $\mathbb{C} \cdot 1 \subset \mathfrak{h}_{\mathrm{poly}}$ on the closed side and $\mathbb{C} \cdot \mathrm{Tr}(1)$ on the open side, with the U(1) center anomaly class $[\bar{c}]$ recording the obstruction to a Lie-algebra splitting. Theorem 2.70, Theorem 3.242, Corollary 3.243, and Proposition 3.222 are all stated on the reduced side; their unreduced refinements are governed by $[\bar{c}]$ via the central extensions of Theorems 2.5 and 2.6.

Quantum-classical equivalence of the U(1) center cocycle

Theorem 2.8 *(Quantum-classical equivalence of the U(1) center class).* Let $\Phi = (\phi_1, \phi_2)$ be the classical brane field on the formal local disk and $\Phi_\hbar = (\phi_1^\hbar, \phi_2^\hbar)$ its Weyl quantization at finite N, with the Moyal product $f * g = m \circ \exp\big(\frac{\hbar}{2}P\big)(f \otimes g)$ for $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$. Then the normal-ordered quantum comoment imposes, in the degree-zero quantum Hamiltonian reduction, the Capelli/canonical relation

$$Y_\hbar X_\hbar - X_\hbar Y_\hbar + \hbar N I \equiv 0, \qquad X_\hbar = \phi_1^\hbar,\ Y_\hbar = \phi_2^\hbar,$$

modulo the quantum moment-map ideal. Its scalar trace part has semiclassical limit $\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$ on the trace generators $X = \mathrm{Tr}\,\phi_1$ and $Y = \mathrm{Tr}\,\phi_2$. In the classical limit $\hbar \to 0$, the bracket $\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$ recovers the central scalar of Theorem 2.6. The Rees quantization of the central extension $\bar{A}_{[\bar{c}]}$ carries the same Lie cocycle $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$. After applying the finite-N trace character $K \mapsto NI$ on the central generator, its order-$\hbar$ scalar contact term is the Capelli shift $\hbar N$.

Proof. The Weyl convention gives $[X_\hbar, Y_\hbar]_* = \hbar\{X, Y\} + O(\hbar^3)$. On the trace generators, Proposition 3.222 at $(k, l, m, n) = (1, 0, 0, 1)$ gives $\{X, Y\} = N$ before scalar reduction. Hence

$$X_\hbar Y_\hbar - Y_\hbar X_\hbar = \hbar N + O(\hbar^3),$$

and at finite N the higher-order terms vanish on these linear generators because P^r requires at least r derivatives in each tensor factor. Rearranging gives the displayed normal-ordered moment relation $Y_\hbar X_\hbar - X_\hbar Y_\hbar + \hbar N\, I \equiv 0$ in the quantum Hamiltonian reduction. The Capelli identity [26, 27] of order one supplies the same relation as the leading nontrivial trace identity, and the Razmyslov–Procesi description [17, 18] identifies the full ideal of trace relations at level N as a separate fact; the argument here uses only the order-one canonical commutator.

For the central-extension comparison, the Rees algebra $U_\hbar(\bar{A}_{[\bar{c}]})$ has special fibre $\widehat{\mathbf{S}}(\bar{A}_{[\bar{c}]})$ at $\hbar = 0$ and generic fibre containing $\bar{A}_{[\bar{c}]}$ as Lie algebra at $\hbar = 1$. The PBW filtration's associated graded carries the Lie 2-cocycle as the primary obstruction to PBW-splitting the deformation. Both sides of the classical-to-quantum reduction therefore record the same scalar-extension class in $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$: classically ω is the obstruction to splitting the central extension $\mathfrak{h}_{\mathrm{poly}} \to \bar{A}$, quantum-mechanically the same ω is the leading-order coefficient of the Moyal star commutator on the central direction. Evaluating the central generator by the trace character $K \mapsto NI$ converts that order-$\hbar$ central term into the scalar Capelli contact $\hbar N$. □

Remark 2.9. The class $[\bar{c}]$ is a physical observable of the brane: the U(1) center of $\mathfrak{gl}_N$ acts on the brane Hilbert space with character Tr, and the Capelli shift $\hbar N$ is the charge of this central direction. Scalar reduction kills the empty-trace scalar value $\mathrm{Tr}(1) = N$. It does not remove the nonempty linear trace observables $\mathrm{Tr}\,\phi_1$, $\mathrm{Tr}\,\phi_2$; their unreduced bracket is the scalar N, which becomes zero only after this projection.

2.6 *The two algebras*

The local model contains two algebras, both *derived* from the canonical mechanics above rather than postulated:

- The algebra $\mathcal{A}_N$ of GL_N-invariant local trace observables on the Dirac-reduced probe stack of (1); equivalently, the algebra of local operators of the open BV theory on the brane line $L = \mathbb{R}_{\text{brane}} \times \{0\} \times \{p\} \subset \mathbb{R}^2_{\text{top}} \times \mathbb{C}^2_{\text{hol}}$.
- The algebra $\mathcal{B}$ of local operators of the closed Hamiltonian BF sector (2); the operator-product expansion in the $\mathbb{R}$-direction equips $\mathcal{B}$ with an associative product, derived from the locally constant factorisation structure of the topological direction.

The product on the closed side in this paragraph is the topological-current avatar used for the local trace comparison. The native holomorphic operator algebra is the $\mathbb{C}^2$-holomorphic factorization algebra $\mathcal{F}^{\text{hol}}_{\text{Ham}}$ of Definition 2.19; no one-dimensional vertex OPE or Zhu algebra is asserted here.

Stable large-N admissibility target. The stable large-N comparison is not the source of the local theorem. It is a later admissibility target, organized by Definition 2.57 and proved only inside the admissible Tate/stable category of Theorem 2.58. The local theorem below supplies the formal Hamiltonian BF/Moyal construction and the coordinate CE/PV central-operation comparison that such a target may use after its own conilpotence, endpoint, and inverse-limit hypotheses have been supplied. The raw finite-N operator algebras are not identified here with the strict formal Hamiltonian envelope.

The Koszul spine is stratified. Layer 0 is the operadic Lie/coComm Koszul input. Layer 1 is the Tate CE/bar-cobar identity under an admissible lower-central Tate-pronilpotent envelope. Layer 2 is the Tate Quillen equivalence. Layer 3 is the coordinate CE/PV reduced P_0 central-operation theorem. Layer 4 is the PBW/Rees special-fibre shadow. Layer 5 is the Matlis/Grothendieck cotangent module. Layer 6 is the brane-line factorization-current comparison. Layer 7 is the stable large-N bulk-boundary horizon, conjectural here and not used in the local Hamiltonian BF/Moyal theorem.

The provenance of this local model in higher-dimensional physics is motivational. Costello's protected sector of perturbative twisted M-theory on $\mathbb{R} \times X$, with X a holomorphic-symplectic surface and a stack of N topological branes on $\mathbb{R} \times p$, suggests a Hamiltonian formal sector at the symplectic disk $\widehat{X}_p$, but this manuscript does not prove such a sector for an arbitrary ambient background. If an ambient background supplies the admissible protected Hamiltonian restriction of Definition 5.2, then the local theorem of §2.15 applies to that restricted sector; §5.1 records the corresponding protected-sector transfer criterion after the local core and the ambient hypotheses have been separated in the theorem-data table.

2.7 *Three notions of locality*

Locality is not one notion in this model. It is three.

(a) Support locality along the brane. For an open interval $I \subset \mathbb{R}$ on the world-line, define the compactly supported Hamiltonian current algebra $\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\widehat{\mathfrak{h}}$ with bracket induced from the de Rham product and the Hamiltonian bracket. In the degree-zero test-function notation used for explicit measurements this reads

$$[a \otimes f,\, b \otimes g] = ab \otimes \{f, g\}_{\widehat{\mathfrak{h}}}.$$

The reduced Hamiltonian open target on I is $A_\partial^{\mathrm{Ham}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$, generated by the scalar-quotiented smeared trace functionals

$$B_f(a) = \int_I a(t)\, \overline{\operatorname{Tr} f\big(\phi_1(t), \phi_2(t)\big)}\, dt.$$

Their bracket in the reduced Hamiltonian target is the canonical equal-time bracket:

$$\big\{B_f(a), B_g(b)\big\}_{A_\partial^{\mathrm{Ham}}} = B_{\{f,g\}_{\widehat{\mathfrak{h}}}}(ab).$$

This paragraph is after the Hamiltonian scalar quotient; before projection the unreduced bracket has the central term $N\omega(f,g)\int_I ab\, dt$. Disjoint supports multiply, since $ab = 0$ outside the intersection; overlapping supports give a delta-function contact term. This is the operational meaning of locality for the open factorisation algebra: disjoint independent measurements multiply, and Poisson brackets are supported where the measurements meet.

(b) Topological locality. In the topological direction (the $\mathbb{R}^2$ factor), translations are Q-exact:

$$L_v = [Q, \iota_v].$$

Equivalently, positive jets in the topological direction vanish in local-operator cohomology by the contraction homotopy $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$ (Lemma 3.191). Therefore the precise position of an insertion is not physical; only the collision pattern of insertions matters. This is the structural reason the factorisation algebra is locally constant in the topological direction.

(c) Formal holomorphic locality. In the holomorphic direction (the $\mathbb{C}^2$ factor), the model has been completed at the brane point $p \in \mathbb{C}^2$. It is a theory of holomorphic jets at p, not a theory of finite propagation through $\mathbb{C}^2$. Antiholomorphic translations are Q-exact:

$$L_{\partial/\partial\bar z_i} = [Q, \iota_{\partial/\partial\bar z_i}].$$

The kinematic differential of the closed sector is $D = d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}$, and on the formal-jet model $\bar\partial$-cohomology is concentrated in degree zero. The cotangent boundary sector is *distributionally local, not polynomially local*: a coadjoint mode is a transverse principal part $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2$, paired with a brane Hamiltonian by residue,

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}.$$

This is exactly what one expects for a momentum dual field attached to a defect: support on the defect, transverse behaviour encoded by residues. The polynomial one-ψ descendants $\Psi_{k,l}$ realise the PBW special fibre of $\mathfrak{h}^\vee[1]$ (Proposition 3.223); Corollary 3.224 records the deformation-level mismatch with the coadjoint module, exactly the ordinary distinction between polynomial and distributional realisations.

Under the support, topological, and formal-holomorphic locality hypotheses in (a)–(c), the factorisation-algebra structure is the resulting local algebra of measurements. Disjoint supports multiply along the brane; topological motion is Q-exact and changes nothing in cohomology; transverse holomorphic dependence is distributional. No external categorical input is imposed.

Proposition 2.10 *(Locality forced by the three differentials).* In the reduced Hamiltonian model the three locality statements above are consequences of three explicit operators:

$$\delta(t-s), \qquad L_v = [Q, \iota_v] \quad (v \in T\mathbb{R}^2_{\mathrm{top}}), \qquad L_{\partial/\partial\bar z_i} = [Q, \iota_{\partial/\partial\bar z_i}].$$

Support locality gives strict vanishing of P_0-brackets for disjoint smeared Hamiltonian currents. Topological locality gives the locally constant brane-line factorization structure after the compact-support density shift. Formal holomorphic locality gives Taylor Hamiltonians and residue-dual principal parts at the brane point. These are independent pieces of structure. None of them supplies the missing unreduced cotangent centrality homotopies.

Proof. Support locality is the calculation of Remark 3.21: after smearing, the factor $\delta(t-s)$ becomes the product ab, so disjoint supports give zero bracket. This proves the strict support-local factorization identity used by Proposition 3.20. Topological locality follows because translation along a topological direction is Q-exact; moving an insertion changes it by a Q-boundary. On compactly supported interval currents the surviving cohomology class is the density line of Lemma 3.19. Formal holomorphic locality follows in the same way for antiholomorphic translations, while holomorphic dependence is completed at the brane point and paired with cotangent modes by the residue pairing. The three operators live in different directions, so their consequences do not imply centrality homotopies for arbitrary unreduced open observables. □

2.8 Unitarity

The model is a twisted BV sector. It is not a Lorentzian unitary quantum mechanics. There is no positive-definite Hilbert space, no Wightman positivity, and no reflection positivity. The fields include ghosts and antifields; topological time has been made cohomologically trivial.

The replacements for unitarity are the consistency conditions of the BV formalism: the classical master equation

$$QI + \tfrac{1}{2}\{I, I\} = 0$$

and the quantum master equation

$$QI + \tfrac{1}{2}\{I, I\} + \hbar\,\Delta I = 0,$$

together with associativity of the resulting factorization algebra; an OPE algebra is present only after the separate one-complex-dimensional reduction datum of Statement 2.42. If the local model arises as a protected sector of a larger unitary theory (for example, of a topological twist of a supersymmetric quantum field theory whose local observables for the local model arise as cohomology with respect to a supercharge), then unitarity is carried by the larger theory, and the local sector retains only the cohomological algebra of protected observables. No unitarity is imposed below, and no Lorentzian claims are made.

THEOREM 2.11 *(BV replacement for unitarity in the protected sector).* For the local mixed holomorphic-topological sector studied here, the consistency conditions are

$$QI + \frac{1}{2}\{I, I\} = 0, \qquad QI + \frac{1}{2}\{I, I\} + \hbar\Delta I = 0$$

where the second equation is imposed only after a BV Laplacian and a renormalization scheme have been constructed, together with associativity and locality of the resulting factorization products. These conditions imply that the cohomology of local observables carries the stated P_0 or quantum factorization algebra structure and that closed insertions act through the appropriate central-operation complex. They do not imply positivity, a Hilbert-space inner product, or a Lorentzian unitary evolution operator. If the sector is obtained as Q-cohomology of a unitary parent theory, the unitarity datum belongs to that parent together with the choice of Q-compatible adjoint; it is not reconstructed from the CE/PV calculation.

Proof. The classical master equation says that the interacting differential $Q_I = Q + \{I, -\}$ squares to zero, since $Q_I^2 = \{QI + \frac{1}{2}\{I, I\}, -\}$. The quantum master equation is the corresponding statement for the BV operator $Q + \{I, -\} + \hbar\Delta$. Factorization associativity is the compatibility of local products with disjoint inclusions and collision limits; locality says these products are supported on the collision loci allowed by the factorization algebra. These are exactly the algebraic structures used below to form P_0-brackets, central operations, and Moyal deformations. The construction never introduces a positive sesquilinear pairing on the BRST complex, and ghosts and antifields have no positivity interpretation. Thus no Lorentzian unitarity conclusion follows from the BV equations. Conversely, if a larger physical theory supplies a Hilbert space, a positive form, and a supercharge Q compatible with adjoints, then the local theory is its protected cohomological algebra. That is extra structure, not a consequence of the reduced Hamiltonian CE/PV theorem. □

2.9 *Mixed holomorphic-topological strings*

By a local string theory we mean the portion of string field theory visible to local operators: closed fields forming a BV theory, local observables forming a factorization algebra, and a brane supplying a cyclic open field theory together with a bulk-to-boundary map. The words *topological* and *holomorphic* describe how those local operators vary in the spacetime directions. This is a local string-field-theoretic use of the word "string", not a definition of a global theory of worldsheet moduli, amplitudes on all backgrounds, or a completed quantum anomaly theory.

Gwilliam–Williams [5] are the precedent for formulating a holomorphic string in the Costello BV/factorization formalism. Their object is the holomorphic bosonic string on a Riemann surface, with Dolbeault fields (γ, β, c, b), a one-loop quantization in the critical complex dimension, and a factorization algebra recovering the BRST cohomology. The object here is not that theory and not a replacement definition. It is a formal local target-space sector: de Rham/Floer directions, a holomorphic symplectic formal disk, Hamiltonian cotangent BF fields, and a brane large-N operator algebra.

The topological word refers to locally constant factorization in the de Rham/Floer directions. The holomorphic word refers to the Dolbeault presentation: the kinetic operator contains $\bar{\partial}$, the fields are holomorphic jets after local formal completion, and anti-holomorphic translations act by cochain-homotopic zero. A holomorphic theory may acquire further topological descent after quantization; that extra descent is not part of the definition used here.

Definition 2.12 *(Working local topological factorization sector).* Let M be a smooth manifold. A local topological closed-string sector on M is a classical BV theory whose local observables form a locally constant factorization algebra on M: for inclusions of sufficiently small disks $U \subset V$, the factorization structure map

$$\mathrm{Obs}^{\mathrm{cl}}(U) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(V)$$

is a quasi-isomorphism. Attaching branes to such a sector means specifying a cyclic open BV theory and the bulk-to-boundary central-operation data it carries. In a de Rham model the closed fields carry the differential d_M plus the BV linearized differential, and the locally constant factorization product records collision of operators up to isotopy.

Remark 2.13. The exact A-model factor used below is a local topological sector in this sense: the self-Floer model for $\mathbb{R} \subset T^*\mathbb{R}$ is represented by the de Rham complex $\Omega^*(\mathbb{R})$, and its local observables are insensitive to the metric geometry of the interval. This is the topological part of the mixed holomorphic-topological example.

Definition 2.14 *(Working local holomorphic factorization sector).* Let Y be a complex manifold. A local holomorphic sector on Y is a classical BV theory whose fields are Dolbeault complexes on Y, whose differential has the form $\bar{\partial}$ plus the BV linearized differential, whose interactions are holomorphic polydifferential operators, and whose local observables form a holomorphic factorization algebra. Here holomorphic factorization means that, on coordinate polydisks, the factorization products and single-open structure maps are represented by holomorphic polydifferential operations and every anti-holomorphic translation $\partial/\partial\bar{z}_i$ acts null-homotopically on observables. In the target-space Kodaira-Spencer examples the fields are Dolbeault-valued polyvectors; in the holomorphic symplectic examples below they are Hamiltonian polyvectors after removing constants. Branes are treated through cyclic open BV theory and the bulk-to-boundary operations constructed below.

Remark 2.15. The active geometry is the formal local mixed space $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$. BCOV/Kodaira-Spencer theory supplies only the source convention [1]: Dolbeault polyvectors, cotangent BV notation, and the divergence determined by a holomorphic volume form. In the present surface model the holomorphic symplectic form $dz_1 \wedge dz_2$ and the volume form $dz_1 dz_2$ are local unimodularity data; they give the BV pairing, the divergence operator, and cyclic trace densities. The resulting divergence-free vector fields are represented locally by Hamiltonians modulo constants. This is not a truncation of threefold BCOV theory and no compact CY_3 BCOV theorem is used below.

Remark 2.16. The classical local observable theory underlying Gwilliam–Williams is an example of a pure holomorphic BV/factorization sector on a Riemann surface, with additional worldsheet ghost, BRST, and critical-dimension quantum structure. Their critical-dimension and BRST-cohomology theorems are not used as transfer theorems here. The model below is a different formal local target-space sector whose holomorphic sector is Hamiltonian BF, or the holomorphic symplectic Kodaira-Spencer analogue, and whose topological sector is de Rham/Floer. In particular, the critical complex dimension in Gwilliam–Williams is a worldsheet ghost-anomaly statement for the holomorphic bosonic string; it is not a constraint on the local $\mathbb{R}^2 \times \mathbb{C}^2$ Hamiltonian sector studied here.

Definition 2.17 *(Working classical local mixed holomorphic-topological sector).* Let M be a smooth manifold and let Y be a complex manifold. A classical local mixed holomorphic-topological closed sector on $M \times Y$ consists of:

(i) a sheaf of cochain complexes of fields $\mathcal{E}$ which, on product coordinate opens $D \times B$, has the form

$$\mathcal{E}(D \times B) = \Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B, \mathcal{V})$$

for a graded holomorphic vector bundle or formal holomorphic vector space $\mathcal{V}$;

(ii) a degree -1 local BV pairing on $\mathcal{E}$ and a differential

$$Q = d_M + \bar{\partial}_Y + Q_0,$$

where Q_0 is holomorphic over Y and is linear over de Rham forms on M;

(iii) a degree-zero local functional $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E})$, built from differential operators in the M directions and holomorphic polydifferential operators in the Y directions, satisfying the classical master equation

$$QI + \frac{1}{2}\{I, I\} = 0;$$

(iv) the associated classical observables

$$\mathrm{Obs}^{\mathrm{cl}}(U) = \left(\widehat{\mathrm{Sym}}(\mathcal{E}^!_c(U)), Q_{\mathrm{obs}}\right), \qquad Q_{\mathrm{obs}} = Q^! + \{I, -\},$$

with the standard factorization product for disjoint supports. Here $\mathcal{E}^!_c(U)$ denotes compactly supported continuous linear observables, i.e. the density-dual cosheaf of fields; extension by zero gives the covariant structure maps used by the factorization algebra.

These data must satisfy the mixed locality condition. For every inclusion of small disks $D \subset D'$ in M and every holomorphic polydisk $B \subset Y$, the structure map

$$\mathrm{Obs}^{\mathrm{cl}}(D \times B) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(D' \times B)$$

is a quasi-isomorphism. In holomorphic coordinates on B, each anti-holomorphic translation $\partial/\partial\bar{z}_i$ acts null-homotopically on observables: there is a degree -1 derivation η_i with

$$[Q_{\mathrm{obs}}, \eta_i] = \mathcal{L}_{\partial/\partial\bar{z}_i}.$$

A mixed brane for such a sector is supported on $N \times Z$, where $N \subset M$ is a smooth submanifold and $Z \subset Y$ is a holomorphic submanifold or a formal holomorphic neighbourhood of one, and is specified by an object $\mathcal{F}$ of the local holomorphic brane category on Z. Its open fields have the derived form

$$\Omega^*(N)\widehat{\otimes}\mathrm{RHom}_Y(\mathcal{F}, \mathcal{F}) \otimes \mathrm{End}(V)[1],$$

with a cyclic trace. For a smooth brane this recovers the Dolbeault model $\Omega^*(N)\widehat{\otimes}\Omega^{0,*}(Z, \mathcal{W})$; for a skyscraper brane it produces the Ext algebra. A full bulk-to-boundary mixed datum would further require a morphism of factorization algebras

$$\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed}}\big|_{N\times Z} \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}\big(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}}\big),$$

from closed observables to the Poisson/BV factorization-centre object of the open brane observables, compatible with the BV brackets and factorization products. The Hamiltonian model below constructs the degree-zero Hamiltonian current shadow of this map; the full closed CE/cotangent factorization-center lift is isolated in Problem 3.60 and treated there only with the reduced, chain-level, truncated, or weighted hypotheses stated in the local theorem.

Remark 2.18. A full quantum mixed holomorphic-topological string datum would be the corresponding $\hbar$-adic factorization algebra obtained from a renormalized BV action satisfying the quantum master equation. The manuscript proves the componentwise quantum coefficient surface: weighted BV kernel convergence, the balanced scalar-contact zero projection on its habitat, reduced weighted Moyal principal-part current brackets, and a degree-zero Hamiltonian trace comparison controlled by the radial/Weyl normal-form gate. The current finite certificates close all non-edge bidegrees of total degree at most 12 and the balanced diagonal through $(6, 6)$, while Proposition F.27 proves the complete one-moving-letter edge strips $\mathfrak{o}_{2,b} = \mathfrak{o}_{a,2} = 0$. The all-interior statement is the exact obstruction problem

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b}.$$

Its positive form is $\Omega^{\mathrm{rad}}_{a,b} = 0$, equivalently the decorated PBW Stokes identity $D^{\square}_{a,b} = C^+_{a,b}\partial_2$; failure is exactly a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. This sharpens the older $C_{a,b}$-left-cokernel language for the obstruction classes $\mathfrak{o}_{a,b}$ of Definition F.21. The finite-window non-scalar closed-open BV criterion is the vanishing problem in $\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$ of Problem 3.271. The full renormalized all-graph realization still also requires the analytic graph-realization data of Problem 3.264.

Definition 2.17 separates three levels. A pure topological sector is locally constant in all real directions; in examples this includes the A-model local/Floer direction and the Chern-Simons-type open field theory on a topological brane. A pure holomorphic sector is Dolbeault in its complex directions and produces a holomorphic factorization algebra; Gwilliam–Williams' holomorphic bosonic string is a worldsheet example, while BCOV/Kodaira-Spencer supplies only the target-space source convention for Dolbeault polyvectors, divergence, and cotangent BV notation. The Hamiltonian BF complex below is the formal local target-space sector used here. A mixed holomorphic-topological sector lives on $M \times Y$ with locally constant factorization in M and holomorphic factorization in Y. The A/B terminology used below is one source of such examples: the A-side supplies the de Rham/Floer direction and the B-side supplies the Dolbeault or formal holomorphic direction.

2.10 *The local mixed model*

The product local model is

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}.$$

The theorem is not a global statement about all twisted M-theory backgrounds. It is the formal local calculation obtained after completing at L and taking the Darboux holomorphic normal disk $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$. The closed sector is the Hamiltonian BF formal-local sector on this mixed space, in the Kodaira–Spencer source convention rather than in a pure A-model or pure B-model. The open sector is the cyclic open string field theory on the mixed brane: topological along $\mathbb{R}$ and formal holomorphic in the normal $\mathbb{C}^2$ directions. Unless a statement is explicitly placed in the matched-conventions firewall, every theorem below is a theorem about this completed $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ model.

Definition 2.19 *(Local Hamiltonian holomorphic factorization algebra).* The central factorization object in this paper is

$$\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$$

the formal Hamiltonian holomorphic factorization algebra on the formal symplectic disk $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$. On a holomorphic polydisk $B \subset \mathbb{C}^2_{\mathrm{hol}}$ its dg Lie input is

$$\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g}, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1], \qquad \mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1.$$

The formal stalk of observables is $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$; its identification with polyvectors is the coordinate/admissible CE/PV theorem below, not an unrestricted strict-product P_0 assertion.

Its mixed topological-current enhancement is

$$\mathcal{F}^{\mathrm{mix}}_{\mathrm{Ham}}(U \times B), \qquad \mathfrak{g}^{\mathrm{mix}}_{U,B} = \Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g},$$

for $U \subset \mathbb{R}^2_{\mathrm{top}}$. This object is locally constant in U and holomorphic in B.

The open quantum brane avatar is the Weyl/Moyal algebra of the formal symplectic disk, together with its finite-N matrix Weyl Hamiltonian reduction and matrix or supermatrix cyclic trace bar-cobar models. The support-local defect/cotangent avatar on an interval $I \subset \mathbb{R}_{\mathrm{brane}}$ is the principal-part current algebra

$$\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}, \qquad \mathcal{P}_I = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}, \qquad \mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1],$$

with reduced target

$$A^{\mathrm{pp}}_\partial(I) = \widehat{\mathbf{S}}(\Omega^0_c(I)\widehat{\otimes}\bar{A})\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1]).$$

Its quantum version replaces the classical coadjoint action by the Moyal coadjoint action.

Thus any use of the word "chiral" in this paper refers to a two-complex-dimensional holomorphic factorization algebra and its mixed topological-current extensions. It does not mean a one-dimensional vertex algebra, an OPE mode algebra, compact-CY chiral homology, CoHA, or BKM data. A one-dimensional VOA/OPE statement requires an additional theorem choosing a complex line or holomorphic defect, a transverse boundary condition or restriction functor, and the finite Laurent OPE expansion on that restricted object.

Proposition 2.20 *(Constructed Hamiltonian CE/factorization observables).* In the formal local mixed model $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$, the Hamiltonian BF fields of Definition 2.17 reduce at a formal holomorphic insertion to the shifted cotangent dg Lie algebra

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1], \qquad \mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1.$$

For a holomorphic polydisk B, the constructed holomorphic factorization algebra is represented by the continuous CE observables of the compactly supported Dolbeault dg Lie algebra

$$\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}(B) = C^\bullet_{\mathrm{CE,cont}}\big(\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g}\big).$$

For a product open $U \times B$, its mixed topological-current enhancement is represented by

$$\mathcal{F}^{\mathrm{mix}}_{\mathrm{Ham}}(U \times B) = C^\bullet_{\mathrm{CE,cont}}\big(\Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g}\big).$$

These assignments are the local holomorphic factorization algebras of the paper. Their formal stalk is $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$, and its identification with the PV model is only the coordinate/admissible theorem cited below.

Proof. Proposition 3.13 identifies the divergence-free holomorphic vector fields in the formal disk with Hamiltonians modulo constants. The Hamiltonian BF action then has connection field valued in $\mathfrak{h}[1]$ and cotangent field valued in $\mathfrak{h}^\vee_{\mathrm{cont}}[2]$. After the de Rham and Dolbeault Poincare contractions at the formal insertion point, Proposition 3.215 identifies the residual closed local operators with $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$, where the CE differential is dual to the semidirect bracket $[f, g] = \{f, g\}$, $[f, \eta_\lambda] = \eta_{\mathrm{ad}^\vee_f \lambda}$, and $[\eta_\lambda, \eta_\mu] = 0$.

On an actual polydisk one keeps the compactly supported Dolbeault factor; on a mixed product open one also keeps the compactly supported de Rham factor. Extension by zero supplies the factorization structure maps, while the local Poincare homotopies give holomorphic factorization in B and local constancy in U, exactly as in Definitions 2.14 and 2.17. The PV, P_0, kernel, and bar-cobar upgrades are the admissible consequences of Theorem 2.48 and Theorem 3.28; no one-dimensional vertex/OPE structure is asserted. □

Proposition 2.21 *(Bochner–Martinelli current data and finite-window obstruction for the native E_2 lane).* Let $B_0 \Subset B_1 \Subset B_2 \Subset B \subset \mathbb{C}^2$ be nested polydisks around the collision point and choose $\chi \in C^\infty_c(B_2)$ with $\chi = 1$ on B_1. Put

$$\mathcal{P} = \bigoplus_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

and

$$\mathcal{P}_{\leq N} = \bigoplus_{0<a+b\leq N} \mathbb{C}\rho_{a,b}.$$

The symbol $\mathfrak{h}_{\leq N} = \mathfrak{m}/\mathfrak{m}^{N+1}$ is used only as a coordinate window; it is not asserted to be a Lie quotient of the full Hamiltonian algebra.

On $(B_\zeta \times B_z) \setminus \Delta$, set

$$\eta = \zeta - z, \qquad r^2 = |\eta_1|^2 + |\eta_2|^2, \qquad d\bar{\eta}_j = d\bar{\zeta}_j - d\bar{z}_j.$$

The Dolbeault homotopy kernel is the Bochner–Martinelli–Koppelman current [33, Ch. III, Def. 1.2, Lem. 1.3]

$$K_{\mathrm{BM}}(\zeta, z) = \frac{1}{(2\pi i)^2} \frac{\overline{\eta_1}\, d\bar{\eta}_2 - \overline{\eta_2}\, d\bar{\eta}_1}{r^4} \wedge d\zeta_1 \wedge d\zeta_2.$$

The external several-complex-variables sources are used here only for this pre-cutoff kernel, its component $\bar{\partial}$-identity away from the diagonal, and the classical Koppelman homotopy formula on bounded C^1 domains for C^1 forms [34, §5, Def. 5.1, Thm. 5.2, Eq. (10)]. They do not prove the cutoff-current limit H_χ, the annular diagonal-current sign, the finite moment projection p_N, the residue-current section i_N, the analytic/pro-Matlis quotient, the Hamiltonian/Matlis arity-two action, or any Costello/QME or native factorization transfer below. The $d\bar{z}_j$-terms in $d\bar{\eta}_j$ are part of the kernel; the $d\bar{z} = 0$ component is only the scalar integral kernel. With a radial cutoff $\vartheta_\epsilon(r)$, equal to 0 for $r \leq \epsilon$ and to 1 for $r \geq 2\epsilon$, put

$$K_{\chi,\epsilon} = \chi(\zeta)\chi(z)\vartheta_\epsilon(|\zeta - z|)K_{\mathrm{BM}}(\zeta, z).$$

The current limit

$$H_\chi \alpha(z) = \lim_{\epsilon \to 0} \int_{\zeta \in B} K_{\chi,\epsilon}(\zeta, z) \wedge \alpha(\zeta)$$

is well-defined on each fixed finite configuration stratum. The singularity is locally integrable in real dimension four ($|K_{\mathrm{BM}}| \sim r^{-3}$), and for separated compact supports its derivatives obey estimates

$$|\nabla^k K_{\mathrm{BM}}(\zeta, z)| \leq C_k |\zeta - z|^{-3-k}.$$

Let $i_N(\rho_{a,b})$ be the support-at-zero current

$$\Delta_{a,b} = \frac{(-1)^{a+b}}{a!\, b!} \partial_{z_1}^a \partial_{z_2}^b \delta_0\, d\bar{z}_1 \wedge d\bar{z}_2, \qquad \langle \Delta_{a,b}, f \rangle = \mathrm{Res}_0(\rho_{a,b} f),$$

and define the finite principal-part moment projection on top Dolbeault degree by

$$p_N(\alpha) = \sum_{0 < a+b \leq N} \left(\int_B z_1^a z_2^b\, \alpha^{0,2} \wedge dz_1 \wedge dz_2 \right) \rho_{a,b}.$$

The data above construct the current-limit operator H_χ, the finite moment map p_N, and the residue-current section i_N. They do not by themselves define a finite-window homotopy $H_{\chi,N}$ on the full compact-support/current complex. The desired identity

$$\bar{\partial} H_{\chi,N} + H_{\chi,N} \bar{\partial} = \mathrm{id} - i_N p_N + E_{\chi,N} \tag{$*_N$}$$

is blocked, in the native full-current habitat, by the finite-window obstruction vector

$$\mathrm{Ob}_N^{\text{BMK-curr}} = \left(\mathrm{ob}_{\mathrm{diag}}, \mathrm{ob}_{\mathrm{an/fin},N}, \mathrm{ob}_{\mathfrak{h},N}, \mathrm{ob}_{\text{collar-fact},N} \right).$$

Here $\mathrm{ob}_{\mathrm{diag}}$ is the sign and scalar comparison between the annular diagonal current $\bar\partial\vartheta_\epsilon(|\zeta - z|) \wedge K_{\mathrm{BM}}$ and the Matlis derivative-delta convention; with the displayed kernel and real outward boundary orientation its angular mass is -1, so a BMK diagonal-current convention must be declared before the sign in $i_N p_N$ is theorem-grade. The component $\mathrm{ob}_{\mathrm{an/fin},N}$ is the quotient between the full analytic moment kernel $\ker(\mathcal{O}(B_1)^\vee \to \mathcal{P}_{\le N})$ and the smaller support-at-zero span generated by $\Delta_{0,0}$ and the $\Delta_{a,b}$ with $a + b > N$. The component $\mathrm{ob}_{\mathfrak{h},N}$ is the Hamiltonian-invariance defect

$$\pi_N\big(\mathfrak{h} \cdot \ker \pi_N\big),$$

already nonzero because

$$z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1},$$

while $\rho_{N+1,0}$ is killed by the window and $\rho_{0,1}$ is retained. Finally $\mathrm{ob}_{\text{collar-fact},N}$ is the failure, still unproved away from one nested pair, that the collar quotient, extension-by-zero maps, factorization products, and the BMK homotopy preserve the same relation subspace.

Consequently there is no strict finite support-local current quotient retaining $\mathcal{P}_{\le N}$, killing the scalar and high support-at-zero currents, and carrying the full Hamiltonian factorization action. The projected finite-window L_∞ escape repair is obstructed as well: the forced return block already has

$$\kappa_N(z_1^{N+2}, \rho_{N+1,0}) = -(N+2)\rho_{0,1},$$

and the projected Jacobiator on $(z_1^{N+2}, z_2, \rho_{N,0})$ is

$$(N+1)(N+2)\rho_{0,1}.$$

Since the escape-return image spans the retained finite Matlis window, absorbing these returns as unary boundaries would collapse $\mathcal{P}_{\le N}$ itself. A one-pair derived analytic quotient does give a differential identity, after the full analytic BMK contraction and diagonal orientation are fixed: quotient the relative current complex by representatives of $\ker(\mathcal{O}(B_1)^\vee \to \mathcal{P}_{\le N})$, normalize the homotopy by

$$(1 - I_{\mathrm{an}} P_{\mathrm{an}}) H_\chi (1 - I_{\mathrm{an}} P_{\mathrm{an}}),$$

and then

$$\bar\partial H_{\mathrm{BM},N} + H_{\mathrm{BM},N}\bar\partial = \mathrm{id} - i_N p_N.$$

This quotient is analytic and one-pair relative; it is not a native support-local factorization-current quotient. The source theorem used here is only the pre-cutoff Koppelman identity; the representative quotient, the normalized contraction, and the compatible pro-Matlis moment map are local constructions of this proposition.

Keeping the entire Taylor tower gives the positive one-pair pro-Matlis replacement. Let

$$\widehat{\mathcal{P}}_\Pi = \prod_{\substack{a,b\ge 0\\ a+b>0}} \mathbb{C}\rho_{a,b}, \qquad \mathcal{P}_{\mathrm{an}}(B_1) = \tau\big(\mathcal{O}(B_1)^\vee_{\mathrm{red}}\big) \subset \widehat{\mathcal{P}}_\Pi,$$

where

$$\tau(\lambda) = \big(\lambda(z_1^a z_2^b)\big)_{a+b>0}.$$

On the single-pair analytic BMK habitat, after the diagonal orientation is fixed and the collar error is quotiented, the normalized contraction $H_\chi^\circ = (1 - I_{\mathrm{an}} P_{\mathrm{an}}) H_\chi (1 - I_{\mathrm{an}} P_{\mathrm{an}})$ satisfies

$$\bar\partial H_\chi^\circ + H_\chi^\circ \bar\partial = \mathrm{id} - I_{\mathrm{an}} P_{\mathrm{an}}.$$

Hence the pro-analytic moment map $P_{\Pi}^{\mathrm{an}} = \tau P_{\mathrm{an}}$ has compatible finite projections

$$\pi_N P_{\Pi}^{\mathrm{an}} = p_N, \qquad \pi_{M,N} p_M = p_N.$$

This is a one-pair analytic pro-Matlis retract. It is not a native support-local factorization-current transfer.

The binary operation

$$\ell_2^{\mathrm{BM},N}(x, y) = p_N[ix, iy]$$

is therefore retained only as an arity-two Hamiltonian/Matlis coefficient comparison induced by p_N and i_N, not as a consequence of a proved native finite-window homotopy. It agrees, coefficient by coefficient and in every fixed output window, with the Hamiltonian semidirect bracket. Thus for Hamiltonians $f, g \in \mathfrak{h}$,

$$\ell_2^{\mathrm{BM},N}(f, g) = \mathrm{pr}_{\leq N}\big(\partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g\big) \pmod{\mathbb{C}},$$

while for $f = z_1^p z_2^q$ and $\rho_{i,j} = z_1^{-i-1} z_2^{-j-1} dz_1 dz_2$,

$$\ell_2^{\mathrm{BM},N}(f, \rho_{i,j}) = -(pj - qi + p - q)\, \mathrm{pr}_{\leq N}\, \rho_{i-p+1, j-q+1},$$

with negative-index targets equal to zero, and

$$\ell_2^{\mathrm{BM},N}(\rho, \sigma) = 0.$$

This is the Matlis coadjoint action of Theorem 3.230.

Only after $\mathrm{Ob}_N^{\mathrm{BMK\text{-}curr}}$ is killed, or after the one-pair analytic quotient is replaced by a genuine native factorization-current habitat, does the first possible obstruction to a strict all-arity E_2 collision operation become the ternary cutoff term

$$\mathrm{Ob}_3^{\chi}(x, y, z) = p[H_{\mathrm{BM}}([ix, iy] - i\ell_2^{\mathrm{BM}}(x, y)), iz] + \text{cyclic Koszul terms}.$$

It vanishes on separated support strata by the estimates above, and its coefficient Jacobiator on the collision stratum is the Jacobiator of $\mathfrak{h} \ltimes \mathcal{P}[1]$, hence zero. Any surviving term is therefore a cutoff/collar contribution supported on the small diagonal. The strict native all-window current E_2 transfer is downstream and is controlled by the six-entry obstruction vector

$$\mathrm{Ob}_{\mathrm{BM}}^{\Pi} = \big(\mathrm{ob}_{\oplus}, \mathrm{ob}_{I_\Pi}, \mathrm{ob}_{\mathfrak{h},\Pi}, \mathrm{ob}_{\mathrm{collar},\Pi}, \mathrm{ob}_{3,\Pi}, \mathrm{ob}_{\mathrm{unif},\Pi}\big).$$

Here $\mathrm{ob}_{\oplus}$ is the nonzero class of the all-window moment sequence in $\widehat{\mathcal{P}}_\Pi/\mathcal{P}$; ob_{I_Π} is the nonexistence of a compact-current section $i_\Pi : \widehat{\mathcal{P}}_\Pi \to \mathcal{D}'^{0,2}_0(B)$ extending the finite derivative-delta sections; and $\mathrm{ob}_{\mathfrak{h},\Pi}$ is the failure of the full formal Hamiltonian action on the pro-product module, detected by the divergent $\rho_{0,1}$-coefficient for $f = \sum_{p\geq 2} z_1^p$ acting on $\lambda = \sum_{p\geq 2} \rho_{p-1,0}$. The remaining entries are the compatible collar primitive class, the first ternary small-diagonal transfer class, and the all-window uniform estimate class. The first, second, third, and sixth entries are nonzero in the native current topology; the collar and ternary entries are exact primitive construction targets.

The principal-part face of the transfer is genuinely two-variable:

$$\frac{1}{(2\pi i)^2} \frac{d\zeta_1 \wedge d\zeta_2}{(\zeta_1 - z_1)(\zeta_2 - z_2)} = \sum_{a,b\geq 0} z_1^a z_2^b \rho_{a,b}(\zeta).$$

Killing z_2 kills the Hamiltonian bracket, and the $b = 0$ principal-part line is not stable because $z_1 \cdot \rho_{a,0} = -\rho_{a,1}$. Hence this proposition proves the BMK current data, the finite-window obstruction theorem, and the arity-two native $\mathbb{C}^2$ coefficient comparison. It does not prove a curve VOA, a Zhu algebra, a strict support-local finite-window quotient, or a strict all-arity transfer.

Proof. The external Koppelman identity for K_{BM} supplies only the smooth-domain, pre-cutoff Dolbeault homotopy [34, §5, Thm. 5.2]. The cutoff derivative, annular diagonal current, finite moments, and Matlis orientation below are checked locally. After multiplication by the cutoff, the load-bearing derivative is

$$\bar\partial(\chi(\zeta)\chi(z)\vartheta_\epsilon K_{\mathrm{BM}}) = \chi\chi\,\vartheta_\epsilon\,\bar\partial K_{\mathrm{BM}} + \bar\partial(\chi\chi)\,\vartheta_\epsilon K_{\mathrm{BM}} + \chi\chi\,\bar\partial\vartheta_\epsilon \wedge K_{\mathrm{BM}}.$$

The second summand is collar-supported. The third summand is an annular diagonal current; it is not a collar term, and its sign and scalar are the first entry of $\mathrm{Ob}_N^{\text{BMK-curr}}$. Applying finite moments gives the formula for p_N. The normalization of i_N follows by integration by parts against a holomorphic test function, giving the residue functional represented by $\rho_{a,b}$, once the diagonal-current orientation is tied to the Matlis convention.

The full-current finite-window identity cannot hold on the native current complex. If it did, every closed top current whose analytic Dolbeault cohomology class lies in $\ker \pi_N$ would be exact after quotienting by the finite selected moments. Pairing with a holomorphic function detecting that class contradicts Stokes. In particular, $T = \Delta_{N+1,0}$ has $p_N(T) = 0$, but $\langle T, z_1^{N+1}\rangle = 1$, whereas $\langle \bar\partial S, z_1^{N+1}\rangle = 0$ for every current S. The scalar current $\Delta_{0,0}$ gives the same obstruction because $\rho_{0,0}$ is excluded from the reduced principal-part module.

The stronger support-local quotient obstruction is Hamiltonian invariance. A native quotient relation must be an $\mathfrak{h}$-submodule relation. The Matlis coadjoint formula gives $z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1}$; hence killing the high mode $\rho_{N+1,0}$ while retaining the finite mode $\rho_{0,1}$ is incompatible with the full Hamiltonian action. This proves the nonexistence of the strict finite support-local quotient stated above.

The one-pair analytic quotient is formal once the full analytic BMK contraction is supplied locally: the normalized homotopy annihilates the chosen representatives of $\ker \pi_N$, so it descends and the analytic projector becomes $i_N p_N$ in the quotient. Without quotienting the Taylor tail, the same normalized contraction gives $\bar\partial H_\chi^0 + H_\chi^0 \bar\partial = \mathrm{id} - I_{\mathrm{an}} P_{\mathrm{an}}$, and Taylor moments give $P_\Pi^{\mathrm{an}} = \tau P_{\mathrm{an}}$ with $\pi_N P_\Pi^{\mathrm{an}} = p_N$. Because this pro-Matlis retract depends on the analytic moment space, a representative map, and a chosen polydisk pair, it does not supply a factorization-current theorem.

The arity-two formula is the residue transpose calculation. On holomorphic coefficient labels this is the Poisson bracket modulo constants, projected after the coefficient is computed. On cotangent labels the residue identity

$$\mathrm{Res}_0((f \cdot \rho)g) = -\,\mathrm{Res}_0(\rho\,\{f, g\})$$

gives exactly the principal-part coadjoint formula already proved in Theorem 3.230. The cotangent summand is abelian, so the (ρ, σ)-bracket is zero.

The displayed ternary expression is therefore only the next obstruction after the finite-current gate is crossed. Away from collisions the kernel is smooth and the separated-support estimates force it to vanish in the factorization product. On the coefficient collision stratum the semidirect Lie algebra satisfies Jacobi. The remaining work is the collar, diagonal, and completed all-window transfer data. The two-variable Cauchy expansion records the same residue basis and shows that no one-variable VOA/Zhu reduction is being used. □

Proposition 2.22 *(Native theorem package from the Darboux disk).* For

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\},$$

with holomorphic Darboux form $dz_1 \wedge dz_2$, the local theorem package consists of the following five native constructions.

(i) The closed Hamiltonian BF coefficient algebra is

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1, \qquad P = \mathfrak{h}^\vee_{\mathrm{cont}}, \qquad \mathfrak{g} = \mathfrak{h} \ltimes P[1].$$

The quotient by constants is forced by the kernel of the Hamiltonian vector-field map.

(ii) The finite-N open probe is the Dirac/Koszul matrix model

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0,$$

obtained from the first-order action $\int_{\mathbb{R}_t} \mathrm{Tr}(\phi_1\, d\phi_2 + A[\phi_1, \phi_2])$. Thus the open algebra is the derived zero-fibre of the conjugation moment map, not an imposed commuting scheme.

(iii) After killing the scalar Hamiltonian and the empty trace, the derived-centre comparison is the coordinate CE/PV map

$$\Phi_{\mathrm{coord}}(c^I) = \theta^I, \qquad \Phi_{\mathrm{coord}}(u_I) = O_I.$$

The c-coordinate gives the Schouten vector-field operation; the u-coordinate gives the boundary Hamiltonian observable.

(iv) The open one-ψ classes are PBW special-fibre labels. The deformation-level cotangent dual is P, realized by Matlis principal parts and acted on by the coadjoint action. The PBW/Rees label bridge is extra Fourier-polarized structure; it is not a same-action identification of $\Psi_{a,b}$ with $\rho_{a,b}$.

(v) The Bochner–Martinelli–Koppelman current limit, finite Matlis moment maps, arity-two coefficient comparison, and one-pair analytic pro-Matlis retract $P^{\mathrm{an}}_{\Pi} = \tau P_{\mathrm{an}}$, $\pi_N P^{\mathrm{an}}_{\Pi} = p_N$, are constructed in Proposition 2.21. A strict support-local finite-window current quotient is obstructed by $\mathrm{Ob}^{\mathrm{BMK\text{-}curr}}_N$. A strict all-arity native E_2 transfer is the downstream theorem target: it requires first crossing this finite-current gate and then solving the six-entry all-window obstruction problem $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$.

This package is internal to the formal local $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ model. Compact Calabi–Yau theory, a curve VOA, CoHA, or BKM data can enter only through a separate matched-conventions theorem.

Proof. The formal Darboux disk is the habitat fixed in the opening of this section. Item (1) is Proposition 3.13 and Proposition 2.20. Item (2) is the constrained-mechanics computation of Lemma 3.6, Lemma 3.5, and Proposition 3.7. Item (3) is Theorem 3.28, with the scalar convention of Remark 2.1 and Theorems 2.5–2.8. Item (4) is Theorem 3.232 together with the principal-part uniqueness theorem 3.230. Item (5) is Proposition 2.21. The final sentence is the firewall already imposed by Definition 2.19 and Proposition 2.25. □

Remark 2.23 *(Reduced chiral-interface target).* A curve chiral algebra attached to this theory is the reduced target of Statement 2.42, not part of the native Darboux package. It must pass through the retained-mode $\mathbb{C} \times \mathbb{R}$ interface, keeping the z_2-coefficient or the full two-index principal-part system. Its admission obstruction is

$$\mathrm{Ob}_{\mathfrak{I}_{\mathrm{ch}}} = (\mathrm{Ob}_{\mathrm{red}}, \mathrm{Ob}_{\mathrm{vert}}, \mathrm{Ob}_{\mathrm{Zhu}}, \mathrm{Ob}_{\mathrm{nat}}),$$

measuring the retained-mode reduction, vertex/OPE package, Zhu multiplicativity, and comparison back to $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$.

Remark 2.24 *(External global-obstruction firewall).* The Hamiltonian identification used here is formal local. On a holomorphic symplectic surface X, a holomorphic vector field v is symplectic when

$$\alpha_v = -\iota_v \omega$$

is a closed holomorphic one-form, and it is Hamiltonian only after choosing f with $df = \alpha_v$, in the convention $\iota_{X_f}\omega = -df$. Thus any external attempt to globalize the Hamiltonian sector must first choose a sheaf of local Hamiltonians or kill the corresponding holomorphic de Rham obstruction. The formal disk used in this paper has no such obstruction by the holomorphic Poincare lemma; a general X is not an input to the local theorem.

Proposition 2.25 *(External formal-stalk Hamiltonian restriction).* Let (X, ω) be a connected holomorphic symplectic surface and let $p \in X$. Put

$$\mathcal{H}_X = \mathcal{O}_X/\mathbb{C}_X, \qquad \widehat{\mathfrak{h}}_{X,p} = \widehat{\mathcal{O}}_{X,p}/\mathbb{C}.$$

This is an external comparison statement, not a hypothesis of Theorem 2.70. The differential identifies $\mathcal{H}_X$ with the sheaf of closed holomorphic one-forms, hence with the sheaf of holomorphic symplectic vector fields by $\alpha = -\iota_v\omega$. For every $p \in X$ there is a Lie algebra restriction

$$H^0(X, \mathcal{H}_X) \longrightarrow \widehat{\mathfrak{h}}_{X,p}.$$

After choosing a formal Darboux coordinate system, the formal neighbourhood $(\widehat{X}_p, \omega)$ has the same completed Hamiltonian Lie algebra as the formal disk used in the local model. The resulting stalk-level boundary map

$$\widehat{\mathfrak{h}}_{X,p} \longrightarrow H^0_{\mathrm{Ham}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{brane}})$$

is independent of this choice up to the natural action of the formal symplectomorphism group. The subspace coming from global Hamiltonian functions is the image of

$$H^0(X, \mathcal{O}_X)/\mathbb{C} \longrightarrow H^0(X, \mathcal{H}_X).$$

The obstruction for a locally Hamiltonian symplectic vector field to come from a global Hamiltonian function is its period class in $H^1(X, \mathbb{C}_X)$.

Proof. The holomorphic Poincare lemma gives an isomorphism

$$d : \mathcal{O}_X/\mathbb{C}_X \xrightarrow{\sim} \Omega^1_{X,\mathrm{cl}}.$$

The holomorphic symplectic form identifies a symplectic vector field v with the closed one-form $\alpha_v = -\iota_v\omega$. Thus a section of $\mathcal{H}_X$ is exactly a local Hamiltonian primitive, modulo local constants, and the Poisson bracket is independent of the local constants. Taking the germ at p and completing gives the displayed restriction to $\widehat{\mathfrak{h}}_{X,p}$.

The formal holomorphic Darboux theorem identifies the completed local symplectic algebra at p with $\mathbb{C}[[z_1, z_2]]$ equipped with $dz_1 \wedge dz_2$. Under this identification Hamiltonians modulo constants become $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}$, and the point-brane Ext algebra is the coordinate-free exterior algebra $\mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p) \cong \Lambda^* T_p X$, with cyclic trace fixed by ω_p. Hence the open BV reduction and the Hamiltonian boundary evaluation are exactly the formal local construction after choosing Darboux coordinates.

A different Darboux coordinate system differs by a formal symplectomorphism, which acts on Hamiltonians and on the normal coordinate algebra; the substitution map $z_i \mapsto \phi_i$ is equivariant for this action.

The exact sequence

$$0 \to \mathbb{C}_X \to \mathcal{O}_X \to \mathcal{H}_X \to 0$$

gives

$$0 \to H^0(X, \mathcal{O}_X)/\mathbb{C} \to H^0(X, \mathcal{H}_X) \xrightarrow{\delta} H^1(X, \mathbb{C}_X) \to H^1(X, \mathcal{O}_X).$$

For a locally Hamiltonian vector field, δ is the period class of the closed one-form $-\iota_v\omega$. Vanishing of this class is exactly the existence of a single global holomorphic Hamiltonian primitive, unique modulo constants. □

Remark 2.26. Proposition 2.25 is the firewall for external holomorphic symplectic surfaces: it concerns restriction to a formal stalk only, not descent for factorization centres or quantum open-closed maps. If $H^1(X, \mathbb{C}) = 0$, global symplectic vector fields which are locally Hamiltonian have global Hamiltonian primitives. If X is Stein, every finite jet needed for a finite local calculation can be lifted to a global holomorphic Hamiltonian, noncanonically: apply Cartan's theorem B to

$$0 \to \mathfrak{m}_p^{k+1} \to \mathcal{O}_X \to \mathcal{O}_X/\mathfrak{m}_p^{k+1} \to 0.$$

This does not imply surjectivity onto $\widehat{\mathcal{O}}_{X,p}$. If X is compact connected, then $H^0(X, \mathcal{O}_X)/\mathbb{C} = 0$; abelian surfaces still have nonzero locally Hamiltonian translations, obstructed by their period classes. Such compact or global examples are comparison data only. A global cochain-level Poisson factorization-center map or a global quantum deformation-quantization statement requires additional descent, anomaly, and background-choice data and is not supplied by the local $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ theorem.

The closed Hamiltonian coefficient Lie algebra is the reduced Tate vector space

$$\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1.$$

We write a class by its unique zero-constant Taylor representative. The bracket is the projected Poisson bracket

$$\{f, g\} = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g \mod \mathbb{C} \cdot 1.$$

Thus $\{z_1, z_2\} = 0$ in $\widehat{\mathfrak{h}}$, although the unreduced Poisson bracket of the chosen representatives is 1. The cotangent BF field contributes the continuous dual, so the residual local closed Lie algebra is

$$\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1].$$

On product opens in $\mathbb{R}^2 \times \mathbb{C}^2$ the closed fields have the mixed form

$$\Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big),$$

with differential $d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ and BF interaction $\frac{1}{2}\int \langle \beta, \{\alpha, \alpha\}\rangle dz_1 dz_2$. Proposition 3.1 checks that this is a local mixed holomorphic-topological BV sector in the sense above.

The open brane is the product of the exact local A-brane $\mathbb{R} \subset T^*\mathbb{R}$ and the B-brane $\mathcal{O}_p$ at the formal holomorphic point. The Yoneda algebra of the skyscraper at the origin is computed in Proposition 3.2; the only Calabi-Yau datum used here is the local residue/cyclic pairing on this two-dimensional Ext

algebra. In the general source convention of cyclic open field theory, the open field of an object $\mathcal{F}$ is $\mathrm{RHom}(\mathcal{F},\mathcal{F})[1]$, and the formal Chern-Simons action

$$S(a) = \sum_{n\geq 1} \frac{1}{n+1}\mathrm{Tr}\big(a\, m_n(a,\dots,a)\big), \qquad a \in \mathrm{RHom}(\mathcal{F},\mathcal{F})[1],$$

is the algebraic core of Witten's open string field theory [4] and of the AKSZ packaging [6]. This is a source convention for the cyclic open BV form, not a compact Calabi-Yau input; the local proof below is the finite Dirac/Koszul matrix calculation. For the local model at hand,

$$\mathrm{RHom}_{\mathbb{C}[z_1,z_2]}(\mathcal{O}_p,\mathcal{O}_p) \cong \Lambda(\varepsilon_1,\varepsilon_2),$$

and the ghost-zero open field is

$$\alpha_0^{\mathrm{coef}} = A_{\mathrm{CS}} + \chi_1\varepsilon_1 + \chi_2\varepsilon_2.$$

The Dirac variables used in the body are $\phi_1 = -\chi_1$, $\phi_2 = \chi_2$, and $A = -A_{\mathrm{CS}}$. In those variables the ghost-zero action is the first-order gauged matrix model of (1), where ϕ_1, ϕ_2 are adjoint scalar normal coordinates and A imposes the moment-map equation $[\phi_1,\phi_2] = 0$.

2.11 *Hamiltonian sector and stable large N*

Everything below is over $\mathbb{C}$. The phrase *formal local model* means the completed Taylor model at the chosen point of $\mathbb{C}^2$; equivalently, one may work polynomially degree by degree and then complete at the augmentation ideal. The symbol $\widehat{\mathbf{S}}$ denotes this completed symmetric algebra, and all Chevalley-Eilenberg cochains are continuous cochains for the same filtration.

Shifts follow the BV convention used in the field descriptions: an element of $V[1]$ has parity reversed and cohomological degree one lower than the corresponding element of V. The notation $\mathbf{S}(V)$ always means the free graded-commutative algebra on the graded vector space V. The CE/PV cotangent block below uses its own coordinate grading: the odd BV cotangent generator $\lambda[1]$ has BV degree -1, while the CE coordinate u_λ dual to it is assigned PV function degree 0. Equivalently, the symbol "dual" in that block means the CE-suspended cotangent coordinate dual, not the ordinary graded dual computed from the BV degree alone.

Let

$$\bar{A} = \mathbb{C}[z_1,z_2]/\mathbb{C}\cdot 1, \qquad \mathfrak{h} = \widehat{\mathfrak{h}} = \mathbb{C}[[z_1,z_2]]/\mathbb{C}\cdot 1, \qquad \mathfrak{h}^\vee = \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}.$$

Constants are removed because a constant Hamiltonian has zero Hamiltonian vector field. We use zero-constant Taylor representatives for calculations, but the Lie bracket is always the Poisson bracket followed by projection modulo constants. On the brane side $\mathrm{Tr}_N(1) = N$ is the empty trace, a central scalar depending on the rank. We do not quotient the associative operator algebra by its unit. The Hamiltonian comparison uses the connected stable trace sector: the unit is retained in the symmetric algebra, but the empty trace is not a primitive generator and constants project to zero in the Hamiltonian Lie/vector quotient. If T is a positive word-degree trace expression, $\overline{T}$ denotes its connected trace class; constants have zero connected class.

At a fixed point of the brane line the reduced BV algebra is generated by two even matrices ϕ_1, ϕ_2 and the odd antifield $\psi = A^*$, with

$$Q\phi_1 = Q\phi_2 = 0, \qquad Q\psi = [\phi_1,\phi_2].$$

Thus the moment-map equation is imposed homologically. Boundary terms on $\mathbb{R}$ are discarded by taking compactly supported variations, or equivalently by working on an interval with fixed boundary values and passing to local operators in the interior.

The term *stable large-N* means the degreewise stable value of the invariant algebras under block restriction as $N \to \infty$. Inside a fixed finite matrix algebra all finite-N trace identities remain present. In the stable limit, Procesi–Razmyslov invariant theory [17, 18] says that invariant functions of two matrices are freely generated by cyclic trace words. The differential $Q\psi = [\phi_1, \phi_2]$ then makes adjacent swaps exact, so ghost-zero cyclic words reduce to ordinary commutative monomials

$$\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \qquad k + l > 0.$$

The primitive one-ψ Koszul homology supplies a second nonconstant family $\Psi_{k,l}$, obtained by inserting one ψ and symmetrising over all words with k copies of ϕ_1 and l copies of ϕ_2. The scalar class $\mathrm{Tr}(\psi)$ is excluded from the Hamiltonian comparison because constants have been removed from $\bar{A}$; higher ψ-count homology lies outside this Hamiltonian sector. The collapse of cyclic words to commutative monomials on $H^0(Q)$ is the algebraic shadow of Dirac reduction: the constraint $Q\psi = [\phi_1, \phi_2]$ is the derived Koszul zero-fibre complex for the moment-map equation. For $N \geq 2$ the commuting-pair equations are not being asserted to form a regular sequence.

Open strings on a stack produce matrix algebras; closed modes appear in the primitive stable part. In this local computation, Procesi–Razmyslov invariant theory controls the stable trace invariant coordinate ring. The Loday–Quillen–Tsygan principle [2, 3]; see also [24, Ch. 13] for the operadic packaging, supplies the cyclic-homology template for primitive large-N classes, not the coordinate-ring invariant theorem itself. The moment-map equation $[\phi_1, \phi_2] = 0$ is the first-class constraint whose Koszul derived zero-fibre complex is exactly the BV reduction $Q\psi = [\phi_1, \phi_2]$. Theorem 2.70 verifies the corresponding connected-trace pattern directly on the Hamiltonian trace sector.

THEOREM 2.27 *(Loday–Quillen–Tsygan, external template).* Let A be a unital associative algebra over a field of characteristic zero and let $\mathfrak{gl}_\infty(A) = \varinjlim_N \mathfrak{gl}_N(A)$. As connected graded-commutative Hopf algebras,

$$H_*^{\mathrm{CE}}\big(\mathfrak{gl}_\infty(A); \mathbb{C}\big) \cong \mathbf{S}\big(HC_{*-1}(A)\big).$$

Equivalently,

$$\mathrm{Prim}\, H_*^{\mathrm{CE}}\big(\mathfrak{gl}_\infty(A); \mathbb{C}\big) \cong HC_{*-1}(A).$$

THEOREM 2.28 *(Stable Eulerian finite-window LQT projection).* Let

$$A_\psi = T(x, y, \psi), \qquad d\psi = xy - yx,$$

with

$$\mathrm{wt}(x) = (1, 0), \qquad \mathrm{wt}(y) = (0, 1), \qquad \mathrm{wt}(\psi) = (1, 1).$$

For a Koszul-weight bound K, let $A_{\psi,K} = A_\psi / F^{>K} A_\psi$, and let L bound CE length. In the stable block-sum Hopf CE window $\mathsf{H}^{\mathrm{st}}_{K,L}(A_\psi)$, with product induced by block diagonal sum and coproduct the CE coproduct, put

$$J = \mathrm{id} - u\epsilon, \qquad P_{1\mathrm{cyc}} = \log^*(\mathrm{id}) = \sum_{m=1}^{L} \frac{(-1)^{m-1}}{m} J^{*m}.$$

Then $P_{1\mathrm{cyc}}$ is a chain idempotent whose image is the stable single-cycle trace span $\mathrm{im}\, \Theta_{N,K,L}$. For

$$N \geq \max(K, L + 2),$$

the finite-window LQT projection is the chain map

$$\lambda_{\mathrm{LQT},K,L} = \Theta_{N,K,L}^{-1} P_{\mathrm{1cyc}} \colon \mathsf{H}^{\mathrm{st}}_{K,L}(A_\psi) \longrightarrow CC_{\mathrm{red},\le L}(A_{\psi,K})[1].$$

If one integer W controls both $K \le W$ and $L \le W$, this means $N \ge W + 2$. The scalar one-ψ line is a chain quotient only after quotienting the two-term scalar-Koszul subcomplex

$$\Theta_1(\psi) \longmapsto \Theta_1(xy - yx), \qquad [\psi] \longmapsto [xy - yx],$$

or after passing to homology.

Proof. The quotient $A_{\psi,K}$ is a dg quotient because $d\psi = xy - yx$ preserves the Koszul bidegree $(1,1)$. The trace cycle map $\Theta_{N,K,L}$ sends a reduced cyclic word $a_0|\cdots|a_r$, $r+1 \le L$, to the usual cyclic matrix tensor

$$\sum_{i_0,\dots,i_r} (E_{i_0 i_1} a_0) \wedge \cdots \wedge (E_{i_r i_0} a_r),$$

and the matrix bracket gives the reduced cyclic differential with the standard graded cyclic signs. In particular $d_{\mathrm{CE}}\Theta_1(\psi) = \Theta_1(xy - yx)$.

The stable CE window is formed before projecting. Block diagonal sum is a chain product, the CE coproduct is a chain coproduct, and $J = \mathrm{id} - u\epsilon$ is a chain map; hence every convolution power J^{*m} and therefore P_{1cyc} is a chain map. Stable invariant CE tensors are indexed by permutations of the CE factors, and a permutation decomposes uniquely into cycles. A single cycle is exactly a trace-cycle tensor in $\operatorname{im}\Theta_{N,K,L}$; a multicycle tensor is a block-sum product of such single cycles. The Hopf logarithm kills decomposable products because on a product of n positive single-cycle generators its coefficient is

$$\sum_{m=1}^{n} (-1)^{m-1}(m-1)!\,S(n,m),$$

the coefficient of $t^n/n!$ in $\log(1 + (e^t - 1)) = t$. This is 1 for $n = 1$ and 0 for $n > 1$, proving idempotence and the stated image. The bound $N \ge K$ removes trace identities in the displayed Koszul window, and $N \ge L + 2$ is the safe CE stable range in length at most L. □

Remark 2.29 *(Same-rank deletion is not the LQT projection).* The operator P_{1cyc} is not the fixed-rank operation that keeps one-cycle permutation tensors and deletes all multicycle tensors inside one exterior CE complex. If P_{del} denotes that operation, then for homogeneous a, b

$$P_{\mathrm{del}} d_{\mathrm{CE}}\big(\Theta_1(a) \wedge \Theta_1(b)\big) = \pm\Theta_1(ab - (-1)^{|a||b|} ba), \qquad d_{\mathrm{CE}} P_{\mathrm{del}}\big(\Theta_1(a) \wedge \Theta_1(b)\big) = 0.$$

Thus same-rank deletion is not a chain map when the graded commutator is nonzero. The theorem uses the stable block-sum Hopf cumulant projection before returning to any finite rank. More precisely, set $B(K,L) = \max(K, L+2)$ and $M = L\,B(K,L)$. Let

$$\beta_{M,K,L} \colon \mathsf{H}^{\mathrm{st}}_{K,L}(A_\psi) \to C_*^{\mathrm{CE}}(\mathfrak{gl}_M(A_{\psi,K}))^{\mathrm{sep}}_{\le L}$$

be the normalized signed average over injections of the connected Hopf factors into the L labeled diagonal $B(K,L)$-blocks, and let

$$\alpha_{M,K,L} \colon C_*^{\mathrm{CE}}(\mathfrak{gl}_M(A_{\psi,K}))^{\mathrm{sep}}_{\le L} \to \mathsf{H}^{\mathrm{st}}_{K,L}(A_\psi)$$

forget the block labels and read back the stable block-sum Hopf word. Thus $\alpha_{M,K,L}\beta_{M,K,L} = \mathrm{id}$. Then

$$P^{\mathrm{sep}}_{M,K,L} = \beta_{M,K,L} P_{1\mathrm{cyc}} \alpha_{M,K,L}$$

is the strict finite-rank chain idempotent on the separated-block subcomplex. This zigzag, not a same-rank endomorphism of one exterior CE complex, is the theorem-grade finite-rank bridge. A same-rank chain-homotopy bridge would have to kill the collision coderivation

$$\Gamma_N = d_{\mathrm{same}} F_N - F_N d_{\oplus},$$

whose first nonzero component is

$$\Gamma_N(\Theta^N_1(a) \wedge \Theta^N_1(b)) = \pm\Theta^N_1(ab - (-1)^{|a||b|} ba).$$

LQT supplies the template; the reduced computation below uses Procesi–Razmyslov invariant theory together with the explicit Koszul differential. No global HC_{*-1} convention is transferred into Theorem 2.70: cyclic homology supplies the stable-trace analogy that organises the comparison, not the proof of the Hamiltonian sector.

2.12 The concrete dictionary

The dictionary uses three disjoint alphabets. Coincident bidegrees are only an indexing convention; they are not an identification of the underlying modules.

Open PBW special fibre. For $k, l \geq 0$ and $k + l > 0$ the open Hamiltonian trace label is

$$O_{k,l} = \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}.$$

The primitive one-antifield PBW special-fibre label is the symmetrised trace

$$\Psi_{k,l} = \sum_{w \in \mathsf{W}_{k,l}} \mathrm{Tr}\big(\psi\, w(\phi_1, \phi_2)\big),$$

where $\mathsf{W}_{k,l}$ is the set of words with k copies of ϕ_1 and l copies of ϕ_2. The symbols $O_{k,l}$ and $\Psi_{k,l}$ are open brane classes in the connected stable PBW special fibre. They are not CE coordinates and they are not principal parts.

Closed CE/PV coordinates. For the reduced Hamiltonian Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, write $H_{k,l} = z_1^k z_2^l$ for the Hamiltonian basis element and $\eta_{k,l} = \delta_{k,l}[1] \in \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ for the shifted cotangent Lie generator, where

$$\delta_{k,l}(z_1^a z_2^b) = \delta_{k,a}\delta_{l,b}.$$

The CE/PV comparison is written in the coordinate alphabet

$$c^{k,l} \quad \text{dual to } H_{k,l}, \qquad u_{k,l} \quad \text{dual to } \eta_{k,l}.$$

The map in the CE/PV theorem sends $c^{k,l}$ to the corresponding Schouten coordinate $\theta^{k,l}$ and sends $u_{k,l}$ to the Hamiltonian central operation represented on the brane by $O_{k,l}$. Thus $u_{k,l}$ is a closed CE coordinate dual to the shifted cotangent generator; it is not the open one-ψ class $\Psi_{k,l}$.

Matlis principal parts. The deformation-compatible cotangent module is the reduced Matlis principal-part module with basis

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2, \qquad a + b > 0,$$

paired with $H_{m,n}$ by residue:

$$\langle \rho_{a,b}, H_{m,n} \rangle = \delta_{a,m}\delta_{b,n}.$$

This realizes $\mathfrak{h}^\vee_{\mathrm{cont}}$ with the principal-part coadjoint action of Proposition 3.227.

The vector-space pairing $e^{\mathrm{PBW}}_{k,l} \mapsto [\Psi_{k,l}]$ used in the PBW special fibre is therefore not the principal-part realization and not an $\mathfrak{h}$-module isomorphism to the Matlis module $\mathcal{P} = \bigoplus \mathbb{C}\rho_{a,b}$. The available passage from the PBW bidegree label carried by $\Psi_{k,l}$ to the principal-part label $\rho_{k,l}$ is the Fourier-polarized Rees $D_\hbar$-module label bridge of Theorem A.10; it is not a same-action $\mathfrak{h}$-module realization and it is not the unreduced quantum boundary observable.

Each later step has a concrete target: construct $O_{k,l}$, $\Psi_{k,l}$ from the open fields; construct $c^{k,l}$, $u_{k,l}$ in the CE/PV closed algebra; construct $\rho_{a,b}$ as the Matlis principal-part basis; prove that commutators of words reduce to commutative monomials in $H^0(Q)$; and verify that the Hamiltonian bracket coefficient is $kn - lm$.

2.13 *Universal cotangent CE/PV spine*

The closed-open comparison is the formal-local instance of a universal cotangent CE/PV construction. The finite identity is purely algebraic: CE cochains on $T^*[1]\mathfrak{l}$ are Schouten polyvectors on $\mathbf{S}(\mathfrak{l})$. Completion at the formal disk keeps the same formula by finite-coordinate tests. Bar-cobar/Koszul duality is the separate admissible enhancement governed by conilpotence and filtered convergence. Thus the boundary Hamiltonian is not a Hamiltonian ghost; it is the function coordinate dual to the cotangent generator. This paragraph uses the cotangent-coordinate grading fixed above: the Schouten/cotangent bracket has degree -1, while the reduced Hamiltonian boundary bracket introduced later is the degree-zero Poisson bracket.

Definition 2.30 *(Reduced Hamiltonian P_0 central-operation complex).* Let A be a complete graded-commutative Poisson algebra with a chosen topology in which the Poisson bivector π and the Schouten contraction bracket act continuously. Its reduced Hamiltonian P_0 central-operation complex is the Poisson cochain algebra

$$Z^{P_0}_{\mathrm{red,Ham}}(A) := \mathrm{PV}(A), \qquad d_\pi = [\pi, -]_S,$$

with the completed Schouten bracket of degree -1. This is a local reduced model for central operations, not a Poisson center in the ordinary sense. For the formal disk this definition is used only on explicitly topologized bracket-admissible completions B, not on the strict ambient coordinate product. It is not the unreduced factorization-centre object, and its Schouten bracket is not the degree-zero reduced Hamiltonian Poisson bracket on boundary currents. No HKR or Hochschild-center identification is asserted here; such an identification would require a continuous HKR theorem compatible with the chosen completion and interval functoriality.

Theorem 2.31 *(Coordinate-free cotangent CE/PV identity).* Let $\mathfrak{l}$ be a finite-dimensional Lie algebra over $\mathbb{C}$, and put

$$T^*[1]\mathfrak{l} = \mathfrak{l} \ltimes \mathfrak{l}^\vee[1].$$

The dual module is the coadjoint module:

$$(x \cdot \lambda)(y) = -\lambda([x, y]).$$

Let $A_{\mathfrak{l}} = \mathbf{S}(\mathfrak{l})$, regarded as functions on $\mathfrak{l}^\vee$, with the Kirillov Poisson bracket. Then there is a canonical isomorphism of dg graded-commutative algebras with their degree-(-1) cotangent/Schouten brackets

$$\Phi_{\mathfrak{l}} : C^\bullet_{\mathrm{CE}}\big(T^*[1]\mathfrak{l}\big) \xrightarrow{\sim} Z^{P_0}_{\mathrm{red,Ham}}(A_{\mathfrak{l}}) = \mathrm{PV}(A_{\mathfrak{l}}).$$

For $x \in \mathfrak{l}$, let O_x be the linear function on $\mathfrak{l}^\vee$ and let u_x be the CE coordinate obtained by pairing the shifted cotangent generator $\mathfrak{l}^\vee[1]$ with x. For $\xi \in \mathfrak{l}^\vee$, let c^ξ be the Hamiltonian CE coordinate and θ^ξ the constant vector field on $\mathfrak{l}^\vee$ in the direction ξ. Then

$$\Phi_{\mathfrak{l}}(u_x) = O_x, \qquad \Phi_{\mathfrak{l}}(c^\xi) = \theta^\xi.$$

Proof. Choose a basis e_i of $\mathfrak{l}$, dual basis ϵ^i, and structure constants $[e_i, e_j] = C^k_{ij} e_k$. Write $O_i = O_{e_i}$, $u_i = u_{e_i}$, $c^i = c^{\epsilon^i}$, and $\theta^i = \theta^{\epsilon^i}$. The CE differential of the shifted cotangent Lie algebra is

$$d_{\mathrm{CE}} c^k = -\frac{1}{2} C^k_{ij} c^i c^j, \qquad d_{\mathrm{CE}} u_k = C^\ell_{kj} u_\ell c^j.$$

The Kirillov Poisson tensor on $A_{\mathfrak{l}}$ is

$$\pi = \frac{1}{2} C^k_{ij} O_k \theta^i \theta^j.$$

The Schouten differential $d_\pi = [\pi, -]_S$ satisfies

$$d_\pi \theta^k = -\frac{1}{2} C^k_{ij} \theta^i \theta^j, \qquad d_\pi O_k = C^\ell_{kj} O_\ell \theta^j.$$

Hence the substitution $c^i \mapsto \theta^i$, $u_i \mapsto O_i$ intertwines the differentials on generators. Both differentials are derivations, so the equality extends to all CE words. The Schouten and degree-(-1) cotangent/Schouten brackets are also identified on generators:

$$\{c^i, u_j\}_{\mathrm{CE}} = \delta^i_j, \qquad [\theta^i, O_j]_S = \delta^i_j,$$

with the same zero brackets among two c's, two u's, two θ's, or two O's. The biderivation rule gives bracket compatibility on the whole algebra. A change of basis acts by the same linear substitution on both sides, so the isomorphism is coordinate-free. □

THEOREM 2.32 *(Formal-disk closed-open CE/PV reconstruction).* Let

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$$

be the reduced Hamiltonian Lie algebra of the formal symplectic disk, and let $A^{\mathrm{st}}_{\partial,H}$ be the stable scalar-reduced Hamiltonian trace algebra. Under the stable trace identification

$$A^{\mathrm{st}}_{\partial,H} \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}),$$

whose connected/indecomposable quotient is $\mathfrak{h}$, the coordinate finite-test construction gives

$$C^\bullet_{\mathrm{CE,coord}}\big(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]\big) \dashrightarrow Z^{P_0}_{\mathrm{red,Ham}}\big(A^{\mathrm{st}}_{\partial,H}\big).$$

This is a coordinate dg-algebra comparison. After a mixed or weighted completion makes the coordinate contraction bracket continuous, the dashed generator map is upgraded to the completed bracketed

comparison on that bracket-admissible realization. In the strict ambient coordinate product no global P_0-bracket is asserted. The boundary evaluation

$$J(f) = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$$

is the image of the shifted-cotangent coordinate u_f. The Hamiltonian CE coordinate c^f maps to the Hamiltonian vector-field operation $\{J(f), -\}$. The coordinate map Φ_{coord} is functorial under formal Darboux coordinate changes. The P_0, kernel, and principal-part enhancements are functorial only for Darboux changes that transport the chosen admissible completion and preserve the diagonal convergence data.

Proof. The finite theorem fixes the generator signs. The formal-disk passage is coefficientwise, not by quotienting $\mathfrak{h}$ to finite Hamiltonian Lie algebras. For each requested output coefficient of $d_{\mathrm{CE}}c$, $d_{\mathrm{CE}}u$, $d_\pi^{\mathrm{coord}}\theta$, or $d_\pi^{\mathrm{coord}}O$, only finitely many monomial labels can contribute after coordinate projection. These finite coefficient checks are compatible under restriction of coordinate sets and determine Φ_{coord}. A completed bracketed limit requires the mixed or weighted continuity condition recorded in Theorem 2.48.

Proposition 3.192, Lemma 3.200, and Corollary 3.202 identify the stable scalar-reduced ghost-zero brane Hamiltonians with the completed symmetric algebra on $\mathfrak{h}$. The matrix Poisson calculation gives

$$\{J(f), J(g)\}_\partial = J(\{f, g\}_{\mathfrak{h}}),$$

so this identification is an isomorphism of reduced Hamiltonian Poisson algebras. Under Theorem 2.31, the linear function coordinate O_f is precisely $\Phi_{\mathrm{coord}}(u_f)$, while $\Phi_{\mathrm{coord}}(c^f) = \theta^f$ acts as the constant vector-field generator whose Schouten differential on O_g is the Hamiltonian action. After choosing a bracket-admissible realization, the closed CE cochains of the shifted cotangent Hamiltonian Lie algebra give the reduced Hamiltonian P_0-central operations on the stable brane algebra.

A formal Darboux change is a formal symplectomorphism. It acts by Lie algebra automorphisms on $\mathfrak{h}$, by Poisson automorphisms on $\widehat{\mathbf{S}}(\mathfrak{h})$, and by substitution on the matrix normal-coordinate evaluation after passing to the scalar-reduced stable trace quotient. The coordinate CE/PV construction is functorial for these Lie algebra automorphisms, so Φ_{coord} does not depend on the chosen Darboux coordinates. A bracket-admissible B, a kernel-admissible tensor product, or a divisor-dependent principal-part completion is extra structure; functoriality of those enhancements requires preservation or explicit transport of that structure. □

2.14 Local theorem dependencies and analytic obstruction

Theorem 2.70 below is the precise local assertion; its constituents are constructed in the subsequent sections. The proof combines: the affine **Ext** calculation (Proposition 3.2); the open BV reduction (Lemma 3.5, Proposition 3.7); the Dirac reduction of the probe phase space (Lemma 3.6); the stable invariant-theory calculation of the Hamiltonian trace algebra (Proposition 3.192); the computation of the primitive one-antifield Koszul homology (Proposition 3.197); the formal Hamiltonian BF model for the closed side (Proposition 3.13); the construction of the bulk-to-boundary Hamiltonian evaluation map (Construction 3.195, Theorem 3.242); the admissible bar-cobar/enveloping computation; and the coordinate-free CE/PV closed-open synthesis of Theorem 2.32; the componentwise quantum coefficient surface of Theorem 3.270; and the first bracket calculation (5).

The formal Weyl/Moyal calculation below is an all-order algebraic deformation-quantisation target for the constant Poisson formal disk; its monomial coefficients are checked by finite exact arithmetic. The quantum coefficient theorem adds the proved weighted surface

$$\mathfrak{S}_{\hbar,w}(I) = \mathcal{K}^{w}_{\mathrm{BV}} \times \{0_{\mathrm{sc}}\} \times A^{\mathrm{pp},w}_{\partial,\hbar}(I) \times T_{\mathrm{Ham},\hbar}.$$

This records weighted coefficient-kernel convergence and graphwise boundedness, the balanced scalar-contact habitat with zero filtered scalar projection, the reduced weighted Moyal principal-part current target, and the radial/Weyl normal-form trace gate. The finite scans certify the balanced diagonal through $(6,6)$, every non-edge bidegree of total degree at most 13, and the total-degree 14 cases

$$(3,11), (11,3), (4,10), (10,4), (5,9), (9,5);$$

the first undecided total-degree 14 systems are $(6,8), (8,6), (7,7)$. The edge strips are no longer only finite evidence: Proposition F.27 proves the PBW telescoping primitive for every $(2,b)$ and $(a,2)$. The remaining uniform radial theorem is the interior dual-potential problem: prove

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^{+}_{a,b})] \in \operatorname{coker} B_{a,b}$$

vanishes for every $a, b \geq 3$, equivalently prove decorated PBW Stokes $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$. A failure is precisely a signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$ with $\lambda(E^{+}_{a,b}) - \phi(T_{a,b}) \neq 0$.

The non-scalar Costello graph/QME surface has been sharpened to the finite-window criterion

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^{1}(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$$

inside $\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$, with compatible finite-window counterterms and a curved bulk-to-defect kernel. The closed finite-window rows α_{sc} and α_{II} vanish; the θ_3 source-face row remains the first named finite-row obstruction until its CE ancestor, local counterterm primitive, or companion-face table is supplied.

For the binary centrality row, the proved statement is exactly the zero-row statement in the reduced coefficient-current and minimal full-equivariant habitat. For

$$x, y \in \{u_{a(t)dt\otimes f}, c_{B,\rho}\}$$

one has

$$R^{\mathrm{cent,coef}}_{x,y,M} = 0, \qquad S^{\mathrm{cent,coef}}_{x,y,M} = 0, \qquad H^{\mathrm{coef},M}_{x,y} = 0.$$

Equivalently, after suppressing zero rows,

$$A^{M}_{\min} : 0 \to 0, \qquad r^{M}_{\min} = 0, \qquad h^{M}_{\min} = 0,$$

with zero Roos primitive class. In a larger Costello local-functional package the same assertion becomes the row problem $A^{M}h^{M} = -r^{M}$; it is not a constructed centrality homotopy until the row values, scalar gate, primitive columns, and transition matrices are supplied.

A closed-string Feynman-integral derivation of the Moyal coefficients is a separate external analytic problem, not an input to the local theorem. It requires the BV/gauge-fixing data of Problem 3.264: an elliptic complex on a chosen analytic avatar of $\mathbb{R}^{2}_{\mathrm{top}} \times \mathbb{C}^{2}_{\mathrm{hol}}$ extending the formal jet model; boundary or defect conditions on $\mathbb{C}^{2}$; a gauge-fixing Q^{GF} and the heat-kernel propagator $P_{\epsilon,L} = \int_{\epsilon}^{L} Q^{\mathrm{GF}} K_t \, dt$; zero-mode treatment and the IR limit; UV renormalisation scheme and local counterterms; quantum master equation or anomaly class; bulk-to-boundary kernel for the brane; trace normalisation and large-N weighting; and the connected/single-trace extraction convention. Costello's renormalized BV formalism supplies the general graph/RG formalism; Costello–Li prove the flat odd-dimensional BCOV/HCS case [13, 14]. The proof of Theorem 2.70 uses the formal local algebra alone and is independent of these analytic choices.

2.15 *The local theorem*

The local theorem has a classical core and a quantum coefficient surface. The classical core has three components. The open calculation identifies the ghost-zero brane theory with the algebra of invariant functions on commuting matrix pairs together with the first Koszul descendants of the commuting-pair equation. The closed calculation constructs the CE/PV coordinates of the Hamiltonian cotangent Lie algebra and realizes its continuous cotangent module by Matlis principal parts. The Hamiltonian boundary map is the local algebraic operation of evaluating a Hamiltonian on the matrix-valued normal coordinates of the brane; the polynomial one-ψ descendants remain PBW special-fibre classes, not principal-part currents.

Equivalently, the theorem constructs and compares the local objects named in Definition 2.19: the formal Hamiltonian holomorphic factorization algebra $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$, its mixed topological-current enhancement over $\mathbb{R}^2_{\mathrm{top}}$, the Weyl/Moyal brane algebra, the defect principal-part current avatars, and the Bochner–Martinelli–Koppelman current data, finite-window obstruction gate, arity-two Matlis comparison, and one-pair analytic pro-Matlis retract

$$P^{\mathrm{an}}_{\Pi} = \tau P_{\mathrm{an}}, \qquad \pi_N P^{\mathrm{an}}_{\Pi} = p_N$$

of Proposition 2.21. The theorem does not construct a one-dimensional VOA or OPE algebra, nor a strict all-arity BM transfer; those require the separate complex-line or holomorphic-defect restriction theorem and, respectively, first the vanishing or replacement of $\mathrm{Ob}^{\mathrm{BMK\text{-}curr}}_N$ and then the six-entry all-window obstruction problem $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$.

The local theorem follows from the component theorems below. Its reduced core is the coordinate CE/PV theorem and admissible Koszul criterion of Theorem 2.69, the brane-line factorization-current lift, the Matlis principal-part cotangent factor after de Rham-current contraction, the scalar-reduced degree-zero Moyal/Weyl deformation, the componentwise quantum coefficient surface, and the U(1) scalar anomaly. The full formal disk is a completed coordinate cochain theorem. The P_0 statement appears only after choosing an explicitly topologized bracket-admissible realization B; boundary Hamiltonian uses require B to contain the relevant O_f's and $d^{\mathrm{coord}}_{\pi} O_f$'s. A kernel appears only after the diagonal converges in a specified kernel-admissible convolution dg Lie algebra. The bar-cobar/enveloping upgrade is an admissibility criterion: admissible Tate, nilpotent, relative, or weighted replacement data promote the coordinate theorem to Koszul duality, and the residual terms are the obstruction data listed in the criterion. The unreduced Tate-coefficient cotangent lift, one-antifield Moyal target, regulator comparison, and mixed brane-defect QME enter through the theorem data of Section 4.

2.15.1 Local theorem dependencies

This section records local hypotheses, asserted scope, and proof inputs. The proof-bearing text is the cited theorem, proposition, lemma, or computation. Each assertion is interpreted with the proof status and admissibility hypotheses recorded in the claim-status table and the stratified Koszul criterion.

Statement 2.33 *(Statement: Dirac brane probe). Status. proved in finite algebraic model.* The proof is confined to the formal local ghost-zero sector of the brane probe.

Exact hypotheses. The transverse formal disk is $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$; the stack size N is finite; fields are $\phi_1, \phi_2, A \in \mathfrak{gl}_N$ on the brane line; the raw shifted-Chern–Simons coefficient fields have first been put in the Dirac normalization of Construction 3.3; variations are compactly supported; and the only residual gauge group retained in this statement is ordinary GL_N-conjugation on degree-zero functions.

Local assertion. For a stack of N branes probing $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$, the ghost-zero open action is the first-order Dirac matrix system

$$S_\partial = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big),$$

and its reduced BV algebra is

$$\mathbb{C}[\phi_1{}^i{}_j, \phi_2{}^i{}_j] \otimes \Lambda(\psi^i{}_j), \qquad Q\psi = [\phi_1, \phi_2], \quad Q\phi_i = 0.$$

The moment map for conjugation is $-\mathrm{Tr}(\epsilon[\phi_1, \phi_2])$. With the paper's convention $\iota_{X_f}\Omega = -df$, its components satisfy $\{\mu_\epsilon, \mu_\eta\} = -\mu_{[\epsilon,\eta]}$; this anti-equivariance affects only the comoment convention, not the zero locus. The BV differential is the derived Koszul zero-fibre differential for the commuting-pair equation, not an underived regular sequence assertion.

Proof inputs. Lemma 3.6 derives the ghost-zero action from the shifted normal directions and fixes the moment-map sign. Proposition 3.7 proves that the ghost-zero truncation is consistent and identifies the residual Koszul complex.

Scope and exclusions. The statement does not assert an untruncated open-string BV model beyond this ghost-zero finite algebraic truncation.

Statement 2.34 *(Statement: unital cyclic Koszul and normal-ordering core). Status. proved algebraically, with an explicit obstruction to the full cyclic contraction*. This is the necklace-level Koszul calculation with the scalar word and one-ψ homology kept explicit.

Exact hypotheses. Work in the unital cyclic dg Lie algebra

$$\mathfrak{p}_{\mathrm{K}} = \mathrm{Cyc}\big(T(\mathbb{C}x \oplus \mathbb{C}y \oplus \mathbb{C}\psi)\big), \qquad Q\psi = xy - yx, \quad Qx = Qy = 0,$$

with the scalar word $[1]$ retained until the bracket has been checked. The scalar-reduced quotient $\bar{\mathfrak{p}}_{\mathrm{K}} = \mathfrak{p}_{\mathrm{K}}/\mathbb{C}[1]$ is taken only after this unital computation. The normal-ordering homotopy requires a chosen cyclic cut and applies to the no-ψ subcomplex selected by that cut.

Local assertion. The cyclic necklace bracket makes $\mathfrak{p}_{\mathrm{K}}$ a dg Lie algebra. The normal-ordering map N is a strict dg Lie map to the unital polynomial Hamiltonian algebra and sends $[1]$ to 1; after quotienting constants it gives the reduced Hamiltonian map $N_{\mathrm{red}} \colon \bar{\mathfrak{p}}_{\mathrm{K}} \to \bar{A}$. On the cut no-$\psi$ sector the straightening homotopy satisfies $QH_c = \mathrm{id} - iN$. There is no cyclic homotopy K with $QK + KQ = \mathrm{id} - iN$ on the full cyclic complex when N kills the primitive one-ψ classes.

Proof inputs. Theorem 3.204 proves the unital dg Lie model and the scalar bracket $\{[x], [y]\} = [1]$. Proposition 3.205 proves the strict normal-ordering quotient. Theorem 3.201 constructs the cut-dependent no-ψ straightening homotopy. Lemma 3.206 proves the obstruction to a full cyclic contraction. Lemma 3.207 records the failure of the naive gauge-paired quotient.

Scope and exclusions. This block is not an unreduced BV/factorization-centre theorem. It does not identify arbitrary open observables with central operations, and polynomial one-ψ descendants do not replace the principal-part cotangent current module.

Statement 2.35 *(Statement: stable trace and one-ψ PBW shadow). Status. proved degreewise stable*. It is a PBW special-fibre trace statement, not a statement about the principal-part cotangent module.

Exact hypotheses. Work at fixed total polynomial degree and then in the stable connected large-N range; pass to scalar-reduced connected traces; use the Koszul differential $Q\psi = [\phi_1, \phi_2]$; and take the PBW special fibre before any filtered or $D_\hbar$-interpolation is introduced.

Local assertion. The Hamiltonian open observables in ghost number zero are generated by scalar-reduced connected trace classes $\overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$. The non-scalar primitive one-ψ classes $\Psi_{a,b}$, $a + b > 0$, are the primitive PBW special-fibre shadow classes. They do not realize the continuous $\mathfrak{h}$-coadjoint module and are not the principal-part currents $\rho_{a,b}$.

Proof inputs. Procesi–Razmyslov stable invariant theory supplies the stable trace generators. The Koszul differential $Q\psi = [\phi_1, \phi_2]$ turns adjacent swaps into boundaries in the connected trace quotient. Proposition 3.197 computes the primitive one-ψ homology. Theorem 3.154 identifies the chain-level primitive shadow, while Theorem 3.232 and Theorem 3.234 separate that PBW shadow from the coadjoint principal-part module. Theorem 3.157 proves the same separation for every $\hbar$-adic deformation satisfying its flatness, PBW-linear-action, Moyal-coadjoint-target, and leading-term hypotheses.

Scope and exclusions. Unstable finite-N trace identities and any ordinary $\mathfrak{h}$-module identification of $\Psi_{a,b}$ with $\rho_{a,b}$ are outside this statement. The Fourier–Rees comparison of Theorem A.10 is only an associated-graded label bridge for the original Hamiltonian action; the legitimate bridge is Fourier-twisted in the Rees $\mathcal{D}_\hbar$-module sense. The corrected quantum target, when needed, is the reduced distributional principal-part target of Construction 3.156, not the PBW one-ψ span.

Statement 2.36 *(Statement: CE/PV reduced Hamiltonian central-operation model). Status. proved degree-wise stable.* The proof is at cochain level for the reduced formal Hamiltonian algebra and its continuous shifted cotangent extension. The bracketed P_0 consequence is used only after a bracket-admissible realization has been chosen, and the bar-cobar consequence is used only under the stated lower-central Tate-pronilpotent hypothesis.

Exact hypotheses. The Hamiltonian algebra is $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ with its m-adic topology; the cotangent direction is $\mathfrak{h}^\vee_{\mathrm{cont}}[1]$, equivalently to $\mathcal{P}[1]$ after the residue scalar is fixed by Theorem 3.230; CE cochains are completed in coordinate products; P_0-brackets require a bracket-admissible subalgebra B; kernels require a kernel-admissible B; and every use of bar-cobar duality is restricted to the lower-central Tate-pronilpotent hypothesis named in the cited lemma.

Local assertion. Let

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$

The coordinate completion of the CE cochains of $\mathfrak{g}$ identifies, windowwise and after the coordinate completion specified in Definition 2.45, with the corresponding Schouten polyvector algebra:

$$\Phi_{\mathrm{coord}} \colon C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}}).$$

On a bracket-admissible $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ the restriction $\Phi^B_{P_0}$ is a dg P_0-isomorphism. On a kernel-admissible B the diagonal tensor Θ_B is the associated Maurer–Cartan kernel. The ambient strict product completion carries neither assertion. The generator dictionary is

$$c^I \mapsto \theta^I, \qquad u_I \mapsto O_I.$$

The $Z^{P_0}_{\mathrm{red,Ham}}$ identification is asserted only on the coordinate-window or mixed/weighted-continuous sector where the Schouten bracket is a continuous operation; the strict ambient product completion is not a P_0-algebra here. Here $|u_I| = 0$, $|c^I| = 1$, and u_I is the CE coordinate dual to the shifted-cotangent generator, not to a Hamiltonian generator.

Proof inputs. Proposition 3.215 identifies closed local operators as continuous CE cochains, which sit inside the coordinate envelope only after the chosen continuity condition is specified. Lemma 2.47 computes the coordinate PV differential. Theorem 3.28 checks $\Phi_{\mathrm{coord}} d_{\mathrm{CE}} = d^{\mathrm{coord}}_\pi \Phi_{\mathrm{coord}}$ on generators

and extends by the completed Leibniz rule. Lemma 3.217 supplies the conilpotent bar-cobar template only in its own hypothesis; the untruncated formal Hamiltonian algebra with linear modes is not used as a lower-central Tate-pronilpotent input.

Scope and exclusions. The statement does not prove an unrestricted factorization-centre theorem or a conilpotence statement for the untruncated formal Hamiltonian algebra. It also does not construct a BV heat-kernel propagator, a BV Laplacian, or a QME-compatible closed-open graph model for the unreduced cotangent sector.

Statement 2.37 *(Statement: local chiral/factorization taxonomy). Status. theorem-bridged local construction, with proved object identifications on the cited local habitats and a definitional firewall against vertex and compact-target overclaims.* This statement fixes what the manuscript may call a chiral object in the local mixed holomorphic-topological theory. It is not a compact target theorem and not a one-dimensional vertex-algebra theorem.

Exact hypotheses. The base geometry is

$$\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}}, \qquad \Omega_{\mathrm{loc}} = dz_1 \wedge dz_2,$$

with Hamiltonian Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ and principal-part cotangent module

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \subset H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2 \cong \mathfrak{h}^\vee_{\mathrm{cont}}.$$

The shifted cotangent algebra may therefore be written either as

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1] \quad \text{or} \quad \mathfrak{g}_{\mathrm{hol}} = \mathfrak{h} \ltimes \mathcal{P}[1],$$

with the residue normalization fixed. Brane-current statements use intervals $I \subset \mathbb{R}_{\mathrm{brane}}$. Weyl/Moyal statements use the Capelli/Weyl convention of the degree-zero quantum theorem. No restriction to a complex line, holomorphic defect curve, compact Calabi–Yau target, or CoHA target is assumed.

Local assertion. The local taxonomy has four objects.

(1) The closed holomorphic object is the formal Hamiltonian holomorphic factorization algebra on the formal $\mathbb{C}^2$-disk:
$$\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}(B) = C^\bullet_{\mathrm{CE,cont}}\big(\Omega^{0,\bullet}_c(B)\widehat{\otimes}(\mathfrak{h} \ltimes \mathcal{P}[1])\big).$$
On a holomorphic polydisk B, its dg Lie input is
$$\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g}_{\mathrm{hol}},$$
and its formal stalk is the completed CE object $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$ only with the coordinate/admissible PV identification supplied by the CE/PV lane.

(2) The mixed topological-current enhancement over $\mathbb{R}^2_{\mathrm{top}}$ is
$$\mathcal{F}^{\mathrm{mix}}_{\mathrm{Ham}}(U \times B) = C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{g}^{\mathrm{mix}}_{U,B}\big),$$
where the coefficient input on a topological open U is
$$\mathfrak{g}^{\mathrm{mix}}_{U,B} = \Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g}.$$
This object is locally constant in U and holomorphic in B. The interval current algebra is its support-local brane-line restriction, not a complex one-dimensional chiral algebra.

(3) The open quantum brane object is the Weyl/Moyal algebra and its matrix or supermatrix cyclic bar-cobar avatars:

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1},$$
$$f * g = m \circ \exp\left(\frac{\hbar}{2} P\right)(f \otimes g),$$
$$\{f, g\}_\hbar = \hbar^{-1}(f * g - g * f),$$

together with the finite-N matrix Weyl relations

$$[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar\, \delta^i_l \delta^k_j.$$

Trace and supertrace cyclic complexes are bookkeeping avatars for the brane algebra and its balanced scalar-contact branch.

(4) The defect/principal-part current object is

$$\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}, \qquad \mathcal{P}_I = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}, \qquad \mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1],$$

with reduced target

$$A^{\mathrm{pp}}_\partial(I) = \widehat{\mathbf{S}}(\Omega^\circ_c(I)\widehat{\otimes}\bar{A})\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1])$$

and brackets

$$\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab),$$
$$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB),$$
$$\{\Theta_\rho(B), \Theta_\sigma(C)\} = 0.$$

Its quantum version replaces $f \cdot \rho$ by $\mathrm{ad}^\vee_{f,\hbar}\,\rho$.

None of these four objects is compact-CY chiral homology, a CoHA, a quintic/GV/OSV object, an Abel–Jacobi normal-function object, an Igusa product, a Borcherds lift, or a BKM algebra. A one-dimensional vertex algebra can enter only after an additional theorem chooses a complex line or holomorphic defect, proves factorization on that line, and constructs the corresponding OPE modes.

Proof inputs. Definition 2.14 and Definition 2.17 supply the holomorphic and mixed factorization meanings. Proposition 2.20 proves the local-field-to-$\mathfrak{g}$-to-CE factorization-observable bridge for $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$ and $\mathcal{F}^{\mathrm{mix}}_{\mathrm{Ham}}$. Theorem 2.48 and Theorem 3.28 identify the formal Hamiltonian CE/PV stalk under their admissibility hypotheses. Theorem 3.22, Theorem 3.237, and Corollary B.8 give the interval current and principal-part brackets. Proposition 3.268 and Theorem 3.251 give the formal Weyl/Moyal coefficients and the radial/Weyl finite-certificate trace gate.

Scope and exclusions. The theorem lanes below may use holomorphic factorization, mixed topological-current factorization, Weyl/Moyal brane algebra, cyclic bar-cobar trace avatars, and principal-part current algebra only in the senses above. They do not assert a vertex algebra, compact-CY chiral homology, CoHA, quintic, OSV/GV, Abel–Jacobi, Igusa, Borcherds, BKM, or global BCOV theorem.

Statement 2.38 *(Statement: native holomorphic E_2 collision lane and Bochner–Martinelli obligation). Status. proved CE/factorization object and proved semidirect collision algebra; BMK current-limit data, finite Matlis moment maps, arity-two output projection, and one-pair analytic pro-Matlis retract proved; strict native all-window current transfer recorded as an exact obstruction*. This is the native holomorphic E_2-type theorem lane on $\mathbb{C}^2$. It is not a curve vertex algebra theorem and not a Costello graph/QME theorem.

Exact hypotheses. Work on a holomorphic polydisk $B \subset \mathbb{C}^2_{\mathrm{hol}}$ in the formal-local model, with

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1, \qquad \mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \cong \mathfrak{h}^\vee_{\mathrm{cont}},$$

where

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \qquad \mathrm{Res}_0(\rho_{a,b} z_1^m z_2^n) = \delta_{a,m}\delta_{b,n}.$$

The topology is the compactly supported Dolbeault topology in B tensored with the m-adic Hamiltonian topology and the continuous principal-part dual. No holomorphic line, transverse vacuum, Laurent-mode expansion, Zhu map, elliptic gauge fixing, BV heat kernel, counterterm scheme, or QME solution is part of this hypothesis.

Native object. The holomorphic algebra is

$$\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}(B) = C^\bullet_{\mathrm{CE,cont}}\big(\Omega^{0,\bullet}_c(B)\widehat{\otimes}(\mathfrak{h} \ltimes \mathcal{P}[1])\big).$$

Equivalently, before passing to compactly supported observables, this is the classical Hamiltonian BF field theory with fields

$$\alpha_{\mathrm{BF}} \in \Omega^{0,\bullet}(B)\widehat{\otimes}\mathfrak{h}[1], \qquad \beta_{\mathrm{BF}} \in \Omega^{0,\bullet}(B)\widehat{\otimes}\mathcal{P}[2],$$

Dolbeault differential $\bar\partial$, and cubic interaction

$$I_{\mathrm{Ham}} = \frac{1}{2}\int_B \langle \beta_{\mathrm{BF}}, \{\alpha_{\mathrm{BF}}, \alpha_{\mathrm{BF}}\}\rangle_{\mathrm{res}}.$$

The compactly supported observable dg Lie input is exactly

$$\Omega^{0,\bullet}_c(B)\widehat{\otimes}(\mathfrak{h} \ltimes \mathcal{P}[1]).$$

The binary collision coefficients are those of the semidirect dg Lie algebra. For compactly supported Dolbeault forms α, β, Hamiltonians $f, g \in \mathfrak{h}$, and principal parts $\rho, \sigma \in \mathcal{P}$,

$$[\alpha \otimes f, \beta \otimes g] = (\alpha \wedge \beta) \otimes \{f, g\},$$

$$[\alpha \otimes f, \beta \otimes \rho[1]] = (\alpha \wedge \beta) \otimes (f \cdot \rho)[1], \qquad [\alpha \otimes \rho[1], \beta \otimes \sigma[1]] = 0,$$

with the ordinary Koszul signs for the tensor product dg Lie algebra. The coadjoint action is fixed by residue integration by parts:

$$\mathrm{Res}_0((f \cdot \rho)g) = -\mathrm{Res}_0(\rho\{f, g\}).$$

In monomial coordinates this gives the current-lane formula

$$z_1^p z_2^q \cdot \rho_{i,j} = -(pj - qi + p - q)\rho_{i-p+1, j-q+1},$$

with negative-index targets equal to zero. Thus binary holomorphic collisions see exactly the Hamiltonian bracket and its coadjoint action; they do not produce a one-variable OPE mode algebra.

Bochner–Martinelli current data and obstruction. On $(\mathbb{C}^2_\zeta \times \mathbb{C}^2_z) \setminus \Delta$, set

$$\eta = \zeta - z, \qquad r^2 = |\eta_1|^2 + |\eta_2|^2, \qquad d\bar\eta_j = d\bar\zeta_j - d\bar z_j.$$

The explicit holomorphic kernel is the Bochner–Martinelli–Koppelman current [33, Ch. III, Def. 1.2, Lem. 1.3]

$$K_{\mathrm{BM}}(\zeta, z) = \frac{1}{(2\pi i)^2} \frac{\overline{\eta_1}\, d\bar{\eta}_2 - \overline{\eta_2}\, d\bar{\eta}_1}{r^4} \wedge d\zeta_1 \wedge d\zeta_2.$$

Source boundary: Laurent–Thiébaut and Ruppenthal support the pre-cutoff BMK kernel, its component $\bar{\partial}$-identity away from the diagonal, and the classical Koppelman homotopy formula on bounded C^1 domains [34, §5, Def. 5.1, Thm. 5.2, Eq. (10)]. They do not supply the cutoff-current limit, the finite Matlis moment projection, the residue-current section, the collar quotient, the Hamiltonian action, the one-pair analytic pro-Matlis retract, or the six-entry native all-window obstruction vector. It gives current-limit data and finite Matlis moment maps, but not a native finite-window homotopy on the full compact-support/current complex. The finite-current obstruction is

$$\mathrm{Ob}_N^{\text{BMK-curr}} = (\mathrm{ob}_{\mathrm{diag}}, \mathrm{ob}_{\mathrm{an/fin},N}, \mathrm{ob}_{\mathfrak{h},N}, \mathrm{ob}_{\text{collar-fact},N}).$$

Its decisive support-local quotient obstruction is

$$z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1}.$$

The projected finite-window L_∞ escape repair is also obstructed: on $(z_1^{N+2}, z_2, \rho_{N,0})$ the projected Jacobiator contributes $(N+1)(N+2)\rho_{0,1}$. The surviving theorem data are:

(1) current-limit existence on fixed finite configuration strata;
(2) finite moment maps p_N and residue-current sections i_N;
(3) the arity-two output projection $\ell_2^{\mathrm{BM},N}(x, y) = p_N[ix, iy]$, identifying the Hamiltonian bracket and Matlis coadjoint action coefficientwise.

The positive all-window statement currently proved is only the one-pair analytic pro-Matlis retract $P_\Pi^{\mathrm{an}} = \tau P_{\mathrm{an}}$, with $\pi_N P_\Pi^{\mathrm{an}} = p_N$. This retract is an internal one-pair analytic construction, not a theorem imported from the BMK/Koppelman sources. The strict native all-window current transfer is obstructed by

$$\mathrm{Ob}_{\mathrm{BM}}^{\Pi} = (\mathrm{ob}_{\oplus}, \mathrm{ob}_{I_\Pi}, \mathrm{ob}_{\mathfrak{h},\Pi}, \mathrm{ob}_{\mathrm{collar},\Pi}, \mathrm{ob}_{3,\Pi}, \mathrm{ob}_{\mathrm{unif},\Pi}),$$

after this finite-current obstruction gate.

Controlled reductions only. A curve chiral algebra, fields $J_f(w)$, Laurent singular products, or a Zhu comparison can enter only after a separate reduction datum

$$(\iota_L, \mathcal{B}_\perp, \mathsf{R}_{L,\mathcal{B}_\perp}, V_L, \mathbf{1}_L, T_L, Y_L, \zeta_{L,\hbar})$$

satisfies the vertex/OPE obligations of Statement 2.39. The support-local real interval current algebra is not such a reduction.

Costello/QME firewall. The Bochner–Martinelli kernel is a Dolbeault homotopy kernel for the holomorphic factorization proof. It is not a BV heat-kernel propagator, does not define a BV Laplacian, and supplies no counterterm, anomaly cancellation, brane-defect graph, or QME solution. Those claims remain in the Costello graph/QME lanes and their stated obstruction complexes.

Statement 2.39 *(Statement: constructed local chiral/factorization algebra and vertex boundary). Status. proved local chiral/factorization construction, with a precise residual boundary for one-dimensional vertex/OPE specialization*. The manuscript has constructed a local Hamiltonian chiral/factorization algebra.

The valid obstruction is narrower: the present data do not yet extract from it a one-complex-dimensional vertex algebra.

Exact hypotheses. The base is

$$\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}, \qquad \Omega_{\mathrm{loc}} = dz_1 \wedge dz_2,$$

with

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$

Via the residue pairing this is equivalently $\mathfrak{h} \ltimes \mathcal{P}[1]$, with $\mathcal{P}$ the principal-part module of Statement 2.38. Brane-current statements use the real brane interval $I \subset \mathbb{R}_{\mathrm{brane}}$. Quantum brane statements use the Weyl/Moyal algebra $(\bar{A}_\hbar, *)$ and the Moyal coadjoint action on principal parts. No compact Calabi–Yau, CoHA, quintic, OSV/GV, Abel–Jacobi, Igusa, Borcherds, or BKM datum is part of the hypothesis.

Local assertion. The constructed local chiral/factorization object is

$$\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}},$$

the formal Hamiltonian holomorphic factorization algebra on $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$. On a holomorphic polydisk B its dg Lie input is

$$\Omega^{0,\bullet}_c(B)\widehat{\otimes}(\mathfrak{h} \ltimes \mathcal{P}[1]).$$

Its formal stalk is the completed CE object $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$, identified with the coordinate/admissible PV model only under the hypotheses of the CE/PV lane. Its mixed enhancement over a topological open $U \subset \mathbb{R}^2_{\mathrm{top}}$ has input

$$\mathfrak{g}^{\mathrm{mix}}_{U,B} = \Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g}.$$

The brane avatars of this local chiral/factorization object are the Weyl/Moyal brane algebra, the matrix or supermatrix cyclic trace avatars, and the support-local principal-part current algebra

$$\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}, \qquad \mathcal{P}_I = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}, \qquad \mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1],$$

with reduced current brackets

$$\begin{aligned} \{B_f(a), B_g(b)\} &= B_{\{f,g\}}(ab), \\ \{B_f(a), \Theta_\rho(B)\} &= \Theta_{f\cdot\rho}(aB), \\ \{\Theta_\rho(B), \Theta_\sigma(C)\} &= 0. \end{aligned}$$

The quantum coefficient surface replaces $f \cdot \rho$ by $\mathrm{ad}^\vee_{f,\hbar}\,\rho$ and uses

$$f * g = m \circ \exp\left(\frac{\hbar}{2}P\right)(f \otimes g).$$

Proof path. Definition 2.19 names the constructed object and its mixed/current/brane avatars. Proposition 2.20 proves the path from the local mixed HT fields through the Hamiltonian reduction to $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ and then to the CE factorization observables $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$ and $\mathcal{F}^{\mathrm{mix}}_{\mathrm{Ham}}$. Definition 2.14 and Definition 2.17 give the holomorphic and mixed factorization habitats. Theorem 2.48 and Theorem 3.28 prove the coordinate CE/PV stalk comparison on the admissible habitats. Theorem 3.22, Theorem 3.237, and Corollary B.8 prove the interval current interface and principal-part brackets. Proposition 3.268,

Theorem 3.251, and Theorem 3.270 give the Weyl/Moyal coefficient surface and the quantum principal-part current action. Statement 2.38 isolates the native two-complex-dimensional binary collision law and the separate Bochner–Martinelli proof obligation.

Vertex/OPE boundary. A one-dimensional vertex algebra is a further specialization of the constructed local chiral/factorization object. It requires all of the following extra data.

(1) A holomorphic embedding
$$\iota_L\colon L = \operatorname{Spf}\mathbb{C}[[w]] \hookrightarrow \widehat{\mathbb{C}^2}_0.$$
The existing brane interval is a real topological line; it is not this complex line.

(2) A transverse datum $\mathcal{B}_\perp$ and a restriction functor
$$\mathsf{R}_{L,\mathcal{B}_\perp}(\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}) = \mathcal{F}_{L,\mathcal{B}_\perp},$$
with independence of normal choices, compatibility with disjoint disks in L, and translation covariance in w.

(3) A factorization-to-vertex comparison producing V_L, a vacuum $\mathbf{1}_L$, and a translation operator T_L.

(4) Finite Laurent OPE modes
$$Y_L(a,w)b = \sum_{n\in\mathbb{Z}}(a_{(n)}b)w^{-n-1}, \qquad a_{(n)}b = 0 \quad (n \gg 0),$$
with vacuum, translation, and locality.

(5) A Weyl/Moyal zero-mode or Zhu comparison
$$\zeta_{L,\hbar}\colon (\bar{A}_\hbar, *) \longrightarrow \mathrm{Zhu}(V_L),$$
or fields $J_f(w)$ whose singular product has specified central level and whose zero modes act on principal parts by $\mathrm{ad}^\vee_{f,\hbar}$.

These five data are not supplied by the present manuscript.

Conclusion. The chiral-algebra construction is present in the precise local sense proved here: the constructed object is the two-complex-dimensional formal Hamiltonian holomorphic factorization algebra and its mixed/current/brane avatars. The manuscript does not yet assert the stronger one-dimensional VOA/OPE theorem. If a tuple
$$\mathfrak{R}_L = (\iota_L, \mathcal{B}_\perp, \mathsf{R}_{L,\mathcal{B}_\perp}, V_L, \mathbf{1}_L, T_L, Y_L, \zeta_{L,\hbar})$$
satisfying the five displayed obligations is supplied, then V_L is a one-dimensional vertex/OPE algebra compatible with the Weyl/Moyal brane algebra through $\zeta_{L,\hbar}$ and the stated zero-mode coadjoint action. This implication is only the formal consequence of the supplied vertex data; it is not needed for the constructed local chiral/factorization algebra. In the $\mathbb{C}\times\mathbb{R}$ interface below, such a generic V_L datum is admissible only after it factors through $\mathcal{R}_{\mathbb{C}\times\mathbb{R}}$ and is identified with V_{red} in $\mathfrak{I}_{\mathrm{ch}}$.

Statement 2.40 *(Statement: controlled $\mathbb{C}\times\mathbb{R}$ reduction with retained z_2-coefficients). Status. theorem-surface with an algebraically stable coefficient system and analytic pushforward obligations.* This is the only admissible route from the native $\mathbb{R}^2_{\mathrm{top}}\times\mathbb{C}^2_{\mathrm{hol}}$ factorization object to a $\mathbb{C}\times\mathbb{R}$ curve shadow. It is not a replacement of the native $\mathbb{C}^2$ holomorphic factorization algebra by a curve vertex algebra.

Exact hypotheses. Let
$$X = \mathbb{R}_t\times\mathbb{R}_s\times\mathbb{C}_{z_1}\times\mathbb{C}_{z_2}, \qquad Y = \mathbb{R}_t\times\mathbb{C}_{z_1}, \qquad \pi(t,s;z_1,z_2) = (t;z_1).$$

A controlled reduction datum is

$$\mathcal{R}_{\mathbb{C}\times\mathbb{R}} = (\pi, \kappa_{\mathrm{top}}, \kappa_{\mathrm{hol}}, B_{z_2}, \pi_!, L_{\mathrm{red}}, \langle -,-\rangle_{\mathrm{red}}, \{-,-\}_{\mathrm{red}}, H_{\mathrm{anom}}).$$

The topological part contains, on each oriented compact-support s-interval J, maps

$$p_s : \Omega_c^\bullet(J) \to \mathbb{C}[-1], \qquad i_s : \mathbb{C}[-1] \to \Omega_c^\bullet(J), \qquad K_s,$$

with

$$d_s K_s + K_s d_s = \mathrm{id} - i_s p_s.$$

This keeps the compact-support orientation line $H_c^1(J) \cong \mathbb{C}[-1]$; it is not an acyclicity assertion. The holomorphic part must either construct a Dolbeault z_2-fiber datum

$$\bar\partial_{z_2} K_{z_2} + K_{z_2}\bar\partial_{z_2} = \mathrm{id} - i_{z_2} p_{z_2}$$

with support and factorization control, or use the equivalent full z_2-principal-part coefficient system.

Reduced coefficient system. The reduced Hamiltonian algebra is

$$\mathfrak{h}_{\mathrm{red}} = \mathbb{C}[[z_1]][[z_2]]/\mathbb{C}\cdot 1,$$

read as a coefficient system over Y. The cotangent module is the full two-index principal-part module

$$\mathcal{P}_{\mathrm{red}} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) \subset H^2_{(z_1,z_2)}(\mathbb{C}[[z_1,z_2]])\, dz_1 dz_2, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2.$$

Its pairing is

$$\langle \rho_{a,b}, z_1^m z_2^n\rangle_{\mathrm{red}} = \delta_{a,m}\delta_{b,n}.$$

The reduced bracket remains the two-variable Hamiltonian bracket

$$\{f,g\}_{\mathrm{red}} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g \pmod{\mathbb{C}},$$

so that

$$\{z_1^a z_2^b, z_1^c z_2^d\}_{\mathrm{red}} = (ad-bc) z_1^{a+c-1} z_2^{b+d-1}.$$

The shifted cotangent algebra is

$$\mathfrak{g}_{\mathrm{red}} = \mathfrak{h}_{\mathrm{red}} \ltimes \mathcal{P}_{\mathrm{red}}[1],$$

with the same residue coadjoint action

$$z_1^p z_2^q \cdot \rho_{i,j} = -(pj - qi + p - q)\rho_{i-p+1, j-q+1},$$

negative-index targets being zero.

Local assertion. Under the datum $\mathcal{R}_{\mathbb{C}\times\mathbb{R}}$, the brane image is

$$L_{\mathrm{red}} = \mathbb{R}_t \times \{z_1 = 0\} \subset Y,$$

but the second normal matrix survives:

$$\mathrm{ev}_N^{\mathrm{red}}(f) = \overline{\mathrm{Tr}\, f(X,Y)}, \qquad X = \phi_1, \quad Y = \phi_2.$$

The support-local reduced currents are

$$B_f(a) = \int_I a(t)\,\overline{\operatorname{Tr} f(X(t), Y(t))}\,dt, \qquad \Theta_\rho(B), \quad B \in \mathcal{D}'_c(I),$$

with brackets

$$\{B_f(a), B_g(b)\} = B_{\{f,g\}_{\mathrm{red}}}(ab),$$
$$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\} = 0.$$

The scalar anomaly survives because $\{z_1, z_2\} = 1$:

$$\mathrm{Ob}^{\mathrm{red}}_{\mathrm{sc}}(\gamma_1; a, f; b, g) = \hbar N\,[\{f, g\}_{\mathrm{red}}]_0 \int_{\mathbb{R}_t} a(t)b(t)\gamma_1(t)\,dt.$$

The same scalar-cancellation alternatives as in the unreduced theorem must therefore be chosen again. In the quantum branch the Moyal product remains

$$f * g = m \circ \exp\left(\frac{\hbar}{2}(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})\right)(f \otimes g).$$

Killing z_2, or keeping only the $b = 0$ principal-part residue line, is not an admissible reduction: it kills the bracket, fails $\mathfrak{h}_{\mathrm{red}}$-equivariance since $z_1 \cdot \rho_{a,0} = -\rho_{a,1}$, and cannot match the matrix Weyl relation $[X^i{}_j, Y^k{}_\ell] = \hbar\delta^i_\ell\delta^k_j$.

Proof inputs and obligations. Statement 2.38 supplies the native two-complex-dimensional object that is being reduced. Statement 2.59 and Statement 2.61 supply the principal-part and interval-current formulas. Statement 2.62 supplies the Weyl/Moyal coefficient convention, while Statement 2.67 supplies the scalar anomaly. What is not yet supplied is the analytic $\pi_!$-theorem: the s-orientation sign, the z_2-Dolbeault or residue pushforward with support control, the reduced BV pairing degrees, factorization compatibility on two and three disks, and the anomaly homotopy H_{anom}.

Obstruction vector. The open datum is

$$\mathrm{Ob}_{\mathbb{C}\times\mathbb{R}} = (\mathrm{ob}_{s,\mathrm{or}}, \mathrm{ob}_{z_2,\mathrm{fib}}, \mathrm{ob}_{\mathrm{BV,pair}}, \mathrm{ob}_{\mathrm{fac}}, \mathrm{ob}_{\mathrm{brane}}, \mathrm{ob}_{\mathrm{anom}}, \mathrm{ob}_{\mathrm{Moyal}}).$$

A curve chiral or Zhu theorem may use this lane only after the relevant components vanish or are carried as explicit hypotheses.

Statement 2.41 (Statement: reduced Dirac BRST curve algebra and Zhu bridge). Status. typed theorem surface after Statement 2.40, with false transfer before it. This statement records the curve vertex and Zhu package that may be targeted after controlled reduction. It is not native to $\mathbb{C}^2$.

Exact hypotheses. Assume a controlled reduction datum $\mathcal{R}_{\mathbb{C}\times\mathbb{R}}$ as in Statement 2.40, including the retained z_2-coefficient or principal-part system and an anomaly convention. Assume also a factorization-to-vertex theorem on the z_1-line, producing a $\mathbb{Z}_{\geq 0}$-graded vertex algebra with vacuum, translation, finite Laurent singular products, and conformal weights

$$\mathrm{wt}(Z_1) = 0, \qquad \mathrm{wt}(Z_2) = 1, \qquad \mathrm{wt}(c) = 0, \qquad \mathrm{wt}(b) = 1.$$

The finite-N gauge convention removes the scalar ghost or treats it as a decoupled acyclic factor, so that the effective reduction is the $\mathfrak{sl}_N$ quantum Hamiltonian reduction.

Reduced vertex data. The free reduced Dirac vertex algebra has matrix-valued fields

$$Z_1{}^i{}_j(z), \quad Z_2{}^i{}_j(z) \quad \text{even}, \qquad c^i{}_j(z), \quad b^i{}_j(z) \quad \text{odd},$$

with singular products

$$Z_1{}^i{}_j(z) Z_2{}^k{}_\ell(w) \sim \frac{\hbar\, \delta^i_\ell \delta^k_j}{z-w}, \qquad b^i{}_j(z) c^k{}_\ell(w) \sim \frac{\delta^i_\ell \delta^k_j}{z-w},$$

all other generating singular products being regular. The BRST current and residue operator are

$$j_{\mathrm{BRST}} = \mathrm{Tr}(c[Z_1, Z_2]) + \frac{1}{2}\mathrm{Tr}(b[c,c]), \qquad Q_{\mathrm{BRST}} = \mathrm{Res}\, j_{\mathrm{BRST}}(z)\, dz.$$

They give

$$QZ_a = [c, Z_a], \qquad Qc = \frac{1}{2}[c,c], \qquad Qb = [Z_1, Z_2] + [b,c].$$

The square-zero check is the finite BRST Jacobi computation: $Q[Z_1, Z_2] = [c, [Z_1, Z_2]]$, and hence $Q^2 Z_a = Q^2 c = Q^2 b = 0$ after the displayed anomaly convention is included.

Zero modes and Zhu relation. With

$$\phi_a = Z_{a,0}, \qquad \psi = b_0,$$

the gauge-invariant Koszul zero-mode sector recovers

$$Q\phi_1 = Q\phi_2 = 0, \qquad Q\psi = [\phi_1, \phi_2].$$

For Zhu zero modes $o(a) = a_{\mathrm{wt}(a)-1}$, use the transpose-compatible convention

$$X^j{}_i = o(Z_1{}^i{}_j), \qquad Y^j{}_i = o(Z_2{}^i{}_j), \qquad C^j{}_i = o(c^i{}_j), \qquad B^j{}_i = o(b^i{}_j).$$

Then

$$[Y^j{}_i, X^\ell{}_k] = \hbar\, \delta^\ell_i \delta^j_k, \qquad [B^j{}_i, C^\ell{}_k]_{\mathrm{super}} = \delta^\ell_i \delta^j_k.$$

The pre-BRST Zhu algebra is

$$W_\hbar(\mathrm{Mat}^2_N) \otimes \mathrm{Cliff}_{bc}(\mathfrak{gl}_N),$$

and the Zhu BRST differential is

$$QX = [C, X], \qquad QY = [C, Y], \qquad QC = \frac{1}{2}[C, C],$$

$$QB = YX - XY + \hbar N I + [B, C].$$

The term $\hbar N I$ is the Capelli contact shift; omitting it contradicts the Weyl/Moyal normalization of Statement 2.62. In degree zero the target is therefore the BRST quantum Hamiltonian reduction with moment relation

$$YX - XY + \hbar N I = 0.$$

Comparison chain. The admissible bridge is the typed chain available only after the reduction and vertex data above are supplied:

$$\mathrm{ChirHoch}^\bullet(V_N^{\mathrm{Dir}}) \longrightarrow HH^\bullet(A(V_N^{\mathrm{Dir}})) \longrightarrow_{\mathrm{gr}_{\hbar=0},\, N\to\infty} HH^\bullet(A_{\mathrm{cl,st}})$$

$$\longrightarrow_{\mathrm{HKR}} \mathrm{PV}^\bullet(\widehat{S}(\mathfrak{h}_{\mathrm{red}})) \longrightarrow C^\bullet_{\mathrm{CE}}(\mathfrak{h}_{\mathrm{red}} \ltimes \mathcal{P}_{\mathrm{red}}[1]).$$

It is not a finite-N, finite-$\hbar$ quasi-isomorphism. The Zhu bridge requires the vertex hypotheses above; HKR is classical, completed, and stable; and the final CE/PV comparison uses the admissible retained-z_2 Hamiltonian completion.

Scope and residuals. The reduced BRST/Zhu package still requires the analytic reduction theorem, the conformal vector and weight proof, the residue normalization giving the OPE coefficient $\hbar$, the anomaly homotopy matching the scalar branch, and continuous HKR in the chosen completed stable algebra. Until then it is a typed theorem surface, not a native conclusion.

Statement 2.42 (Statement: constructed chiral algebra interface). Status. theorem-surface-with-missing-construction after controlled reduction; false transfer before it. A constructed curve chiral algebra enters the present local mixed HT theory only as the vertex factor V_{red} in the interface

$$\mathfrak{I}_{\mathrm{ch}} = (\mathcal{R}_{\mathbb{C}\times\mathbb{R}}, \mathcal{F}_{\mathrm{red}}, V_{\mathrm{red}}, Q_{\mathrm{BRST}}, \mathbf{1}, T, Y, \mathrm{wt}, \zeta_\hbar, H_{\mathrm{anom}}).$$

Here $\mathcal{R}_{\mathbb{C}\times\mathbb{R}}$ is the retained-mode reduction of Statement 2.40; it keeps $\mathfrak{h}_{\mathrm{red}} = \mathbb{C}[[z_1]][[z_2]]/\mathbb{C}\cdot 1$ and the full two-index principal-part module $\mathcal{P}_{\mathrm{red}}$. $\mathcal{F}_{\mathrm{red}}$ is the pushed-forward factorization algebra on $\mathbb{R}_t \times \mathbb{C}_{z_1}$ with this retained coefficient system; before the reduction obstruction vanishes it is a theorem target, not native data. Thus the chiral algebra is a reduced shadow of $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$, not the native $\mathbb{C}^2$ holomorphic E_2 object.

The vertex component must construct finite Laurent singular products, vacuum, translation, locality, conformal weights, and a BRST operator Q_{BRST} with $Q^2_{\mathrm{BRST}} = 0$ after the selected scalar anomaly branch. The Zhu component is a continuous zero-mode map

$$\zeta_\hbar \colon (\bar{A}_\hbar, *) \longrightarrow \mathrm{Zhu}(V_{\mathrm{red}})$$

compatible with the Weyl/Moyal product, the Capelli shift $\hbar N I$, and the principal-part action

$$[J_{f,(0)}, \Theta_\rho] = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\,\rho}.$$

The exact obstruction vector is

$$\mathrm{Ob}_{\mathfrak{I}_{\mathrm{ch}}} = (\mathrm{Ob}_{\mathrm{red}}, \mathrm{Ob}_{\mathrm{vert}}, \mathrm{Ob}_{\mathrm{Zhu}}, \mathrm{Ob}_{\mathrm{nat}}).$$

The entries record, respectively, the s, z_2 pushforward, BV pairing, factorization, brane image, anomaly and Moyal compatibility; the factorization-to-vertex and BRST data; the zero-mode/Zhu multiplicativity and Capelli/HKR comparison; and the comparison back to the native $\mathbb{C}^2$ factorization source. Vanishing of this vector is the admission theorem for the constructed chiral algebra.

Cochain-level CE/PV–Koszul block.

Definition 2.43 (Cotangent CE/PV datum). A cotangent CE/PV datum is a Lie algebra $\mathfrak{l}$ in a complete linear category, a continuous dual module $\mathfrak{l}^\vee$, and the shifted cotangent Lie algebra

$$T^*[1]\mathfrak{l} = \mathfrak{l} \ltimes \mathfrak{l}^\vee[1], \qquad [x, \lambda[1]] = (x\cdot\lambda)[1], \quad [\lambda[1], \mu[1]] = 0.$$

The dual action is the coadjoint action:

$$(x\cdot\lambda)(y) = -\lambda([x, y]).$$

Choose a basis e_i of $\mathfrak{l}$, dual basis ϵ^i, and structure constants $[e_i, e_j] = C_{ij}^k e_k$. The CE coordinates are c^i, dual to e_i, and u_i, dual to $\epsilon^i[1]$, with

$$|c^i| = 1, \qquad |u_i| = 0.$$

This is the cotangent-coordinate grading: u_i is the CE-suspended coordinate dual to the odd cotangent generator, not the ordinary graded dual computed from the BV degree of $\epsilon^i[1]$. The polyvector model is

$$\mathrm{PV}\big(\mathbf{S}(\mathfrak{l})\big) = \mathbf{S}(O_i) \otimes \Lambda(\theta^i), \qquad |O_i| = 0, \quad |\theta^i| = 1,$$

equipped with the Lichnerowicz differential for the linear Poisson tensor

$$\pi = \frac{1}{2} C_{ij}^k O_k \theta^i \theta^j.$$

The completed version is obtained by replacing tensor, symmetric, and exterior powers by their continuous completed analogues.

THEOREM 2.44 *(Finite cotangent CE/PV identity).* Let $\mathfrak{l}$ be finite-dimensional. In the notation of Definition 2.43, the assignment

$$\Phi(c^i) = \theta^i, \qquad \Phi(u_i) = O_i$$

extends uniquely to an isomorphism of dg graded-commutative algebras with their degree-(-1) cotangent/Schouten brackets

$$\Phi\colon C_{\mathrm{CE}}^\bullet\big(\mathfrak{l} \ltimes \mathfrak{l}^\vee[1]\big) \xrightarrow{\sim} \mathrm{PV}\big(\mathbf{S}(\mathfrak{l})\big).$$

The differentials are

$$d_{\mathrm{CE}} c^k = -\frac{1}{2} C_{ij}^k c^i c^j, \qquad d_{\mathrm{CE}} u_k = C_{kj}^\ell u_\ell c^j,$$

and

$$d_\pi \theta^k = -\frac{1}{2} C_{ij}^k \theta^i \theta^j, \qquad d_\pi O_k = C_{kj}^\ell O_\ell \theta^j.$$

Proof. The underlying graded algebras are freely generated by the same even and odd coordinate sets after the displayed substitution. Hence Φ is an isomorphism of graded-commutative algebras.

The CE formula for c^k is the ordinary Chevalley–Eilenberg differential dual to $[e_i, e_j] = C_{ij}^k e_k$. The formula for u_k is the dual of the coadjoint action in the semidirect product: $[e_k, e_j] = C_{kj}^\ell e_\ell$ makes e_j act on the coordinate u_k by $C_{kj}^\ell u_\ell$. On the PV side, $d_\pi = [\pi, -]_S$ and the Schouten identities

$$[\theta^i, O_k]_S = \delta_k^i, \qquad [\theta^i, \theta^j]_S = [O_i, O_j]_S = 0$$

give exactly the two displayed Lichnerowicz formulas. Thus $\Phi d_{\mathrm{CE}} = d_\pi \Phi$ on generators. Both differentials are degree-one derivations, so equality propagates to all words.

The degree-(-1) cotangent/Schouten brackets are also checked on generators:

$$\{c^i, u_j\}_{\mathrm{CE}} = \delta_j^i, \qquad [\theta^i, O_j]_S = \delta_j^i,$$

with all c, c, u, u, θ, θ, and O, O brackets equal to zero. The biderivation rule gives the full bracket identity. The square-zero check on the cotangent coordinate is the coadjoint representation identity in coordinates:

$$d_{\mathrm{CE}}^2 u_k = C_{kj}^\ell C_{\ell r}^m u_m c^r c^j - \frac{1}{2} C_{kj}^\ell C_{rs}^j u_\ell c^r c^s = 0,$$

because $[\mathrm{ad}_{e_r}^\vee, \mathrm{ad}_{e_s}^\vee] = \mathrm{ad}_{[e_r, e_s]}^\vee$. $\square$

Definition 2.45 *(Formal-disk coordinate CE/PV completion).* Let $H_d = \mathrm{Span}\{z_1^a z_2^b : a + b = d\}$ and

$$\mathfrak{h} = \prod_{d \geq 1} H_d = (z_1, z_2)\mathbb{C}[[z_1, z_2]].$$

Its continuous dual is $\mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d \geq 1} H_d^\vee$. Write $c^{a,b}$ for the CE coordinate dual to $H_{a,b} = z_1^a z_2^b$, and $u_{a,b}$ for the CE coordinate dual to the shifted cotangent generator $(H_{a,b})^\vee[1]$. When $I = (a, b)$ with $a + b > 0$, write

$$H_I = z_1^a z_2^b, \qquad \eta^I = (H_I)^\vee[1] \in \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$

For $M, L \geq 0$, let $\mathcal{B}^{\mathrm{CE}}_{M,L}$ be the finite set of graded-commutative monomials in the letters $u_{a,b}$, $c^{a,b}$ of word length at most L, all of whose labels satisfy $1 \leq a + b \leq M$. The coordinate product algebra used here is the product over all finite words, equivalently the inverse limit of finite coordinate projections

$$C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g}) := \widehat{\mathbf{S}}_{\mathrm{coord}}(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]) = \varprojlim_{M,L} \prod_{w \in \mathcal{B}^{\mathrm{CE}}_{M,L}} \mathbb{C}w, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$

The PV completion is defined by the same rule with $u_{a,b}$ replaced by $O_{a,b}$ and $c^{a,b}$ by $\theta^{a,b}$:

$$A_{\partial,\mathrm{coord}} := \widehat{\mathbf{S}}_{\mathrm{coord}}(\mathfrak{h}), \qquad \mathrm{PV}_{\mathrm{coord}}(\mathfrak{h}) = \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}}) := \widehat{\mathbf{S}}_{\mathrm{coord}}(O_{a,b})\widehat{\otimes}\widehat{\Lambda}_{\mathrm{coord}}(\theta^{a,b}).$$

These product algebras are coordinate enlargements. They strictly contain the literal continuous CE cochains: for example an infinite length-one sum in the $c^{a,b}$-coordinates is a vector in the coordinate product but not a continuous linear functional on $\mathfrak{h} = \prod_d H_d$. A genuine completed P_0-object is the mixed continuous subspace in which the c/θ-linear directions have continuous-dual support and the contraction pairing is a continuous biderivation. Without this mixed condition the construction is only an ambient pro-coordinate product. The projections are finite coordinate tests for the displayed differential formulas, not Lie-algebra quotients of the full formal Hamiltonian algebra. A map out of either coordinate product is *coordinate-continuous* when every output coefficient depends on finitely many input coefficients after projection to any finite set of word labels. Multiplication is defined by the usual finite decompositions of a fixed output word into two subwords; hence each output coefficient of a product is a finite sum. The infinite behavior enters through the differential and contraction formulas, where coordinate-continuity or bracket-admissibility is the required summability condition.

Definition 2.46 *(Bracket- and kernel-admissible CE/PV realizations).* Let $A_{\partial,\mathrm{coord}}$ and $\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ be as in Definition 2.45. A *bracket-admissible realization* is a complete Hausdorff topological subalgebra $i_B\colon B \hookrightarrow \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ with continuous inclusion such that multiplication $B\widehat{\otimes}B \to B$, the restricted differential $d_\pi^{\mathrm{coord}}\colon B \to B$, and the Schouten contraction bracket $B\widehat{\otimes}B \to B[-1]$ are continuous. The source $\Phi^{-1}_{\mathrm{coord}}(B)$ is given the pullback topology; it is required to be closed under d_{CE}, multiplication, and the transported cotangent bracket, with these operations continuous. For such B, write

$$\Phi^B_{P_0}\colon \Phi^{-1}_{\mathrm{coord}}(B) \xrightarrow{\sim} B$$

for the restricted completed dg P_0-map.

If $S \subset \mathfrak{h}$ is a class of Hamiltonians, B is *S-Hamiltonian-spanning* if $O_f \in B$ and $d_\pi^{\mathrm{coord}} O_f \in B$ for every $f \in S$. Boundary Hamiltonian statements inside $\Phi^B_{P_0}$ are made only for such f.

A *kernel-admissible datum* is a bracket-admissible B together with a specified completed tensor product, hence a completed convolution dg Lie algebra $\mathcal{K}_B$, in which the diagonal sums

$$\sum_I H_I \otimes \theta^I, \qquad \sum_I \eta^I \otimes O_I$$

converge and the differential and convolution bracket are continuous. In that case set, as an element of $\mathcal{K}_B$,

$$\Theta_B = \sum_I H_I \otimes \theta^I + \sum_I \eta^I \otimes O_I.$$

The strict ambient product/direct-sum completion is not kernel-admissible unless this diagonal convergence is supplied as additional structure.

The finite-support subalgebra B_{fin} is a finite-test model in every finite coordinate window; it is not a global bracket-admissible B for the full formal disk unless the chosen completion is also stable under the infinite coadjoint differential. For $q > 1$ the weighted rapidly decreasing test target E_q is defined by the seminorms

$$p_n(x) = \sup_{a,b} |x_{a,b}|\, q^{a+b}(1 + a + b)^n, \qquad n \geq 0,$$

and by the analogous finite-word seminorms on polyvectors. The monomial estimate

$$|ad - bc| \leq (a + b)(c + d)$$

is only the first local convolution bound needed to absorb the Poisson structure constants into the polynomial factor of the seminorms. Thus a Köthe completion built from these seminorms is a candidate bracket-admissible B only after the chosen finite-word tensor convention proves multiplication, the coordinate differential, and the Schouten contraction continuous; it is kernel-admissible only if the two diagonal sums above converge in that completed tensor product.

LEMMA 2.47 *(Formal-disk coordinate differential formulas).* In the coordinate product completion of Definition 2.45, the formulas

$$[H_{a,b}, H_{c,d}] = (ad - bc)H_{a+c-1,b+d-1}$$

with $H_{r,s} = 0$ for $r < 0$, $s < 0$, or $r + s = 0$, determine degree-one coordinate-continuous derivations d_{CE} and d_π^{coord}, and these derivations satisfy $d_{\mathrm{CE}}^2 = (d_\pi^{\mathrm{coord}})^2 = 0$. On generators,

$$d_{\mathrm{CE}} c^{r,s} = -\frac{1}{2} \sum_{\substack{a+c-1=r \\ b+d-1=s}} (ad - bc) c^{a,b} c^{c,d},$$

$$d_{\mathrm{CE}} u_{r,s} = \sum_{c,d} (rd - sc)\, u_{r+c-1,s+d-1} c^{c,d},$$

and the coordinate PV differential d_π^{coord} has the identical formulas after $c^{a,b} \mapsto \theta^{a,b}$ and $u_{a,b} \mapsto O_{a,b}$. Terms whose output label is the reduced constant $H_{0,0}$ are omitted.

Proof. For $d_{\mathrm{CE}} c^{r,s}$, the equations $a + c - 1 = r$ and $b + d - 1 = s$ have finitely many non-negative solutions, hence the displayed sum is finite.

The coadjoint term $d_{\mathrm{CE}} u_{r,s}$ is infinite before projection. Its projection to the finite coordinate window $\mathcal{B}_{M,L}^{\mathrm{CE}}$ keeps only terms with $c + d \leq M$ and $r + s + c + d - 2 \leq M$. There are finitely many such pairs (c, d). The word length increases by one, so the length projection is finite as well. Thus every finite coordinate projection of $d_{\mathrm{CE}} u_{r,s}$ is finite, which is exactly continuity in the product topology.

A completed monomial has finite word length before passage to the product completion, so the derivation rule applies to finitely many factors. Continuity extends the derivation uniquely to arbitrary

products over the finite-word basis. The PV coordinate formulas have the same coefficient support, with O, θ in place of u, c, so the same argument proves continuity of d_π^{coord}.

The square-zero assertions are checked before completion and then extended by continuity. The CE formulas are the continuous dual of the genuine Hamiltonian Lie bracket on $\mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, equivalently on the zero-constant representatives $(z_1, z_2)\mathbb{C}[[z_1, z_2]]$, whose Jacobi identity gives $d_{\mathrm{CE}}^2 c^{r,s} = 0$. On the cotangent coordinates,

$$d_{\mathrm{CE}}^2 u_{r,s} = C^L_{(r,s)J} C^M_{LR} u_M c^R c^J - \frac{1}{2} C^L_{(r,s)J} C^J_{RS} u_L c^R c^S = 0,$$

where the constants C^K_{IJ} are those of $[H_I, H_J] = C^K_{IJ} H_K$, and the equality is exactly $[\mathrm{ad}^\vee_{H_R}, \mathrm{ad}^\vee_{H_S}] = \mathrm{ad}^\vee_{[H_R, H_S]}$. On the PV side the displayed coordinate formulas define a derivation d_π^{coord}; its square on generators is the same Jacobi and coadjoint identity with O, θ replacing u, c. This avoids asserting that the infinite linear Poisson bivector is a globally summable Schouten element in the strict ambient product. Since both differentials are continuous coordinate derivations, generator-level square-zero implies square-zero on the completed coordinate product algebras. □

Theorem 2.48 *(Coordinate formal-disk CE/PV theorem and admissible P_0 realizations).* Let

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1 \cong (z_1, z_2)\mathbb{C}[[z_1, z_2]]$$

where the isomorphism chooses the zero-constant representative, with the projected Poisson bracket modulo constants

$$\{f, g\} = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g.$$

Complete the coordinate CE and PV algebras as in Definition 2.45. Then

$$\Phi_{\mathrm{coord}} \colon C^\bullet_{\mathrm{CE,coord}}\big(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]\big) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(\mathfrak{h}), \qquad c^{a,b} \mapsto \theta^{a,b}, \quad u_{a,b} \mapsto O_{a,b},$$

is an isomorphism of completed coordinate dg commutative algebras. If $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ is bracket-admissible in the sense of Definition 2.46, then

$$\Phi^B_{P_0} \colon \Phi^{-1}_{\mathrm{coord}}(B) \xrightarrow{\sim} B$$

is an isomorphism of completed dg P_0-algebras. If B is kernel-admissible as part of a datum $(B, \mathcal{K}_B)$, then the diagonal element

$$\Theta_B = \sum_I H_I \otimes \theta^I + \sum_I \eta^I \otimes O_I$$

satisfies the Maurer–Cartan equation $d\Theta_B + \frac{1}{2}[\Theta_B, \Theta_B] = 0$ in $\mathcal{K}_B$. The full product coordinate algebra is not asserted to carry the global continuous contraction bracket or a strict kernel; the missing infinite diagonal is the strict Casimir obstruction of Theorem 2.52. The admissible formal-local interface is

$$\mathfrak{U}_{\mathfrak{h}} = (\Phi_{\mathrm{coord}}, \{\Phi^B_{P_0}\}_B, \{\Theta_B\}_B),$$

with B ranging over bracket-admissible realizations in the middle term and kernel-admissible data in the last term. At the coordinate dg-algebra level, and inside $\Phi^B_{P_0}$ for every Hamiltonian-spanning B, the stable boundary evaluation of Theorem 3.242 sends the degree-zero cotangent coordinate u_f to the boundary Hamiltonian

$$B_f = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)},$$

while c^f maps to the Hamiltonian vector field of B_f. This is the cochain-level bulk-boundary Koszul shadow; it does not use lower-central Tate-pronilpotence of the full Hamiltonian Lie algebra.

Proof. The completed coordinate source and target are products over the same finite-word coordinate set after the substitutions $c^{a,b} \leftrightarrow \theta^{a,b}$ and $u_{a,b} \leftrightarrow O_{a,b}$. Hence Φ_{coord} is a continuous isomorphism of completed graded-commutative algebras, with continuous inverse given by the reverse substitution.

Lemma 2.47 gives coordinate-continuous derivations d_{CE} and d_π^{coord}. On the generators $c^{r,s}$ and $u_{r,s}$, the displayed CE formulas are the duals of the Hamiltonian bracket and coadjoint action in $\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$. The PV coordinate formulas are the coefficientwise Lichnerowicz formulas for the linear Poisson tensor

$$\pi = \frac{1}{2} C^K_{IJ} O_K \theta^I \theta^J.$$

Therefore $\Phi_{\mathrm{coord}} d_{\mathrm{CE}} = d_\pi^{\mathrm{coord}} \Phi_{\mathrm{coord}}$ on $c^{r,s}$ and $u_{r,s}$. Both sides are continuous degree-one derivations, so the equality extends from generators to every finite word and then, by the product completion, to the whole completed algebra.

On the finite-support mixed subalgebra the cotangent/Schouten bracket is checked on generators:

$$\{c^{a,b}, u_{r,s}\}_{\mathrm{CE}} = \delta_{a,r}\delta_{b,s}, \qquad [\theta^{a,b}, O_{r,s}]_S = \delta_{a,r}\delta_{b,s},$$

and all c, c, u, u, θ, θ, and O, O brackets vanish. This gives bracket compatibility on finite words and on any mixed topology in which the c/θ-directions have finite label support. It does not extend to the arbitrary product completion: the bracket of $\sum_I c^I$ with $\sum_I u_I$, which are both admitted by the coordinate product, would require the divergent diagonal coefficient $\sum_I 1$. Thus the full product statement above is a dg coordinate reconstruction, not a global completed P_0-algebra theorem.

If B is bracket-admissible, the preceding generator calculation restricts to B, and all infinite contractions that occur in the P_0 bracket converge by definition of B. Hence the restricted map $\Phi^B_{P_0}$ is a completed dg P_0-isomorphism. If B is kernel-admissible, the two diagonal sums defining Θ_B converge in the specified convolution dg Lie algebra $\mathcal{K}_B$. The Maurer–Cartan equation is then checked coefficientwise in the completed convolution complex: every requested coefficient sees only the finitely many labels contributing to that coefficient, and the equality is exactly the Jacobi and coadjoint identity in the full Hamiltonian Lie algebra. Convergence in B passes these coefficient identities to the completed tensor product. No finite coordinate window is being used as a Hamiltonian Lie quotient. Without diagonal convergence this sentence has no target object, which is precisely the strict Casimir obstruction.

The coordinate moment-map shadow is the restriction of Φ_{coord} to the cotangent coordinates: $u_f \mapsto O_f$. The Kirillov–Poisson bracket gives $\{O_f, O_g\} = O_{\{f,g\}}$. Theorem 3.242 sends O_f to $\overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ and preserves the Hamiltonian bracket in the stable connected sector. Inside a bracket-admissible P_0-realization this sentence applies only for the f's in the Hamiltonian-spanning set of B. Thus u_f is the bulk source of the boundary Hamiltonian, while $c^f \mapsto \theta^f$ records the corresponding Hamiltonian vector field. □

Statement 2.49 *(Statement: equivariant CE/PV refined bracket). Status. proved as a coordinate dg-algebra formula after adjoining the symplectic-character parameter; bracketed, kernel, QME, and physical-trace upgrades remain obstruction data*. This is the T_Ω-equivariant refinement of Theorem 2.48.

Exact hypotheses. Let

$$T_\Omega = \mathbb{C}^*_{\epsilon_s} \times \mathbb{C}^*_{\epsilon_1} \times \mathbb{C}^*_{\epsilon_2}, \qquad \chi_\omega = \epsilon_1 + \epsilon_2.$$

Put $w(z_i) = \epsilon_i$, so $w(dz_1 \wedge dz_2) = \chi_\omega$ and the Poisson tensor has weight $-\chi_\omega$. For $H_{a,b} = z_1^a z_2^b$, $a + b > 0$, set $W_{a,b} = a\epsilon_1 + b\epsilon_2$. Then

$$w(H_{a,b}) = W_{a,b}, \qquad w(\rho_{a,b}) = -W_{a,b}.$$

The coordinate-dual weights are

$$w(c^{a,b}) = w(\theta^{a,b}) = -W_{a,b}, \qquad w(u_{a,b}) = w(O_{a,b}) = W_{a,b}.$$

The second equality is load-bearing: $u_{a,b}$ is dual to the shifted cotangent generator, not itself a cotangent generator.

Refined bracket. Off the self-dual locus, the ordinary Poisson bracket is line-valued:

$$\{-,-\}\colon \wedge^2 \mathfrak{h} \longrightarrow \mathfrak{h} \otimes L_\omega^{-1}.$$

Equivalently, after adjoining a central parameter $\hbar_\omega$ of weight χ_ω, one uses the scalar bracket

$$[f,g]_\Omega = \hbar_\omega \{f,g\}.$$

In monomial notation,

$$[H_I, H_J]_\Omega = \hbar_\omega C_{IJ}^K H_K, \qquad C_{IJ}^{I+J-(1,1)} = ad - bc$$

for $I = (a,b)$, $J = (c,d)$. This is weight-homogeneous because $W_K + \chi_\omega = W_I + W_J$. Jacobi is inherited from the ordinary Poisson bracket since $\hbar_\omega$ is central.

Coordinate CE/PV assertion. In the coordinate completion of Definition 2.45, define

$$d_{\mathrm{CE},\Omega} c^K = -\frac{1}{2}\hbar_\omega C_{IJ}^K c^I c^J, \qquad d_{\mathrm{CE},\Omega} u_K = \hbar_\omega C_{KJ}^L u_L c^J.$$

Equivalently,

$$d_{\mathrm{CE},\Omega} c^{r,s} = -\frac{1}{2}\hbar_\omega \sum_{\substack{a+c-1=r\\ b+d-1=s}} (ad-bc) c^{a,b} c^{c,d},$$

and

$$d_{\mathrm{CE},\Omega} u_{r,s} = \hbar_\omega \sum_{c,d} (rd - sc) u_{r+c-1,s+d-1} c^{c,d},$$

omitting negative labels and the reduced constant. On the PV side set

$$\pi_\Omega = \frac{1}{2}\hbar_\omega C_{IJ}^K O_K \theta^I \theta^J.$$

Then

$$d_{\pi,\Omega}\theta^K = -\frac{1}{2}\hbar_\omega C_{IJ}^K \theta^I \theta^J, \qquad d_{\pi,\Omega} O_K = \hbar_\omega C_{KJ}^L O_L \theta^J.$$

The generator assignment

$$\Phi_\Omega(c^{a,b}) = \theta^{a,b}, \qquad \Phi_\Omega(u_{a,b}) = O_{a,b}$$

is T_Ω-equivariant and satisfies

$$\Phi_\Omega d_{\mathrm{CE},\Omega} = d_{\pi,\Omega}\Phi_\Omega$$

on generators, hence on the completed coordinate dg algebra by the same finite-coordinate derivation argument as Theorem 2.48. The cotangent and Schouten contractions may either remain the unscaled cochain-level contractions or be scaled simultaneously by $\hbar_\omega$:

$$\{c^I, u_J\}_{\mathrm{CE},\Omega} = \hbar_\omega \delta_J^I, \qquad [\theta^I, O_J]_{S,\Omega} = \hbar_\omega \delta_J^I.$$

The latter is the L_ω-weighted P_0 convention.

Parameter convention. The symbol $\hbar_\omega$ is the equivariant scalarizer of the symplectic-character line. The Weyl/Moyal deformation parameter and the BV/QME loop parameter are independent until a branch datum identifies them. A Weyl-equivariant branch is a ring map sending $\hbar_\omega$ to the Weyl parameter $\hbar_{\mathrm{W}}$, with

$$w(\hbar_{\mathrm{W}}) = \epsilon_1 + \epsilon_2.$$

The physical large-N branch uses $\hbar_{\mathrm{W},N} = \lambda/N$. It is not identified with the BV loop parameter $\hbar_{\mathrm{QME}}$ unless that identification is included in the QME normalization datum. Thus Q_Ω-notation and the scalarizer $\hbar_\omega$ do not by themselves solve any QME.

Self-dual branch. On

$$\epsilon_1 + \epsilon_2 = 0$$

the Poisson tensor has weight zero. The ordinary Hamiltonian bracket is scalar, and $\hbar_\omega$ has weight zero; it may be retained as a central formal parameter or specialized to 1. No theorem may invert $\epsilon_1 + \epsilon_2$ and then specialize to this branch.

Obstruction boundary. The coordinate dg formula above does not construct the mixed Cartan complex, a bracket-admissible global completion, diagonal kernel convergence, residue/Euler normalization, a Costello graph package, or a physical trace state. The exact obstruction vector is

$$\mathrm{Ob}_{\Omega,\mathrm{CE/PV}} = (\mathrm{ob}_{\Omega,\mathrm{Cartan}}, \mathrm{ob}_{\Omega,\mathrm{top}}, \mathrm{ob}_{\Omega,\mathrm{br}}, \mathrm{ob}_{\Omega,\mathrm{diag}}, \mathrm{ob}_{\Omega,\mathrm{res}}, \mathrm{ob}_{\Omega,\mathrm{norm}}, \mathrm{ob}_{\Omega,\mathrm{QME}}).$$

The first term records $Q_\Omega^2 = L_{V_\Omega}$ before invariants; the middle terms record topology, biderivation, diagonal, resonance, and normalization data; and the last term records the separate brane-defect QME construction.

THEOREM 2.50 *(Abstract bracket-compatible CE/PV recognition criterion).* Let $L = \varprojlim_\alpha L_\alpha$ be a complete Hausdorff filtered dg Lie algebra over $\mathbb{C}$, with finite-dimensional bracket-compatible dg Lie quotients L_α, and let $D = L^\vee_{\mathrm{cont}} = \varinjlim_\alpha L_\alpha^\vee$. Put $G = L \ltimes D[1]$. Form the completed coordinate CE and PV algebras by taking products over finite words and finite coordinate windows. Assume:

(U1) the internal differential, Lie bracket on L, and coadjoint action on D are continuous, equivalently every finite coordinate projection of $d_L x$, $[x, y]$, and $x \cdot \lambda$ factors through a finite Lie quotient of the inputs;

(U2) the linear Poisson bivector

$$\pi_L = \frac{1}{2} C_{ij}^k O_k \theta^i \theta^j$$

determined by the structure constants of L is a continuous Schouten bivector on $\widehat{\mathbf{S}}_{\mathrm{coord}}(L)$;

(U3) the source cotangent bracket $\{c^i, u_j\}_{\mathrm{CE}} = \delta^i_j$ and the target Schouten contraction bracket $[\theta^i, O_j]_S = \delta^i_j$ are continuous biderivations on the chosen mixed or weighted completion. This is a summability hypothesis, not a consequence of the generator formula on the ambient product.

Then the generator assignment

$$\Phi_L \colon C^\bullet_{\mathrm{CE,cont}}(G) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{coord}}(L)\big), \qquad c^i \mapsto \theta^i, \quad u_i \mapsto O_i,$$

is an isomorphism of completed dg algebras with the degree-(-1) cotangent/Schouten contraction bracket. Without the summability condition in (U3), the conclusion is only the compatible pro finite-window dg-commutative CE/PV identity and the global contraction bracket is obstruction data. This is a

formal-local cochain theorem. It does not require lower-central Tate-pronilpotence and it does not assert that the coordinate pairing is represented by a strict kernel $C_L \in L\widehat{\otimes}D$. The strict formal Hamiltonian disk is not an instance of this theorem unless such a bracket-compatible finite quotient system is supplied; for the disk itself one uses the coordinate finite-test theorem above.

Proof. The completed CE algebra of $G = L \ltimes D[1]$ and the completed PV algebra of the linear formal space L are products over the same finite-word coordinate monomials after $c^i \leftrightarrow \theta^i$ and $u_i \leftrightarrow O_i$. Thus Φ_L is a continuous graded-commutative algebra isomorphism with continuous inverse.

Assumption (U1) makes the CE differential continuous: every finite output coordinate of the internal differential, of $d_{\mathrm{CE}}c^k$, and of $d_{\mathrm{CE}}u_i$ is computed from finitely many input coordinates. The internal differential is carried to the identical linear vector field on the PV side by the coordinate substitution. Assumption (U2) makes $d_{\pi_L} = [\pi_L, -]_S$ a continuous differential on the PV side. On generators the bracket terms are the same structure-constant formulas:

$$d_{\mathrm{CE}}c^k = -\frac{1}{2}C^k_{ij}c^i c^j, \qquad d_{\pi_L}\theta^k = -\frac{1}{2}C^k_{ij}\theta^i\theta^j,$$

and

$$d_{\mathrm{CE}}u_i = C^\ell_{ij}u_\ell c^j, \qquad d_{\pi_L}O_i = C^\ell_{ij}O_\ell\theta^j.$$

Jacobi for L gives $d^2_{\mathrm{CE}} = 0$ and $[\pi_L, \pi_L]_S = 0$, hence $d^2_{\pi_L} = 0$. Since both differentials are continuous derivations, generator-level compatibility gives $\Phi_L d_{\mathrm{CE}} = d_{\pi_L}\Phi_L$ on the completed algebras.

Assumption (U3) gives the same result for the contraction brackets:

$$\{c^i, u_j\}_{\mathrm{CE}} = \delta^i_j \quad \text{and} \quad [\theta^i, O_j]_S = \delta^i_j$$

on generators, with all same-type brackets zero. Continuity and the biderivation rule extend this equality to finite words and then to the chosen mixed or weighted completion. The last sentence records the boundary of the theorem: a strict Casimir kernel is stronger than the coordinate biderivation, while the unweighted ambient product may fail even to carry that biderivation globally. □

Proposition 2.51 *(Shifted-cotangent coordinate is the boundary Hamiltonian).* In the completed formal-disk identity of Theorem 2.48, the boundary Hamiltonian O_f is the image of the shifted-cotangent CE coordinate u_f:

$$\Phi_{\mathrm{coord}}(u_f) = O_f.$$

The Hamiltonian CE coordinate c^f maps to the vector-field generator θ^f, not to O_f. Equivalently,

$$d^{\mathrm{coord}}_\pi O_f = X_{O_f} = \Phi_{\mathrm{coord}}(d_{\mathrm{CE}}u_f).$$

After stable boundary evaluation, O_f becomes $\overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$, the moment-map shadow of the Hamiltonian f on the matrix probe.

Proof. This is the u-generator part of the defining formula for Φ_{coord}. For $f = H_k$, the CE differential is $d_{\mathrm{CE}}u_k = C^\ell_{kj}u_\ell c^j$. Applying Φ_{coord} gives $C^\ell_{kj}O_\ell\theta^j$, which is exactly the Hamiltonian vector field $d^{\mathrm{coord}}_\pi O_k$; on a bracket-admissible B this is the Schouten bracket $[\pi, O_k]_S$. Thus the cotangent coordinate, not the Hamiltonian ghost coordinate, carries the boundary Hamiltonian. The last assertion is Theorem 3.242 applied to the degree-zero Kirillov map $f \mapsto O_f$. □

THEOREM 2.52 *(Strict product/direct-sum Casimir obstruction).* Let

$$\mathfrak{h}_\Pi = \prod_{d\geq 1} H_d, \qquad D_\oplus = \bigoplus_{d\geq 1} H_d^\vee$$

be the strict product/direct-sum coefficient pair of the formal Hamiltonian disk. There is no tensor $K \in \mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$ whose projection to every finite window $d \leq M$ is the finite Casimir

$$C_{\leq M} = \sum_{1\leq d\leq M} \sum_i e_{d,i} \otimes e_d^i.$$

The same statement holds with tensor factors reversed.

Proof. In the strict tensor convention for the product/direct-sum Tate pair, an element of $\mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$ has finite support in the direct-sum factor $D_\oplus$. Thus there is a degree M_0 such that all $H_d^\vee$-components of K vanish for $d > M_0$. Projecting to the window $d \leq M_0 + 1$ then gives zero in the $H_{M_0+1} \otimes H_{M_0+1}^\vee$ component, while $C_{\leq M_0+1}$ has the nonzero identity tensor in that component. This contradiction proves the claim. Reversing the tensor factors is identical. □

THEOREM 2.53 *(Primitive-label obstruction to Matlis cotangent-current equivariance).* Let V_Ψ be the span of primitive PBW special-fibre labels $\widetilde{\Psi}_{a,b}$, $a + b > 0$, with the reduced scalar target killed and with the linear Hamiltonian action

$$O_{1,0} \cdot \widetilde{\Psi}_{a,b} = b\, \widetilde{\Psi}_{a,b-1}, \qquad O_{0,1} \cdot \widetilde{\Psi}_{a,b} = -a\, \widetilde{\Psi}_{a-1,b}.$$

Let $\mathcal{P}$ be the principal-part module with

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

There is no $\mathfrak{h}$-module map $F\colon V_\Psi \to \mathcal{P}$ satisfying $F(\widetilde{\Psi}_{a,b}) = \lambda_{a,b}\rho_{a,b}$ with all $\lambda_{a,b} \neq 0$.

Proof. Apply equivariance for z_1 to $\widetilde{\Psi}_{a,0}$, $a > 0$. On the PBW special-fibre side, $O_{1,0} \cdot \widetilde{\Psi}_{a,0} = 0$. Hence $F(O_{1,0} \cdot \widetilde{\Psi}_{a,0}) = 0$. On the principal-part side,

$$z_1 \cdot F(\widetilde{\Psi}_{a,0}) = z_1 \cdot (\lambda_{a,0}\rho_{a,0}) = -\lambda_{a,0}\rho_{a,1},$$

which is nonzero. This contradicts equivariance. The z_2-action gives the analogous contradiction at $a = 0$. □

PROPOSITION 2.54 *(Low-degree obstruction to a pronilpotent full-disk splitting).* The subspace $\mathfrak{m}^3 \subset \mathfrak{h} = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ is not an ideal of the full reduced Hamiltonian Lie algebra $\mathfrak{h}$. Consequently the vector-space decomposition

$$\mathfrak{h} = (H_1 \oplus H_2) \oplus \mathfrak{m}^3, \qquad H_d = \mathrm{Span}\{z_1^a z_2^b : a + b = d\},$$

does not make $\mathfrak{h}$ a semidirect product of the low-degree sector with a pronilpotent ideal.

Proof. The linear Hamiltonian $z_1 \in H_1$ lowers $z_2^3 \in \mathfrak{m}^3$:

$$\{z_1, z_2^3\} = 3z_2^2 \in H_2.$$

Since $H_2 \cap \mathfrak{m}^3 = 0$, the bracket does not lie in $\mathfrak{m}^3$. Thus $\mathfrak{m}^3$ is not an ideal in $\mathfrak{h}$. A semidirect-product presentation with pronilpotent radical $\mathfrak{m}^3$ would require that radical to be an ideal, so no such presentation exists. □

Theorem 2.55 *(Relative filtered CE/PV reconstruction criterion).* Let $\mathfrak{h}_{\mathrm{rel}}$ be a complete filtered dg Lie algebra with a finite-dimensional reductive degree-zero subalgebra $\mathfrak{a}$, a complete pronilpotent dg ideal $\mathfrak{n}$, and a filtered splitting

$$\mathfrak{h}_{\mathrm{rel}} \cong \mathfrak{a} \ltimes \mathfrak{n}.$$

Let D_{rel} be a continuous $\mathfrak{h}_{\mathrm{rel}}$-module with a perfect filtered pairing with $\mathfrak{h}_{\mathrm{rel}}$ on every finite quotient, and assume the coevaluation tensor belongs to $\mathfrak{h}_{\mathrm{rel}} \widehat{\otimes} D_{\mathrm{rel}}$. Put

$$\mathfrak{g}_{\mathrm{rel}} = \mathfrak{h}_{\mathrm{rel}} \ltimes D_{\mathrm{rel}}[1], \qquad A_{\mathrm{rel}} = \widehat{\mathbf{S}}(\mathfrak{h}_{\mathrm{rel}}).$$

Then the generator assignment $c \mapsto \theta$, $u \mapsto O$ defines a filtered isomorphism

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{rel}}) \xrightarrow{\sim} \mathrm{PV}(A_{\mathrm{rel}})$$

of completed dg algebras with the degree-(-1) cotangent/Schouten contraction bracket. If, in addition, the relative bar-cobar functor is taken in the category of $\mathfrak{a}$-modules and the $\mathfrak{n}$-coalgebra is conilpotent, this cochain identity is promoted to relative Lie-Koszul duality:

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{rel}}, \mathfrak{a}) \simeq \big(\widehat{U}_{\mathfrak{a}}(\mathfrak{g}_{\mathrm{rel}})\big)^{!}.$$

Proof. The cochain-level statement is the completed version of Theorem 2.44: finite quotients of the filtration carry the finite CE/PV identity, the transition maps preserve $c \mapsto \theta$, $u \mapsto O$, and completeness identifies the inverse limit with the displayed completed complexes. The coevaluation hypothesis is the exact condition needed to keep the cotangent generators inside the same completed tensor category.

For the Koszul-dual statement, work relative to the reductive subalgebra $\mathfrak{a}$. The coalgebra generated by the pronilpotent ideal $\mathfrak{n}$ is conilpotent, so the Lie/commutative bar-cobar equivalence applies to the relative CE chains. Reductivity of $\mathfrak{a}$ makes passage to $\mathfrak{a}$-modules exact on finite quotients; in the filtered inverse limit this gives the completed relative equivalence. Dualizing yields the displayed cochain Koszul-dual form.

This is a criterion, not a statement about the unmodified formal disk: Proposition 2.54 shows that the actual decomposition $\mathfrak{h} = (H_1 \oplus H_2) \oplus \mathfrak{m}^3$ has no pronilpotent ideal $\mathfrak{m}^3$ for the full bracket, so the criterion applies only after an independent relative model supplies the displayed semidirect-product hypotheses. □

Lemma 2.56 *(Filtered cobar acceptance criterion).* Let C be a coaugmented conilpotent dg coalgebra and B a complete augmented filtered A_∞-algebra in the same complete linear category. For a continuous degree-one map $\tau\colon C \to B$, define

$$\mathrm{MC}(\tau) = d_B\tau + \tau d_C + \sum_{r\geq 2} m_r^B(\tau^{\otimes r})\Delta_C^{(r)} \in \mathrm{Hom}^2_{\mathrm{cont}}(C, B),$$

where $\Delta_C^{(r)}$ is the r-fold reduced coproduct. The equation $\mathrm{MC}(\tau) = 0$ is equivalent to the existence of the adjoint complete cobar morphism

$$\kappa_\tau\colon \Omega C \longrightarrow B.$$

Assume the filtrations on ΩC and B are bounded below, complete, separated, and strongly convergent, with finite-dimensional associated graded pieces in every total degree. Then κ_τ is an admissible filtered quasi-isomorphism if and only if

$$H^{\bullet}\big(\mathrm{Cone}(\mathrm{gr}\,\kappa_\tau)\big) = 0$$

and the two finite-window Milnor defects

$$\varprojlim_M{}^1 H^{n-1}((\Omega C)_{\le M}), \qquad \varprojlim_M{}^1 H^{n-1}(B_{\le M})$$

vanish for every n.

Proof. The Maurer–Cartan formula is the bar-cobar adjunction written in the continuous convolution complex. For a dg target it reduces to $d_B\tau + \tau d_C + \mu_B(\tau\otimes\tau)\Delta_C = 0$, which says exactly that $\Omega C \to B$ commutes with differentials and multiplication. The displayed A_∞ formula is the same statement with all target multiplications retained.

Filter source and target by tensor length and by finite coordinate windows. Exactness of the associated-graded cone is equivalent to $\operatorname{gr}\kappa_\tau$ being a quasi-isomorphism. The two $\varprojlim^1$-terms are the Milnor errors in computing the cohomology of the completed source and target from their finite windows. With those terms zero, strong convergence gives the spectral-sequence comparison theorem in both directions. Necessity is the definition of admissible filtered quasi-isomorphism in this envelope: it is detected on associated graded and on the exact finite-window towers. □

Definition 2.57 (Universal open-trace A_∞-Koszul acceptance datum). A universal open-trace A_∞-Koszul acceptance datum consists of:

(K1) a finite-window tower of one-object cyclic open dg categories whose endomorphism complexes are

$$\Omega^\bullet_c(I)\otimes\Lambda(\varepsilon_1,\varepsilon_2)\otimes\operatorname{End}(\mathbb{C}^N),$$

with cyclic trace

$$\int_I \eta\,[\varepsilon_1\varepsilon_2]\,\operatorname{Tr}_N(M)$$

and Dirac/Koszul reduction $Q\psi = [\phi_1,\phi_2]$;

(K2) the reduced cyclic dg algebra $A_\psi = T(x,y,\psi)$, $d\psi = xy - yx$, together with $CC_{\mathrm{red}}(A_\psi)$ and the necklace H_0-shadow $\operatorname{Cyc}(A_\psi)$;

(K3) a finite-window Loday–Quillen–Tsygan comparison on the corrected dg finite-window habitat. Give $A_\psi = T(x,y,\psi)$ the Koszul weights

$$\operatorname{wt}(x) = (1,0), \qquad \operatorname{wt}(y) = (0,1), \qquad \operatorname{wt}(\psi) = (1,1),$$

and put $A_{\psi,K} = A_\psi/F^{>K}A_\psi$, where $F^{>K}$ is the two-sided dg ideal generated by words of total Koszul weight $> K$. The stable single-cycle image is

$$\operatorname{Prim}^{\mathrm{1cyc}}_{N,K,L} := \operatorname{im}\Theta_{N,K,L} \subset C^{\mathrm{CE}}_{*,\le L}(\mathfrak{gl}_N(A_{\psi,K}))^{GL_N}.$$

The actual finite-window map is the inverse of the trace-cycle map on this image,

$$\lambda_{\mathrm{LQT},K,L}\colon\ \operatorname{Prim}^{\mathrm{1cyc}}_{N,K,L} \longrightarrow CC_{\mathrm{red},\le L}(A_{\psi,K})[1]$$

in the stable range in which N dominates both the Koszul weight window and CE length. Equivalently, $\lambda_{\mathrm{LQT},K,L} = \Theta^{-1}_{N,K,L}P_{\mathrm{1cyc}}$ on stable invariant CE chains, where in the stable block-sum Hopf CE window

$$P_{\mathrm{1cyc}} = \log^*(\mathrm{id}) = \sum_{m=1}^{L}\frac{(-1)^{m-1}}{m}(\mathrm{id} - u\epsilon)^{*m}.$$

This is theorem-grade in the range $N \geq \max(K, L+2)$, or $N \geq W+2$ when one bound W controls both windows, in which case $A_{\psi,W}$, $\Theta_{N,W}$, and $\lambda_{\mathrm{LQT},W}$ are abbreviations for the specialization $K, L \leq W$. The raw same-rank operation that deletes multicycle tensors is not a chain map, since $d_{\mathrm{CE}}(\Theta_1(a) \wedge \Theta_1(b))$ has a single-cycle commutator component while deletion before the differential gives zero. The finite-rank bridge is the separated-block stabilization zigzag: with $B(K,L) = \max(K, L+2)$ and $M = L\,B(K,L)$, let $\beta_{M,K,L}$ be the normalized signed average over injections of connected Hopf factors into the L labeled diagonal $B(K,L)$-blocks, read separated blocks back by $\alpha_{M,K,L}$, so that $\alpha_{M,K,L}\beta_{M,K,L} = \mathrm{id}$, and use

$$P^{\mathrm{sep}}_{M,K,L} = \beta_{M,K,L} P_{1\mathrm{cyc}} \alpha_{M,K,L}.$$

A same-rank chain-homotopy bridge is blocked by the collision coderivation

$$\Gamma_N = d_{\mathrm{same}} F_N - F_N d_{\oplus}, \qquad \Gamma_N(\Theta_1^N(a) \wedge \Theta_1^N(b)) = \pm\Theta_1^N(ab - (-1)^{|a||b|} ba).$$

The scalar quotient is a chain quotient only after killing the scalar-Koszul subcomplex $\Theta_1(\psi) \mapsto \Theta_1(xy - yx)$, or after passing to homology; killing $\Theta_1(\psi)$ alone is not a chain map;

(K4) a completed augmented filtered A_∞-algebra A_{op} in the scalar-reduced stable trace/Koszul/BV sector, with associated graded

$$\mathrm{gr}\, A_{\mathrm{op}} \cong \widehat{\mathbf{S}}(V_H), \qquad \mathrm{Indec}\,\mathrm{gr}\, A_{\mathrm{op}} = \mathfrak{m}/\mathfrak{m}^2 \cong V_H,$$

where the connected/primitive expression

$$\pi_{\mathrm{Ham}}\, \mathrm{Conn}_{N\to\infty}\, \mathrm{Prim}_{\mathrm{st}}\big(\mathcal{R}_N^{GL_N}, Q\big)$$

is only this linear associated-graded symbol, not the algebra object itself; The target indecomposables must be either the principal-part model $\bar{A} \oplus \mathcal{P}[1]$, or the descendant model $\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]$ together with a proved Fourier–Rees/Moyal bridge to $\mathcal{P}[1]$; the raw same-label map $\Psi_{a,b} \mapsto \rho_{a,b}$ is not accepted.

(K5) the completed closed target

$$B_{\mathrm{CE/PV}} = C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) \cong \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{coord}}(\mathfrak{h}_{\mathrm{adm}})\big),$$

with the generator dictionary $c^f \mapsto \theta^f$, $u_f \mapsto O_f$;

(K6) the open bar coalgebra $C_{\mathrm{op}} = \mathrm{Bar}(A_{\mathrm{op}})$, equipped with a complete bounded-below filtration compatible with the bar differential, coproduct, and finite coordinate windows;

(K7) a continuous degree-one map

$$\tau_{\mathrm{op}} \colon C_{\mathrm{op}} \longrightarrow B_{\mathrm{CE/PV}}$$

satisfying the explicit Maurer–Cartan equation

$$d_{B_{\mathrm{CE/PV}}} \tau_{\mathrm{op}} + \tau_{\mathrm{op}} b_{\mathrm{op}} + \mu_{B_{\mathrm{CE/PV}}}(\tau_{\mathrm{op}} \otimes \tau_{\mathrm{op}})\Delta = 0;$$

(K8) the induced completed cobar morphism

$$\kappa_{\tau_{\mathrm{op}}} \colon \Omega C_{\mathrm{op}} \longrightarrow B_{\mathrm{CE/PV}}$$

whose associated-graded linear symbol is

$$B_f \longmapsto O_f, \qquad \Theta_{\rho_{a,b}} \longmapsto \theta^{a,b};$$

(K9) the acceptance obstruction complex

$$\begin{aligned}\mathfrak{C}_{\tau_{\mathrm{op}}} = \mathrm{Cone}(\mathrm{gr}\,\kappa_{\tau_{\mathrm{op}}}) &\\ \oplus \bigoplus_n \varprojlim_M{}^1 H^{n-1}\big((\Omega C_{\mathrm{op}})_{\le M}\big)[-n] &\\ \oplus \bigoplus_n \varprojlim_M{}^1 H^{n-1}\big((B_{\mathrm{CE/PV}})_{\le M}\big)[-n]. &\end{aligned}$$

(K10) exact continuous dualization on the same finite-window tower;

(K11) a cyclicity and BV-pairing package: the scalar word is retained before scalar reduction, graded cyclicity uses $\mathrm{Tr}(ab) = (-1)^{|a||b|}\,\mathrm{Tr}(ba)$, two ψ-letters skew, the open trace is continuous, and the BV pairing has degree -1 after the de Rham, Ext, and field-shift conventions.

THEOREM 2.58 *(Universal open-side filtered A_∞-Koszul acceptance).* For a datum in the sense of Definition 2.57, the following are equivalent:

(a) τ_{op} is an accepted open-side twisting cochain, i.e. $\kappa_{\tau_{\mathrm{op}}}$ is an admissible filtered quasi-isomorphism of complete augmented A_∞-algebras with the displayed associated-graded symbol;

(b) the Maurer–Cartan equation in (K7) holds and the obstruction complex $\mathfrak{C}_{\tau_{\mathrm{op}}}$ from (K9) is exact.

Under these equivalent conditions,

$$\Omega\mathrm{Bar}(A_{\mathrm{op}}) \xrightarrow{\sim} B_{\mathrm{CE/PV}}$$

and hence $A_{\mathrm{op}} \simeq B_{\mathrm{CE/PV}}$ in the completed filtered A_∞ homotopy category. If the CE-side cobar map

$$\Omega C_*^{\mathrm{CE}}(\mathfrak{g}_{\mathrm{adm}}) \longrightarrow A_{\mathrm{op}}$$

satisfies the same exact cone and $\varprojlim^1$ test, exact continuous dualization gives the Koszul-dual cochain form

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) \simeq A^!_{\mathrm{op}}.$$

Proof. Apply Lemma 2.56 with $C = C_{\mathrm{op}}$ and $B = B_{\mathrm{CE/PV}}$. The Maurer–Cartan equation in (K7) is precisely the condition that $\kappa_{\tau_{\mathrm{op}}}$ is a morphism. Exactness of $\mathfrak{C}_{\tau_{\mathrm{op}}}$ says simultaneously that $\mathrm{gr}\,\kappa_{\tau_{\mathrm{op}}}$ is a quasi-isomorphism and that both finite-window towers compute completed cohomology. The filtered comparison theorem then gives the accepted quasi-isomorphism, and the converse is the admissible-filtered definition.

The bar-cobar counit $\Omega\mathrm{Bar}(A_{\mathrm{op}}) \to A_{\mathrm{op}}$ is a filtered quasi-isomorphism under the same boundedness and convergence hypotheses, so the accepted $\kappa_{\tau_{\mathrm{op}}}$ identifies A_{op} with $B_{\mathrm{CE/PV}}$ in the homotopy category. The final displayed equivalence is the CE-side specialization of the same criterion, followed by exact continuous dualization. □

Statement 2.59 (Statement: Matlis principal-part cotangent module). Status. proved degreewise stable. The proof uses residue calculus and Matlis local-cohomology duality for the m-adic topology on $R = \mathbb{C}[[z_1, z_2]]$, after quotienting constants.

Exact hypotheses. Work over $R = \mathbb{C}[[z_1, z_2]]$ with maximal ideal $\mathfrak{m} = (z_1, z_2)$; put $\mathfrak{h} = R/\mathbb{C}\cdot 1$; take continuous duals for the m-adic topology; give $\mathcal{P}$ the locally finite direct-sum topology transported from $\mathfrak{h}^\vee_{\mathrm{cont}}$; normalize the residue pairing only up to one nonzero scalar; and, for the Fourier–Rees comparison

only, distinguish the Fourier delta Rees lattice $\mathcal{D}_\hbar/(z_1, z_2)$ from the full Čech local-cohomology lattice before $\hbar$ is inverted.

Local assertion. The continuous dual of the reduced Hamiltonian Lie algebra is the reduced Matlis-local-cohomology module

$$\mathfrak{h}_{\mathrm{cont}}^\vee \cong \mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) \subset H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2,$$

with basis $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2$, $a + b > 0$, and coadjoint action

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with negative-index targets equal to zero. Among continuous total-degree-zero pairings, the residue pairing is determined up to one nonzero scalar, and this scalar is the only normalization ambiguity in this statement.

Proof inputs. Appendix A centralizes the reduced Matlis calculation in the notation of this statement. Theorem 3.228 proves the residue pairing and its total-degree-zero scalar ambiguity. Proposition 3.227 computes the explicit coadjoint action in the body. The upgraded appendix proof is Proposition A.5, and its finite-jet continuity is Lemma A.6. The stronger uniqueness statement is Theorem A.8: a perfect continuous equivariant pairing is unique up to scalar, and the displayed residue normalization fixes that scalar. Theorem 3.230 remains the body-level total-degree-zero formulation. Theorem 3.232, Theorem 3.234, and Theorem A.9 state the no-go result for replacing $\mathcal{P}$ by polynomial descendants. Theorem A.10 supplies the limited Fourier–Rees label bridge through the Rees lattice $\bigoplus \hbar^{a+b}\mathbb{C}[[\hbar]]\rho_{a,b}$ and proves that it is not an $\mathfrak{h}$-module bridge. Its flatness is the PBW normal-form flatness of the left cyclic $\mathcal{D}_\hbar$-modules $\mathcal{D}_\hbar/(\partial_1, \partial_2)$ and $\mathcal{D}_\hbar/(z_1, z_2)$.

Scope and exclusions. A same-action $D_\hbar$, Fourier, or Rees comparison identifying the PBW descendant module with the Matlis coadjoint module is ruled out on the generic fibre by local nilpotence unless the Hamiltonian action or the boundary target is changed. The Fourier-twisted label bridge does not supply such a same-action comparison.

Statement 2.60 *(Statement: brane-line factorization currents). Status. proved degreewise stable.* The proof is for compactly supported de Rham Hamiltonian currents and compactly supported distributional principal-part currents on intervals. This statement is a strict support-local P_0 current statement in the reduced Hamiltonian sector; local constancy is asserted only after the compact-support de Rham contraction of the Hamiltonian density sector.

Exact hypotheses. Intervals lie in the brane line $\mathbb{R}$; $\mathfrak{h}_I = \Omega_c^\bullet(I)\widehat{\otimes}\mathfrak{h}$; $\mathcal{P}_I = \mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P}$; extension by zero is the functoriality; and source coordinates in the displayed boundary operation are shifted-cotangent CE coordinates $u_{a(t)dt\otimes f}$, not Hamiltonian ghost coordinates. The CE source is cofactorization data: extension by zero on current Lie algebras induces pullback/coproduct on CE cochains, while covariant factorization belongs to the compactly supported current target.

Local assertion. For each interval $I \subset \mathbb{R}$,

$$\mathfrak{h}_I = \Omega_c^\bullet(I)\widehat{\otimes}\mathfrak{h}, \qquad \mathcal{P}_I = \mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P}, \qquad \mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1].$$

The Hamiltonian current sector admits an interval-wise locally constant P_0 factorization equivalence. On length-one shifted-cotangent CE coordinates the source is

$$u_{a(t)dt\otimes f} \longmapsto B_f(a) = \int_I a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt.$$

In the degree-zero notation used here $a \in \Omega_c^0(I)$, the paired density is $a(t)\,dt$, $|u_{a(t)dt\otimes f}| = 0$, and the de Rham-current pairing is

$$\langle B \otimes \rho,\ a \otimes f\rangle = B(a)\,\mathrm{Res}(\rho f).$$

For a regular current $B = b(t)\,dt$ this is $\int_I a(t)b(t)\,dt\ \mathrm{Res}(\rho f)$. Orientation reversal contributes only the ordinary sign of the one-form density dt; no extra sign is inserted in this statement.

Proof inputs. Appendix B fixes the interval source, current dual, shifted-cotangent CE coordinate, and orientation convention used in this statement. Construction 3.18 fixes the de Rham-current objects and extension-by-zero functoriality. Proposition 3.20 proves disjoint-support locality. Theorem 3.25 supplies the smeared Hamiltonian central operation. Theorem 3.22 identifies the interval-wise reduced Hamiltonian current central-operation map and its compatibility with factorization products.

Scope and exclusions. The strict current statement does not include the unreduced BV/factorization-centre morphism or its centrality homotopies against arbitrary open observables.

Statement 2.61 *(Statement: reduced principal-part defect currents). Status. proved degreewise stable*. The proof applies only after the de Rham-current correction and only in the reduced P_0 model after contraction. It is not an unreduced BV/factorization theorem and does not supply centrality homotopies against arbitrary open observables.

Exact hypotheses. The algebra is the reduced defect-current model on intervals after de Rham-current contraction; the Hamiltonian factor is $\widehat{\mathbf{S}}(\Omega_c^0(I)\widehat{\otimes}\bar{A})$; the cotangent factor is $\widehat{\mathbf{S}}(\mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P}[1])$; brackets are strict P_0-brackets; the completed symmetric algebra is interpreted degreewise in the current category; and the primitive one-ψ comparison is taken only after passing to indecomposables.

Local assertion. The principal-part cotangent factor extends the Hamiltonian brane-line algebra to the reduced defect-current model

$$A_\partial^{\mathrm{pp}}(I) = \widehat{\mathbf{S}}(\Omega_c^0(I)\widehat{\otimes}\bar{A})\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P}[1])$$

with

$$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\} = 0.$$

The chain-level primitive projection gives only the primitive indecomposable P_0 shadow of this reduced comparison. Its quantum analogue is the reduced Moyal principal-part target of Construction 3.156, where $f\cdot\rho$ is replaced by $\mathrm{ad}^\vee_{f,\hbar}\,\rho$.

Proof inputs. Theorem 3.237 proves the reduced current model and its P_0 bracket. Theorem 3.240 records the boundary local-dual realization in the same reduced setting. Remark 3.239 and Remark 3.241 state the unreduced obligations. Theorem 3.154, Corollary 3.161, and Remark 3.162 supply only the primitive shadow status. Proposition 3.155 and Theorem 3.157 prove that the polynomial primitive label projection does not quantize to the Moyal coadjoint target.

Scope and exclusions. The unreduced cotangent factorization-centre morphism and its centrality homotopies remain outside this statement. The reduced Moyal principal-part target is constructed, but a quantum lift from polynomial one-ψ primitives is ruled out under the leading-term assumptions above. The pointwise classical obstruction is the mismatch between the polynomial one-ψ action and the principal-part coadjoint action:

$$O_{1,0} : \widetilde{\Psi}_{a,b} \mapsto b\,\widetilde{\Psi}_{a,b-1}, \qquad z_1 : \rho_{a,b} \mapsto -(b+1)\rho_{a,b+1},$$

and similarly

$$O_{0,1} : \widetilde{\Psi}_{a,b} \mapsto -a\,\widetilde{\Psi}_{a-1,b}, \qquad z_2 : \rho_{a,b} \mapsto (a+1)\rho_{a+1,b}.$$

Therefore the primitive projection records labels only; it does not supply centrality homotopies against arbitrary unreduced open observables.

Statement 2.62 *(Statement: degree-zero Moyal/Weyl sector and graph boundary). Status. proved finite algebraic model plus exact radial/Weyl obstruction theorem target*. The formal Weyl/Moyal coefficient algebra, Capelli triangular normalization, reduced open-line Wick normalization, free indexed normal-ordering formula, stable trace-diagram injectivity, edge PBW telescoping strip, $K_{4,4}$ correction, and named finite non-edge certificates are proved in the cited local habitats. The all-N ordinary Weyl-trace comparison is not closed by a generic radial-parts appeal: it is reduced to the moment-ideal primitive, equivalently to the free normal-diagram classes $\mathfrak{o}_{a,b} \in \operatorname{coker} C_{a,b}$. Its sharpened theorem target is the dual-potential obstruction theorem: with $B_{a,b}(W) = (\partial W, C^{+}_{a,b}(W))$ and $\eta_{a,b} = (T_{a,b}, E^{+}_{a,b})$, the same gate is

$$\Omega^{\mathrm{rad}}_{a,b} = [\eta_{a,b}] \in \operatorname{coker} B_{a,b};$$

failure is exactly a signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$ with $\lambda(E^{+}_{a,b}) - \phi(T_{a,b}) \neq 0$. The ordinary square-cell necklace cycle mechanism is proved; what remains is the all-bidegree decorated PBW Stokes identity for $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$. The mixed QME, unweighted regulator endpoint, descendant one-ψ lift, and Costello brane-defect graph realization are separate residual statements and are not hypotheses for the formal Moyal coefficient theorem.

Exact hypotheses. Work in the scalar-reduced degree-zero Hamiltonian sector at finite N; use the Capelli/Weyl normalization of the Weyl algebra and the formal Moyal product on $\bar{A}_\hbar$; exclude one-antifield descendants and graph-theoretic BV data from the proved comparison; and, when comparing ordinary Weyl traces $J_N(f)$ with the Cartan radial image, impose the radial-parts package of Statement F.8. Its matched-normalization clause $J_N(f) = J^{\mathrm{rad}}_N(f)$ is exactly the all-N algebraic gate. By Lemma F.15, this gate is the moment-ideal primitive

$$E_{a,b,N} = \sum_{j=0}^{b} \operatorname{Tr}(Y^j X^a Y^{b-j}) - J_N\Big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\Big) = \sum_{i,j} A^j{}_i(a,b,N)\widehat{\mu}^i{}_j$$

with $A^j{}_i(a,b,N) \in \mathcal{W}_N$. Equivalently, after choosing a cyclic lift $c^{\mathrm{cl}}_{a,b}$, it is the vanishing of

$$\mathfrak{o}_{a,b} = [R^{\mathrm{free}}_{a,b}] \in \operatorname{coker}\big(C_{a,b}\colon \ \ker\partial \longrightarrow \mathsf{Diag}_{a,b}\big)$$

together with functorial lifts $K_{a,b}$. The edge classes $\mathfrak{o}_{2,b}$ and $\mathfrak{o}_{a,2}$ vanish by the PBW telescoping theorem. The first interior correction $K_{4,4}$ and the listed low non-edge certificates are all-N free normal-diagram certificates; they are not finite-rank trace identities.

Local assertion. On the scalar-reduced degree-zero Hamiltonian sector the formal algebraic target is closed: the commutator is the formal Moyal commutator on $\bar{A}_\hbar$ under the Capelli/Weyl normalization, and the reduced open-line Wick product has the same coefficients. After the radial/Weyl gate is proved in all bidegrees, namely after $\Omega^{\mathrm{rad}}_{a,b} = 0$ or equivalently $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$, with failure certified by a signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$ satisfying $\lambda(E^{+}_{a,b}) - \phi(T_{a,b}) \neq 0$, the stable connected trace map satisfies

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f,g\}_\hbar).$$

The first and third reduced open-line coefficients are $\hbar P(f,g)$ and $\hbar^3 P^3(f,g)/24$. A Costello-theoretic graph lift of those coefficients requires the sector, propagator, counterterm, anomaly, and QME data named in the analytic graph problem. Here "first" and "third" refer to raw star-commutator powers $\hbar^1$

and $\hbar^3$. After division by $\hbar$, these become the normalized Hamiltonian-bracket coefficients in powers $\hbar^0$ and $\hbar^2$, namely $P(f,g)$ and $P^3(f,g)/24$. In the descendant direction the reduced target is not polynomial: Construction 3.156 gives the Moyal principal-part coadjoint complex, and Theorem 3.157 rules out any strict $\hbar$-adic deformation of the classical $\Psi_{a,b} \mapsto \rho_{a,b}$ projection satisfying its flatness, PBW-linear-action, Moyal-coadjoint-target, and leading-term hypotheses.

Proof inputs. Appendix F separates the formal Moyal/Capelli calculation from the radial/Weyl trace gate. Proposition F.14 and Lemma F.15 reduce the unproved all-N trace normalization to the displayed moment-ideal primitive. Definition F.17, Theorem F.18, Lemma F.19, Definition F.21, and Statement F.22 identify the free normal-diagram obstruction class; the controlling dual-potential form is Proposition F.25, namely $\Omega^{\mathrm{rad}}_{a,b} = 0$, equivalently $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$, with signed $B^{*}_{a,b}$-rows as the exact failure criterion. Proposition F.27 proves the all-edge PBW telescoping theorem. Proposition F.20 supplies the first interior all-N kernel correction, and Proposition F.28 supplies the named finite non-edge certificates. Theorem 3.251 uses that input to obtain the finite-N radial-parts identity and the stable degree-zero commutator. Corollary 3.252 and Theorem 3.269 assemble the stable Capelli-renormalized map. Proposition 3.268 proves the Moyal coefficient formula, with an independent finite exact arithmetic sweep as verification evidence. Proposition 3.275 and Theorem 3.276 compute the reduced open-line first and third coefficients. Theorem 3.278 records the implication from the analytic graph-realization data of Problem 3.264, and Problem 3.87 carries the mixed brane-defect QME obstruction. Theorem 3.123 gives the finite nilpotent obstruction to an all-order descendant lift. Proposition 3.155, Construction 3.156, and Theorem 3.157 isolate the one-ψ quantum target and the no-go for the polynomial primitive projection.

Scope and exclusions. Before the all-bidegree dual-potential signed-row theorem is supplied, the ordinary all-N trace identity is closed only on the proved certificate surface: formal coefficients, Capelli triangular normalization, open-line Wick weights, edge bidegrees, $K_{4,4}$, the listed finite non-edge free-diagram certificates, the rank-two Cartan shadow, and the direct $N = 2, 3$ radial restrictions. The sharpened obstruction is a nonzero signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$ with $\lambda(E^{+}_{a,b}) - \phi(T_{a,b}) \neq 0$, equivalently a nonzero class $\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^{+}_{a,b})] \in \operatorname{coker} B_{a,b}$. This would be the obstruction theorem for the original all-bidegree statement. The all-order Costello graph realization, the mixed brane-defect QME, the unweighted regulator problem, and an unreduced one-antifield Moyal lift are not consequences of this algebraic statement. As a statement about this degree-zero sector alone, the reduced principal-part Moyal target is not constructed from polynomial one-ψ descendants. Its weighted current-bracket version is one component of the separate quantum coefficient surface in Theorem 3.270; the non-scalar Costello/QME construction problem remains Problem 3.271.

Statement 2.63 *(Statement: componentwise quantum coefficient surface). Status. proved componentwise; balanced scalar-contact projection proved on its habitat; minimal full-equivariant finite-window non-scalar branch proved all orders; general graph-habitat extension and larger non-scalar obstruction problems open*. The theorem constructs a product coefficient surface. It does not construct unrestricted graph-level non-scalar quantum P_0-operation data.

Exact hypotheses. Fix the exponential weight $w(d) = q^d$, $q > 1$, and work over $\mathbb{C}[[\hbar]]$. Use the weighted Tate coefficient pair $(\mathfrak{h}_w, \mathfrak{h}^{\vee,w}_{\mathrm{cont}})$, the Moyal Hamiltonian algebra $\mathfrak{h}_{w,\hbar} = \mathfrak{h}_w[[\hbar]]$, and the weighted continuous coefficient module $\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}} = \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[[\hbar]]$. The Moyal coadjoint action is used only in the vertexwise polynomial-degree-bounded finite-window sector. The scalar QME factor is the balanced $\mathfrak{gl}(N|N)$ supertrace branch, read on the balanced scalar-contact habitat of Definition E.15. The degree-zero Hamiltonian trace comparison is read through the radial/Weyl gate of Statement 2.62: formal

coefficients, Capelli normalization, open-line Wick weights, edge bidegrees, $K_{4,4}$, and named finite non-edge certificates are proved, while the remaining interior bidegrees require $\Omega^{\mathrm{rad}}_{a,b} = 0$, equivalently the decorated PBW Stokes identity $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$, with the normalized signed-row criterion as the exact obstruction theorem.

Local assertion. The proved surface is the product

$$\mathfrak{S}_{\hbar,w}(I) = \mathcal{K}^{w}_{\mathrm{BV}} \times \{0_{\mathrm{sc}}\} \times A^{\mathrm{pp},w}_{\partial,\hbar}(I) \times T_{\mathrm{Ham},\hbar}.$$

Its four components are: weighted BV coefficient-kernel convergence and graphwise weighted boundedness; the zero filtered balanced scalar-contact projection on its habitat; reduced weighted Moyal principal-part current brackets; and the radial/Weyl finite-certificate-relative degree-zero Moyal trace comparison. The current brackets are

$$\{B_f(a), B_g(b)\}_\hbar = B_{\{f,g\}_\hbar}(ab), \qquad \{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$

$$\{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0,$$

with $\mathrm{ad}^\vee_{f,\hbar}$ the full Moyal coadjoint action on the weighted continuous coefficient module, not the PBW one-ψ action.

Proof inputs. The weighted kernel and graphwise boundedness are Theorem 3.85. The scalar zero is Theorem E.14; the filtered zero projection and vanishing scalar-lift classes on the balanced scalar-contact habitat are Proposition E.16. The ordinary scalar alternatives are recorded in Theorem 3.265; their source-replacement branches include the unreduced central character and the opposite central extension $\bar{A}^{\mathrm{cl}}_{-\hbar N\bar{c}}$, where the central cochain $J_{\mathrm{op}} = -k^\vee$ satisfies $d_{\mathrm{CE}}J_{\mathrm{op}} = -\hbar N\bar{c}$. This is not a same-branch Costello-local closed exchange. The current brackets use Theorem 3.237, Corollary B.8, and Construction 3.156, together with Proposition 3.155 and Theorem 3.157 to exclude the polynomial descendant lift. The trace comparison is Theorem 3.269, whose remaining radial/Weyl gate is the moment-ideal primitive of Lemma F.15, sharpened by Proposition F.25 and the decorated PBW Stokes problem $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$, together with Definition F.21, Statement F.22, and Proposition F.27. Lemma E.4 is not an input to this theorem except on the balanced habitat, where Proposition E.16 kills the scalar-lift classes. It becomes the extension obstruction for promoting the scalar result to the full graph/counterterm habitat.

Next obligation. The next theorem starts with the filtered scalar-lift problem in the local-functional complex

$$\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I) = \Big(O^{\mathrm{bd}}_{\mathrm{loc}}\big(\mathcal{E}_w|_I;\ A^{\mathrm{pp},w}_{\partial,\hbar}(I)\big), Q + \{I^w_0, -\}_\hbar\Big).$$

On the balanced scalar-contact habitat, the choice $\widehat{\sigma}_{\mathrm{bal,sc}} = 0$ is already a filtered chain projection by Proposition E.16. For the full graph/counterterm habitat one must either prove the same superdimension factorization or kill the successive chain-lift classes

$$o^{(r)}_{\sigma,w}(I) \in H^1\Big(\mathrm{Hom}\big(\mathrm{gr}_F\,\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I), C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]]\big)_{-r}\Big), \qquad r > 0.$$

Only on such a habitat does the filtered chain projection $\widehat{\sigma}_{\mathrm{sc}}$ exist. Then the residual complex is $\ker\widehat{\sigma}_{\mathrm{sc}}$, and the next obstruction is to construct the bulk-to-defect curvature cocycle with zero scalar projection and kill the class

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker\widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$$

by compatible finite-window counterterms. Windowwise, this is the inverse-limit H^1 obstruction and the $\varprojlim^1 H^0$ primitive-compatibility obstruction of Theorem 3.98. At the first non-scalar order n_0, a

closed-exchange repair is typed by finite-window complexes $\mathcal{X}^{\bullet}_{w,M}(I)$, cochain maps $\Xi_M : \mathcal{X}^{\bullet}_{w,M}(I) \to \mathcal{Q}^{\bullet}_{w,M}(I)$, and the equation

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + \Xi_M(W_{n_0,M}) + d_M C_{n_0,M} = 0.$$

The compatible branch is equivalent to vanishing of

$$\left(\beta^{\mathrm{cl}}_{n_0}, \mu^{\mathrm{cl}}_{n_0}, \lambda^{\mathrm{cl}}_{n_0}\right),$$

where $\beta^{\mathrm{cl}}_{n_0}$ is the cokernel class of $\varprojlim H^1(\Xi_M)$, while $\mu^{\mathrm{cl}}_{n_0} \in \varprojlim^1 H^0(\mathcal{X}^{\bullet}_{w,M}(I))$ and $\lambda^{\mathrm{cl}}_{n_0} \in \varprojlim^1 H^0(\mathcal{Q}^{\bullet}_{w,M}(I))$ are the closed-kernel and counterterm primitive-compatibility classes. The bulk-to-defect kernel must be the continuous curved kernel of Definition B.14, with generator-level bracket defects killed by compatible d_{QME}-homotopies as in Proposition B.21. The minimal full-equivariant finite-window branch is the named exception: Theorem E.42 proves $R^{\mathrm{ns}}_{n,M} = 0$, $\mathfrak{o}^{\mathrm{ns}}_{n,M} = 0$, and $C_{n,M} = 0$ for every $n \geq 1$ only in the finite marked list generated by the strict current-interface rows, α_{sc}, and its subset-deletion closure, with no additional positive-order seed graph, CE-source face, graph counterterm face, or nonzero scalar-zero row. Outside that habitat, until the scalar-contact extension classes, the closed-exchange obstruction triple, and the reduced non-scalar class all vanish in the relevant package, no larger non-scalar Costello graph/QME construction is asserted.

Statement 2.64 *(Statement: normal Ω weighted kernels and QME boundary). Status. finite-window weighted kernel admissibility under the stated character hypotheses; Costello brane-defect QME remains an obstruction criterion*. Normal Ω-localization supplies weights, denominators, and candidate contractions. It does not by itself supply propagators, scalar gates, non-scalar primitives, counterterms, centrality homotopies, or a physical state.

Exact hypotheses. The brane-preserving torus acts on

$$N_L X = \mathbb{R}_s \oplus \mathbb{C}_{z_1} \oplus \mathbb{C}_{z_2}$$

with t fixed. Its vector field and Cartan differential are

$$V_\Omega = \epsilon_s s\partial_s + \epsilon_1 z_1 \partial_{z_1} + \epsilon_2 z_2 \partial_{z_2}, \qquad Q_\Omega = Q + \iota_{V_\Omega}, \qquad Q_\Omega^2 = L_{V_\Omega}.$$

Hence every dg assertion is made on the invariant/basic complex, or it keeps the L_{V_Ω}-curvature explicitly.

For a finite normal/Hamiltonian window

$$\mathsf{W}_{N,M} = \{(n,a,b) \mid 0 \leq n \leq N,\ a,b \geq 0,\ 1 \leq a+b \leq M\},$$

the Hamiltonian and residue-dual characters are

$$\chi^H_{n,a,b} = n\epsilon_s + a\epsilon_1 + b\epsilon_2, \qquad \chi^\vee_{n,a,b} = n\epsilon_s - a\epsilon_1 - b\epsilon_2.$$

Let $\mathsf{X}_{N,M}$ be the finite set of nonzero formal characters among $\chi^H_{n,a,b}$ and $\chi^\vee_{n,a,b}$ for the moving summands contracted in the window. The finite-window coefficient ring is

$$R^{N,M}_\Omega = \mathbb{C}[\epsilon_s, \epsilon_1, \epsilon_2, \hbar_\omega][\chi^{-1} \mid \chi \in \mathsf{X}_{N,M}], \qquad w(\hbar_\omega) = \epsilon_1 + \epsilon_2.$$

Finite-window nonresonance means that every moving character inverted in the window is nonzero. Analytic all-window use requires the stronger small-denominator bound

$$|\chi_{n,a,b}|^{-1} \leq AQ^{n+a+b}$$

for some A, Q in the chosen specialization.

Weighted kernel assertion. With exponential weight $w(d) = q^d$, $q > 1$, the finite-window diagonal coefficient tensor is

$$C^{N,M}_{\Omega,w} = \sum_{\alpha \in \mathsf{A}_{N,M}} e_{\alpha,w} \otimes e^{\alpha}_{w^{-1}}, \qquad e_{\alpha,w} = w(d(\alpha))e_\alpha, \quad e^{\alpha}_{w^{-1}} = w(d(\alpha))^{-1}e^{\alpha}.$$

If the normal Koszul numerator satisfies

$$Q_\Omega k^{\mathrm{Kos}}_\alpha + k^{\mathrm{Kos}}_\alpha Q_\Omega = \chi_\alpha \ \mathrm{id}_\alpha,$$

then on the nonresonant moving summands the localized normal contracting kernel is

$$H^{N,M}_{\Omega,w} = \sum_{\alpha \in \mathsf{A}^{\mathrm{mov}}_{N,M}} \chi_\alpha^{-1} e_{\alpha,w} \otimes e^{\alpha}_{w^{-1}} \otimes k^{\mathrm{Kos}}_\alpha,$$

and it obeys

$$Q_\Omega H^{N,M}_{\Omega,w} + H^{N,M}_{\Omega,w} Q_\Omega = \mathrm{id} - \mathrm{pr}_{\chi=0}.$$

Zero characters are resonant summands; they are not removable denominators.

Normalization. The coefficient Casimir is unchanged by the choice of residue or Euler convention, but scalar-contact and trace rows are not. The two branches are

$$\nu^{\mathrm{res}}_\Omega = 1, \qquad \nu^{\mathrm{Eu}}_\Omega = \sigma_s(\epsilon_s\epsilon_1\epsilon_2)^{-1},$$

with σ_s the real s-orientation sign. A theorem using both conventions must supply the comparison constant and its sign.

QME boundary. A normal weighted kernel promotes to an equivariant Costello brane-defect QME solution only after the finite-window graph package, scalar projection, non-scalar kernel complex, and primitive tower have been built. The scalar projection starts with the relative Hom defect

$$\delta^{(0)}_{\Omega,\sigma,M} = d_{\mathrm{CE},\Omega}\sigma_{\mathrm{sc},\Omega,M} - \sigma_{\mathrm{sc},\Omega,M} d_{\mathrm{gr},\Omega},$$

and higher classes

$$\mathfrak{o}^{(r)}_{\Omega,\sigma,M} \in H^1\Big(\mathrm{Hom}_{T_\Omega,-r}(\mathrm{gr}_F\, \mathfrak{Q}^\bullet_{\mathcal{G},\Omega,M}, S^\bullet_{\Omega,M})\Big).$$

Only after these vanish is

$$\mathcal{K}^\bullet_{\mathrm{ns},\Omega,M} = \ker \widehat{\sigma}_{\mathrm{sc},\Omega,M}$$

a $d_{M,\Omega}$-complex. The order-n residual and scalar gate are

$$\begin{aligned} R^{\mathrm{ns}}_{n,\Omega,M} = [\hbar^n_{\mathrm{QME}}]\big(&\mathrm{Curv}_{\mathbf{K},\Omega,M} + d_{M,\Omega} C_{<n,\Omega,M} \\ &+ \tfrac{1}{2}\{C_{<n,\Omega,M}, C_{<n,\Omega,M}\}_{\hbar_{\mathrm{QME}},M,\Omega}\big), \end{aligned}$$

$$\mathfrak{s}_{n,\Omega,M} = \widehat{\sigma}_{\mathrm{sc},\Omega,M}(R^{\mathrm{ns}}_{n,\Omega,M}).$$

If $\mathfrak{s}_{n,\Omega,M} = 0$, the non-scalar class is

$$\mathfrak{o}^{\mathrm{ns}}_{n,\Omega,M} = [R^{\mathrm{ns}}_{n,\Omega,M}] \in H^1(\mathcal{K}^\bullet_{\mathrm{ns},\Omega,M}),$$

and compatible counterterms also require the corresponding $\varprojlim^1 H^0$ primitive-compatibility class to vanish.

Finite-row primitive interface. Once a scalar-zero finite row array is supplied, the primitive search is the linear system

$$A^M c = -r^M.$$

A left-null covector $\ell A^M = 0$, $\ell(r^M) \neq 0$, is a finite-row H^1-certificate. Windowwise primitive compatibility is the Roos class

$$[P^{M,N} c_M - c_N] \in H^0(\mathcal{K}^\bullet_{\mathrm{ns},N}(I)).$$

The θ_3 one-row obstruction is theorem-grade only for the displayed finite subcomplex in which $\mathcal{K}^0_{\theta_3,M} = 0$ and $\mathcal{K}^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}$. It does not rule out a primitive in a larger local-functional complex.

Proof inputs. The finite normal windows and kernel admissibility are Definition 3.80, Theorem 3.81, and Corollary 3.82. The equivariant QME data and criterion are Definition E.36, Theorem E.37, and Proposition E.45. The algebraic Ω-weighted current brackets on the stable brane target are Theorem B.10. The finite-row linear algebra is Proposition 3.180 and Proposition 3.182.

Obstruction vector. The exact unresolved QME datum is

$$\mathrm{Ob}^{\mathrm{QME}}_{\Omega,w} = \Big(\mathcal{R}_\Omega, \mathrm{ob}_{\Omega,\nu}, \{\mathfrak{s}_{n,\Omega}\}_{n\geq 1}, \{(([\mathfrak{v}^{\mathrm{ns}}_{n,\Omega,M}])_M, \lambda_{n,\Omega})\}_{n\geq 1}\Big).$$

Its common zero locus, together with invariant centrality primitives, is the theorem surface for an equivariant brane-defect QME solution.

Statement 2.65 (Statement: stratified trace and physical Ω-large-N criterion). Status. theorem-surface with exact data and obstruction vectors. Algebraic stable trace invariant theory supplies candidate single-trace labels. It does not construct a stratified brane factorization algebra, a state, a topology, Ward identities, or a genus expansion.

Stratified datum. Let

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}.$$

For a chosen finite normal window (N_{win}, M), the stronger brane theorem is a functor

$$\mathcal{F}^{\mathrm{str}}_{\Omega,N} : \mathrm{Disk}^{\mathrm{str}}_{X,L} \longrightarrow \mathrm{Ch}_{R^{N_{\mathrm{win}},M}_\Omega[[\hbar_{\mathrm{QME}}]]},$$

whose values include bulk mixed HT disks in $X \setminus L$, brane intervals in L, and collars meeting the lower stratum. The finite brane branch is

$$A^q_{N,\Omega} = \mathrm{Weyl}_{\hbar_{\mathrm{W}}}(\mathrm{Mat}^2_N)//_{\hbar_{\mathrm{W}}} GL_N, \qquad YX - XY + \hbar_{\mathrm{W}} NI = 0,$$

with

$$w(X) = \epsilon_1, \qquad w(Y) = \epsilon_2, \qquad w(\hbar_{\mathrm{W}}) = \epsilon_1 + \epsilon_2.$$

The stable brane branch is the weighted principal-part current target $A^{\mathrm{pp},w}_{\partial,\Omega,\hbar_{\mathrm{W}}}(I)$. On its algebraic current subtarget, the Ω-weighted brackets are those of Theorem B.10; off the self-dual locus they are either L^{-1}_ω-valued or scalarized by $\hbar_\omega$. These current brackets are algebraic current data and do not supply Costello graph kernels for arbitrary distributional labels. Choosing one branch or constructing a comparison between them is part of the theorem. Replacing the displayed finite-window ring by an all-window coefficient completion requires the separate q-moderate small-denominator theorem.

On the reduced product basis whose basics are product bulk disks $U \times B \subset X \setminus L$, brane intervals $I \subset L$, and finite disjoint unions, extension by zero gives a reduced prefactorization datum with

$$\mathcal{F}^{\mathrm{red}}_{\Omega,X}(U \times B) = C^{\bullet}_{\mathrm{CE,cont}}\big(\Omega^{\bullet}_{c}(U)\widehat{\otimes}\Omega^{0,\bullet}_{c}(B)\widehat{\otimes}(\mathfrak{h} \ltimes \mathcal{P}[1])\big)\widehat{\otimes}R^{N_{\mathrm{win}},\mathcal{M}}_{\Omega},$$

and

$$\mathcal{F}^{\mathrm{red}}_{\Omega,L}(I) = A^{\mathrm{pp},w}_{\partial,\Omega,\hbar_{\mathrm{W}}}(I).$$

Thus $\delta_{\mathrm{pref}} = \delta_{\mathrm{assoc}} = 0$ only on this reduced product-basis datum. The collar module $\mathcal{M}_{\Omega,N}(U;I)$, products beyond the reduced basis, and stratified Weiss descent must still be supplied. The internal obstruction vector is

$$\mathrm{Ob}^{\mathrm{int}}_{\mathrm{str},\Omega,N} = (\delta_{\mathrm{pref}}, \delta_{\mathrm{assoc}}, \delta^{\check{C}/R}_{\mathrm{Weiss}}, \mathrm{ob}_{\mathrm{collar}}, \mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}, \mathrm{ob}^{P_0}_{\mathrm{cent}}, \mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}, \mathrm{ob}_{\Omega-\mathrm{basic}}, \mathrm{ob}_{\mathrm{tr-top}}).$$

Here $\mathrm{ob}^{P_0}_{\mathrm{cent}}$ is the $P_{0,\hbar_{\mathrm{QME}}}$-centrality obstruction in the brane-defect local-functional complex, not the reduced Weyl/Moyal current bracket over $\hbar_{\mathrm{W}}$. The source-primary class $\mathrm{ob}^{\mathrm{src}}_{\mathrm{CFG}}$ is an external matched-conventions gate, not an internal component of $\mathrm{Ob}^{\mathrm{int}}_{\mathrm{str},\Omega,N}$. Here

$$\delta^{\check{C}/R}_{\mathrm{Weiss}} = \Big(\{H^{\bullet}(\Delta^{N,\mathcal{M}}_{V,\mathcal{U}})\}_{V,\mathcal{U},N,\mathcal{M}}, \mathfrak{w}^{\check{C}/R}_{V,\Omega,N,\mathcal{M}}\Big).$$

For a collared stratified Weiss cover $\mathcal{U}$ of V, the precise descent defect is the Cech cone

$$\Delta^{N,\mathcal{M}}_{V,\mathcal{U}} = \mathrm{Cone}\Big(\mathcal{F}^{\mathrm{str}}_{\Omega,N,\mathcal{M}}(V) \to \mathrm{Tot}\,\check{C}^{\bullet}(\mathcal{U};\mathcal{F}^{\mathrm{str}}_{\Omega,N,\mathcal{M}})\Big).$$

Full Weiss descent requires $H^{\bullet}(\Delta^{N,\mathcal{M}}_{V,\mathcal{U}}) = 0$ for every such cover and vanishing of the associated Roos compatibility class $\mathfrak{w}^{\check{C}/R}_{V,\Omega,N,\mathcal{M}}$ under refinements, finite windows, interval inclusions, and admissible weight transports. The centrality terms require product and $P_{0,\hbar_{\mathrm{QME}}}$-homotopies against unreduced brane observables:

$$d_{\mathrm{QME}}H^{\mathrm{prod}}_{x,a} = \rho(x)a - (-1)^{|x||a|}a\rho(x),$$

$$d_{\mathrm{QME}}H^{P_0}_{x,a} = \{\rho(x), a\}_{P_{0,\hbar_{\mathrm{QME}}}} - \rho_{\mathrm{mod}}(x;a).$$

Reduced current brackets alone do not supply these homotopies.

Physical trace-state criterion. A physical Ω-large-N theorem fixes the 't Hooft scaling

$$\lambda = N\hbar_{\mathrm{W},N}, \qquad \hbar_{\mathrm{W},N} = \lambda/N,$$

and a continuous cyclic state

$$\omega_{N,\Omega}\colon A^{q}_{N,\Omega,\lambda} \longrightarrow R^{N_{\mathrm{win}},\mathcal{M}}_{\Omega}$$

or equivalently a state on $\int_{L\subset X}\mathcal{F}^{\mathrm{str}}_{\Omega,N}$, with a BRST/BV lift annihilating the quantum moment ideal. Its normalization datum

$$\nu_{N,\Omega} = (\omega_{N,\Omega}(1) = 1,\ O^{(N)}_{0,0} = N,\ \lambda = N\hbar_{\mathrm{W},N},\ \nu^{\mathrm{res}}_{\Omega} \text{ or } \nu^{\mathrm{Eu}}_{\Omega})$$

fixes the empty trace and the residue/Euler branch. The two localization conventions differ by the inverse normal Euler factor $(\epsilon_3\epsilon_1\epsilon_2)^{-1}$, up to the real orientation sign; they cannot be mixed silently.

The trace-test algebra is

$$\mathscr{T}_+ = R_\Omega^{N_{\mathrm{win}},\mathcal{M}}[[p_{a,b} \mid a,b \geq 0,\ a+b > 0]], \qquad p_{a,b} \mapsto O_{a,b}^{(N)} = \mathrm{Tr}(X^a Y^b).$$

Connected cumulants are

$$\langle F_1, \dots, F_k \rangle_{N,\Omega}^c = [t_1 \cdots t_k] \log \omega_{N,\Omega}\left(\exp\left(\sum_i t_i F_i\right)\right).$$

In the unnormalized single-trace convention the physical genus expansion is

$$\langle O_{a_1,b_1}^{(N)}, \dots, O_{a_k,b_k}^{(N)} \rangle_{N,\Omega}^c \sim \sum_{g \geq 0} N^{2-2g-k} F_{g;k}^\Omega((a_1,b_1), \dots, (a_k,b_k))$$

in the coefficientwise N^{-2}-Poincare topology $\mathcal{T}_\Omega$. With smeared insertions on L, the coefficients lie in $\mathcal{D}'(L^k)\widehat{\otimes} R_\Omega^{N_{\mathrm{win}},\mathcal{M}}$.

Ward identities. If

$$D_{N,\Omega} = Q_\Omega + \{I_{N,\Omega}, -\}_{\hbar_{\mathrm{W},N}} + \hbar_{\mathrm{QME}} \Delta_{N,\Omega}$$

is the renormalized brane-defect differential and $D_{N,\Omega}^2 = 0$, the state must satisfy

$$\omega_{N,\Omega}(D_{N,\Omega} F) = 0, \qquad \omega_{N,\Omega}(Q_\Omega F) = 0,$$

$$\omega_{N,\Omega}(\widehat{\mu}(\xi) F) = 0, \qquad \omega_{N,\Omega}(\mathrm{Curv}(K_{\Omega,N}) F) = 0.$$

If $\mathrm{Curv}(K_{\Omega,N}) \neq 0$, the last expression is a Ward anomaly, not a protected trace theorem.

Exact obstruction vector. The physical theorem surface is closed only when

$$\mathrm{Ob}_{\Omega,\mathrm{phys}} = (\mathrm{ob}_{\Omega,\mathrm{Cartan}}, \mathrm{ob}_{\Omega,\mathrm{contr}}, \mathrm{ob}_{\mathrm{red}}, \mathrm{ob}_{\mathrm{state}}, \mathrm{ob}_{\mathrm{norm}}, \mathrm{ob}_{\mathrm{QME}}^{\mathrm{sc}},$$
$$\mathrm{ob}_{\mathrm{QME}}^{\mathrm{ns}}, \mathrm{ob}_{\mathrm{cent}}, \mathrm{ob}_{\mathrm{Ward}}, \mathrm{ob}_{\mathrm{cum}}, \mathrm{ob}_{\mathrm{asymp}}, \mathrm{ob}_{\mathrm{src}})$$

vanishes. The scalar contact entry in the ordinary scalar-reduced branch is $\hbar_{\mathrm{W},N} N[\bar{c}] = \lambda[\bar{c}]$; the BV/QME loop parameter remains $\hbar_{\mathrm{QME}}$ unless the QME normalization datum explicitly identifies it with $\hbar_{\mathrm{W},N}$. The non-scalar entry is the class in $H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$ together with the finite-window obstruction pairs. External factorization-homology, Chern–Simons, Costello, CFG, compact Calabi–Yau, Vol III, or physical large-N analogies enter only after a primary-source fixture and matched-conventions morphism to (X, L) are supplied.

Statement 2.66 (Statement: local graph/QME branch catalogue). Status. proved on the named local habitats, relative to supplied finite-window data where stated, and obstructed by the displayed classes otherwise. This is the theorem-surface ledger for the present local graph/QME branches. It is not a compact target or global comparison statement.

Exact hypotheses. Work only in the formal-local mixed HT model on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$, with the weighted finite-window brane-defect local-functional complexes, scalar-contact filtrations, Costello finite-window graph packages, and principal-part current targets fixed in the cited definitions. All graph claims are finite-window or compatible inverse-limit statements; no summation over all graph topologies is included unless a separate graph-summability datum is supplied.

On a chosen Weyl branch, the native finite-window Costello habitat is the completed local-functional complex

$$\mathfrak{Q}_{\mathcal{G},\mathcal{M}}^\bullet(I) = \left(\mathcal{O}_{\mathrm{loc},\mathcal{G},\mathcal{M}}^{\mathrm{bd}}(\mathcal{E}_w^{\mathcal{M}}|_I; \mathcal{A}_{\partial,\hbar_{\mathrm{W}}}^{\mathrm{pp},w,\mathcal{M}}(I)), d_{\mathcal{M}}\right), \qquad d_{\mathcal{M}} = Q + \{I_{0,\mathcal{M}}^w, -\}_{\hbar_{\mathrm{W}},\mathcal{M}},$$

together with the Hom-valued kernel complex

$$\mathbf{K}^{\bullet}_{\mathcal{G},M}(I) = \mathrm{Hom}_{\mathrm{cont}}(C^{\bullet}_{\mathrm{lab}}(\mathcal{G}_M), \mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I)).$$

The source-face differential is part of this habitat. Closed rows, scalar gates, θ_3-source rows, and primitive compatibility are therefore finite-window row data, not evidence for an all-graph QME.

Local assertion. The branches have the following exact strength.

(1) *Balanced scalar-contact projection.* Proved on every balanced scalar-contact habitat, with $\widehat{\sigma}_{\mathrm{bal,sc}} = 0$, zero chain-map defect, and vanishing $o^{(r)}_{\sigma,w,\mathrm{bal}}(I)$. The same assertion is proved on a balanced graph/counterterm subhabitat once the identity-loop supertrace factorization is part of the habitat data. Outside these habitats the obstruction is the first nonzero class $o^{(r)}_{\sigma,w}(I)$ of the scalar-contact lift tower.

(2) *Full-equivariant scalar shadow.* Proved on complete finite-window inverse-limit habitats generated windowwise by codimension-two closed finite-window marked Costello lists and full-equivariant Brauer marker tensors. For $V = \mathbb{C}^{N|N}$, every cyclic supertrace of a transported full-equivariant marker tensor is zero, so the zero scalar projection is a filtered chain map there. If the marked boundary table, incidence signs, marker transports, or codimension-two closure are not supplied, the missing entry is the exact obstruction to forming this scalar-shadow theorem.

(3) *Finite closed graph list.* Proved as a finite-window closure criterion. A seed graph set and its boundary operations give a legitimate finite graph list only when every field-differential, BV-edge, collision, counterterm, and source face remains in the list with compatible signs and marker transports. Failure of the codimension-two closure equations is the obstruction to the finite marked differential $d^2_{\mathrm{mk}} = 0$.

(4) *Finite-window arrays and the minimal full-equivariant branch.* The computed rows are theorem-grade: the length-one current-interface row has $R^{\mathrm{row}}_{0,M} = S_{0,M} = 0$, and the one-cross two-Hamiltonian scalar-contact row has scalar gate $N\gamma_{\mathbf{1}}$, $N(\lambda_0 - \lambda_1)\gamma_{\mathbf{1}}$, or 0 in the ordinary, even-block, or full-equivariant branch. In the full-equivariant branch this row is zero as a local functional. The minimal finite marked list generated by the strict current-interface rows, this scalar-contact row, and its subset-deletion closure therefore satisfies, for every $n \geq 1$,

$$R^{\mathrm{ns}}_{n,M} = 0, \qquad \mathfrak{v}^{\mathrm{ns}}_{n,M} = 0, \qquad C_{n,M} = 0.$$

If zero rows are suppressed, the positive-order minimal arrays are empty; if labelled zero rows are retained, every retained positive-order row has zero row value, zero scalar gate, and zero truncation row. In particular the binary centrality row in the reduced coefficient-current/minimal full-equivariant habitat has $R^{\mathrm{cent,coef}}_{x,y,M} = 0$, $S^{\mathrm{cent,coef}}_{x,y,M} = 0$, and $H^{\mathrm{coef},M}_{x,y} = 0$; equivalently $A^M_{\min} : 0 \to 0$, $r^M_{\min} = 0$, and $h^M_{\min} = 0$, with zero Roos class. For any larger package the first obstruction is not supplied by this theorem. The enlarged binary row is the matrix problem $A^M h^M = -r^M$ after the row values, scalar gate, primitive columns, and transition matrices have been supplied. The first supplied open source-face row has

$$R^{\mathrm{row}}_{\theta_3,M} = -K^M_{\Theta_3}(\zeta_{M,\nu_3}) =: \mathsf{E}_{\theta_3,M}, \qquad S_{\theta_3,M} = 0.$$

In the displayed one-row finite subcomplex

$$\mathcal{K}^0_{\theta_3,M} = 0, \qquad \mathcal{K}^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, \qquad d_{\theta_3,M} = 0,$$

the covector $\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$ certifies a finite-row obstruction. This does not rule out a primitive in a larger local-functional complex or after Hom-valued companion rows are added. The tested finite enlargements are now part of the obstruction theorem. In the ordinary pure two-u CE source envelope, in every finite Hamiltonian degree, the covector

$$q_{2u} = e^*_{((H_{0,1},H_{2,0}),c_{\rho_{2,1}})} - \frac{1}{2}e^*_{((H_{0,1},H_{0,1}),c_{\rho_{0,2}})}$$

satisfies

$$q_{2u}d_{\mathrm{CE}} = 0, \qquad q_{2u}(\zeta_{M,\nu_3}) = 1.$$

Thus no ordinary pure two-u degree enlargement contains a CE ancestor for ζ_{M,ν_3}. This is an exact obstruction theorem for that source envelope, not a universal no-go for larger local-functional, marked-source, or Hom-valued companion habitats. The tested eight-face companion row has coordinates

$$(2, -2, 3, 2, -1, 1, -2, -3)$$

on

$$E_{\nu_{02}}, E_{\nu_{20}}, E_{\nu_{03}}, E_{\nu_{12}^{(1)}}, E_{\nu_{12}^{(2)}}, E_{\theta_3}, E_{\nu_{21}^{(2)}}, E_{\nu_{30}},$$

and diagonal transport to the $N = 2$ window leaves $2E^2_{\nu_{02}} - 2E^2_{\nu_{20}}$. The same eight-face row has q_{2u}-value zero, so it is not the missing marked source column. The lower-window functional $\lambda^{(2)}_{\nu_{02}} = \frac{1}{2}(E^2_{\nu_{02}})^*$ detects the transported residual. The next possible repair is a controlled enlargement of the row complex, not reuse of either tested finite route. The exact missing data are: a counterterm column $A^M_{\theta_3,C} = -1$ with $d_M C^{\mathrm{ct}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$, scalar-zero value, and zero Roos class; a marked source generator $\eta^{\mathrm{mark}}_{\theta_3,M}$ outside the ordinary pure two-u algebra, whose new boundary column has q_{2u}-value 1 and satisfies codimension-two closure, Hom-valued Bianchi cancellation, scalar-zero value, and compatible transport; or a marked companion relation $R^{\mathrm{cand}}_{\Theta_3,M} = 0$ together with lower-window transport $v^{M,N}R^{\mathrm{cand}}_{\Theta_3,M} = 0$, including the $N = 2$ relation killing $2E^2_{\nu_{02}} - 2E^2_{\nu_{20}}$ by a derived non-diagonal transport, lower primitive, or relation $E^2_{\nu_{02}} = E^2_{\nu_{20}}$. The canonical source-coordinate lower primitive candidate is

$$\zeta^0_2 = u_{a,H_{1,0}}u_{b,H_{0,1}}, \qquad D^2_{02,20} = K^2_{\Theta_3}(\zeta^0_2), \qquad d_{\mathrm{CE}}\zeta^0_2 = 2\zeta_{\nu_{02}} - 2\zeta_{\nu_{20}}.$$

This is source exactness. In the local-functional complex

$$d_{\mathrm{ns},2}D^2_{02,20} = -2E^2_{\nu_{02}} + 2E^2_{\nu_{20}} + \mathfrak{b}^2_{02,20},$$

where

$$\mathfrak{b}^2_{02,20} = d_{\mathrm{ns},2}K^2_{\Theta_3}(\zeta^0_2) - K^2_{\Theta_3}(d_{\mathrm{CE}}\zeta^0_2).$$

In the Bianchi-exposed lower basis $(E^2_{\nu_{02}}, E^2_{\nu_{20}}, \mathfrak{b}^2_{02,20})$, the column and residual are

$$A^2_D = (-2, 2, 1)^T, \qquad r_2 = (2, -2, 0)^T.$$

The cokernel

$$\widetilde{\lambda}_{02,\mathfrak{b}} = \frac{1}{2}(E^2_{\nu_{02}})^* + (\mathfrak{b}^2_{02,20})^*$$

kills A^2_D and evaluates to 1 on r_2. Thus the lower source primitive becomes a local-functional primitive only after $\mathfrak{b}^2_{02,20}$ is killed, or after adding a scalar-zero local counterterm $B^2_{02,20}$ with $d_{\mathrm{ns},2}B^2_{02,20} =$

$-\mathfrak{b}^2_{02,20}$ and compatible transport. In the current finite-row habitat no such $B^2_{02,20}$ exists: the target $(0,0,-1)^T = -\mathfrak{b}^2_{02,20}$ is detected by

$$\widetilde{\lambda}_{02,\mathfrak{b}}((0,0,-1)^T) = -1,$$

while $\widetilde{\lambda}_{02,\mathfrak{b}}$ kills the only available degree-zero column. The controlled enlargement must add a genuine scalar-zero local-functional column $A^2_B = (0,0,-1)^T$, with locality or wavefront admissibility and finite-window transport. For a tower the Bianchi transport cocycle is

$$\Delta^1_{M,N} = -\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N.$$

A compatible counterterm family B^M kills it exactly when

$$\Delta^1_{M,N} = d_{\mathrm{ns},N}(\pi_{M,N}B^M - B^N).$$

Hence the controlled enlargement must also kill the H^1 class of $\Delta^1_{M,N}$ and the secondary $\varprojlim^1 H^0$ primitive-compatibility class; otherwise the Bianchi row remains obstructed. For any larger package the first undefined obstruction is the unfilled row sum

$$R^{\mathrm{ns}}_{1,M} = \sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{1,M}\setminus\{\alpha_{\mathrm{sc}}\}} R^{\mathrm{row}}_{\alpha,M},$$

or, when no nonzero order-one row is supplied, the corresponding order-two and order-three row sums. Those larger rows require their graph type, finite-window labels, marker tensor, incidence signs, row values, scalar gates, and truncation coefficients before the H^1 class and the Roos finite-window compatibility class are defined.

(5) *Regular-density curvature certificate.* Proved under the regular-density graph package hypotheses: all cotangent defect labels are smooth compactly supported densities, the finite graph list is closed under the relevant d_M- and BV-bracket terms, the scalar projection is zero on the chosen subhabitat, and the lower counterterms are compatible and regular-density. Then the evaluated curvature residual lies in $\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$, is killed by $W_{n_0,M} = -\mathfrak{o}^{\mathrm{ns,reg}}_{n_0,M}$ and $C_{n_0,M} = 0$, and the classes

$$\eta^{\mathrm{reg}}_{n_0},\quad \kappa^{\mathrm{reg}}_{n_0},\quad \beta^{\mathrm{cl,reg}}_{n_0},\quad \mu^{\mathrm{cl}}_{n_0},\quad \lambda^{\mathrm{cl}}_{n_0}$$

vanish. Without this certificate the exact obstruction is the quotient class $\eta^{\mathrm{reg}}_{n_0,M}$ and, after its vanishing, the Roos lift class $\kappa^{\mathrm{reg}}_{n_0}$.

(6) *Wavefront eta obstruction.* Proved as an obstruction theorem, with a positive graph branch on the tuple relation $\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M)$ and on the two sign-conic finite-scaling habitats $\mathcal{D}'_{c,\Lambda_\pm,\mathrm{fs}}(I)$ for ordinary collision-conormal graph kernels. For non-smooth wavefront-admissible labels the singular quotient $\mathcal{S}_c(I) = \mathcal{D}'_c(I)/\mathcal{D}^{\mathrm{reg}}_c(I)$ is nonzero. The finite-window obstruction class is

$$\eta^{\mathrm{reg}}_{n_0,M} = [\mathfrak{s}^{\mathrm{WF}}_{n_0,M}] \in H^1(C^{\bullet,\mathrm{reg}}_{w,M}(I)).$$

It is killed in the original complex exactly when a degree-zero distributional counterterm $C^{\mathrm{sing}}_{n_0,M}$ satisfies $\bar{d}_M q_M(C^{\mathrm{sing}}_{n_0,M}) = -\mathfrak{s}^{\mathrm{WF}}_{n_0,M}$. If not, the regular-density branch is obstructed. The formal primitive cone adjoins a new generator and is an extension theorem, not a smoothing theorem.

(7) *Radial trace-diagram theorem lane.* The formal Moyal coefficient calculation, Capelli triangular normalization, free indexed normal-ordering formula, stable trace-diagram injectivity, and reduced open-line first/third weights are proved. The all-N ordinary Weyl-trace comparison is reduced to the moment-ideal membership of

$$E_{a,b,N} = \sum_{j=0}^{b} \mathrm{Tr}(Y^j X^a Y^{b-j}) - J_N\Big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\Big)$$

in the chosen quantum moment-map quotient. It holds exactly when one constructs $A^j{}_i(a,b,N)$ with $E_{a,b,N} = \sum_{i,j} A^j{}_i(a,b,N)\widehat{\mu}^i{}_j$. This condition is equivalently the free normal-diagram equation

$$\mathfrak{v}_{a,b} = [R^{\mathrm{free}}_{a,b}] \in \mathrm{coker}\big(C_{a,b}\colon \ker\partial \longrightarrow \mathsf{Diag}_{a,b}\big)$$

plus a functorial lift $K_{a,b}$. In the dual-potential form put

$$B_{a,b}(W) = (\partial W, C^+_{a,b}(W)), \qquad \eta_{a,b} = (T_{a,b}, E^+_{a,b}),$$

and

$$\Omega^{\mathrm{rad}}_{a,b} = [\eta_{a,b}] \in \mathrm{coker}\, B_{a,b}.$$

The all-N ordinary trace statement is equivalent to $\Omega^{\mathrm{rad}}_{a,b} = 0$, equivalently to the decorated PBW Stokes identity $D^{\square}_{a,b} = C^+_{a,b}\partial_2$. Failure is exactly a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. The edge classes $\mathfrak{v}_{2,b}$ and $\mathfrak{v}_{a,2}$ vanish by the PBW telescoping theorem. The $K_{4,4}$ correction and the named finite non-edge certificates are all-N free normal-diagram certificates. For the remaining interior bidegrees the exact theorem target is a natural contraction of the positive-contraction two-colour necklace complex, equivalently the dual-potential signed-row theorem above. This obstruction is algebraic in the local Weyl/radial comparison and does not provide a Costello graph/QME construction.

Proof inputs. The balanced scalar claims are Theorem E.14, Proposition E.16, and Proposition E.19. The full-equivariant branch is Lemma E.29 and Theorem E.30, with the finite-list and differential data of Definition E.25, Definition E.26, Lemma E.27, and Proposition E.28. The native Hom-valued Costello/QME realization habitat and its row criterion are Definition E.32, Theorem E.33, and Proposition E.34. The finite-window array is Definition 3.173, Proposition 3.174, and Definition E.31. The minimal full-equivariant non-scalar branch is Theorem E.38 and Theorem E.42; larger finite-window packages are governed by Proposition E.39, Proposition E.40, and Proposition E.43. The regular-density and wavefront branches are Proposition 3.105, Theorem 3.106, and Theorem 3.107. The radial/Weyl lane uses Lemma F.15, Definition F.17, Theorem F.18, Lemma F.19, Definition F.21, Statement F.22, Proposition F.27, Proposition F.20, Proposition F.28, and Theorem 3.251.

Scope and exclusions. None of these branch statuses imports or implies compact Calabi–Yau, CoHA, quintic, OSV/GV, Abel–Jacobi, Igusa, Borcherds/BKM, or global BCOV claims.

Statement 2.67 *(Statement: scalar $U(1)$ anomaly). Status. proved in finite algebraic model.* The statement records the scalar-reduction obstruction to lifting the reduced Hamiltonian comparison back to the unreduced constant direction.

Exact hypotheses. The closed algebra is the unreduced polynomial Hamiltonian algebra $\mathfrak{h}_{\mathrm{poly}} = \mathbb{C}[z_1, z_2]$ and its quotient $\bar{A} = \mathfrak{h}_{\mathrm{poly}}/\mathbb{C}\cdot 1$; the open side is the finite-N trace algebra before and after

quotienting $\mathbb{C} \cdot \mathrm{Tr}(1)$; and the quantum comparison uses the same Capelli/Weyl normalization as the degree-zero Moyal theorem.

Local assertion. The scalar-reduction axis carries the distinguished class

$$[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}),$$

represented by the constant Taylor coefficient of the Poisson bracket. The same class appears on the open side as the centre-of-mass $\mathfrak{gl}_N$ anomaly; quantum mechanically its finite-N trace-character representative is the Capelli contact $\hbar N$. All reduced Hamiltonian comparisons are made after killing constants; the unreduced scalar line is governed by this anomaly class.

Proof inputs. Lemma 2.4 identifies the closed Poisson cocycle. Theorem 2.5 proves the closed-side central extension statement. Theorem 2.6 computes the open $U(1)$ centre. Theorem 2.8 identifies the classical cocycle with the Rees central extension whose finite-N trace-character value is the quantum Capelli shift. Corollary 3.243 gives the explicit local central term in the smeared boundary bracket.

Scope and exclusions. The statement does not lift the reduced comparison to the unreduced scalar direction and does not imply compact Calabi–Yau, CoHA, quintic, OSV/GV, Abel–Jacobi, Igusa, Borcherds/BKM, or sister-volume anomaly statements.

Remark 2.68 *(Finite-window status of the BM lane).* Statement 2.38 is used below together with Proposition 2.21. The Bochner–Martinelli–Koppelman obligation is discharged there as current-limit data, finite Matlis moments, and an arity-two coefficient comparison. The finite-window differential identity is blocked, in the native full-current habitat, by $\mathrm{Ob}_N^{\mathrm{BMK\text{-}curr}}$. The strict all-window native E_2 transfer remains downstream of that gate and then of the one-pair/pro-product split recorded by $\mathrm{Ob}_{\mathrm{BM}}^{\Pi}$. The positive all-window datum in this remark is only the one-pair analytic retract $P_{\Pi}^{\mathrm{an}} = \tau P_{\mathrm{an}}$ with $\pi_N P_{\Pi}^{\mathrm{an}} = p_N$, not a native factorization-current transfer.

Theorem 2.69 *(Coordinate CE/PV theorem and admissible Koszul criterion).* In the formal local Hamiltonian disk model the cotangent CE/PV construction extends through the following exact ladder.

(i) For every finite-dimensional Lie algebra $\mathfrak{l}$, the assignment $c \mapsto \theta$, $u \mapsto O$ gives

$$C^{\bullet}_{\mathrm{CE}}\big(T^*[1]\mathfrak{l}\big) \cong \mathrm{PV}\big(\mathbf{S}(\mathfrak{l})\big)$$

as dg graded-commutative algebras with the degree-(-1) cotangent/Schouten bracket.

(ii) For the completed formal symplectic disk $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, the same generator assignment gives the pro coordinate-window cochain isomorphism

$$\Phi_{\mathrm{coord}} \colon C^{\bullet}_{\mathrm{CE,coord}}\big(\mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1]\big) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(\mathcal{A}_{\partial,\mathrm{coord}}).$$

After fixing the residue scalar, $\mathfrak{h}^{\vee}_{\mathrm{cont}} \cong \mathcal{P}$ by Theorem 3.230. If $B \subset \mathrm{PV}_{\mathrm{coord}}(\mathcal{A}_{\partial,\mathrm{coord}})$ is bracket-admissible, then

$$\Phi^{B}_{P_0} \colon \Phi^{-1}_{\mathrm{coord}}(B) \xrightarrow{\sim} B$$

is a dg P_0-isomorphism. If B is Hamiltonian-spanning for a class $S \subset \mathfrak{h}$, then for every $f \in S$ the boundary Hamiltonian B_f is the image of the shifted cotangent coordinate u_f, while the Hamiltonian CE coordinate c^f maps to the vector-field operation. If a kernel-admissible datum $(B, \mathcal{K}_B)$ is supplied, the associated diagonal tensor Θ_B satisfies the Maurer–Cartan equation in $\mathcal{K}_B$. The strict product/direct-sum endpoint carries none of the P_0, boundary-Hamiltonian, or kernel conclusions without these admissibility hypotheses.

(iii) If an admissible Tate, nilpotent, relative, or weighted replacement supplies conilpotent CE chains, exact continuous duality, a filtered twisting cochain, strong convergence, and exact finite-window inverse limits, then the closed-side cochain identity upgrades only to the admissible bar-cobar/enveloping statement

$$\left(C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}})\right)^{!} \simeq \widehat{U}(\mathfrak{g}_{\mathrm{adm}}).$$

If, in addition, an open-trace A_∞-Koszul acceptance datum $(A_{\mathrm{op}}, C_{\mathrm{op}}, B_{\mathrm{CE/PV}}, \tau_{\mathrm{op}})$ of Definition 2.57 is supplied and its obstruction complex is exact, then

$$\Omega\mathrm{Bar}(A_{\mathrm{op}}) \xrightarrow{\sim} B_{\mathrm{CE/PV}}, \qquad A_{\mathrm{op}} \simeq B_{\mathrm{CE/PV}}.$$

If the CE-side cobar comparison also passes the same finite-window and $\varprojlim^1$ tests, then

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) \simeq A^{!}_{\mathrm{op}}.$$

(iv) The obstruction to promoting the strict full formal-disk coefficient pair to this bar-cobar statement is the named obstruction datum: the strict Casimir support obstruction $C_{\mathfrak{h}}$, the low-degree linear translation term $\{z_1, z_2^{N+1}\} = (N+1)z_2^N$, the $H_2 \cong \mathfrak{sl}_2$ lower-central obstruction to conilpotence, the associated-graded cone $H^{\bullet}(\mathrm{Cone}(\mathrm{gr}\,\kappa_\tau))$, and the Milnor $\varprojlim^1$ defect. Quantum or graph-level extensions add the mixed brane-defect QME obstruction class.

Proof. The first item is Theorem 2.44; the second is Theorem 2.48 together with Proposition 2.51. The third item separates two inputs: Lemma 3.217 supplies the closed Lie bar-cobar/enveloping comparison when its lower-central Tate-pronilpotence hypothesis holds, while Lemma 2.56 and Theorem 2.58 supply the additional open A_∞-Koszul comparison only after the open-side acceptance datum is given. The obstruction terms in the fourth item are Theorem 2.52, Remark 3.128, Remark 3.131, and the obstruction clause of Lemma 2.56; the QME class is the residual class of Problem 3.87. Thus the formal-disk pro coordinate-window cochain identity is already a theorem, bracket and kernel realizations are exactly the admissible B clauses of that theorem, and the enveloping/open-Koszul upgrades are precisely the two stated admissibility criteria. □

THEOREM 2.70 *(Formal local Hamiltonian BF/Dirac-brane comparison).* In the formal Darboux local model

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\},$$

let

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1, \qquad P = \mathfrak{h}^{\vee}_{\mathrm{cont}}, \qquad \mathfrak{g} = \mathfrak{h} \ltimes P[1].$$

The manuscript constructs the native package of Proposition 2.22:

(i) the Dirac matrix probe with $Q\psi = [\phi_1, \phi_2]$ and scalar-reduced connected trace observables;

(ii) the closed Hamiltonian BF factorization algebra with fields valued in $\mathfrak{h}[1] \oplus P[2]$;

(iii) the coordinate derived-centre CE/PV map

$$c^I \mapsto \theta^I, \qquad u_I \mapsto O_I,$$

after killing constants on the Hamiltonian side and the empty trace on the brane side;

(iv) the separation of the PBW one-ψ shadow from the deformation-level Matlis/coadjoint cotangent module;

(v) the Bochner–Martinelli–Koppelman current data, the finite-window current obstruction $\mathrm{Ob}_N^{\text{BMK-curr}}$, the one-pair analytic pro-Matlis retract, and the arity-two Matlis comparison.

Section 2.15.1 supplies the local proof dependencies: stable trace and one-ψ PBW shadow, coordinate CE/PV theorem with admissible Koszul criterion, Matlis principal parts, brane-line factorization currents, reduced principal-part defect currents, BMK current-limit data with finite-window obstruction gate and one-pair analytic pro-Matlis retract, scalar-reduced degree-zero Moyal/Weyl deformation, the componentwise quantum coefficient surface, and the U(1) scalar anomaly.

Consequently the coordinate envelope

$$C^{\bullet}_{\mathrm{CE,coord}}(\mathfrak{g})$$

gives the CE/PV Hamiltonian central-operation model attached to the stable open Hamiltonian brane algebra. The finite cotangent identity, coordinate formal-disk reconstruction, and admissible Tate Koszul-duality criterion are the formal-local spine of this coordinate identification. This is a coordinate CE/PV and reduced-current statement. It is not a morphism from closed BV states to the unreduced open factorization centre, and the principal-part current sector is the added reduced local-dual factor of Theorem 3.237. The cotangent direction is realized by the Matlis principal-part module; the polynomial one-ψ descendants give the PBW special-fibre labels and the primitive Koszul homology. The proved quantum coefficient surface is the product

$$\mathfrak{S}_{\hbar,w}(I) = \mathcal{K}^{w}_{\mathrm{BV}} \times \{0_{\mathrm{sc}}\} \times A^{\mathrm{pp},w}_{\partial,\hbar}(I) \times T_{\mathrm{Ham},\hbar}$$

of Theorem 3.270: weighted BV kernel convergence and graphwise boundedness, the balanced scalar-contact zero projection on its habitat, reduced weighted Moyal principal-part current brackets, and the degree-zero Moyal trace comparison controlled by the radial/Weyl normal-form gate. The finite certificate range closes the balanced diagonal through $(6,6)$, all non-edge bidegrees of total degree at most 13, and the total-degree 14 cases

$$(3,11),\ (11,3),\ (4,10),\ (10,4),\ (5,9),\ (9,5);$$

the first undecided total-degree 14 systems are $(6,8)$, $(8,6)$, $(7,7)$. The edge theorem closes all one-moving-letter strips by Proposition F.27. The all-interior theorem is not a conditional slogan: it is the remaining dual-potential obstruction problem. In the form of Proposition F.25, this is the vanishing of

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^{+}_{a,b})] \in \operatorname{coker} B_{a,b},$$

equivalently the assertion that every signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$ satisfies $\lambda(E^{+}_{a,b}) = \phi(T_{a,b})$. The ordinary square-cell necklace cycle mechanism is proved; the missing theorem is the decorated PBW Stokes identity for $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$. The non-scalar Costello graph/QME construction has a theorem-grade finite-window criterion: it is the vanishing problem for

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$$

inside $\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$ of Problem 3.271.

The analytic graph, regulator, unreduced cotangent, non-scalar QME, and cross-volume extensions are the obstruction and theorem data listed in Theorem 2.69 and Section 4.

Proof. Proposition 2.22 fixes the Darboux habitat, the Dirac probe, the Hamiltonian BF coefficient algebra, the scalar quotient, and the PBW/deformation separation. Theorem 2.69 supplies the CE/PV–Koszul spine. Taking the common Hamiltonian consequence of that criterion with the brane-line, Matlis, Moyal/Weyl, quantum-coefficient, and scalar-anomaly statements, together with the BMK current data, finite Matlis moment maps, one-pair analytic pro-Matlis retract, and arity-two output projection of Proposition 2.21, gives the displayed reduced Hamiltonian CE/PV central-operation statement. □

Remark 2.71 *(Admissible CE/PV criterion and specialization).* The CE/PV cochain identity in 2.69 is the full formal-disk statement in the reduced Hamiltonian sector. Its enveloping/Koszul enhancement is the admissible criterion: one must supply the coefficient category in which CE chains are conilpotent, continuous duality is exact, and the filtered cobar spectral sequence converges. Nilpotent, relative, and weighted Tate replacements are mechanisms for satisfying that criterion; the strict product/direct-sum endpoint is governed by the obstruction datum, not by an absent theorem. The other proved statements are dimension-two specific (they use the residue pairing on $\mathbb{C}^2$, the matrix Weyl algebra of pairs (X, Y), and the U(1) center of $\mathfrak{gl}_N$). Higher-dimensional analogues require the same cochain identity together with the appropriate residue pairing, matrix Weyl data, and a separate verification of the required conilpotent, relative, weighted, or completed coefficient hypotheses.

Remark 2.72 *(Dirac probe principle in the reduced Hamiltonian sector).* Theorem 2.70 makes precise the Dirac thesis sentence for the reduced Hamiltonian sector:

> *A stack of branes is a system of probes for the formal symplectic normal disk. Dirac reduction of the probe phase space gives the open trace algebra. Hamiltonian BF theory is the shifted cotangent theory of canonical transformations. In the reduced formal Hamiltonian sector, the coordinate CE cochains of the shifted-cotangent Hamiltonian Lie algebra identify, after a bracket-admissible realization, with reduced P_0 Hamiltonian central operations on the stable open brane algebra.*

This follows from 2.70: the brane stack is the open BV theory of N matrix-valued probes (ϕ_1, ϕ_2) for the formal symplectic disk; Dirac first-class reduction along the effective traceless moment-map ideal $I_N = \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$ produces the stable connected open trace algebra $H^0_{\mathrm{Ham,conn}}(O^{\mathrm{q}}_{\mathrm{brane},\infty})$; the closed Hamiltonian BF theory has fields valued in the shifted-cotangent semidirect product $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ with $\mathfrak{h}$ the Hamiltonian Lie algebra of canonical transformations of the formal disk; and the closed observables $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$ are identified, via Φ, with the reduced CE/PV Hamiltonian central-operation model $Z^{P_0}_{\mathrm{red,Ham}}(A^{\mathrm{Ham}}_\partial)$ on the brane line after de Rham-current contraction. The degree-zero Hamiltonian sector has the radial-parts quantum comparison stated later; the full unreduced factorization centre is the obstruction datum of Problem 3.60.

The proof uses the formal local model alone. Its portable content is not confined to one coefficient envelope: the cochain CE/PV identification is formal-disk level, while the enveloping/Koszul upgrade is the criterion of Theorem 2.69. The unreduced Tate-coefficient cotangent lift is isolated in 3.60 and separated into its precise chain-level, nilpotent, weighted, associated-graded, inverse-limit, and QME-obstruction components. The stronger analytic realization requires a Costello renormalized BV graph construction on an elliptic parent with brane-defect kernel and a vanishing mixed brane-defect QME obstruction class.

The protected twisted-M-theory origin is not used as a proof of the local theorem. It is recorded in §5.1 as a protected-sector transfer criterion: once an ambient background supplies the admissible protected Hamiltonian restriction, the local model applies to that protected sector and no further.

3 The Local Model

The local theorem is built from a single geometric setup: a topological line carrying a stack of probes, crossed with a holomorphic-symplectic disk to which they couple. Quantizing this setup gives the data on which the component theorems are verified.

3.1 Setup

Let

$$M = \mathbb{R}^2, \qquad Y = \mathbb{C}^2, \qquad X = M \times Y,$$

with M the topological factor, Y the holomorphic factor, and all Hamiltonian calculations completed at the formal neighbourhood of $p = 0 \in Y$. The exact A-brane is the standard line $\mathbb{R}_{\text{brane}} \times \{0\} \subset M$ with trivial brane data; the B-brane is the skyscraper $\mathcal{O}_p \in \operatorname{Perf}(Y)$. Their product is the mixed brane

$$L = \mathbb{R}_{\text{brane}} \times \{0\} \times \{p\} \hookrightarrow \mathbb{R}^2_{\text{top}} \times \mathbb{C}^2_{\text{hol}}, \qquad p = 0.$$

The holomorphic factor carries the standard holomorphic symplectic form

$$\omega = dz_1 \wedge dz_2,$$

so the model is topological along $\mathbb{R}^2$ and holomorphic Hamiltonian along $\mathbb{C}^2$.

Proposition 3.1 *(Local mixed model).* The formal local Hamiltonian data used here define a closed mixed holomorphic-topological BV sector and a compatible cyclic open brane BV theory. The constructed operation is the degree-zero Hamiltonian boundary evaluation. The full Poisson factorization-center morphism is Problem 3.60, not part of this proposition.

Proof. On product opens $D \times B$ in the completed formal neighbourhood of $\mathbb{R}^2 \times \{p\} \subset \mathbb{R}^2 \times \mathbb{C}^2$, the closed fields are

$$\mathcal{E}_{\text{cl}}(D \times B) = \Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\text{cont}}[2]\big).$$

Degreewise polynomial calculations use the dense subspace $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The differential is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, so $Q_0 = 0$ in Definition 2.17. The BV pairing is the local pairing on compactly supported fields, obtained by multiplying the form components, integrating over $D \times B$ against $dz_1\, dz_2$, and using the continuous evaluation pairing $\widehat{\mathfrak{h}}^\vee_{\text{cont}} \otimes \widehat{\mathfrak{h}} \to \mathbb{C}$. If $\alpha \in \Omega^p\widehat{\otimes}\widehat{\mathfrak{h}}[1]$ and $\beta \in \Omega^q\widehat{\otimes}\widehat{\mathfrak{h}}^\vee_{\text{cont}}[2]$, then $|\alpha| = p - 1$ and $|\beta| = q - 2$; the top-form condition $p + q = 4$ makes the pairing degree -1.

The free quadratic term

$$S_{\text{free}} = \int_{D\times B} \langle \beta, D\alpha \rangle dz_1\, dz_2$$

is the BV Hamiltonian for the differential D. The interaction in Definition 2.17 is only

$$I_{\text{BF}} = \frac{1}{2}\int_{D\times B} \langle \beta, \{\alpha, \alpha\} \rangle dz_1\, dz_2.$$

The total BF action is $S_{\text{free}} + I_{\text{BF}}$. The Poisson bracket on $\bar{A}$ is the holomorphic bracket determined by $dz_1 \wedge dz_2$. Write

$$F_\alpha = D\alpha + \frac{1}{2}\{\alpha, \alpha\}, \qquad D_\alpha = D + \{\alpha, -\}.$$

Since D is a derivation of the bracket and the bracket satisfies Jacobi, $D_\alpha F_\alpha = 0$. With compact support in the local calculation and with the coadjoint-invariant evaluation pairing, the BV coadjoint transpose is fixed by

$$\int_{D\times B} \langle D_\alpha^\vee \beta, \xi\rangle \, dz_1 dz_2 = -(-1)^{|\beta|} \int_{D\times B} \langle \beta, D_\alpha \xi\rangle \, dz_1 dz_2.$$

variation of $S = \int \langle \beta, F_\alpha\rangle dz_1 dz_2$ gives

$$\frac{1}{2}\{S, S\}_{\mathrm{BV}} = \int_{D\times B} \left\langle D_\alpha^\vee \beta, F_\alpha \right\rangle dz_1 dz_2 = -(-1)^{|\beta|} \int_{D\times B} \left\langle \beta, D_\alpha F_\alpha \right\rangle dz_1 dz_2 = 0.$$

Thus the total BF action satisfies the classical master equation without double-counting the free differential.

Mixed locality is the Cartan homotopy calculation on the interacting observables. For a constant vector field v in the $\mathbb{R}^2$ direction, $[D, \iota_v] = \mathcal{L}_v$; for an anti-holomorphic coordinate vector field $\partial/\partial\bar{z}_i$, $[D, \iota_{\partial/\partial\bar{z}_i}] = \mathcal{L}_{\partial/\partial\bar{z}_i}$. The interaction is built from the de Rham wedge product and holomorphic polydifferential operators, so the contractions are derivations of the local observable algebra and commute with $\{I_{\mathrm{BF}}, -\}$ up to the displayed Lie derivative. Thus $[Q_{\mathrm{obs}}, \iota_v] = \mathcal{L}_v$ and $[Q_{\mathrm{obs}}, \iota_{\partial/\partial\bar{z}_i}] = \mathcal{L}_{\partial/\partial\bar{z}_i}$ on local observables. The de Rham Poincare lemma in the D directions gives the quasi-isomorphism for small disk inclusions, and the Dolbeault Poincare lemma gives the holomorphic dependence in the B direction.

The brane is $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$, with $p \in Z$ understood as the formal holomorphic neighbourhood. Its open fields are

$$\Omega^*(\mathbb{R}) \otimes \mathbf{Ext}^*_{\mathbb{C}[z_1,z_2]}(\mathcal{O}_p, \mathcal{O}_p) \otimes \mathfrak{gl}_N[1],$$

and Proposition 3.2 identifies the Ext algebra with $\Lambda(\varepsilon_1, \varepsilon_2)$. This is the derived $\mathrm{RHom}_Y(\mathcal{O}_p, \mathcal{O}_p)$ allowed in Definition 2.17, not an ordinary Dolbeault algebra on the point. The trace on $\mathfrak{gl}_N$ and the Berezin trace on this exterior algebra give the cyclic open string field theory. Construction 3.195 and Theorem 3.242 construct the degree-zero Hamiltonian boundary component and prove its bracket compatibility. The full Poisson factorization-center morphism is Problem 3.60, outside the proved local theorem. □

3.2 Open mixed brane states

The open string states are the tensor product of the topological brane direction and the formal holomorphic normal direction.

- *Topological factor.* In the exact self-Floer model for the graded, spin zero-section $\mathbb{R} \subset T^*\mathbb{R} \cong \mathbb{R}^2$, we use the local exact Fukaya model near a compact interval in the brane. A small compactly supported Hamiltonian perturbation identifies $CF^*(\mathbb{R}, \mathbb{R})$ with the Morse complex on that interval, and passage to local operators in the interior is represented by the de Rham complex $\Omega^*(\mathbb{R})$ with compactly supported variations. No wrapped generators at infinity are included in this local model.
- *Holomorphic factor.* Put affine coordinates at $p = 0$ and write $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $\mathcal{O}_p = R/\mathfrak{m}$. The skyscraper sheaf has open string states

  $$\mathbf{Ext}^*_R(R/\mathfrak{m}, R/\mathfrak{m}),$$

  computed in Proposition 3.2.

Proposition 3.2 *(Affine skyscraper self-Ext).* Let $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $k = R/\mathfrak{m}$. Then the Yoneda algebra of the skyscraper sheaf at the origin is

$$\mathbf{Ext}^*_R(k, k) \cong \Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2),$$

where $|\varepsilon_i| = 1$. In particular this is the free graded-commutative algebra on two odd generators.

Proof. The pair (z_1, z_2) is a regular sequence in the polynomial ring R, so the affine Koszul complex

$$K = \left(0 \to R\, e_{12} \xrightarrow{d_2} R\, e_1 \oplus R\, e_2 \xrightarrow{d_1} R \to 0\right)$$

is a free resolution of k, with

$$d_1(ae_1 + be_2) = az_1 + bz_2, \qquad d_2(e_{12}) = z_2 e_1 - z_1 e_2.$$

Applying $\mathrm{Hom}_R(-, k)$ gives

$$0 \to k \xrightarrow{0} k\,\varepsilon_1 \oplus k\,\varepsilon_2 \xrightarrow{0} k\,\varepsilon_{12} \to 0,$$

because z_1 and z_2 act by zero on k. Hence the graded vector space of Ext groups is k in degree 0, $k\varepsilon_1 \oplus k\varepsilon_2$ in degree 1, and $k\varepsilon_{12}$ in degree 2.

The product is the Yoneda product. The Koszul resolution is the exterior dg-algebra $R \otimes \Lambda(e_1, e_2)$ with $d(e_i) = z_i$; its diagonal sends e_i to $e_i \otimes 1 + 1 \otimes e_i$. Dualising this diagonal gives $\varepsilon_i^2 = 0$ and $\varepsilon_1\varepsilon_2 = -\varepsilon_2\varepsilon_1$, while $\varepsilon_1\varepsilon_2$ spans the top Ext group. Thus the Yoneda algebra is $\Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2)$. □

3.3 *Open mixed brane field theory*

The open theory is the minimal local field theory which remembers the brane's two normal holomorphic directions. The two Ext generators are odd because they are derived normal directions of a point brane; after the BV shift their ghost-zero coefficients become the two ordinary matrix coordinates ϕ_1, ϕ_2. The one-form coefficient is the gauge field A along the brane. Thus the formal Chern-Simons action is the mechanism which turns the Ext algebra into first-order matrix mechanics.

The total space of fields (in the Batalin-Vilkovisky formalism) is

$$\alpha \in \Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1]$$

with action

$$S(\alpha) = \int_{\mathbb{R}\times\mathbb{C}^{0|2}} \frac{1}{2}\,\mathrm{Tr}(\alpha\, d\alpha) + \frac{1}{3}\,\mathrm{Tr}\,\alpha^3.$$

Here $d = d_{\mathrm{dR}}$; the Ext algebra $\Lambda(\varepsilon_1, \varepsilon_2)$ has zero differential. The integral over $\mathbb{C}^{0|2}$ is the Berezin integral $\int d\varepsilon_1\, d\varepsilon_2$, which picks out the top component in the odd directions.

The cyclic trace is

$$\mathrm{Tr}_{\mathrm{open}}(\eta \otimes u \otimes M) = \int_{\mathbb{R}} \eta\, [\varepsilon_1\varepsilon_2]u\ \mathrm{Tr}_N(M),$$

where $[\varepsilon_1\varepsilon_2]u$ denotes the coefficient of the top Ext class. The top Ext class is the residue orientation dual to $dz_1 \wedge dz_2$. The de Rham integral has degree -1 and the Ext residue has degree -2; after the field shift by $[1]$, the induced BV pairing has degree -1.

Construction 3.3 *(The open field from the Ext basis).* Write $\varepsilon_{12} = \varepsilon_1\varepsilon_2$. Before imposing ghost number, an open field is a finite sum

$$\alpha = a_0 + a_1\varepsilon_1 + a_2\varepsilon_2 + a_{12}\varepsilon_{12},$$

where each a_I is a de Rham form on $\mathbb{R}$ with values in $\mathfrak{gl}_N$, shifted by [1]. The ghost-number-zero components are exactly

$$A_{\mathrm{CS}} \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N, \qquad \chi_1, \chi_2 \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

and in coefficient coordinates they appear as

$$\alpha_0^{\mathrm{coef}} = A_{\mathrm{CS}} + \chi_1\varepsilon_1 + \chi_2\varepsilon_2.$$

The Dirac variables used in the rest of the manuscript are the linearly normalized variables

$$\phi_1 = -\chi_1, \qquad \phi_2 = \chi_2, \qquad A = -A_{\mathrm{CS}}.$$

This normalization is chosen once and for all so that the symplectic potential is $\mathrm{Tr}(\phi_1 d\phi_2)$ and the multiplier term is $A[\phi_1, \phi_2]$. The Berezin integral kills every monomial except the coefficient of ε_{12}, so every term in the ghost-zero action is obtained by multiplying the displayed three summands and keeping exactly one real one-form and exactly the product $\varepsilon_1\varepsilon_2$.

Remark 3.4. The shift [1] acts on the total cohomological degree: a form of de Rham degree p and Grassmann degree q has total degree $p + q$, which after the shift sits in ghost number $p + q - 1$. In particular, the 1-form component of α (de Rham degree 1, Grassmann degree 0) sits in ghost number 0 and is the bosonic gauge field. The scalar components proportional to ε_i (de Rham degree 0, Grassmann degree 1) also sit in ghost number 0 and represent the motion of the brane in the two normal directions.

3.3.1 Ghost-zero open field content

The ghost-zero coefficient truncation α_0^{coef} of Construction 3.3, written after the Dirac-normalization $(A_{\mathrm{CS}}, \chi_1, \chi_2) \mapsto (A, \phi_1, \phi_2)$, is the physical sector used below. Reducing the Chern-Simons action to it produces the first-order matrix mechanics governing the brane:

LEMMA 3.5. With the Berezin convention $\int_{\mathbb{C}^{0|2}} \varepsilon_1\varepsilon_2 = 1$, the restriction of the BV action to the ghost-zero coefficient fields, written in the Dirac variables of Construction 3.3, is, up to a boundary term on $\mathbb{R}$,

$$S(\alpha_0) = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Proof. Only monomials containing one real one-form and the factor $\varepsilon_1\varepsilon_2$ survive integration over $\mathbb{R} \times \mathbb{C}^{0|2}$. In coefficient coordinates $\alpha_0^{\mathrm{coef}} = A_{\mathrm{CS}} + \chi_1\varepsilon_1 + \chi_2\varepsilon_2$, the tensor-product Koszul sign gives

$$\frac{1}{2}\int_{\mathbb{R}} \mathrm{Tr}\big(-\chi_1\, d\chi_2 + \chi_2\, d\chi_1\big) \equiv -\int_{\mathbb{R}} \mathrm{Tr}(\chi_1\, d\chi_2),$$

modulo an exact one-form. In the cubic term the six orderings of $A_{\mathrm{CS}}, \chi_1\varepsilon_1, \chi_2\varepsilon_2$ reduce, by cyclicity of the trace and $\varepsilon_2\varepsilon_1 = -\varepsilon_1\varepsilon_2$, to

$$3\,\mathrm{Tr}(A_{\mathrm{CS}}\chi_1\chi_2 - A_{\mathrm{CS}}\chi_2\chi_1).$$

The coefficient $\frac{1}{3}$ in the Chern-Simons action therefore produces $\mathrm{Tr}(A_{\mathrm{CS}}[\chi_1, \chi_2])$. Substituting $\phi_1 = -\chi_1$, $\phi_2 = \chi_2$, and $A = -A_{\mathrm{CS}}$ gives the displayed Dirac-normalized action. □

Lemma 3.6 *(Dirac reduction of the probe phase space).* Let

$$\mathcal{P}_N = \mathfrak{gl}_N \oplus \mathfrak{gl}_N$$

with coordinates (ϕ_1, ϕ_2) and symplectic form

$$\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2).$$

With the convention $\iota_{X_f}\Omega_N = -df$, simultaneous conjugation by $\mathfrak{gl}_N$ is Hamiltonian with moment map components

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big), \qquad \epsilon \in \mathfrak{gl}_N.$$

They satisfy the first-class identity

$$\{\mu_\epsilon, \mu_\eta\} = -\mu_{[\epsilon,\eta]}.$$

Thus this comoment is anti-equivariant for the displayed Poisson convention, equivalently equivariant for the opposite Lie algebra; the zero locus and the generated constraint ideal are unchanged. The action of Lemma 3.5 is the corresponding Dirac first-order action

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2) - \mu_A,$$

and varying A imposes the constrained phase space

$$\mu^{-1}(0)/GL_N = \{[\phi_1, \phi_2] = 0\}/GL_N.$$

Proof. The symplectic form gives

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j.$$

For $v_\epsilon(\phi_i) = [\epsilon, \phi_i]$, cyclicity of the trace gives

$$d\mu_\epsilon(\delta\phi_1, \delta\phi_2) = -\mathrm{Tr}\big(\epsilon([\delta\phi_1, \phi_2] + [\phi_1, \delta\phi_2])\big) = -\Omega_N(v_\epsilon, (\delta\phi_1, \delta\phi_2)).$$

Thus $X_{\mu_\epsilon} = v_\epsilon$. The moment-map bracket is therefore the anti-equivariance identity $\{\mu_\epsilon, \mu_\eta\} = -\mu_{[\epsilon,\eta]}$. Substituting $\epsilon = A(t)$ gives

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2) - \mu_A = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big),$$

and the A-equation is $[\phi_1, \phi_2] = 0$. □

Proposition 3.7 *(Classical BV truncation).* The subspace of ghost-number-zero coefficient fields

$$\alpha_0^{\mathrm{coef}} = A_{\mathrm{CS}} + \chi_1\varepsilon_1 + \chi_2\varepsilon_2$$

with all nonzero ghost-number components set to zero is a consistent classical truncation of the open BV theory. After the linear Dirac-normalization of Construction 3.3, the induced classical action is the first-order gauged matrix model of Lemma 3.5.

Proof. Decompose the BV field as $\alpha = \sum_r \alpha^{(r)}$ by ghost number. The differential $d_{\mathbb{R}}$ and the product on $\Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N$ preserve total degree, and the BV action has ghost number zero. Hence every monomial in the Euler-Lagrange expression for a component of ghost number $r \neq 0$ contains at least one field component of nonzero ghost number when evaluated at $\alpha = \alpha_0^{\mathrm{coef}}$. Setting all such components to zero therefore makes the discarded equations vanish. The remaining equation is exactly the variation of the ghost-zero action computed in Lemma 3.5.

Concretely, the full Euler-Lagrange expression is the curvature

$$F(\alpha) = d\alpha + \alpha^2.$$

For $\alpha_0^{\mathrm{coef}} = A_{\mathrm{CS}} + \chi_1\varepsilon_1 + \chi_2\varepsilon_2$ on a line,

$$F(\alpha_0^{\mathrm{coef}}) = (d\chi_1 + [A_{\mathrm{CS}}, \chi_1])\varepsilon_1 + (d\chi_2 + [A_{\mathrm{CS}}, \chi_2])\varepsilon_2 + [\chi_1, \chi_2]\varepsilon_1\varepsilon_2.$$

The BV pairing keeps exactly real-form degree one and $\varepsilon_1\varepsilon_2$. Hence the displayed three components pair exactly with the coefficient variations of $A_{\mathrm{CS}}, \chi_2, \chi_1$ respectively, while variations of nonzero ghost number see complementary curvature components that are zero for α_0^{coef} by real-form degree or ε-degree. This is a physical ghost-zero truncation. The Dirac-normalized variables give the ordinary GL_N-conjugation action used in the reduced local-operator calculation. □

The truncated action is invariant under

$$\delta_\varepsilon \phi_i = [\varepsilon, \phi_i], \qquad \delta_\varepsilon A = d\varepsilon + [A, \varepsilon], \qquad \varepsilon \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N :$$

the $d\varepsilon$ variation of $\mathrm{Tr}(\phi_1\, d\phi_2)$ produces $-\mathrm{Tr}(d\varepsilon[\phi_1, \phi_2])$, which cancels the $d\varepsilon$ variation of $\mathrm{Tr}(A[\phi_1, \phi_2])$; the remaining terms cancel by invariance of the trace pairing.

Remark 3.8. The same local open model is obtained from the affine $\mathbb{C}^2$ point-brane residue calculation by tensoring the skyscraper object with $\mathbb{C}^N$ and passing to endomorphisms. This is the local two-dimensional residue pairing fixed by $dz_1 dz_2$, not a global CY-to-chiral input. Global transfer would require the corresponding negative cyclic Calabi-Yau class, framing data, and descent; none of that is used in the proof below.

Remark 3.9. In AKSZ language this is the one-dimensional gauged sigma model with Hamiltonian target $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$ and symplectic potential $\mathrm{Tr}(\phi_1\, d\phi_2)$. We use the resulting Poisson bivector

$$\Pi = \sum_{i,j} \frac{\partial}{\partial(\phi_1)^i{}_j} \wedge \frac{\partial}{\partial(\phi_2)^j{}_i},$$

so $\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j$. Equivalently, with the convention $\iota_{X_f}\omega = -df$ used throughout, the symplectic form is $\mathrm{Tr}(d\phi_1 \wedge d\phi_2)$. For $\epsilon \in \mathfrak{gl}_N$, the Hamiltonian

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big)$$

generates $\delta_\epsilon \phi_i = [\epsilon, \phi_i]$. Its zero locus is the commuting-pair equation $[\phi_1, \phi_2] = 0$, and the classical reduced phase space is the stack of commuting pairs of matrices modulo conjugation.

Remark 3.10. Equivalently, take Chern-Simons theory on $\mathbb{R} \times T^2$ and restrict to the harmonic forms $1, \eta_1, \eta_2, \eta_1\eta_2$ on T^2. Writing $A_{3d} = A_t\, dt + \phi_1\eta_1 + \phi_2\eta_2$ and integrating over T^2 gives the same action; the odd variables ε_i represent the degree-one cohomology classes η_i.

3.4 *Closed mixed Hamiltonian sector*

The closed field is the infinitesimal geometry of the formal normal surface. Because the surface is holomorphic symplectic, an infinitesimal canonical transformation is a Hamiltonian function modulo constants. The BF partner is the cotangent variable which makes the theory local and BV. This is the closed counterpart of the open first-order form: both sides are governed by the same symplectic normal plane.

The closed sector used here is a mixed field theory, not a tensor product of two independent closed-string state spaces. Its fields have a topological de Rham factor along $\mathbb{R}^2$ and a holomorphic Dolbeault factor along $\mathbb{C}^2$:

$$\Omega^*(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,*}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big).$$

The coefficient $\widehat{\mathfrak{h}}$ is the completed Lie algebra of nonconstant Hamiltonian functions on the formal holomorphic symplectic disk. The continuous dual copy $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]$ is the cotangent/BF field. Thus the closed theory used here is the local Hamiltonian Kodaira-Spencer/BF analogue in the mixed holomorphic-topological model.

Remark 3.11. This is a source-comparison remark. The original BCOV analysis [1] is written for Calabi-Yau threefolds; the higher-genus B-model on Calabi–Yau manifolds is taken up at the quantum level in [13]. On the formal holomorphic symplectic surface used here, the local cotangent theory is instead the BF theory for the dg Lie algebra of Hamiltonian vector fields. The holomorphic symplectic form supplies the Poisson bracket, while the local volume form $dz_1 dz_2$ supplies the divergence operator on polyvectors.

We consider the space of polyvector fields

$$\mathbf{PV}^{*,*}(\mathbb{C}^2) \cong \Omega^{0,*}(\mathbb{C}^2)[\partial_{z_1}, \partial_{z_2}]$$

where ∂_{z_i} are odd generators of poly-vector degree 1 (corresponding to the two holomorphic vector fields on $\mathbb{C}^2$). The operator

$$\partial : \mathbf{PV}^{i,*} \longrightarrow \mathbf{PV}^{i-1,*}$$

is the divergence operator determined by the holomorphic volume form $\Omega = dz_1 \wedge dz_2$:

$$\partial_\Omega = \iota_\Omega^{-1} \circ \partial_{\mathrm{hol}} \circ \iota_\Omega, \qquad \iota_\Omega : \mathbf{PV}^{i,*} \xrightarrow{\sim} \Omega^{2-i,*}.$$

Thus for $X = a_1\partial_{z_1} + a_2\partial_{z_2}$,

$$\partial_\Omega X = (\partial_{z_1} a_1 + \partial_{z_2} a_2),$$

while ∂_Ω vanishes on functions.

We work in the formal polynomial local model at $0 \in \mathbb{C}^2$, so Dolbeault cohomology is represented by holomorphic Taylor polynomials. Here ∂_Ω is the divergence constraint selecting the Hamiltonian BF sector. The kinematic differential of the closed theory below is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$. Thus the Poincare lemma identifies divergence-free vector fields with Hamiltonian potentials. The quotient by ∂_Ω-exact Hamiltonians is absent. Analysing degree by degree:

- $\mathbf{PV}^{0,*}(\mathbb{C}^2) = \Omega^{0,*}(\mathbb{C}^2) \subset \mathbf{Ker}\,\partial$ trivially.
- $\mathbf{PV}^{1,*}(\mathbb{C}^2) \cong \Omega^{1,*}(\mathbb{C}^2)$ via $\iota_{(-)}\omega$. Under this identification, with the Hamiltonian convention $\iota_{X_f}\omega = -df$ used below, $\mathbf{Ker}\,\partial_\Omega$ is the space of closed holomorphic one-forms. Lemma 3.12 identifies these with df, uniquely modulo constants.

- The constant Hamiltonians act trivially, since their Hamiltonian vector fields vanish. We therefore use the reduced polynomial algebra
$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$$
for nontrivial Hamiltonian fields.

LEMMA 3.12 *(Polynomial Poincare primitive).* Let
$$\theta = a(z_1, z_2)\, dz_1 + b(z_1, z_2)\, dz_2$$
be a polynomial one-form on $\mathbb{C}^2$. If $\partial_{z_2} a = \partial_{z_1} b$, then there is a polynomial $f \in \mathbb{C}[z_1, z_2]$, unique modulo constants, such that $\theta = df$.

Proof. Define
$$f(z_1, z_2) = \int_0^1 \big(z_1 a(tz_1, tz_2) + z_2 b(tz_1, tz_2)\big)\, dt.$$
Since a and b are polynomials, the integral is termwise division of each homogeneous term by its degree plus one, hence f is polynomial. Differentiate under the integral sign. Closedness says $\partial_{z_2} a = \partial_{z_1} b$, so
$$\begin{aligned}
\partial_{z_1} f &= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_1} b(tz)\big)\, dt \\
&= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_2} a(tz)\big)\, dt \\
&= \int_0^1 \frac{d}{dt}\big(t\, a(tz)\big)\, dt = a(z).
\end{aligned}$$
The same calculation, with a and b interchanged, gives $\partial_{z_2} f = b(z)$. Therefore $df = \theta$. If $df = dg$, then $d(f - g) = 0$, hence $f - g$ is constant. □

PROPOSITION 3.13 *(Hamiltonian polyvector reduction).* In the formal polynomial model on $(\mathbb{C}^2, \omega = dz_1 \wedge dz_2)$, the divergence-closed polyvector-one fields reduce to Hamiltonian vector fields:
$$\mathbf{Ker}(\partial_\Omega : \mathbf{PV}^1 \to \mathbf{PV}^0) \cong \{df : f \in \mathbb{C}[z_1, z_2]\} \cong \bar{A}.$$
Under the convention
$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \iota_{X_f}\omega = -df,$$
the commutator of vector fields is
$$[X_f, X_g] = X_{\{f,g\}_{\bar{A}}}.$$

Proof. Let $X = a_1 \partial_{z_1} + a_2 \partial_{z_2}$. The divergence operator is $\partial_\Omega X = \partial_{z_1} a_1 + \partial_{z_2} a_2$. Contracting with ω gives
$$\iota_X \omega = a_1\, dz_2 - a_2\, dz_1.$$
The equation $\partial_\Omega X = 0$ is exactly the closedness of this one-form:
$$d(\iota_X \omega) = (\partial_{z_1} a_1 + \partial_{z_2} a_2)\, dz_1 \wedge dz_2 = 0.$$

By Lemma 3.12, there is a polynomial f, unique modulo constants, such that $\iota_X \omega = -df$. Therefore

$$a_1 = -\partial_{z_2} f, \qquad a_2 = \partial_{z_1} f,$$

so $X = X_f$. Constants are exactly the kernel of $f \mapsto X_f$, giving $\bar{A}$.

Finally, for any polynomial h,

$$\begin{aligned}[X_f, X_g](h) &= \{f, \{g, h\}\} - \{g, \{f, h\}\} \\ &= \{\{f, g\}, h\}\end{aligned}$$

by the Jacobi identity for the constant Poisson bracket. Thus $[X_f, X_g] = X_{\{f,g\}}$, and projecting constants gives the bracket on $\bar{A}$. □

The Dolbeault-valued upgrade is immediate. The Hamiltonian sector of the closed mixed field theory replaces the divergence kernel $\mathbf{Ker}\,\partial_\Omega$ on poly-1-vectors with its Dolbeault enhancement: a Hamiltonian potential $f \in \Omega^{0,*}(\mathbb{C}^2) \widehat{\otimes} \bar{A}$ generates the Hamiltonian vector field

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \iota_{X_f}\omega = -df, \qquad \{f, g\}_\omega = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

Constants drop because $X_f = 0$ if and only if f is constant; this is 3.13 read on Dolbeault-valued coefficients.

3.5 *Hamiltonian BF action*

3.5.1 The action functional

The closed BF action for the dg Lie algebra $\widehat{\mathfrak{h}}$ of formal Hamiltonians, with kinematic differential $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, has fields

$$\alpha \in \Omega^*(\mathbb{R}^2) \widehat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \widehat{\otimes} \widehat{\mathfrak{h}}[1], \qquad \beta \in \Omega^*(\mathbb{R}^2) \widehat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \widehat{\otimes} \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2],$$

the Hamiltonian connection α and the cotangent partner β. The form-valued Hamiltonian bracket wedges the form factors and takes the Poisson bracket on the Hamiltonian entries with the tensor-product Koszul sign,

$$[\eta \otimes f, \theta \otimes g] = (-1)^{|f||\theta|}(\eta \wedge \theta) \otimes \{f, g\},$$

which equals +1 on the ghost-zero (even-coefficient) sector used below. The action is the BF Hamiltonian

$$S = \int_{\mathbb{R}^2 \times \mathbb{C}^2} \left\langle \beta,\ D\alpha + \tfrac{1}{2}\{\alpha, \alpha\} \right\rangle dz_1\, dz_2,$$

with the pairing the continuous evaluation $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}} \otimes \widehat{\mathfrak{h}} \to \mathbb{C}$; the dense polynomial submodel pairs $\bar{A}^\vee$ with $\bar{A}$ in each finite degree. On ghost-zero fields, α has total real/Dolbeault form degree 1 and β has total degree 2, so $\beta \wedge D\alpha$ and $\beta \wedge \{\alpha, \alpha\}$ have total degree 4; wedge with $dz_1 \wedge dz_2$ produces a top form on $\mathbb{R}^2 \times \mathbb{C}^2$.

Remark 3.14. The open mixed brane theory lives on the brane line; the closed Hamiltonian BF theory lives on the full mixed space-time $\mathbb{R}^2 \times \mathbb{C}^2$.

3.5.2 Classical field content

The BF action exhibits β as a Lagrange multiplier whose equation of motion enforces flatness of the Hamiltonian connection,

$$F_\alpha = D\alpha + \tfrac{1}{2}\{\alpha, \alpha\} = 0,$$

where α is read as a connection on a bundle of holomorphic-symplectic surfaces over $\mathbb{R}^2$.

Consider the space-time $\mathbb{R}^2_{x_i} \times \mathbb{C}^2_{z_i}$. The ghost-number-zero components of α and β are:

$$\alpha = \alpha_{x_i}\, dx_i + \alpha_{\bar z_j}\, d\bar z_j, \qquad \beta = \beta_{x_1 x_2}\, dx_1\, dx_2 + \beta_{x_i \bar z_j}\, dx_i\, d\bar z_j + \beta_{\bar z_1 \bar z_2}\, d\bar z_1\, d\bar z_2.$$

Here the coefficients of α are Hamiltonian functions modulo constants. The gauge symmetries are the BF-type transformations

$$\delta_\varepsilon \alpha = D\varepsilon + \{\alpha, \varepsilon\}, \qquad \delta_\varepsilon \beta = \mathrm{ad}^\vee_\varepsilon\, \beta,$$

and

$$\delta_\lambda \alpha = 0, \qquad \delta_\lambda \beta = D_\alpha \lambda,$$

where $D_\alpha = D + \{\alpha, -\}$ and λ is a one-form valued in $\bar{A}^\vee$. The second symmetry is reducible on shell: if $\lambda = D_\alpha \mu$, then $D_\alpha \lambda = D_\alpha^2 \mu = \{F_\alpha, \mu\}$, which vanishes when $F_\alpha = 0$.

Analysing α as a Hamiltonian function on $\mathbb{C}^2$: the component $\alpha_{\bar z_j} d\bar z_j$ is the Hamiltonian potential whose vector field $X_{\alpha_{\bar z_j}} d\bar z_j$ is the Beltrami differential encoding a deformation of the complex structure of $\mathbb{C}^2$; the component $\alpha_{x_i} dx_i$ similarly defines a connection on $\mathbb{R}^2$ valued in Hamiltonian vector fields on $\mathbb{C}^2$, encoding a bundle of $\mathbb{C}^2$s over $\mathbb{R}^2$. Together, the equation of motion $F_\alpha = 0$ says this data constitutes an integrable deformation of complex structure with flat holonomy. The field β appears linearly in the action and enforces exactly this equation.

Remark 3.15. This is the BF specialization to the dg Lie algebra of formal Hamiltonians: the multiplier β enforces $F_\alpha = 0$, and the classical phase space is $T^*\mathcal{M}_{\mathrm{flat}}(\widehat{\mathfrak{h}})$.

3.5.3 Phase space

For an analytic model adequate to propagators, circle-regularize one real topological direction to obtain $\mathbb{R} \times S^1 \times Y_{\mathrm{hol}}$ with Y_{hol} holomorphic symplectic, completed at the trivial solution: an analytic avatar of the formal local model, not a compact Calabi-Yau target and not a renaming of $X = M \times Y$. The equations of motion $F_\alpha = 0$ cut out the Maurer-Cartan problem for Hamiltonian vector fields on $S^1 \times Y_{\mathrm{hol}}$, with the analytic data of 3.264 fixed in this compact or formal setting (elliptic complex, brane-defect or boundary condition, gauge fixing, propagator, renormalization scheme, anomaly cancellation). The perturbative phase space is the derived cotangent stack

$$T^*_{\mathrm{der}}\mathrm{MC}\big(\Omega^*(S^1)\widehat{\otimes}\Omega^{0,*}(Y_{\mathrm{hol}}, \mathrm{Ham}(Y_{\mathrm{hol}})), D, \{-,-\}\big),$$

collapsing on the underived tangent at the trivial solution to the ordinary cotangent of the linearized deformation complex.

3.6 Reduced Hamiltonian brane coupling

The brane sees a closed Hamiltonian by evaluating it on its matrix-valued normal coordinates and tracing over Chan-Paton indices. For a brane along x_1 and a closed field $dx_1 \otimes z_1^k z_2^l$, this gives the local boundary functional

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big) + \int_{\mathbb{R}} dx_1\, \mathrm{Tr}\, \phi_1^k \phi_2^l$$

inside the formal local algebraic model: the closed Hamiltonian is pulled back along the matrix-valued normal-coordinate map $z_i \mapsto \phi_i$ and traced. 3.195 is the algebraic form of this bulk-to-boundary prescription; 3.242 below proves that the resulting Hamiltonian evaluation preserves the classical bracket. This is the degree-zero Hamiltonian component of the reduced CE/PV central-operation map; the full Poisson factorization-center morphism is left to 3.60.

Construction 3.16 *(Reduced line-defect current kernel).* Let

$$X = \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}, \qquad L = \mathbb{R}_{x_1} \times \{x_2 = 0\} \times \{p\} \subset X,$$

and let $i : L \hookrightarrow X$ be the inclusion. This is a line defect, not a spacetime boundary. In the reduced formal local model, a compactly supported test density $a(t)\,dt$ on L defines the defect-supported current $i_*(a(t)\,dt)$, locally written as $a(x_1)\delta(x_2)\delta_p\,dx_1dx_2$ together with the formal holomorphic delta-current at p.

The degree-zero Hamiltonian part of the reduced PBW/Rees defect-current vertex is

$$K_L^{\mathrm{cl}}(a \otimes f) = \int_L a(t)\,\overline{\operatorname{Tr} f(\phi_1(t), \phi_2(t))}\,dt, \qquad f \in \bar{A}.$$

For the cotangent principal-part source, the reduced vertex sends

$$B \otimes \rho[1] \longmapsto \Theta_\rho(B), \qquad B \in \mathcal{D}'_c(L), \quad \rho \in \mathcal{P},$$

using Theorem 3.237. These formulas define a reduced PBW/Rees defect-current kernel. They are not a cochain-level map from closed CE observables, and the quantum Costello brane-defect kernel is a separate construction target: it must supply the Ward identity, anomaly cancellation, finite-scale propagator, and quantum moment-map compatibility.

Remark 3.17. Geometrically, specifying a closed Hamiltonian field on $\mathbb{C}^2$ amounts to equipping $\mathbb{C}^2$ with a connection, so that the brane can parallel-transport its internal degrees of freedom as it moves along $\mathbb{R}$.

Construction 3.18 *(Interval Hamiltonian and factorization algebras).* For an interval $I \subset \mathbb{R}$, define

$$\begin{aligned} \mathfrak{h}_I &= \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}, \\ \mathcal{P}_I &= \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}, \\ \mathfrak{g}_I &= \mathfrak{h}_I \ltimes \mathcal{P}_I[1], \end{aligned}$$

with brackets

$$\begin{aligned} [a \otimes f, b \otimes g]_{\mathfrak{h}_I} &= ab \otimes \{f, g\}_{\bar{A}}, \\ [a \otimes f, B \otimes \rho[1]]_{\mathfrak{g}_I} &= aB \otimes (f \cdot \rho)[1], \\ [B \otimes \rho[1], C \otimes \sigma[1]]_{\mathfrak{g}_I} &= 0, \end{aligned}$$

where products in $\Omega^\bullet_c(I)$ are de Rham products and, in the displayed degree-zero formulas, aB is multiplication of a compactly supported current by a smooth compactly supported function and $f \cdot \rho = \pi_{\mathrm{pp}}\{f, \rho\}$ is the principal-part coadjoint action of 3.227. Inclusions $I' \subset I$ act by extension by zero on compactly supported de Rham forms and currents; disjoint unions act by sum. This makes $I \mapsto \mathfrak{h}_I, \mathfrak{g}_I$ cosheaves of (graded) Lie algebras on $\mathbb{R}$. For $j : I \hookrightarrow J$, the map $j_!$ is defined on $\Omega^\bullet_c(I)$ because

supports are compactly contained in I; equivalently, smooth representatives vanish near the boundary of I inside J. It satisfies

$$j_!(\alpha \wedge \beta) = j_!\alpha \wedge j_!\beta, \qquad j_!(aB) = (j_!a)(j_!B),$$

in the degree-zero current notation. On CE cochains the same extension-by-zero maps induce pullback/coproduct maps; they do not make the CE cochain source a covariant restriction functor.

Define the boundary P_0 factorization algebras on $\mathbb{R}$ in the formal stable Hamiltonian sector by

$$\begin{aligned} A_\partial^{\mathrm{Ham}}(I) &= \widehat{\mathbf{S}}(\mathfrak{h}_I), \\ A_\partial^{\mathrm{pp}}(I) &= \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{P}_I[1]), \end{aligned}$$

with degree-zero reduced Hamiltonian P_0 bracket extended as a biderivation from

$$\begin{aligned} \{a \otimes f, b \otimes g\}_{A_\partial} &= ab \otimes \{f, g\}_{\bar{A}}, \\ \{a \otimes f, B \otimes \rho[1]\}_{A_\partial} &= aB \otimes (f \cdot \rho)[1], \\ \{B \otimes \rho[1], C \otimes \sigma[1]\}_{A_\partial} &= 0. \end{aligned}$$

The factorization product on disjoint intervals is the multiplication in $\widehat{\mathbf{S}}$ after extension by zero. These are the reduced targets after de Rham-current contraction for the reduced Hamiltonian current central-operation map. The Hamiltonian summand $A_\partial^{\mathrm{Ham}}$ is the connected component of the formal Hamiltonian sector of the open brane factorization algebra; the principal-part summand $A_\partial^{\mathrm{pp}}/A_\partial^{\mathrm{Ham}}$ is the reduced current object of 3.237.

LEMMA 3.19 *(Compact-support interval contraction).* Let $I \subset \mathbb{R}$ be a nonempty open interval, oriented by the coordinate t. Choose a compactly supported one-form $\lambda_I(t)\,dt \in \Omega_c^1(I)$ with $\int_I \lambda_I(t)\,dt = 1$. Define

$$p_I : \Omega_c^\bullet(I) \longrightarrow \mathbb{C}[-1], \qquad p_I(\alpha) = \int_I \alpha,$$

with p_I zero on degree 0, and

$$i_I : \mathbb{C}[-1] \longrightarrow \Omega_c^\bullet(I), \qquad i_I(1) = \lambda_I(t)\,dt.$$

Extend compactly supported forms by zero to $\mathbb{R}$. For a one-form $\alpha = f(t)\,dt$, set

$$K_I(\alpha)(t) = \int_{-\infty}^{t} \left(f(s) - \lambda_I(s) \int_I f(r)\,dr \right) ds, \qquad K_I(\varphi) = 0 \quad (\varphi \in \Omega_c^0(I)).$$

Then $K_I : \Omega_c^1(I) \to \Omega_c^0(I)$ is well-defined and compactly supported, and

$$dK_I + K_I d = \mathrm{id}_{\Omega_c^\bullet(I)} - i_I p_I.$$

Consequently $\Omega_c^\bullet(I)$ contracts onto its compact-support cohomology line $\mathbb{C}[-1]$. No acyclicity of $\Omega_c^\bullet(I)$ is asserted: a compactly supported primitive of a compactly supported one-form exists precisely on the kernel of the integral map, and the integral class is the surviving compact-support cohomology generator.

Proof. The one-form inside the integral defining $K_I(\alpha)$ has compact support and total integral zero. Its primitive therefore vanishes near both ends of I, after extension by zero to $\mathbb{R}$, so $K_I(\alpha) \in \Omega_c^0(I)$.

For $\alpha = f(t)\,dt$,

$$dK_I(\alpha) = \left(f(t) - \lambda_I(t)\int_I f(r)\,dr\right)dt = \alpha - i_I p_I(\alpha).$$

For a compactly supported function φ,

$$K_I(d\varphi)(t) = \int_{-\infty}^{t} \varphi'(s)\,ds = \varphi(t),$$

because φ vanishes near the left end, and $p_I(\varphi) = 0$. These two identities are exactly $dK_I + K_I d = \mathrm{id} - i_I p_I$ in degrees 1 and 0. □

Proposition 3.20 *(Locality of the brane P_0 bracket).* Let $a \in \Omega_c^0(I_1)$, $b \in \Omega_c^0(I_2)$ with $I_1 \cap I_2 = \emptyset$. For all $f, g \in \bar{A}$,

$$\{B_f(a), B_g(b)\}_{A_\partial^{\mathrm{Ham}}} = 0.$$

In particular, $A_\partial^{\mathrm{Ham}}$ has the strict factorization product property: the bracket between sections supported on disjoint intervals vanishes.

Proof. By 3.18, $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$. The pointwise product $ab \equiv 0$ since the supports are disjoint, so $B_{\{f,g\}}(ab) = 0$. □

Remark 3.21 *(Locality from the canonical equal-time bracket).* The locality input is the canonical equal-time bracket of the first-order open BV theory:

$$\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta^i_l \delta^k_j\, \delta(t-s), \qquad \{(\phi_a)^i{}_j(t), (\phi_a)^k{}_l(s)\} = 0 \ \ (a = 1, 2).$$

Tracing $f(\phi_1, \phi_2)(t)$ and $g(\phi_1, \phi_2)(s)$, applying the Leibniz rule, and integrating against $a(t)b(s)\delta(t-s)$ produces exactly the smeared identity $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$. This is the unsmeared content recorded in 3.243. In the classical reduced, de Rham-contracted sector this delta-function locality produces a strict support-local P_0 factorization product. The unreduced BV current complex still requires its own homotopies, contact-term choices, and QME data.

Theorem 3.22 *(Interval-wise reduced Hamiltonian CE/PV central-operation map, Hamiltonian sector).* Let $\mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N}$ be the reduced open-brane factorization algebra of 3.25. Let $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H,\mathrm{Ham}}|_L$ denote the formal classical Hamiltonian-mode subfactorization algebra of the cotangent BF theory on the formal neighbourhood of L obtained by restricting to the connected Hamiltonian central-operation sector. Define the interval-wise CE cochain cofactorization object, equivalently a factorization object on the opposite interval category, by

$$\mathcal{F}_{\mathrm{cl}}(I) = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_I), \qquad \mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1],$$

where extension by zero on $\Omega_c^\bullet$ and on compactly supported currents induces the CE coproduct. Then:

(i) there is a continuous interval-wise morphism of P_0 cofactorization source data to the reduced Hamiltonian central-operation target

$$\Phi^{\mathrm{Ham}}_{\mathrm{fact}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H,\mathrm{Ham}}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{red},\mathrm{Ham}}\big(A^{\mathrm{Ham}}_{\partial}\big),$$

whose underlying CE/PV map on each interval I is given on length-one coordinates by

$$B_f(a) := \Phi_I(u_{a(t)dt\otimes f}) = \int_I a(t)\,\overline{\mathrm{Tr}\, f(\phi_1(t),\phi_2(t))}\,dt,$$

and

$$\Phi_I(c_{B\otimes\rho}) = \theta_{B\otimes\rho}.$$

The reduced principal-part current representative $\Theta_\rho(B)$ belongs to the boundary local-dual model of Theorem 3.237; it is not identified with the CE/PV vector-field coordinate before that reduction.

(ii) this morphism is an isomorphism of P_0-cdg algebras in each fixed interval, after the stated reduced current model is chosen, and is compatible with the CE coproduct on the source and disjoint factorization products on the target;

(iii) after applying the compact-support de Rham contraction $p_I : \Omega^\bullet_c(I) \to \mathbb{C}[-1]$, the induced Hamiltonian density sector is locally constant in the topological direction $\mathbb{R}$ in the sense of Lurie, *Higher Algebra*, §5.5; Costello–Gwilliam [15, Vol. I §6.4] supplies the published factorization-algebra terminology. This does not assert that extension by zero $\mathcal{D}'_c(I') \to \mathcal{D}'_c(I)$ is a quasi-isomorphism for arbitrary compactly supported distributional currents;

(iv) the factorization product $A^{\mathrm{Ham}}_{\partial}(I_1) \otimes A^{\mathrm{Ham}}_{\partial}(I_2) \to A^{\mathrm{Ham}}_{\partial}(V)$ for disjoint $I_1 \sqcup I_2 \subset V$ corresponds under $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ to the CE coproduct on the closed side induced by extension by zero $\Omega^\bullet_c(I_j) \hookrightarrow \Omega^\bullet_c(V)$;

(v) at the stalk $I \to \{t_0\}$, $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ restricts to the stalkwise central operation

$$f \longmapsto L_{B_f} \in C^0_{\mathrm{Hoch}}\big(A^{\mathrm{pt}}_N, A^{\mathrm{pt}}_N\big)$$

of 3.209.

Proof. Step 1. Identify the interval-wise reduced P_0 model. In the formal stable Hamiltonian sector, the boundary factorization algebra $A^{\mathrm{Ham}}_{\partial}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$ is graded-commutative, with strict P_0 bracket inherited from $\mathfrak{h}_I$ as a biderivation. In this theorem $\mathrm{Z}^{P_0}_{\mathrm{red},\mathrm{Ham}}$ means the reduced interval-wise Poisson-CE model $\mathrm{PV}^\bullet\big(\widehat{\mathbf{S}}(\mathfrak{h}_I)\big) = \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathfrak{h}^\vee_{I,\mathrm{cont}}[-1])$ with the Schouten bracket. A genuine Hochschild/factorization-centre identification for the completed Tate algebra would require a continuous HKR theorem with interval functoriality; it is not used as an input here. Restricting to the polynomial subalgebra $\mathbf{S}(\mathfrak{h}_I) \subset \widehat{\mathbf{S}}(\mathfrak{h}_I)$ and to the constant polyvectors gives the central multiplication operations, whose boundary names are the Hamiltonian currents $B_f(a)$. The CE coordinates c dual to Hamiltonian generators map instead to constant polyvectors. This is the interval version of the generator dictionary $c^I \mapsto \theta^I$, $u_I \mapsto O_I$ in 3.28.

Step 2. Construct the CE source. By 3.217, applied to $\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$ in the stated completed category,

$$C^\bullet_{\mathrm{CE},\mathrm{cont}}(\mathfrak{g}_I) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{I,\mathrm{cont}}[-1]\big),$$

and its CE differential is the dual of the semidirect Lie bracket. The Hamiltonian part is controlled by the de Rham-current pairing on the $\Omega_c^\bullet(I)$ factor and the residue-Taylor pairing on $\bar{A}$. In degree-zero notation this is the familiar pairing between compactly supported test functions and compactly supported currents; the polynomial subspace $\bar{A} \subset \widehat{\bar{A}}$ pairs with the discrete Taylor dual generated by the $\delta_{a,b}$. The shifted-cotangent CE coordinate represented by $a(t)dt \otimes f$ is dual to the cotangent generator determined by that current. Its boundary image is $B_f(a)$.

Step 3. Define $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}(I)$ on generators. By 3.215, the closed Hamiltonian cotangent BF observables on the formal neighbourhood of $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ in the Hamiltonian sector are $\widehat{\mathbf{S}}(\mathfrak{h}^\vee_{I,\mathrm{cont}}[-1])$, with CE differential dual to the Hamiltonian Lie bracket on $\mathfrak{g}_I$. The interval-wise map sends a length-one shifted-cotangent CE coordinate represented by $a(t)dt \otimes f$ to the element of the central operations factorization algebra given by the multiplication operator

$$L_{B_f(a)} \in \mathrm{Hoch}^0\big(A^{\mathrm{Ham}}_\partial(I)\big) \subset \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}\big(A^{\mathrm{Ham}}_\partial\big)(I),$$

with $B_f(a) = \Phi_I(u_{a(t)dt\otimes f})$ as displayed. By 3.209, this operator is closed (Hochschild 0-cocycle on the graded-commutative open stalk). Extend $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}(I)$ as a continuous algebra morphism on $\widehat{\mathbf{S}}$.

Step 4. Verify CE differential compatibility. On length two, the CE differential of a shifted-cotangent generator represented by $a(t)dt \otimes f$ is the coadjoint term dual to the Lie bracket on $\mathfrak{h}_I$. By 3.243 (its sharpened form below),

$$\{L_{B_f(a)}, L_{B_g(b)}\}_{A^{\mathrm{Ham}}_\partial} = L_{B_{\{f,g\}}(ab)}.$$

Pulling back through $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ matches this with the shifted-cotangent CE differential dual to $[a \otimes f, b \otimes g]_{\mathfrak{h}_I} = ab \otimes \{f, g\}$. Inductive extension by length and the biderivation property propagates this to all CE lengths.

Step 5. Quasi-isomorphism on each interval. Both sides are free graded-commutative algebras on the same chosen compact-support current reduced Tate vector space after the generator dictionary $c \mapsto \theta$, $u \mapsto O$, with the same CE differential induced from the semidirect Lie bracket of $\mathfrak{g}_I$ (closed side: by definition; brane side: by the reduced interval-wise PV model of Step 1 and the Schouten bracket). The map respects the symmetric-algebra generators by construction. Thus it is an isomorphism of underlying graded vector spaces and intertwines the differentials. The map is therefore an isomorphism of P_0-cdg algebras on each interval.

Step 6. Factorization product compatibility. For disjoint $I_1, I_2 \subset V$, the CE source extension by zero gives $\mathfrak{h}_{I_1} \oplus \mathfrak{h}_{I_2} \hookrightarrow \mathfrak{h}_V$. Thus the CE source is cofactorization: if $j_{I_k,V,!} : \mathfrak{g}_{I_k} \to \mathfrak{g}_V$ denotes extension by zero, the source structure map is

$$\Delta_{I_1,I_2;V} : C^\bullet_{\mathrm{CE}}(\mathfrak{g}_V) \to C^\bullet_{\mathrm{CE}}(\mathfrak{g}_{I_1}) \widehat{\otimes} C^\bullet_{\mathrm{CE}}(\mathfrak{g}_{I_2}).$$

On the brane side, $A^{\mathrm{Ham}}_\partial(I_1) \otimes A^{\mathrm{Ham}}_\partial(I_2) \to A^{\mathrm{Ham}}_\partial(V)$ is the multiplication map under the inclusion of $\mathfrak{h}_{I_1} \oplus \mathfrak{h}_{I_2} \hookrightarrow \mathfrak{h}_V$. By 3.20, the bracket between the two factors vanishes on disjoint supports, so the product is strictly multiplicative. The induced map between central-operation factorization algebras is the corresponding tensor product of multiplication operators. The compatibility is the mixed-variance equation

$$\Phi_V = m_{I_1,I_2;V} \circ \big(\Phi_{I_1} \otimes \Phi_{I_2}\big) \circ \Delta_{I_1,I_2;V},$$

for the operations whose support lies in $I_1 \sqcup I_2 \subset V$. Equivalently, the diagram

$$\begin{array}{ccc} \mathcal{F}_{\mathrm{cl}}(I_1) \otimes \mathcal{F}_{\mathrm{cl}}(I_2) & \xrightarrow{\Phi(I_1)\otimes\Phi(I_2)} & \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(A^{\mathrm{Ham}}_{\partial})(I_1) \otimes \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(A^{\mathrm{Ham}}_{\partial})(I_2) \\ \uparrow & & \downarrow \\ \mathcal{F}_{\mathrm{cl}}(V) & \xrightarrow{\Phi(V)} & \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(A^{\mathrm{Ham}}_{\partial})(V) \end{array}$$

commutes after vertical reversal of the left arrow (CE coproduct = dual of extension by zero).

Step 7. Locally constant structure. For each interval I, Lemma 3.19 gives the explicit contraction

$$\Omega^{\bullet}_c(I) \underset{i_I}{\overset{p_I}{\rightleftarrows}} \mathbb{C}[-1], \qquad i_I(1) = \lambda_I(t)\,dt, \quad p_I(\alpha) = \int_I \alpha,$$

with $dK_I + K_I d = \mathrm{id} - i_I p_I$. The maps p_I are natural for extension by zero; the choice of i_I and K_I is a choice of density representative for the compact-support cohomology generator, not a point-evaluation map. The degree-zero boundary notation $B_f(a)$ used above already includes the shifted-cotangent degree, so the density representative appears as a degree-zero CE coordinate after the BV shift. Translation is homotopically trivial by the same de Rham contraction, hence the interval-wise objects are locally constant in the topological direction after this density contraction.

Step 8. Stalk recovery. Under the density contraction, a length-one shifted-cotangent CE coordinate represented by $a(t)dt \otimes f$ maps to the boundary current $B_f(a)$ and hence to the multiplication operator $L_{B_f(a)}$. Stalk recovery is obtained by any cofinal family of compactly supported unit densities supported near t_0. This represents the compact-support cohomology generator under the locally constant factorization-algebra quasi-isomorphism; it is not a literal degree-zero point evaluation. □

Remark 3.23 *(Locally constant equivalence and E_1-algebra translation).* In the locally constant setting on $\mathbb{R}$, factorization algebras are equivalent to homotopy associative algebras up to translation invariance by Lurie, *Higher Algebra*, §5.5.4, in the Costello–Gwilliam vocabulary of locally constant factorization algebras [15, Vol. I §6.4]. The Hochschild cochain complex of the E_1 algebra computes the factorization-center stalk. The present reduced Hamiltonian theorem uses the explicit PV model $\widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathfrak{h}_I^{\vee}[-1])$ and its strict P_0 subalgebra of multiplication operators. Identifying this PV model with continuous Hochschild cochains for the completed Tate algebra is a separate HKR/topology theorem. The CE source identification of 3.22 is exactly the closed dual of this PV picture in the connected Hamiltonian sector.

Remark 3.24 *(Presentations of one-dimensional factorization algebras).* The precise equivalence used here is Lurie's identification of locally constant factorization algebras on the line with E_1-algebras [37, Thm. A.3.7.5]. The Costello–Gwilliam model-categorical language is used only as a compatible presentation convention unless the cited corollary is separately anchored.

For a factorization algebra $\mathcal{A}$ on the line, write $\mathrm{Z}_{\mathrm{fact}}(\mathcal{A})$ for the factorization algebra of E_1 central operations. In the locally constant one-dimensional case its stalk is the Hochschild cochain complex of the corresponding E_1 algebra. The notation $\mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathcal{A})$ means the same central operations with the additional compatibility homotopies for the classical Poisson/BV bracket. This is the local convention used below; no unrestricted Costello–Gwilliam central-operations theorem is used in the reduced Hamiltonian proof. The proposition below records the degree-zero current shadow as a corollary of 3.22.

Theorem 3.25 *(Hamiltonian-current central operation, smeared).* 3.22 restricted to the length-one CE component on each interval recovers the degree-zero Hamiltonian current shadow: the assignment

$$\Phi_{I,N}(u_{a(t)dt\otimes f})(u) = B_f(a)\cdot u = \Big(\int_I a(t)\,\overline{\operatorname{Tr} f(\phi_1(t),\phi_2(t))}\,dt\Big)u$$

is a closed Hochschild 0-cochain, hence a degree-zero central operation, in $\mathrm{Z}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N})(I)$, the assignments are compatible with interval inclusions and disjoint factorization products, and the induced operations satisfy

$$\{\Phi(u_{a(t)dt\otimes f}),\Phi(u_{b(t)dt\otimes g})\}_{A^{\mathrm{Ham}}_\partial} = \Phi(u_{a(t)b(t)dt\otimes\{f,g\}_{\bar{A}}}).$$

The degree-zero current shadow is the length-one component of the reduced CE/PV central-operation comparison: its content is the strict P_0 identity, one of the structure equations defining $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ of 3.22.

Proof. In the reduced open model $Q\phi_1 = Q\phi_2 = 0$. Hence $\overline{\operatorname{Tr} f(\phi_1,\phi_2)}$ is Q-closed for every Hamiltonian f, and the compactly supported integral is Q-closed. Since the reduced open observable algebra is graded-commutative, multiplication by this closed element is a Hochschild 0-cocycle; this is the interval-wise version of 3.209. Extension by zero of the compactly supported test density is compatible with enlargement to a larger interval through the extension map $j_!$. For disjoint intervals 3.20 gives vanishing of the P_0 bracket between the two factors, so the factorization product of two such operations is the strict multiplication by $I_\partial(a,f)I_\partial(b,g)$; this is the multiplicative extension from $\mathfrak{h}_I$ to $A^{\mathrm{Ham}}_\partial(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$, not the linear current bracket.

The Poisson statement is local on the line. For compactly supported test functions a, b, 3.243 (the strict P_0 identity on the brane factorization algebra) gives

$$\{I_\partial(a,f), I_\partial(b,g)\}_{A^{\mathrm{Ham}}_\partial} = I_\partial(ab,\{f,g\}_{\bar{A}}).$$

Passing to the connected stable Hamiltonian quotient removes only the scalar Hamiltonian line, which is central for the bracket. Thus the current bracket is preserved. Taking a and b supported in a small coordinate interval and then passing to the locally constant stalk recovers 3.242. □

Proposition 3.26 *(CE-source obstruction for perfect Hamiltonian Lie algebras).* Let $\mathfrak{h}$ be a perfect Lie algebra in **TateVec** (i.e. $[\mathfrak{h},\mathfrak{h}] = \mathfrak{h}$). Then no length-one CE coordinate c^I dual to a Hamiltonian generator $H_I \in \mathfrak{h}$ is closed under d_{CE}:

$$d_{\mathrm{CE}}c^K = -\tfrac{1}{2}\,C^K_{IJ}\,c^I c^J \neq 0.$$

Equivalently, $H^1(\mathfrak{h};\mathbb{C}) = 0$.

This obstruction is the structural reason that the boundary observable $O_I = J(H_I) \in A_\partial$ cannot be the image under Φ_{coord} of c^I. Instead, under 3.28,

$$O_I = \Phi_{\mathrm{coord}}(u_I),$$

the image of the CE coordinate dual to the shifted cotangent generator $\eta_{\partial_I} \in \mathfrak{h}^\vee[1]$. The boundary observable lives in the cotangent direction of the CE cochain algebra, not in the Hamiltonian direction. Its CE differential is not zero but $d_{\mathrm{CE}}u_K = C^L_{KJ}u_L c^J$, mapping under Φ_{coord} to the Hamiltonian vector field $d^{\mathrm{coord}}_\pi O_K = C^L_{KJ}O_L\theta^J$ of O_K in the Schouten algebra.

This is exactly what the bracket-admissible P_0-map produces: a function whose Schouten differential is its Hamiltonian vector field.

Proof. $\mathfrak{h}$ perfect means each basis vector has nonzero projection from some bracket combination; equivalently, for every length-one Hamiltonian CE coordinate c^K there are indices I, J with $C^K_{IJ} c^I c^J \neq 0$. Hence no c^K is closed, and H^1 vanishes. Identification $O_I = \Phi_{\mathrm{coord}}(u_I)$ is the content of 3.55 combined with $\Phi_{\mathrm{coord}}(u_I) = O_I$ from 3.28. Differentials match on generators by 3.219. □

Remark 3.27. The reduced Hamiltonian Lie algebra $\mathfrak{h} = \widehat{A}/\mathbb{C} \cdot 1 = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ of the formal symplectic disk in dimension two is perfect: for every monomial one has, before quotienting constants, $z_1^a z_2^b = \left\{z_1, \frac{1}{b+1} z_1^a z_2^{b+1}\right\}$, so $[\mathfrak{h}, \mathfrak{h}] = \mathfrak{h}$. In particular $[H_{1,0}, H_{m,n}] = n\, H_{m,n-1}$ and $[H_{0,1}, H_{m,n}] = -m\, H_{m-1,n}$. The hypothesis of 3.26 is met. The CE-length-one cochain $\xi_{a,b} = c^{a,b}$ dual to $H_{a,b} = z_1^a z_2^b$ is therefore not closed, and no closed CE cochain has linear Hamiltonian part $c^{a,b}$; this is the source obstruction.

THEOREM 3.28 *(Admissible reduced Hamiltonian CE/PV central-operation map, cochain level).* Let

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$$

be the reduced Hamiltonian Lie algebra of the formal symplectic disk, equipped with the coordinate completion of Definition 2.45. Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ be the shifted-cotangent extension. Let $A_{\partial,\mathrm{coord}}$ be the coordinate completed symmetric algebra on the stable scalar-reduced Hamiltonian trace labels.

Then there is a canonical isomorphism of completed dg graded-commutative coordinate algebras

$$\Phi_{\mathrm{coord}} : C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}}),$$

determined on generators by

$$\Phi_{\mathrm{coord}}(c^I) = \theta^I, \qquad \Phi_{\mathrm{coord}}(u_I) = O_I,$$

where c^I is the CE coordinate dual to $H_I \in \mathfrak{h}$ and u_I is the CE coordinate dual to the shifted cotangent generator $\eta_{\delta_I} \in \mathfrak{h}^\vee[1]$.

If $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ is a bracket-admissible realization of the reduced Hamiltonian central-operation model, then the restriction

$$\Phi^B_{P_0} : \Phi^{-1}_{\mathrm{coord}}(B) \xrightarrow{\sim} B$$

is the cochain-level open-closed reduced Hamiltonian central-operation isomorphism in the formal/local sector. The strict ambient product completion is not used here as a global P_0 algebra. No Tate bar-cobar or enveloping conclusion is part of this coordinate statement; those require the separate admissible conilpotent hypotheses of Lemma 3.217.

Proof. Generators and coordinate extension. By Definition 2.45, $C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g})$ is the product over finite graded-commutative words in the letters u_I, c^I, with $|u_I| = 0$ and $|c^I| = 1$. The target $\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ is the identical coordinate product with O_I, θ^I in place of u_I, c^I. The assignment $c^I \mapsto \theta^I$, $u_I \mapsto O_I$ therefore extends uniquely to a continuous bijection of completed coordinate dg-commutative algebras. This is a coordinate envelope statement. Literal continuous CE cochains form the continuous subspace inside this product when the coefficient support is continuous for the chosen topology; they are not identified with the whole coordinate product.

Cochain identity on generators. The CE differential is

$$d_{\mathrm{CE}} c^K = -\tfrac{1}{2} C^K_{IJ} c^I c^J, \qquad d_{\mathrm{CE}} u_K = C^L_{KJ} u_L c^J.$$

By Lemma 2.47,

$$d_\pi^{\mathrm{coord}}\theta^K = -\tfrac{1}{2}C_{IJ}^K\theta^I\theta^J, \qquad d_\pi^{\mathrm{coord}}O_K = C_{KJ}^L O_L\theta^J.$$

Therefore

$$\Phi_{\mathrm{coord}}(d_{\mathrm{CE}}c^K) = -\tfrac{1}{2}C_{IJ}^K\theta^I\theta^J = d_\pi^{\mathrm{coord}}\theta^K = d_\pi^{\mathrm{coord}}\Phi_{\mathrm{coord}}(c^K),$$
$$\Phi_{\mathrm{coord}}(d_{\mathrm{CE}}u_K) = C_{KJ}^L O_L\theta^J = d_\pi^{\mathrm{coord}}O_K = d_\pi^{\mathrm{coord}}\Phi_{\mathrm{coord}}(u_K).$$

Cochain identity on the full coordinate algebra. Both d_{CE} and d_π^{coord} are degree-1 derivations of the completed graded-commutative algebra structures on each side. They satisfy the graded Leibniz rule with the same Koszul sign convention. Φ_{coord} is an algebra homomorphism, so

$$\begin{aligned}\Phi_{\mathrm{coord}}(d_{\mathrm{CE}}(xy)) &= \Phi_{\mathrm{coord}}\big((d_{\mathrm{CE}}x)\,y + (-1)^{|x|}x\,d_{\mathrm{CE}}y\big)\\ &= \Phi_{\mathrm{coord}}(d_{\mathrm{CE}}x)\,\Phi_{\mathrm{coord}}(y) + (-1)^{|x|}\Phi_{\mathrm{coord}}(x)\,\Phi_{\mathrm{coord}}(d_{\mathrm{CE}}y),\end{aligned}$$

while

$$\begin{aligned}d_\pi^{\mathrm{coord}}\Phi_{\mathrm{coord}}(xy) &= d_\pi^{\mathrm{coord}}\big(\Phi_{\mathrm{coord}}(x)\Phi_{\mathrm{coord}}(y)\big)\\ &= (d_\pi^{\mathrm{coord}}\Phi_{\mathrm{coord}}(x))\,\Phi_{\mathrm{coord}}(y) + (-1)^{|x|}\Phi_{\mathrm{coord}}(x)\,d_\pi^{\mathrm{coord}}\Phi_{\mathrm{coord}}(y).\end{aligned}$$

Generator-level agreement plus Leibniz/algebra-homomorphism gives agreement on every word in $\{c^I, u_J\}$ at fixed total length; coordinate-product completeness of both sides plus continuity of $d_{\mathrm{CE}}, d_\pi^{\mathrm{coord}}, \Phi_{\mathrm{coord}}$ for finite coordinate projections extends the equality to the completed coordinate algebras.

Inverse. The assignment $\theta^I \mapsto c^I, O_I \mapsto u_I$ extends to a continuous algebra homomorphism in the opposite direction. By the same Leibniz argument, it intertwines d_π^{coord} and d_{CE}. The two compositions fix generators, hence fix the completed algebras. So Φ_{coord} is an isomorphism of complete dg graded-commutative algebras.

Cotangent/Schouten bracket compatibility. On finite words the CE side carries the canonical (-1)-shifted Poisson structure on $C^\bullet_{\mathrm{CE,coord}}(T^*[1]\mathfrak{h})$ with

$$\{c^I, u_J\}_{\mathrm{CE}} = \delta^I_J, \quad \{c^I, c^J\}_{\mathrm{CE}} = 0, \quad \{u_I, u_J\}_{\mathrm{CE}} = 0.$$

These are the degree-(-1) cotangent brackets on the cochain algebra of a shifted cotangent Lie algebra. The bracket compatibility used here is the displayed generator computation; it is separate from the degree-zero reduced Hamiltonian P_0 bracket on boundary currents. The finite-word Schouten bracket on $\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ satisfies

$$[\theta^I, O_J]_S = \delta^I_J, \quad [\theta^I, \theta^J]_S = 0, \quad [O_I, O_J]_S = 0,$$

the last because O_I, O_J are functions on the formal affine space $\mathfrak{h}^\vee$ and the Schouten bracket of two functions is zero. Generator-level matching gives $\Phi_{\mathrm{coord}}\big(\{c^I, u_J\}_{\mathrm{CE}}\big) = \Phi_{\mathrm{coord}}(\delta^I_J) = \delta^I_J = [\theta^I, O_J]_S = [\Phi_{\mathrm{coord}}(c^I), \Phi_{\mathrm{coord}}(u_J)]_S$, and similarly for the other two pairs. This proves bracket compatibility on finite words. On a bracket-admissible B, continuity and the biderivation rule extend the equality to $\Phi^{-1}_{\mathrm{coord}}(B)$ and B. No bracket statement is made on the strict ambient product completion.

Identification with the reduced P_0 Hamiltonian central-operation model. For the reduced constant-stalk convention used here, the reduced P_0 Hamiltonian central-operation model of a Poisson algebra A

is by definition represented by the Poisson cochain complex $\mathrm{PV}(A)$ with the Lichnerowicz differential $[\pi, -]_S$. This is the local Poisson-CE model; it is not an import of an unreduced Costello–Gwilliam central-operations theorem. Therefore any bracket-admissible B representing this local Poisson-CE model identifies with $Z^{P_0}_{\mathrm{red,Ham}}(A_{\partial,\mathrm{coord}}; B)$. Therefore $\Phi^B_{P_0}$, not the ambient product coordinate map by itself, is the reduced Hamiltonian central-operation isomorphism in this local stalk sense. □

PROPOSITION 3.29 *(Reduced coordinate coupling equations).* Let A be a bracket-admissible dg P_0-target and let

$$\kappa\colon C^\bullet_{\mathrm{CE,coord}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]) \longrightarrow A$$

be a continuous graded-commutative algebra map. Write $X^I = \kappa(c^I)$ and $\mu_I = \kappa(u_I)$. Then κ is a dg map exactly when

$$d_A X^K + \frac{1}{2} C^K_{IJ} X^I X^J = 0, \qquad d_A \mu_K - C^L_{KJ} \mu_L X^J = 0.$$

It is a P_0-map exactly when, in addition, the generator contraction identities are satisfied:

$$\{X^I, \mu_J\}_A = \delta^I_J, \qquad \{X^I, X^J\}_A = 0, \qquad \{\mu_I, \mu_J\}_A = 0.$$

The Dirac CE/PV solution is $X^I = \theta^I$, $\mu_I = O_I$. For an unreduced factorization target the same equations are replaced by the curvature condition

$$\mathrm{Curv}(\kappa) = d\kappa + \frac{1}{2}[\kappa, \kappa] = 0$$

in the completed convolution complex, together with the descent and QME data of Section 4.

Proof. The source is freely generated, as a completed graded-commutative coordinate algebra, by c^I and u_I. A continuous algebra map is therefore determined by the two coordinate families X^I and μ_I. Intertwining the differentials is exactly the two displayed equations, because $d_{\mathrm{CE}} c^K = -\frac{1}{2} C^K_{IJ} c^I c^J$ and $d_{\mathrm{CE}} u_K = C^L_{KJ} u_L c^J$. The P_0 condition is likewise the generator-level cotangent/Schouten contraction identity; bracket-admissibility is the hypothesis that extends this identity from generators to the completed target. The last sentence is the same condition written in the convolution dg Lie algebra where an unreduced factorization lift would have to live. □

3.7 Tate Quillen equivalence and the ∞-categorical upgrade of the cochain bar-cobar identity

3.7.1 CATEGORICAL HOME

Let **TateVec** be the closed symmetric monoidal category of complete filtered Tate $\mathbb{C}$-vector spaces of 3.217. The filtration on every object is taken *complete and Hausdorff*: $V = \varprojlim_w V_{\leq w}$ and $\bigcap_w F^w V = 0$. Concrete morphisms are filtration-preserving continuous chain maps. The working ambient category is the bounded-below filtered subcategory $\mathbf{TateChain}_{\geq 0}$ of non-negatively filtered chain complexes in **TateVec**, large enough to contain every CE chain coalgebra and completed enveloping algebra used here. The exact completion used below is constructed first. The model-categorical statements add transfer, smallness, and monoid hypotheses to this exact Mittag-Leffler habitat; they are not statements about the raw category of all Tate spaces.

Definition 3.30 *(Strict finite-window Mittag-Leffler Tate habitat).* Let $\mathbf{MLTateChain}^{\mathrm{str}}_{\geq 0}$ be the category whose objects are inverse systems

$$V = (V_{\leq M}, \pi^V_{M,N})_{M \geq N \geq 1}$$

of bounded-below finite-dimensional chain complexes over $\mathbb{C}$, with degreewise surjective transition maps. The represented complete object is

$$|V| = \varprojlim_M V_{\leq M}, \qquad F^M|V| = \ker(|V| \to V_{\leq M-1}).$$

Morphisms are compatible families of chain maps $f_{\leq M} : V_{\leq M} \to W_{\leq M}$. The completed tensor product is defined on paired windows by

$$V \widehat{\otimes}_{\mathrm{ML}} W = (V_{\leq M} \otimes W_{\leq M})_{M \geq 1},$$

with the induced surjective transition maps, and

$$|V \widehat{\otimes}_{\mathrm{ML}} W| = \varprojlim_M (V_{\leq M} \otimes W_{\leq M}).$$

The continuous dual attached to this strict tower is

$$V^\vee_{\mathrm{ML}} = \varinjlim_M V^\vee_{\leq M}.$$

A morphism is an admissible filtered weak equivalence when each $f_{\leq M}$ and the induced associated-graded map are quasi-isomorphisms.

The only universal property asserted for this habitat is the ordinary inverse-limit one: a compatible family $x_M \in V_{\leq M}$ determines a unique element $x \in |V|$, and every map out of a test strict tower is exactly a compatible family of finite-window maps. No initial or terminal property among all completions of a Tate vector space is asserted.

Proposition 3.31 *(Exactness and kernels in the strict ML habitat).* In $\mathbf{MLTateChain}^{\mathrm{str}}_{\geq 0}$:

(i) the filtration on $|V|$ is complete and Hausdorff;

(ii) inverse limits of strict surjective towers are exact, so $H^\bullet(|V|) = \varprojlim_M H^\bullet(V_{\leq M})$ and $\varprojlim^1_M H^{\bullet-1}(V_{\leq M}) = 0$;

(iii) completed tensor products and continuous duals are computed by the displayed finite-window formulas;

(iv) if $H = (H_{\leq M})$ and $D = (D_{\leq M})$ are paired strict ML systems and finite tensors $C_M \in H_{\leq M} \otimes D_{\leq M}$ are compatible under transition, then

$$C = \varprojlim_M C_M \in |H| \widehat{\otimes}_{\mathrm{ML}} |D|$$

exists and is the unique strict ML kernel with those finite-window projections.

Proof. Completeness is the definition of $|V|$ as an inverse limit; the intersection of the kernels $F^M|V|$ is zero because an element whose image vanishes in every finite quotient is the zero compatible family. Surjectivity of the transition maps gives the Mittag-Leffler condition. The Milnor exact sequence

then has zero $\varprojlim^1$-term, proving exactness and the cohomology formula. The tensor and dual formulas are definitions inside the strict habitat and are finite-dimensional on each window. Finally, a compatible tensor family $(C_M)_M$ is precisely an element of the inverse limit $\varprojlim_M (H_{\leq M} \otimes D_{\leq M})$, which is the completed tensor product by definition. The uniqueness is the inverse-limit universal property of Definition 3.30. □

Statement 3.32 *(Admissible Tate model envelope).* An admissible Tate model envelope is a bounded-below, locally presentable symmetric monoidal model category $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ equipped with a fully faithful exact symmetric-monoidal embedding of the strict habitat

$$\mathbf{MLTateChain}^{\mathrm{str}}_{\geq 0} \hookrightarrow \mathbf{TateChain}^{\mathrm{adm}}_{\geq 0},$$

and containing the CE chains, completed enveloping algebras, completed symmetric algebras, and finite-window inverse systems used in this manuscript. Weak equivalences are admissible filtered quasi-isomorphisms: quasi-isomorphisms on every finite quotient and on associated graded. The tensor product restricts to $\widehat{\otimes}_{\mathrm{ML}}$ on strict Mittag-Leffler systems, and filtered inverse limits with surjective transition maps are exact by Proposition 3.31. The additional hypotheses are transfer and smallness for the bar-cobar adjunction, the monoid axiom for completed tensor products, and compatibility of $\widehat{\otimes}$ with the finite-window model structures. All Quillen statements below are relative to such an envelope.

LEMMA 3.33 *(Mittag-Leffler dictionary for closed Tate windows).* Let

$$\{V_{\leq M}\}_{M\geq 1} \quad \text{and} \quad \{W_{\leq M}\}_{M\geq 1}$$

be inverse systems of finite-dimensional complexes with surjective transition maps, and put

$$V = \varprojlim_M V_{\leq M}, \qquad W = \varprojlim_M W_{\leq M}.$$

In the strict ML habitat, and hence in any admissible envelope containing it, the following assertions hold:

(i) $\varprojlim$ is exact on these systems, so $H^\bullet(V) = \varprojlim_M H^\bullet(V_{\leq M})$;
(ii) continuous dualization changes the inverse system into the direct system of finite duals: $V^\vee_{\mathrm{cont}} = \varinjlim_M V^\vee_{\leq M}$;
(iii) completed tensor products are computed windowwise: $V \widehat{\otimes} W = \varprojlim_M (V_{\leq M} \otimes W_{\leq M})$ whenever the transition maps are the paired finite-window truncations;
(iv) a map $f : V \to W$ is an admissible filtered quasi-isomorphism iff each finite-window map $f_{\leq M} : V_{\leq M} \to W_{\leq M}$ is a quasi-isomorphism and the induced associated-graded maps are quasi-isomorphisms.

Proof. Surjectivity of the transition maps gives the Mittag-Leffler condition, hence $\varprojlim^1 = 0$. The cohomology assertion is the standard Milnor exact sequence with the $\varprojlim^1$-term killed. Continuous linear functionals on V factor through a finite quotient $V_{\leq M}$, giving the displayed direct limit of duals. The completed tensor product in Definition 3.30 is defined by the same finite-window inverse system, so item (iii) is tautological inside the strict habitat. Item (iv) is the definition of weak equivalence in Definition 3.30 and Statement 3.32. □

Definition 3.34 *(Closed-window obstruction terms).* Let $\mathfrak{h} = \varprojlim_M \mathfrak{h}_{\leq M}$ be a complete filtered Lie algebra with finite-dimensional windows and surjective transition maps $\pi_{M,N}$. Write $D_{\leq M}$ for a finite dual module paired with $\mathfrak{h}_{\leq M}$, with surjective transition maps $\pi^\vee_{M,N}$, and put $D_{\mathrm{cont}} = \varprojlim_M D_{\leq M}$. For $M \geq N$ define:

$$\mathfrak{o}^{\mathrm{br}}_{M,N}(x, y) = \pi_{M,N}[x, y]_M - [\pi_{M,N}x, \pi_{M,N}y]_N,$$

$$\mathfrak{o}^{\mathrm{coad}}_{M,N}(x, \lambda) = \pi^\vee_{M,N}(x \cdot_M \lambda) - (\pi_{M,N}x) \cdot_N (\pi^\vee_{M,N}\lambda),$$

and

$$\mathfrak{o}^{\mathrm{Cas}}_{M,N} = (\pi_{M,N} \otimes \pi^\vee_{M,N})C_M - C_N.$$

The closed window system is *cochain-CE/PV-admissible* if $\mathfrak{o}^{\mathrm{br}}$ and $\mathfrak{o}^{\mathrm{coad}}$ vanish for all $M \geq N$. It is *kernel-admissible* only after the additional Casimir term $\mathfrak{o}^{\mathrm{Cas}}$ vanishes in the chosen completed tensor category. The Casimir condition is not part of the coordinate CE/PV cochain identity.

Lemma 3.35 *(Closed CE/PV finite-window compatibility and variance boundary).* Assume the closed window system of Definition 3.34 is cochain-CE/PV-admissible and each finite window has the finite cotangent CE/PV identity

$$\Phi_M \colon C^\bullet_{\mathrm{CE}}\big(\mathfrak{h}_{\leq M} \ltimes D_{\leq M}[1]\big) \xrightarrow{\sim} \mathrm{PV}\big(\mathbf{S}(\mathfrak{h}_{\leq M})\big).$$

Then the Φ_M are compatible as finite coordinate tests under the contravariant CE pullbacks and the corresponding PV coordinate maps. They do not by themselves form an inverse system of dg cochain projections. A completed admissible filtered quasi-isomorphism

$$\Phi \colon C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{h} \ltimes D_{\mathrm{cont}}[1]\big) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big)$$

follows only after one supplies either a chain-side inverse limit with exact continuous dualization or a topology whose cochain transition maps are dg maps. If the kernel-admissible Casimir condition is also supplied, the same finite windows carry compatible P_0-pairings and kernels in that chosen tensor category.

Proof. The finite map Φ_M is determined on generators by $c \mapsto \theta$ and $u \mapsto O$. Vanishing of $\mathfrak{o}^{\mathrm{br}}_{M,N}$ says the CE differential on the Hamiltonian ghosts commutes with transition maps. Vanishing of $\mathfrak{o}^{\mathrm{coad}}_{M,N}$ says the cotangent-coordinate part of the CE differential, and equivalently the Lichnerowicz differential on O-coordinates, commutes with transition maps. Therefore the finite squares with Φ_M and Φ_N commute in the variance appropriate to cochains. A quotient $\mathfrak{h}_{\leq M} \twoheadrightarrow \mathfrak{h}_{\leq N}$ induces CE cochain maps by pullback, not by projection from M to N. Thus inverse-limit language on cochains requires the additional continuous-dualization or topology hypothesis stated above. Vanishing of $\mathfrak{o}^{\mathrm{Cas}}_{M,N}$ is the separate statement that the finite P_0-brackets and BV coefficient pairings are compatible under transition in the chosen kernel-admissible tensor category. □

3.7.2 Direct admissibility recognition

Definition 3.36 *(Open-side twisting cochain datum).* Fix a candidate admissible filtered-cobar datum

$$\mathfrak{h} = \varprojlim_M \mathfrak{h}_{\leq M}, \qquad D = \varinjlim_M \mathfrak{h}^\vee_{\leq M}, \qquad \mathfrak{g} = \mathfrak{h} \ltimes D[1],$$

a complete augmented filtered open A_∞-algebra A_{op}, and the completed CE/PV target

$$B_{\mathrm{cl}} = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \cong \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{coord}}(\mathfrak{h})\big).$$

Put $C_{\mathrm{op}} = \mathrm{Bar}(\mathcal{A}_{\mathrm{op}})$, with reduced bar differential b_{op}, coproduct Δ, and iterated reduced coproducts $\Delta^{(r)}$. The bar coalgebra is required to be locally conilpotent in the completed sense: for every finite output window of B_{cl} and every finite coaugmentation stage of C_{op}, the iterated reduced coproducts $\Delta^{(r)}$ vanish for all sufficiently large r. Equivalently, the conilpotence obstruction in Definition 3.37 is zero. Without this condition the Maurer–Cartan curvature below is not a continuous map and no twisting-cochain datum has been constructed.

An open-side twisting cochain is a continuous degree-one map

$$\tau\colon C_{\mathrm{op}} \longrightarrow B_{\mathrm{cl}}$$

vanishing on the coaugmentation and satisfying the Maurer–Cartan equation

$$d_{B_{\mathrm{cl}}}\tau + \tau b_{\mathrm{op}} + \sum_{r\geq 2} m_r^{B_{\mathrm{cl}}}(\tau^{\otimes r})\Delta^{(r)} = 0.$$

For the dg P_0-algebra B_{cl}, the higher multiplications vanish and this reads

$$d_{B_{\mathrm{cl}}}\tau + \tau b_{\mathrm{op}} + \mu_{B_{\mathrm{cl}}}(\tau\otimes\tau)\Delta = 0.$$

Equivalently,

$$d_{\mathrm{conv}}\tau + \frac{1}{2}[\tau,\tau]_{\mathrm{conv}} = 0, \qquad d_{\mathrm{conv}}f = d_{B_{\mathrm{cl}}}f - (-1)^{|f|} f\, b_{\mathrm{op}},$$

in the continuous convolution dg Lie algebra $\mathrm{Hom}_{\mathrm{cont}}(C_{\mathrm{op}}, B_{\mathrm{cl}})$.

The twisting cochain induces the completed cobar morphism

$$\kappa_\tau\colon \Omega C_{\mathrm{op}} \longrightarrow B_{\mathrm{cl}}.$$

Its associated-graded symbol is the map

$$\mathrm{gr}\,\kappa_\tau\colon\ \mathrm{gr}\,\Omega\mathrm{Bar}(\mathcal{A}_{\mathrm{op}}) \longrightarrow \mathrm{gr}\,B_{\mathrm{cl}}.$$

In the Hamiltonian trace presentation the required linear symbol is

$$\mathrm{gr}\,\kappa_\tau(B_f) = O_f, \qquad \mathrm{gr}\,\kappa_\tau(\Theta_{\rho_{a,b}}) = \theta^{a,b},$$

after the scalar trace is removed and the cotangent generator is read in the principal-part local-dual basis.

Definition 3.37 (Open-side Koszul obstruction complex). In the notation of Definition 3.36, put $D_{\leq M} = \mathfrak{h}^\vee_{\leq M}$, $\mathrm{PV}_{\leq M} = \mathrm{PV}(\mathbf{S}(\mathfrak{h}_{\leq M}))$, and $B_{\mathrm{cl},\leq M}$ for the corresponding finite CE/PV window. The obstruction datum is

$$O_{\mathrm{Kos}}(\mathfrak{h}, D, \mathcal{A}_{\mathrm{op}}, \tau) = O_{\mathrm{win}} \oplus O_{\mathrm{top}} \oplus O_{\mathrm{conilp}} \oplus O_{\mathrm{MC}}(\tau) \oplus O_{\mathrm{cobar}}(\tau).$$

The window and conilpotence summands are degree-zero:

$$O_{\mathrm{win}} = \prod_{M\geq N} (\mathrm{Hom}(\Lambda^2\mathfrak{h}_{\leq M}, \mathfrak{h}_{\leq N}) \oplus \mathrm{Hom}(\mathfrak{h}_{\leq M}\otimes D_{\leq M}, D_{\leq N})) \oplus Q_{\mathrm{Cas}},$$

where

$$Q_{\mathrm{Cas}} = \frac{\varprojlim_M (\mathfrak{h}_{\leq M}\otimes D_{\leq M})}{\mathrm{im}(\mathfrak{h}\widehat{\otimes} D)}.$$

The distinguished window class is

$$\omega_{\mathrm{win}} = \big(\mathfrak{v}^{\mathrm{br}}_{M,N}, \mathfrak{v}^{\mathrm{coad}}_{M,N}, [(C_{\leq M})_M]\big),$$

where the first two terms are those of Definition 3.34 and the last term is the compatible finite-window Casimir family modulo strict completed tensors. The topology summand is

$$O^n_{\mathrm{top}} = \varprojlim_M{}^1 H^{n-1}\big(C^\bullet_{\mathrm{CE}}(\mathfrak{g}_{\leq M})\big) \oplus \varprojlim_M{}^1 H^{n-1}(\mathrm{PV}_{\leq M}).$$

The conilpotence summand is

$$O_{\mathrm{conilp}} = \left(C^{\mathrm{CE}}_*(\mathfrak{g})/\bigcup_r F^{(r)}_{\mathrm{coaug}} C^{\mathrm{CE}}_*(\mathfrak{g})\right) \oplus \bigcap_{r\geq 1} \mathfrak{g}^{(r)}.$$

The Maurer–Cartan summand is the explicit curvature

$$O_{\mathrm{MC}}(\tau) = \mathrm{MC}(\tau) = d_{\mathrm{conv}}\tau + \frac{1}{2}[\tau,\tau]_{\mathrm{conv}} \in \mathrm{Hom}^2_{\mathrm{cont}}(C_{\mathrm{op}}, B_{\mathrm{cl}}).$$

The cobar obstruction complex is

$$O_{\mathrm{cobar}}(\tau) = \mathrm{Cone}(\mathrm{gr}\,\kappa_\tau) \oplus \bigoplus_n \varprojlim_M{}^1 H^{n-1}\big((\Omega C_{\mathrm{op}})_{\leq M}\big)[-n] \oplus \bigoplus_n \varprojlim_M{}^1 H^{n-1}\big(B_{\mathrm{cl},\leq M}\big)[-n].$$

Theorem 3.38 *(Filtered-cobar admissibility criterion).* Fix a candidate datum as in Definitions 3.36 and 3.37. Assume the filtrations on ΩC_{op} and B_{cl} are bounded below, complete, separated, and strongly convergent, with finite-dimensional associated graded pieces in every total degree. Assume moreover that the local conilpotence condition in Definition 3.36 holds and continuous dualization is exact on the finite-window towers. Then the following conditions are equivalent.

(A1) The datum is accepted: the generator substitution $c \mapsto \theta$, $u \mapsto O$ is the completed CE/PV identity, τ is a twisting cochain, and

$$\kappa_\tau \colon \Omega\mathrm{Bar}(A_{\mathrm{op}}) \longrightarrow B_{\mathrm{cl}}$$

is an admissible filtered quasi-isomorphism with the linear symbol of Definition 3.36.

(A2) The obstruction equations are exact:

$$\omega_{\mathrm{win}} = 0, \qquad O_{\mathrm{conilp}} = 0, \qquad O_{\mathrm{MC}}(\tau) = 0, \qquad H^\bullet\big(O_{\mathrm{top}} \oplus O_{\mathrm{cobar}}(\tau)\big) = 0.$$

When these equivalent conditions hold,

$$B_{\mathrm{cl}} \simeq A_{\mathrm{op}}$$

in the homotopy category of complete augmented filtered A_∞-algebras. If the CE coalgebra is also used as the closed Koszul source, exact continuous duality identifies the dual form as

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \simeq A^!_{\mathrm{op}}$$

precisely when the CE-side cobar map $\Omega C^{\mathrm{CE}}_*(\mathfrak{g}) \to A_{\mathrm{op}}$ has the same exact associated-graded cone and $\varprojlim^1$ obstruction.

Proof. The equality $\omega_{\mathrm{win}} = 0$ is equivalent to strict finite-window compatibility of the CE differential and the Schouten differential, with P_0-pairings added only in the kernel-admissible tensor category; this is exactly Lemma 3.35 and Theorem 2.50. The vanishing of O_{top} is the Milnor and variance condition identifying cohomology of the completed CE/PV objects with the admissible finite-window calculation.

The local conilpotence hypothesis makes the Maurer–Cartan sum finite on each finite output window, hence a continuous curvature. The equation $O_{\mathrm{MC}}(\tau) = 0$ is then the Maurer–Cartan equation in the continuous convolution dg Lie algebra. It is equivalent to the statement that the adjoint map κ_τ commutes with the completed cobar differential and the A_∞-operations. Without this equality there is no morphism to test.

After the Maurer–Cartan equation holds, filter ΩC_{op} and B_{cl} by tensor length and by the finite coordinate windows. The mapping-cone summand in $O_{\mathrm{cobar}}(\tau)$ is acyclic exactly when $\mathrm{gr}\,\kappa_\tau$ is a quasi-isomorphism. The two $\varprojlim^1$-summands are exactly the Milnor defects for the completed source and target towers. Strong convergence and completeness then make the spectral-sequence comparison theorem equivalent to admissible filtered quasi-isomorphism of κ_τ. Necessity follows from the definition of admissible filtered quasi-isomorphism in the envelope: it is detected on the associated graded and on the exact finite-window towers.

The bar-cobar counit $\Omega\mathrm{Bar}(A_{\mathrm{op}}) \to A_{\mathrm{op}}$ is an admissible filtered quasi-isomorphism under the same boundedness and convergence hypotheses. Composing its inverse in the homotopy category with κ_τ gives $A_{\mathrm{op}} \simeq B_{\mathrm{cl}}$. The final statement is the same argument applied to the CE coalgebra $C_*^{\mathrm{CE}}(\mathfrak{g})$, followed by exact continuous dualization. □

Definition 3.39 (Tate cofibrantly generated model structure in the envelope). Define generating cofibrations I and acyclic cofibrations J by

$$I = \{F^w S^{n-1} \hookrightarrow F^w D^n \mid n \in \mathbb{Z}_{\geq 0},\ w \in \mathbb{N}\}, \qquad J = \{0 \hookrightarrow F^w D^n \mid n \in \mathbb{Z}_{\geq 0},\ w \in \mathbb{N}\},$$

where S^{n-1}, D^n are the standard sphere and disk chain complexes in **Vec** and the Tate window $F^w(-)$ takes the w-th piece of the filtration. A morphism $f : V \to W$ in $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ is:

(i) a *weak equivalence* if it is an admissible filtered quasi-isomorphism in the sense of 3.217: a quasi-isomorphism on every quotient $V/F^wV \to W/F^wW$ and on the associated graded;

(ii) a *cofibration* if it is in the class generated by I under transfinite composition, retracts, and pushouts;

(iii) a *fibration* if it has the right lifting property against J.

THEOREM 3.40 *(Admissible-envelope model-structure existence criterion).* Under Statement 3.32, the triple $(\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}, I, J)$ is a cofibrantly generated model category. The class of weak equivalences is closed under retracts and satisfies the two-out-of-three property; cofibrations and fibrations form complementary lifting classes; the small-object argument terminates after countably many steps because every Tate window is finite-dimensional and every generating cofibration changes only one window.

Proof. Quillen's small-object argument ([38, §II.3]) applies in the admissible envelope once the following are verified:

(i) I and J permit the small-object argument: the domains and codomains of the generating maps are small with respect to relative I-cell complexes because each generator lives in one finite-dimensional Tate window $V_{\leq w}$, hence is $\aleph_0$-small in the admissible envelope.

(ii) Every map in J-cell is a weak equivalence: pushouts of acyclic generating cofibrations are levelwise acyclic by finite-dimensional standard chain-complex theory in each window, and the filtered colimit of windowwise acyclic cofibrations is acyclic by Mittag-Leffler exactness in the Tate filtration (3.217 item 1).

(iii) Every map with the right lifting property against I is a trivial fibration: standard, levelwise.

The two-out-of-three property for filtered quasi-isomorphisms follows from the long exact sequence of the filtration: a composite $V \to W \to U$ has filtered E_1-page agreeing with the composite of E_1-pages of $V \to W$ and $W \to U$ in each window, so two-out-of-three on each window propagates to the global filtered class. This proves the model structure in the chosen envelope only. No assertion is made for arbitrary unbounded Tate complexes outside it. □

Lemma 3.41 *(Tail quotient obstruction invisible to finite-window weak equivalence).* Let $\{V_d\}_{d\geq 1}$ be nonzero finite-dimensional vector spaces with zero differential, and put

$$V_\oplus = \bigoplus_{d\geq 1} V_d, \qquad V_\Pi = \prod_{d\geq 1} V_d.$$

Filter both complexes by degree tails. The inclusion $V_\oplus \hookrightarrow V_\Pi$ is an admissible filtered weak equivalence in the sense of Statement 3.32: it is an isomorphism on every finite quotient and on associated graded pieces. It is not an ordinary quasi-isomorphism of chain complexes. Its cofiber has

$$H^0(V_\Pi/V_\oplus) = V_\Pi/V_\oplus \neq 0.$$

Proof. The quotient by the tail $d > w_0$ is

$$V_\oplus/\bigoplus_{d>w_0} V_d \cong \bigoplus_{d\leq w_0} V_d \cong \prod_{d\leq w_0} V_d \cong V_\Pi/\prod_{d>w_0} V_d.$$

Thus the inclusion is an isomorphism on every finite quotient. The associated graded in degree d is V_d on both sides, so the associated graded map is also an isomorphism. This is precisely an admissible filtered weak equivalence.

Since the differential is zero, ordinary cohomology is the underlying vector space. A sequence with one nonzero component in every degree defines a nonzero class of $V_\Pi/V_\oplus$. Hence the cofiber has nonzero H^0, and the inclusion is not an ordinary quasi-isomorphism. □

Remark 3.42 *(Boundedness restriction is harmless).* Restriction to $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ avoids inverse-limit pathologies in unbounded filtrations. Every CE coalgebra $C^{\mathrm{CE}}_*(\mathfrak{g})$ and every completed enveloping algebra $\widehat{U}(\mathfrak{g})$ that appears here is non-negatively filtered (CE length and PBW length are ≥ 0). A model structure on the raw unbounded category **TateChain** would require the Spaltenstein semi-model modification ([37, §7.5]) and is not needed for the application here.

3.7.3 Operad-algebra transfer

Let $\mathrm{Lie}_{\mathbf{TateVec}}$ be the Lie operad in **TateVec**, regarded as a Σ-cofibrant Σ-module (by the standard Koszul resolution of the Σ-cofibrant replacement), and similarly $\mathrm{Ass}^+_{\mathbf{TateVec}}$ for augmented associative. A $\mathrm{Lie}_{\mathbf{TateVec}}$-algebra is a Tate Lie algebra in **TateVec**; an $\mathrm{Ass}^+_{\mathbf{TateVec}}$-algebra is a Tate complete augmented associative algebra. Write

$$\mathrm{TateLie} = \mathrm{Lie}_{\mathbf{TateVec}}\text{-alg}_{\geq 0}, \qquad \mathrm{TateAssocAlg}^+ = \mathrm{Ass}^+_{\mathbf{TateVec}}\text{-alg}_{\geq 0}$$

in $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$.

LEMMA 3.43 *(Transferred model structures on Tate operad-algebras).* In the admissible envelope, the free-forget adjunction $\mathrm{Free}_{\mathcal{P}} \dashv \mathrm{Forget}$ for $\mathcal{P} \in \{\mathrm{Lie}_{\mathbf{TateVec}}, \mathrm{Ass}^{+}_{\mathbf{TateVec}}\}$ transfers the cofibrantly generated model structure of 3.40 to $\mathcal{P}\text{-alg}_{\geq 0}$ in the sense of Hinich [31, §2.2]: the weak equivalences and fibrations in $\mathcal{P}\text{-alg}_{\geq 0}$ are the maps whose underlying chain map in $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ is one, and the cofibrations are determined by lifting.

Proof. Hinich's transfer theorem requires:

(1) a path object on every algebra of the form $\mathrm{Free}_{\mathcal{P}}(F^w D^n)$;

(2) that pushouts along generating acyclic cofibrations be weak equivalences.

Item (1) follows because $\mathcal{P} = \mathrm{Lie}$ or Ass^{+} is a Σ-cofibrant operad in **TateVec**; the path object is constructed from the cylinder object on chain complexes via the Berger-Moerdijk path-functor lift [29], valid whenever the operad is admissible and the ambient category is cofibrantly generated and proper. Item (2) follows because $\mathcal{P}$ acts levelwise on each Tate window, and the windowwise finite-dimensional pushout is a quasi-isomorphism by Schwede-Shipley [39, Lemma 2.3]. Filtered-colimit assembly preserves quasi-isomorphism by Mittag-Leffler exactness (3.217 item 1). □

The dual coalgebra side requires more care: Hinich's transfer applies to algebra categories, not coalgebra categories, because the cofree coalgebra functor is not a left adjoint in general. The substitute is the Aubry-Lemaire-Smirnov [24] §11.4 model structure on conilpotent dg coalgebras transferred through the bar-cobar adjunction itself.

Definition 3.44 *(Conilpotent Tate coalgebras).* Let

$$\begin{aligned}\mathrm{TateConilpCoAlg} = \big\{\, (C, \Delta_C, d_C) \;:\; & C \in \mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}, \\ & \Delta_C \colon C \to C \widehat{\otimes} C \text{ continuous coproduct}, \\ & \text{coaugmentation filtration exhaustive} \,\big\}.\end{aligned}$$

with morphisms continuous coalgebra maps. Conilpotency means the coaugmentation filtration $F^{(n)}_{\mathrm{coaug}} C = \ker(\overline{\Delta}^{(n)})$ is exhaustive: $C = \bigcup_n F^{(n)}_{\mathrm{coaug}} C$.

LEMMA 3.45 *(Transferred model structure on Tate conilpotent coalgebras).* TateConilpCoAlg admits, in the admissible envelope, a model structure with

- weak equivalences = maps $f \colon C \to C'$ such that $\Omega(f) \colon \Omega(C) \to \Omega(C')$ is a weak equivalence in $\mathrm{TateAssocAlg}^{+}$ (3.43);
- cofibrations = monomorphisms of underlying filtered chain complexes;
- fibrations = right lifting against acyclic cofibrations.

Equivalently, this is the Lefevre-Hasegawa [36, Th. 1.3.1.2] transferred model structure in the admissible Tate envelope.

Proof. Lefevre-Hasegawa, with proof sketched in Loday-Vallette §11.4.10, constructs the conilpotent-coalgebra transferred model structure in the symmetric monoidal category of $\mathbb{Q}$- or $\mathbb{C}$-vector spaces. The proof passes through the cobar-bar adjunction $\Omega \colon \mathrm{ConilpCoAlg} \rightleftarrows \mathrm{AugAssocAlg} \colon \mathrm{Bar}$ and uses that

(i) Ω takes coalgebra cofibrations (= mono with filtered cokernel) to algebra cofibrations;

(ii) Ω inverts coalgebra weak equivalences;

(iii) the unit $\eta_C\colon C \to \mathrm{Bar}\,\Omega(C)$ is a weak equivalence for every conilpotent dg coalgebra.

Items (i)-(iii) are checked windowwise (each Tate window is finite-dimensional, so the standard Lefevre-Hasegawa proof applies) and assemble by the transfer, smallness, monoid, completed tensor, and Mittag-Leffler exactness hypotheses built into Statement 3.32. The unit weak-equivalence (iii) is precisely 3.217 item 3 (admissible filtered qiso) inside that envelope. □

3.7.4 Bar-cobar Quillen equivalence in the admissible envelope

Theorem 3.46 *(Tate bar-cobar Quillen equivalence in the admissible envelope).* The bar-cobar adjunction

$$\Omega\colon \mathrm{TateConilpCoAlg} \rightleftarrows \mathrm{TateAssocAlg}^{+}\colon \mathrm{Bar}$$

is a Quillen equivalence with respect to the admissible-envelope model structures of 3.43 and 3.45. In particular:

(i) Ω is left Quillen: it preserves cofibrations and acyclic cofibrations.

(ii) Bar is right Quillen: it preserves fibrations and acyclic fibrations.

(iii) For every cofibrant conilpotent coalgebra C and every fibrant augmented associative algebra A, the unit $\eta_C\colon C \to \mathrm{Bar}\,\Omega(C)$ and the counit $\varepsilon_A\colon \Omega\,\mathrm{Bar}(A) \to A$ are weak equivalences.

(iv) The induced derived adjunction $L\Omega\colon \mathrm{Ho}(\mathrm{TateConilpCoAlg}) \rightleftarrows \mathrm{Ho}(\mathrm{TateAssocAlg}^{+})\colon R\,\mathrm{Bar}$ is an equivalence of homotopy categories.

Proof. Items (1) and (2) are immediate from 3.45: weak equivalences in TateConilpCoAlg are defined as maps whose Ω is a weak equivalence, and fibrations on the coalgebra side are defined by lifting.

For item (3), the unit η_C: filter both sides by symmetric / tensor length on the underlying **TateVec** object, plus the Tate window filtration F^w. The associated graded gives, windowwise, a finite-dimensional dg coalgebra and the standard bar-cobar resolution $C^{\mathrm{gr}} \to \mathrm{Bar}\,\Omega(C^{\mathrm{gr}})$. By Loday-Vallette [24, §2.2.5, §11.1.5], this is a quasi-isomorphism on each window. Conilpotency (3.44) ensures the coaugmentation filtration on C is exhaustive and Hausdorff, so the abutment of the spectral sequence of the filtration computes the global qiso. Mittag-Leffler exactness in the Tate filtration (3.217 item 1) lifts the windowwise qiso to a global filtered qiso.

For the counit $\varepsilon_A\colon \Omega\,\mathrm{Bar}(A) \to A$, filter both sides by augmentation/tensor length and by the Tate window. On every finite window and every finite length cap this is the ordinary bar-cobar counit for an augmented dg associative algebra over $\mathbb{C}$, hence a quasi-isomorphism by the standard normalized bar resolution [24, §2.2.5, §11.1.5]. The filtrations are bounded below, complete, and Hausdorff in the admissible envelope, so the spectral sequence comparison and Mittag-Leffler exactness lift the windowwise quasi-isomorphism to the completed Tate object.

In the Lie specialization $A = \widehat{U}(\mathfrak{g})$, the same counit can also be read through the PBW filtration: $\mathrm{gr}_{\mathrm{PBW}}\widehat{U}(\mathfrak{g}) \cong \widehat{S}_{\mathrm{Tate}}(\mathfrak{g})$, and the associated graded comparison is the symmetric Koszul resolution of Loday-Vallette [24, §3.4.13]. This is the closed-side PBW form used later, but the Quillen counit above is the associative bar-cobar counit for general A.

Item (4) is formal from (1)-(3) by Quillen's adjoint-functor theorem ([35, Th. 1.3.10]). □

3.7.5 Promotion to the associated ∞-categories

Corollary 3.47 *(Infinity-categorical Tate bar-cobar equivalence).* Apply the Dwyer-Kan / Hammock-localization functor to the Quillen equivalence of 3.46:

$$N_\Delta\big(\mathrm{TateConilpCoAlg}^{\mathrm{cf}}, W\big) \simeq N_\Delta\big(\mathrm{TateAssocAlg}^{+,\mathrm{cf}}, W\big)$$

as the ∞-categories associated to the admissible model envelope in the sense of Lurie's *Higher Algebra* §A.3, where cf denotes the full subcategory of cofibrant-fibrant objects and W is the class of weak equivalences. The equivalence is induced by $L\Omega$ and $R\,\mathrm{Bar}$ at the cofibrant-fibrant level. Presentability is an attribute of the chosen locally presentable envelope, not of the raw Tate category.

Proof. Lurie [37, A.3.7.6, A.3.7.10]: a Quillen equivalence of model categories induces an equivalence of associated ∞-categories. The presentability of both sides, inside the envelope, is inherited from the cofibrant generation of the model structures (3.40, 3.43, 3.45) by Lurie A.2.6.8. □

3.7.6 Identification with the Lie operadic Koszul duality

The Quillen equivalence of 3.46 is *associative* bar-cobar. Specialization to the Lie operad gives the cochain-level equivalence used in the CE/PV theorem.

COROLLARY 3.48 *(Tate Lie bar-cobar / CE Koszul duality).* For $\mathfrak{g}$ a lower-central Tate-pronilpotent graded Lie algebra in the sense of 3.217, the bar-cobar Quillen equivalence applied to $A = \widehat{U}(\mathfrak{g}) \in \mathrm{TateAssocAlg}^{+}$ yields the equivalence in the homotopy category

$$L\Omega\big(C_*^{\mathrm{CE}}(\mathfrak{g})\big) \xrightarrow{\sim} \widehat{U}(\mathfrak{g}), \qquad R\,\mathrm{Bar}\big(\widehat{U}(\mathfrak{g})\big) \xrightarrow{\sim} C_*^{\mathrm{CE}}(\mathfrak{g}),$$

and hence, after continuous dualization, a Lie-Koszul-dual equivalence of cochain algebras

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) \simeq \big(\widehat{U}(\mathfrak{g})\big)^{!}$$

in the corresponding homotopy category of complete augmented Tate algebras.

Inside the admissible envelope, by the symmetric-monoidal-functoriality of the bar-cobar Quillen equivalence [24, §13.2] and Lurie [37, §5.5], this lifts to an equivalence of symmetric monoidal ∞-categories,

$$N_\Delta\big(\mathrm{TateConilpCoAlg}^{\mathrm{cf}}_{\mathrm{Lie}}, W\big) \simeq N_\Delta\big(\mathrm{TateAssocAlg}^{+,\mathrm{cf}}_{\mathrm{Lie}}, W\big)$$

in the framework of Lurie *Higher Algebra* §5.5 (locally constant factorization algebras over $\mathbb{R}$, applied pointwise on the line; or directly the formal-disk version).

Proof. Apply 3.46 to the Lie-operadic Koszul-dual pair (Lie, CoComm) in **TateVec**. Loday-Vallette [24, Th. 13.4.6] identifies the operadic bar-cobar with the Lie-bar-cobar of CE chains and completed enveloping algebra. The symmetric-monoidal extension is direct because the operad Lie is a symmetric-monoidal Hopf operad in **Vec**, and base change to **TateVec** via the strong symmetric-monoidal functor **Vec** $\to$ **TateVec** preserves operadic Koszul duality ([24, §13.5]). □

LEMMA 3.49 *(Closed-side Tate formalism package).* Let $\mathfrak{h} = \varprojlim_M \mathfrak{h}_{\le M}$ be a closed Hamiltonian Tate Lie algebra whose finite windows are cochain-CE/PV-admissible, and kernel-admissible when a P_0 kernel is used, in the sense of Definition 3.34. Assume:

(K1) the shifted cotangent extension $\mathfrak{g} = \mathfrak{h} \ltimes D_{\mathrm{cont}}[1]$ is an object of $\mathbf{TateChain}^{\mathrm{adm}}_{\ge 0}$;

(K2) the CE coalgebra $C_*^{\mathrm{CE}}(\mathfrak{g})$ is conilpotent for the coaugmentation filtration;

(K3) the completed enveloping algebra $\widehat{U}(\mathfrak{g}) = \varprojlim_M \widehat{U}(\mathfrak{g}_{\le M})$ is complete and separated for the PBW filtration;

(K4) the finite-window PBW associated graded maps assemble to

$$\mathrm{gr}_{\mathrm{PBW}}\widehat{U}(\mathfrak{g}) \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{g}).$$

Then the closed-side formalism has two compatible outputs:

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) \xrightarrow{\Phi} \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big)$$

is the admissible completed CE/PV identity obtained from the finite-window family of Lemma 3.35 together with the exact continuous-dualization and variance data in (K1)–(K4), and

$$L\Omega C^{\mathrm{CE}}_{*}(\mathfrak{g}) \simeq \widehat{U}(\mathfrak{g}), \qquad R\,\mathrm{Bar}\,\widehat{U}(\mathfrak{g}) \simeq C^{\mathrm{CE}}_{*}(\mathfrak{g})$$

is the Tate Lie bar-cobar / CE Koszul duality of Corollary 3.48. The two outputs are compatible with the same finite-window transition maps.

Proof. The first output is Lemma 3.35 plus the exact continuous-dualization and variance hypotheses included above. The obstruction terms $\mathfrak{o}^{\mathrm{br}}$ and $\mathfrak{o}^{\mathrm{coad}}$ are the cochain CE/PV transition obstructions; $\mathfrak{o}^{\mathrm{Cas}}$ is the additional kernel/P_0-pairing obstruction. Their vanishing does not by itself make $\Phi = \varprojlim_M \Phi_M$ a cochain inverse limit; the admissible envelope supplies that extra datum.

For the Koszul output, hypotheses (K1)–(K4) put $C^{\mathrm{CE}}_{*}(\mathfrak{g})$ and $\widehat{U}(\mathfrak{g})$ in the admissible model envelope with complete filtrations. Conilpotency is the coalgebra hypothesis needed for Bar Ω, and PBW completeness identifies the associated graded of the enveloping algebra with the completed symmetric algebra. Corollary 3.48 therefore applies. Both constructions are obtained by first applying the finite-dimensional CE/PV and bar-cobar statements on $\mathfrak{g}_{\leq M}$, then passing through the same Mittag-Leffler inverse limit; hence the transition maps are the same. □

3.7.7 Application to the local component theorems

Theorem 3.50 *(Admissible-envelope module reconstruction from the CE/PV identity).* Let $\mathfrak{h}_{\mathrm{adm}}$ be a pro-finite-dimensional Lie algebra in **TateVec** whose shifted-cotangent extension $\mathfrak{g}_{\mathrm{adm}} = \mathfrak{h}_{\mathrm{adm}} \ltimes \mathfrak{h}^{\vee}_{\mathrm{adm,cont}}[1]$ satisfies the lower-central Tate-pronilpotent admissible-envelope hypotheses of 3.46. Assume further that:

(M1) the strict cochain map

$$\Phi\colon C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) \longrightarrow \mathrm{PV}\big(\widehat{S}_{\mathrm{Tate}}(\mathfrak{h}_{\mathrm{adm}})\big)$$

is the admissible CE/PV quasi-isomorphism obtained from the finite-window family of Lemma 3.35 after the required variance and exact continuous-dualization data are supplied;

(M2) the completed dg P_0-algebras on both sides lie in an admissible transferred P_0-algebra model structure inside $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$, with weak equivalences detected on underlying admissible filtered quasi-isomorphisms;

(M3) the module categories are formed in the same locally presentable monoidal envelope, and the two algebras are replaced by cofibrant-fibrant representatives when needed.

Then Φ induces an equivalence, inside that admissible Tate model envelope, of module ∞-categories over the two identified completed dg P_0-algebras

$$\widetilde{\Phi}\colon N_{\Delta}\big(C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}})\text{-mod}, W\big) \xrightarrow{\sim} N_{\Delta}\big(\mathrm{PV}\big(\widehat{S}_{\mathrm{Tate}}(\mathfrak{h}_{\mathrm{adm}})\big)\text{-mod}, W\big)$$

in the framework of Lurie's *Higher Algebra* §5.5. This is a reconstruction criterion from a strict CE/PV weak equivalence in an already fixed admissible P_0-algebra envelope. It is not a construction of a completed dg P_0 model structure and not a factorization-centre theorem for the full untruncated formal Hamiltonian algebra. The equivalence itself is the admissible-envelope algebraic one.

Proof. Assumption (M1) gives a strict admissible filtered quasi-isomorphism of completed dg P_0-algebras. The strictness is the generator-level identity $c \mapsto \theta,\ u \mapsto O$, together with $d_\pi \Phi = \Phi d_{\mathrm{CE}}$ and bracket compatibility from the finite-window CE/PV calculation.

By (M2), this strict weak equivalence is a weak equivalence in the transferred P_0-algebra model structure. Replace the two P_0-algebras by cofibrant-fibrant representatives if necessary. Extension and restriction of scalars along a weak equivalence of admissible algebra objects give a Quillen equivalence of module model categories; this is the standard Schwede–Shipley/Lurie mechanism for module categories over weakly equivalent algebra objects [39] and [37, Th. 7.1.4.6]. Applying Dwyer–Kan localization gives the displayed equivalence of module ∞-categories.

Functoriality in the Lie algebra gives interval-wise compatibility for admissible interval objects $\mathfrak{h}_{\mathrm{adm},I} = \Omega_c^\bullet(I) \otimes \mathfrak{h}_{\mathrm{adm}}$ when these interval objects remain in the same admissible envelope. This is interval-wise categorical compatibility, not an independent Weiss descent theorem and not an unreduced open-BV centrality theorem. □

Remark 3.51 *(Relation to Lurie's higher-algebraic Koszul duality).* Lurie [37, §5.5.5] constructs a higher-algebraic Koszul duality between augmented ∞-operad-algebras and their Koszul-dual coalgebras. For the operad $\mathrm{Lie}_{\mathbf{TateVec}}$, Koszul self-duality (with shift) gives the equivalence of 3.48 in the lower-central Tate-pronilpotent setting used here. The advantage of the model- categorical proof is concrete: weak equivalences are spelled out in terms of admissible filtered qiso of 3.217, not abstractly via the Hammock localization. The two constructions agree by Lurie A.3.7 plus the Hinich transfer of 3.43.

3.7.8 Compatibility with the descendant residual

The Tate Quillen equivalence of 3.46 operates on the bar-cobar duality between $\widehat{U}(\mathfrak{g})$ and $C_*^{\mathrm{CE}}(\mathfrak{g})$. It does not directly use the Casimir element $C_{\mathfrak{h}} = \sum_d e_{d,i} \otimes e_d^{i,\vee}$ of 3.217 or its convergence in the Tate completion.

Proposition 3.52 *(Bar-cobar conilpotence separated from Casimir convergence).* The Quillen equivalence of 3.46 and the resulting ∞-categorical equivalence of 3.50, when their admissible-envelope hypotheses hold, are independent of the convergence of the Tate Casimir element. The cochain complexes $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$ and $\mathrm{PV}\big(\widehat{S}_{\mathrm{Tate}}(\mathfrak{h})\big)$ are equivalent in the model category of 3.43 only after the same conilpotent admissibility hypotheses are imposed. Casimir convergence is not one of those hypotheses; conilpotence is.

Proof. The bar-cobar comparison uses $\widehat{S}^c(\mathfrak{g}[1])$ and $T(\mathfrak{g})$ as underlying graded objects. Both are well-defined and conilpotent in the admissible envelope under 3.217 items 1, 2 (continuous dual exact, conilpotent CE coalgebra). The formal Casimir $\sum_d e_{d,i} \otimes e_d^{i,\vee}$ is a kernel hypothesis only when it converges in the chosen coefficient category; in the strict product/direct-sum Tate endpoint it is precisely the obstruction isolated in Lemma 3.263. Such a kernel is required for the Costello [12] heat-kernel propagator construction at the analytic / factorization-algebra-with-non-strict-bracket level. It is *not* required for the algebraic bar-cobar comparison.

Therefore the cochain-level Φ promotes, in the admissible model envelope and only there, to an ∞-categorical equivalence. What remains outside this algebraic promotion is the Weiss-cosheaf upgrade and the heat-kernel quantum lift, flagged in 3.54 below. □

Proposition 3.53 *(No universal strict product/direct-sum kernel habitat).* Let $\mathcal{C}_{\mathrm{ker}}$ be the category whose objects are complete coefficient pairs (H, D), equipped with finite quotients $(H_{\leq M}, D_{\leq M})$, compatible transition maps, and a tensor $K \in H\widehat{\otimes}D$ whose image in $H_{\leq M} \otimes D_{\leq M}$ is the finite

Casimir C_M for every M. Morphisms preserve the finite quotients and carry K to K'. The original strict product/direct-sum pair

$$H_\Pi = \prod_{d\geq 1} H_d, \qquad D_\oplus = \bigoplus_{d\geq 1} H_d^\vee$$

is not an object of $\mathcal{C}_{\mathrm{ker}}$. More strongly, no object of $\mathcal{C}_{\mathrm{ker}}$ can map to the strict pair by a morphism inducing the identity on every finite window.

Hence the strict product/direct-sum pair has no initial, terminal, or representing universal property for kernel-admissible completions. A weighted product completion is a constructed regulator habitat, not a universal endpoint for tail-sensitive ordinary cohomology classes.

Proof. If the strict pair were an object, it would contain a tensor $K \in H_\Pi \widehat{\otimes} D_\oplus$ whose image in every finite window is C_M. If a kernel-admissible object mapped to the strict pair by a finite-window identity morphism, the image of its kernel would give the same forbidden tensor. But $D_\oplus$ has finite support in the degree direction, while the compatible family $(C_M)_M$ has a nonzero $H_d^\vee$-component for every d. This is exactly the support obstruction of Theorem 3.109. Therefore the strict pair is outside $\mathcal{C}_{\mathrm{ker}}$, and no universal property through it is available. □

Proposition 3.54 *(Residual Weiss-cosheaf condition).* The ∞-categorical equivalence of 3.50 on stalks and on the locally-constant interval-factorization $\mathcal{F}_{\mathrm{cl}}$ of 3.22 does not, by itself, give the Weiss-cosheaf condition for the *unreduced* factorization algebra $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}$ on $\mathbb{R}^2 \times \mathbb{C}^2$. The Weiss / descent condition for an unreduced factorization algebra on a higher-dimensional manifold requires checking Čech descent for Weiss covers, which is a separate cosheaf-theoretic statement. Costello-Gwilliam Vol. I §§6.1, 6.5, 7.2 provide the published factorization, cosheaf, and basis-extension vocabulary; no theorem from those sections is used here to close the unreduced descent obligation. The admissible Tate Quillen-equivalence promotion closes the cochain-level ∞-categorical promotion; the Weiss upgrade is recorded as the standing residual on the unreduced-factorization side of the local theorem.

Proof. Lurie [37, §5.5.4], with the same terminology as Costello-Gwilliam [15, Vol. I §6.4]: a locally constant factorization algebra on $\mathbb{R}$ determines an E_1-algebra up to the standard contractible choice; the Quillen equivalence of 3.46 realizes this. On a higher-dimensional manifold, the Weiss cover $\{U_\alpha\}$ has structure-map descent independent of the algebraic bar-cobar; in particular the heat-kernel propagator on $\mathbb{R}^2 \times \mathbb{C}^2$ enters the analytic check. Without that propagator the bar-cobar identity is consistent but does not trivialize Weiss descent. □

3.7.9 Summary diagram

$$\begin{array}{ccc}
\begin{array}{c}3.217\\ \text{(filtered qiso of cochain complexes)}\\ \downarrow \\ \text{cochain identity } \Phi\, d_{\mathrm{CE}} = d_\pi\, \Phi\end{array} & \xrightarrow{\text{admissible Tate promotion}} & \begin{array}{c}\text{Quillen equivalence}\\ \text{of model cats}\\ \downarrow \\ \text{equivalence of } \infty\text{-cats}\\ \downarrow \text{Lurie 5.5}\\ \text{equivalence in admissible } \infty\text{-cats of}\\ \text{completed dg } P_0\text{-algebras}\end{array}\\
 & \xrightarrow{\text{Lurie A.3.7}} &
\end{array}$$

The chain on the left is Φ as a strict cochain map. The chain on the right is the admissible Tate promotion: model-categorical, then ∞-categorical, then P_0-algebra-categorical. The promotion does not

depend on the Tate Casimir convergence (3.52); the Weiss-cosheaf descent on the unreduced factorization algebra is a separate residual (3.54).

COROLLARY 3.55 *(Moment-map shadow).* In the situation of 3.28, the degree-zero specialization of Φ_{coord} on the cotangent direction is the moment-map / Kirillov assignment

$$J : \mathfrak{h} \longrightarrow A_\partial = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}), \qquad J(f) = O_f.$$

The closed-side Lie bracket $\{f, g\}_{\mathfrak{h}}$ matches the Kirillov–Poisson bracket on A_∂:

$$\{J(f), J(g)\}_\partial = J\big(\{f, g\}_{\mathfrak{h}}\big).$$

Proof. J is the restriction of Φ_{coord} to the cotangent direction $u_I \mapsto O_I$ followed by linear extension along $\mathfrak{h} \hookrightarrow A_\partial$ (the canonical embedding of $\mathfrak{h}$ as the linear functions on $\mathfrak{h}^\vee$). Bracket compatibility is the (c^J, u_K)-versus-(θ^J, O_K) identity from 3.28 read off in degree zero. Equivalently, this is the standard Kirillov–Kostant–Souriau theorem for linear Poisson structures. □

COROLLARY 3.56 *(CE/PV generator identity, formal symplectic disk).* Let $\mathfrak{h} = \widehat{A}/\mathbb{C} \cdot 1 = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ be the reduced Hamiltonian Lie algebra of the formal symplectic disk on $\mathbb{C}^2$, with bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ modulo constants. Choose the Taylor basis $H_{a,b} = z_1^a z_2^b$ for $a, b \geq 0$, $a + b > 0$. Structure constants:

$$[H_{a,b}, H_{c,d}] = (ad - bc)\, H_{a+c-1, b+d-1},$$

with the convention that $H_{m,n} = 0$ if $\min(m, n) < 0$ or $m + n = 0$.

Set $A_\partial = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$, with linear coordinates $O_{a,b}$ on $\mathfrak{h}^\vee$ and Kirillov–Poisson bracket $\{O_{a,b}, O_{c,d}\}_\partial = (ad - bc) O_{a+c-1, b+d-1}$. The generator calculation underlying 3.28 gives the cochain-level coordinate CE/PV identity, read in the pro coordinate-window sense,

$$\Phi_{\mathrm{coord}} : C^\bullet_{\mathrm{CE,coord}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big), \qquad c^{a,b} \mapsto \theta^{a,b}, \quad u_{a,b} \mapsto O_{a,b}.$$

Its moment-map shadow is the boundary evaluation map $J(f) = O_f = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ of 3.242 after identifying A_∂ with the stable scalar-reduced Hamiltonian trace algebra of 3.210. The P_0 and factorization central-operation interpretation is imposed only after the coordinate-window or mixed/weighted continuity hypotheses in the theorem-data table; the strict product coordinate completion is not used as a global P_0-algebra.

Proof. The structure constants are an instance of 3.13 read on monomials. The CE/PV differential identity is checked directly on the monomial generators $c^{a,b}$, $u_{a,b}$ and then extended by the completed Leibniz rule; no lower-central Tate-pronilpotence of the full Hamiltonian algebra is used. The moment-map shadow follows from 3.55 together with 3.242. □

Example 3.57 *(Low-degree monomial verification).* We verify Φ_{coord} on small monomials of the formal symplectic disk.

- *Degree-1 generators.* Take $H_{1,0} = z_1$, $H_{0,1} = z_2$. The bracket $[H_{1,0}, H_{0,1}] = 1 \cdot 1 - 0 \cdot 0 = 1$ lifts to the constant $H_{0,0}$, which is set to zero in $\mathfrak{h}$. Hence $C^{0,0}_{(1,0),(0,1)} = 0$ in $\mathfrak{h}$. Consequently $d_{\mathrm{CE}} u_{0,0}$ is not present (no such generator); the $u_{0,0}$ degree of freedom does not appear, consistent with constants being removed.

- *Degree-2 calculation.* Take $K = (1,1)$. The differential of u_K uses outgoing brackets with H_K, not brackets landing in H_K:
$$d_{\mathrm{CE}} u_{1,1} = \sum_{c,d} (d-c)\, u_{c,d} c^{c,d}.$$
Hence the first low-degree terms are
$$d_{\mathrm{CE}} u_{1,1} = 2u_{0,2}c^{0,2} - 2u_{2,0}c^{2,0} - u_{1,0}c^{1,0} + u_{0,1}c^{0,1} + \cdots .$$
Each term $C^L_{(1,1),J}\, u_L\, c^J$ arises from $\{H_{1,1}, H_J\} = \sum_L C^L_{(1,1),J}\, H_L$. For example, $\{H_{1,1}, H_{1,0}\} = \partial_{z_1}(z_1 z_2)\cdot 0 - \partial_{z_2}(z_1 z_2)\cdot 1 = -z_1 = -H_{1,0}$, so $C^{1,0}_{(1,1),(1,0)} = -1$, contributing $-u_{1,0}\, c^{1,0}$ to $d_{\mathrm{CE}} u_{1,1}$. On the open/PV side,
$$d^{\mathrm{coord}}_{\pi} O_{1,1} = 2O_{0,2}\theta^{0,2} - 2O_{2,0}\theta^{2,0} - O_{1,0}\theta^{1,0} + O_{0,1}\theta^{0,1} + \cdots ,$$
the same expression with $u \to O$, $c \to \theta$. Hence $\Phi_{\mathrm{coord}}(d_{\mathrm{CE}} u_{1,1}) = d^{\mathrm{coord}}_{\pi}\Phi_{\mathrm{coord}}(u_{1,1})$. An independent exact arithmetic check verifies the same structure-constant input on every monomial pair with exponents up to 6 via the first-order Moyal coefficient $kn - lm$ at $\hbar^1$, which equals C^K_{IJ} for these indices.
- *Higher-order word.* For $K = (1,1)$, $J = (1,0)$, $J' = (0,1)$, the cochain element $u_{1,1}c^{1,0}c^{0,1}$ has degree $0+1+1 = 2$. Its CE differential, by graded Leibniz, is
$$d_{\mathrm{CE}}\big(u_{1,1}c^{1,0}c^{0,1}\big) = (d_{\mathrm{CE}} u_{1,1})c^{1,0}c^{0,1} - u_{1,1}(d_{\mathrm{CE}} c^{1,0})c^{0,1} + u_{1,1}c^{1,0}(d_{\mathrm{CE}} c^{0,1}),$$
with each summand applying the generator formulas above. The same word on the PV side is $O_{1,1}\theta^{1,0}\theta^{0,1}$, and d^{coord}_{π} applied with the same Leibniz convention produces the corresponding triple sum with $u \to O$ and $c \to \theta$. Term-by-term equality follows from the generator identities; the full word equality is the algebra-homomorphism Leibniz argument from 3.28.

Construction 3.58 *(Compact current PBW/Rees source).* Let $\mathcal{P}$ denote the principal-part module
$$\mathcal{P} = \bigoplus_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\,\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$
identified with $\mathfrak{h}^\vee_{\mathrm{cont}}$ by the residue pairing of Proposition 3.227. Put
$$\mathfrak{g}_{\mathrm{pp}} = \mathfrak{h} \ltimes \mathcal{P}[1],$$
with zero differential and bracket
$$[f,g] = \{f,g\}, \qquad [f,\rho[1]] = (f\cdot\rho)[1], \qquad [\rho[1],\sigma[1]] = 0,$$
where $f\cdot\rho = \pi_{\mathrm{pp}}\{f,\rho\}$. On monomials this action is
$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$
with negative-index targets zero.

For an interval $I \subset \mathbb{R}$, define the compactly supported current Lie algebra
$$\mathfrak{J}_{\mathfrak{g}}(I) = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{g}_{\mathrm{pp}}, \qquad [\alpha\otimes x, \beta\otimes y] = \alpha\beta\otimes[x,y].$$

Its PBW/Rees algebra is

$$U_\hbar(\mathfrak{J}_\mathfrak{g}(I)) = \widehat{T}(\mathfrak{J}_\mathfrak{g}(I))[[\hbar]] / \langle xy - (-1)^{|x||y|} yx - \hbar[x, y] \rangle.$$

The PBW filtration is the tensor-word filtration. Its special fibre is

$$U_\hbar(\mathfrak{J}_\mathfrak{g}(I))/(\hbar) \cong \widehat{\mathbf{S}}(\mathfrak{J}_\mathfrak{g}(I)),$$

equipped with the semiclassical P_0 bracket

$$\{\bar{X}, \bar{Y}\}_{P_0} = \left(\hbar^{-1}(XY - (-1)^{|X||Y|} YX)\right) \bmod \hbar.$$

This is the PBW/Rees source for a Layer 4 comparison involving Hamiltonian and principal-part modes. It is not a closed CE cochain algebra map; the CE comparison must be made separately by bar-cobar twisting or by a completed pairing.

Problem 3.59 *(PBW-source P_0-centre action criterion and CE comparison datum).* Construct a filtered factorization-center action of the compactly supported current PBW/Rees source $U_\hbar(\mathfrak{J}_\mathfrak{g})$ of Construction 3.58. The target cannot be the polynomial open observable algebra alone if the cotangent generators are included; it must use the enlarged principal-part boundary object of Theorem 3.237. In the classical special fibre the action must send

$$f \longmapsto B_f = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}, \qquad \rho_{a,b}[1] \longmapsto \Theta_{a,b},$$

and satisfy the stalkwise bracket identities

$$\{B_f, B_g\}_{\text{open}} = B_{\{f,g\}}, \qquad \{B_f, \Theta_\rho\}_{\text{open}} = \Theta_{f \cdot \rho}, \qquad \{\Theta_\rho, \Theta_\sigma\}_{\text{open}} = 0.$$

This special-fibre statement is the natural P_0 theorem. A strict action of the full filtered Rees algebra on the same commutative reduced target is not a consequence of these identities: in the source

$$XY - (-1)^{|X||Y|} YX = \hbar[X, Y],$$

while multiplication operators on a commutative classical target commute. Thus the Rees-level theorem requires either a deformed boundary target or an explicit homotopical quantization of the central operations. Consequently the compact current PBW/Rees source supplies a P_0-central-operation action only after passing to the classical special fibre, or after replacing the boundary target by a genuinely deformed/homotopical central-operation object. The primitive one-ψ projection is only an indecomposable label shadow and is not the filtered Rees action datum. The target data must include differential compatibility, factorization centrality homotopies, coherent P_0 bracket homotopies, and compact-support naturality for interval inclusions and disjoint unions before the de Rham contraction.

Separately compare this PBW/Koszul-dual source with the closed CE observable algebra by an explicit twisting or pairing construction. Without this comparison, the result is a PBW/Koszul-dual boundary action, not a cochain-level map from closed CE observables.

Problem 3.60 *(Unreduced Tate-coefficient cotangent lift datum).* Construct a Tate or weighted-coefficient extension of the BV heat-kernel calculus on $\mathbb{R}^2 \times \mathbb{C}^2$ in which:

(i) the formal Casimir $C_\mathfrak{h} = \sum_{d \geq 1} \sum_i e_{d,i} \otimes e_d^i$ converges as an element of $\widehat{\mathfrak{h}} \widehat{\otimes} \widehat{\mathfrak{h}}^\vee_{\text{cont}}$ this is proved by 3.85(a) via the weighted Tate completion of 3.74;

(ii) the Costello propagator $P_{\epsilon,L}$ extends to that coefficient category this is proved by 3.85(b);

(iii) the primitive indecomposable shadow of the principal-part current sector

$$\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \,\widehat{\otimes}\, \widehat{\mathbf{S}}\big(\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I)\big)$$

admits a chain-level primitive projection π_{prim} with verified Q-compatibility and factorization compatibility on the indecomposable quotient the primitive shadow is supplied by 3.154; the cotangent centrality homotopies remain part of item (4);

(iv) the reduced principal-part defect-current theorem of 2.70 lifts to the unreduced factorization-algebra model and the degree-zero Moyal/Weyl theorem extends to the cotangent (one-antifield) sector at all orders in $\hbar$ after the primitive shadow of 3.161; unreduced cotangent centrality and the one-antifield quantum lift require vanishing of the QME obstruction class of 3.87; the truncated nilpotent sector $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ closed classically by 3.122 and obstructed quantum-mechanically by 3.123.

Equivalently, lift the cochain-level 3.28 and the Hamiltonian-sector factorization-algebra 3.22 to a cochain-level morphism of factorization algebras

$$\Phi_{\mathrm{cl}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}\big(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}\big)$$

on the unreduced (cotangent-included) sector, with all P_0-center homotopies, factorization centrality homotopies, the stable connected quotient, and a boundary realization of the shifted cotangent generators. Any global version must also pass through the local-Hamiltonian descent and period obstruction of 2.25; a stalkwise formal disk map is not a Weiss/Ran descent theorem for factorization centers.

Four mathematical entries separate the residual (§3.8): the Costello–Li 2012 BCOV/HCS theorem is a negative source-comparison and does not transfer to the present mixed defect geometry; weighted Tate completion, which repairs the coefficient kernel and finite-scale propagator and gives graphwise RG control; nilpotent truncation to $\mathfrak{m}^3/\mathfrak{m}^{N+1}$, classical only, with an explicit Moyal obstruction at all orders; and direct chain-level primitive projection, closing the classical cosheaf and factorization compatibilities of the indecomposable shadow without analytic input.

3.8 Obstruction calculus for the unreduced Tate-coefficient cotangent lift

The mechanisms in 3.8 analyzing 3.60 are separated below into proved mechanisms and the residual obstruction. The Costello–Li 2012 BCOV/HCS theorem is recorded as a negative applicability check: it does not transfer to the geometry studied here, and the failure mode forces the replacement mechanisms below. The weighted Tate completion closes the Casimir convergence and the finite-scale heat-kernel propagator extension. The nilpotent truncation closes a conformal-symplectic-quotient sector classically and exhibits the all-order Moyal obstruction. The chain-level primitive projection closes the cosheaf and factorization compatibilities of the indecomposable shadow without analytic input. The weighted Hadamard/Mittag-Leffler theorem below reduces the remaining Costello-locality compatibility of 3.87 to the mixed brane-defect QME obstruction class.

BV-cohomology vanishing and the unreduced cotangent lift

Costello–Li 2012 hypotheses, restated

Statement 3.61 *(Costello–Li 2012 BCOV package, restated).* Let X be a compact Calabi–Yau manifold of complex dimension d, with holomorphic volume form Ω. Let $\mathrm{PV}(X)[[t]]$ be the polyvector field

complex with the Tate parameter t (the worldsheet S^1 rotation), with differential $Q = \bar{\partial} + t\,\partial_\Omega$ and (-1)-shifted symplectic pairing

$$\langle \alpha, \beta \rangle = \int_X (\alpha \wedge \beta)\big|_{\mathrm{PV}^{d,d}} \cdot \Omega^2.$$

The classical BCOV interaction I_{BCOV} is the cubic vertex on $\mathrm{PV}(X)[[t]]$ supplied by [13]. The source package supplies the generalized classical BCOV complex, its local-functional obstruction complex, and the renormalization framework in which quantization obstructions are stated. The quantization vanishing theorems used as primary-source precedent here are only those with their own stated hypotheses. No general all-dimensional compact-Calabi–Yau theorem is imported saying that every higher-loop obstruction vanishes in the mixed brane-defect sector.

The 2015 sequel [14] extends to open-closed BCOV with open gauge data $\mathfrak{gl}(N \mid N)$, anomaly cancellation by the supertrace, on flat $\mathbb{C}^d$, d odd. That is the precedent for defects in the Costello–Li line, but on $\mathbb{C}^d$, not on $\mathbb{R}^2 \times \mathbb{C}^2$ with topological factor.

Applicability check on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ and $\mathfrak{gl}_N$

PROPOSITION 3.62 *(Costello–Li 2012 hypothesis separation for the local model).* The hypotheses of 3.61 differ from the formal local model on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ and gauge data $\mathfrak{gl}_N$ along five independent axes.

(S1) *Compact-Hodge input.* The 2012 compact Hodge-theoretic input does not apply directly to the formal $\mathbb{C}^2$ jet model. The 2015 theorem shows that flat non-compact geometry can be admissible under its own odd-dimensional $\mathfrak{gl}(N \mid N)$ hypotheses; those are not the mixed topological-holomorphic defect hypotheses used here.

(S2) *Topological–holomorphic mixing.* The local theory has a topological factor $\mathbb{R}^2$ (de Rham) crossed with a holomorphic factor $\mathbb{C}^2$ (Dolbeault). Costello–Li 2012 is pure BCOV/HCS on a single holomorphic Calabi–Yau; there is no pure BCOV statement covering the mixed kinematics. The protected sector of twisted M-theory [25] has the right shape as motivation, but no theorem from that source is used here without a separate local anchor, and Costello–Li 2012 is not stated for it.

(S3) *Brane defect.* The brane L is one-real-dimensional in the topological direction and point-supported in the holomorphic direction. Costello–Li 2012 has no brane sector. Costello–Li 2015 [14] has a brane sector but on flat $\mathbb{C}^d$ (not $\mathbb{R}^2 \times \mathbb{C}^2$) with $\mathfrak{gl}(N \mid N)$ open gauge data.

(S4) *Gauge data.* Ordinary $\mathfrak{gl}_N$ is not $\mathfrak{gl}(N \mid N)$. Costello–Li 2015 cancel the U(1) center anomaly by the supertrace; this cancellation is unavailable for ordinary $\mathfrak{gl}_N$. The U(1) center class $[\bar{c}]$ in part (f) of 2.70 is the local trace-line avatar of that Costello–Li anomaly, not a graph-level identification with it.

(S5) *Tate direction.* Costello–Li's t-Tate completion is along the worldsheet S^1 rotation, not along the coefficient system. The Tate completion required by 3.60 is along the *coefficient* direction $\widehat{\mathfrak{h}} \widehat{\otimes} \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$ in which the Casimir $C_{\mathfrak{h}} = \sum_{d \geq 1} \sum_i e_{d,i} \otimes e^i_d$ must converge. By 3.263, the direct-sum dual fixed in the parent calculation obstructs convergence, and the coefficient category itself must change. Costello–Li provide no convergence theorem in this direction.

Hypothesis (S5) is the coefficient endpoint obstruction: even if (S1)–(S4) were supplied by an Omega-background or twisted M-theory extension, the coefficient-system Tate completion would still be unprovided.

Proof. Each clause is a direct comparison of the hypotheses of 3.61 (and its 2015 sequel) with the data of 2.70 and the topology declared in the linear-heat / coefficient-Casimir analysis (3.262, 3.263). The non-compactness of $\mathbb{C}^2$ blocks the harmonic-projection step in Costello–Li's one-loop anomaly cancellation; the topological factor $\mathbb{R}^2$ is not BCOV kinematics; the brane L is a topological line transverse to a holomorphic point, not an open boundary on a flat $\mathbb{C}^d$; ordinary $\mathfrak{gl}_N$ lacks the Costello–Li supertrace cancellation (whose local trace-line avatar here is the class $[\bar{c}]$ of part (f)); and the worldsheet-t Tate completion is orthogonal to the coefficient-Casimir Tate direction. □

Corollary 3.63 *(Costello–Li input and the unreduced coefficient endpoint).* The Costello–Li 2012 BCOV package supplies a pure closed BCOV template. The unreduced Tate-coefficient cotangent lift for the present brane-defect model additionally requires the endpoint coefficient data of Definition 3.110. The independent chain-level route 3.154 closes the classical primitive indecomposable shadow without analytic input. The unreduced cotangent factorization-centre morphism and the quantum one-antifield extension of part (e) require the weighted coefficient category and the mixed brane-defect locality/QME input isolated in Problem 3.87 and 3.166.

Proof. By Proposition 3.62, the BCOV package of [13] is stated for different geometric and coefficient data. Its conclusion, in the form used here as source precedent, contains neither the coefficient-system Tate completion in which the Casimir converges nor a brane-defect extension of the propagator. The closures cited in the statement supply the classical chain-level lift and the weighted finite-scale coefficient kernel by other routes. The endpoint data remain exactly those listed in Definition 3.110. □

Theorem 3.64 *(Costello–Li endpoint coefficient criterion).* The Costello–Li 2012 BCOV package [13] and its 2015 open-closed extension [14], in the forms cited above, can yield an endpoint-admissible coefficient module for Definition 3.110 only after three independent data are added:

(E2) coefficient Casimir kernel,

(E3) mixed brane-defect transport,

(E5) endpoint RG preservation.

Here (E1) and (E4) are the inherited finite-window and Mittag–Leffler envelope conditions of Definition 3.110. A quantum endpoint additionally requires vanishing of the separate QME component $\mathfrak{o}_{\mathrm{QME}}$ in Definition 3.65.

Proof. An endpoint-admissible coefficient module requires a coefficient kernel $C_* \in \mathfrak{h}_* \widehat{\otimes} D_*$ whose projection to every finite Hamiltonian window is the finite Casimir, together with mixed brane-defect transport of the bracket, propagator, interaction, counterterms, and RG recursion. By Proposition 3.62, the cited Costello–Li theorems are pure closed BCOV statements on the polyvector t-Tate direction, with a separate $\mathfrak{gl}(N \mid N)$ open-closed setting in the sequel. Their stated data are orthogonal to the coefficient pair $\widehat{\mathfrak{h}} \widehat{\otimes} \widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}$, its Casimir kernel, and the topological line defect with ordinary $\mathfrak{gl}_N$ boundary observables. Thus the endpoint conditions (E2), (E3), and (E5) of Definition 3.110 are not consequences of the cited theorems. □

Definition 3.65 *(BV endpoint obstruction tuple).* For the mixed local geometry $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$, the obstruction tuple for a BV-cohomology vanishing theorem to construct an endpoint-admissible coefficient module is

$$\mathfrak{o}_{\mathrm{BV,end}} = \left(\mathfrak{o}_{\mathrm{Cas}}, \mathfrak{o}_{\mathrm{def}}, \mathfrak{o}_{\mathrm{RG}}, \mathfrak{o}_{\mathrm{QME}}\right).$$

Its components are:

(O1) $\mathfrak{o}_{\mathrm{Cas}}$: the class of the compatible finite Casimirs $(C_{\mathfrak{h}}^{M})_{M\geq 1}$ in the quotient by actual coefficient kernels $\mathfrak{h}_{*}\widehat{\otimes}D_{*}$;

(O2) $\mathfrak{o}_{\mathrm{def}}$: the obstruction to extending the bulk propagator and bracket transport to the topological line defect with ordinary $\mathfrak{gl}_N$ open observables;

(O3) $\mathfrak{o}_{\mathrm{RG}}$: the obstruction to preserving the same strict finite-window Mittag-Leffler tower under Costello's recursion $W(P_{\epsilon,L}, I)$;

(O4) $\mathfrak{o}_{\mathrm{QME}}$: the mixed brane-defect local functional class in

$$H^{1}\big(O_{\mathrm{loc}}(\mathcal{E}_{*}), Q + \{I_{*}, -\}\big)$$

produced by the quantum master equation at the endpoint.

PROPOSITION 3.66 *(Endpoint construction is equivalent to vanishing of the tuple).* A Costello–Li type BV-cohomology theorem closes the coefficient endpoint for this local brane-defect problem only if all four components of $\mathfrak{o}_{\mathrm{BV,end}}$ vanish and a compatible choice of primitives is made. Under such a choice, the endpoint satisfies Definition 3.110; conversely an endpoint-admissible coefficient module kills $\mathfrak{o}_{\mathrm{Cas}}$, $\mathfrak{o}_{\mathrm{def}}$, and $\mathfrak{o}_{\mathrm{RG}}$, and the QME solution kills $\mathfrak{o}_{\mathrm{QME}}$.

Proof. Definition 3.110(E2) is precisely the existence of an actual coefficient kernel whose finite-window projections are the $C_{\mathfrak{h}}^{M}$; this is the vanishing of $\mathfrak{o}_{\mathrm{Cas}}$. Condition (E3) requires transport of bracket, propagator contraction, interaction, and counterterm representatives through the brane-defect geometry; this is exactly $\mathfrak{o}_{\mathrm{def}} = 0$ together with the chosen transport primitives. Condition (E5) is $\mathfrak{o}_{\mathrm{RG}} = 0$. The endpoint QME is the vanishing of the displayed degree-one local-functional class $\mathfrak{o}_{\mathrm{QME}}$. These conditions are also necessary: a nonzero component removes one of the endpoint data just listed. □

Costello–Li extension criterion

THEOREM 3.67 *(Costello–Li extension criterion for the mixed coefficient endpoint).* Any Costello–Li type theorem that proves an endpoint-admissible coefficient module for the setup here must supply the following four components simultaneously:

(i) *Non-compact / Omega-background BV-cohomology vanishing.* A version of 3.61 on $\mathbb{R}^2 \times \mathbb{C}^2$, either via a separately anchored Omega-background route or via a compactification with Lagrangian boundary at infinity.

(ii) *Topological–holomorphic mixed kinematics.* The vanishing must hold on the kinematic differential $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, with the linearized complex of 3.262, and on the Hamiltonian sector with cotangent [1] extension.

(iii) *Brane-defect extension.* The vanishing must allow the transverse one-real-dimensional defect L on which open gauge data $\mathfrak{gl}_N$ live; not the flat-$\mathbb{C}^d$ $\mathfrak{gl}(N \mid N)$ setup of [14]. It must also decide the scalar contact class of Theorem E.12: closed exchange cancellation, supertrace removal, central-character normalization, or renormalized boundary generators.

(iv) *Coefficient-system Tate completion.* The coefficient direction $\widehat{\mathfrak{h}}\,\widehat{\otimes}\,\widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}$ must support the Casimir as a convergent kernel. By 3.263 this is a category-level change, not a damping factor.

These four inputs are the coefficient package of 3.60. They are the necessary data for turning the Costello–Li convention template into the mixed brane-defect coefficient theorem used here.

Proof. Component (1) supplies the compactness/Hodge-theoretic replacement needed in (S1). Component (2) supplies the mixed topological–holomorphic complex required by (S2). Component (3) supplies the line-defect transport and ordinary $\mathfrak{gl}_N$ scalar anomaly decision required by (S3)–(S4). Component (4) is precisely condition (E2) of Definition 3.110; together with the transport and RG requirements in (E3) and (E5), it is the coefficient endpoint datum. Proposition 3.66 identifies these requirements with vanishing of the obstruction tuple. □

Remark 3.68 *(Matched-conventions boundary).* The local Hamiltonian BF sector here supplies the formal-disk coefficient normalization. Compact Calabi–Yau outputs of a Φ_d functor, κ_{ch} or κ_{BKM} consequences, compact CY_3 BCOV anomaly cancellation, the Vol III CY-to-chiral frontier, and the Igusa Δ_5 / Borcherds–BKM frontier require their own matched-conventions data. Costello–Li 2012 supplies the convention template used to state the coefficient problem for the unreduced cotangent lift.

Weighted Tate completion of the coefficient pair

Theorem lane. The weighted coefficient category carries the Casimir, finite-window Mittag-Leffler, heat-kernel, and classical cochain statements below. Renormalization-flow compatibility is graphwise at finite scale for polynomial-degree-bounded vertices, and the passage to an all-order QME solution is exactly the acceptance gate 3.93. For $q, q' > 1$, admissible exponential weights define canonically equivalent finite-degree local observables on the same strict Mittag-Leffler tower. The symbol $q \to 1^+$ means finite-window stabilization after unweighting, and an actual endpoint exists precisely under Definition 3.110. The original product/direct-sum Tate pair carries the endpoint obstruction of Theorem 3.109.

Diagnosis

Remark 3.69 *(Strict Tate Casimir obstruction).* In the parent manuscript the Hamiltonian Lie algebra is the product/direct-sum Tate pair

$$\mathfrak{h} = \prod_{d\geq 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d\geq 1} H_d^\vee, \tag{6}$$

with $H_d = \mathrm{Span}\{z_1^a z_2^b : a + b = d\}$ and Hamiltonian bracket satisfying $\{H_p, H_q\} \subset H_{p+q-2}$. By Lemma 3.217 item (1), the continuous dual of a Tate product is a strict direct sum, and *vice versa*. The formal Casimir

$$C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_i e_{d,i} \otimes e_d^i$$

has a nonzero summand in every degree d, so its second tensor factor lies in $\prod_d H_d^\vee$, not in $\bigoplus_d H_d^\vee = \mathfrak{h}^\vee_{\mathrm{cont}}$. Thus $C_{\mathfrak{h}} \notin \mathfrak{h}\widehat{\otimes}\mathfrak{h}^\vee_{\mathrm{cont}}$. This is Lemma 3.263 of the parent.

The obstruction is categorical, not analytic: no rescaling of the bases $\{e_{d,i}\}$ or $\{e_d^i\}$ *within* the strict pair (6) can move the Casimir from the product $\prod_d (H_d \otimes H_d^\vee)$ to the strict sum-with-product $(\prod_d H_d)\widehat{\otimes}(\bigoplus_d H_d^\vee)$. The cure must change the coefficient *category*: the second factor must itself be a product, with the change of dualization inserted by an intermediate weight system that is itself a Mittag-Leffler inverse-limit object.

Remark 3.70 *(Costello's coefficient requirements).* In Costello's heat-kernel BV calculus, the BV propagator $P_{\epsilon,L}$ on a free elliptic complex $(\mathcal{E}, Q, \langle\,,\,\rangle)$ is the kernel of the operator $\int_\epsilon^L Q^{\mathrm{GF}} e^{-tD} dt$ in $\mathcal{E}\widehat{\otimes}\mathcal{E}$ [12, Ch. 5]. When the field complex factors as

$$\mathcal{E} = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big),$$

with the Tate coefficient pair acting passively on the spacetime, the propagator factors as

$$P_{\epsilon,L} = P^{\text{base}}_{\epsilon,L} \widehat{\otimes} C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1]}, \tag{7}$$

where the spacetime piece $P^{\text{base}}_{\epsilon,L}$ on $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$ is the standard heat-kernel BV inverse on the bulk (Costello, op. cit., Ch. 5), and the coefficient piece is the BV inverse pairing on $\widehat{\mathfrak{h}} \oplus \widehat{\mathfrak{h}}^\vee_{\text{cont}}[1]$. By Remark 3.262, the spacetime piece exists as an actual continuous operator. The whole kernel exists as an element of $\mathcal{E}\widehat{\otimes}\mathcal{E}$ *iff* the coefficient piece, the bare Casimir $C_{\mathfrak{h}}$ together with its dual coevaluation, exists as a coefficient kernel.

Costello's heat-kernel BV requires three properties of the coefficient module:

(C1) an actual distributional inverse pairing, i.e. a coefficient coevaluation $C_{\mathfrak{h}} \in \mathfrak{h}\widehat{\otimes}\mathfrak{h}^\vee_{\text{cont}}$ realizing the bilinear form with kernel zero in the cohomology of the heat operator;

(C2) regularity in the cutoff parameters ϵ and L, in the sense that the heat-kernel propagator $P_{\epsilon,L}$ is a continuous family in the coefficient topology and converges as $L \to \infty, \epsilon \to 0^+$ to a distributional kernel after counterterm subtraction;

(C3) compatibility with the Costello RG flow $W(P_{\epsilon,L}, I)$: the weight or topology defining the coefficient module is preserved by the renormalization flow on effective interactions induced by varying (ϵ, L), in the precise sense that the source of the flow lies in the same completed coefficient category as its target.

The strict Tate pair (6) has the obstruction to (C1); the subsection below constructs the coefficient kernel and the finite-scale propagator by a weighted Mittag-Leffler completion. Requirement (C3) is not a formal consequence of the coefficient change; it is the residual locality/QME problem isolated in Problem 3.87.

Weighted Tate completion: definition

Definition 3.71 *(Polynomial Hamiltonian degree weight).* A *degree weight* is a function $w\colon \mathbb{Z}_{\geq 1} \to \mathbb{R}_{>0}$ satisfying:

(W1) *Bracket-tame:* there exists a constant $C_w \geq 0$ such that $w(p+q-2) \leq C_w\, w(p)w(q)$ for every $p, q \geq 2$;

(W2) *Windowwise coadjoint-continuity:* for every Hamiltonian degree bound R and every output Tate window w_0, the principal-part coadjoint shifts $\rho_{a,b} \mapsto \rho_{a-p+1,b-q+1}$ with $p+q \leq R$ and $a+b-p-q+2 \leq w_0$ involve only the finite input window $a+b \leq w_0 + R - 2$, and the finitely many corresponding weight ratios have a maximum;

(W3) *Mittag-Leffler coevaluation:* for each w_0 the finite-window Casimir $\sum_{d\leq w_0}\sum_i w(d)e_{d,i} \otimes w(d)^{-1}e^i_d$ is compatible under the surjective truncation maps $w_0' \to w_0$. Equivalently, the coefficient kernel is a strict Mittag-Leffler inverse system of finite-dimensional kernels.

LEMMA 3.72 *(Existence of weights).* Every exponential weight $w(d) = q^d$ with $q > 1$ satisfies (W1)–(W3). The proof uses only finite-window maxima for the coadjoint action; there is no uniform bound in the unbounded principal-part degree.

Proof. For $w(d) = q^d$, (W1) reads $q^{p+q-2} \leq C_w q^{p+q}$, i.e. $C_w = q^{-2}$ suffices. For (W2), fix R and w_0. If $p+q \leq R$ and the output degree is $a+b-p-q+2 \leq w_0$, then $a+b \leq w_0 + R - 2$. Hence the set of possible triples $(p, q, a+b)$ is finite, and the corresponding ratios of powers of q admit a finite maximum. No bound independent of $a+b$ is asserted; a uniform inequality without this finite-window

restriction is false. Condition (W3) is the strict Mittag-Leffler compatibility of finite-window Casimirs under truncation. The transition maps are surjective, hence the system is Mittag-Leffler. □

Remark 3.73 *(Choice of exponential weight).* The exponential weight $w(d) = q^d$ is the working choice, with $q > 1$ formally an element of $\mathbb{R}_{>0}$ and concretely set to $q = 2$ for the proofs below. The choice is a grading weight on the coefficient module, not a symplectic automorphism of the formal disk. Thus the Poisson bracket and the counterterm representatives are compared between two weights only after transport by the diagonal map $R_{w,w'}$ below. In coordinates, the bracket constants in the basis $w(d)e_{d,i}$ are conjugated by the finite ratios $w(p)w(q)/w(p+q-2)$, and condition (W1) is exactly the boundedness condition needed for this transported bracket on each finite graph. The weight is therefore a controlled coefficient regulator, not an assertion that rescaling the variables z_1, z_2 preserves the original Hamiltonian bracket.

Definition 3.74 *(Weighted Tate coefficient pair).* Given a degree weight w as in Definition 3.71, write $H_d^{(w)}$ for the rescaled copy of the finite-dimensional degree piece H_d with basis $w(d)e_{d,i}$, and $(H_d^\vee)^{(w^{-1})}$ for the rescaled dual basis $w(d)^{-1}e_d^i$. Define

$$\mathfrak{h}_w = \varprojlim_{w_0 \in \mathbb{Z}_{\geq 1}} \prod_{d=1}^{w_0} H_d^{(w)}$$

with the Tate filtration $F^{w_0}\mathfrak{h}_w = \ker(\mathfrak{h}_w \to \prod_{d \leq w_0} H_d^{(w)})$, and define the *weighted coefficient dual*

$$\mathfrak{h}_{\mathrm{cont}}^{\vee,w} = \varprojlim_{w_0 \in \mathbb{Z}_{\geq 1}} \prod_{d=1}^{w_0} (H_d^\vee)^{(w^{-1})}.$$

The pairing is fixed degreewise by $\langle w(d)^{-1}e_d^i, w(d)e_{d,j}\rangle = \delta_j^i$. The weighted Tate coefficient pair is $(\mathfrak{h}_w, \mathfrak{h}_{\mathrm{cont}}^{\vee,w})$. The second factor is a product coefficient module, not the strict continuous dual of the first product Tate space.

Remark 3.75 *(Topological content).* Both $\mathfrak{h}_w$ and $\mathfrak{h}_{\mathrm{cont}}^{\vee,w}$ are Mittag-Leffler inverse limits of finite-dimensional Tate windows, and both are products in the degree direction. The two are no longer in strict Tate duality (one product, one direct sum); the point is precisely to replace the direct-sum dual by a symmetric Mittag-Leffler product coefficient module. This is an additional coefficient category for the BV kernel. It does not identify $\mathfrak{h}_{\mathrm{cont}}^{\vee,w}$ with the ordinary continuous dual of $\mathfrak{h}_w$.

Proposition 3.76 *(Casimir convergence).* In the weighted Tate coefficient pair $(\mathfrak{h}_w, \mathfrak{h}_{\mathrm{cont}}^{\vee,w})$ of Definition 3.74, the formal Casimir

$$C_{\mathfrak{h},w} = \sum_{d \geq 1} \sum_i \left(w(d)e_{d,i}\right) \otimes \left(w(d)^{-1}e_d^i\right) \tag{8}$$

converges as an element of $\mathfrak{h}_w \widehat{\otimes} \mathfrak{h}_{\mathrm{cont}}^{\vee,w}$. The pairing of $C_{\mathfrak{h},w}$ with any test functional in the dual is the identity on the corresponding finite-dimensional Tate window, so $C_{\mathfrak{h},w}$ is the BV coefficient inverse pairing in the weighted category.

Proof. The displayed Casimir, restricted to the Tate window $d \leq w_0$, is the finite sum $\sum_{d \leq w_0} \sum_i (w(d)e_{d,i}) \otimes (w(d)^{-1}e_d^i)$, an element of $\prod_{d \leq w_0} w(d)H_d \otimes \prod_{d \leq w_0} w(d)^{-1}H_d^\vee$. This is a tensor in a finite-dimensional vector space because each window is finite-dimensional. The

tower of finite-window Casimirs is compatible: enlarging w_0 extends the sum by the terms in degrees $w_0 + 1, \dots, w_0'$. The inverse limit in **TateVec** exists by Definition 3.74 and (W3) of Definition 3.71.

Pairing the Casimir with a test functional $\phi \in \mathfrak{h}^{\vee,w}_{\mathrm{cont}}$, supported in finitely many Tate windows, gives a finite sum of identity pairings, so the result is the canonical evaluation. Hence $C_{\mathfrak{h},w}$ realizes the inverse pairing on the weighted Tate pair, satisfying Costello requirement (C1). □

Remark 3.77 (Independence of basis). Definition (8) is written in a basis $\{e_{d,i}\}$ of H_d, but does not depend on the choice. The dual basis $\{e^i_d\}$ is fixed by $\langle e_{d,i}, e^j_d\rangle = \delta^j_i$. An orthogonal change of basis $e_{d,i} \mapsto \sum_j O^j_i e_{d,j}$ with $O \in GL(H_d)$ shifts the dual by the inverse transpose, yielding the same Casimir. Thus $C_{\mathfrak{h},w} = \sum_d \mathrm{id}_{H_d \otimes H_d^\vee}$ is the basis-independent identity element of $\bigoplus_d H_d \otimes H_d^\vee$ extended to the weighted product.

Heat-kernel propagator on the weighted pair

PROPOSITION 3.78 *(Heat-kernel propagator extension to the weighted pair).* Let w be a degree weight as in Definition 3.71. The Costello heat-kernel propagator $P^{\mathrm{op}}_{\epsilon,L}$ of Remark 3.262 extends from the strict Tate field complex to the weighted field complex

$$\mathcal{E}_w = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\mathfrak{h}_w[1] \oplus \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[2]\big), \tag{9}$$

to give a continuous family of BV propagators $P^w_{\epsilon,L} \in \mathcal{E}_w\widehat{\otimes}\mathcal{E}_w$ for $0 < \epsilon \le L < \infty$, satisfying:

(1) the BV identity $[Q, P^w_{\epsilon,L}] = K^w_\epsilon - K^w_L$, where $K^w_t = K^{\mathrm{base}}_t\widehat{\otimes}C_{\mathfrak{h},w}$ is the weighted heat kernel of the BV inverse pairing;

(2) Costello regularity (C2): the family is continuous in (ϵ, L) in the weighted Tate topology;

(3) the limit $L \to \infty$, $\epsilon \to 0^+$ exists as a distributional kernel after the standard spacetime counterterm subtraction in Costello's heat-kernel RG package [12, Chs. 5–7]; the coefficient direction introduces no additional spacetime singular support. Locality and QME counterterm compatibility for the mixed brane-defect theory remain the content of Problem 3.87.

Proof. By the factorization (7), the propagator splits as a tensor product of the spacetime kernel $P^{\mathrm{base}}_{\epsilon,L}$ on $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$ and the coefficient Casimir $C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}$ on the weighted Tate pair. The spacetime piece is unchanged by the weight on the coefficient direction; it is the Costello heat-kernel inverse on the bulk free elliptic complex $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$, constructed in [12, Ch. 5]. The BV identity for $P^{\mathrm{base}}_{\epsilon,L}$ is standard: with $Q^{\mathrm{base}} = d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}$, gauge-fixing $Q^{\mathrm{GF}} = d^*_{\mathbb{R}^2} + \bar\partial^*_{\mathbb{C}^2}$, and the associated Laplacian $\Delta = [Q^{\mathrm{base}}, Q^{\mathrm{GF}}]$, one has

$$[Q^{\mathrm{base}}, P^{\mathrm{base}}_{\epsilon,L}] = \int_\epsilon^L [Q^{\mathrm{base}}, Q^{\mathrm{GF}}e^{-t\Delta}]\,dt = \int_\epsilon^L \partial_t(-e^{-t\Delta})\,dt = e^{-\epsilon\Delta} - e^{-L\Delta} = K^{\mathrm{base}}_\epsilon - K^{\mathrm{base}}_L.$$

The coefficient piece is the Casimir of Proposition 3.76, which is annihilated by the trivial differential on $\mathfrak{h}_w[1] \oplus \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[2]$. Therefore $[Q, P^w_{\epsilon,L}] = K^w_\epsilon - K^w_L$, and the family is continuous in (ϵ, L) because $C_{\mathfrak{h},w}$ is a fixed element of the weighted pair and the bulk family is continuous. Item (3), at the level asserted here, is the Costello bulk regularity package [12, Chs. 5–7] tensored with a fixed coefficient kernel. This proves finite-scale kernel regularity and absence of new coefficient wave-front directions. The brane-defect QME obstruction classes are separate local-functional classes. □

Remark 3.79 *(Brane defect).* The brane defect $L = \mathbb{R}_{\text{brane}} \times \{0\} \times \{p\} \subset \mathbb{R}^2 \times \mathbb{C}^2$ pulls the weighted field complex $\mathcal{E}_w$ to its restriction along the defect:

$$\mathcal{E}_w^{\partial} = \Omega^\bullet(\mathbb{R})\widehat{\otimes}\big(\mathfrak{h}_w[1] \oplus \mathfrak{h}_{\text{cont}}^{\vee,w}[2]\big).$$

The boundary propagator $P_{\epsilon,L}^{w,\partial} \in \mathcal{E}_w^{\partial}\widehat{\otimes}\mathcal{E}_w^{\partial}$ is the bulk-to-boundary restriction of $P_{\epsilon,L}^{w}$ followed by the zero-mode-removed midpoint propagator $G(t,s) = \frac{1}{2}\operatorname{sgn}(t-s)$ of Proposition 3.275. The weighted Casimir $C_{\mathfrak{h},w}$ is the same on the boundary as in the bulk; the weight w does not interact with the coordinate t.

Normal Ω-weights and equivariant kernel admissibility

Definition 3.80 *(Equivariant finite normal window).* Fix normal equivariant parameters $\epsilon_s, \epsilon_1, \epsilon_2$, with t fixed and s, z_1, z_2 carrying weights $\epsilon_s, \epsilon_1, \epsilon_2$. For a finite normal/Hamiltonian window (N, M), write

$$\mathsf{W}_{N,M} = \{(n,a,b) \mid 0 \le n \le N,\ a,b \ge 0,\ 1 \le a+b \le M\}.$$

The integer n is the total s-weight of the homogeneous normal summand; if a de Rham ds-factor is present, its weight is included in n. For the Hamiltonian monomial $s^n z_1^a z_2^b$ the normal character is

$$\chi_{n,a,b}^{H} = n\epsilon_s + a\epsilon_1 + b\epsilon_2 \tag{10}$$

For the residue-dual principal-part label with the same s-weight, the holomorphic part has the opposite character:

$$\chi_{n,a,b}^{\vee} = n\epsilon_s - a\epsilon_1 - b\epsilon_2 \tag{11}$$

Let $\mathsf{X}_{N,M}$ be the finite set of nonzero formal characters among $\chi_{n,a,b}^{H}$ and $\chi_{n,a,b}^{\vee}$ for those homogeneous summands which the normal Koszul contraction is required to move. The finite-window equivariant coefficient ring is

$$R_{\Omega}^{N,M} = \mathbb{C}[\epsilon_s, \epsilon_1, \epsilon_2, \hbar_\omega][\chi^{-1} \mid \chi \in \mathsf{X}_{N,M}], \qquad \operatorname{wt}(\hbar_\omega) = \epsilon_1 + \epsilon_2$$

A scalar specialization of $(\epsilon_s, \epsilon_1, \epsilon_2)$ is (N, M)-nonresonant if every $\chi \in \mathsf{X}_{N,M}$ remains nonzero. It is all-window nonresonant if this holds for every (N, M). If one imposes an analytic Kothe norm rather than the algebraic Mittag–Leffler product topology, the stronger q-moderate nonresonance condition is required: there are constants A, Q such that

$$|\chi_{n,a,b}|^{-1} \le A\, Q^{n+a+b}$$

for every inverted character. Without this quantitative bound, nonresonance still gives every finite-window algebraic kernel, but not an analytic all-window bounded homotopy in a fixed normed scale.

Theorem 3.81 *(Equivariant weighted-kernel admissibility).* Let w be a degree weight satisfying Definition 3.71. Work in a finite normal window (N, M) over $R_{\Omega}^{N,M}$, and let $\mathsf{A}_{N,M}$ be a homogeneous basis of the corresponding Hamiltonian-plus-cotangent coefficient pair. For $\alpha = (n,a,b,\tau) \in \mathsf{A}_{N,M}$, put $d(\alpha) = a+b$, let $e_{\alpha,w} = w(d(\alpha))e_\alpha$, let $e_{w^{-1}}^{\alpha} = \big(w(d(\alpha))\big)^{-1}e^{\alpha}$, and let χ_α be the character (10) for Hamiltonian labels and (11) for cotangent labels.

Then the equivariant diagonal coefficient Casimir

$$C_{\Omega,w}^{N,M} = \sum_{\alpha \in \mathsf{A}_{N,M}} e_{\alpha,w} \otimes e_{w^{-1}}^{\alpha}$$

is a finite-window kernel, and these kernels are compatible under the normal/Hamiltonian truncation maps. Thus the inverse system of $C^{N,M}_{\Omega,w}$'s is a strict Mittag–Leffler diagonal Casimir in the weighted coefficient category.

Suppose moreover that on each moving homogeneous normal summand there is a finite-dimensional Koszul numerator $k^{\mathrm{Kos}}_{\alpha}$ with

$$Q_\Omega k^{\mathrm{Kos}}_{\alpha} + k^{\mathrm{Kos}}_{\alpha} Q_\Omega = \chi_\alpha \ \mathrm{id}_\alpha$$

before division by the character. Then the normal contracting kernel

$$H^{N,M}_{\Omega,w} = \sum_{\alpha \in \mathsf{A}^{\mathrm{mov}}_{N,M}} \chi_\alpha^{-1} \, e_{\alpha,w} \otimes e^{\alpha}_{w^{-1}} \otimes k^{\mathrm{Kos}}_{\alpha} \tag{12}$$

is a well-defined element of the finite-window weighted kernel module and satisfies

$$Q_\Omega H^{N,M}_{\Omega,w} + H^{N,M}_{\Omega,w} Q_\Omega = \mathrm{id} - \mathrm{pr}_{\chi=0}$$

on that window. A character with specialized value zero is not a small denominator to be regularized; it is a genuine resonant summand. It must be retained in $\mathrm{pr}_{\chi=0}$, or the contraction does not exist.

The residue and Euler normalizations do not change the diagonal Casimir. They change the brane pushforward/scalar-contact normalization. With orientation sign $\sigma_s = \pm 1$,

$$\nu^{\mathrm{res}}_{\Omega} = 1, \qquad \nu^{\mathrm{Eu}}_{\Omega} = \sigma_s \big(\epsilon_s \epsilon_1 \epsilon_2\big)^{-1}.$$

A normalized equivariant kernel row is therefore $K^{N,M}_{\Omega,w,\nu} = \nu_\Omega K^{N,M}_{\Omega,w}$. The comparison between residue and Euler conventions is the scalar map

$$K^{N,M}_{\Omega,w,\mathrm{res}} \longmapsto \nu^{\mathrm{Eu}}_{\Omega} K^{N,M}_{\Omega,w,\mathrm{res}}.$$

Using both conventions in one scalar-contact or trace statement without this factor is exactly the normalization obstruction.

Proof. The Casimir is finite in the window (N, M). The two diagonal weights $w(d(\alpha))$ and $\big(w(d(\alpha))\big)^{-1}$ cancel in the evaluation pairing, so the finite sum is the identity tensor on the corresponding finite-dimensional coefficient quotient. Truncating from (N, M) to a smaller window simply deletes the omitted homogeneous labels, hence sends $C^{N,M}_{\Omega,w}$ to the smaller-window Casimir. This proves strict Mittag–Leffler compatibility.

For a moving homogeneous summand, the displayed Koszul identity gives

$$Q_\Omega(\chi_\alpha^{-1} k^{\mathrm{Kos}}_{\alpha}) + (\chi_\alpha^{-1} k^{\mathrm{Kos}}_{\alpha}) Q_\Omega = \mathrm{id}_\alpha$$

after adjoining χ_α^{-1} to the coefficient ring. Summing over the finite set of moving labels gives (12) and the homotopy identity with the unmoved zero-character projection. If a specialized character vanishes, χ_α^{-1} is not an element of the specialized coefficient ring; the same equation cannot be written. Thus resonance is precisely the obstruction to moving that summand.

The last paragraph is bookkeeping in the one-dimensional scalar localization line. The residue convention fixes the formal residue density and inserts no smooth normal Euler factor. The Euler convention pushes along the normal bundle with inverse equivariant Euler class $\sigma_s \big(\epsilon_s \epsilon_1 \epsilon_2\big)^{-1}$. The coefficient Casimir pairs dual bases before this scalar pushforward is applied, so it is unchanged. Scalar-contact rows and trace rows are changed by multiplication by the chosen ν_Ω, which is exactly the displayed comparison map. □

Corollary 3.82 *(Equivariant denominators do not decide the Costello QME).* Theorem 3.81 proves the coefficient Casimir and the localized normal contracting kernel in the weighted finite-window category. It gives the input needed to form an equivariant Costello kernel complex; it is not the vanishing theorem for that complex.

Once an equivariant brane-defect Costello package is supplied, in the sense of Definition E.36, the exact remaining obstruction vector is

$$\mathrm{Ob}^{\mathrm{QME}}_{\Omega,w} = \Big(\mathcal{R}_\Omega, \mathrm{ob}_{\Omega,\nu}, \{\mathfrak{s}_{n,\Omega}\}_{n\geq 1}, \{(([\mathfrak{o}^{\mathrm{ns}}_{n,\Omega,M}])_M, \lambda_{n,\Omega})\}_{n\geq 1}\Big).$$

Here

$$\mathcal{R}^{N,M}_\Omega = \{\alpha \in \mathsf{A}^{\mathrm{mov}}_{N,M} \mid \chi_\alpha = 0\}$$

is the resonance obstruction to the normal contraction, $\mathrm{ob}_{\Omega,\nu}$ is the failure to use one named residue/Euler normalization, $\mathfrak{s}_{n,\Omega,M}$ is the scalar gate

$$\widehat{\sigma}_{\mathrm{sc},\Omega,M}(R^{\mathrm{ns}}_{n,\Omega,M}),$$

and

$$([\mathfrak{o}^{\mathrm{ns}}_{n,\Omega,M}])_M \in \varprojlim_M H^1(\mathcal{K}^\bullet_{\mathrm{ns},\Omega,M}(I)), \qquad \lambda_{n,\Omega} \in \varprojlim_M{}^1 H^0(\mathcal{K}^\bullet_{\mathrm{ns},\Omega,M}(I))$$

are the non-scalar cohomology and primitive-compatibility obstructions. The equivariant weighted kernel package promotes to a Costello brane-defect QME solution exactly on the common zero locus of this vector, and only there.

Proof. Theorem 3.81 supplies only $C_{\Omega,w}$, the normal denominators, and the normalization line. Costello's QME is the Maurer–Cartan equation for the Hom-valued brane-defect graph kernel plus local counterterms. Therefore, after the equivariant package is present, the scalar projection must first be a chain projection and its scalar gate must vanish. Only then is the residual a class in the non-scalar kernel complex, and the finite-window counterterm criterion is the same inverse-limit H^1 and $\varprojlim^1 H^0$ criterion as Theorem 3.98, with $Q, d_M, \widehat{\sigma}_{\mathrm{sc}}$ replaced by their equivariant versions. Resonant normal modes and normalization mismatch occur before this cohomology class is even formed. Hence the displayed vector is necessary and sufficient for the equivariant weighted-kernel construction to become a QME solution. □

Renormalization-flow compatibility

The Costello RG flow on effective interactions is the assignment $I_L = W(P_{\epsilon,L}, I)$ that converts a local interaction I at scale ϵ into the effective interaction at scale L, where W is the homotopy renormalization group action on functionals [12, Ch. 6]. For the weighted mechanism to satisfy Costello requirement (C3), the RG flow on $\mathcal{E}_w$-valued interactions must preserve the weighted coefficient module: starting from a polynomial-degree-bounded interaction in $\mathcal{O}(\mathcal{E}_w)$, the effective interaction I_L must remain in the same weighted category.

Proposition 3.83 *(Graphwise RG-tame interactions).* Let $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ be a local interaction polynomial in the fields, of total polynomial-degree bounded by D_0 at each vertex in the coefficient direction. For every fixed Costello graph Γ and every $0 < \epsilon \leq L < \infty$, the graph contribution $W_\Gamma(P^w_{\epsilon,L}, I)$ has finite coefficient degree bounded by $D_0 V(\Gamma) - 2E(\Gamma)$ and lies in the weighted completed coefficient category. The bracket-tame estimate (W1), together with the finite-window coadjoint condition (W2), bounds this single graph by a finite constant times the corresponding spacetime graph weight.

Proof. The interaction I has bounded polynomial degree D_0 at each vertex in the coefficient direction by hypothesis. A connected Costello graph Γ with V vertices, each of polynomial degree $\leq D_0$, and E internal edges, contributes a coefficient pairing of external length at most $V \cdot D_0 - 2E$ in the source factor and a matching length in the target. By the bracket-tame estimate (W1), each internal edge contraction $C_{\mathfrak{h},w} : \mathfrak{h}_w \widehat{\otimes} \mathfrak{h}^{\vee,w}_{\mathrm{cont}} \to \mathbb{C}$ changes the total weight by at most the finite constant attached to the degrees present in that graph. Coadjoint legs are controlled windowwise by (W2). Hence the graph contribution is a well-defined element of the weighted completion. No assertion is made that the full effective interaction preserves the original degree bound D_0; Costello's expansion is a completed sum over graph degrees. □

THEOREM 3.84 *(Weighted RG flow, graphwise compatibility).* Let w be a degree weight as in Definition 3.71 and let $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ be a local interaction satisfying a vertexwise polynomial-degree bound D_0. Each fixed graph contribution to the Costello RG flow

$$I_L = W(P^w_{\epsilon,L}, I)$$

lies in the weighted Tate completion and is continuous in (ϵ, L). The theorem is graphwise. The completed sum over counterterms and the QME obstruction class require the locality input of Problem 3.87.

Proof. By Proposition 3.83, every Feynman diagram considered individually remains in the weighted Tate category. Its finite coefficient-degree bound is $D_0 V(\Gamma) - 2E(\Gamma)$, and continuity in (ϵ, L) follows from Proposition 3.78 (2). Passing from graphwise well-definedness to a renormalized QME solution is precisely the residual analytic problem. □

Cotangent-lift theorem

THEOREM 3.85 *(Cotangent-lift inside the weighted coefficient category).* Let $w(d) = q^d$ with $q > 1$ be the exponential degree weight of Remark 3.73. Then in the weighted Tate coefficient pair $(\mathfrak{h}_w, \mathfrak{h}^{\vee,w}_{\mathrm{cont}})$ of Definition 3.74 the following internal weighted-category statements hold:

(a) the Casimir $C_{\mathfrak{h},w} = \sum_d \sum_i w(d) e_{d,i} \otimes w(d)^{-1} e^i_d$ converges as an element of $\mathfrak{h}_w \widehat{\otimes} \mathfrak{h}^{\vee,w}_{\mathrm{cont}}$ (Proposition 3.76);

(b) the Costello heat-kernel propagator $P^w_{\epsilon,L}$ extends to the weighted field complex $\mathcal{E}_w$ as a continuous family in (ϵ, L) satisfying the BV identity (Proposition 3.78);

(c) each fixed Costello graph contribution preserves the weighted category on vertexwise polynomial-degree-bounded interactions (Theorem 3.84);

(d) the cochain-level Φ of Theorem 3.28 extends to the weighted cotangent extension $\mathfrak{g}_w = \mathfrak{h}_w \ltimes \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[1]$ as a continuous cochain map in the weighted coefficient envelope

$$\Phi^w : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_w) \to \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}_w)\big);$$

Proof. Items (a)–(c) are Propositions 3.76, 3.78, and Theorem 3.84. For item (d): the classical Φ of Theorem 3.28 acts on continuous CE cochains by the cochain identity $\Phi d_{\mathrm{CE}} = d_\pi \Phi$, which is a degree-by-degree algebraic identity on the polynomial Hamiltonian Lie algebra $\mathfrak{h}$ together with its dual $\mathfrak{h}^\vee[1]$. The identity is preserved under any rescaling of the bases by an invertible weight, so it descends to the weighted pair. The P_0-algebra reading is available only after the weighted envelope is treated as a bracket-admissible realization in the sense of Definition 2.46; this is not the strict continuous duality statement of Lemma 3.217 for a product/direct-sum Tate pair. The needed dual generators are inserted

by the symmetric Mittag-Leffler coefficient pairing of Remark 3.75. On the target, the Schouten bracket descends to the weighted symmetric algebra by (W2), subject to the same bracket-admissibility reading.

The all-order QME and regulator-independence assertions are separate theorems; the quantum assertion is isolated in Proposition 3.86. □

Proposition 3.86 *(Weighted QME lift under obstruction vanishing).* Assume the Costello-locality/QME compatibility identity of Problem 3.87: the finite-window counterterm system for the mixed brane-defect interaction has vanishing obstruction class and is compatible with the weighted coefficient category. Under this assumption, the formal quantum CE/PV central-operation map $\Phi_{\hbar}^{(0)}$ governed by Theorem 3.269 has a weighted-continuous cotangent extension

$$\Phi^{w}_{\hbar,\mathrm{cond}}\colon C^{\bullet}_{\mathrm{CE,cont}}\big(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]\big)[[\hbar]] \xrightarrow[\text{QME obstruction=0}]{\sim} Z^{P_{0,\hbar}}_{\mathrm{fact}}\big(\mathrm{Obs}^{\hbar}_{\mathrm{open}}\big)|_{\mathfrak{g}_w}.$$

The displayed isomorphism is a theorem in the weighted-continuous sector with vanishing QME obstruction. The original unweighted Tate pair is governed by the endpoint obstruction of Theorem 3.109.

Proof. The algebraic cochain identity is supplied by Theorem 3.85. The BV graph realization decomposes the cotangent action $\mathrm{ad}^{\vee}$ into Costello graphs in $\mathcal{E}_w$, and Theorem 3.84 keeps each fixed graph inside the weighted category. What is not formal is the cancellation between Costello-locality counterterms and the bulk-boundary kernel at the brane defect. The proposition assumes exactly that weighted QME compatibility and the vanishing of the obstruction class named in Problem 3.87. With that assumption in force, the formal $\hbar$-adic construction of Theorem 3.269 applies in the weighted coefficient envelope. The graphwise finite-scale assertions above are the part proved before this obstruction is evaluated and before the principal-part current extension is adjoined; they do not decide compatibility with $\mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1]$. □

Problem 3.87 *(Costello-locality compatibility for the weighted RG flow).* Verify that the Costello QME counterterms at the brane defect $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ in the weighted field complex $\mathcal{E}_w$ remain local-functional classes $O \in H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$, i.e. that the bulk Costello locality package [12, Chs. 5–7] extends to the weighted pair. The exact published theorem numbering is not an input to this problem. Concretely: a counterterm contribution $C_n \in \mathcal{O}^{\hbar^n}_{\mathrm{loc}}(\mathcal{E})$ at order $\hbar^n$ in the standard pair is a polynomial functional of bounded degree; one must check that the same counterterm, rescaled by the weight w on each coefficient leg, remains a local functional of the same bounded polynomial degree in the weighted pair. The estimate (W1) bounds vertex weights, and (W2) bounds coadjoint legs graph by graph. The endpoint question is whether the compatible finite-window counterterm system obtained from Costello's Hadamard locality theorem has vanishing obstruction class in $H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ for the mixed brane-defect interaction. Theorem 3.166 controls the wave-front and finite-window locality estimates. The obstruction class displayed above is the exact further datum.

Corollary 3.88 *(Descendant cochain comparison under weighted QME vanishing).* Assume the obstruction class in Problem 3.87 vanishes, so that Proposition 3.86 applies. Then the Tate-coefficient lift of the descendant cochain comparison of Problem 3.273 exists in the weighted Tate completion only under this hypothesis: the deformation

$$\Phi^{(0,1)}_{\hbar,\mathrm{cond}}\colon C^{\bullet}_{\mathrm{CE,cont}}\big(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]\big)^{(0)+(1)}[[\hbar]] \xrightarrow[\text{QME obstruction=0}]{\sim} Z^{P_{0,\hbar}}_{\mathrm{fact}}\big(\mathrm{Obs}^{\hbar}_{\mathrm{open}}\big)^{(0)+(1)}|_{\mathfrak{g}_w}$$

is an isomorphism on the degree-zero plus one-antifield sector in the weighted theory with vanishing QME obstruction.

Remark 3.89 *(Sector restriction).* The weighted mechanism restricts the cotangent direction from the ambient polynomial Hamiltonian dual to the weighted dual $\mathfrak{h}^{\vee,w}_{\mathrm{cont}} = \prod_d w(d)^{-1} H^\vee_d$. The strict polynomial Hamiltonian dual $\mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_d H^\vee_d$ is a *dense subspace* of $\mathfrak{h}^{\vee,w}_{\mathrm{cont}}$ for every w with $w(d) \to \infty$, since the strict direct sum is finite-degree truncations and the weighted product is the weighted Mittag-Leffler inverse-limit completion. Hence the weighted lift extends the cochain map continuously from the strict polynomial sector to the weighted product completion. This recovers only the weighted-continuous Hamiltonian identities, not an unweighted regulator-independence statement and not an identification of the weighted product with the original strict continuous dual.

Regulator independence and its boundary

Definition 3.90 *(Regulator-admissible finite-window sector).* A local functional, interaction, obstruction cocycle, or graph coefficient in the weighted theory is *regulator-admissible* if it is represented by a strict Mittag-Leffler family on the finite Tate windows

$$\{\mathcal{E}^{w_0}_w\}_{w_0 \geq 1}$$

and satisfies the following four conditions.

(A1) The transition maps in the coefficient and local-functional directions are the truncations to degrees $d \leq w_0$, or the induced restriction obtained by setting higher-degree coefficient fields to zero. They are degreewise surjective maps of complexes.

(A2) At each fixed $\hbar$-order and for each fixed Costello graph, the coefficient expression uses only vertex-wise polynomial-degree-bounded Hamiltonian and cotangent legs.

(A3) If w, w' are two degree weights satisfying Definition 3.71, the diagonal finite-window rescaling

$$R^{w_0}_{w,w'}(w(d)e_{d,i}) = w'(d)e_{d,i}, \qquad R^{w_0}_{w,w'}(w(d)^{-1}e^i_d) = w'(d)^{-1}e^i_d$$

transports the interaction, differential, bracket, propagator contraction, and counterterm representative from the w-window to the w'-window. The bracket is the transported bracket; comparison of two weights is made after this transport, not with the raw unweighted structure constants held fixed.

(A4) The cohomology being computed is the admissible filtered cohomology detected on finite quotients and associated graded pieces in the envelope of Statement 3.32; no observable is allowed to detect an infinite product tail invisible in every finite quotient.

Definition 3.91 *(Exponential finite-window regulator tower).* For $q > 1$ put $w_q(d) = q^d$. For a finite Hamiltonian window M set

$$\mathfrak{h}_{q,\leq M} = \prod_{1 \leq d \leq M} q^d H_d, \qquad D_{q,\leq M} = \prod_{1 \leq d \leq M} q^{-d} H^\vee_d,$$

and

$$\mathcal{E}^M_q = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\mathfrak{h}_{q,\leq M}[1] \oplus D_{q,\leq M}[2]\big).$$

For $M \geq N$, let

$$\pi^q_{M,N} \colon (\mathfrak{h}_{q,\leq M}, D_{q,\leq M}) \twoheadrightarrow (\mathfrak{h}_{q,\leq N}, D_{q,\leq N})$$

be degree truncation. The weight transport and the unweighting map on the window M are

$$R^M_{q,q'}(q^d e_{d,i}) = (q')^d e_{d,i}, \qquad R^M_{q,q'}(q^{-d} e^i_d) = (q')^{-d} e^i_d,$$

and

$$U_q^M(q^d e_{d,i}) = e_{d,i}, \qquad U_q^M(q^{-d} e_d^i) = e_d^i.$$

The finite-window unweighted endpoint is the finite-dimensional pair

$$(\mathfrak{h}_{\le M}, D_{\le M}) = \Big(\prod_{d\le M} H_d, \prod_{d\le M} H_d^\vee\Big).$$

Its transition map for $M \ge N$ is denoted $\pi^{\circ}_{M,N}$.

Lemma 3.92 *(Finite-window transport squares).* For $q, q' > 1$ and $M \ge N$, the maps of Definition 3.91 satisfy

$$\pi^{q'}_{M,N} R^M_{q,q'} = R^N_{q,q'} \pi^q_{M,N}, \qquad \pi^{\circ}_{M,N} U_q^M = U_q^N \pi^q_{M,N},$$

and

$$U_{q'}^M R^M_{q,q'} = U_q^M, \qquad R^M_{q',q''} R^M_{q,q'} = R^M_{q,q''}.$$

The truncated Casimir

$$C_q^M = \sum_{d\le M} \sum_i (q^d e_{d,i}) \otimes (q^{-d} e_d^i)$$

is transported to $C_{q'}^M$ by $R^M_{q,q'}$ and to the ordinary finite Casimir $C_\circ^M$ by U_q^M. Hence every Costello graph whose coefficient legs lie in $d \le M$ has the same unweighted coefficient after applying U_q^M, independently of q.

Proof. All four identities are degreewise identities on the two bases $q^d e_{d,i}$, $q^{-d} e_d^i$. Truncation forgets degrees $d > N$, while $R^M_{q,q'}$ and U_q^M multiply only the surviving degree-d basis vector by a scalar. Thus truncation and transport commute.

The Casimir statement follows term by term:

$$R^M_{q,q'}\big((q^d e_{d,i}) \otimes (q^{-d} e_d^i)\big) = (q')^d e_{d,i} \otimes (q')^{-d} e_d^i,$$

and U_q^M sends the same term to $e_{d,i} \otimes e_d^i$. A fixed graph has finitely many coefficient contractions and external coefficient legs. If all of them lie in the window M, applying U_q^M to each leg removes every diagonal weight and each internal Casimir becomes $C_\circ^M$. □

Statement 3.93 *(Acceptance gate for weighted RG transport).* A weighted mixed brane-defect interaction enters the regulator theorem only with the following data.

(G1) Its coefficient and local-functional restrictions to $d \le w_\circ$ form strict Mittag-Leffler towers with degreewise surjective transition maps.

(G2) At every fixed $\hbar$-order and for every fixed Costello graph, the coefficient expression is vertexwise bounded in polynomial Hamiltonian and cotangent degree.

(G3) For any admissible degree weights w, w', the diagonal finite-window maps $R^{w_\circ}_{w,w'}$ transport the differential, bracket, propagator contraction, interaction, and counterterm representatives.

(G4) The cohomology theory is the admissible filtered cohomology of Statement 3.32.

(G5) The Costello RG recursion $W(P_{\epsilon,L}, I)$ preserves the same strict Mittag-Leffler tower: every RG step has source and target in the weighted coefficient envelope.

Proof. Conditions (G1)–(G4) are the four clauses of Definition 3.90. Condition (G5) is the coefficient part of Costello requirement (C3). Proposition 3.83 proves (G2) for a fixed graph. Theorem 3.166 proves that the finite-window Hadamard estimates pass through the Mittag-Leffler limit when the windowwise counterterms are compatible. The all-order datum beyond this graphwise statement is the obstruction class of Problem 3.87. □

THEOREM 3.94 *(Transport groupoid of admissible weights).* On the class of interactions satisfying 3.93, admissible degree weights form a connected transport groupoid. Its objects are the weighted finite-window local-functional complexes

$$C_w^{w_\circ} = \big(O_{\mathrm{loc}}(\mathcal{E}_w^{w_\circ}), Q + \{I_\circ^w, -\}\big),$$

and its arrows are the diagonal maps $R_{w,w'}^{w_\circ}$. For every w, w', w'' and every finite window $w_\circ$,

$$R_{w',w''}^{w_\circ} R_{w,w'}^{w_\circ} = R_{w,w''}^{w_\circ}, \qquad R_{w,w}^{w_\circ} = \mathrm{id},$$

and these arrows commute with truncation in $w_\circ$, Costello graph contractions, and RG counterterm representatives.

Proof. The formula defining $R_{w,w'}^{w_\circ}$ is diagonal:

$$w(d)e_{d,i} \mapsto w'(d)e_{d,i}, \qquad w(d)^{-1}e_d^i \mapsto w'(d)^{-1}e_d^i.$$

Composition multiplies the scalars degree by degree, hence gives the displayed groupoid law. Since the maps are diagonal in d, they commute with truncation to $d \le w_\circ$. By 3.93(G3), the transported bracket, interaction, propagator, and counterterm representatives are the corresponding w'-objects. Therefore $R_{w,w'}^{w_\circ}$ is an isomorphism of the finite-window RG complexes and intertwines every fixed graph contraction. □

THEOREM 3.95 *(Canonical equivalence of finite-degree local observables).* Let $w(d) = q^d$ and $w'(d) = (q')^d$, with $q, q' > 1$. Let O^w be a regulator-admissible local observable, graph coefficient, or obstruction cocycle represented on a finite window $d \le M$. Then $R_{q,q'}^M$ is the canonical identification of the q-weighted and q'-weighted representatives on that finite window:

$$O_M^{q'} = R_{q,q'}^M O_M^q.$$

After writing both representatives in the common unweighted basis,

$$U_q^M(O_M^q) = U_{q'}^M(O_M^{q'}).$$

If O_M^q is a cocycle for $Q + \{I_M^q, -\}$, then the common unweighted representative is a cocycle for the unweighted finite-window transported differential. If $\mathrm{Ob}_{q,M}$ is the finite-window obstruction cocycle produced by the Costello recursion, then

$$U_q^M(\mathrm{Ob}_{q,M}) = U_{q'}^M(\mathrm{Ob}_{q',M}), \qquad \pi_{M,N}^\circ U_q^M(\mathrm{Ob}_{q,M}) = U_q^N(\mathrm{Ob}_{q,N})$$

for $M \ge N$. Thus a finite-degree local observable, fixed graph coefficient, or obstruction class is a single object of the finite-window transported complex, presented in different weighted bases by the choices $q > 1$.

Proof. The observable lies in the finite complex C_q^M. By Theorem 3.94, $R_{q,q'}^M$ is an isomorphism from C_q^M to $C_{q'}^M$, carrying the differential, bracket, propagator contraction, interaction, counterterm representative, and obstruction cocycle. Lemma 3.92 gives $U_{q'}^M R_{q,q'}^M = U_q^M$. Therefore both weighted representatives have the same coefficient array after unweighting.

Since $R_{q,q'}^M$ is a cochain isomorphism, cocycles go to cocycles and coboundaries go to coboundaries. Applying U_q^M merely writes the same finite complex in the common unweighted basis; the transported differential is $U_q^M(Q + \{I_M^q, -\})(U_q^M)^{-1}$. Hence the cocycle condition is preserved. The obstruction equalities are the same statement for the obstruction cochain produced at the relevant RG step. The transition identity follows from $\pi_{M,N}^{\circ} U_q^M = U_q^N \pi_{M,N}^q$ and strict Mittag-Leffler compatibility of the finite-window obstruction system. □

The next theorem is a finite-window weight-transport statement. It is not a $q \to 1^+$ endpoint theorem for the strict product/direct-sum Tate pair, and it is not a full regulator-independence theorem for classes detecting the quotient $D_{\Pi,w}/D_{\oplus}$. That endpoint is excluded by Theorem 3.109.

Theorem 3.96 *(Regulator independence inside the admissible finite-window sector).* Let w, w' be degree weights satisfying Definition 3.71, and let $I_{\circ}^w$ be a regulator-admissible mixed brane-defect interaction satisfying the acceptance gate 3.93. Put $I_{\circ}^{w'} = R_{w,w'} I_{\circ}^w$. Then the diagonal rescalings $R_{w,w'}^{w_\circ}$ assemble to an isomorphism of strict Mittag-Leffler local-functional complexes

$$R_{w,w'} : \varprojlim_{w_\circ} \big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_\circ}), Q + \{I_{\circ}^w, -\}\big) \xrightarrow{\cong} \varprojlim_{w_\circ} \big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_{w'}^{w_\circ}), Q + \{I_{\circ}^{w'}, -\}\big).$$

Under this isomorphism:

(i) truncated Casimirs, heat-kernel propagators, and fixed graphwise Costello contractions are identified;

(ii) the obstruction class
$$\mathrm{Ob}_w \in H^1_{\mathrm{adm}}(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_{\circ}^w, -\})$$
maps to $\mathrm{Ob}_{w'}$;

(iii) vanishing or nonvanishing of the mixed brane-defect QME obstruction is independent of the admissible weight.

These isomorphisms are the arrows of the transport groupoid of Theorem 3.94; they are functorial in triples of admissible weights.

Proof. On the finite window $d \le w_\circ$, the map $R_{w,w'}^{w_\circ}$ is an invertible linear map on the coefficient space and the identity on the spacetime factor. The bracket and the interaction in the w'-window are, by admissibility, the transported bracket and transported interaction. The map sends

$$\sum_{d \le w_\circ} \sum_i (w(d) e_{d,i}) \otimes (w(d)^{-1} e_d^i)$$

to the corresponding w'-weighted truncated Casimir. Hence it sends $P_{\epsilon,L}^{w,w_\circ}$ to $P_{\epsilon,L}^{w',w_\circ}$, commutes with the differential Q, and carries the transported BV bracket and every finite-window graph contraction to the w'-window contraction. Condition (A3) gives the same statement for $I_\circ$, the counterterm representatives, and the obstruction cocycle.

The maps $R^{w_0}_{w,w'}$ commute with truncation in w_0. The finite-window complexes therefore form an isomorphic pair of strict Mittag-Leffler inverse systems. By (A1), these are degreewise surjective towers of complexes. Hence the inverse limit is exact in the admissible envelope of Statement 3.32; equivalently $\varprojlim^1 = 0$ for this tower. Passing to the inverse limit gives the displayed isomorphism of local-functional complexes and identifies their admissible filtered cohomology classes. In particular the obstruction class vanishes for w if and only if the transported obstruction class vanishes for w'.

Functoriality in triples is the groupoid identity proved in Theorem 3.94. □

Definition 3.97 *(Finite-window non-scalar QME tower).* Fix an interval I, a degree weight w, and a regulator-admissible interaction satisfying Statement 3.93. For a finite Hamiltonian window M, put

$$D_{w,\le M} := \prod_{d\le M} w(d)^{-1} H_d^\vee,$$

and define the truncated principal-part target

$$A^{\mathrm{pp},w,M}_{\partial,\hbar}(I) = \widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}\big(\Omega^\bullet_c(I) \otimes \prod_{d\le M} w(d) H_d[[\hbar]]\big)\widehat{\otimes}\widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}\big((\mathcal{D}'_c(I)\widehat{\otimes} D_{w,\le M}[[\hbar]])[1]\big).$$

Assume that the scalar-contact projection of Problem 3.271 has been chosen windowwise, commutes with truncation, and is transported by $R^M_{w,w'}$. The finite-window non-scalar obstruction complex is

$$Q^\bullet_{w,M}(I) = \ker(\widehat{\sigma}_{\mathrm{sc},M}) \subset \Big(O^{\mathrm{bd}}_{\mathrm{loc}}\big(\mathcal{E}^M_w|_I; A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)\big), d_M\Big),$$

where $d_M = Q + \{I^w_{0,M}, -\}_{\hbar,M}$. The bracket, interaction, and differential are the transported finite-window structures supplied by the acceptance gate; they are not naive projected Hamiltonian brackets. The inverse system $\{Q^\bullet_{w,M}(I)\}_{M\ge 1}$ uses degree truncation $\pi_{M,N}$ for $M \ge N$, together with the $\hbar$-adic filtration, the scalar-contact filtration, and the finite-window coefficient filtration.

THEOREM 3.98 *(Finite-window counterterm criterion for the non-scalar QME).* Work in the tower of Definition 3.97. Suppose that a counterterm system has been chosen through order $\hbar^{n-1}$, and let

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} \in Z^1\big(Q^\bullet_{w,M}(I)\big)$$

be the order-n non-scalar residual after those lower-order choices. Equivalently, in the curved Maurer–Cartan form,

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} = [\hbar^n]\Big(\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar,M} + d_M C_{<n,M} + \frac{1}{2}\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}\Big),$$

with the quadratic term omitted for the linearized equation $\mathrm{Ob}^{\mathrm{red}} + d_M C = 0$. Compatible order-$n$ finite-window counterterms

$$C_{n,M} \in Q^0_{w,M}(I), \qquad d_M C_{n,M} = -\mathfrak{o}^{\mathrm{ns}}_{n,M}, \qquad \pi_{M,N} C_{n,M} = C_{n,N},$$

exist if and only if the following two obstructions vanish:

$$\big([\mathfrak{o}^{\mathrm{ns}}_{n,M}]\big)_M \in \varprojlim_M H^1\big(Q^\bullet_{w,M}(I)\big),$$

and, after choosing any windowwise primitives $c_{n,M}$ with $d_M c_{n,M} = -\mathfrak{o}^{\mathrm{ns}}_{n,M}$, the Milnor primitive-compatibility class

$$\lambda_n \in \varprojlim{}^1_M H^0\big(Q^\bullet_{w,M}(I)\big)$$

represented, after passing to cohomology, by

$$\big[\pi_{M,N} c_{n,M} - c_{n,N}\big] \in H^0\big(Q^\bullet_{w,N}(I)\big)$$

is zero. If the inverse system $\{H^0(Q^\bullet_{w,M}(I))\}_M$ is Mittag–Leffler, then $\lambda_n = 0$ automatically, so windowwise vanishing of the H^1-classes is sufficient.

Iterating this criterion for all $n \geq 1$ constructs an $\hbar$-adic compatible counterterm $C_{\hbar,w} = \sum_{n\geq 1} \hbar^n C_{n,w}$ in the admissible finite-window inverse limit. The transport maps $R^M_{w,w'}$ identify the obstruction pair

$$\mathfrak{O}^{\mathrm{ns}}_n = \big(([\mathfrak{o}^{\mathrm{ns}}_{n,M}])_M, \lambda_n\big)$$

for all admissible weights. Therefore a nonzero $\mathfrak{O}^{\mathrm{ns}}_n$, at the first order where it occurs, is the precise finite-window obstruction to killing the non-scalar QME class by compatible weighted counterterms.

Proof. The lower-order counterterms solve the QME modulo $\hbar^n$. Applying d_M to the displayed order-n residual and using $d_M^2 = 0$, the graded Jacobi identity, and the lower-order equations gives $d_M \mathfrak{o}^{\mathrm{ns}}_{n,M} = 0$. The scalar-contact projection is a chain map by Definition 3.97, and the balanced scalar branch has zero scalar symbol, so the residual lies in $Q^\bullet_{w,M}(I)$.

Necessity is immediate. A compatible counterterm $C_{n,M}$ is a primitive of $-\mathfrak{o}^{\mathrm{ns}}_{n,M}$, hence each H^1-class vanishes. Its truncation identities make $\pi_{M,N} C_{n,M} - C_{n,N} = 0$, so the associated Milnor $\varprojlim{}^1 H^0$-class is also zero.

Conversely, assume the two displayed obstructions vanish. The first condition gives primitives $c_{n,M}$ window by window. For $M \geq N$, the difference $\pi_{M,N} c_{n,M} - c_{n,N}$ is d_N-closed because the residuals commute with truncation. The cohomology classes of these closed degree-zero differences form the standard Roos 1-cocycle computing $\varprojlim{}^1_M H^0(Q^\bullet_{w,M}(I))$. Vanishing of λ_n, together with the degreewise Mittag–Leffler property of the finite-window truncation tower, means that the primitives can be adjusted by degree-zero cocycles and degree-zero boundaries so that the adjusted primitives commute with every truncation map. Those adjusted primitives are the desired $C_{n,M}$. Passing to the inverse limit gives $C_{n,w}$.

The assertion about the H^0-Mittag–Leffler hypothesis is the standard vanishing of $\varprojlim{}^1$ for a Mittag–Leffler inverse system. Weight invariance follows from Theorem 3.96: the maps $R^M_{w,w'}$ are cochain isomorphisms, commute with truncation, and carry residuals, primitives, and their difference cocycles to the corresponding data for w'. □

Definition 3.99 (Regular-density closed-exchange finite-window complex). Work in the tower of Definition 3.97, and assume the regular-density Costello graph habitat of Proposition B.17 on every finite Hamiltonian window. Thus the cotangent current labels are restricted to

$$\mathcal{D}^{\mathrm{reg}}_c(I) = \{\, b(t)dt \mid b \in C^\infty_c(I) \,\},$$

and the finite-window graph weights

$$W^{R,\mathrm{reg},M}_{\Gamma,L}(a_\bullet, b_\bullet; f_\bullet, \rho_\bullet)$$

are renormalized brane-defect local functionals compatible with interval extension, degree truncation, and admissible weight transport. Let $\mathcal{G}^{\bullet,\mathrm{reg}}_{w,M}(I)$ be the complete filtered $\mathbb{C}[[\hbar]]$-span, inside $Q^\bullet_{w,M}(I)$,

of the scalar-zero members among these graph weights and their regular-density local counterterms. The zero-internal-edge generators are the coefficient-current terms

$$\iota_I(B_f^w(a)), \qquad \iota_I(\Theta_\rho^w(b(t)dt)),$$

whenever they have zero scalar-contact projection.

Define

$$\mathcal{X}_{w,M}^{\bullet,\mathrm{reg}}(I) \subset \mathcal{Q}_{w,M}^{\bullet}(I)$$

to be the smallest complete filtered d_M-stable subspace containing $\mathcal{G}_{w,M}^{\bullet,\mathrm{reg}}(I)$. Its differential is

$$d_M^{X,\mathrm{reg}} = d_M|_{\mathcal{X}_{w,M}^{\bullet,\mathrm{reg}}(I)}.$$

This is a finite-window graph complex, not the naive finite-dimensional Hamiltonian Lie quotient ruled out by Lemma 3.261. The window is a coefficient-support and graph-locality condition on already constructed regular-density local functionals.

PROPOSITION 3.100 *(Regular-density construction of X and Ξ).* The complexes of Definition 3.99 form an inverse system

$$\rho_{M,N}^{X,\mathrm{reg}} : \mathcal{X}_{w,M}^{\bullet,\mathrm{reg}}(I) \longrightarrow \mathcal{X}_{w,N}^{\bullet,\mathrm{reg}}(I), \qquad M \geq N,$$

by restricting the finite-window truncation maps $\pi_{M,N}$ of $\mathcal{Q}_{w,M}^{\bullet}(I)$. The closed-exchange insertion map is the degree-zero inclusion

$$\Xi_M^{\mathrm{reg}} : \mathcal{X}_{w,M}^{\bullet,\mathrm{reg}}(I) \hookrightarrow \mathcal{Q}_{w,M}^{\bullet}(I).$$

It is a cochain map and satisfies

$$\pi_{M,N}\Xi_M^{\mathrm{reg}} = \Xi_N^{\mathrm{reg}}\rho_{M,N}^{X,\mathrm{reg}}.$$

For admissible weights w, w', the diagonal transports restrict to cochain isomorphisms

$$R_{w,w'}^{M,X} : \mathcal{X}_{w,M}^{\bullet,\mathrm{reg}}(I) \xrightarrow{\cong} \mathcal{X}_{w',M}^{\bullet,\mathrm{reg}}(I)$$

and intertwine the inclusions:

$$R_{w,w'}^{M}\Xi_M^{\mathrm{reg}} = \Xi_{M,w'}^{\mathrm{reg}} R_{w,w'}^{M,X}$$

Consequently the abstract closed-exchange criterion of Theorem 3.102 applies to the regular-density tower $\mathcal{X}_{w,M}^{\bullet,\mathrm{reg}}(I)$ with $\Xi_M = \Xi_M^{\mathrm{reg}}$.

Proof. Proposition B.17 gives the finite-window identities

$$\pi_{M,N}W_{\Gamma,L}^{R,\mathrm{reg},M} = W_{\Gamma,L}^{R,\mathrm{reg},N}\pi_{M,N}^{\mathrm{CE}}, \qquad R_{w,w'}^{M}W_{\Gamma,L}^{R,\mathrm{reg},M} = W_{\Gamma,L,w'}^{R,\mathrm{reg},M}R_{w,w'}^{M,\mathrm{CE}}$$

and the same compatibility for the regular-density counterterms. Since $\widehat{\sigma}_{\mathrm{sc},M}$ is a chain map in Definition 3.97, its kernel $\mathcal{Q}_{w,M}^{\bullet}(I)$ is a subcomplex. Therefore scalar-zero graph weights and their d_M-closures stay in $\mathcal{Q}_{w,M}^{\bullet}(I)$.

Truncation $\pi_{M,N}$ and weight transport $R_{w,w'}^{M}$ are cochain maps on the non-scalar tower. They carry the generating regular-density graph weights and counterterms to the corresponding generators in the lower window or transported weight. By minimality of the complete d_M-stable subcomplex generated by these elements, they restrict to the displayed maps on $\mathcal{X}^{\bullet,\mathrm{reg}}$.

The map Ξ_M^{reg} is inclusion of a subcomplex of $\mathcal{Q}_{w,M}^{\bullet}(I)$, hence $d_M\Xi_M^{\mathrm{reg}} = \Xi_M^{\mathrm{reg}}d_M^{X,\mathrm{reg}}$. The compatibility with $\rho_{M,N}^{X,\mathrm{reg}}$ and $R_{w,w'}^{M,X}$ is precisely the restriction of the same identities on $\mathcal{Q}^{\bullet}$. □

Proposition 3.101 *(Raw kernel obstruction to a stronger cochain map).* Let $\kappa^{\mathrm{graph,reg},M}_{\hbar,w,I}$ be the finite-window regular-density graph kernel of Proposition B.17, viewed as a degree-zero element of the convolution Hom from the regular-density CE source to the target $\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$. If one tries to take the regular-density CE source itself as the closed-exchange complex and $\Xi_M = \kappa^{\mathrm{graph,reg},M}_{\hbar,w,I}$, then the exact obstruction to an ordinary cochain map into $\mathcal{Q}^{\bullet}_{w,M}(I)$ is the pair

$$\widehat{\sigma}_{\mathrm{sc},M}\kappa^{\mathrm{graph,reg},M}_{\hbar,w,I}, \qquad d_{\mathbf{K}}\kappa^{\mathrm{graph,reg},M}_{\hbar,w,I} = d_M\kappa^{\mathrm{graph,reg},M}_{\hbar,w,I} - \kappa^{\mathrm{graph,reg},M}_{\hbar,w,I}d_{\mathrm{CE}}$$

The first term is the obstruction to landing in $\ker \widehat{\sigma}_{\mathrm{sc},M}$; the second is the obstruction to commuting with differentials. If the desired closed-exchange datum is required to be a $P_{0,\hbar}$-central-operation kernel, the ordinary cochain obstruction is strengthened to the Maurer–Cartan curvature

$$\mathrm{Curv}_{\mathbf{K}}(\kappa^{\mathrm{graph,reg},M}_{\hbar,w,I}) = d_{\mathbf{K}}\kappa^{\mathrm{graph,reg},M}_{\hbar,w,I} + \frac{1}{2}[\kappa^{\mathrm{graph,reg},M}_{\hbar,w,I}, \kappa^{\mathrm{graph,reg},M}_{\hbar,w,I}]_{\mathbf{K}}.$$

Thus the regular-density graph layer supplies a theorem-grade closed-exchange tower by the subcomplex construction above. The stronger assertion that the raw bulk-to-defect CE kernel itself is the cochain map is equivalent to vanishing of the displayed scalar and convolution defects, with finite-window and weight-transport compatibility.

Proof. The convolution differential is $d_{\mathbf{K}}T = d_M T - (-1)^{|T|}T d_{\mathrm{CE}}$. Since $\kappa^{\mathrm{graph,reg},M}_{\hbar,w,I}$ has degree zero, the displayed formula is exactly the commutator of the proposed map with the two differentials. Hence it vanishes if and only if the raw kernel is a cochain map into the ambient target. It lands in the non-scalar complex precisely when its scalar-contact projection vanishes.

The bracket term is the completed convolution bracket of Definition B.15. A $P_{0,\hbar}$-central-operation kernel is a Maurer–Cartan element in that convolution dg Lie algebra, so its exact obstruction is $d_{\mathbf{K}}\kappa + \frac{1}{2}[\kappa,\kappa]_{\mathbf{K}}$. The generator-level form of this statement is the obstruction list in Proposition B.21. The compatibility clauses follow from the same truncation and weight-transport identities used in Proposition 3.100. □

Theorem 3.102 *(Closed-exchange plus counterterm criterion).* Work in the non-scalar tower of Definition 3.97. Let

$$\rho^{X}_{M,N}\colon \mathcal{X}^{\bullet}_{w,M}(I) \longrightarrow \mathcal{X}^{\bullet}_{w,N}(I), \qquad M \geq N,$$

be an inverse system of closed-exchange local-functional complexes, and let

$$\Xi_M\colon \mathcal{X}^{\bullet}_{w,M}(I) \longrightarrow \mathcal{Q}^{\bullet}_{w,M}(I)$$

be degree-zero cochain maps satisfying

$$\pi_{M,N}\Xi_M = \Xi_N\rho^{X}_{M,N}.$$

Assume also that the closed-exchange tower and the maps Ξ_M are transported by $R^{M}_{w,w'}$, in the same sense as the non-scalar tower. Let

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} \in Z^1(\mathcal{Q}^{\bullet}_{w,M}(I)), \qquad \pi_{M,N}\mathfrak{o}^{\mathrm{ns}}_{n,M} = \mathfrak{o}^{\mathrm{ns}}_{n,N},$$

be the order-n residual after the lower-order choices.

For a fixed window M, the branch equation

$$\mathfrak{v}^{\mathrm{ns}}_{n,M} + \Xi_M(W_{n,M}) + d_M C_{n,M} = 0, \qquad W_{n,M} \in Z^1(\mathcal{X}^\bullet_{w,M}(I)), \quad C_{n,M} \in \mathcal{Q}^0_{w,M}(I),$$

has a solution if and only if

$$[\mathfrak{v}^{\mathrm{ns}}_{n,M}] \in -\operatorname{im}\Big(H^1(\Xi_M)\colon H^1(\mathcal{X}^\bullet_{w,M}(I)) \to H^1(\mathcal{Q}^\bullet_{w,M}(I))\Big).$$

In the inverse limit, put

$$\beta^{\mathrm{cl}}_n = \Big[\big([\mathfrak{v}^{\mathrm{ns}}_{n,M}]\big)_M\Big] \in \operatorname{coker}\Bigg(\varprojlim_M H^1(\mathcal{X}^\bullet_{w,M}(I)) \xrightarrow{\varprojlim H^1(\Xi_M)} \varprojlim_M H^1(\mathcal{Q}^\bullet_{w,M}(I))\Bigg).$$

Compatible closed-exchange cocycles and compatible counterterms solving

$$\mathfrak{v}^{\mathrm{ns}}_{n,M} + \Xi_M(W_{n,M}) + d_M C_{n,M} = 0, \qquad \rho^X_{M,N} W_{n,M} = W_{n,N}, \quad \pi_{M,N} C_{n,M} = C_{n,N},$$

exist if and only if $\beta^{\mathrm{cl}}_n = 0$ and one can choose the following secondary data with zero Roos classes. Vanishing of β^{cl}_n is equivalent to the existence of a compatible cohomology class

$$\alpha = (\alpha_M)_M \in \varprojlim_M H^1(\mathcal{X}^\bullet_{w,M}(I))$$

with

$$H^1(\Xi_M)(\alpha_M) = -[\mathfrak{v}^{\mathrm{ns}}_{n,M}] \quad \text{for every } M.$$

For such a choice of α, the representative-compatibility class

$$\mu^{\mathrm{cl}}_n(\alpha) \in \varprojlim\nolimits^1_M H^0(\mathcal{X}^\bullet_{w,M}(I))$$

vanishes. Explicitly, choose cocycles $w_M \in Z^1(\mathcal{X}^\bullet_{w,M}(I))$ representing α_M. Since α is compatible, for the cofinal successor tower one can choose $u_M \in \mathcal{X}^0_{w,M}(I)$ with

$$\rho^X_{M+1,M} w_{M+1} - w_M = d^X_M u_M.$$

With the Roos convention

$$\Delta_X((a_M)_M) = \big(\rho^X_{M+1,M} a_{M+1} - a_M\big)_M,$$

the class of $([u_M])_M$ in $\operatorname{coker}\Delta_X = \varprojlim^1_M H^0(\mathcal{X}^\bullet_{w,M}(I))$ is $\mu^{\mathrm{cl}}_n(\alpha)$. Its vanishing is exactly the condition that the representatives w_M can be changed by degree-zero coboundaries to compatible cocycles $W_{n,M}$.

After choosing such compatible $W_{n,M}$, the counterterm compatibility class

$$\lambda^{\mathrm{cl}}_n(W) \in \varprojlim\nolimits^1_M H^0(\mathcal{Q}^\bullet_{w,M}(I))$$

vanishes. Namely, set

$$r_M = \mathfrak{v}^{\mathrm{ns}}_{n,M} + \Xi_M(W_{n,M}).$$

The class of r_M in $H^1(Q^\bullet_{w,M}(I))$ is zero, so choose $c_M \in Q^0_{w,M}(I)$ with $d_M c_M = -r_M$. The differences

$$\pi_{M+1,M} c_{M+1} - c_M \in Z^0(Q^\bullet_{w,M}(I))$$

define, with the analogous Roos operator Δ_Q, the class

$$\lambda_n^{\mathrm{cl}}(W) = \left[\left([\pi_{M+1,M} c_{M+1} - c_M]\right)_M\right] \in \operatorname{coker} \Delta_Q$$

This class vanishes exactly when the primitives c_M can be changed by degree-zero cocycles and coboundaries to compatible counterterms $C_{n,M}$.

If the inverse system $\{H^0(X^\bullet_{w,M}(I))\}_M$ is Mittag–Leffler, then $\mu_n^{\mathrm{cl}}(\alpha) = 0$ for every admissible lift α. If $\{H^0(Q^\bullet_{w,M}(I))\}_M$ is Mittag–Leffler, then $\lambda_n^{\mathrm{cl}}(W) = 0$ for every compatible closed-exchange representative W. Under both Mittag–Leffler hypotheses, the compatible closed-exchange branch is obstructed only by β_n^{cl}. The counterterm-only criterion of Theorem 3.98 is the special case $X^\bullet_{w,M}(I) = 0$. The admissible weight transports $R^M_{w,w'}$ identify the fixed-window image condition, β_n^{cl}, and the secondary Roos classes for w with the corresponding data for w'.

Proof. The fixed-window statement is the cohomology calculation. If the branch equation holds, then passing to H^1 gives $[\mathfrak{v}^{\mathrm{ns}}_{n,M}] = -H^1(\Xi_M)[W_{n,M}]$. Conversely, if the displayed image condition holds, choose a cocycle $W_{n,M}$ in the closed-exchange complex whose class maps to $-[\mathfrak{v}^{\mathrm{ns}}_{n,M}]$. Then $\mathfrak{v}^{\mathrm{ns}}_{n,M} + \Xi_M(W_{n,M})$ is a d_M-boundary, so a degree-zero $C_{n,M}$ gives the branch equation.

The compatibility of the residuals and the identity $\pi_{M,N}\Xi_M = \Xi_N \rho^X_{M,N}$ make $([\mathfrak{v}^{\mathrm{ns}}_{n,M}])_M$ a point of $\varprojlim_M H^1(Q^\bullet_{w,M}(I))$. Its cokernel class is zero exactly when a compatible cohomology class α in the closed-exchange tower maps to its negative. This proves the first inverse-limit obstruction.

Fix such an α. The differences $\rho^X_{M+1,M} w_{M+1} - w_M$ are closed degree-one elements with zero cohomology class, hence have the form $d^X_M u_M$. Changing the representatives w_M by degree-zero coboundaries changes $([u_M])_M$ by the Roos coboundary Δ_X. Thus the displayed class $\mu_n^{\mathrm{cl}}(\alpha)$ is independent of the choices. In the two-term Roos complex computing the first derived inverse limit of the tower $H^0(X^\bullet_{w,M}(I))$, its vanishing is exactly the equation needed to absorb these differences by changing w_M. This gives compatible cocycles $W_{n,M}$, and no weaker condition gives such a family.

With compatible $W_{n,M}$, the cochains $r_M = \mathfrak{v}^{\mathrm{ns}}_{n,M} + \Xi_M(W_{n,M})$ are compatible d_M-cocycles whose windowwise H^1-classes vanish. Choose primitives c_M. Since $\pi_{M+1,M} r_{M+1} = r_M$, the differences $\pi_{M+1,M} c_{M+1} - c_M$ are d_M-closed of degree zero. Changing the primitives by closed degree-zero cochains, and by boundaries, changes their cohomology classes by the Roos coboundary Δ_Q. Hence the displayed $\lambda_n^{\mathrm{cl}}(W)$ is independent of primitive choices and vanishes exactly when the primitives can be adjusted to a compatible system $C_{n,M}$. This system is precisely the desired compatible local counterterm family.

The last paragraph is the standard vanishing of $\varprojlim^1$ for Mittag–Leffler inverse systems of vector spaces. Therefore an H^0-Mittag–Leffler hypothesis on the closed-exchange tower kills the representative obstruction μ_n^{cl}, and the corresponding hypothesis on the non-scalar counterterm tower kills λ_n^{cl}. If both hold, only the cokernel H^1-class remains. If the closed-exchange tower is zero, the cokernel condition reduces to vanishing of $([\mathfrak{v}^{\mathrm{ns}}_{n,M}])_M$, and the only secondary class is the λ_n of Theorem 3.98. The weight-transport assertion follows because $R^M_{w,w'}$ is a compatible cochain isomorphism on both towers and intertwines Ξ_M, residuals, truncation maps, cocycle representatives, and primitive-difference cocycles. □

PROPOSITION 3.103 *(Regular-density quotient obstruction for the first residual).* Work in the regular-density tower of Definition 3.99 and Proposition 3.100. Let n_0 be the first $\hbar$-order at which the lower-order choices leave a compatible non-scalar residual

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} \in Z^1\big(Q^\bullet_{w,M}(I)\big), \qquad \pi_{M,N}\mathfrak{o}^{\mathrm{ns}}_{n_0,M} = \mathfrak{o}^{\mathrm{ns}}_{n_0,N}.$$

Put

$$\begin{aligned} C^{\bullet,\mathrm{reg}}_{w,M}(I) &= Q^\bullet_{w,M}(I)/\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I), \\ \bar{d}_M &= d_M \bmod \mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I), q_M : Q^\bullet_{w,M}(I) \twoheadrightarrow C^{\bullet,\mathrm{reg}}_{w,M}(I). \end{aligned}$$

The maps $\pi_{M,N}$ and $R^M_{w,w'}$ descend to quotient cochain maps on $C^{\bullet,\mathrm{reg}}$. Define the finite-window quotient class

$$\eta^{\mathrm{reg}}_{n_0,M} = \big[q_M(\mathfrak{o}^{\mathrm{ns}}_{n_0,M})\big] \in H^1\big(C^{\bullet,\mathrm{reg}}_{w,M}(I)\big).$$

Then, for a fixed M,

$$[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}] \in -\operatorname{im} H^1(\Xi^{\mathrm{reg}}_M) \quad\Longleftrightarrow\quad \eta^{\mathrm{reg}}_{n_0,M} = 0.$$

Equivalently, the regular-density closed-exchange branch at window M is solvable exactly when the image of the residual is exact in $Q^\bullet_{w,M}(I)/\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$.

The compatible classes

$$\eta^{\mathrm{reg}}_{n_0} = \big(\eta^{\mathrm{reg}}_{n_0,M}\big)_M \in \varprojlim_M H^1\big(C^{\bullet,\mathrm{reg}}_{w,M}(I)\big)$$

are the quotient shadow of the closed-exchange obstruction

$$\beta^{\mathrm{cl,reg}}_{n_0} \in \operatorname{coker}\left(\varprojlim_M H^1(\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)) \xrightarrow{\varprojlim H^1(\Xi^{\mathrm{reg}}_M)} \varprojlim_M H^1(Q^\bullet_{w,M}(I))\right).$$

If $\eta^{\mathrm{reg}}_{n_0} \neq 0$, then $\beta^{\mathrm{cl,reg}}_{n_0} \neq 0$. On the zero locus of $\eta^{\mathrm{reg}}_{n_0}$, choose windowwise classes

$$\alpha_M \in H^1(\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)), \qquad H^1(\Xi^{\mathrm{reg}}_M)(\alpha_M) = -[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}].$$

Let

$$K^{\mathrm{reg}}_M = \ker\Big(H^1(\Xi^{\mathrm{reg}}_M) \colon H^1(\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)) \to H^1(Q^\bullet_{w,M}(I))\Big).$$

For the cofinal successor tower the differences

$$\rho^{X,\mathrm{reg}}_{M+1,M}\alpha_{M+1} - \alpha_M \in K^{\mathrm{reg}}_M$$

define a Roos class

$$\kappa^{\mathrm{reg}}_{n_0} \in \varprojlim\nolimits^1_M K^{\mathrm{reg}}_M.$$

It is independent of the chosen windowwise lifts up to Roos coboundary, and

$$\beta^{\mathrm{cl,reg}}_{n_0} = 0 \quad\Longleftrightarrow\quad \eta^{\mathrm{reg}}_{n_0} = 0 \text{ and } \kappa^{\mathrm{reg}}_{n_0} = 0.$$

In particular, suppose the branch is restricted to a complete regular-density subhabitat

$$Q^{\bullet,\mathrm{rd}}_{w,M}(I) \subset Q^\bullet_{w,M}(I)$$

on which

$$Q^{\bullet,\mathrm{rd}}_{w,M}(I) = \mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$$

compatibly with truncation and admissible weight transport, and assume $\mathfrak{o}^{\mathrm{ns}}_{n_0,M} \in Z^1(Q^{\bullet,\mathrm{rd}}_{w,M}(I))$ for all M. Then

$$\beta^{\mathrm{cl,reg}}_{n_0} = 0.$$

The canonical compatible choice $W_{n_0,M} = -\mathfrak{o}^{\mathrm{ns}}_{n_0,M}$ and $C_{n_0,M} = 0$ has

$$\mu^{\mathrm{cl}}_{n_0} = 0, \qquad \lambda^{\mathrm{cl}}_{n_0} = 0.$$

Thus a regular-density equality $\mathcal{X}^{\bullet,\mathrm{reg}} = Q^{\bullet,\mathrm{rd}}$ kills the first residual only for residuals lying in that specified subhabitat. Outside it, the exact obstruction is the quotient/Roos pair $(\eta^{\mathrm{reg}}_{n_0}, \kappa^{\mathrm{reg}}_{n_0})$ above.

Proof. Proposition 3.100 says that $\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$ is a complete d_M-stable subcomplex of $Q^{\bullet}_{w,M}(I)$, and that truncation and admissible weight transport preserve this subcomplex. Therefore the quotient complexes and their transition maps are defined. Since the residuals commute with truncation, the classes $\eta^{\mathrm{reg}}_{n_0,M}$ form an inverse-system element; weight transport gives the same statement for w'.

For a fixed window, the short exact sequence of complexes

$$0 \longrightarrow \mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I) \xrightarrow{\Xi^{\mathrm{reg}}_M} Q^{\bullet}_{w,M}(I) \xrightarrow{q_M} C^{\bullet,\mathrm{reg}}_{w,M}(I) \longrightarrow 0$$

gives the long exact cohomology sequence. Exactness at $H^1(Q^{\bullet}_{w,M}(I))$ gives

$$\operatorname{im} H^1(\Xi^{\mathrm{reg}}_M) = \ker H^1(q_M).$$

The sign is irrelevant over $\mathbb{C}$. Hence the image criterion is exactly the vanishing of $[q_M(\mathfrak{o}^{\mathrm{ns}}_{n_0,M})]$. Equivalently, if $q_M(\mathfrak{o}^{\mathrm{ns}}_{n_0,M}) = -\bar{d}_M \bar{C}_M$, choose any lift C_M of $\bar{C}_M$. Then $\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + d_M C_M$ lies in $\mathcal{X}^{1,\mathrm{reg}}_{w,M}(I)$ and is closed, so $W_{n_0,M} = -(\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + d_M C_M)$ solves the branch equation. The converse follows by applying q_M to the branch equation.

Passing to the inverse limit, the quotient maps induce a map from the displayed cokernel to $\varprojlim_M H^1(C^{\bullet,\mathrm{reg}}_{w,M}(I))$, and the image of $\beta^{\mathrm{cl,reg}}_{n_0}$ is precisely $\eta^{\mathrm{reg}}_{n_0}$. Thus nonvanishing of $\eta^{\mathrm{reg}}_{n_0}$ forces nonvanishing of $\beta^{\mathrm{cl,reg}}_{n_0}$.

Assume now $\eta^{\mathrm{reg}}_{n_0} = 0$. Windowwise exactness gives the classes α_M. Since both $\rho^{X,\mathrm{reg}}_{M+1,M}\alpha_{M+1}$ and α_M map to $-[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}]$, their difference lies in K^{reg}_M. Changing the windowwise lift α_M by an element of K^{reg}_M changes the difference family by the Roos coboundary. Hence the class $\kappa^{\mathrm{reg}}_{n_0}$ is well defined. It vanishes exactly when the α_M can be adjusted to a compatible element of $\varprojlim_M H^1(\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I))$ mapping to $-([\mathfrak{o}^{\mathrm{ns}}_{n_0,M}])_M$. This is exactly the vanishing of the cokernel class $\beta^{\mathrm{cl,reg}}_{n_0}$.

On a subhabitat $Q^{\bullet,\mathrm{rd}} = \mathcal{X}^{\bullet,\mathrm{reg}}$, the inclusion is the identity. If the residual lies in that subhabitat, then $W_{n_0,M} = -\mathfrak{o}^{\mathrm{ns}}_{n_0,M}$ is a compatible d_M-cocycle in $\mathcal{X}^{1,\mathrm{reg}}_{w,M}(I)$, and $C_{n_0,M} = 0$ gives

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + \Xi^{\mathrm{reg}}_M(W_{n_0,M}) + d_M C_{n_0,M} = 0.$$

The representative differences defining $\mu^{\mathrm{cl}}_{n_0}$ are zero because the residuals are compatible, and the primitive differences defining $\lambda^{\mathrm{cl}}_{n_0}$ are zero because every primitive is chosen to be 0. This proves the saturated regular-density claim. □

PROPOSITION 3.104 *(Actual first residual and the regular-density decision).* The first non-scalar residual used in the closed-exchange criterion is not an additional symbol. For the chosen scalar-contact chain projection, the chosen lower-order counterterms, and the finite window M, it is the curvature coefficient

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} = [\hbar^{n_0}]\Big(\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar,M} + d_M C_{<n_0,M} + \frac{1}{2}\{C_{<n_0,M}, C_{<n_0,M}\}_{\hbar,M}\Big) \in Z^1\big(Q^{\bullet}_{w,M}(I)\big),$$

where n_0 is the first order at which this compatible family is not killed by the lower-order choices. Thus the residual lies in the saturated regular-density subhabitat

$$Q^{\bullet,\mathrm{rd}}_{w,M}(I) = X^{\bullet,\mathrm{reg}}_{w,M}(I)$$

if and only if the following regular-density residual certificate is supplied for every finite window M, compatibly with $\pi_{M,N}$ and $R^{M}_{w,w'}$.

(R1) The displayed curvature coefficient is represented, in $Q^1_{w,M}(I)$, by a complete filtered sum whose image in every finite filtration quotient is finite:

$$\sum_{\Gamma,\ell} a_{\Gamma,\ell}\, W^{R,\mathrm{reg},M}_{\Gamma,L,w}(a_{\Gamma,\ell,\bullet}, b_{\Gamma,\ell,\bullet}dt; f_{\Gamma,\ell,\bullet}, \rho_{\Gamma,\ell,\bullet}) \;+\; d_M B^{\mathrm{reg}}_{n_0,M},$$

where all defect cotangent labels are smooth compactly supported densities $b_{\Gamma,\ell,j}(t)dt$, all coefficient labels lie in degree $\leq M$, and $B^{\mathrm{reg}}_{n_0,M}$ is a regular-density local counterterm in the graph/counterterm habitat.

(R2) The representative has zero scalar-contact projection:

$$\widehat{\sigma}_{\mathrm{sc},M}(\mathfrak{o}^{\mathrm{ns}}_{n_0,M}) = 0$$

(R3) The equality with the curvature coefficient above holds in the local-functional complex $O^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}^{M}_{w}|_I; A^{\mathrm{pp},w,M}_{\partial,\hbar}(I))$, not merely after applying cohomology or scalar reduction.

(R4) The representatives commute with finite-window truncation and admissible weight transport:

$$\pi_{M,N}\mathfrak{o}^{\mathrm{ns}}_{n_0,M} = \mathfrak{o}^{\mathrm{ns}}_{n_0,N}, \qquad R^{M}_{w,w'}\mathfrak{o}^{\mathrm{ns}}_{n_0,M,w} = \mathfrak{o}^{\mathrm{ns}}_{n_0,M,w'}$$

If this certificate is supplied, then

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} \in Z^1\big(X^{\bullet,\mathrm{reg}}_{w,M}(I)\big)$$

for every M. The closed-exchange equation is solved by the explicit compatible representatives

$$W_{n_0,M} = -\mathfrak{o}^{\mathrm{ns}}_{n_0,M}, \qquad C_{n_0,M} = 0,$$

and hence

$$\eta^{\mathrm{reg}}_{n_0} = 0, \qquad \kappa^{\mathrm{reg}}_{n_0} = 0, \qquad \beta^{\mathrm{cl,reg}}_{n_0} = 0, \qquad \mu^{\mathrm{cl}}_{n_0} = 0, \qquad \lambda^{\mathrm{cl}}_{n_0} = 0$$

for this saturated branch.

If the certificate is not supplied, the regular-density branch is not decided. The exact computable obstruction is

$$\eta^{\mathrm{reg}}_{n_0,M} = \Big[q_M[\hbar^{n_0}]\Big(\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar,M} + d_M C_{<n_0,M} + \frac{1}{2}\{C_{<n_0,M}, C_{<n_0,M}\}_{\hbar,M}\Big)\Big] \in H^1\big(C^{\bullet,\mathrm{reg}}_{w,M}(I), \bar{d}_M\big).$$

Computing this class requires the finite-window coefficient array of $\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar,M}$, the lower-order counterterms $C_{<n_0,M}$, the scalar-contact chain projection $\widehat{\sigma}_{\mathrm{sc},M}$, and the reduction of the displayed degree-one cochain modulo the regular-density graph/counterterm subcomplex. If all these quotient classes vanish, choose quotient primitives $\bar{C}_{n_0,M}$ and lifts $C_{n_0,M} \in \boldsymbol{Q}^0_{w,M}(I)$ with

$$q_M(\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + d_M C_{n_0,M}) = 0.$$

Then

$$x_{n_0,M} := \mathfrak{o}^{\mathrm{ns}}_{n_0,M} + d_M C_{n_0,M} \in Z^1\big(\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)\big), \qquad \alpha_M = [-x_{n_0,M}]$$

is a windowwise lift of $-[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}]$. The remaining cohomology-lift obstruction is precisely

$$\kappa^{\mathrm{reg}}_{n_0} = \Big[\big(\rho^{X,\mathrm{reg}}_{M+1,M}\alpha_{M+1} - \alpha_M\big)_M\Big] \in \varprojlim_M{}^1 \ker H^1(\Xi^{\mathrm{reg}}_M).$$

Only after this class vanishes does the regular-density branch have a compatible cohomology lift; the representative and primitive compatibility classes $\mu^{\mathrm{cl}}_{n_0}$ and $\lambda^{\mathrm{cl}}_{n_0}$ are then the secondary Roos classes of Theorem 3.102.

Proof. The formula for $\mathfrak{o}^{\mathrm{ns}}_{n_0,M}$ is the order n_0 specialization of Theorem 3.98. The Bianchi identity in the QME complex gives $d_M\mathfrak{o}^{\mathrm{ns}}_{n_0,M} = 0$, and the chosen scalar-contact chain projection places it in $\boldsymbol{Q}^{\bullet}_{w,M}(I)$.

Conditions (R1)–(R4) are exactly the definition of membership in the complete regular-density graph/counterterm subcomplex constructed in Definition 3.99 and Proposition 3.100. Hence the certificate is equivalent to $\mathfrak{o}^{\mathrm{ns}}_{n_0,M} \in Z^1(\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I))$ compatibly in the finite-window tower. Since Ξ^{reg}_M is inclusion on the saturated subhabitat, $W_{n_0,M} = -\mathfrak{o}^{\mathrm{ns}}_{n_0,M}$ and $C_{n_0,M} = 0$ give

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + \Xi^{\mathrm{reg}}_M(W_{n_0,M}) + d_M C_{n_0,M} = 0$$

Compatibility of the residual family makes the Roos differences for W vanish, and the zero primitives make the counterterm Roos differences vanish. This proves the displayed vanishing of η, κ, β, μ, and λ.

Without the certificate, the only invariant operation supplied by the constructed regular-density tower is quotienting by $\mathcal{X}^{\bullet,\mathrm{reg}}$. Applying Proposition 3.103 to the explicit curvature coefficient gives the displayed $\eta^{\mathrm{reg}}_{n_0,M}$. Its computation needs precisely the coefficient array of the curvature coefficient, the lower-order counterterms entering it, the chain projection defining $\boldsymbol{Q}$, and the quotient reduction modulo $\mathcal{X}^{\mathrm{reg}}$. If $\eta^{\mathrm{reg}}_{n_0,M} = 0$, a quotient primitive gives $q_M(\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + d_M C_{n_0,M}) = 0$, hence $x_{n_0,M} \in \mathcal{X}^{1,\mathrm{reg}}_{w,M}(I)$. Closedness follows from $d_M\mathfrak{o}^{\mathrm{ns}}_{n_0,M} = 0$ and $d_M^2 = 0$. The Roos differences of the resulting cohomology classes lie in $\ker H^1(\Xi^{\mathrm{reg}}_M)$ because both terms map to $-[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}]$. This is exactly the $\kappa^{\mathrm{reg}}_{n_0}$ obstruction of Proposition 3.103. □

Proposition 3.105 *(Regular-density certificate for the evaluated curvature).* Work at a fixed finite window M. Assume the finite-window graph/QME package of Definition 3.170 is the regular-density package: every cotangent defect label is $B_j = b_j(t)dt \in \mathcal{D}^{\mathrm{reg}}_c(I)$, the finite graph list is closed under the d_M-boundary and BV-bracket terms appearing in the package, and the scalar-contact projection is zero on the chosen subhabitat. Let

$$\mathrm{Ob}^{\mathrm{red,reg}}_{w,\partial,\hbar,M} = \mathrm{ev}_{\mathcal{G}_M}\,\mathrm{Curv}_{\mathbf{K}}\big(\kappa^{\mathrm{graph,reg},M}_{\hbar,w,I}\big) \in \boldsymbol{Q}^1_{w,M}(I)$$

be the evaluated reduced curvature of the regular-density graph kernel. Suppose the lower-order counterterms are compatible and regular-density:

$$C^{\mathrm{reg}}_{<n_0,M} \in \boldsymbol{Q}^{0,\mathrm{rd}}_{w,M}(I) = \mathcal{X}^{0,\mathrm{reg}}_{w,M}(I), \qquad \pi_{M,N}C^{\mathrm{reg}}_{<n_0,M} = C^{\mathrm{reg}}_{<n_0,N}.$$

Put

$$\mathfrak{o}^{\mathrm{ns,reg}}_{n_0,M} = [\hbar^{n_0}]\Big(\mathrm{Ob}^{\mathrm{red,reg}}_{w,\partial,\hbar,M} + d_M C^{\mathrm{reg}}_{<n_0,M} + \frac{1}{2}\{C^{\mathrm{reg}}_{<n_0,M}, C^{\mathrm{reg}}_{<n_0,M}\}_{\hbar,M}\Big).$$

Then

$$\mathfrak{o}^{\mathrm{ns,reg}}_{n_0,M} \in Z^1\big(\mathcal{Q}^{\bullet,\mathrm{rd}}_{w,M}(I)\big) = Z^1\big(\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)\big)$$

compatibly with finite-window truncation and admissible weight transport.

More explicitly, after expanding the finite graph package at the coefficient of $\hbar^{n_0}$, there is a finite indexing set $\mathcal{A}^{\mathrm{reg}}_{n_0,M}$ and coefficients $\varepsilon_{\alpha,M} \in \mathbb{C}$ such that

$$\mathfrak{o}^{\mathrm{ns,reg}}_{n_0,M} = \sum_{\alpha\in\mathcal{A}^{\mathrm{reg}}_{n_0,M}} \varepsilon_{\alpha,M}\, G^M_{\alpha,w} + [\hbar^{n_0}] d_M C^{\mathrm{reg}}_{<n_0,M} + \frac{1}{2}[\hbar^{n_0}]\{C^{\mathrm{reg}}_{<n_0,M}, C^{\mathrm{reg}}_{<n_0,M}\}_{\hbar,M},$$

where each graph generator is of the form

$$G^M_{\alpha,w} = W^{R,\mathrm{reg},M}_{\Gamma_\alpha,L,w}(a_{\alpha,\bullet}, b_{\alpha,\bullet}dt; f_{\alpha,\bullet}, \rho_{\alpha,\bullet}), \qquad b_{\alpha,j} \in C^\infty_c(I),$$

or is one of the regular-density local counterterm generators attached to such a labelled graph. The last two displayed terms also lie in $\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$: the first by d_M-stability, the second by the bracket-closure clause in the finite-window graph/QME package.

Hence the regular-density residual certificate of Proposition 3.104 is supplied for this actual evaluated curvature. The closed-exchange solution is

$$W_{n_0,M} = -\mathfrak{o}^{\mathrm{ns,reg}}_{n_0,M}, \qquad C_{n_0,M} = 0,$$

and

$$\eta^{\mathrm{reg}}_{n_0} = \kappa^{\mathrm{reg}}_{n_0} = \beta^{\mathrm{cl,reg}}_{n_0} = \mu^{\mathrm{cl}}_{n_0} = \lambda^{\mathrm{cl}}_{n_0} = 0$$

on this regular-density branch.

If the chosen curvature contains non-regular labelled data, choose any linear splitting modulo $\mathcal{Q}^{\bullet,\mathrm{rd}}_{w,M}(I)$ and write

$$\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar,M} = \mathrm{Ob}^{\mathrm{red,reg}}_{w,\partial,\hbar,M} + \mathrm{Ob}^{\mathrm{red},\neg\mathrm{rd}}_{w,\partial,\hbar,M}, \qquad C_{<n_0,M} = C^{\mathrm{reg}}_{<n_0,M} + C^{\neg\mathrm{rd}}_{<n_0,M}.$$

The resulting quotient class is independent of the splitting. The non-regular summand is generated exactly by graph weights

$$W^{R,\mathrm{WF},M}_{\Gamma,L,w}(a_\bullet, B_\bullet; f_\bullet, \rho_\bullet) \quad \text{with some } B_j \notin \mathcal{D}^{\mathrm{reg}}_c(I),$$

by the corresponding distributional current counterterms, and by d_M- or BV-bracket terms involving such counterterms. Its quotient obstruction is the explicit component

$$\begin{aligned}\eta^{\mathrm{reg}}_{n_0,M} = \Big[q_M[\hbar^{n_0}]\Big(&\mathrm{Ob}^{\mathrm{red},\neg\mathrm{rd}}_{w,\partial,\hbar,M} + d_M C^{\neg\mathrm{rd}}_{<n_0,M} \\ &+ \{C^{\mathrm{reg}}_{<n_0,M}, C^{\neg\mathrm{rd}}_{<n_0,M}\}_{\hbar,M} + \frac{1}{2}\{C^{\neg\mathrm{rd}}_{<n_0,M}, C^{\neg\mathrm{rd}}_{<n_0,M}\}_{\hbar,M}\Big)\Big]\end{aligned}$$

in

$$H^1\big(\mathcal{Q}^\bullet_{w,M}(I)/\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)\big).$$

Thus a single wavefront-admissible but non-smooth cotangent label, or a distributional current counterterm forced by such a label, is the exact datum that leaves $Q^{\mathrm{rd}} = X^{\mathrm{reg}}$.

Proof. Proposition B.17 constructs $W^{R,\mathrm{reg},M}_{\Gamma,L,w}$ with smooth compactly supported density labels and proves compatibility with interval extension, finite-window truncation, and admissible weight transport. Definition 3.99 then puts all scalar-zero regular graph weights and their regular local counterterms in $\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$, and closes this space under d_M. The finite-window graph/QME package includes the BV-bracket terms forced by the finite graph differential, so the quadratic counterterm term remains in the same complete regular-density graph/counterterm habitat. Hence every summand in the displayed $\hbar^{n_0}$-coefficient lies in $\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$.

The Bianchi identity in the convolution dg Lie algebra gives closedness of the first unsolved coefficient after the lower-order equations have been imposed. The scalar projection is zero on the chosen regular-density package, so this closed coefficient lies in $\mathcal{Q}^{\bullet}_{w,M}(I)$. The same truncation and weight-transport identities hold term by term for the graph weights, counterterms, differential, and bracket; therefore the residuals are compatible in the finite-window tower.

Since $\mathcal{Q}^{\bullet,\mathrm{rd}}_{w,M}(I) = \mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$ on this branch and Ξ^{reg}_M is inclusion, the displayed $W_{n_0,M}$ and $C_{n_0,M} = 0$ give

$$\mathfrak{o}^{\mathrm{ns,reg}}_{n_0,M} + \Xi^{\mathrm{reg}}_M(W_{n_0,M}) + d_M C_{n_0,M} = 0.$$

Compatibility of the residual family makes the closed-exchange Roos differences vanish, and the primitive family is identically zero. The quotient class $\eta^{\mathrm{reg}}_{n_0}$, the lift class $\kappa^{\mathrm{reg}}_{n_0}$, and the three closed-exchange and counterterm obstruction classes therefore vanish.

For a non-regular label package, the quotient map q_M kills exactly the regular-density subcomplex. Expanding the same curvature expression into regular and non-regular summands therefore leaves only the four non-regular terms displayed in the formula for $\eta^{\mathrm{reg}}_{n_0,M}$: the non-regular graph curvature, the d_M-image of non-regular lower counterterms, the mixed bracket with a regular counterterm, and the purely non-regular bracket. The wavefront-admissible graph layer of Proposition B.19 is theorem-grade as a distributional graph layer, but it is not a regular-density layer when some $B_j \notin \mathcal{D}^{\mathrm{reg}}_c(I)$. Such a term can enter the regular-density branch only if its displayed quotient class is killed by an explicit degree-zero primitive in the ambient distributional quotient complex. This proves the obstruction formula. □

Theorem 3.106 *(Wavefront singular-current obstruction).* Fix a finite window M and a wavefront-admissible finite graph/QME package as in Proposition B.19. Put

$$\mathcal{S}_c(I) = \mathcal{D}'_c(I)/\mathcal{D}^{\mathrm{reg}}_c(I).$$

A non-smooth label $B \in \mathcal{D}'_c(I)$ with non-empty wavefront has nonzero class in $\mathcal{S}_c(I)$. Hence no smoothing or finite-part prescription can give a canonical regular-density representative of B modulo $\mathcal{D}^{\mathrm{reg}}_c(I)$: if $B - B^{\mathrm{sm}} \in \mathcal{D}^{\mathrm{reg}}_c(I)$ for some $B^{\mathrm{sm}} \in \mathcal{D}^{\mathrm{reg}}_c(I)$, then B was already regular.

Let the first non-scalar wavefront residual be written as

$$\mathfrak{o}^{\mathrm{ns,WF}}_{n_0,M} = [\hbar^{n_0}]\Big(\mathrm{Ob}^{\mathrm{red,WF}}_{w,\partial,\hbar,M} + d_M C^{\mathrm{WF}}_{<n_0,M} + \frac{1}{2}\{C^{\mathrm{WF}}_{<n_0,M}, C^{\mathrm{WF}}_{<n_0,M}\}_{\hbar,M}\Big) \in Z^1\big(\mathcal{Q}^{\bullet}_{w,M}(I)\big),$$

with at least one cotangent label in $\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M) \setminus \mathcal{D}^{\mathrm{reg}}_c(I)$ or at least one distributional current counterterm. Its singular-current component is

$$\begin{aligned}\mathfrak{s}^{\mathrm{WF}}_{n_0,M} = q_M[\hbar^{n_0}]\Big(&\mathrm{Ob}^{\mathrm{red},\neg\mathrm{rd}}_{w,\partial,\hbar,M} + d_M C^{\neg\mathrm{rd}}_{<n_0,M} \\ &+ \{C^{\mathrm{reg}}_{<n_0,M}, C^{\neg\mathrm{rd}}_{<n_0,M}\}_{\hbar,M} + \frac{1}{2}\{C^{\neg\mathrm{rd}}_{<n_0,M}, C^{\neg\mathrm{rd}}_{<n_0,M}\}_{\hbar,M}\Big) \in C^{1,\mathrm{reg}}_{w,M}(I),\end{aligned} \tag{13}$$

where

$$C^{\bullet,\mathrm{reg}}_{w,M}(I) = Q^{\bullet}_{w,M}(I)/\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I).$$

The obstruction class is exactly

$$\eta^{\mathrm{reg}}_{n_0,M} = [\mathfrak{s}^{\mathrm{WF}}_{n_0,M}] \in H^1\big(C^{\bullet,\mathrm{reg}}_{w,M}(I)\big).$$

More concretely, for a graph summand $W^{R,\mathrm{WF},M}_{\Gamma,L,w}(a_\bullet, B_\bullet; f_\bullet, \rho_\bullet)$, choose arbitrary smooth densities $B^{\mathrm{sm}}_j \in \mathcal{D}^{\mathrm{reg}}_c(I)$ and put $S_j = B_j - B^{\mathrm{sm}}_j$. By multilinearity,

$$q_M W^{R,\mathrm{WF},M}_{\Gamma,L,w}(a_\bullet, B_\bullet; f_\bullet, \rho_\bullet) = q_M \sum_{\emptyset \neq J \subset \{1,\dots,r\}} W^{R,\mathrm{WF},M}_{\Gamma,L,w}(a_\bullet, B^J_\bullet; f_\bullet, \rho_\bullet),$$

where $B^J_j = S_j$ for $j \in J$ and $B^J_j = B^{\mathrm{sm}}_j$ for $j \notin J$. This expression is independent of the chosen smooth densities after passing to $C^{\bullet,\mathrm{reg}}_{w,M}(I)$. The zero-internal-edge case, whenever the generator lies in the non-scalar kernel, is the current class

$$q_M \iota_I(\Theta^w_\rho(B)) = [B] \otimes \rho \in \mathcal{S}_c(I)\widehat{\otimes} D_{w,\leq M},$$

so a singular cotangent current with nonzero coefficient is already a nonzero regular-density quotient generator.

The wavefront of $\mathfrak{s}^{\mathrm{WF}}_{n_0,M}$ is contained in the finite union obtained from the graph pushforward of

$$\mathrm{WF}(K^M_{\Gamma,\alpha}) + \mathrm{WF}(B_\Gamma)$$

over the Hadamard coefficients appearing at order n_0, together with the local counterterm wavefronts supported on

$$\bigcup_{\Delta \in \mathrm{Diag}(\Gamma)} \Delta \cap \mathrm{supp}(B_\Gamma).$$

Thus the component in (13) is the exact wavefront/counterterm component left after quotienting by the regular-density graph complex.

A finite-window counterterm kills the non-regular branch if and only if there is a degree-zero distributional counterterm $C^{\mathrm{sing}}_{n_0,M} \in Q^0_{w,M}(I)$ such that

$$\bar{d}_M\, q_M(C^{\mathrm{sing}}_{n_0,M}) = -\mathfrak{s}^{\mathrm{WF}}_{n_0,M}.$$

In that case $\eta^{\mathrm{reg}}_{n_0,M} = 0$, and the remaining compatibility obstruction is the Roos class $\kappa^{\mathrm{reg}}_{n_0}$ of Proposition 3.103. If no such distributional primitive exists, then

$$\eta^{\mathrm{reg}}_{n_0,M} \neq 0,$$

and the regular-density closed-exchange branch cannot be repaired by a smoothing prescription. The only honest enlargement is a new wavefront-labelled closed-exchange complex; it is not $\mathcal{X}^{\bullet,\mathrm{reg}}$.

Proof. A compactly supported smooth density has empty wavefront set. Therefore a compactly supported distribution with non-empty wavefront cannot differ from a smooth density by another smooth density. This proves the label-level obstruction to canonical smoothing. The zero-internal-edge generator

shows that this obstruction is already visible in the current target: $\Theta_\rho^w(B)$ records the distribution B linearly in $\mathcal{D}'_c(I)\widehat{\otimes}D_{w,\le M}$, and the quotient by $\mathcal{X}^{\bullet,\mathrm{reg}}$ kills only the smooth-density subspace.

Proposition B.19 constructs the finite-window wavefront graph weights and the corresponding distributional current counterterms. The regular-density subcomplex of Definition 3.99 contains exactly the smooth-label graph weights, their regular counterterms, and their d_M-closures. Hence applying q_M to the first residual removes precisely the regular-density terms and leaves the four non-regular terms displayed in Formula (13). Changing the auxiliary smooth densities B_j^{sm} changes the expansion only by graph terms with all labels smooth; these lie in $\mathcal{X}^{\bullet,\mathrm{reg}}$, so the quotient expression is independent of the splitting.

Closedness follows from the same Bianchi identity used in Proposition 3.105, after passing to the quotient complex. The wavefront and support containment is exactly the Hörmander product and extension statement in Proposition B.19: graph terms have wavefront contained in the pushforward of the product wavefronts, and local extension ambiguities are supported on the collision diagonals meeting $\mathrm{supp}(B_\Gamma)$.

Finally, $H^1(C^{\bullet,\mathrm{reg}}_{w,M}(I),\bar{d}_M)$ is the cohomology of the quotient complex. Therefore the class represented by $\mathfrak{s}^{\mathrm{WF}}_{n_0,M}$ vanishes exactly when it is a $\bar{d}_M$-boundary, equivalently exactly when a degree-zero distributional counterterm $C^{\mathrm{sing}}_{n_0,M}$ with the displayed equation exists. If it is not a boundary, the quotient obstruction $\eta^{\mathrm{reg}}_{n_0,M}$ is nonzero and the regular-density branch is obstructed. □

Theorem 3.107 *(Distributional primitive cone for the wavefront quotient).* Keep the hypotheses and notation of Theorem 3.106, and assume that the chosen wavefront graph/QME packages are compatible under finite-window truncation and admissible weight transport. Thus

$$\pi_{M,N}\mathfrak{o}^{\mathrm{ns,WF}}_{n_0,M}=\mathfrak{o}^{\mathrm{ns,WF}}_{n_0,N},\qquad R^M_{w,w'}\mathfrak{o}^{\mathrm{ns,WF}}_{n_0,M,w}=\mathfrak{o}^{\mathrm{ns,WF}}_{n_0,M,w'}.$$

Let

$$\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M}:=\mathfrak{o}^{\mathrm{ns,WF}}_{n_0,M}\in Z^1\big(Q^\bullet_{w,M}(I)\big)$$

be the closed residual lift of $\mathfrak{s}^{\mathrm{WF}}_{n_0,M}$, so that $q_M\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M}=\mathfrak{s}^{\mathrm{WF}}_{n_0,M}$.

Define the local wavefront-primitive extension by

$$\widetilde{Q}^{\bullet,\mathrm{WFpr}}_{w,M}(I)=Q^\bullet_{w,M}(I)\widehat{\oplus}\mathbb{C}[[\hbar]]\cdot p^{\mathrm{WF}}_{n_0,M},\qquad |p^{\mathrm{WF}}_{n_0,M}|=0,$$

completed in the same $\hbar$-adic, scalar-contact, and finite-window filtrations as $Q^\bullet_{w,M}(I)$. The new generator is placed in $\hbar$-order n_0, in the scalar-contact filtration of $\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M}$, and with local support and wavefront label equal to the support and wavefront cone displayed in Theorem 3.106. Set

$$\widetilde{d}_M|_Q=d_M,\qquad \widetilde{d}_M p^{\mathrm{WF}}_{n_0,M}=-\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M}.$$

Let

$$\widetilde{C}^{\bullet,\mathrm{reg}}_{w,M}(I)=\widetilde{Q}^{\bullet,\mathrm{WFpr}}_{w,M}(I)/\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I),$$

and write again q_M for the quotient map and $\bar{d}_M$ for the induced differential. Then $\widetilde{d}_M^2=0$, the inclusion $Q^\bullet_{w,M}(I)\hookrightarrow\widetilde{Q}^{\bullet,\mathrm{WFpr}}_{w,M}(I)$ is a filtered cochain map, and the element

$$C^{\mathrm{sing}}_{n_0,M}:=p^{\mathrm{WF}}_{n_0,M}$$

satisfies the desired quotient primitive equation

$$\bar{d}_M q_M(C^{\mathrm{sing}}_{n_0,M}) = -\mathfrak{s}^{\mathrm{WF}}_{n_0,M}.$$

The extension is strict Mittag–Leffler on compatible finite-window towers: define

$$\widetilde{\pi}_{M,N} p^{\mathrm{WF}}_{n_0,M} = p^{\mathrm{WF}}_{n_0,N}$$

and let $\widetilde{\pi}_{M,N}$ be $\pi_{M,N}$ on $Q^\bullet_{w,M}(I)$. For admissible weights set

$$\widetilde{R}^M_{w,w'} p^{\mathrm{WF}}_{n_0,M,w} = p^{\mathrm{WF}}_{n_0,M,w'}$$

and let $\widetilde{R}^M_{w,w'}$ be $R^M_{w,w'}$ on $Q^\bullet_{w,M}(I)$. These maps are continuous filtered cochain maps and commute with the quotient to $\widetilde{C}^{\bullet,\mathrm{reg}}$.

This theorem is an extension theorem. It does not say that $p^{\mathrm{WF}}_{n_0,M}$ belongs to the original Costello counterterm complex, and it does not identify a singular current with a smooth density. Inside the original complex $Q^\bullet_{w,M}(I)$, existence of a primitive remains exactly the vanishing condition $\eta^{\mathrm{reg}}_{n_0,M} = 0$ of Theorem 3.106.

Proof. The residual $\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M}$ is d_M-closed by the Bianchi identity used in Theorem 3.106. Hence

$$\widetilde{d}^2_M p^{\mathrm{WF}}_{n_0,M} = -d_M \widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M} = 0,$$

and $\widetilde{d}^2_M = 0$ on the old summand because $d^2_M = 0$. The image of the new generator has the same local support, wavefront cone, $\hbar$-order, scalar-contact order, and finite coefficient window assigned to the generator. Therefore the differential is continuous for the completed filtration topology.

Since $q_M \widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M} = \mathfrak{s}^{\mathrm{WF}}_{n_0,M}$, applying the quotient map to the defining differential gives

$$\bar{d}_M q_M(p^{\mathrm{WF}}_{n_0,M}) = q_M \widetilde{d}_M p^{\mathrm{WF}}_{n_0,M} = -\mathfrak{s}^{\mathrm{WF}}_{n_0,M}.$$

This is the displayed primitive equation.

The truncation and weight-transport formulas are cochain identities because the closed residual lifts are compatible:

$$\widetilde{\pi}_{M,N} \widetilde{d}_M p^{\mathrm{WF}}_{n_0,M} = -\pi_{M,N} \widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M} = -\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,N} = \widetilde{d}_N \widetilde{\pi}_{M,N} p^{\mathrm{WF}}_{n_0,M},$$

and similarly

$$\widetilde{R}^M_{w,w'} \widetilde{d}_M p^{\mathrm{WF}}_{n_0,M,w} = -R^M_{w,w'} \widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M,w} = -\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M,w'} = \widetilde{d}_M \widetilde{R}^M_{w,w'} p^{\mathrm{WF}}_{n_0,M,w}.$$

On the old summand these are the already proved identities for $Q^\bullet_{w,M}(I)$. The transition maps are surjective on the single new generator and are the old strict Mittag–Leffler maps on Q, so the extended tower is again strict Mittag–Leffler. The same formulas descend to the quotient because $\mathcal{X}^{\bullet,\mathrm{reg}}$ is preserved by truncation and weight transport.

The last assertion is formal. The construction adjoins a new degree-zero generator whose differential is the closed wavefront residual. It leaves the quotient $\mathcal{D}'_c(I)/\mathcal{D}^{\mathrm{reg}}_c(I)$ unchanged on the original current labels, and therefore does not smooth a singular distribution. If the new generator is not adjoined, the primitive equation is precisely the statement that $\mathfrak{s}^{\mathrm{WF}}_{n_0,M}$ is a boundary in the original quotient complex, i.e. that $\eta^{\mathrm{reg}}_{n_0,M}$ vanishes. □

Corollary 3.108 *(Finite-window machinery does not by itself kill the non-scalar class).* The weighted Tate construction supplies the coefficient kernel, graphwise Hadamard control, Mittag–Leffler transport, and regulator-independence inside the admissible finite-window sector. It constructs non-scalar QME counterterms exactly on the vanishing locus of the obstruction vector $\{\mathfrak{O}_n^{\mathrm{ns}}\}_{n\geq 1}$ of Theorem 3.98. If some $\mathfrak{O}_n^{\mathrm{ns}}$ is nonzero, the non-scalar class in

$$H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$$

remains. The obstruction is not removed by changing admissible weight and is not an endpoint effect of the strict product/direct-sum Tate pair.

Proof. The first sentence is the content of Theorems 3.85, 3.166, and 3.96. The second and third sentences are exactly the equivalence and weight-transport statement of Theorem 3.98. The strict endpoint obstruction is separate and is treated in Theorem 3.109. □

Theorem 3.109 *(Strict unweighted endpoint obstruction).* Let

$$\mathfrak{h}_\Pi = \prod_{d\geq 1} H_d, \qquad D_\oplus = \bigoplus_{d\geq 1} H_d^\vee$$

be the original product/direct-sum Tate pair. There is no tensor $K \in \mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$, and no tensor in $D_\oplus \widehat{\otimes} \mathfrak{h}_\Pi$, whose projection to every finite window $d \leq w_0$ is the finite Casimir

$$C_{\mathfrak{h}}^{w_0} = \sum_{d\leq w_0} \sum_i e_{d,i} \otimes e_d^i.$$

Consequently the strict unweighted pair cannot be the endpoint of a finite-window Costello regulator theory satisfying the coefficient kernel requirement (C1) and the windowwise compatibility used in Theorem 3.96. Equivalently, no $q \to 1^+$ family of weighted finite-window kernels has a strict endpoint in $\mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$ whose projections are the $C_{\mathfrak{h}}^{w_0}$.

Moreover, for any admissible weight w, the inclusion $D_\oplus \hookrightarrow D_{\Pi,w} := \prod_{d\geq 1} w(d)^{-1} H_d^\vee$ is only an admissible filtered weak equivalence. It is not an ordinary quasi-isomorphism of chain complexes with zero differential. Thus any observable or obstruction class detecting the quotient $D_{\Pi,w}/D_\oplus$ is outside the proved regulator-independent sector.

Proof. The direct-sum factor $D_\oplus$ is coefficient-finite by Lemma 3.217 item (1): every element of a completed tensor product with $D_\oplus$ has support in only finitely many $H_d^\vee$-degrees. If K projected to $C_{\mathfrak{h}}^{w_0}$ for all w_0, then for every degree d the $H_d^\vee$-component of K would be nonzero. This is infinite support in the direct-sum direction, a contradiction. The same argument applies after reversing the tensor factors.

Costello's heat-kernel BV propagator for the cotangent coefficient field is $P_{\epsilon,L}^{\mathrm{base}} \widehat{\otimes} K$. Since no such K exists in the strict product/direct-sum pair, the strict pair has an operator-level heat homotopy but no coefficient distributional inverse pairing and hence no graphwise regulator endpoint satisfying the displayed finite-window Casimir condition.

Suppose a $q \to 1^+$ endpoint existed in the strict pair. Its finite-window projection would have to be the limit of the weighted finite-window kernels. After the weighted bases are identified with the unweighted basis, every such finite-window kernel is exactly $C_{\mathfrak{h}}^{w_0}$, independent of q. The endpoint would therefore give a tensor K with the forbidden projections above. This proves the endpoint assertion.

Finally, with zero differential,

$$H^0(D_{\Pi,w}/D_\oplus) = D_{\Pi,w}/D_\oplus.$$

The class of any sequence with one nonzero covector in every degree is nonzero in this quotient. Therefore the completion changes ordinary cohomology even though finite quotients and associated graded pieces cannot see the tail. This proves the asserted boundary of regulator independence. □

Definition 3.110 *(Endpoint-admissible coefficient module).* An endpoint for the weighted regulators $w_q(d) = q^d$, $q > 1$, is *admissible* if it consists of a complete Hausdorff coefficient pair $(\mathfrak{h}_*, D_*)$, finite quotients $(\mathfrak{h}_*, D_*)^{w_0}$, and diagonal transport maps from every weighted window to the endpoint window, satisfying:

(E1) the finite quotients form strict Mittag-Leffler towers with surjective transition maps and endpoint quotients $H_{\leq w_0} \oplus H^\vee_{\leq w_0}$;

(E2) there is a coefficient kernel $C_* \in \mathfrak{h}_* \widehat{\otimes} D_*$ whose projection to $d \leq w_0$ is $C_{\mathfrak{h}}^{w_0}$ for every w_0;

(E3) the endpoint transport maps carry differential, Poisson bracket, coadjoint action, propagator contraction, interaction, counterterm representatives, and the CE/PV central-operation generator assignment $c \mapsto \theta$, $u \mapsto O$ to their endpoint versions;

(E4) local observables compute the admissible filtered cohomology of Statement 3.32, so exactness of the strict Mittag-Leffler limit applies;

(E5) the Costello RG recursion $W(P_{\epsilon,L}, I)$ preserves the endpoint tower and the endpoint counterterms.

THEOREM 3.111 *(Endpoint stabilization criterion).* Let $w_q(d) = q^d$, $q > 1$. If there is an endpoint-admissible coefficient module in the sense of Definition 3.110, then the transport maps assemble to isomorphisms

$$R_{q,*}\colon \varprojlim_{w_0}\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{w_0}_{w_q}), Q + \{I_0^{w_q}, -\}\big) \xrightarrow{\cong} \varprojlim_{w_0}\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{w_0}_{*}), Q + \{I_0^{*}, -\}\big)$$

for all $q > 1$, functorially in q. Hence finite-degree local observables, fixed-graph Costello contractions, RG counterterm representatives, and the mixed brane-defect QME obstruction class stabilize at the endpoint. More precisely, if $\mathcal{O}^q_M$ is represented in the finite window $d \leq M$, then its endpoint representative is

$$\mathcal{O}^*_M = R^M_{q,*}\mathcal{O}^q_M$$

and is independent of q after the common finite-window identification with $H_{\leq M} \oplus H^\vee_{\leq M}$. The notation $q \to 1^+$ is therefore stabilization of this finite-window representative, not convergence in the strict product/direct-sum topology. In particular

$$R_{q,*}(\mathrm{Ob}_q) = \mathrm{Ob}_*, \qquad \mathrm{Ob}_q = 0 \Longleftrightarrow \mathrm{Ob}_* = 0.$$

Proof. Replace the second admissible weight w' in Theorem 3.96 by the endpoint object. Conditions (E1)–(E5) are exactly the hypotheses used there: strict Mittag-Leffler exactness, existence of the coefficient kernel, transport of the algebraic and graphwise structures, admissible filtered cohomology, and preservation by RG recursion. The same finite-window diagonal maps therefore give isomorphisms of strict inverse systems. Passing to the exact inverse limit identifies cohomology and sends the obstruction

cocycle to the endpoint obstruction cocycle. Functoriality follows from the composition law of the transport maps on each finite window.

For a representative supported in $d \leq M$, the endpoint statement is already finite-dimensional: $R^M_{q,*}$ is the diagonal map from the q-basis to the endpoint basis, and the endpoint transport commutes with the unweighting maps by condition (E3). Thus the finite-window coefficient array has no remaining dependence on q. The inverse-limit assertion adds exactly the strict Mittag-Leffler compatibility required by (E1) and exactness in the admissible envelope. □

Theorem 3.112 *(Finite-window $q \to 1^+$ stabilization of the CE/PV generator map).* Let $w_q(d) = q^d$, $q > 1$, and let

$$\Phi^q_M \colon C^\bullet_{\mathrm{CE}}\big(\mathfrak{h}_{q,\leq M} \ltimes D_{q,\leq M}[1]\big) \longrightarrow \mathrm{PV}\big(\mathbf{S}(\mathfrak{h}_{q,\leq M})\big)$$

be the finite-window restriction of the weighted CE/PV generator map. Let $\Phi^{0,M}$ be the finite-dimensional unweighted CE/PV map on $(\mathfrak{h}_{\leq M}, D_{\leq M})$. Then

$$U^M_q\, \Phi^q_M = \Phi^{0,M}\, U^M_q$$

as cochain P_0-algebra maps. Consequently the unweighted finite-window representative $U^M_q \Phi^q_M (U^M_q)^{-1}$ is independent of q and has $q \to 1^+$ limit $\Phi^{0,M}$.

If an endpoint-admissible coefficient module exists, the same identity passes to the strict Mittag-Leffler inverse limit: $R_{q,*}\Phi^q = \Phi^* R_{q,*}$. For the original product/direct-sum endpoint the obstruction term is precisely

$$\mathfrak{o}_{\mathrm{Cas}} = \big(C^M_{\mathfrak{h}}\big)_{M\geq 1} \notin \mathfrak{h}_\Pi \widehat{\otimes} D_\oplus,$$

equivalently the nonexistence of a tensor whose finite-window projections are all $C^M_{\mathfrak{h}}$.

Proof. On generators, Φ^q_M sends the CE coordinate dual to $q^d e_{d,i}$ to the Schouten vector coordinate dual to $q^d e_{d,i}$, and sends the CE coordinate dual to $q^{-d} e^i_d[1]$ to the Hamiltonian coordinate $O^q_{d,i}$. The map U^M_q removes the same diagonal weight on both the CE and PV sides. Therefore the square commutes on the generators c, u, θ, O. Since both CE and Schouten differentials are derivations and their structure constants are transported by the same diagonal conjugation, the generator check gives the equality of dg P_0-algebra maps.

The displayed equality says that the finite-window map, after being written in the unweighted basis, is exactly the finite CE/PV map $\Phi^{0,M}$; it has no remaining q-dependence. Hence the $q \to 1^+$ limit exists on every finite window.

Under Definition 3.110, the endpoint maps commute with finite-window transition maps and carry the CE/PV generator assignment to the endpoint assignment. Exactness of the strict Mittag-Leffler inverse limit then passes the finite-window identity to the endpoint. In the original product/direct-sum pair, condition (E2) is violated by Theorem 3.109; the named obstruction is the compatible family of finite Casimirs with no global tensor in $\mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$. □

Corollary 3.113 *(Strict direct-sum endpoint obstruction criterion).* The strict product/direct-sum pair $(\mathfrak{h}_\Pi, D_\oplus)$ violates the endpoint-admissibility criterion exactly at condition (E2): the compatible family of finite Casimirs has no representing coefficient kernel in $\mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$.

Proof. Condition (E2) is exactly the coefficient-kernel condition ruled out by Theorem 3.109. □

PROPOSITION 3.114 *(The completion changes ordinary cohomology).* Let

$$D_{\oplus} = \bigoplus_{d \geq 1} H_d^{\vee}, \qquad D_{\Pi,w} = \prod_{d \geq 1} w(d)^{-1} H_d^{\vee}$$

with zero differential. The finite-window inclusion $D_{\oplus} \hookrightarrow D_{\Pi,w}$ induces an isomorphism on every quotient by the tail $d > w_0$, and therefore is a weak equivalence in the admissible filtered sense of Statement 3.32. It is not a quasi-isomorphism in the ordinary category of chain complexes:

$$H^0(D_{\Pi,w}/D_{\oplus}) = D_{\Pi,w}/D_{\oplus} \neq 0,$$

for example the class of any sequence with one nonzero covector in every degree is nonzero. Thus the weighted product contains tail states absent from the strict continuous dual. Any observable detecting such a tail is outside the regulator-admissible sector of Definition 3.90.

Example 3.115 *(Super-exponential bracket obstruction).* The super-exponential function $w(d) = \exp(d^2)$ is not a degree weight. Indeed

$$\frac{w(p+q-2)}{w(p)w(q)} = \exp\big((p+q-2)^2 - p^2 - q^2\big)$$

is unbounded, already along $p = q \to \infty$. Hence condition (W1) is violated. The Poisson bracket is not bracket-tame in that coefficient topology, and the graphwise constants in Proposition 3.83 need not be finite uniformly in the degrees used by the interaction. The regulator-independence theorem above does not compare such a topology with the admissible weighted theory.

COROLLARY 3.116 *(Maximality of the weighted regulator theorem).* The weighted regulator theorem is maximal for the present construction. It compares all admissible weights $q, q' > 1$ on the finite-window Mittag-Leffler sector. It extends to a $q \to 1^+$ endpoint exactly under Definition 3.110. The original strict product/direct-sum Tate pair carries the obstruction to that endpoint condition.

Proof. Pairwise independence for $q, q' > 1$ is Theorem 3.96. The endpoint implication is Theorem 3.111. The obstruction at the strict endpoint is Corollary 3.113. □

Matched-conventions boundary

Remark 3.117 *(Local-disk regulator datum).* The weighted Tate completion is a local-disk operation on the formal symplectic plane $\mathbb{C}^2$ with brane defect $L = \mathbb{R}_{\text{brane}} \times \{0\} \times \{p\}$ and bulk $\mathbb{R}^2 \times \mathbb{C}^2$. Compact Calabi-Yau data, a Φ_d functor, κ_{ch} or κ_{BKM} consequences, $K3 \times E$ specializations, Igusa cusp forms, Borcherds input forms, BKM denominator identities, and Stage-1 or Stage-2 Φ_d specializations require a matched-conventions theorem. This regulator datum addresses the analytic obstruction named in Lemma 3.263 and Problem 3.273 on the formal local model.

NILPOTENT TRUNCATION OF THE COTANGENT SHADOW

The main theorem 2.70 is unconditional for the cochain-level reduced CE/PV central-operation, principal-part, brane-line factorization, formal degree-zero Weyl/Moyal coefficient algebra, and scalar-anomaly statements. The finite-N and stable trace comparison through radial parts remains conditional on the external radial-parts input named in the degree-zero Moyal theorem. Part (d) is proved at the reduced principal-part level after de Rham-current contraction; its unreduced primitive

shadow and part (e) on the descendant (one-antifield) sector are governed by 3.60, the unreduced Tate-coefficient cotangent lift. The deciding obstructions are 3.263 (Casimir support fails in the direct-sum dual) and 3.261 (no naive finite-dimensional Lie truncation of the full sector). Two routes occur in 3.60: a weighted Tate completion and a nilpotent truncation. The argument below closes the classical nilpotent-truncation route on a precisely specified truncation, records what is lost, and records the quantum Moyal obstruction that prevents the same finite truncation from being an all-order descendant theorem.

The truncation

Let $\mathfrak{h} = (z_1, z_2)\mathbb{C}[[z_1, z_2]] = \mathfrak{m}$ be the reduced Hamiltonian Lie algebra of the formal symplectic disk on $\mathbb{C}^2$, with bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ modulo constants. Let $\mathfrak{m}^k = (z_1, z_2)^k \mathbb{C}[[z_1, z_2]]$ for $k \geq 1$ be the k-th power of the maximal ideal. Fix integers $N \geq k \geq 3$ and define the *nilpotent truncation*

$$\mathfrak{h}^{(k)}_{\leq N} := \mathfrak{m}^k / \mathfrak{m}^{N+1},$$

with $\mathfrak{m}$-adic Poisson bracket inherited from the ambient $\mathfrak{h}_{\text{poly}}$. The standard choice is $k = 3$:

$$\mathfrak{h}_{\leq N} := \mathfrak{h}^{(3)}_{\leq N} = \mathfrak{m}^3 / \mathfrak{m}^{N+1}.$$

Concretely, $\mathfrak{h}_{\leq N}$ is the finite-dimensional vector space spanned by monomials $z_1^a z_2^b$ with $3 \leq a + b \leq N$.

Remark 3.118 *(Why degree three and not degree two).* The naive choice $k = 2$ fails to give a nilpotent Lie algebra: the degree-2 piece $H_2 = \operatorname{span}\{z_1^2, z_1 z_2, z_2^2\}$ is closed under the Poisson bracket because $\{H_2, H_2\} \subset H_{2+2-2} = H_2$, and the structure is exactly $\mathfrak{sl}_2$:

$$\{z_1^2, z_2^2\} = 4 z_1 z_2, \quad \{z_1^2, z_1 z_2\} = 2 z_1^2, \quad \{z_1 z_2, z_2^2\} = 2 z_2^2.$$

Thus $\mathfrak{m}^2/\mathfrak{m}^{N+1}$ contains a nonabelian semisimple subalgebra and is not nilpotent, hence does not satisfy the lower-central Tate-pronilpotence hypothesis of 3.217. The truncation $k = 3$ removes the H_2 direction; the bracket on $\mathfrak{m}^3$ raises $\mathfrak{m}$-adic degree strictly, making the truncation nilpotent.

Lemma 3.119 *(Truncation is a finite-dimensional nilpotent Lie algebra).* For every $N \geq 3$:

(i) $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ is a finite-dimensional Lie algebra of dimension $\binom{N+2}{2} - 6$ (the number of monomials of degree between 3 and N inclusive).

(ii) $\mathfrak{h}_{\leq N}$ is nilpotent of class at most $N - 2$ in the lower central series.

(iii) $\mathfrak{h}^\vee_{\leq N}$ is finite-dimensional, hence the formal Casimir $C_{\mathfrak{h}_{\leq N}} = \sum_{3 \leq d \leq N} \sum_i e_{d,i} \otimes e^i_d$ is a convergent element of $\mathfrak{h}_{\leq N} \otimes \mathfrak{h}^\vee_{\leq N}$ (a finite sum, no Tate completion required).

(iv) The linear Hamiltonians $z_1, z_2 \in \mathfrak{m} \setminus \mathfrak{m}^3$, the conformal-symplectic generators $H_2 = \operatorname{span}\{z_1^2, z_1 z_2, z_2^2\} \cong \mathfrak{sl}_2$, and the constant Hamiltonian $1 \notin \mathfrak{m}$ are absent from $\mathfrak{h}_{\leq N}$ by construction.

Proof. (i) The space $\mathfrak{m}^3$ is a Lie subalgebra of the polynomial Hamiltonian Lie algebra $\mathfrak{h}_{\text{poly}}$, and $\mathfrak{m}^{N+1}$ is a Lie ideal inside $\mathfrak{m}^3$: for $f \in \mathfrak{m}^p$ and $g \in \mathfrak{m}^q$ one has $\{f, g\} \in \mathfrak{m}^{p+q-2}$, since the Poisson bracket on $\mathbb{C}[z_1, z_2]$ lowers total polynomial degree by exactly two, and each partial derivative lowers $\mathfrak{m}$-adic degree by exactly one. For $p, q \geq 3$ the bracket lands in $\mathfrak{m}^{p+q-2} \subset \mathfrak{m}^4 \subset \mathfrak{m}^3$, making $\mathfrak{m}^3$ a Lie subalgebra; for $p \geq 3$ and $q \geq N + 1$ the bracket lands in $\mathfrak{m}^{p+q-2} \subset \mathfrak{m}^{N+2} \subset \mathfrak{m}^{N+1}$, making $\mathfrak{m}^{N+1}$ a Lie ideal of $\mathfrak{m}^3$. The quotient $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ inherits a well-defined Lie bracket. Dimension count: monomials

$z_1^a z_2^b$ with $3 \le a+b \le N$ number $\binom{N+2}{2} - 6$ (the total number with $0 \le a+b \le N$ minus the six of degrees 0, 1, 2).

(ii) The Poisson bracket of two elements of $\mathfrak{m}^3$ raises $\mathfrak{m}$-adic degree by at least 1: $\{H_p, H_q\} \subset H_{p+q-2} \subset \mathfrak{m}^{p+q-2}$, and for $p, q \ge 3$, $p+q-2 \ge 4 > p$ (since $q \ge 3$). Thus the descending Lie series

$$\mathfrak{h}_{\le N}^{(\ge m)} := \underbrace{[\mathfrak{h}_{\le N}, [\mathfrak{h}_{\le N}, \dots]]}_{m \text{ brackets}} \subset \left(\mathfrak{m}^{m+3}/\mathfrak{m}^{N+1}\right)$$

satisfies $\mathfrak{h}_{\le N}^{(\ge m)} = 0$ for $m+3 \ge N+1$, i.e. for $m \ge N-2$. Hence $\mathfrak{h}_{\le N}$ is nilpotent of class at most $N-2$.

(iii) Immediate from finite-dimensionality.

(iv) Immediate from the $\mathfrak{m}^3$-adic cutoff and the exclusion $a+b \le 2$. □

Remark 3.120 *(Why the maximal-ideal-adic obstruction does not apply).* 3.261 shows that $\mathfrak{h}/\mathfrak{m}^{N+1}$ is *not* a Lie quotient of the full Hamiltonian algebra $\mathfrak{h}_{\text{poly}}$: the linear Hamiltonian z_1 acting on $z_2^{N+1} \in \mathfrak{m}^{N+1}$ produces $\{z_1, z_2^{N+1}\} = (N+1)z_2^N \notin \mathfrak{m}^{N+1}$. The nilpotent truncation $\mathfrak{h}_{\le N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ *kills the linear Hamiltonian* and the conformal $\mathfrak{sl}_2$ direction from the algebra. The obstruction lemma is then irrelevant: no element of $\mathfrak{h}_{\le N}$ has $\mathfrak{m}$-adic degree below 3, so the displayed counterexample $\{z_1, z_2^{N+1}\}$ never occurs in $\mathfrak{h}_{\le N}$.

Remark 3.121 *(Why the Tate Casimir obstruction does not apply).* The product/direct-sum support obstruction of 3.263 is a feature of the unbounded expansion $\mathfrak{h} = \prod_{d \ge 1} H_d$: the Casimir $C_{\mathfrak{h}} = \sum_d \sum_i e_{d,i} \otimes e_d^i$ has nonzero component in every $H_d^\vee$, and $\bigoplus_d H_d^\vee$ admits no infinite-support element. On the truncation $\mathfrak{h}_{\le N}$ the Casimir is the finite sum $C_{\mathfrak{h}_{\le N}} = \sum_{d=3}^N \sum_i e_{d,i} \otimes e_d^i$ in the finite-dimensional vector space $\mathfrak{h}_{\le N} \otimes \mathfrak{h}_{\le N}^\vee$. There is no convergence question.

The classical truncated cotangent CE/PV shadow

THEOREM 3.122 *(Classical truncated cotangent CE/PV shadow).* Set $\mathfrak{g}_{\le N} = \mathfrak{h}_{\le N} \ltimes \mathfrak{h}_{\le N}^\vee[1]$. For every interval $I \subset \mathbb{R}$ let $\mathfrak{g}_{\le N,I} = \Omega_c^\bullet(I) \widehat{\otimes} \mathfrak{h}_{\le N} \ltimes \mathcal{D}_c'(I) \widehat{\otimes} \mathfrak{h}_{\le N}^\vee[1]$. Then, after the Poisson, residue, and interval-smearing conventions fixed above, there is a filtered cochain isomorphism of completed dg graded-commutative P_0-algebras

$$\Phi_{\le N} : C^\bullet_{\text{CE,cont}}(\mathfrak{g}_{\le N,I}) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}(\mathfrak{h}_{\le N,I})\big)$$

on every interval $I \subset \mathbb{R}$. This is the finite nilpotent CE/PV cotangent shadow of the reduced Hamiltonian model; it is not an unreduced open BV factorization-centre morphism. The isomorphism is determined on generators by $\Phi_{\le N}(c^I) = \theta^I$, $\Phi_{\le N}(u_I) = O_I$, with the same Schouten/Lichnerowicz convention as in 3.28. It is compatible with extension by zero and with disjoint-interval products in this reduced finite model, and on the Hamiltonian generators it recovers the stalkwise operation $f \mapsto L_{B_f}$. The cotangent generators are finite principal-part labels in the CE/PV shadow; their unreduced boundary centrality homotopies remain part of Problem 3.60.

Proof. The Lie algebra $\mathfrak{g}_{\le N}$ is finite-dimensional ($\dim \mathfrak{g}_{\le N} = 2 \dim \mathfrak{h}_{\le N} = 2\binom{N+2}{2} - 12$), hence pro-finite-dimensional in the trivial constant filtration. It is lower-central Tate-pronilpotent in the sense of 3.217: by 3.119(ii) the descending Lie powers of $\mathfrak{h}_{\le N}$ vanish after at most $N-2$ brackets. The cotangent summand is abelian, but this alone would not imply nilpotence of the semidirect product. The needed point is the degree-lowering coadjoint action: since $[H_p, H_q] \subset H_{p+q-2}$, an element $f \in H_p$, $p \ge 3$,

sends $H_d^\vee$ to $H_{d-p+2}^\vee$, hence lowers dual degree by at least one. Repeated action therefore kills every finite quotient $\mathfrak{h}_{\leq N}^\vee$. Together with nilpotence of $\mathfrak{h}_{\leq N}$, the descending Lie powers of $\mathfrak{g}_{\leq N} = \mathfrak{h}_{\leq N} \ltimes \mathfrak{h}_{\leq N}^\vee[1]$ also terminate. Hence $\bigcap_m \mathfrak{g}_{\leq N}^{(\geq m)} = 0$ in any Tate filtration, and the lower-central Tate-pronilpotence hypothesis of 3.217 is satisfied trivially.

Apply 3.28 (the cochain-level reduced Hamiltonian CE/PV central-operation theorem) to $\mathfrak{h}_{\leq N}$. The conclusion is a canonical isomorphism of completed dg graded-commutative P_0-algebras

$$\Phi_{\leq N}^{\mathrm{stalk}} : C_{\mathrm{CE,cont}}^\bullet(\mathfrak{g}_{\leq N}) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}(\mathfrak{h}_{\leq N})\big),$$

with the explicit generator-level data of 3.28.

To pass from the stalk identification to the factorization-algebra identification on intervals $I \subset \mathbb{R}$, the Hamiltonian-sector factorization upgrade of 3.22 applies verbatim with $\mathfrak{h}$ replaced by $\mathfrak{h}_{\leq N}$: the locality input $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta_l^i \delta_j^k \partial(t-s)$ is unchanged, the smearing $\mathfrak{h}_{\leq N,I} = \Omega_c^\bullet(I) \widehat{\otimes} \mathfrak{h}_{\leq N}$ is canonical because $\mathfrak{h}_{\leq N}$ is finite-dimensional (no Tate completion), and the strict P_0 bracket follows from the delta-function locality.

In this finite truncated classical sector, the cotangent factor has no Tate-Casimir convergence problem. The principal-part sub-factorization algebra

$$\widetilde{\mathrm{Obs}}_{\partial,\leq N}^{\mathrm{cl,pp}}(I) = \mathrm{Obs}_{\mathrm{open},\leq N}^{\mathrm{cl}}(I) \widehat{\otimes} \widehat{\mathbf{S}}\big(\mathfrak{h}_{\leq N,I}^\vee[1]\big)$$

admits a chain-level primitive projection $\pi_{\mathrm{prim}}^{\leq N}$ because the Tate Casimir $C_{\mathfrak{h}_{\leq N}} = \sum_{d=3}^N \sum_i e_{d,i} \otimes e_d^i$ is a finite sum, hence a literal element of the (uncompleted) tensor product $\mathfrak{h}_{\leq N} \otimes \mathfrak{h}_{\leq N}^\vee$. The Q-compatibility, factorization compatibility, and P_0-bracket homotopies are then formal, since at no step does one need an infinite-support tensor in the direct-sum dual.

The bracket compatibility and graded Leibniz extension to the completed finite CE/PV algebra is the same argument as in 3.28, with the additional simplification that all completions are trivial because $\mathfrak{h}_{\leq N}$ is finite-dimensional. Compatibility with extension by zero, disjoint-interval factorization, and the stalkwise limit follow as in 3.22. □

Quantum obstruction for the finite truncation

Theorem 3.123 *(Quantum obstruction for the nilpotent truncation).* The classical finite truncation 3.122 is unconditional. The same finite Lie algebra $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ does *not* carry an all-order Moyal deformation for $N \geq 4$: it is not closed under the reduced Moyal bracket. Consequently the nilpotent truncation supplies no all-order quantum descendant theorem on the strict finite truncation unless an additional Moyal-flat replacement or a projected bracket with Jacobi and cochain compatibility is supplied.

What remains true at quantum level is the restriction of the full-sector degree-zero map 3.269 to the finite set of generators of $\mathfrak{h}_{\leq N}$, viewed inside the full $\mathfrak{h}_\hbar$. This is a restriction of a proved map on the full Hamiltonian source, not a closed CE theorem for a finite quantum Lie subalgebra.

Proof. The Moyal bracket is

$$\{f, g\}_\hbar = \sum_{j \geq 0} \frac{\hbar^{2j}}{(2j+1)!\, 2^{2j}} P^{2j+1}(f, g), \qquad P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

If $f \in \mathfrak{m}^p$ and $g \in \mathfrak{m}^q$, the order-$\hbar^{2j}$ summand lies in $\mathfrak{m}^{p+q-2(2j+1)}$ when nonzero. The Poisson term $j = 0$ preserves $\mathfrak{m}^3$ as a Lie subalgebra, but the higher Moyal terms need not. For $N \geq 4$,

$$P^3(z_1^4, z_2^4) = 24z_1 \cdot 24z_2 = 576z_1z_2,$$

so

$$\{z_1^4, z_2^4\}_\hbar = 16z_1^3z_2^3 + 24\hbar^2 z_1z_2 + \cdots .$$

The term $24\hbar^2 z_1z_2$ has $\mathfrak{m}$-adic degree 2. It is neither in $\mathfrak{m}^3$ nor a constant killed by the reduction modulo $\mathbb{C} \cdot 1$. Thus $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ is not a Moyal Lie subalgebra for $N \geq 4$.

Therefore the CE differential for a hypothetical finite quantum source $\mathfrak{h}_{\leq N,\hbar} \ltimes \mathfrak{h}^\vee_{\leq N,\hbar}[1]$ is not defined by restriction of the Moyal bracket. Any projected bracket would require a new Jacobi proof; the projected-degree warning of 3.261 shows that this is a real condition, not a formality. The all-order theorem 3.269 remains valid on the full Hamiltonian algebra $\mathfrak{h}_\hbar$, and one may evaluate its map on generators of degrees $3, \ldots, N$. That operation does not make those generators a finite quantum CE source. The descendant cotangent sector at $\hbar = 0$ is exactly 3.122; at $\hbar \neq 0$ it remains a named Moyal-flat truncation problem. □

What is lost

PROPOSITION 3.124 *(Lost sectors).* The truncation $\mathfrak{h} \rightsquigarrow \mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ removes the following five data from the local theorem 2.70:

(i) The linear Hamiltonians $z_1, z_2 \in \mathfrak{m} \setminus \mathfrak{m}^2$, which generate translations in the symplectic plane and the center-of-mass mode of the brane.

(ii) The conformal-symplectic $\mathfrak{sl}_2$ generators $H_2 = \operatorname{span}\{z_1^2, z_1z_2, z_2^2\}$, which generate rotations and dilations of the symplectic plane. This is the semisimple sector that prevents $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ from being nilpotent.

(iii) The constant Hamiltonian $1 \notin \mathfrak{m}$, on which the $\hbar$-normalization of the Moyal product depends. The central element $\mathbb{C} \cdot 1 \subset \mathfrak{h}_{\mathrm{poly}}$ is absent from $\mathfrak{m}^3$ a fortiori, so the $\hbar$-normalization in the truncated theorem is conventional.

(iv) The U(1) center anomaly cohomology class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ of 2.70(f), since the cocycle

$$\omega(z_1^k z_2^l, z_1^m z_2^n) = (kn - lm)\, \mathbf{1}[(k+m, l+n) = (1,1)]$$

is concentrated on multidegree $(1,1)$, with both arguments of total $\mathfrak{m}$-adic degree 1, hence the cocycle pairs only elements outside $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$. There is no central anomaly to anchor on the truncation.

(v) The stable connected limit $N \to \infty$ on the full $\mathfrak{h}$. The inverse system $\{\mathfrak{h}_{\leq N}\}_{N \geq 3}$ has structure maps the natural surjections $\mathfrak{h}_{\leq N+1} \twoheadrightarrow \mathfrak{h}_{\leq N}$ induced by the $\mathfrak{m}$-adic projection $\mathfrak{m}^{N+1} \subset \mathfrak{m}^{N+2}$. The inverse limit is $\varprojlim_N \mathfrak{h}_{\leq N} = \mathfrak{m}^3$, the $\mathfrak{m}^3$-truncation of the Hamiltonian Lie algebra, not the full $\mathfrak{h} = \mathfrak{m}$.

What remains classically: the third-order and higher canonical transformations, which form a nilpotent Lie subalgebra $\mathfrak{m}^3 \subset \mathfrak{h}$, a Lie ideal in $\mathfrak{m}^2$, admitting the classical finite cotangent CE/PV shadow unconditionally. The unreduced open BV factorization-centre morphism is not constructed by this truncation. The all-order Moyal descendant lift is excluded by 3.123 unless extra quantum truncation data are added.

Proof. (i)–(iv) are immediate from the definitions and from the $\mathfrak{m}$-adic concentration of ω. (v) is the explicit inverse limit calculation: a compatible family $\{f_N \in \mathfrak{m}^3/\mathfrak{m}^{N+1}\}_N$ assembles to a unique formal series in $\mathfrak{m}^3 \subset \mathbb{C}[[z_1, z_2]]$, and the $\mathfrak{m}$-adic Lie structure is preserved at every level. □

Cubic inverse-limit survivor

Proposition 3.125 *(Cubic coordinate-window survivor and variance boundary).* The finite Lie quotients

$$\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$$

recover the complete cubic Lie algebra $\varprojlim_N \mathfrak{h}_{\leq N} = \mathfrak{m}^3$. The finite cotangent maps $\Phi_{\leq N}$ of 3.122 form a compatible family of coordinate tests. They do not, however, form an inverse system of dg CE cochain projections. A Lie quotient $\mathfrak{h}_{\leq M} \twoheadrightarrow \mathfrak{h}_{\leq N}$ induces CE cochain maps by pullback in the opposite direction. Consequently $\Phi_{\mathfrak{m}^3}$ denotes the coordinate-window family $\{\Phi_{\leq N}\}_N$ together with the chain-side inverse limit and an exact continuous-dualization datum; it is not an unconditional formula $\varprojlim_N \Phi_{\leq N}$ on CE cochains.

This is the maximal survivor of the nilpotent tower only in that precise sense: the Lie-side tower recovers $\mathfrak{m}^3$, while a completed cochain P_0-statement requires the additional variance and bracket-admissibility data named above.

Proof. The Lie-side inverse limit is the usual $\mathfrak{m}$-adic completion of $\mathfrak{m}^3$. The variance warning is forced already at $M = 4$, $N = 3$. In $\mathfrak{h}_{\leq 4}$,

$$\{z_1^3, z_2^3\} = 9z_1^2 z_2^2, \qquad d_{\mathrm{CE}} c^{2,2} \supset -9c^{3,0}c^{0,3}.$$

The coordinate projection to the $N = 3$ cochain window kills $c^{2,2}$ but not $c^{3,0}c^{0,3}$, so

$$p(d_{\mathrm{CE}} c^{2,2}) = -9c^{3,0}c^{0,3} \neq 0 = d_{\mathrm{CE}} p(c^{2,2}).$$

Thus projection of CE cochains is not a dg map. The correct finite functoriality is contravariant pullback along Lie quotients, while a completed cochain theorem must be obtained either from a chain-side inverse limit followed by exact continuous dualization or from a separately supplied topology whose cochain transition maps are dg maps.

The Lie algebra $\mathfrak{m}^3$ is lower-central Tate-pronilpotent in the $\mathfrak{m}$-adic Tate filtration: its descending Lie powers satisfy $(\mathfrak{m}^3)^{(\geq m)} \subset \mathfrak{m}^{m+3}$ because each iterated bracket raises the $\mathfrak{m}$-adic degree by at least 1 when both arguments lie in $\mathfrak{m}^3$ (since $\{H_p, H_q\} \subset \mathfrak{m}^{p+q-2}$ and $p, q \geq 3$ implies $p + q - 2 \geq p + 1$). Hence $\bigcap_m (\mathfrak{m}^3)^{(\geq m)} = 0$, and the lower-central Tate-pronilpotence hypothesis of 3.217 holds. The Tate Casimir $C_{\mathfrak{m}^3} = \sum_{d\geq 3} \sum_i e_{d,i} \otimes e_d^i$ is governed by the same product/direct-sum convergence question as in 3.263, but on the smaller direct-sum dual $\bigoplus_{d\geq 3} H_d^\vee \subset \bigoplus_{d\geq 1} H_d^\vee$. The finite maps $\Phi_{\leq N}$ at every finite N use only the finite Casimir $\sum_{3\leq d\leq N} \sum_i e_{d,i} \otimes e_d^i$. Passing from these finite identities to a completed cochain identity is exactly the continuous-dualization and variance datum isolated in the statement, not a formal inverse-limit consequence. □

Definition 3.126 *(Low-degree obstruction datum for the nilpotent tower).* For $N \geq 3$ let

$$\pi_N : \mathfrak{m}^3 \twoheadrightarrow \mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$$

be the nilpotent window projection. The two obstruction terms for adjoining the removed low-degree sector are:

$$\mathfrak{o}_N^{\mathrm{lin}}(x; \tau) = \{x, \tau\} \bmod \mathfrak{m}^{N+1}, \qquad x \in H_1,\ \tau \in \mathfrak{m}^{N+1}/\mathfrak{m}^{N+2},$$

and

$$\mathfrak{o}^{\mathrm{ss}}\colon \Lambda^2 H_2 \to H_2, \qquad \mathfrak{o}^{\mathrm{ss}}(x, y) = \{x, y\}.$$

Concretely

$$\mathfrak{o}_N^{\mathrm{lin}}(z_1; z_2^{N+1}) = (N+1)z_2^N \neq 0 \quad \text{in } \mathfrak{m}^N/\mathfrak{m}^{N+1},$$

and

$$\{z_1^2, z_2^2\} = 4z_1z_2, \qquad \{z_1^2, z_1z_2\} = 2z_1^2, \qquad \{z_1z_2, z_2^2\} = 2z_2^2.$$

The compatible Casimir missing from the nilpotent tower is

$$\mathfrak{o}_{\leq 2}^{\mathrm{Cas}} = \sum_i e_{1,i} \otimes e_1^i + \sum_i e_{2,i} \otimes e_2^i.$$

LEMMA 3.127 *(Obstruction-datum criterion for enlarging the tower).* A finite $\mathfrak{m}$-adic tower extending $\{\mathfrak{h}_{\leq N}\}_{N\geq 3}$ to a Lie tower containing $H_1 \oplus H_2$ and using the same quotient maps exists only if $\mathfrak{o}_N^{\mathrm{lin}} = 0$ for all N and $\mathfrak{o}^{\mathrm{ss}}$ is nilpotent in the lower-central filtration. A cotangent CE/PV extension of the inverse limit to the full $\mathfrak{h} = \mathfrak{m}$ also requires a coefficient category in which the compatible tensor $\mathfrak{o}_{\leq 2}^{\mathrm{Cas}} + \sum_{d\geq 3}\sum_i e_{d,i} \otimes e_d^i$ converges.

Proof. If H_1 is included and the window N is still the quotient by $\mathfrak{m}^{N+1}$, then the tail $\mathfrak{m}^{N+1}$ must be a Lie ideal for the enlarged source. The displayed value $\{z_1, z_2^{N+1}\} = (N+1)z_2^N \notin \mathfrak{m}^{N+1}$ shows that this ideal condition is exactly the vanishing of $\mathfrak{o}_N^{\mathrm{lin}}$, and it fails.

If H_2 is included, then its bracket remains inside H_2 and equals the nonabelian $\mathfrak{sl}_2$ bracket displayed in Definition 3.126. The image of $\mathfrak{o}^{\mathrm{ss}}$ is all of H_2, so the descending Lie powers never leave H_2. This is exactly the obstruction to nilpotence and to the conilpotent bar-cobar hypothesis.

The finite cotangent windows for $\mathfrak{h}_{\leq N}$ use the finite Casimir $\sum_{3\leq d\leq N}\sum_i e_{d,i} \otimes e_d^i$. Extending to $\mathfrak{h}$ adds the missing $d = 1, 2$ components and then the whole compatible family over all $d \geq 1$. Thus the full cotangent lift requires precisely the convergence of the displayed full Casimir family. □

THEOREM 3.128 *(Low-degree obstruction datum for full Hamiltonian recovery).* The coordinate-window family $\Phi_{\mathfrak{m}^3}$ of 3.125 is the classical cotangent lift datum on $\mathfrak{m}^3$, not on $\mathfrak{h} = \mathfrak{m}$. To extend to all of $\mathfrak{h}$ one must include the low-degree sectors $\mathfrak{m}/\mathfrak{m}^3 = H_1 \oplus H_2 = \mathbb{C}\cdot z_1 \oplus \mathbb{C}\cdot z_2 \oplus \mathrm{span}\{z_1^2, z_1z_2, z_2^2\}$, which by 3.261(iii) and 3.127 is not stable under any $\mathfrak{m}$-adic cutoff: the bracket $\{z_1, z_2^{N+1}\} = (N+1)z_2^N$ is the explicit obstruction. Equivalently, the obstruction datum is

$$(\mathfrak{o}_N^{\mathrm{lin}},\ \mathfrak{o}^{\mathrm{ss}},\ C_{\mathfrak{h}}),$$

where

$$\mathfrak{o}_N^{\mathrm{lin}}(z_1; z_2^{N+1}) = (N+1)z_2^N \neq 0, \qquad \mathrm{im}\,\mathfrak{o}^{\mathrm{ss}} = H_2 \cong \mathfrak{sl}_2,$$

and $C_{\mathfrak{h}}$ is the full infinite Casimir family below. Equivalently, the Tate Casimir $C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_i e_{d,i} \otimes e_d^i$ has low-degree components $\sum_i e_{1,i} \otimes e_1^i \in H_1 \otimes H_1^\vee$ and $\sum_i e_{2,i} \otimes e_2^i \in H_2 \otimes H_2^\vee$ that the truncation removes; the degree-1 component is the source of the linear-Hamiltonian translation, and the degree-2 component is the source of the conformal-symplectic $\mathfrak{sl}_2$ direction on which the truncation fails to be nilpotent. Recovering the full $\mathfrak{h}$ requires a coefficient category in which $C_{\mathfrak{h}}$ converges *including* both low-degree components, which is exactly the Casimir support obstruction of 3.263. The truncation route does not solve that obstruction; it removes the sector on which the obstruction is felt. A full recovery theorem must therefore supply both a coefficient category in which $C_{\mathfrak{h}}$ converges and a relative or reductive model for the H_2-sector.

Proof. The tower used in 3.125 is $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$. Since the transition maps are the usual quotient maps,

$$\varprojlim_N \mathfrak{h}_{\leq N} = \varprojlim_N \mathfrak{m}^3/\mathfrak{m}^{N+1} = \mathfrak{m}^3$$

as a complete filtered Lie algebra. The homogeneous pieces H_1 and H_2 have zero image in every term of this tower; hence no element of the inverse limit can recover them.

The failure is not an artefact of the chosen notation. If one tries to include H_1 while keeping the same $\mathfrak{m}$-adic finite windows, the tails are not Lie ideals: for every N,

$$z_2^{N+1} \in \mathfrak{m}^{N+1}, \qquad \{z_1, z_2^{N+1}\} = (N+1)z_2^N \notin \mathfrak{m}^{N+1}.$$

Therefore the quotient by the tail does not carry the full Hamiltonian bracket. If one includes H_2, the degree-two Hamiltonians contain the $\mathfrak{sl}_2$ bracket $[H_2, H_2] = H_2$, so the finite quotient is not nilpotent. Thus a finite nilpotent $\mathfrak{m}$-adic tower cannot contain the low-degree sector $H_1 \oplus H_2$.

Finally, the full Casimir $C_{\mathfrak{h}}$ has nonzero components in $H_1 \otimes H_1^\vee$, $H_2 \otimes H_2^\vee$, and in every higher $H_d \otimes H_d^\vee$. The nilpotent tower sees only the $d \geq 3$ part. Hence the inverse-limit cotangent lift on $\mathfrak{m}^3$ cannot be promoted to the full Hamiltonian sector without an additional coefficient category in which the full Casimir converges. □

Corollary 3.129 *(Full maximal-ideal universal obstruction class).* For the full reduced formal Hamiltonian algebra $\mathfrak{h} = \mathfrak{m} = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$, the direct formal-local CE/PV recognition theorem remains valid in the finite-coordinate product completion. The universal filtered-cobar Koszul recognition criterion of Theorem 3.38, however, has a nonzero obstruction class in O_{Kos} of Definition 3.37. Its visible components are

$$\mathfrak{o}_N^{\mathrm{lin}}(z_1; z_2^{N+1}) = (N+1)z_2^N \neq 0, \qquad \operatorname{im} \mathfrak{o}^{\mathrm{ss}} = H_2 \cong \mathfrak{sl}_2,$$

$$[C_{\mathfrak{h}}] = \left[\sum_{d\geq 1}\sum_i e_{d,i} \otimes e_d^i\right] \in Q_{\mathrm{Cas}}, \qquad H_2 \subset \bigcap_{r\geq 1} \mathfrak{h}^{(r)}.$$

Therefore a universal full-$\mathfrak{h}$ bar-cobar/enveloping theorem requires, before any open-side A_∞ comparison, all of the following: a coefficient topology representing $C_{\mathfrak{h}}$, continuity of the finite-window bracket and coadjoint transition maps, vanishing $\varprojlim^1$ for the completed CE/PV and cobar towers, a relative or reductive treatment of the H_2-sector, and a twisting cochain whose associated-graded cobar map has acyclic cone.

Proof. The formal-local CE/PV part is Theorem 2.48, equivalently the formal-disk instance of Theorem 2.50: the bracket and coadjoint formulas are continuous for finite coordinate projections, so the generator substitution $c \mapsto \theta$, $u \mapsto O$ is a completed dg commutative cochain isomorphism. The P_0-algebra reading is finite-level or bracket-admissible data; it is not supplied by an unqualified inverse limit of ambient coordinate products.

The filtered-cobar recognition theorem asks for more. The linear obstruction is the same computation as in Definition 3.126: adjoining H_1 to the $\mathfrak{m}$-adic nilpotent windows makes the tail non-invariant because $\{z_1, z_2^{N+1}\} = (N+1)z_2^N \notin \mathfrak{m}^{N+1}$. The semisimple obstruction is the displayed $\mathfrak{sl}_2$ bracket on H_2, hence $H_2 \subset \bigcap_r \mathfrak{h}^{(r)}$. This gives a nonzero lower-central component in O_{conilp}.

The Casimir component is nonzero in the strict product/direct-sum topology by the support argument of Theorem 2.52: the compatible finite-window Casimirs have nonzero identity component in

every $H_d \otimes H_d^\vee$, while a strict tensor with a direct-sum dual has only finitely many dual-degree components. Finally, even after changing topology and handling H_2 relatively, the universal criterion still demands the cobar cone and the Milnor $\varprojlim^1$ terms in Definition 3.37 to vanish. These are precisely the listed remaining requirements. □

Problem 3.130 *(Low-degree-sector lift from the cubic ideal to the maximal ideal).* Extend the coordinate-window cotangent lift datum $\Phi_{\mathfrak{m}^3}$ of 3.125 from $\mathfrak{m}^3$ to all of $\mathfrak{h} = \mathfrak{m}$. Equivalently, supply a Tate or weighted-coefficient extension in which the degree-1 and degree-2 Casimir components converge as kernels for the BV propagator, and in which the conformal-symplectic $\mathfrak{sl}_2$ direction H_2 is included by a relative or reductive replacement for the lower-central Tate-pronilpotence hypothesis of 3.217. By 3.128 this is the low-degree-sector failure; by 3.126 its obstruction terms are $\mathfrak{o}_N^{\mathrm{lin}}$, $\mathfrak{o}^{\mathrm{ss}}$, and $\mathfrak{o}_{\leq 2}^{\mathrm{Cas}}$; by 3.263 this is the same coefficient-category problem as 3.60. Thus the nilpotent-truncation route closes the classical nilpotent cotangent shadow on $\mathfrak{m}^3$, and the residual moves from 3.60 as stated to its restricted form on the low-degree sector $\mathfrak{m}/\mathfrak{m}^3 = H_1 \oplus H_2$ alone.

THEOREM 3.131 *(Semisimple obstruction to lower-central Tate-pronilpotence).* A second obstruction to extending $\Phi_{\mathfrak{m}^3}$ to all of $\mathfrak{h}$ is structural rather than analytic: the inclusion of $H_2 = \mathfrak{sl}_2$ violates the lower-central Tate-pronilpotence route used in Lemma 3.217 because $[H_2, H_2] = H_2$ is its own descending Lie series. Even if the degree-1 and degree-2 Casimir components converged in some weighted Tate completion, the pronilpotent bar-cobar route would not apply directly to the resulting Lie algebra. This does not say that ordinary finite-dimensional bar-cobar or PBW formalism fails for $\mathfrak{sl}_2$; it says that the manuscript's chosen lower-central Tate-pronilpotent criterion must be replaced by a relative or reductive model that builds in semisimple subalgebras separately.

Proof. Let L be any filtered Lie algebra containing the degree-two Hamiltonians H_2 as a Lie subalgebra with the usual Poisson bracket. The descending Lie powers of L satisfy $L^{(n)} \cap H_2 \supset H_2$ for every n, because $[H_2, H_2] = H_2$. Hence

$$H_2 \subset \bigcap_{n \geq 1} L^{(n)}.$$

The intersection of the descending Lie powers is therefore nonzero. This contradicts the lower-central Tate-pronilpotence condition used in Lemma 3.217. The bar-cobar argument may be applied to the nilpotent radical $\mathfrak{m}^3$, but it cannot be applied unchanged to a Lie source containing the semisimple H_2-sector. □

Costello–Li precedent

Remark 3.132 *(Costello–Li balanced matrix truncation as structural model).* Costello and Li [13] prove the quantum perturbative BCOV theorem on flat Calabi–Yau by an $N \to \infty$ stable connected limit through the open gauge algebra $\mathfrak{gl}(N \mid N)$, with the directed Lie embedding $\mathfrak{gl}(N \mid N) \hookrightarrow \mathfrak{gl}(N + k \mid N + k)$ and the BV-cohomology vanishing $H^*_{\mathrm{BV}}\big(\mathfrak{gl}(N \mid N)\big) = 0$ (supertrace identity $\mathrm{str}(I) = N - N = 0$ kills the would-be U(1)-anomaly). The structural analogy with the truncation here:

(i) Each window $\mathfrak{gl}(N \mid N)$ is finite-dimensional, BV-cohomology vanishes, the universal CE/Schouten identification and the BV-propagator coefficient kernel are well-defined on each window.

(ii) The inverse / direct system is the Lie embedding $\mathfrak{gl}(N \mid N) \hookrightarrow \mathfrak{gl}(N + 1 \mid N + 1)$ on the Costello–Li side, the Lie surjection $\mathfrak{h}_{\leq N+1} \twoheadrightarrow \mathfrak{h}_{\leq N}$ here.

(iii) The U(1)-anomaly is killed by the structure of each window on the Costello–Li side (supertrace), and is removed by the $\mathfrak{m}^3$-truncation here (the cocycle ω pairs only multidegree-(1, 0) with multidegree-(0, 1), both in $H_1 \subset \mathfrak{m} \setminus \mathfrak{m}^2$, hence outside $\mathfrak{m}^3$).

(iv) The colimit $N \to \infty$ converges on the Costello–Li side to the full $\mathfrak{gl}(\infty \mid \infty)$, providing the stable connected single-trace algebra. Here the inverse limit converges to $\mathfrak{m}^3 \subsetneq \mathfrak{h}$, not to all of $\mathfrak{h}$, by 3.128.

The structural difference: the $\mathfrak{gl}(N \mid N)$ system is graded by matrix size, with trivial action of one window on the next-window orthogonal complement; the $\mathfrak{h} = \mathfrak{m}$ system is graded by $\mathfrak{m}$-adic degree, with the linear and quadratic sectors acting non-trivially on every higher-degree window, which is precisely the finite-window obstruction of 3.261(iii) plus the $\mathfrak{sl}_2 \subset H_2$ semisimple obstruction to lower-central Tate-pronilpotence.

Summary

The nilpotent truncation $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ closes the classical finite nilpotent cotangent CE/PV shadow on the truncated sector (degree-zero Hamiltonian plus finite cotangent labels). It does not construct the unreduced open BV factorization-centre morphism. The quantum all-order version on the same finite source is obstructed by 3.123: the Moyal bracket leaves $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ for $N \geq 4$. The family $\Phi_{\mathfrak{m}^3} = \{\Phi_{\leq N}\}_N$ of 3.125 is the classical coordinate-window cotangent lift datum on the full classical $\mathfrak{m}^3$ sector; a completed cochain inverse-limit statement requires the variance and continuous-dualization datum named there. What is lost: the low-degree sector $\mathfrak{m}/\mathfrak{m}^3 = H_1 \oplus H_2$ (linear translations and the conformal-symplectic $\mathfrak{sl}_2$), the constant-Hamiltonian $\hbar$-normalization, the U(1) center anomaly $[\bar{c}]$, and the stable connected limit on the full $\mathfrak{h}$. Recovering the low-degree sector requires the coefficient-category change of 3.60 together with a relative or reductive replacement for the lower-central Tate-pronilpotent route of 3.217, so that the semisimple $\mathfrak{sl}_2 \subset H_2$ is treated as semisimple data rather than as a pronilpotent radical. Thus the nilpotent truncation closes only the classical $\mathfrak{m}^3$ truncation; its all-order quantum version is the separate Moyal-flat truncation problem above.

This is the same structural truncation as Costello–Li $\mathfrak{gl}(N \mid N) \to \mathfrak{gl}(\infty \mid \infty)$, with the key difference that the symplectic-plane $\mathfrak{m}$-adic system does not admit a free directed exhaustion by nilpotent Lie subalgebras containing $H_1 \oplus H_2$; the truncation removes the low-degree sector by hypothesis.

3.9 Primitive indecomposable P_0-shadow

The primitive indecomposable P_0-shadow map

$$\pi_{\mathrm{prim}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \longrightarrow \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$$

required by Problem 3.60, item (3), is constructed below without recourse to any heat-kernel propagator or finite-scale regularization. The construction relies only on the chain structure of the reduced BV algebra (Construction 3.190), the de Rham contraction (Lemma 3.191), and the stable trace invariant theory of Proposition 3.192. The principal ingredient is the cumulant projection on the free symmetric algebra generated by primitive cyclic-word trace classes.

Construction 3.133 *(Trace-count filtration on the open observables).* Let $\mathcal{R}_N$ be as in Construction 3.190. In the stable connected limit $N \to \infty$, Proposition 3.192 identifies

$$\mathcal{R}_\infty^{GL_\infty,\mathrm{st}} \cong \mathbf{S}\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}}\big)$$

with $F = \mathbb{C}\langle x_1, x_2\rangle$ and the cyclic words being primitive generators (Lemma 3.193 extends this to the super-cyclic case with one displayed odd ψ). The *trace-count filtration* on $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) = \Omega^\bullet_c(I)\widehat{\otimes}\mathcal{R}_N^{GL_N}$ is the symmetric-algebra filtration $F_k Obs$ generated by products of at most k primitive trace classes (cyclic words):

$$F_0 \subset F_1 \subset F_2 \subset \cdots \subset \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I).$$

F_0 is the algebra of compactly supported de Rham forms on I with scalar coefficients (the empty-trace stratum); F_1/F_0 is the *primitive* or *single-trace* component, isomorphic as a graded vector space to $\Omega^\bullet_c(I)\widehat{\otimes}\mathrm{Cyc}(F)_{\mathrm{nonempty}}$ in the stable limit.

LEMMA 3.134 *(Trace-count filtration is preserved by Q and by the open P_0 bracket).* In the stable connected limit, the BV differential Q and the open P_0 bracket $\{-,-\}_{\mathrm{open}}$ both preserve the trace-count filtration of Construction 3.133.

Proof. *Q preserves F_k.* By Construction 3.190, $Q\psi = [\phi_1, \phi_2]$, $Q\phi_i = 0$. Q is an interior derivation on $\mathcal{R}_N$, so on a product of k traces it acts by the Leibniz rule and produces another sum of k-trace products (each application replaces a single ψ inside a trace by $[\phi_1, \phi_2]$ inside the same trace; the total trace count is preserved). Smearing by a de Rham form on I does not change the trace count. Hence $QF_k \subset F_k$.

The open P_0 bracket preserves F_k. By the canonical equal-time bracket of Remark 3.21, $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta^i_l\delta^k_j\delta(t-s)$. On two traces $\mathrm{Tr}\, w_1$ and $\mathrm{Tr}\, w_2$ this bracket produces a single trace $\mathrm{Tr}\, w_3$ obtained by Wick-contracting one ϕ_1 in w_1 with one ϕ_2 in w_2 and joining the two cycles into one. This is the standard merging-of-cycles formula of Procesi–Razmyslov invariant theory. In particular, the bracket of two single-trace classes is a single-trace class, exactly Theorem 3.242. By the Leibniz rule for the Poisson bracket, the bracket of a k_1-trace product with a k_2-trace product is a sum of products of total trace count exactly $(k_1 + k_2 - 1)$, that is, it stays in $F_{k_1+k_2-1} \subset F_{k_1+k_2}$. Hence the bracket preserves the filtration. □

Construction 3.135 *(Primitive indecomposable P_0-shadow).* In the stable connected limit, the symmetric-algebra structure $\mathcal{R}_\infty^{GL_\infty,\mathrm{st}} \cong \mathbf{S}(\mathrm{Cyc}(F)_{\mathrm{nonempty}})$ has the natural cumulant projection onto the linear span of generators (the Hopf-algebraic primitive). Tensored with the identity on $\Omega^\bullet_c(I)$, this gives

$$\pi_{\mathrm{prim}} : \Omega^\bullet_c(I)\widehat{\otimes}\mathbf{S}(\mathrm{Cyc}(F)_{\mathrm{nonempty}}) \twoheadrightarrow \Omega^\bullet_c(I)\widehat{\otimes}\mathrm{Cyc}(F)_{\mathrm{nonempty}}.$$

Composing this shadow map with the identification $\mathrm{Cyc}(F)_{\mathrm{nonempty}} \twoheadrightarrow \bar{A}$ on the Hamiltonian-trace subspace (the abelianization of cyclic words: the trace of a cyclic word in the commuting variables ϕ_1, ϕ_2, taken modulo Q-exact reordering, is exactly the polynomial Hamiltonian $f \in \bar{A}$, a stable cyclic-word/polynomial dictionary recorded by Construction 3.195 and Lemma 3.221), gives

$$\pi_{\mathrm{prim}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \twoheadrightarrow \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}.$$

In this construction π_{prim} is the zero-ψ Hamiltonian shadow: primitive traces containing at least one displayed ψ are killed here and are treated separately by the one-ψ label shadow of Construction 3.153.

For factorization products one must retain the empty-trace stratum until the final linear projection. Define the unit-retaining shadow

$$\pi^{\mathrm{Ham}}_{\leq 1} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \longrightarrow \Omega^\bullet_c(I)\widehat{\otimes}(\mathbb{C}\cdot 1 \oplus \bar{A})$$

as projection to the scalar stratum F_0 together with the zero-ψ Hamiltonian summand of F_1/F_0, killing $F_{\geq 2}$ and the positive-ψ primitive summands. Then π_{prim} is $\pi^{\mathrm{Ham}}_{\leq 1}$ followed by projection to the $\bar{A}$-summand. At finite N this is a degreewise stable construction, not a global map $\mathcal{R}_N \to \mathcal{R}^{\mathrm{st}}_\infty$. For a fixed word-degree d and $N \geq d$, Proposition 3.192 identifies the degree-d trace invariants with the stable cyclic-word span, and the above shadow map is applied on that stable range. Outside the stable range the finite-N trace identities are not discarded; the statement is made after passing to the stable connected limit.

Lemma 3.136 *(Kernel of the Hamiltonian primitive shadow).* On the zero-ψ Hamiltonian subalgebra the kernel of π_{prim} is the closed ideal $F_{\geq 2}$ generated by products of at least two primitive trace classes, together with the empty-trace stratum F_0, and the primitive abelianization kernel

$$\ker\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}} \to \bar{A}\big) \subset F_1/F_0.$$

On the full reduced BV algebra before the cotangent label shadow is added, the kernel also contains the positive ψ-primitive summands. Equivalently, the unit-retaining map $\pi^{\mathrm{Ham}}_{\leq 1}$ has kernel $F_{\geq 2}$, the primitive abelianization kernel, and the non-Hamiltonian primitive summands, while π_{prim} further kills F_0.

Proof. By construction π_{prim} is the projection onto zero-ψ Hamiltonian generators of the symmetric algebra, which is the standard cumulant projection followed by the abelianization $\mathrm{Cyc}(F)_{\mathrm{nonempty}} \to \bar{A}$. The cumulant projection kills the augmentation ideal squared $(\mathbf{S}^{\geq 1})^2 = \mathbf{S}^{\geq 2}$. The abelianization then kills primitive reordering differences, for instance the class of $[xyxy]_{\mathrm{cyc}} - [xxyy]_{\mathrm{cyc}}$, and the final projection to $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$ kills the constants $\mathbb{C} \cdot 1 \subset F_0$. Primitive classes with positive ψ-count are not in the zero-ψ Hamiltonian target and therefore lie in the kernel of this shadow. Tensoring with the de Rham cosheaf $\Omega^\bullet_c(I)$ preserves these direct summands because the symmetric-algebra split is over the discrete coefficient ring on cyclic-word generators. □

Theorem 3.137 *(Primitive indecomposable P_0-shadow is a Q-chain map).* With total differential $Q_{\mathrm{tot}} = d_{\mathrm{dR}} + Q_{\mathrm{BV}}$ on the source and the de Rham differential on $\Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$, π_{prim} commutes strictly with Q_{tot}:

$$\pi_{\mathrm{prim}} \circ Q_{\mathrm{tot}} = Q_{\mathrm{tot}} \circ \pi_{\mathrm{prim}}.$$

Proof. The de Rham part commutes with π_{prim} because the projection acts only on the trace label. It remains to check the BV part. By Lemma 3.134, Q_{BV} preserves the trace-count filtration $F_\bullet$. On F_0 the BV differential is zero on the matrix factor, and π_{prim} kills the scalar Hamiltonian line. On a zero-ψ primitive trace B_f, $Q_{\mathrm{BV}} B_f = 0$ because $Q_{\mathrm{BV}}\phi_i = 0$, while primitive reordering differences remain in the abelianization kernel. On a primitive one-ψ trace $\mathrm{Tr}(\psi w(x, y))$,

$$Q_{\mathrm{BV}}\ \mathrm{Tr}(\psi w) = \mathrm{Tr}((xy - yx)w).$$

Its image in $\bar{A}$ is zero, since the two cyclic words become the same commutative monomial after abelianization. Primitive terms with at least two ψ's remain outside the zero-ψ Hamiltonian target after one application of Q_{BV}, and multi-trace terms remain multi-trace by Lemma 3.134. Thus the two composites agree on every summand. □

Remark 3.138 *(The cumulant target as the indecomposable part).* The target $\Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$ is the *indecomposable* or *linear primitive* part of the brane P_0 factorization algebra $A^{\mathrm{Ham}}_\partial(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$ of Construction 3.18: it is the generating Lie subspace $\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$. Equivalently, it is the quotient $\widehat{\mathbf{S}}(\mathfrak{h}_I)/\widehat{\mathbf{S}}^{\geq 2}(\mathfrak{h}_I) \cong \mathfrak{h}_I$ in

the augmentation-ideal grading, after removal of constants. The factorization product on the cumulant target side is therefore the linear (Lie) bracket of $\mathfrak{h}_I$, not a commutative algebra product; the multiplication of two distinct primitive elements lands in $\mathbf{S}^{\geq 2}$, which is the zero space at the indecomposable level.

THEOREM 3.139 *(Linearized factorization compatibility of the indecomposable shadow).* For disjoint intervals $I_1 \sqcup I_2 \subset V$ in $\mathbb{R}$, let

$$\lambda_{V;I_1,I_2} : \big(\Omega_c^\bullet(I_1)\widehat{\otimes}(\mathbb{C}\cdot 1 \oplus \bar{A})\big) \otimes \big(\Omega_c^\bullet(I_2)\widehat{\otimes}(\mathbb{C}\cdot 1 \oplus \bar{A})\big) \longrightarrow \Omega_c^\bullet(V)\widehat{\otimes}\bar{A}$$

be the exactly-one-primitive part of the brane factorization product after extension to V: it keeps the scalar–primitive and primitive–scalar summands and kills the scalar–scalar and primitive–primitive summands. Then

$$\pi_{\mathrm{prim}} \circ \mu = \lambda_{V;I_1,I_2} \circ \big(\pi^{\mathrm{Ham}}_{\leq 1} \otimes \pi^{\mathrm{Ham}}_{\leq 1}\big).$$

Thus the indecomposable shadow is compatible with the factorization product only after the unit-retaining linear projection; it is not a multiplicative map to $\bar{A}$.

Proof. Work by trace-count strata. If both inputs have nonzero primitive trace count, their factorization product has at least two nonempty traces and is killed by π_{prim}. The right-hand side is also zero because $\lambda_{V;I_1,I_2}$ kills the primitive–primitive summand. If both inputs lie in F_0, the product remains scalar and is killed by the final projection to $\bar{A}$; $\lambda_{V;I_1,I_2}$ kills the scalar–scalar summand.

The only surviving cases are scalar–primitive and primitive–scalar. The unit-retaining projections remember precisely those two summands. Applying the brane factorization product and then projecting to the exactly-one-primitive Hamiltonian component gives the same class as first applying $\pi^{\mathrm{Ham}}_{\leq 1}$ on both inputs and then applying $\lambda_{V;I_1,I_2}$. Positive-ψ primitive factors are killed on both sides of this zero-ψ Hamiltonian statement; their separate label shadow is treated below. Hence the displayed identity holds strictly. □

THEOREM 3.140 *(P_0 bracket compatibility, with explicit homotopy).* For $F, G \in \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I)$ in the stable connected limit,

$$\pi_{\mathrm{prim}}\big(\{F, G\}_{\mathrm{open}}\big) - \big\{\pi_{\mathrm{prim}}F, \pi_{\mathrm{prim}}G\big\}_{\bar{A}} = (Q \circ h_{P_0} + h_{P_0} \circ Q)(F \otimes G),$$

with the explicit homotopy $h_{P_0} \equiv 0$ on every stratum $F_{k_1} \otimes F_{k_2}$. The chain-level P_0-bracket compatibility square commutes strictly.

Proof. We compute on each stratum separately.

Stratum $(k_1, k_2) = (1, 1)$. Both inputs are single traces $B_f(a)$ and $B_g(b)$ for $f, g \in \bar{A}$ and $a, b \in \Omega_c^0(I)$. By Lemma 3.134, the open P_0 bracket of two single traces is again a single trace (cycle merging in Procesi-Razmyslov), and by Theorem 3.242, in the connected Hamiltonian quotient, $\{B_f(a), B_g(b)\}_{\mathrm{open}} = B_{\{f,g\}_{\bar{A}}}(ab)$. Apply π_{prim}: the result is the single-trace $ab \otimes \{f, g\}_{\bar{A}}$ in $\Omega_c^\bullet(I)\widehat{\otimes}\bar{A}$.

Independently, $\pi_{\mathrm{prim}}B_f(a) = a \otimes f$ and $\pi_{\mathrm{prim}}B_g(b) = b \otimes g$. The bracket on $\Omega_c^\bullet(I)\widehat{\otimes}\bar{A}$ is the P_0 bracket of Construction 3.18: $\{a \otimes f, b \otimes g\}_{\bar{A}} = ab \otimes \{f, g\}_{\bar{A}}$. This matches. Strict compatibility on this stratum.

Stratum $(k_1, k_2) = (1, k)$ *with* $k \geq 2$. $F = B_f(a)$ is a single trace; G is a k-trace product $B_{w_1}(b_1) \cdots B_{w_k}(b_k)$. By the Leibniz rule,

$$\{F, G\}_{\text{open}} = \sum_{i=1}^{k} (\pm) B_{w_1}(b_1) \cdots \{B_f(a), B_{w_i}(b_i)\}_{\text{open}} \cdots B_{w_k}(b_k).$$

Each term is a k-trace product (cycle merging joined to a remaining $(k-1)$-trace tail). Hence the result lies in F_k, and by Lemma 3.136, π_{prim} of this is zero whenever $k \geq 2$.

Independently, $\pi_{\text{prim}} G = 0$ because $G \in F_{\geq 2}$, hence $\{\pi_{\text{prim}} F, \pi_{\text{prim}} G\}_{\bar{A}} = 0$.

Both sides are zero on this stratum. Strict.

Stratum (k_1, k_2) *with* $k_1 \geq 2$ *or* $k_2 \geq 2$. By symmetry the same argument: both sides vanish.

Stratum $(0, k)$ *or* $(k, 0)$. One factor is in F_0, that is, a scalar de Rham form. The bracket of a scalar de Rham form with anything is zero (the canonical equal-time bracket only acts on matrix fields). So $\{F, G\}_{\text{open}} = 0$ and $\{\pi_{\text{prim}} F, \pi_{\text{prim}} G\}_{\bar{A}} = 0$. Strict.

In every stratum the difference is zero; setting $h_{P_0} \equiv 0$ realizes the homotopy formula trivially. □

Remark 3.141 *(Why the homotopy is zero).* The strictness in Theorem 3.140 is not an accident of small word degree; it is a structural consequence of the trace-count filtration of Lemma 3.134. The bracket of two single traces is a single trace by Procesi–Razmyslov cycle merging; the bracket of a single trace with a multi-trace is a multi-trace by the Leibniz rule; cumulant projection projects to the single-trace component. Hence the failure of the bracket compatibility, if any, must lie in the multi-trace stratum, where both sides are zero.

Construction 3.142 *(Primitive one-antifield chain complex).* Put $x = \phi_1$, $y = \phi_2$, and let $\mathsf{W}_{k,l}$ be the set of ordinary words with k letters x and l letters y. Let $\mathsf{N}_{a,b}$ be the set of cyclic words with a letters x and b letters y. The primitive one-ψ chain model in bidegree (k, l) is

$$C_2(k, l) \xrightarrow{\partial_2} C_1(k, l) \xrightarrow{\partial_1} C_0(k, l),$$

where

$$C_1(k, l) = \mathbb{C}\{\mathsf{W}_{k,l}\}, \qquad C_0(k, l) = \mathbb{C}\{\mathsf{N}_{k+1,l+1}\}.$$

The basis vector $w \in \mathsf{W}_{k,l}$ represents $\mathrm{Tr}(\psi w)$, with the graded cyclic rule $\mathrm{Tr}(ab) = (-1)^{|a||b|} \mathrm{Tr}(ba)$, $|x| = |y| = 0$, $|\psi| \equiv 1 \pmod 2$. The first boundary is

$$\partial_1(w) = [xyw]_{\text{cyc}} - [yxw]_{\text{cyc}}.$$

The two-ψ source is the alternating quotient generated by pairs of ordinary words u, v, of total bidegree $(k-1, l-1)$, with

$$[u, v] = -[v, u], \qquad [u, u] = 0.$$

In this quotient

$$C_2(k, l) = \big(\mathbb{C}\{(u, v) : \deg u + \deg v = (k-1, l-1)\}\big)_{\text{alt}},$$

and $C_2(k, l) = 0$ if $k = 0$ or $l = 0$. Thus the displayed complex is literally the cellular chain complex in dimensions 2, 1, 0: C_0 is the necklace vertex set, C_1 is the one-moving-chip edge set, and C_2 is the oriented two-moving-chip square set.

LEMMA 3.143 *(Two-antifield boundary sign).* In the chain model of Construction 3.142, the boundary of the generator represented by $\mathrm{Tr}(\psi u \psi v)$ is

$$\partial_2(u, v) = vxyu - vyxu - uxyv + uyxv.$$

This formula is well-defined on the alternating quotient and satisfies $\partial_1\partial_2 = 0$.

Proof. The differential is the odd derivation determined by $Q\psi = xy - yx$, $Qx = Qy = 0$. Since the second ψ is preceded by one odd letter, replacing it carries the Koszul sign -1:

$$Q\ \mathrm{Tr}(\psi u \psi v) = \mathrm{Tr}\big((xy - yx)u\psi v\big) - \mathrm{Tr}\big(\psi u(xy - yx)v\big).$$

In the first term the even word $(xy - yx)u$ is moved cyclically past the final displayed ψv, producing no additional sign. Removing the common front ψ gives $vxyu - vyxu - uxyv + uyxv$. Interchanging u and v reverses this expression:

$$uxyv - uyxv - vxyu + vyxu = -(vxyu - vyxu - uxyv + uyxv),$$

so the formula descends to the alternating quotient $C_2(k, l)$.

Applying ∂_1 to the four summands gives the signed sum of cyclic words obtained by inserting two adjacent commutators. Each final cyclic word occurs twice with opposite signs, according to the order in which the two displayed ψ's are replaced. Equivalently $Q^2 = 0$ for the odd derivation Q. Hence $\partial_1\partial_2 = 0$. □

THEOREM 3.144 *(Primitive one-antifield homology).* For every $k, l \geq 0$,

$$H_1\big(C_2(k, l) \to C_1(k, l) \to C_0(k, l)\big) \cong \mathbb{C} \cdot [\Psi_{k,l}].$$

The Hamiltonian comparison uses the nonconstant sum $k + l > 0$; the scalar class $\Psi_{0,0} = \mathrm{Tr}(\psi)$ is not part of $\bar{A}_{\mathrm{desc}}$.

Proof. If $k = 0$ or $l = 0$, then $C_1(k, l)$ is spanned by the single word x^k or y^l, $\partial_1 = 0$, and $C_2(k, l) = 0$. The class is represented by $\Psi_{k,l}$.

Assume $k, l > 0$. Form the graph $\Gamma_{k,l}$ with vertices $\mathsf{N}_{k+1,l+1}$ and oriented edges $w \in \mathsf{W}_{k,l}$,

$$w : [yxw]_{\mathrm{cyc}} \longrightarrow [xyw]_{\mathrm{cyc}}.$$

Its cellular boundary is ∂_1. Attach one square for each alternating pair (u, v) with boundary $vxyu - vyxu - uxyv + uyxv$. By Lemma 3.143, this square boundary is exactly ∂_2; denote the resulting two-complex by $X_{k,l}$.

Put $R = k + 1$ and $S = l + 1$. $X_{k,l}$ is the two-skeleton of the cyclic chip-moving state complex $Y_{R,S}$. Before cyclic relabelling, a p-cell is a configuration of $R - p$ stationary indistinguishable chips on the vertices of an S-edge oriented circle and p moving chips in distinct edge interiors. The orientation is the increasing order of the moving edge labels. Traversing the labelled circle writes x for stationary chips, y for an empty edge, and ψ for a moving edge. Hence vertices are $\mathsf{N}_{R,S}$, one-cells are the words $w \in \mathsf{W}_{k,l}$, and two-cells are exactly the alternating pairs (u, v) above.

The Abel map sends a labelled configuration to the sum of its chip positions in $\mathbb{R}/S\mathbb{Z}$ and is affine on every cell. After lifting the target to $\mathbb{R}$ and fixing a cyclic order of the lifted chips, a fibre is the convex polytope cut out by

$$t_1 \leq \cdots \leq t_R \leq t_1 + S, \qquad \sum_i t_i = \text{constant},$$

together with the chip-cell inequalities. The chip-moving cells subdivide this polytope. Thus the labelled Abel map is a homology equivalence with S^1.

Cyclic relabelling by $r \in C_S$ adds r to every chip position, so the Abel coordinate is translated by Rr. The action on the target circle is orientation-preserving and therefore trivial on H_0 and H_1. Since $\mathbb{C}[C_S]$ is semisimple, homology commutes with coinvariants. If a stabilizer $G \subset C_S$ fixes an Abel value, it acts affinely on each lifted convex fibre; averaging gives a G-fixed point and straight-line contraction to it descends to a contraction of the quotient fibre. Therefore the cyclic quotient still has the homology of S^1. Finally, attaching cells of dimension at least three cannot change H_1, because H_1 only sees $C_2 \to C_1 \to C_0$. Hence $H_1(X_{k,l}; \mathbb{C}) \cong \mathbb{C}$.

The symmetrised trace

$$\Psi_{k,l} = \sum_{w \in \mathsf{W}_{k,l}} \mathrm{Tr}(\psi w)$$

is a cycle: the coefficient of any zero-ψ necklace in $Q\Psi_{k,l}$ counts cyclic xy-transitions minus cyclic yx-transitions, and these two numbers are equal on a circle. The functional $\epsilon(\mathrm{Tr}(\psi w)) = 1$ kills every two-ψ boundary, because the formula in Lemma 3.143 has two positive and two negative summands, while

$$\epsilon(\Psi_{k,l}) = \binom{k+l}{k} \neq 0.$$

Each one-cell occurring in $\Psi_{k,l}$ is oriented in the positive Abel direction. Hence its Abel image is $\binom{k+l}{k}$ times the fundamental circle class. The unique H_1-line is generated by $[\Psi_{k,l}]$. □

Definition 3.145 *(Full primitive ψ-complex).* Let

$$A_\psi = \mathbb{C}\langle x, y, \psi \rangle, \qquad |x| = |y| = 0, \quad |\psi| \equiv 1 \pmod 2,$$

with $d\psi = xy - yx$ and $dx = dy = 0$. In the graded cyclic quotient $\mathrm{Cyc}(A_\psi) = A_\psi / [A_\psi, A_\psi]_{\mathrm{gr}}$, put

$$\mathrm{wt}(x) = (1, 0), \qquad \mathrm{wt}(y) = (0, 1), \qquad \mathrm{wt}(\psi) = (1, 1).$$

For fixed (R, S), define

$$K^p_{R,S} = \mathbb{C}\{[w]_{\mathrm{cyc}} : \#\psi(w) = p,\ \#x(w) = R - p,\ \#y(w) = S - p\},$$

with $0 \le p \le \min(R, S)$. The boundary is

$$\partial[w] = \sum_{i : a_i = \psi} (-1)^{\#\{\psi \text{ before } i\}} [a_1 \cdots a_{i-1} x y\, a_{i+1} \cdots a_n - a_1 \cdots a_{i-1} y x\, a_{i+1} \cdots a_n]_{\mathrm{cyc}}.$$

The sign is the Koszul sign for replacing the i-th odd letter by the even commutator $xy - yx$; with this sign $\partial^2 = 0$.

Lemma 3.146 *(All-ψ face sign).* Let $u_1, \ldots, u_p$ be ordinary words in x, y. In $K^{p-1}_{R,S}$,

$$\partial[\psi u_1 \psi u_2 \cdots \psi u_p] = \sum_{j=1}^{p} (-1)^{j-1} [\psi u_1 \cdots \psi u_{j-1} x y\, u_j \psi u_{j+1} \cdots \psi u_p - \psi u_1 \cdots \psi u_{j-1} y x\, u_j \psi u_{j+1} \cdots \psi u_p]_{\mathrm{cyc}}.$$

Under the chip-moving cellular model below, this is exactly the oriented cubical boundary of the corresponding p-cell.

Proof. The j-th displayed ψ has precisely $j-1$ odd letters before it. Replacing it by the even commutator $xy - yx$ therefore contributes the Koszul sign $(-1)^{j-1}$ and no further sign. The cellular p-cell is oriented by the increasing order of its p moving chip-edges. Its j-th coordinate has terminal face xy and initial face yx, with cubical sign $(-1)^{j-1}$. These two formulae agree term by term. If a cyclic rotation moves a block containing a odd letters past a block containing b odd letters, the cellular orientation changes by $(-1)^{ab}$, the graded cyclic sign. Hence the formula descends to the cyclic quotient. □

Construction 3.147 *(Chip-moving cellular model).* Assume $R, S > 0$. Let $\widetilde{Y}_{R,S}$ be the labelled state complex on an oriented S-edge circle. A p-cell consists of $R-p$ stationary indistinguishable chips on vertices and p moving chips in the interiors of p distinct edges. Traversing the labelled circle writes x for each stationary chip at a vertex and writes y, respectively ψ, for an empty, respectively moving, following edge. Thus p-cells are labelled words with $R-p$ x's, $S-p$ y's, and p ψ's.

The cyclic group C_S rotates the labelled edge set. The quotient chain complex is taken with the orientation character already encoded by the graded cyclic rule $[uv] = (-1)^{|u||v|}[vu]$.

LEMMA 3.148 *(Cellular chain identification).* For $R, S > 0$, the cellular chains of Construction 3.147, after cyclic relabelling, identify with the full primitive cyclic Koszul complex:

$$C_p(\widetilde{Y}_{R,S};\mathbb{C})_{C_S} \cong K^p_{R,S}.$$

Under this identification the cellular boundary is the boundary of Definition 3.145.

Proof. A labelled p-cell has a unique expression $x^{a_1}m_1 \cdots x^{a_S}m_S$, where $m_i \in \{y, \psi\}$, exactly p markers are ψ, and $\sum a_i = R-p$. Every graded cyclic word admits such a representative after moving the cut to an edge marker; choosing the labelled starting edge makes the representative unique. Forgetting the labelled starting edge is exactly passage to graded cyclic words, with no relation other than cyclic relabelling. The sign check is Lemma 3.146; cyclic relabelling by a rotation that moves a moving-edge coordinates past b moving-edge coordinates changes orientation by $(-1)^{ab}$, which is the same sign as moving the corresponding odd ψ-blocks in the graded cyclic quotient. □

LEMMA 3.149 *(Abel map and cyclic stabilizers).* The quotient state complex $Y_{R,S} = \widetilde{Y}_{R,S}/C_S$, with the orientation local system of Lemma 3.148, has the homology of S^1. The surviving H_1-line is represented by the Abel-positive cycle $\Psi_{R-1,S-1}$.

Proof. Work first before cyclic relabelling. In the universal cover of the S-edge circle, a lifted configuration is described by weakly ordered chip positions

$$t_1 \le \cdots \le t_R \le t_1 + S.$$

The Abel map sends the configuration to $\sum_i t_i \in \mathbb{R}/S\mathbb{Z}$ and is affine on every cell. After lifting the target circle to $\mathbb{R}$, each fibre is the convex polytope cut out by the displayed inequalities and by the fixed-sum equation. The chip-moving cells subdivide this polytope. Hence every lifted fibre is a contractible finite polyhedral complex, and the labelled Abel map is a proper PL map with acyclic fibres. Vietoris–Begle gives a cellular homology equivalence

$$\widetilde{Y}_{R,S} \simeq_{\mathrm{H}} S^1.$$

A cyclic relabelling by $r \in C_S$ adds r to each chip position, so the Abel coordinate is translated by Rr on $\mathbb{R}/S\mathbb{Z}$. Thus relabelling acts on the target by orientation-preserving rotations and acts trivially on H_0 and H_1. Since $\mathbb{C}[C_S]$ is semisimple, taking coinvariants commutes with homology:

$$H_\bullet(C_\bullet(\widetilde{Y}_{R,S})_{C_S}) \cong H_\bullet(\widetilde{Y}_{R,S})_{C_S} \cong H_\bullet(S^1).$$

The fibre statement survives stabilizers. If a subgroup $G \subset C_S$ fixes a target Abel value, choose the lifted fibre before decomposing it into chip cells. It is a convex polytope preserved by the affine G-action. Averaging any point of the fibre gives a G-fixed point, and the straight-line contraction to that fixed point is G-equivariant and respects the chip-cell subdivision. Therefore the stabilizer quotient of the fibre is contractible.

Finally, $\Psi_{R-1,S-1} = \sum_w [\psi w]_{\mathrm{cyc}}$ is closed by the same transition count used in Theorem 3.144. Each of its one-cells is oriented in the positive Abel direction, so its image is a nonzero multiple of the fundamental class of the target circle. Hence it spans the unique H_1-line. □

Theorem 3.150 *(Full primitive higher-ψ Koszul homology).* For every $R, S \geq 0$, the full primitive cyclic Koszul complex of Definition 3.145 has homology

$$H_p(K_{R,S}) \cong \begin{cases} \mathbb{C} \cdot [x^R y^S], & p = 0, \\ \mathbb{C} \cdot [\Psi_{R-1,S-1}], & p = 1,\ R, S \geq 1, \\ 0, & p \geq 2. \end{cases}$$

Equivalently, the stable primitive single-trace Koszul homology is

$$\mathbb{C}[x, y] \oplus \mathbb{C}[x, y]\theta, \qquad |\theta| = 1, \qquad \mathrm{wt}(\theta) = (1, 1),$$

with $x^a y^b \theta$ represented by $\Psi_{a,b}$.

Proof. If $R = 0$ or $S = 0$, no positive ψ-degree term exists and $K^0_{R,S}$ is the one-dimensional span of the necklace x^R or y^S, including the empty word when $R = S = 0$.

Assume $R, S > 0$. Lemma 3.148 identifies $K^\bullet_{R,S}$ with the cyclic-coinvariant cellular chains of $\widetilde{Y}_{R,S}$. Lemma 3.149 computes their homology as $H_\bullet(S^1; \mathbb{C})$. Thus only degrees 0 and 1 survive. The H_0-line is the commutative necklace class $[x^R y^S]$. The H_1-line is the Abel-positive fundamental class, represented by $\Psi_{R-1,S-1}$. The vanishing for $p \geq 2$ is the vanishing of $H_{\geq 2}(S^1; \mathbb{C})$; the H_1 calculation is already visible in the three-term skeleton, since higher cells do not alter $\ker(C_1 \to C_0)/\mathrm{im}(C_2 \to C_1)$. □

Theorem 3.151 *(Hamiltonian sector extracted from full psi homology).* The Hamiltonian comparison uses exactly the nonconstant summands

$$\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1] = \bigoplus_{R+S>0} \mathbb{C} \cdot [x^R y^S] \oplus \bigoplus_{a+b>0} \mathbb{C} \cdot [\Psi_{a,b}].$$

The sectors outside that comparison are precisely: the scalar degree-zero class $[1]$, the scalar odd class $\Psi_{0,0} = \mathrm{Tr}(\psi)$, and the absent higher primitive ψ-sectors $H_p(K_{R,S}) = 0$ for $p \geq 2$. Identifying the remaining $\Psi_{a,b}$ line with principal parts $\rho_{a,b}$ as an $\mathfrak{h}$-module, or after Moyal quantization, is not part of this homology theorem.

Proof. Constants are removed from the closed Hamiltonian algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$, and the open empty trace is the scalar $\mathrm{Tr}(1) = N$. This removes the $H_0(K_{0,0})$ line. The class $\Psi_{0,0}$ is a genuine odd primitive class, but it is the descendant of the removed constant Hamiltonian and therefore has no closed Hamiltonian partner in $\bar{A}_{\mathrm{desc}}[1]$. By Theorem 3.150, all primitive $p \geq 2$ homology groups vanish. The only surviving positive Hamiltonian labels are therefore the displayed nonconstant H_0 and H_1 lines. The final sentence is exactly the module-structure separation proved later by Proposition 3.155 and Theorem 3.157. □

THEOREM 3.152 *(Universal primitive descendant envelope).* The stable primitive source for the reduced cotangent comparison is forced to be

$$\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1] = \bigoplus_{R+S>0} \mathbb{C} \cdot [x^R y^S] \oplus \bigoplus_{a+b>0} \mathbb{C} \cdot [\Psi_{a,b}].$$

More precisely, every stable primitive chain-level construction generated by x, y, ψ, with $Q\psi = xy - yx$ and with only graded-cyclic trace relations, factors on homology through the two displayed summands. Thus a Koszul-dual cotangent theory in this statement has no available hidden higher-ψ correction term: the entire descendant side is the one-ψ line $\bar{A}_{\mathrm{desc}}[1]$.

After the Matlis normalization

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n},$$

the only flat $D_\hbar$-bridge constructed from the polynomial descendant labels is the Fourier-twisted Rees bridge of Theorem 3.158. If the original Hamiltonian action of z_1, z_2 is required to be preserved, the exact obstruction is the local-nilpotence mismatch of Theorem 3.159.

Proof. The first assertion is Theorem 3.150 together with the scalar removal in Theorem 3.151. The cellular proof identifies the universal primitive complex with the chip-moving complex and proves $H_p(K_{R,S}) = 0$ for $p \geq 2$. Therefore any stable primitive construction satisfying the same trace and differential hypotheses has no homology in which an additional universal cotangent correction could live.

The Matlis pairing fixes the dual coefficient normalization up to the single scalar allowed by residue-pairing rigidity; the displayed formula chooses that scalar. With this normalization, the positive flat Rees construction is exactly the Fourier-twisted cyclic $D_\hbar$-module $\mathcal{D}_\hbar/(z_1, z_2)$ embedded into the principal-part Čech lattice with the explicit factor $(-1)^{a+b}\hbar^{a+b}a!b!$. If one removes the Fourier twist and asks for the same Hamiltonian action, local nilpotence of the linear Hamiltonians is preserved by flat generic-fibre isomorphism but differs on the two sides. This is precisely the obstruction theorem cited in the statement. □

Construction 3.153 *(Primitive one-ψ label shadow).* Extend π_{prim} to the one-ψ primitive shadow by recording the labels of the symmetrised traces $\Psi_{a,b}$ of Construction 3.195. The domain here is the enlarged principal-part shadow object $\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I)$: the Hamiltonian primitive sector is smeared by $\Omega^\bullet_c(I)$, and the principal-part label sector is smeared by $\mathcal{D}'_c(I)$. Define

$$\pi^{\mathrm{cot}}_{\mathrm{prim}} : \widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) \longrightarrow \big(\Omega^\bullet_c(I)\widehat{\otimes}\bar{A}\big) \oplus \big(\mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1]\big)$$

by:

- on the zero-ψ Hamiltonian sector, $\pi^{\mathrm{cot}}_{\mathrm{prim}} = \pi_{\mathrm{prim}}$ composed with the inclusion into the $\Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$ summand;
- on a primitive one-ψ word $\mathrm{Tr}(\psi w)$ of bidegree (a, b), $a + b > 0$, smeared by a distributional test current $B \in \mathcal{D}'_c(I)$,
$$\pi^{\mathrm{cot}}_{\mathrm{prim}}\big(B \otimes \mathrm{Tr}(\psi w)\big) = \binom{a+b}{a}^{-1} B \otimes \rho_{a,b}[1].$$
Hence
$$\pi^{\mathrm{cot}}_{\mathrm{prim}}\big(B \otimes \Psi_{a,b}\big) = B \otimes \rho_{a,b}[1] = \Theta_{a,b}(B).$$

- the scalar class $\Psi_{0,0} = \mathrm{Tr}(\psi)$ maps to the removed odd scalar line, equivalently to zero in the Hamiltonian principal-part target $\mathcal{P}[1]$;
- on primitive single-trace chains with at least two displayed ψ's, $\pi^{\mathrm{cot}}_{\mathrm{prim}} \equiv 0$;
- on multi-trace products of one-ψ classes (or mixed products of zero-ψ and one-ψ traces), $\pi^{\mathrm{cot}}_{\mathrm{prim}} \equiv 0$ (cumulant on the multi-trace stratum).

This is a shadow map to the reduced principal-part label target, not a polynomial realization of the coadjoint module. The no-polynomial theorem 3.234 says precisely that the open polynomial one-ψ descendants and the principal-part currents do not carry the same $\mathfrak{h}$-module structure. The map above is therefore an indecomposable shadow map after imposing the reduced current target $\mathcal{P}[1]$; it is not an unreduced boundary observable realizing $\mathcal{P}$ inside the polynomial BV/Koszul algebra.

Theorem 3.154 *(Primitive indecomposable P_0-shadow and one-ψ label shadow).* On every interval I, in the stable connected limit, the Hamiltonian zero-ψ primitive indecomposable P_0-shadow π_{prim} of Construction 3.135 and the one-ψ shadow $\pi^{\mathrm{cot}}_{\mathrm{prim}}$ of Construction 3.153 give a surjection to the generator space, equivalently the indecomposable quotient, of the boundary factorization algebra

$$\pi^{\mathrm{cot}}_{\mathrm{prim}} : \widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) \twoheadrightarrow \mathfrak{h}_I \oplus \mathcal{P}_I[1]$$

in the sense of Construction 3.18, with the following compatibilities:

(i) *(BV chain map.)* $\pi^{\mathrm{cot}}_{\mathrm{prim}} \circ Q_{\mathrm{tot}} = Q_{\mathrm{tot}} \circ \pi^{\mathrm{cot}}_{\mathrm{prim}}$ strictly at the chain level (Theorem 3.137 and Theorem 3.144 extending it to the cotangent sector via the closed cocycle property of $\Psi_{a,b}$);

(ii) *(Factorization product.)* The zero-ψ Hamiltonian-current compatibility is the unit-retaining linearized statement of Theorem 3.139. The one-ψ label shadow satisfies the analogous unit-retaining linearized statement: scalar–one-ψ and one-ψ–scalar products extend the one-ψ label by zero to the larger interval, while products with two positive primitive factors are killed in the indecomposable quotient;

(iii) *(P_0 bracket.)* The P_0-bracket compatibility square commutes strictly modulo a zero homotopy on the Hamiltonian zero-ψ indecomposable quotient (Theorem 3.140). On the one-ψ shadow the map records labels only; it does not intertwine the open polynomial descendant PBW action with the principal-part coadjoint action. The reduced target bracket $\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB)$ is supplied separately by Theorem 3.237;

(iv) *(Cosheaf.)* $\pi^{\mathrm{cot}}_{\mathrm{prim}}$ is compatible with extension by zero: on the Hamiltonian summand this is extension by zero in $\Omega^\bullet_c$, and on the principal-part cotangent summand it is extension by zero in $\mathcal{D}'_c$.

The target is the nonunital indecomposable quotient: multiplication of two positive primitive generators is zero after projection, while the Lie/P_0 bracket remains the Hamiltonian bracket. Thus the statement is not an algebra map to $\bar{A}$ with its ordinary commutative multiplication.

Equivalently, $\pi^{\mathrm{cot}}_{\mathrm{prim}}$ is the chain-level realization of the unreduced-to-reduced primitive shadow required by Problem 3.60, item (3). It is not a proof of the unreduced cotangent factorization-center lift.

Proof. Items (1), (2), (3) on the Hamiltonian zero-ψ sector are Theorems 3.137, 3.139, 3.140 above.

One-ψ shadow. The functional on one-ψ words used in Construction 3.153 sends every word of bidegree (a, b) to the same scalar $\binom{a+b}{a}^{-1}\rho_{a,b}$. It kills the two-$\psi$ boundary because the boundary formula

of Lemma 3.143 has two positive and two negative one-ψ summands of the same bidegree. It sends the indecomposable label represented by $\Psi_{a,b}$ to the imposed reduced target basis vector $\rho_{a,b}$; this is not an $\mathfrak{h}$-equivariant realization of the polynomial one-ψ module. It kills the scalar $\Psi_{0,0}$. Primitive chains with $p \geq 2$ displayed ψ's are killed by definition; after one application of Q_{BV} they land in the kernel just described, since the all-ψ boundary is a signed sum of one fewer ψ-chains and the equal-weight functional cancels the two-ψ faces while higher faces still have positive ψ-count. With the current differential on $\mathcal{D}'_c(I)$, this proves strict Q-compatibility on the one-ψ label summand.

The factorization-product compatibility on one-ψ labels is linearized in the same sense as Theorem 3.139. A scalar factor acts as the unit and extends the one-ψ current by zero to the larger interval. A product of a one-ψ primitive with any positive primitive trace has at least two primitive factors and is killed in the indecomposable quotient. Thus the unit-retaining exactly-one-positive projection agrees on the two routes. The P_0-bracket compatibility asserted here is the zero-ψ Hamiltonian statement already proved above. The one-ψ shadow deliberately does not assert compatibility between the open descendant PBW action and the principal-part coadjoint action; their difference is Corollary 3.224. The bracket of two cotangent generators is zero on both sides because $\mathcal{P}[1]$ is an abelian odd ideal in $\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$.

Cosheaf compatibility is automatic on each summand: extension by zero on the $\Omega^\bullet_c$ Hamiltonian summand and on the $\mathcal{D}'_c$ principal-part current factor commute with the fixed trace-label projections.

The chain-level surjectivity is the surjectivity of the cumulant projection onto generators in the stable symmetric algebra (each generator is hit by itself), tensored with the identity on the compactly supported de Rham or distributional current factor on the same interval.

This establishes all four items. □

PROPOSITION 3.155 *(Quantum obstruction to the primitive one-ψ label map).* Let $\mathfrak{h}_\hbar = \bar{A}_\hbar$ carry the Weyl/Moyal bracket

$$\{f, g\}_\hbar = \sum_{j \geq 0} \frac{\hbar^{2j}}{(2j+1)!\, 2^{2j}} P^{2j+1}(f, g), \qquad P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

The chain-level one-ψ primitive label map $\Psi_{a,b} \mapsto \rho_{a,b}$ is not compatible with the $\mathfrak{h}_\hbar$-coadjoint action. The failure already occurs in the indecomposable target; it is not a decomposable multi-trace correction.

Proof. Compatibility would require the polynomial PBW descendant action of Proposition 3.223, after the label replacement $\widetilde{\Psi}_{a,b} \mapsto \rho_{a,b}$, to agree with the Moyal coadjoint action

$$(\mathrm{ad}^\vee_{f,\hbar}\, \delta)(g) = -\delta(\{f, g\}_\hbar).$$

Test this on the single Hamiltonian $f = z_1^4$ and the single descendant label $(a, b) = (1, 1)$.

On the polynomial one-ψ side, Proposition 3.223 gives

$$z_1^4 : \widetilde{\Psi}_{1,1} \longmapsto 4\, \widetilde{\Psi}_{4,0}.$$

On the Moyal coadjoint side the classical Poisson term has negative first target index and vanishes, but the P^3 term does not:

$$P^3(z_1^4, z_2^4) = 576\, z_1 z_2, \qquad \frac{1}{3!\, 2^2} P^3(z_1^4, z_2^4) = 24\, z_1 z_2.$$

Hence

$$\mathrm{ad}^\vee_{z_1^4,\hbar}\, \delta_{1,1} = -24\, \hbar^2 \delta_{0,4} + \cdots,$$

equivalently $z_1^4 \cdot_\hbar \rho_{1,1} = -24\,\hbar^2 \rho_{0,4} + \cdots$ in the principal-part realization. The two answers have different $\hbar$-order and different primitive label. Since both sides have already been projected to the linear indecomposable summand, adding decomposable multi-trace terms cannot repair this square after applying $\pi^{\mathrm{cot}}_{\mathrm{prim}}$.

The finite exact-arithmetic attack verifies the same obstruction through exponents ≤ 5 and Moyal order ≤ 5: 1200 of 1225 nonzero test cases disagree under the naive label map, and 958 cases have genuine higher Moyal coadjoint terms. The same finite computation also checks the corrected principal-part target below in 3375 finite representation identities with exponents ≤ 3, Moyal order ≤ 5, and $\hbar$-power ≤ 4, and checks the linear-generator leading-term obstruction to every $\hbar$-adic deformation of the classical label shadow in 70 tests with labels ≤ 5. The displayed z_1^4 calculation is already a proof that the classical primitive one-ψ projection does not itself quantize to the descendant coadjoint sector. □

Construction 3.156 *(Reduced Moyal principal-part descendant target).* The corrected quantum descendant target is obtained by changing the target, not by correcting the polynomial one-ψ representatives. For an interval $I \subset \mathbb{R}$ put

$$\mathfrak{h}_{\hbar,I} = \Omega_c^\bullet(I)\widehat{\otimes}\bar{A}_\hbar, \qquad \mathcal{P}_{\hbar,I} = \mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P}[[\hbar]].$$

Define the reduced quantum principal-part descendant complex

$$A^{\mathrm{pp}}_{\partial,\hbar}(I) = \widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}(\mathfrak{h}_{\hbar,I})\widehat{\otimes}\widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}(\mathcal{P}_{\hbar,I}[1]).$$

The differential is the de Rham/current differential on the interval factors and zero on the coefficient modules $\bar{A}_\hbar$ and $\mathcal{P}[[\hbar]]$. The bracket is the Moyal semidirect bracket

$$\{B_f(a), B_g(b)\}_\hbar = B_{\{f,g\}_\hbar}(ab), \qquad \{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$

with

$$\{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0.$$

Here $a, b \in \Omega_c^0(I)$, $B, C \in \mathcal{D}_c'(I)$, and aB denotes multiplication of a compactly supported distribution by a smooth test function.

In monomial coordinates the full coadjoint action is as follows. Put

$$C_r(p,q;m,n) = \sum_{\alpha=0}^{r}(-1)^\alpha\binom{r}{\alpha}(p)_{r-\alpha}(q)_\alpha(m)_\alpha(n)_{r-\alpha}.$$

Then

$$z_1^p z_2^q \cdot_\hbar \rho_{a,b} = -\sum_{\substack{r\geq 1\\ r\text{ odd}}} \frac{\hbar^{r-1}}{2^{r-1}r!}\, C_r(p,q;a-p+r,b-q+r)\,\rho_{a-p+r,b-q+r},$$

with the convention that negative indices and the scalar target $(0,0)$ are zero. This is a finite sum for each monomial $z_1^p z_2^q$, because the falling factorials vanish for $r > p+q$. The $r = 1$ term is the classical Matlis coadjoint action of Theorem 3.230; the $r \geq 3$ terms are the genuine Moyal corrections.

This construction supplies a reduced distributional principal-part target for the quantum descendant sector. It does not make the polynomial PBW classes $\Psi_{a,b}$ into that target.

Theorem 3.157 *($\hbar$-adic obstruction for the one-ψ label shadow).* Let $\mathcal{M}_{\Psi,\hbar}$ be a $\mathbb{C}[[\hbar]]$-flat complex generated over $\mathbb{C}[[\hbar]]$ by lifts of the primitive polynomial one-ψ classes $\widetilde{\Psi}_{a,b}$, $a+b>0$. Assume:

(i) the source action of the linear Hamiltonians reduces modulo $\hbar$ to the PBW descendant action

$$z_1 \cdot_0 \widetilde{\Psi}_{a,b} = b\,\widetilde{\Psi}_{a,b-1}, \qquad z_2 \cdot_0 \widetilde{\Psi}_{a,b} = -a\,\widetilde{\Psi}_{a-1,b};$$

(ii) the target coefficient module is $\mathcal{P}[[\hbar]][1]$, equivalently any fixed interval-current summand of $\mathcal{P}_{\hbar,I}[1]$ from Construction 3.156, with the full Moyal coadjoint action;

(iii) a candidate comparison $T_\hbar : \mathcal{M}_{\Psi,\hbar} \to \mathcal{P}[[\hbar]][1]$ has the classical one-ψ label shadow as leading term:

$$T_\hbar(\widetilde{\Psi}_{a,b}) = \rho_{a,b}[1] + O(\hbar).$$

Then $T_\hbar$ cannot intertwine the $\mathfrak{h}_\hbar$-action. The obstruction already appears after reduction modulo $\hbar$ and for the linear Hamiltonian z_1. Consequently no strict quantum descendant lift of the classical primitive label shadow exists under these assumptions. The only surviving quantum target in this statement is the reduced principal-part complex itself, not the polynomial one-ψ source.

Proof. Reduce a hypothetical intertwiner modulo $\hbar$. For the class $\widetilde{\Psi}_{1,0}$, the PBW source action gives

$$z_1 \cdot_0 \widetilde{\Psi}_{1,0} = 0.$$

Hence the source-then-target route gives

$$T_0(z_1 \cdot_0 \widetilde{\Psi}_{1,0}) = 0.$$

The target-then-action route gives, by the $r = 1$ term of Construction 3.156,

$$z_1 \cdot_0 T_0(\widetilde{\Psi}_{1,0}) = z_1 \cdot \rho_{1,0}[1] = -\rho_{1,1}[1].$$

The principal-part basis vector $\rho_{1,1}$ is nonzero in $\mathcal{P}$. This contradiction is independent of all higher $\hbar$-corrections to $T_\hbar$, because those corrections vanish after taking the leading term. It is also independent of the higher Moyal terms: the linear Hamiltonian has no $r \geq 3$ Moyal correction.

The same computation with z_2 gives $z_2 \cdot_0 \widetilde{\Psi}_{0,1} = 0$ on the PBW side but $z_2 \cdot \rho_{0,1} = \rho_{1,1}$ on the principal-part side. Thus the mismatch is not a special feature of one coordinate. Since the coefficient differential on $\mathcal{P}[[\hbar]]$ is zero, this leading coefficient is not a boundary in the reduced target. A higher homotopy can remove it only after changing the target complex or the classical source action, which is outside the stated assumptions. □

THEOREM 3.158 *(Universal Fourier-twisted Rees D-hbar construction).* Let

$$D_\hbar = \mathbb{C}[[\hbar]]\langle z_1, z_2, \partial_1, \partial_2\rangle/([\partial_i, z_j] = \hbar\delta_{ij},\ [z_i, z_j] = [\partial_i, \partial_j] = 0)$$

be the Rees Weyl algebra. Let

$$\mathcal{O}_\hbar = D_\hbar/(\partial_1, \partial_2), \qquad \Delta^{\mathsf{F}}_\hbar = D_\hbar/(z_1, z_2),$$

and let e_0 be the image of 1 in $\Delta^{\mathsf{F}}_\hbar$. The quotients are left cyclic $D_\hbar$-modules, with left ideals generated by ∂_1, ∂_2 and by z_1, z_2. PBW normal form gives $\mathcal{O}_\hbar$ the free $\mathbb{C}[[\hbar]]$-basis $z_1^a z_2^b$ and gives $\Delta^{\mathsf{F}}_\hbar$ the free basis $\partial_1^a \partial_2^b e_0$, so both modules are $\mathbb{C}[[\hbar]]$-flat. The Fourier automorphism

$$\mathsf{F}(z_i) = \partial_i, \qquad \mathsf{F}(\partial_i) = -z_i$$

identifies the Fourier twist of $\mathcal{O}_\hbar$ with $\Delta_\hbar^{\mathsf{F}}$. In the Čech local-cohomology lattice generated by $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$, the map $e_0 \mapsto \rho_{0,0}$ sends

$$\partial_1^a \partial_2^b e_0 \longmapsto (-1)^{a+b} \hbar^{a+b} a!b!\, \rho_{a,b}.$$

This is a flat $D_\hbar$-module bridge in the Fourier-conjugated sense: $P \in D_\hbar$ acts on the polynomial source as P, and on the delta target as $\mathsf{F}(P)$. After tensoring with $\mathbb{C}((\hbar))$, it gives a filtered Fourier–Rees label bijection after this polarization change between the nonconstant polynomial label span and $\mathcal{P}((\hbar))$. It is not an $\mathfrak{h}$-module identification for the original Hamiltonian action. Before $\hbar$ is inverted, the image is the explicit Rees sublattice

$$\bigoplus_{a,b \geq 0} \mathbb{C}[[\hbar]]\, (-1)^{a+b} \hbar^{a+b} a!b!\, \rho_{a,b}.$$

Proof. The formulas for F preserve the Rees Weyl relation:

$$[\mathsf{F}(\partial_i), \mathsf{F}(z_j)] = [-z_i, \partial_j] = \hbar \delta_{ij}.$$

The polynomial module $\mathcal{O}_\hbar$ is generated by 1 killed by ∂_1, ∂_2. After applying F, the cyclic generator is killed by z_1, z_2, which is exactly the defining cyclic vector of $\Delta_\hbar^{\mathsf{F}}$.

In the Čech model ∂_i acts as $\hbar \partial_{z_i}$. Hence

$$\begin{aligned} \partial_1^a \partial_2^b \rho_{0,0} &= \hbar^{a+b} \partial_{z_1}^a \partial_{z_2}^b (z_1^{-1} z_2^{-1} dz_1 dz_2) \\ &= (-1)^{a+b} \hbar^{a+b} a!b! \rho_{a,b}. \end{aligned}$$

The factorials and signs are units over $\mathbb{C}$; the nontrivial Rees datum is the power $\hbar^{a+b}$. Inverting $\hbar$ turns this sublattice into the whole principal-part label span. □

Theorem 3.159 *(Exact same-action Rees obstruction).* There is no $\mathbb{C}[[\hbar]]$-flat Rees module $M_\hbar$ with all three properties:

(i) its special fibre is the polynomial primitive descendant module generated by the $\widetilde{\Psi}_{a,b}$, $a + b > 0$, with linear Hamiltonian action

$$z_1 \cdot \widetilde{\Psi}_{a,b} = b\, \widetilde{\Psi}_{a,b-1}, \qquad z_2 \cdot \widetilde{\Psi}_{a,b} = -a\, \widetilde{\Psi}_{a-1,b};$$

(ii) on the generic fibre $M_\hbar((\hbar))$, the same linear Hamiltonians z_1, z_2 act locally nilpotently, as they do in the polynomial Rees Weyl module;

(iii) the generic fibre is isomorphic, as a module for the same Hamiltonian Lie algebra, to $\mathcal{P}((\hbar))$ with the Matlis coadjoint action

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Equivalently, matching the local-nilpotence profile of z_1, z_2 on the generic fibre is a necessary condition for any flat same-action Rees bridge from polynomial one-ψ descendants to principal parts, and that necessary condition fails. Thus the universal bridge exists only after the Fourier twist of polarization in Theorem 3.158; it cannot be a same-action Rees family.

Proof. Local nilpotence of a fixed operator is invariant under module isomorphism over $\mathbb{C}((\hbar))$. On the polynomial descendant side, the displayed formulas lower the b-index for z_1 and the a-index for z_2; hence every finite vector is killed by a high enough power of each linear Hamiltonian. The same is true for the polynomial Rees Weyl module after inverting $\hbar$, because multiplying the operators by the invertible scalar $\hbar$ does not change local nilpotence.

On $\mathcal{P}((\hbar))$, no nonzero finite Laurent vector is locally nilpotent. Indeed, for $v = \sum c_{a,b}\rho_{a,b} \neq 0$,

$$z_1^N v = \sum c_{a,b}(-1)^N (b+1)(b+2)\cdots(b+N)\rho_{a,b+N}.$$

The basis vectors $\rho_{a,b+N}$ remain distinct for distinct (a,b), and every rising factorial displayed above is a nonzero element of $\mathbb{C}((\hbar))$. Thus $z_1^N v \neq 0$ for all N. The same argument using z_2 shifts a upward and proves non-local-nilpotence for z_2. Hence $\mathcal{M}_\hbar((\hbar))$ cannot be isomorphic to $\mathcal{P}((\hbar))$ under the same Hamiltonian action. □

Remark 3.160 *(Comparison with the heat-kernel propagator route).* The chain-level construction above avoids any heat-kernel propagator $P_{\epsilon,L}$ on $\mathbb{R}^2 \times \mathbb{C}^2$. In the Costello–Gwilliam BV propagator approach, the primitive indecomposable shadow is realized as a homotopy retraction onto the cohomology of the free open observable algebra, with the homotopy given by $G_L = \int_0^L K_t\, dt$ (the partial heat kernel). Here the role of G_L is played by the symmetric-algebra splitting, and the key analytic fact (positive jets are Q-exact, Lemma 3.191) is needed only to identify the E_1-stalk of the brane factorization algebra with its C^∞-smearing. The two routes agree on the classical cocycle level by uniqueness of the cumulant projection on a symmetric algebra.

The chain-level construction has the advantage of being independent of the analytic input of the heat kernel, and therefore independent of the Tate-completion convergence question recorded in Problem 3.60, item (1). It does not, however, by itself produce the renormalized BV homotopies required for the quantum (one-loop and higher) lift; that requires Costello renormalization of the brane-defect propagator, recorded separately in Problem 3.60, items (1)–(2).

COROLLARY 3.161 *(Primitive indecomposable shadow of the reduced cotangent comparison).* Combining Theorem 3.154 with the cochain-level reduced CE/PV central-operation comparison (Theorem 3.28) and the Hamiltonian factorization-algebra equivalence (Theorem 3.22), one obtains a chain-level primitive indecomposable shadow of the reduced cotangent comparison:

$$\begin{aligned}\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L &\xrightarrow{\Phi_{\mathrm{cl}}} \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}\big(A^{\mathrm{pp}}_\partial\big)\\ &\xrightarrow{\pi^{\mathrm{cot}*}_{\mathrm{prim}}} \text{primitive indecomposable shadow of the open boundary sector.}\end{aligned}$$

This corollary does not construct the unreduced cotangent factorization-center morphism. The missing data are the centrality homotopies for principal-part currents against arbitrary unreduced open observables, together with the Tate/weighted coefficient kernel and the brane-defect propagator extension recorded in Problem 3.60.

Proof. Φ_{cl} is constructed at the cochain level by Theorem 3.28 on the reduced sector after de Rham-current contraction. Theorem 3.22 promotes this to a P_0 factorization algebra equivalence in the Hamiltonian sector. Theorem 3.154 gives the primitive indecomposable P_0-shadow and one-ψ label shadow. The composition therefore exists only at the indecomposable-shadow level. Since the one-ψ projection does not intertwine the descendant PBW action with the principal-part coadjoint action, the asserted result is exactly the shadow statement and nothing stronger. □

Remark 3.162 *(Status of the primitive indecomposable shadow).* The primitive indecomposable P_0-shadow map of Theorem 3.154 supplies the local primitive indecomposable P_0-shadow component of Problem 3.60, item (3), and the primitive indecomposable shadow component of Theorem 2.70. It does not by itself close Problem 3.60, because the Tate-completion of the Casimir ($\bigoplus_{d\geq 1} H_d^\vee \to$ a topological coefficient in which the Casimir converges) and the propagator extension are separate analytic obligations not addressed here. Nor does it supply the missing centrality homotopies for principal-part currents against arbitrary unreduced open observables. Proposition 3.155 further shows that the one-ψ label map is not compatible with the Moyal coadjoint action after quantization. The chain-level construction is therefore sufficient only for the classical primitive indecomposable shadow.

Remark 3.163 *(On the role of the de Rham contraction).* The de Rham contraction Lemma 3.191 enters Construction 3.135 only through the compactly supported density class on the brane line, not through any analytic propagator. Specifically, $H^\bullet(\Omega_c^\bullet(I)) = H_c^\bullet(I;\mathbb{C})$ is one-dimensional in degree 1 for an open interval I, represented by a compactly supported density $\lambda_I(t)\,dt$ with integral 1. The degree-zero notation in the primitive shadow comes from the shifted BV/cotangent coordinate paired with this density; it is not ordinary point-evaluation in $H_c^0(I)$. The cosheaf structure is then transparent. No further analytic input is required; in particular no smoothing kernel, mollifier, or finite-scale BV gauge fix appears.

Hadamard parametrix on the Mittag-Leffler coefficient module

Remark 3.164 *(Spacetime/coefficient factorization of the Hadamard parametrix).* In Costello's heat-kernel BV calculus [12, Ch. 5], the Hadamard parametrix enters as the asymptotic expansion of the heat kernel K_t^{base} on the bulk free elliptic complex $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$, combined with Hörmander wave-front-set tracking [32, Ch. 7] to control the singular support of the propagator $P_{\epsilon,L}^{\mathrm{base}}$ along the diagonal. Costello's locality package [12, Chs. 5–7] is used here only as the heat-kernel/RG vocabulary for free elliptic complexes: local functionals admit Hadamard-parametrix expansions whose counterterms stay in the local-functional class $\mathcal{O}_{\mathrm{loc}}$ under the hypotheses of that formalism. The exact published theorem number is not used as a local input.

In the weighted setting of 7, the propagator factors as $P_{\epsilon,L}^{w} = P_{\epsilon,L}^{\mathrm{base}}\widehat{\otimes}C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}$. The bulk piece $P_{\epsilon,L}^{\mathrm{base}}$ is independent of the coefficient weight; the coefficient piece $C_{\mathfrak{h},w}$ is purely algebraic, with no scale parameter and no Laplacian. The Hadamard parametrix construction therefore lives entirely on the bulk factor; the coefficient factor enters only by tensor.

Lemma 3.165 *(Windowwise parametrix compatibility).* For each $w_0 \in \mathbb{Z}_{\geq 1}$, let

$$C_{\mathfrak{h},w}^{w_0} = \sum_{d\leq w_0}\sum_i (w(d)e_{d,i})\otimes(w(d)^{-1}e_d^i) \in \Big(\prod_{d\leq w_0} w(d)H_d\Big)\otimes\Big(\prod_{d\leq w_0} w(d)^{-1}H_d^\vee\Big)$$

denote the truncated weighted Casimir. Define the windowwise field complex

$$\mathcal{E}_w^{w_0} = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\Big(\Big(\prod_{d\leq w_0} w(d)H_d\Big)[1]\oplus\Big(\prod_{d\leq w_0} w(d)^{-1}H_d^\vee\Big)[2]\Big)$$

and the windowwise propagator $P_{\epsilon,L}^{w,w_0} = P_{\epsilon,L}^{\mathrm{base}}\widehat{\otimes}C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}^{w_0}$. Then:

(1) For each w_0, $\mathcal{E}_w^{w_0}$ is a free elliptic complex with finite-rank coefficient bundle, and Costello's Hadamard parametrix construction [12, Ch. 5] applies verbatim to $P_{\epsilon,L}^{w,w_0}$.

(2) The truncations $\{P^{w,w_0}_{\epsilon,L}\}_{w_0\geq 1}$ form a strict Mittag-Leffler inverse system in the coefficient direction, with surjective transition maps $P^{w,w_0'}_{\epsilon,L} \twoheadrightarrow P^{w,w_0}_{\epsilon,L}$ for $w_0' \geq w_0$ given by truncation to degrees $d \leq w_0$.

(3) The wave-front set $\mathrm{WF}(P^{w,w_0}_{\epsilon,L})$ is contained in the bulk wave-front set $\mathrm{WF}(P^{\mathrm{base}}_{\epsilon,L}) \subset T^*(\mathbb{R}^2 \times \mathbb{C}^2) \setminus 0$, independently of w_0, since the coefficient factor $C^{w_0}_{\mathfrak{h},w}$ is a finite-rank algebraic tensor with no spacetime singularities.

Proof. (1) For fixed w_0, the coefficient bundle $\left(\prod_{d\leq w_0} w(d)H_d\right)[1] \oplus \left(\prod_{d\leq w_0} w(d)^{-1}H_d^\vee\right)[2]$ is a finite-rank $\mathbb{Z}$-graded vector space (finite product of finite-dimensional H_d's). Costello's Hadamard parametrix construction [12, Ch. 5] applies verbatim to any free elliptic complex with finite-rank coefficient bundle, so it applies to $\mathcal{E}^{w_0}_w$. The bulk propagator $P^{\mathrm{base}}_{\epsilon,L}$ is unchanged; the coefficient factor $C^{w_0}_{\mathfrak{h},w}$ is a fixed finite-rank tensor.

(2) Surjectivity of the transition maps is the truncation $\prod_{d\leq w_0'} w(d)H_d \twoheadrightarrow \prod_{d\leq w_0} w(d)H_d$ on the coefficient direction; the bulk factor is unchanged. An inverse system with surjective transition maps satisfies the Mittag-Leffler condition trivially, so the inverse limit is exact in the strict finite-window habitat of Definition 3.30, and Proposition 3.31 gives $\varprojlim^1 = 0$ for this tower. Compatibility with $P^{w,w_0'}_{\epsilon,L} \mapsto P^{w,w_0}_{\epsilon,L}$ follows from the tensor factorization and the truncation of the Casimir.

(3) The Hörmander wave-front set of a tensor product distribution $T_1 \otimes T_2$ on $X_1 \times X_2$ satisfies $\mathrm{WF}(T_1 \otimes T_2) \subseteq \big(\mathrm{WF}(T_1) \times \{(x_2, 0) : x_2 \in X_2\}\big) \cup \big(\{(x_1, 0) : x_1 \in X_1\} \times \mathrm{WF}(T_2)\big) \cup \big(\mathrm{WF}(T_1) \times \mathrm{WF}(T_2)\big)$ ([32, Thm. 8.2.9]). Here $T_1 = P^{\mathrm{base}}_{\epsilon,L}$ has wave-front set in the conormal of the diagonal of $\mathbb{R}^2 \times \mathbb{C}^2$, and $T_2 = C^{w_0}_{\mathfrak{h},w}$ is a finite-rank algebraic tensor with no point dependence, hence smooth, so $\mathrm{WF}(T_2) = \emptyset$. Hence $\mathrm{WF}(P^{w,w_0}_{\epsilon,L}) = \mathrm{WF}(P^{\mathrm{base}}_{\epsilon,L}) \times \{0\}$, which we identify with $\mathrm{WF}(P^{\mathrm{base}}_{\epsilon,L})$ on the spacetime, independent of w_0. □

THEOREM 3.166 *(Hadamard control on the Mittag-Leffler coefficient module).* Let $w(d) = q^d$ with $q > 1$ be the exponential degree weight of Remark 3.73, and let $\mathcal{E}_w$ be the weighted field complex (9). The Hadamard parametrix construction [12, Ch. 5] extends from each finite-window restriction $\mathcal{E}^{w_0}_w$ (Lemma 3.165) to the Mittag-Leffler limit $\mathcal{E}_w = \varprojlim_{w_0} \mathcal{E}^{w_0}_w$, preserving the Hadamard wave-front and finite-window locality estimates graph by graph. Specifically:

(i) The propagator $P^w_{\epsilon,L}$ admits a Hadamard parametrix expansion as the inverse limit

$$P^w_{\epsilon,L} = \varprojlim_{w_0} P^{w,w_0}_{\epsilon,L},$$

with bulk wave-front set independent of w_0.

(ii) For every local interaction $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ of bounded polynomial-degree D_0 at each vertex in the coefficient direction, every fixed Costello graph Γ of order $\hbar^n$ contributes a compatible finite-window counterterm system whose weighted coefficient size is bounded, on every finite output window, by a finite graph-dependent constant times $|C^{\mathrm{base}}_n[\Gamma]|$, where $|C^{\mathrm{base}}_n[\Gamma]|$ is the bulk spacetime counterterm of Costello.

(iii) For every fixed graph whose finite-window counterterms are degreewise surjective and compatible under the Mittag-Leffler transition maps, the corresponding inverse-limit representative lies in the local-functional class $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$. The associated endpoint obstruction is the class in $H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ for the mixed brane-defect interaction.

(iv) The construction has the ordinary strict inverse-limit universal property and no stronger one: a compatible graphwise counterterm family $\{C_n[\Gamma]^{w_0}\}_{w_0}$ determines a unique element $C_n[\Gamma] \in \varprojlim_{w_0} \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_0})$, and every finite-window projection of this element is the given counterterm.

Thus the Hadamard and wave-front part of Problem 3.87 is exactly the finite-window locality plus Mittag-Leffler compatibility statement above. The theorem's hypotheses are graphwise; summability over graph topologies and QME vanishing are the separate obstruction data recorded in the displayed cohomology group.

Proof. (i) Each truncation $P_{\epsilon,L}^{w,w_0}$ admits Costello's Hadamard parametrix expansion by Lemma 3.165 (1). The Mittag-Leffler inverse system of (2) of that lemma is strict (surjective transition maps), so the inverse limit

$$P_{\epsilon,L}^{w} = \varprojlim_{w_0} P_{\epsilon,L}^{w,w_0}$$

exists in the weighted Tate category and inherits the wave-front bound (3) of that lemma uniformly in w_0. The inverse limit of the Hadamard expansions is itself a Hadamard expansion: the asymptotic Hadamard coefficients $a_k(x, y)$ on the bulk diagonal are independent of the coefficient direction, so the windowwise Hadamard expansion of $P_{\epsilon,L}^{w,w_0}$ is

$$\Big(\sum_{k\geq 0} a_k(x, y)\, t^k\Big)\widehat{\otimes} C_{\mathfrak{h},w}^{w_0},$$

and the inverse limit in the coefficient direction commutes with the asymptotic expansion in the bulk parameter t.

(ii) Fix a graph Γ with $V(\Gamma)$ vertices and $E(\Gamma)$ internal edges. Each vertex carries an interaction of polynomial degree $\leq D_0$ in the coefficient direction (hypothesis on I). Each internal edge contracts one source leg with one target leg via the Casimir $C_{\mathfrak{h},w}$. By (W1), each such contraction multiplies the weight by a factor $\leq C_w$ on Hamiltonian legs, and the cotangent legs are bounded on the finite windows appearing in the graph by (W2). The total coefficient size after $E(\Gamma)$ contractions is bounded by a finite graph-dependent constant times the bulk counterterm norm. The bulk counterterm norm $|C_n^{\mathrm{base}}[\Gamma]|$ is finite by Costello's bulk locality theorem [12, Chs. 5–7].

This graph-dependent constant is finite for the fixed graph Γ and the fixed finite output window under consideration. A summability statement over all graph topologies at a fixed $\hbar$-order is not one of the hypotheses. Such a statement belongs to the renormalized QME problem, together with the usual stability, counterterm, and locality hypotheses of Costello's graph expansion. The present theorem needs only the graphwise bound: for a fixed graph, the coefficient degrees and the corresponding weight ratios lie in a finite set, so the compatible finite-window counterterm system has a Mittag-Leffler limit when the windowwise representatives are themselves compatible.

(iii) The bulk Costello locality theorem at fixed window w_0 is used only as a finite-window input: the windowwise counterterm $C_n[\Gamma]^{w_0} \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_0})$ is a local functional with kernel supported on the diagonal of $\mathbb{R}^2 \times \mathbb{C}^2$. By the wave-front bound of Lemma 3.165 (3), the support is the bulk diagonal independently of w_0. The transition map $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_0'}) \to \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_0})$ for $w_0' \geq w_0$ is the restriction of polynomial functionals to truncated coefficient indices, equivalently setting higher coefficient fields to zero; in the regulator-admissible sector these maps are degreewise surjective. This preserves locality (the support of the truncated kernel is the same diagonal). The inverse limit $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w) = \varprojlim_{w_0} \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_0})$ exists in the weighted Tate category and is the candidate local-functional class on $\mathcal{E}_w$. The windowwise counterterms

$\{C_n[\Gamma]^{w_0}\}_{w_0}$ form a compatible inverse system, and their inverse limit $C_n[\Gamma] = \varprojlim_{w_0} C_n[\Gamma]^{w_0}$ lies in $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ with bound (ii). Its universal property is precisely the one proved in Proposition 3.31: it is the unique compatible strict ML element with these finite-window projections.

The endpoint obstruction class $[C_n[\Gamma]] \in H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ is represented by the inverse limit of the compatible windowwise classes when that compatibility holds. The theorem identifies its analytic support and coefficient topology. Its vanishing is an additional algebraic/BV condition on the mixed brane-defect interaction. □

Remark 3.167 *(Why the Hörmander commutation works).* The deciding technical point is whether the Mittag-Leffler limit in the coefficient direction commutes with Hörmander's wave-front-set analysis on the bulk. Three facts make this work:

(a) The propagator factorization (7) is exact (not asymptotic), so the bulk wave-front set is computed once and tensored.

(b) The coefficient factor $C_{\mathfrak{h},w}$ is not a distribution on any spacetime: it is an algebraic tensor in the weighted Tate category. Wave-front analysis on a smooth/algebraic tensor is trivial ($\mathrm{WF} = \emptyset$).

(c) The Hadamard parametrix coefficients $a_k(x, y)$ are universal in the bulk (depend only on the bulk Riemannian/Hermitian data of $\mathbb{R}^2 \times \mathbb{C}^2$); they do not see the coefficient direction.

Consequently, the wave-front bound of Lemma 3.165 (3) is exact and w_0-uniform; passing to the Mittag-Leffler limit preserves the local-functional class graph-by-graph at every order $\hbar^n$ once the windowwise counterterm system is compatible.

COROLLARY 3.168 *(Hadamard coefficient criterion for the weighted QME lift).* In the graphwise Mittag-Leffler sector of Theorem 3.166, Proposition 3.86 is equivalent to the vanishing of the mixed brane-defect QME obstruction class. The all-order quantum derived-center comparison

$$\begin{aligned}\Phi^w_\hbar \colon C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]\big)[[\hbar]] \\ \longrightarrow Z^{P_{0,\hbar}}_{\mathrm{fact}}\big(\mathrm{Obs}^\hbar_{\mathrm{open}}\big)\big|_{\mathfrak{g}_w}\end{aligned}$$

extends to the weighted cotangent direction at all orders in $\hbar$ as a continuous $\mathbb{C}[[\hbar]]$-linear cochain map precisely when the obstruction class in $H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ vanishes. Equivalently, this vanishing statement is the datum required by Corollary 3.88.

Proof. Proposition 3.86 assumes the Costello-locality/QME compatibility identity of Problem 3.87. By Theorem 3.166, the Hadamard wave-front-set and Mittag-Leffler coefficient-topology pieces of that identity are controlled. The proposition therefore asks for exactly the vanishing of the mixed brane-defect QME obstruction class. □

THEOREM 3.169 *(Weighted RG locality obstruction criterion).* Before adjoining the principal-part current extension, in the graphwise finite-window sector of Theorem 3.166, the analytic part of Problem 3.87 is exactly the following finite-window coefficient criterion:

$$\begin{aligned}P^w_{\epsilon,L} &= P^{\mathrm{base}}_{\epsilon,L} \widehat{\otimes} C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w},\\ \mathrm{WF}(C^{w_0}_{\mathfrak{h},w}) &= \emptyset,\\ \varprojlim{}^1_{w_0} P^{w,w_0}_{\epsilon,L} &= 0.\end{aligned}$$

These conditions hold by Lemma 3.165. After they are imposed, the endpoint datum in the weighted QME lift is the class

$$\mathrm{Ob}_w \in H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\}).$$

Proof. The propagator factorization is (7); its finite-window form is the definition of $P^{w,w_0}_{\epsilon,L}$ in Lemma 3.165. The same lemma proves that the coefficient Casimir is an algebraic tensor with empty spacetime wave-front set and that the finite-window propagators form a strict Mittag-Leffler system. Thus the Hadamard singular support, asymptotic coefficients, and locality estimates are exactly the bulk Costello data tensored with a compatible coefficient kernel. The only term of Problem 3.87 not decided by these analytic statements is the local-functional QME class displayed above. This does not decide the $\mathcal{D}'_c(I) \otimes \mathcal{P}[1]$ defect-current compatibility clause of Problem 3.264. □

Finite-window graph/QME assembly

Definition 3.170 *(Finite-window graph/QME package).* Fix an interval I, an admissible weight w, a finite Hamiltonian window M, and a finite set $\mathcal{G}_M$ of connected Costello graphs. A finite-window graph/QME package over $(I, w, M, \mathcal{G}_M)$ consists of the following data.

(F1) For every $\Gamma \in \mathcal{G}_M$ with r_Γ cotangent external labels, a tuple $B^\Gamma_\bullet = (B^\Gamma_1, \dots, B^\Gamma_{r_\Gamma})$ satisfying one of the two proved label conditions:

$$B^\Gamma_\bullet \in \big(\mathcal{D}^{\mathrm{reg}}_c(I)\big)^{r_\Gamma} \quad \text{or} \quad B^\Gamma_\bullet \in \mathcal{D}'_{c,\mathrm{WF}}(I; \Gamma, M).$$

The second condition is the graphwise relation of Definition B.18. If one wants an honest CE source from a linear subspace $\mathcal{V} \subset \mathcal{D}'_c(I)$, every finite tuple from $\mathcal{V}$ must satisfy the corresponding graphwise relation. Otherwise the source is only the finite CE span of the specified labelled inputs. Denote the resulting source CE object by $C^\bullet_{\mathrm{lab}}(\mathcal{G}_M)$.

(F2) The finite-scale regulator package

$$P^{w,M}_{\epsilon,L}, \qquad \Delta^{w,M}_{\epsilon,L}, \qquad I^w_{0,M},$$

together with graph weights $W^M_{\Gamma,\epsilon,L}(B^\Gamma_\bullet)$, local counterterms $C^M_{\Gamma,w}(\epsilon; B^\Gamma_\bullet)$, and renormalized weights

$$W^{R,M}_{\Gamma,L,w} = \lim_{\epsilon\to 0}\big(W^M_{\Gamma,\epsilon,L} - C^M_{\Gamma,w}(\epsilon; B^\Gamma_\bullet)\big).$$

These are the regular-density weights of Proposition B.17, or the wavefront-admissible weights of Proposition B.19.

(F3) A complete filtered local-functional subcomplex

$$\mathfrak{Q}^\bullet_{\mathcal{G},M}(I) = \Big(\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc},\mathcal{G},M}\big(\mathcal{E}^M_w|_I;\ \mathcal{A}^{\mathrm{pp},w,M}_{\partial,\hbar}(I)\big), d_M\Big), \qquad d_M = Q + \{I^w_{0,M}, -\}_{\hbar,M},$$

generated, and then completed in the $\hbar$-adic and scalar-contact filtrations, by the displayed renormalized graph weights, their local counterterms, and the d_M-boundaries and BV brackets forced by the finite graph differential on $\mathcal{G}_M$. This is a subcomplex only after those boundary and bracket terms have been included.

(F4) A filtered chain projection on this subcomplex,

$$\widehat{\sigma}_{\mathrm{sc},M} : \mathfrak{Q}^\bullet_{\mathcal{G},M}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{\mathcal{A}}; \mathbb{C}\gamma_1)[1][[\hbar]],$$

whose associated graded is the scalar-symbol projection. Equivalently, either the graph package lies in the balanced graph/counterterm subhabitat of Definition E.18, or the scalar-extension obstructions

$$\delta^{(0)}_{\mathrm{Cost},\sigma}(I), \qquad \mathfrak{o}^{(r)}_{\mathrm{Cost},\sigma}(I), \quad r > 0,$$

of Theorem E.20 have been killed on the chosen finite-window graph/counterterm subcomplex.

(F5) A degree-zero graph kernel

$$\kappa^{M}_{\mathcal{G},w,I} \in \mathrm{Hom}_{\mathrm{cont}}\Big(C^{\bullet}_{\mathrm{lab}}(\mathcal{G}_M), \mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I)\Big),$$

restricted to the chosen label package, whose zero-internal-edge part is $\kappa^{\mathrm{coef},M}_{\hbar,w,I}$ of Proposition B.16 and whose positive graph components are the renormalized weights $W^{R,M}_{\Gamma,L,w}$.

No summation over graph topologies is part of the datum unless a separate graph-summability hypothesis is supplied.

THEOREM 3.171 *(Finite-window graph/QME assembly).* Let $(I, w, M, \mathcal{G}_M)$ carry a finite-window graph/QME package in the sense of Definition 3.170. Put

$$S^{\bullet} = C^{\bullet}_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]], \qquad \mathcal{Q}^{\bullet}_{\mathcal{G},M}(I) = \ker(\widehat{\sigma}_{\mathrm{sc},M}) \subset \mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I).$$

Let

$$\mathrm{Ob}^{\mathrm{graph}}_{\mathcal{G},M} = \mathrm{ev}_{\mathcal{G}}\,\mathrm{Curv}_{\mathbf{K}}(\kappa^{M}_{\mathcal{G},w,I}) \in \mathfrak{Q}^{1}_{\mathcal{G},M}(I)$$

be the graph curvature evaluated on the chosen finite labelled CE cycles; before evaluation it is the corresponding Hom-valued curvature. The source-differential term is included in $\mathrm{Curv}_{\mathbf{K}}$. Here $\mathrm{Curv}_{\mathbf{K}} = d_{\mathbf{K}} + \frac{1}{2}[-,-]_{\mathbf{K}}$ as in Definition B.15. Suppose counterterms have been chosen through order $\hbar^{n-1}$, and write $C_{<n,M} \in \mathfrak{Q}^{0}_{\mathcal{G},M}(I)$ for their sum. The order-n residual is

$$\mathfrak{o}^{\mathrm{fw}}_{n,M} = [\hbar^n]\Big(\mathrm{Ob}^{\mathrm{graph}}_{\mathcal{G},M} + d_M C_{<n,M} + \frac{1}{2}\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}\Big).$$

Then the following statements hold.

(1) The analytic graph weights in the package are theorem-grade finite-window local functionals. For regular density labels this is Proposition B.17; for wavefront-admissible labels this is Proposition B.19. No statement for arbitrary $\mathcal{D}'_c(I)$-labels follows.

(2) The first obstruction is scalar. If

$$\mathfrak{s}_{n,M} := [\hbar^n]\,\widehat{\sigma}_{\mathrm{sc},M}\Big(\mathrm{Ob}^{\mathrm{graph}}_{\mathcal{G},M} + d_M C_{<n,M} + \frac{1}{2}\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}\Big)$$

is nonzero in S^1, the order-n graph/QME equation cannot be solved inside the non-scalar kernel. In the balanced graph/counterterm subhabitat this scalar term is zero by Proposition E.19.

(3) If the scalar term vanishes through order n and the lower-order QME equations hold, then

$$\mathfrak{o}^{\mathrm{fw}}_{n,M} \in Z^1\big(\mathcal{Q}^{\bullet}_{\mathcal{G},M}(I)\big).$$

An order-n finite-window counterterm $C_{n,M} \in \mathcal{Q}^{0}_{\mathcal{G},M}(I)$ satisfying

$$d_M C_{n,M} = -\mathfrak{o}^{\mathrm{fw}}_{n,M}$$

exists if and only if

$$[\mathfrak{o}^{\mathrm{fw}}_{n,M}] = 0 \quad \text{in} \quad H^1\big(Q^\bullet_{\mathcal{G},M}(I), d_M\big).$$

If this class is nonzero, it is the precise finite-window non-scalar obstruction to the graph/QME realization at order $\hbar^n$.

Proof. The first assertion is exactly the analytic input supplied by the regular-density and wavefront-admissible kernel propositions. The finite-window hypothesis makes every coefficient factor finite rank, so Theorem 3.166 supplies the Hadamard counterterm control for each fixed graph. The label restriction is essential: outside $\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M)$, the product of the graph Hadamard coefficient with the pushed-forward current label is not a canonical distribution.

The scalar statement is formal. A filtered chain projection $\widehat{\sigma}_{\mathrm{sc},M}$ is part of the package. If the displayed scalar projection of the residual is nonzero, then the residual does not lie in $\ker \widehat{\sigma}_{\mathrm{sc},M}$, so no counterterm inside that kernel can solve the equation. In the balanced graph/counterterm subhabitat the projection is the zero chain map, hence the scalar branch vanishes before the non-scalar class is formed.

Finally, the completed convolution Hom is a filtered dg Lie algebra. The Bianchi identity

$$d_{\mathbf{K}} \operatorname{Curv}_{\mathbf{K}}(\kappa) + [\kappa, \operatorname{Curv}_{\mathbf{K}}(\kappa)]_{\mathbf{K}} = 0$$

and the lower-order QME equations imply that the first unsolved $\hbar^n$-coefficient is closed in the convolution differential. After evaluation on labelled CE cycles this is d_M-closed. Since the scalar projection is a chain map and its order-n value is zero, this coefficient lies in the kernel complex $Q^\bullet_{\mathcal{G},M}(I)$. Adding a degree-zero counterterm changes the first unsolved coefficient by the d_M-boundary of that counterterm; quadratic counterterm terms are already included in the displayed residual. Therefore the finite-window equation is solvable exactly when the displayed degree-one cohomology class vanishes. □

Proposition 3.172 *(Finite-window-to-limit obstruction).* Suppose that finite-window graph/QME packages $\{(I, w, M, \mathcal{G}_M)\}_{M\geq 1}$ are given and satisfy the following additional hypotheses.

(L1) The graph package is compatible under truncation: $\pi_{M,N}\mathfrak{Q}^\bullet_{\mathcal{G},M}(I) \subset \mathfrak{Q}^\bullet_{\mathcal{G},N}(I)$, the kernels, counterterms, and scalar projections commute with $\pi_{M,N}$, and the same compatibility holds under admissible weight transport $R^M_{w,w'}$.

(L2) At the fixed $\hbar$-order under consideration the graph sum is finite in each window, or a separate Costello graph-summability datum has been supplied. The finite-window theorem does not create this datum.

(L3) The residuals $\mathfrak{o}^{\mathrm{fw}}_{n,M}$ of Theorem 3.171 are compatible:

$$\pi_{M,N}\mathfrak{o}^{\mathrm{fw}}_{n,M} = \mathfrak{o}^{\mathrm{fw}}_{n,N}, \qquad M \geq N.$$

Put

$$\mathfrak{O}^{\mathrm{fw}}_n = \Big(\big([\mathfrak{o}^{\mathrm{fw}}_{n,M}]\big)_M,\ \lambda^{\mathrm{fw}}_n\Big),$$

where

$$\big([\mathfrak{o}^{\mathrm{fw}}_{n,M}]\big)_M \in \varprojlim_M H^1\big(Q^\bullet_{\mathcal{G},M}(I)\big),$$

and, after choosing windowwise primitives $c_{n,M}$ with $d_M c_{n,M} = -\mathfrak{o}^{\mathrm{fw}}_{n,M}$,

$$\lambda^{\mathrm{fw}}_n \in \varprojlim{}^1_M H^0\big(Q^\bullet_{\mathcal{G},M}(I)\big)$$

is represented by the Roos cocycle

$$\left[\pi_{M,N} c_{n,M} - c_{n,N}\right] \in H^0\big(Q^\bullet_{\mathcal{G},N}(I)\big).$$

Then compatible order-n counterterms $C_{n,M}$ with $\pi_{M,N} C_{n,M} = C_{n,N}$ exist if and only if $\mathfrak{O}^{\mathrm{fw}}_n = 0$. If the tower $\{H^0(Q^\bullet_{\mathcal{G},M}(I))\}_M$ is Mittag–Leffler, then $\lambda^{\mathrm{fw}}_n = 0$, and the inverse-limit obstruction is only the compatible H^1-class above.

Without (L1)–(L3), no inverse-limit graph/QME theorem is asserted. If (L2) fails, the graph sum itself is not an element of the tower, so the obstruction class is not formed.

Proof. Windowwise, Theorem 3.171 says that the only non-scalar obstruction is the class of $\mathfrak{o}^{\mathrm{fw}}_{n,M}$ in H^1. Necessity of the two displayed inverse-limit conditions is immediate from any compatible primitive $C_{n,M}$. Conversely, vanishing of the compatible H^1-class gives primitives $c_{n,M}$ window by window. The differences $\pi_{M,N} c_{n,M} - c_{n,N}$ are closed degree-zero elements, and their cohomology classes are the standard Roos 1-cocycle computing $\varprojlim^1_M H^0(Q^\bullet_{\mathcal{G},M}(I))$. The vanishing of λ^{fw}_n is exactly the statement that the $c_{n,M}$'s can be adjusted by closed degree-zero terms and boundaries to become compatible under all truncation maps. The Mittag–Leffler sufficient condition is the standard vanishing of $\varprojlim^1$ for a Mittag–Leffler tower. The weight-transport compatibility follows from the same diagonal transport squares used in Theorem 3.96. □

Definition 3.173 *(Finite-window graph array).* Fix a finite-window graph/QME package $(I, w, M, \mathcal{G}_M)$, a codimension-two closed marked Costello list $\mathcal{G}^{\mathrm{mk}}_M$, and a labelled CE cycle ζ_M. A finite-window graph array at order n is a finite set

$$\mathsf{A}^{\mathrm{fw}}_{n,M} = \{\alpha\}$$

together with the following row data.

(A1) *Graph type and order.* A row has a type

$$\tau_\alpha \in \{\text{dK, CE, bracket, lower-}d\text{, lower-bracket}\}.$$

For dK, CE, and bracket rows it specifies, respectively, a marked graph Γ_α, a CE-source face of such a graph, or an ordered pair $(\Gamma_{\alpha,1}, \Gamma_{\alpha,2})$, all in the finite marked list. The order is

τ_α	$\lvert\alpha\rvert_\hbar$
dK, CE	$\lvert\Gamma_\alpha\rvert_\hbar$
bracket	$\lvert\Gamma_{\alpha,1}\rvert_\hbar + \lvert\Gamma_{\alpha,2}\rvert_\hbar$
lower-d, lower-bracket	n,

and only rows with $\lvert\alpha\rvert_\hbar = n$ enter $\mathsf{A}^{\mathrm{fw}}_{n,M}$.

(A2) *Edge labels.* Each internal edge of the graph carries one of the finite-window edge kernels

$$P^{w,M}_{\epsilon,L}, \qquad \Delta^{w,M}_{\epsilon,L}, \qquad E_{\mathrm{Weyl}},$$

where E_{Weyl} is the boundary Weyl edge whose scalar two-Hamiltonian part is $(f, g) \mapsto \omega(f, g)\gamma_1$. The two half-edge coefficient labels lie in

$$\mathfrak{h}_{w,\le M} \cup D^w_{\le M}, \qquad D^w_{\le M} = \prod_{d \le M} w(d)^{-1} H^\vee_d,$$

and the edge record includes the source/target orientation used by the BV bracket sign.

(A3) *Vertex labels.* Each vertex is labelled by one of the finite package labels

$$u_{a(t)dt\otimes f}, \quad c_{B,\rho}, \quad I^{w}_{\mathrm{o},M}, \quad W^{R,M}_{\Gamma,L,w}(B^{\Gamma}_{\bullet}), \quad C^{M}_{\Gamma,w}(\epsilon; B^{\Gamma}_{\bullet}), \quad \gamma_1,$$

with f, ρ in degree $\leq M$ and $B^{\Gamma}_{\bullet}$ either regular-density or graphwise wavefront-admissible as in Definition 3.170.

(A4) *Markers and incidence coefficient.* The row contains a marker tensor

$$\mathfrak{m}_{\alpha} = (\mathfrak{m}_{\alpha,L})_L, \qquad \mathfrak{m}_{\alpha,L} \in \mathcal{M}^{\mathrm{full}}_{m_L}(\mathbb{C}^{N|N})$$

or the explicitly chosen smaller marker habitat. If the row comes from an elementary boundary operation $\beta_i \in \mathsf{B}_M(\Gamma_\alpha)$, its incidence coefficient is

$$\iota_{\alpha,M} = (-1)^{i-1}\varepsilon_{\beta_i}.$$

For CE-source and convolution-bracket rows, $\iota_{\alpha,M}$ is the corresponding CE or BV Koszul sign, including the factor $1/2$ in the bracket row.

(A5) *Weight and filtrations.* The row records

$$d_{\max}(\alpha) = \max\{d : H_d \text{ or } H^{\vee}_d \text{ occurs in the row}\} \leq M,$$

its scalar-contact filtration $p_\alpha = \#\{\text{scalar-contact open loops}\}$, and, for a boundary operation β_i,

$$r_\alpha = p_\alpha - p(\partial_{\beta_i}\Gamma_\alpha, \mu_{\beta_i}(\mathfrak{m}_\alpha)).$$

(A6) *Curvature value and scalar gate.* The row evaluates to a local functional

$$R^{\mathrm{row}}_{\alpha,M} \in \mathfrak{Q}^{1}_{\mathcal{G},M}(I).$$

For the graph-curvature part these are

τ_α	$R^{\mathrm{row}}_{\alpha,M}$
dK	$\iota_{\alpha,M}\, d_M K^{M}_{\Gamma_\alpha}$
CE	$\iota_{\alpha,M}\, K^{M}_{\Gamma_\alpha}$
bracket	$\iota_{\alpha,M}\{K^{M}_{\Gamma_{\alpha,1}}, K^{M}_{\Gamma_{\alpha,2}}\}_{\hbar,M}$.

The two lower-counterterm row types are the corresponding homogeneous summands of $[\hbar^n]d_M C_{<n,M}$ and $\frac{1}{2}[\hbar^n]\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}$. The scalar-gate entry is

$$S_{\alpha,M} = \widehat{\sigma}_{\mathrm{sc},M}(R^{\mathrm{row}}_{\alpha,M}).$$

If α is a first filtration-lowering two-Hamiltonian scalar contact row, this specializes to

$$S_{\alpha,M} = \iota_{\alpha,M}\ \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m}_\alpha)\big)\ \omega(f_{\partial_{\beta_i}\Gamma_\alpha}, g_{\partial_{\beta_i}\Gamma_\alpha})\gamma_1.$$

The non-scalar reduced-curvature contribution is not an independent entrywise projection. It is the scalar-zero sum

$$R^{\mathrm{ns}}_{n,M} = \sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} R^{\mathrm{row}}_{\alpha,M} \quad \text{provided} \quad \sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} S_{\alpha,M} = 0.$$

If an entrywise non-scalar summand is wanted, the array must also supply a chain splitting of $\widehat{\sigma}_{\mathrm{sc},M}$; without such a splitting only the displayed scalar-zero sum is canonical.

(A7) *Truncation in M.* For $M \geq N$ the array contains truncation coefficients $t^{M,N}_{\alpha\alpha'} \in \mathbb{C}$ with $\alpha' \in \mathsf{A}^{\mathrm{fw}}_{n,N}$, defined by

$$\pi_{M,N} R^{\mathrm{row}}_{\alpha,M} = \sum_{\alpha'} t^{M,N}_{\alpha\alpha'} R^{\mathrm{row}}_{\alpha',N}, \qquad \pi_{M,N} S_{\alpha,M} = \sum_{\alpha'} t^{M,N}_{\alpha\alpha'} S_{\alpha',N}.$$

A row whose coefficient labels have no degree-$\leq N$ component has all $t^{M,N}_{\alpha\alpha'} = 0$. If this equation cannot be filled for a row, the lower-window theorem is missing exactly that truncation entry.

(A8) *Order-two boundary compositions.* For an order-2 closure check the array also records the ordered pairs of distinct first scalar-contact boundary operations

$$(\beta_i, \beta_j), \qquad r(\beta_i) = r(\beta_j) = 1, \qquad |\beta_i|_\hbar = |\beta_j|_\hbar = 1,$$

whose two composites occur in the codimension-two closure table of Definition E.25. If $\mathrm{pos}_i(j)$ denotes the position of the induced face $\beta_j|\beta_i$ after deleting β_i, the composite incidence coefficient is

$$\iota_{\beta_j|\beta_i,\beta_i} = (-1)^{i-1}(-1)^{\mathrm{pos}_i(j)-1} \varepsilon_{\beta_i} \varepsilon_{\beta_j|\beta_i}.$$

The corresponding two-contact scalar gate is the row

$$S^{(2)}_{\beta_j|\beta_i,M} = \iota_{\beta_j|\beta_i,\beta_i} \left(\prod_{L \in \mathcal{L}_{\mathrm{sc}}(\partial_{\beta_j|\beta_i} \partial_{\beta_i} \Gamma)} \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}'_L) \right) \omega(f_{\beta_i}, g_{\beta_i}) \omega(f_{\beta_j|\beta_i}, g_{\beta_j|\beta_i}) \gamma^{(2)}_{\mathbf{1}}.$$

Here $\mathfrak{m}'_L$ is the transported marker on the output scalar loop L, and $\gamma^{(2)}_{\mathbf{1}}$ denotes the ordered two-contact scalar symbol. This notation records the associated graded scalar-contact symbol; it is not a separate assertion that a product $\gamma^2_{\mathbf{1}}$ survives in the target algebra. If the equality of output graphs, coefficients, and marker transports in (E.25) is not supplied, the missing order-2 datum is exactly that composite face.

The array is *valid* if the row sum reproduces the order-n residual:

$$R_{n,M} = \sum_{\alpha \in \mathsf{A}^{\mathrm{fw}}_{n,M}} R^{\mathrm{row}}_{\alpha,M}, \qquad \mathfrak{s}_{n,M} = \sum_{\alpha \in \mathsf{A}^{\mathrm{fw}}_{n,M}} S_{\alpha,M},$$

and if the truncation coefficients satisfy the displayed equations for every $M \geq N$.

PROPOSITION 3.174 *(First finite-window array rows).* The universal rows that are presently computed are the reduced current interface row in order 0 and the two-Hamiltonian scalar-contact row in order 1.

On the length-one reduced current interface, Theorem B.7 gives strict CE/PV compatibility. Thus the order-0 row has

$$R^{\mathrm{row}}_{0,M} = 0, \qquad S_{0,M} = 0.$$

For the ordered Hamiltonian inputs (a, z_1), (b, z_2), the one-cross boundary row has graph type dK, one boundary Weyl edge E_{Weyl}, two external Hamiltonian vertices, one scalar-contact output vertex $\gamma_{\mathbf{1}}$, and

$$\omega(z_1, z_2) = 1, \qquad |\alpha_{\mathrm{sc}}|_\hbar = 1.$$

Its scalar gate is

$$S_{\alpha_{\mathrm{sc}},M} = \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_{\mathrm{sc}}}(\mathfrak{m})\big)\gamma_{\mathbf{1}},$$

after extracting the $\hbar^1$-coefficient and using the orientation convention in which the ordered pair (z_1, z_2) has positive sign. Therefore:

branch	$S_{\alpha_{sc},M}$	$R^{ns}_{1,M}$ from this row
$\mathfrak{gl}_N$ ordinary	$N\gamma_{\mathbf{1}}$	not formed
even block marker $\lambda_0 P_{\bar{0}} + \lambda_1 P_{\bar{1}}$	$N(\lambda_0 - \lambda_1)\gamma_{\mathbf{1}}$	not formed unless $\lambda_0 = \lambda_1$
$\mathfrak{gl}(N\|N)$ full-equivariant	0	0

Consequently, for the minimal full-equivariant closed array containing only the strict current-interface rows and this scalar-contact row, the first order residual is zero and

$$C_{1,M} = 0$$

is the compatible order-one non-scalar counterterm. For any larger graph package, the first non-scalar residual is

$$R^{ns}_{1,M} = \sum_{\alpha \in \mathsf{A}^{fw}_{1,M} \setminus \{\alpha_{sc}\}} R^{row}_{\alpha,M}$$

after the scalar gate has vanished. Its primitive cannot be decided without the remaining order-one rows of Definition 3.173.

Proof. The order-0 assertion is exactly the strict current-source/target identity of Theorem B.7 on the length-one generators: the CE differential and target P_0 bracket use the same semidirect product constants.

The order-1 scalar-contact row is the computation of Proposition E.21, written in the row notation of Definition 3.173. The monomial value is $\omega(z_1, z_2) = 1$. In the ordinary branch the open index loop contributes N. In the even block-diagonal marker branch the cyclic supertrace is $N(\lambda_0 - \lambda_1)$. In the full-equivariant branch, Lemma E.29 kills the cyclic supertrace of every transported marker tensor in $\mathbb{C}^{N|N}$; the empty marker word is killed by the same $\operatorname{sdim} \mathbb{C}^{N|N} = 0$ calculation. Thus the row contributes zero as a local functional in the full-equivariant branch, not merely zero after cohomology.

If no further order-one rows are present, the order-one residual is zero, so the zero counterterm is a primitive and is compatible under all truncation maps. If further order-one rows are present, the displayed sum is the definition of the remaining scalar-zero residual. A primitive for it is a theorem exactly when the missing row values, signs, scalar gates, and truncation coefficients are supplied and the resulting H^1-class vanishes. □

Proposition 3.175 *(Universal scalar-contact bracket rows).* Work in the local full-equivariant marked branch with $V = \mathbb{C}^{N|N}$, and let α_{sc} be the one-cross two-Hamiltonian scalar-contact row of Proposition 3.174. Write $K^M_{\alpha_{sc}}$ for the local functional carried by this row. Then

$$K^M_{\alpha_{sc}} = 0$$

in the finite-window local-functional complex.

Consequently every order-2 convolution-bracket row containing α_{sc} has the following entries. For any order-1 component β for which the bracket row is present,

row	$R^{row}_{b,M}$
$b(\alpha_{sc}, \beta)$	$\frac{1}{2}\varepsilon_{\alpha_{sc},\beta}\{K^M_{\alpha_{sc}}, K^M_{\beta}\}_{\hbar,M} = 0$
$b(\beta, \alpha_{sc})$	$\frac{1}{2}\varepsilon_{\beta,\alpha_{sc}}\{K^M_{\beta}, K^M_{\alpha_{sc}}\}_{\hbar,M} = 0$
$b(\alpha_{sc}, \alpha_{sc})$	$\frac{1}{2}\varepsilon_{\alpha_{sc},\alpha_{sc}}\{K^M_{\alpha_{sc}}, K^M_{\alpha_{sc}}\}_{\hbar,M} = 0$

Their scalar gates are

$$S_{b,M} = 0.$$

For every $M \geq N$, the truncation row may be chosen as

$$t^{M,N}_{bb'} = 0 \qquad \text{for every lower-window bracket row } b',$$

since $\pi_{M,N} R^{\mathrm{row}}_{b,M} = 0$ and $\pi_{M,N} S_{b,M} = 0$. These rows therefore contribute no order-2 obstruction and require no counterterm.

This proposition closes only the bracket family containing α_{sc}. Bracket rows $b(\beta_1, \beta_2)$ with $\beta_1, \beta_2 \neq \alpha_{\mathrm{sc}}$ remain part of the chosen graph package and must be supplied row by row.

Proof. Proposition 3.174 proves the stronger full-equivariant fact that the scalar-contact row is zero as a local functional, because every transported scalar-loop marker has zero cyclic supertrace on $\mathbb{C}^{N|N}$. The BV/convolution bracket is bilinear and continuous on the finite-window completed local-functional complex, hence any bracket with $K^M_{\alpha_{\mathrm{sc}}} = 0$ is zero before passing to cohomology. Applying the chain projection $\widehat{\sigma}_{\mathrm{sc},M}$ to zero gives the displayed scalar gates. Truncation sends zero to zero, so the zero truncation matrix row is compatible with the defining equations of Definition 3.173. □

PROPOSITION 3.176 *(Genuine order-two graph-weight rows).* Work in the local $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ full-equivariant marked branch with $V = \mathbb{C}^{N|N}$. Let $\mathcal{G}^{\mathrm{gen}}_{2,M}$ be the part of a finite marked Costello list consisting of genuine seed graph weights of $\hbar$-order 2: these are not the codimension-two composites of two first scalar-contact boundary faces and not the bracket rows containing α_{sc} closed in Proposition 3.175. For $\Gamma \in \mathcal{G}^{\mathrm{gen}}_{2,M}$, write

$$K^M_\Gamma = W^{R,M}_{\Gamma,L,w}(B^\Gamma_\bullet)$$

for the normalized renormalized finite-window graph weight, and let ε_Γ be the coefficient with which this component occurs in $d_M \kappa^M_{\mathcal{G},w,I}$, including the orientation, Koszul, and symmetry factors supplied by the finite graph package.

The genuine order-two rows have the following schema.

(G1) The graph-differential row $d\Gamma$ has

$$\tau_{d\Gamma} = \mathrm{dK}, \qquad |d\Gamma|_\hbar = 2,$$

inherits its edge labels, vertex labels, marker tensor, and coefficient degree from Γ, and has

$$R^{\mathrm{row}}_{d\Gamma,M} = \varepsilon_\Gamma\, d_M K^M_\Gamma, \qquad S_{d\Gamma,M} = \varepsilon_\Gamma\, \widehat{\sigma}_{\mathrm{sc},M}(d_M K^M_\Gamma).$$

(G2) Before evaluation on a strict CE cycle, or when the Hom-valued kernel table is retained, each source face is a separate row. Write the relevant part of the source differential as

$$d_{\mathrm{CE}} \zeta_M = \sum_\nu \varepsilon^{\mathrm{CE}}_{\Gamma,\nu} \zeta_{M,\nu},$$

where $\varepsilon^{\mathrm{CE}}_{\Gamma,\nu}$ includes the CE Koszul sign and the graph coefficient attached to the Γ-component. The source row is

$$R^{\mathrm{row}}_{\mathrm{CE}(\Gamma,\nu),M} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu}),$$

and its scalar gate is

$$S_{\mathrm{CE}(\Gamma,\nu),M} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} \widehat{\sigma}_{\mathrm{sc},M}\big(K^M_\Gamma(\zeta_{M,\nu})\big).$$

If the curvature has already been evaluated on a strict labelled CE cycle, this source-face family is empty.

If these row values and all their d_M-faces lie in the full-equivariant marked habitat of Theorem E.30, then

$$S_{d\Gamma,M} = 0, \qquad S_{\mathrm{CE}(\Gamma,\nu),M} = 0.$$

This is only a scalar-gate statement. It does not imply $K_\Gamma^M = 0$ or $d_M K_\Gamma^M = 0$.

After scalar-gate vanishing, the genuine seed-graph contribution to the order-2 reduced curvature is

$$\begin{aligned} R_{2,M}^{\mathrm{ns,gen}} = & \sum_{\Gamma \in \mathcal{G}_{2,M}^{\mathrm{gen}}} \varepsilon_\Gamma \, d_M K_\Gamma^M \\ & - \sum_{\Gamma,\nu} \varepsilon_{\Gamma,\nu}^{\mathrm{CE}} K_\Gamma^M(\zeta_{M,\nu}) \in \mathcal{K}_{\mathrm{ns},M}^1(I). \end{aligned}$$

The second sum is absent after strict CE-cycle evaluation. A genuine order-2 counterterm exists precisely when

$$[R_{2,M}^{\mathrm{ns,gen}}] = 0 \quad \text{in} \quad H^1(\mathcal{K}_{\mathrm{ns},M}^\bullet(I), d_{\mathrm{ns},M}).$$

If this class is nonzero, it is the explicit finite-window obstruction carried by the genuine seed graphs.

Truncation is part of the row data. For $M \geq N$ one must supply normalized coefficients

$$\pi_{M,N}(\varepsilon_\Gamma K_\Gamma^M) = \sum_{\Gamma'} u_{\Gamma\Gamma'}^{M,N} \, \varepsilon_{\Gamma'} K_{\Gamma'}^N,$$

and then the graph-differential truncation entries are

$$t_{d\Gamma,d\Gamma'}^{M,N} = u_{\Gamma\Gamma'}^{M,N}.$$

For source rows one must likewise supply

$$\pi_{M,N}\big(-\varepsilon_{\Gamma,\nu}^{\mathrm{CE}} K_\Gamma^M(\zeta_{M,\nu})\big) = \sum_{\Gamma',\nu'} v_{\Gamma,\nu;\Gamma',\nu'}^{M,N} \, \big(-\varepsilon_{\Gamma',\nu'}^{\mathrm{CE}} K_{\Gamma'}^N(\zeta_{N,\nu'})\big),$$

and set $t_{\mathrm{CE}(\Gamma,\nu),\mathrm{CE}(\Gamma',\nu')}^{M,N} = v_{\Gamma,\nu;\Gamma',\nu'}^{M,N}$. Rows whose coefficient labels have no degree-$\leq N$ component have zero truncation coefficients. If either displayed truncation identity is not supplied, the lower-window theorem is missing exactly that row entry.

In the minimal full-equivariant local package of Proposition 3.177, $\mathcal{G}_{2,M}^{\mathrm{gen}} = \emptyset$ and the source-face set is empty. Hence

$$R_{2,M}^{\mathrm{ns,gen}} = 0, \qquad C_{2,M}^{\mathrm{gen}} = 0.$$

In any larger package, the present manuscript does not yet decide vanishing, counterterms, or obstruction. The exact missing data are: the finite seed list $\mathcal{G}_{2,M}^{\mathrm{gen}}$; the renormalized local weights K_Γ^M with regular-density or wavefront-admissible current labels; the full $d_M K_\Gamma^M$ expansion with orientation signs; the CE-source decomposition and signs unless strict CE-cycle evaluation has removed them; proof that every row lies in the full-equivariant marked habitat, or else its scalar projection; and the truncation matrices displayed above.

Proof. This is the order-2 part of the convolution-curvature formula in Theorem 3.185, with the scalar-contact bracket family removed by Proposition 3.175. The differential term contributes $\varepsilon_\Gamma d_M K_\Gamma^M$. The source term in $d_{\mathbf{K}} = d_M - (-1)^{|\cdot|}(\cdot) d_{\mathrm{CE}}$ contributes the displayed minus sign for a degree-zero kernel. After evaluation on a strict CE cycle, that source term is zero.

The scalar-gate assertion follows from Theorem E.30: on the full-equivariant marked habitat the filtered scalar projection is the zero chain map. Since this projection only kills the scalar shadow, it gives no vanishing theorem for the non-scalar local functional K_Γ^M.

The counterterm criterion is exactly Definition E.31: once the scalar gate is zero, the row sum is a degree-one element of the non-scalar kernel complex, and a degree-zero counterterm exists exactly when this element is a $d_{\mathrm{ns},M}$-boundary. The truncation formulas are the row identities required in Definition 3.173, written for normalized graph and source rows. In the minimal package no such genuine seed row is present, so the sum is empty. □

PROPOSITION 3.177 *(Order-two finite-window boundary rows).* Work in the local $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ full-equivariant marked branch with $V = \mathbb{C}^{N|N}$. Let the minimal array be generated by the strict current-interface rows and the one-cross scalar-contact row of Proposition 3.174. Assume no genuine $|\Gamma|_\hbar = 2$ seed graph, no order-2 CE-source face, and no order-2 graph counterterm face have been added to the finite marked Costello list, and use $C_{1,M} = 0$.

Then the only order-2 boundary data forced by codimension-two closure are the two ordered composites of two first scalar-contact boundary operations. For $o = \beta_1 \wedge \beta_2$ they have incidence signs

$$\iota_{\beta_2|\beta_1,\beta_1} = \varepsilon_{\beta_1}\varepsilon_{\beta_2|\beta_1}, \qquad \iota_{\beta_1|\beta_2,\beta_2} = -\varepsilon_{\beta_2}\varepsilon_{\beta_1|\beta_2}.$$

Their scalar gates are

$$\begin{aligned} S^{(2)}_{\beta_2|\beta_1,M} &= \varepsilon_{\beta_1}\varepsilon_{\beta_2|\beta_1} \prod_L \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}'_L)\, \omega(f_{\beta_1}, g_{\beta_1})\omega(f_{\beta_2|\beta_1}, g_{\beta_2|\beta_1})\gamma_{\mathbf{1}}^{(2)}, \\ S^{(2)}_{\beta_1|\beta_2,M} &= -\varepsilon_{\beta_2}\varepsilon_{\beta_1|\beta_2} \prod_L \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}''_L)\, \omega(f_{\beta_2}, g_{\beta_2})\omega(f_{\beta_1|\beta_2}, g_{\beta_1|\beta_2})\gamma_{\mathbf{1}}^{(2)}. \end{aligned}$$

The codimension-two closure equations identify the two output graphs, the two products of coefficients, and the transported marker tensors. Therefore the pair sum is zero. In the full-equivariant branch it is stronger: every factor $\mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}'_L)$ and $\mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}''_L)$ is zero by Lemma E.29; hence each two-contact scalar row already vanishes as a local scalar row.

Under these minimal hypotheses the order-2 reduced non-scalar row sum is

$$R^{\mathrm{ns}}_{2,M} = 0, \qquad C_{2,M} = 0$$

is a compatible order-2 counterterm.

For any larger finite-window package, the order-2 residual is not decided by the present local data. After scalar-gate vanishing its exact row sum is

$$\begin{aligned} R^{\mathrm{ns}}_{2,M} = &\sum_{|\Gamma|_\hbar=2} \varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{|\Gamma'|_\hbar=2} \varepsilon^{\mathrm{CE}}_{\Gamma'} K^M_{\Gamma'} \\ &+ \frac{1}{2} \sum_{|\Gamma_1|_\hbar=|\Gamma_2|_\hbar=1} \varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M} \\ &+ [\hbar^2]\Big(d_M C_{<2,M} + \frac{1}{2}\{C_{<2,M}, C_{<2,M}\}_{\hbar,M}\Big). \end{aligned}$$

Thus a nonzero order-2 obstruction or its counterterm can be asserted only after supplying the genuine two-cross/contact graph weights K^M_Γ, the CE-source faces, the BV bracket rows between order-1 graph components not containing α_{sc} or another retained zero local-functional row such as α_{II}, the lower-counterterm rows if a nonminimal $C_{1,M}$ was chosen, the scalar projection of any non-full-equivariant two-contact symbol, and the truncation coefficients $t^{M,N}_{\alpha\alpha'}$.

Proof. The signs are those of Lemma E.27. Starting with $o = \beta_1 \wedge \beta_2$, the face β_1 followed by $\beta_2|\beta_1$ contributes +1, while the face β_2 followed by $\beta_1|\beta_2$ contributes −1. The codimension-two closure equations Equation (E.25) identifies the output graph, coefficient, and marker transport in the two composites, so the two boundary-composition rows cancel pairwise.

The scalar-gate formula is the order-2 instance of the last row datum in Definition 3.173: one scalar-contact factor is contributed by each first-lowering boundary face. In the full-equivariant marked habitat, marker transport preserves the signed Brauer span. Lemma E.29 gives zero cyclic supertrace for every transported scalar-loop marker, including the empty marker word. Hence the two-contact scalar rows vanish before passing to cohomology.

Under the stated minimal hypotheses there are no genuine order-2 graph rows, no CE-source rows, no order-1 non-scalar primitive to feed a lower-counterterm row, and no nonzero scalar-contact row left after the full-equivariant trace calculation. The possible bracket row between two one-cross scalar-contact components is also zero, because each such component has zero local-functional value in the full-equivariant branch by Proposition 3.174. The reduced non-scalar order-2 row sum is therefore zero, and the zero counterterm is compatible under finite-window truncation.

Removing any minimal hypothesis reintroduces exactly the displayed row types from the curvature formula in Theorem 3.185. The bracket rows containing α_{sc} are the universal zero rows of Proposition 3.175; bracket rows containing a retained α_{II}-row vanish by the same bilinearity argument and Proposition E.40. Without the remaining graph weights, signs, scalar projections, and truncation coefficients, the cochain $R^{\mathrm{ns}}_{2,M}$ is not defined, so no order-2 obstruction class or counterterm can be claimed. □

Proposition 3.178 *(Truncation matrices for the computed finite-window rows).* Work with the finite-window graph arrays of Definition 3.173. For $M \geq N$, use the following label-retaining convention: if a labelled row has $d_{\max} \leq N$, the same label denotes its lower-window row; if $d_{\max} > N$, the row has no lower-window representative and all entries from it are zero. With this convention the computed rows have the following truncation matrices.

(T1) The one-cross scalar-contact row α_{sc} of Proposition 3.174 has $d_{\max}(\alpha_{\mathrm{sc}}) = 1$. For $M \geq N \geq 1$,

$$t^{M,N}_{\alpha_{\mathrm{sc}}\alpha_{\mathrm{sc}}} = 1, \qquad t^{M,N}_{\alpha_{\mathrm{sc}}\alpha'} = 0 \quad (\alpha' \neq \alpha_{\mathrm{sc}}).$$

This applies to the ordinary, even-block, and full-equivariant branches. In the full-equivariant branch the row value is already zero, so the zero matrix would also satisfy the row equations; the displayed matrix is the chosen transport of the retained row label.

(T2) The labelled scalar-zero row α_{II} of Proposition E.40, with ordered inputs (z_1, z_1), has

$$R^{\mathrm{row}}_{\alpha_{\mathrm{II}},M} = 0, \qquad S_{\alpha_{\mathrm{II}},M} = 0, \qquad d_{\max}(\alpha_{\mathrm{II}}) = 1.$$

If the row convention retains this labelled zero row, then for $M \geq N \geq 1$

$$t^{M,N}_{\alpha_{\mathrm{II}}\alpha_{\mathrm{II}}} = 1, \qquad t^{M,N}_{\alpha_{\mathrm{II}}\alpha'} = 0 \quad (\alpha' \neq \alpha_{\mathrm{II}}).$$

If zero rows are suppressed, the α_{II}-row and its truncation matrix are empty.

(T3) Let $\gamma_{21} = (\beta_2|\beta_1, \beta_1)$ and $\gamma_{12} = (\beta_1|\beta_2, \beta_2)$ denote the two ordered codimension-two scalar-contact composites in Proposition 3.177. When the degree-$\leq N$ window contains the two Hamiltonian labels, the composite rows are transported diagonally:

$$t^{M,N}_{\gamma_{21}\gamma_{21}} = 1, \qquad t^{M,N}_{\gamma_{12}\gamma_{12}} = 1, \qquad t^{M,N}_{\gamma_{ij}\gamma'} = 0 \quad (\gamma' \neq \gamma_{ij}).$$

The codimension-two pair cancellation is therefore stable under truncation. In the full-equivariant branch the stronger rowwise vanishing is also stable.

(T4) Let b be any order-2 bracket row containing a retained zero local-functional row

$$\alpha_0 \in \{\alpha_{\mathrm{SC}}, \alpha_{\mathrm{II}}\}.$$

Thus $b = b(\alpha_0, \beta)$, $b = b(\beta, \alpha_0)$, or $b = b(\alpha_0, \alpha_0)$, whenever the displayed row belongs to the chosen finite marked list. Then

$$R^{\mathrm{row}}_{b,M} = 0, \qquad S_{b,M} = 0,$$

and one may take the zero truncation row

$$t^{M,N}_{bb'} = 0 \qquad \text{for every lower-window bracket row } b'.$$

(T5) If genuine order-2 seed rows are absent, as in the minimal full-equivariant package, their truncation matrices are empty. If a larger package supplies genuine seeds $\Gamma \in \mathcal{G}^{\mathrm{gen}}_{2,M}$, then the graph-differential rows are transported by the normalized matrix $u^{M,N}$:

$$\pi_{M,N}(\varepsilon_\Gamma K^M_\Gamma) = \sum_{\Gamma'} u^{M,N}_{\Gamma\Gamma'}\, \varepsilon_{\Gamma'} K^N_{\Gamma'}, \qquad t^{M,N}_{d\Gamma,d\Gamma'} = u^{M,N}_{\Gamma\Gamma'}.$$

Source-face rows, when present, are transported by the normalized matrix $v^{M,N}$:

$$\pi_{M,N}\big(-\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu})\big) = \sum_{\Gamma',\nu'} v^{M,N}_{\Gamma,\nu;\Gamma',\nu'}\, \big(-\varepsilon^{\mathrm{CE}}_{\Gamma',\nu'} K^N_{\Gamma'}(\zeta_{N,\nu'})\big),$$

$$t^{M,N}_{\mathrm{CE}(\Gamma,\nu),\mathrm{CE}(\Gamma',\nu')} = v^{M,N}_{\Gamma,\nu;\Gamma',\nu'}.$$

Under the projection-closed row-basis hypotheses of Proposition 3.179, these are the coordinate entries of the finite-window projection. Without those hypotheses the lower-window theorem is missing exactly the entries $u^{M,N}_{\Gamma\Gamma'}$ and $v^{M,N}_{\Gamma,\nu;\Gamma',\nu'}$ for those supplied genuine rows.

(T6) Bracket rows $b(\beta_1, \beta_2)$ with $\beta_i \notin \{\alpha_{\mathrm{SC}}, \alpha_{\mathrm{II}}\}$ are not computed by the present local data. After a lower-window bracket row basis, or an equivalent row presentation with chosen coordinates, is fixed, their required truncation matrix is the coordinate family $q^{M,N}$ satisfying

$$\pi_{M,N}\Big(\frac{1}{2}\varepsilon_{\beta_1,\beta_2}\{K^M_{\beta_1}, K^M_{\beta_2}\}_{\hbar,M}\Big) = \sum_{\beta'_1,\beta'_2} q^{M,N}_{\beta_1,\beta_2;\beta'_1,\beta'_2} \frac{1}{2}\varepsilon_{\beta'_1,\beta'_2}\{K^N_{\beta'_1}, K^N_{\beta'_2}\}_{\hbar,N}.$$

These q-entries, together with the underlying order-1 row values, signs, and lower-window bracket-basis reduction, are still missing larger-package data.

The nonempty displayed matrices are stable under further truncation: for $M \geq N \geq P$,

$$t^{M,P} = t^{M,N} t^{N,P}$$

on the scalar-contact, scalar-zero, codimension-two, and zero-bracket families. For genuine seed rows and nonzero added bracket rows, stability is exactly the cocycle condition

$$u^{M,P}_{\Gamma\Gamma''} = \sum_{\Gamma'} u^{M,N}_{\Gamma\Gamma'} u^{N,P}_{\Gamma'\Gamma''},$$

$$v^{M,P}_{\Gamma,\nu;\Gamma'',\nu''} = \sum_{\Gamma',\nu'} v^{M,N}_{\Gamma,\nu;\Gamma',\nu'} v^{N,P}_{\Gamma',\nu';\Gamma'',\nu''},$$

and the analogous equation for q.

Proof. For the one-cross scalar-contact rows the coefficient labels are z_1, z_2, and for α_{11} the labels are z_1, z_1. These all have window degree 1. Therefore $\pi_{M,N}$ preserves the labelled row for $N \geq 1$, and kills it only if the lower window does not contain the degree-one labels. The diagonal matrices in (T1) and (T2) are exactly this statement. The row α_{11} is zero because $\omega(z_1, z_1) = 0$.

The codimension-two rows use the same two degree-one Hamiltonian labels and the same transported marker words before and after projection. The finite-window marked-habitat compatibility preserves the output graph, coefficient product, marker transport, and incidence signs. Hence the two ordered composites truncate to the same labelled composites in the lower window, and their pair cancellation survives.

If a bracket row contains α_{sc} or α_{11}, one bracket input is zero as a local functional in the retained full-equivariant row family. Bilinearity of the BV/convolution bracket gives $R^{\mathrm{row}}_{b,M} = 0$, and applying the scalar projection gives $S_{b,M} = 0$. The zero truncation row is then compatible with the defining equations of Definition 3.173.

For genuine seeds and for nonzero added bracket rows, the displayed formulas are not consequences of the cutoff alone; they are the required decompositions of the projected normalized row values in the chosen lower-window row basis. Applying $\pi_{N,P}\pi_{M,N} = \pi_{M,P}$ gives the displayed cocycle equations. Conversely, those equations are exactly the statement that the supplied matrices define a compatible finite-window row system. □

Proposition 3.179 *(Projection formulas for the u, v, and q rows).* Work in a larger finite-window graph package in the local $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ full-equivariant marked branch. Fix $M \geq N$. Suppose the package supplies ordered lower-window row bases, or equivalently row presentations with chosen coordinate maps, for the following three finite-dimensional spans:

$$\begin{aligned}
\mathcal{U}_M &= \mathrm{span}\{\mathbf{K}^M_\Gamma\}_{\Gamma \in \mathcal{G}^{\mathrm{gen}}_{2,M}}, \qquad \mathbf{K}^M_\Gamma = \varepsilon_\Gamma K^M_\Gamma, \\
\mathcal{V}_M &= \mathrm{span}\{\mathbf{E}^M_{\Gamma,\nu}\}_{\Gamma,\nu}, \qquad \mathbf{E}^M_{\Gamma,\nu} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu}), \\
\mathcal{B}_M &= \mathrm{span}\{\mathbf{B}^M_{\beta_1,\beta_2}\}_{\beta_i \notin \{\alpha_{sc},\alpha_{11}\}}, \qquad \mathbf{B}^M_{\beta_1,\beta_2} = \frac{1}{2}\varepsilon^M_{\beta_1,\beta_2}\{K^M_{\beta_1}, K^M_{\beta_2}\}_{\hbar,M}.
\end{aligned}$$

Assume these spans are projection-closed:

$$\pi_{M,N}\mathcal{U}_M \subset \mathcal{U}_N, \qquad \pi_{M,N}\mathcal{V}_M \subset \mathcal{V}_N, \qquad \pi_{M,N}\mathcal{B}_M \subset \mathcal{B}_N,$$

and that $\pi_{M,N}$ is the chain and BV-bracket projection on the chosen finite-window subcomplex:

$$\pi_{M,N} d_M = d_N \pi_{M,N}, \qquad \pi_{M,N}\{x, y\}_{\hbar,M} = \{\pi_{M,N}x, \pi_{M,N}y\}_{\hbar,N}$$

for the rows under consideration.

Then the missing entries isolated above are defined by coordinates of finite-window projection:

$$\pi_{M,N}\mathbf{K}^M_\Gamma = \sum_{\Gamma'} u^{M,N}_{\Gamma,\Gamma'}\mathbf{K}^N_{\Gamma'},$$

$$\pi_{M,N}\mathbf{E}^M_{\Gamma,\nu} = \sum_{\Gamma',\nu'} v^{M,N}_{\Gamma,\nu;\Gamma',\nu'}\mathbf{E}^N_{\Gamma',\nu'},$$

and

$$\pi_{M,N}\mathbf{B}^M_{\beta_1,\beta_2} = \sum_{\beta'_1,\beta'_2} q^{M,N}_{\beta_1,\beta_2;\beta'_1,\beta'_2}\mathbf{B}^N_{\beta'_1,\beta'_2}.$$

These equations are the definitions of $u^{M,N}$, $v^{M,N}$, and $q^{M,N}$ relative to the chosen lower-window coordinates. If the displayed lower-window families are honest bases, the entries are unique. If the package supplies only a spanning family with relations, the coordinates are an additional presentation datum; without it there is no canonical matrix entry.

The u-entries also transport the differential rows:

$$\pi_{M,N}(\varepsilon_\Gamma d_M K^M_\Gamma) = \sum_{\Gamma'} u^{M,N}_{\Gamma,\Gamma'}\ \varepsilon_{\Gamma'} d_N K^N_{\Gamma'}.$$

This is just the first coordinate equation after applying $d_N \pi_{M,N} = \pi_{M,N} d_M$.

For the bracket rows there is a useful factorized form. Suppose the added nonzero order-1 rows have transport

$$\pi_{M,N} K^M_\beta = \sum_\eta r^{M,N}_{\beta,\eta} K^N_\eta, \qquad \eta \notin \{\alpha_{\mathrm{SC}}, \alpha_{\mathrm{II}}\},$$

and write the lower-window ordered raw bracket reduction as

$$\frac{1}{2}\{K^N_{\eta_1}, K^N_{\eta_2}\}_{\hbar,N} = \sum_{\beta'_1,\beta'_2} \Theta^N_{\eta_1,\eta_2;\beta'_1,\beta'_2} \mathbf{B}^N_{\beta'_1,\beta'_2}.$$

Then

$$q^{M,N}_{\beta_1,\beta_2;\beta'_1,\beta'_2} = \varepsilon^M_{\beta_1,\beta_2} \sum_{\eta_1,\eta_2} r^{M,N}_{\beta_1,\eta_1} r^{M,N}_{\beta_2,\eta_2} \Theta^N_{\eta_1,\eta_2;\beta'_1,\beta'_2}.$$

If the lower package retains all ordered normalized bracket rows and no further row relation, then

$$\Theta^N_{\eta_1,\eta_2;\beta'_1,\beta'_2} = \frac{\delta_{\eta_1,\beta'_1}\delta_{\eta_2,\beta'_2}}{\varepsilon^N_{\beta'_1,\beta'_2}},$$

and hence

$$q^{M,N}_{\beta_1,\beta_2;\beta'_1,\beta'_2} = \frac{\varepsilon^M_{\beta_1,\beta_2}}{\varepsilon^N_{\beta'_1,\beta'_2}} r^{M,N}_{\beta_1,\beta'_1} r^{M,N}_{\beta_2,\beta'_2}.$$

For $M \geq N \geq P$, these coordinate matrices satisfy

$$u^{M,P}_{\Gamma,\Gamma''} = \sum_{\Gamma'} u^{M,N}_{\Gamma,\Gamma'} u^{N,P}_{\Gamma',\Gamma''},$$

$$v^{M,P}_{\Gamma,\nu;\Gamma'',\nu''} = \sum_{\Gamma',\nu'} v^{M,N}_{\Gamma,\nu;\Gamma',\nu'} v^{N,P}_{\Gamma',\nu';\Gamma'',\nu''},$$

and

$$q^{M,P}_{\beta_1,\beta_2;\theta_1,\theta_2} = \sum_{\beta'_1,\beta'_2} q^{M,N}_{\beta_1,\beta_2;\beta'_1,\beta'_2} q^{N,P}_{\beta'_1,\beta'_2;\theta_1,\theta_2}.$$

Consequently the larger-package order-2 non-scalar residual is a theorem-grade element of the compatible finite-window tower only after the following data are supplied: the genuine seed weights, source-face values, nonzero order-1 rows, all orientation/Koszul/symmetry signs, CE-source transport, lower-window row bases or presentations, scalar-gate vanishing or scalar projections, and the projection-closure identities above. With this package the obstruction is exactly the class of the resulting row sum in $H^1(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M})$, together with the Roos compatibility class in $\varprojlim^1 H^0(\mathcal{K}^\bullet_{\mathrm{ns},M}(I))$. Without this package the non-scalar QME obstruction is not yet a defined class.

Proof. The first three displayed equations are simply the coordinate expansion of the projected normalized row values in the chosen lower-window row bases. Projection-closure is exactly the hypothesis that these coordinates exist. If the lower rows are linearly independent, the coordinate expansion is unique. If there are row relations, uniqueness is lost unless the package supplies a presentation or a coordinate splitting.

The $d\Gamma$-row formula follows by applying the chain-map identity to $\pi_{M,N}\mathbf{K}_{\Gamma}^{M}$. The q-formula follows by applying the BV-bracket identity:

$$\begin{aligned}\pi_{M,N}\mathbf{B}^{M}_{\beta_1,\beta_2} &= \frac{1}{2}\varepsilon^{M}_{\beta_1,\beta_2}\{\pi_{M,N}K^{M}_{\beta_1}, \pi_{M,N}K^{M}_{\beta_2}\}_{\hbar,N}\\ &= \frac{1}{2}\varepsilon^{M}_{\beta_1,\beta_2}\sum_{\eta_1,\eta_2} r^{M,N}_{\beta_1,\eta_1} r^{M,N}_{\beta_2,\eta_2}\{K^{N}_{\eta_1}, K^{N}_{\eta_2}\}_{\hbar,N},\end{aligned}$$

and then reducing each lower raw bracket by the matrix Θ^{N}. The ordered normalized-row specialization is the case where the lower raw bracket is already one basis row after division by its incidence coefficient.

The cocycle identities are forced by $\pi_{N,P}\pi_{M,N} = \pi_{M,P}$. Expanding first directly from M to P, and then through N, gives two coordinate expansions of the same projected row in the same lower-window coordinates. Equality of coordinates gives the three displayed matrix product formulas. These compatibility identities are precisely the truncation part of Definition 3.173. Once they are supplied, the obstruction criterion is the non-scalar kernel criterion of Definition E.31 and Proposition 3.172. □

Proposition 3.180 *(Finite-row primitive search interface).* Let $\mathsf{A}^{\mathrm{fw}}_{n,M}$ be a valid finite-window graph array at order n, and suppose its scalar gate has vanished:

$$\sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} S_{\alpha,M} = 0.$$

Choose an ordered row presentation

$$\rho^{M}_{1},\dots,\rho^{M}_{a}$$

for the degree-one non-scalar rows retained at this order: graph differentials, CE-source faces, nonzero bracket rows, and the lower counterterm rows. Write the residual as

$$R_{n,M} = \sum_{i} r^{M}_{i}\,\rho^{M}_{i}.$$

Choose also an ordered finite list of degree-zero counterterm candidates

$$\eta^{M}_{1},\dots,\eta^{M}_{b} \in \mathcal{K}^{0}_{\mathrm{ns},M}(I)$$

supplied by the same finite package, and write

$$d_{\mathrm{ns},M}\eta^{M}_{j} = \sum_{i} A^{M}_{ij}\,\rho^{M}_{i}.$$

Then a counterterm in the supplied candidate span, $C_{n,M} = \sum_j c_j\eta^{M}_{j}$, exists if and only if the finite linear system

$$A^{M}c = -r^{M}$$

is solvable. If it is not solvable, any covector $\ell \in (\mathbb{C}^{a})^{\vee}$ satisfying

$$\ell A^{M} = 0, \qquad \ell(r^{M}) \neq 0$$

is a certificate that the class of $R_{n,M}$ is nonzero in the cohomology of the displayed finite row complex.

For a tower $M \geq N$, let $T^{M,N}$ be the row-truncation matrix on the ρ-presentation and $P^{M,N}$ the chosen transport matrix on the primitive candidates. The row block $T^{M,N}$ is the direct sum of the already computed identity or zero blocks, the u-block for genuine seed rows, the v-block for CE-source faces, and the q-block for nonzero bracket rows. A compatible primitive search is defined only when

$$T^{M,N} r^M = r^N, \qquad T^{M,N} A^M = A^N P^{M,N}.$$

After windowwise solutions c_M have been chosen, the remaining compatibility obstruction is the Roos class represented by

$$\left[P^{M,N} c_M - c_N\right] \in H^0(\mathcal{K}^\bullet_{\mathrm{ns},N}(I)).$$

Thus the primitive search has three gates, in this order: scalar gate, fixed-window linear solvability, and Roos $\varprojlim^1$ compatibility. The last gate is not reached when a fixed-window H^1-obstruction is already certified by such a covector ℓ.

Proof. Once the scalar gate vanishes, the residual lies in the non-scalar kernel complex. Restricting the possible counterterm to the supplied degree-zero span turns the equation $d_{\mathrm{ns},M} C_{n,M} = -R_{n,M}$ into the displayed matrix equation. Failure of solvability is equivalent to $-r^M \notin \operatorname{im} A^M$. In finite dimension this is certified by a linear functional annihilating $\operatorname{im} A^M$ and not annihilating r^M, exactly as written.

The truncation identities say that the row residual and the boundaries of candidate primitives commute with finite-window projection. The computed blocks are those of Propositions 3.178 and 3.179. If c_M and c_N solve the two fixed-window equations, then $P^{M,N} c_M - c_N$ is closed in degree zero by $T^{M,N} A^M = A^N P^{M,N}$ and $T^{M,N} r^M = r^N$. Its cohomology class is the Roos 1-cocycle of Proposition 3.172. □

PROPOSITION 3.181 *(Concrete order-three larger-package row).* Work in the full-equivariant marked branch with $V = \mathbb{C}^{N|N}$, and let $M \geq 3$. Enlarge the minimal finite-window package by one Hom-valued CE-source face row

$$\theta_3 = \mathrm{CE}(\Theta_3, \nu_3), \qquad |\Theta_3|_\hbar = 3,$$

where Θ_3 is a genuine marked Costello seed, not a scalar-contact closure composite and not a bracket row. The row data are as follows.

The graph type and order are

$$\tau_{\theta_3} = \mathrm{CE}, \qquad |\theta_3|_\hbar = 3.$$

Its finite labels are

$$\begin{aligned} e_{\mathrm{pp}} &: \quad P^{w,M}_{\epsilon,L}, \quad h_{2,1} = z_1^2 z_2,\ \rho_{2,1}, \\ e_{\mathrm{ct}} &: \quad \Delta^{w,M}_{\epsilon,L}, \quad h_{1,1} = z_1 z_2,\ \rho_{1,1}, \\ e_{\mathrm{W}} &: \quad E_{\mathrm{Weyl}}, \quad z_1^2,\ z_2 \end{aligned}$$

with $B_\theta \in \mathcal{D}_c^{\mathrm{reg}}(I)$ on the cotangent external leg. The vertex labels are

$$u_{a(t)dt\otimes z_1^2}, \qquad u_{b(t)dt\otimes z_2}, \qquad c_{B_\theta,\rho_{2,1}}, \qquad I^w_{0,M}, \qquad W^{R,M}_{\Theta_3,L,w}(B^\Theta_\bullet).$$

The marker tensor is a full $\mathfrak{gl}(N|N)$-equivariant signed Brauer marker

$$\mathfrak{m}_{\theta_3} \in \mathcal{M}^{\mathrm{full}}(\mathbb{C}^{N|N}), \qquad \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}_{\theta_3}) = 0.$$

Choose the source-face coefficient

$$d_{\mathrm{CE}}\zeta_M = +\zeta_{M,\nu_3} + \text{other faces}$$

on the Θ_3-component. Since the kernel has degree zero, the incidence sign of this CE row in $-\kappa^M_{\mathcal{G},w,I}(d_{\mathrm{CE}}\zeta_M)$ is -1.

Write

$$K^M_{\Theta_3} = W^{R,M}_{\Theta_3,L,w}(B^{\Theta}_{\bullet}).$$

The row value is the nonzero symbolic local functional

$$R^{\mathrm{row}}_{\theta_3,M} = -K^M_{\Theta_3}(\zeta_{M,\nu_3}) =: \mathsf{E}_{\theta_3,M} \neq 0.$$

Its scalar gate is

$$S_{\theta_3,M} = \widehat{\sigma}_{\mathrm{sc},M}(\mathsf{E}_{\theta_3,M}) = 0.$$

Indeed, the row lies in the full-equivariant marked habitat, so Theorem E.30 kills its scalar projection; also the displayed Weyl half-edge has $\omega(z_1^2, z_2) = 0$.

The class-forming closure check is part of the row datum:

$$d_{\mathrm{ns},M}\mathsf{E}_{\theta_3,M} = 0.$$

If this equation is not supplied, the row is not a valid obstruction row and no H^1-class is formed. With it supplied, the row defines

$$\mathfrak{o}^{\mathrm{ns}}_{\theta_3,M} = [\mathsf{E}_{\theta_3,M}] \in H^1\Big(\mathcal{K}^{\bullet}_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}\Big).$$

The truncation matrix is explicit. For $M \geq N$,

$$\pi_{M,N}\mathsf{E}_{\theta_3,M} = \begin{cases} \mathsf{E}_{\theta_3,N}, & N \geq 3, \\ 0, & N < 3, \end{cases}$$

hence

$$t^{M,N}_{\theta_3\theta_3} = 1 \quad (N \geq 3), \qquad t^{M,N}_{\theta_3\alpha'} = 0 \quad (\alpha' \neq \theta_3),$$

and every entry from θ_3 is zero for $N < 3$. These entries satisfy $t^{M,P} = t^{M,N}t^{N,P}$.

Therefore this larger-package row does not vanish as a local-functional cochain, but its scalar gate vanishes. It is an explicit symbolic obstruction datum. A fixed-window order-three counterterm exists precisely when

$$\mathfrak{o}^{\mathrm{ns}}_{\theta_3,M} = 0, \qquad d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$$

for some $C_{\theta_3,M} \in \mathcal{K}^0_{\mathrm{ns},M}(I)$. No such primitive is supplied by this row, so the manuscript may name the class but may not claim the larger order-three QME branch is solved.

Proposition 3.182 *(θ_3 finite-row obstruction certificate).* Keep the one-row order-three package of Proposition 3.181. In the finite row complex actually supplied there, set

$$\mathcal{K}^0_{\theta_3,M} = 0, \qquad \mathcal{K}^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, \qquad d_{\theta_3,M} = 0,$$

and take residual $R_{\theta_3,M} = \mathsf{E}_{\theta_3,M}$. The scalar gate has already vanished:

$$\widehat{\sigma}_{\mathrm{sc},M}(\mathsf{E}_{\theta_3,M}) = 0.$$

The primitive-search matrix is the 1×0 zero matrix and the target vector is $-\mathsf{E}_{\theta_3,M}$. Hence the equation $A^M c = -r^M$ has no solution in the displayed finite row complex. The covector

$$\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$$

annihilates $\operatorname{im} d_{\theta_3,M} = 0$ and satisfies $\ell_{\theta_3}(R_{\theta_3,M}) = 1$. Therefore

$$[\mathsf{E}_{\theta_3,M}] \neq 0 \quad \text{in} \quad H^1(\mathcal{K}^\bullet_{\theta_3,M}).$$

This is an exact obstruction statement for the finite row subcomplex presently supplied. It does not exclude a primitive in a larger local-functional complex. To heal the row one must add either a degree-zero candidate $C_{\theta_3,M}$ with $d_{\mathrm{ns},M} C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$, or additional Hom-valued companion rows whose sum changes the residual vector.

If such a candidate $C_{\theta_3,M}$ is supplied and truncates by

$$\pi_{M,N} C_{\theta_3,M} = C_{\theta_3,N} \quad (N \geq 3), \qquad \pi_{M,N} C_{\theta_3,M} = 0 \quad (N < 3),$$

then the fixed-window search is solved and the Roos $\varprojlim^1$-class of this one-row tower is zero. If the truncation identity fails, the residual fixed-window obstruction is gone but the Roos class is represented by

$$[\pi_{M,N} C_{\theta_3,M} - C_{\theta_3,N}] \in H^0(\mathcal{K}^\bullet_{\mathrm{ns},N}(I)).$$

For the minimal order-two full-equivariant rows and for bracket rows containing α_{sc} or α_{II}, the same primitive search has residual vector 0, so the zero counterterm is the unique supplied solution and its Roos class is zero.

Proof. This is Proposition 3.180 applied to the one-row presentation. Since no degree-zero row is supplied, the image of the finite differential is zero. The residual vector is the basis vector $\mathsf{E}_{\theta_3,M}$, so the displayed covector is a cokernel certificate. The Roos statement is the same proposition applied after a hypothetical primitive has been added. The minimal order-two and zero-bracket rows have zero residual before cohomology, hence the zero primitive is compatible under truncation. □

PROPOSITION 3.183 *(θ_3 payload acceptance gate).* Keep the row $\theta_3 = \mathrm{CE}(\Theta_3, \nu_3)$ and the notation of Propositions 3.181 and 3.182. A proposed degree-zero healing payload for this row is accepted by the finite-row theorem only after the row array supplies one of the following two primitive entries.

(P1) A CE-ancestor entry

$$\eta_{\theta_3,M}, \qquad d_{\mathrm{CE}} \eta_{\theta_3,M} = \zeta_{M,\nu_3},$$

together with the Hom-valued Bianchi and companion-face terms needed to prove

$$C^{\mathrm{CE}}_{\theta_3,M} = K^M_{\Theta_3}(\eta_{\theta_3,M}) \in \mathcal{K}^0_{\mathrm{ns},M}(I), \qquad d_{\mathrm{ns},M} C^{\mathrm{CE}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}.$$

(P2) A local counterterm entry $C^{\mathrm{ct}}_{\theta_3,M} \in \mathcal{K}^0_{\mathrm{ns},M}(I)$, with its locality proof, scalar projection, and differential expansion, satisfying

$$d_{\mathrm{ns},M} C^{\mathrm{ct}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}.$$

Equivalently, in the ordered one-row presentation

$$\rho_1^M = \mathsf{E}_{\theta_3,M}, \qquad r^M = (1),$$

the supplied candidate column $C_{\theta_3,M}$ must have boundary-matrix coefficient

$$A^M_{\theta_3,C} = -1, \qquad d_{\mathrm{ns},M} C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}.$$

A row-data payload with no displayed differential entry, or with any coefficient other than -1 for the named candidate, is not this accepted payload; it must first be replaced by an explicitly supplied candidate whose differential column is the one above.

The scalar and tower gates are part of the same datum:

$$\widehat{\sigma}_{\mathrm{sc},M}(C_{\theta_3,M}) = 0, \qquad \pi_{M,N} C_{\theta_3,M} = \begin{cases} C_{\theta_3,N}, & N \geq 3, \\ 0, & N < 3. \end{cases}$$

With the row transport $T^{M,N}_{\theta_3\theta_3} = 1$ for $N \geq 3$ and 0 for $N < 3$, these equations give

$$T^{M,N} A^M = A^N P^{M,N}, \qquad [P^{M,N} c_M - c_N] = 0 \quad \text{for the solution } c_M = 1.$$

Thus such a payload kills the fixed-window row and has zero Roos compatibility class. The present manuscript data do not contain this payload: in the supplied one-row complex $K^{\circ}_{\theta_3,M} = 0$, so Proposition 3.182 remains the active claim.

A complete companion-face table is a different gate. It does not supply a primitive. Instead it replaces the one-face residual by a full CE-source-face row sum

$$\sum_{\nu \in F_{\Theta_3,M}} R^{\mathrm{row}}_{\mathrm{CE}(\Theta_\nu,\nu),M}, \qquad \nu_3 \in F_{\Theta_3,M},$$

whose ν_3-summand is $\mathsf{E}_{\theta_3,M}$. Such a table is accepted only if it supplies every face, every CE coefficient, every scalar gate, the $v^{M,N}$-truncation matrices with their cocycle identity, and the codimension-two closure table, and if

$$\sum_{\nu \in F_{\Theta_3,M}} R^{\mathrm{row}}_{\mathrm{CE}(\Theta_\nu,\nu),M} = 0, \qquad \sum_{\nu \in F_{\Theta_3,M}} S_{\mathrm{CE}(\Theta_\nu,\nu),M} = 0.$$

The current one-face table is only an interface for this possibility: its residual is $\mathsf{E}_{\theta_3,M}$, not zero. Hence it does not solve the θ_3 row and does not form a primitive Roos class. The first source-algebraic companion table containing ν_3 is recorded in Theorem 3.184; it produces an obstruction datum, not a cancellation theorem.

Proof. For a primitive payload, apply Proposition 3.180 with degree-one basis $\rho_1^M = \mathsf{E}_{\theta_3,M}$ and residual vector $r^M = (1)$. The equation $A^M c = -r^M$ is solved by the supplied candidate with $c = 1$ exactly when the candidate column is $A^M_{\theta_3,C} = -1$. The CE-ancestor and local-counterterm alternatives are precisely two ways of producing such a degree-zero column; without the displayed target differential, scalar projection, and support proof, the finite row complex has no boundary matrix entry to use.

The truncation assertion is the identity/zero transport already attached to the row $\mathsf{E}_{\theta_3,M}$. With the same identity/zero rule for $C_{\theta_3,M}$, the matrix equation $TA = AP$ is tautological: above degree 3 it

reads $(-1) = (-1)$, and below degree 3 both sides are 0. The chosen coefficients $c_M = 1$ are therefore compatible, so the Roos representative vanishes.

For a companion-face table no degree-zero candidate is chosen. The residual vector itself is replaced by the signed row sum of all CE-source faces. The finite-row theorem can test that new row sum only after the table supplies its scalar gates, truncation coordinates, and codimension-two closure data. If the signed sum is not zero, the row remains a residual; if it is zero, cancellation occurs before the primitive search and no primitive Roos class is formed. □

THEOREM 3.184 *(θ_3 companion-face construction obstruction).* Work in the order-three full-equivariant marked branch of Proposition 3.181, and put $H_{p,q} = z_1^p z_2^q$. Consider the two-u source-algebra ancestor

$$\zeta_M^0 = u_{a,H_{1,0}} u_{b,H_{0,1}}, \qquad [H_{1,0}, H_{2,1}] = H_{2,0}.$$

Its monomial CE differential is the first source-algebraic table of the same external shape whose source faces contain the displayed ν_3-row $u_{a,H_{2,0}} u_{b,H_{0,1}} c_{B_\theta,\rho_{2,1}}$. After truncation to Hamiltonian degree ≤ 3, removal of constants, and cancellation of the two $\rho_{1,1}$-terms, the nonzero candidate face set is the following eight-face set. With

$$E_\nu^M = -K_{\Theta_3}^M(\zeta_{M,\nu}), \qquad E_{\nu_3}^M = \mathsf{E}_{\theta_3,M},$$

the table is

ν	$\epsilon^{\mathrm{CE}}_{\Theta_3,\nu}$	$\zeta_{M,\nu}$	$\epsilon^{\mathrm{CE}}_{\Theta_3,\nu} E_\nu^M$
ν_{02}	$+2$	$u_{a,H_{0,1}} u_{b,H_{0,1}} c_{B_\theta,\rho_{0,2}}$	$2E_{\nu_{02}}^M$
ν_{20}	-2	$u_{a,H_{1,0}} u_{b,H_{1,0}} c_{B_\theta,\rho_{2,0}}$	$-2E_{\nu_{20}}^M$
ν_{03}	$+3$	$u_{a,H_{0,2}} u_{b,H_{0,1}} c_{B_\theta,\rho_{0,3}}$	$3E_{\nu_{03}}^M$
$\nu_{12}^{(1)}$	$+2$	$u_{a,H_{1,1}} u_{b,H_{0,1}} c_{B_\theta,\rho_{1,2}}$	$2E_{\nu_{12}^{(1)}}^M$
$\nu_{12}^{(2)}$	-1	$u_{a,H_{1,0}} u_{b,H_{0,2}} c_{B_\theta,\rho_{1,2}}$	$-E_{\nu_{12}^{(2)}}^M$
$\nu_3 = \nu_{21}^{(1)}$	$+1$	$u_{a,H_{2,0}} u_{b,H_{0,1}} c_{B_\theta,\rho_{2,1}}$	$\mathsf{E}_{\theta_3,M}$
$\nu_{21}^{(2)}$	-2	$u_{a,H_{1,0}} u_{b,H_{1,1}} c_{B_\theta,\rho_{2,1}}$	$-2E_{\nu_{21}^{(2)}}^M$
ν_{30}	-3	$u_{a,H_{1,0}} u_{b,H_{2,0}} c_{B_\theta,\rho_{3,0}}$	$-3E_{\nu_{30}}^M$

Hence the source-algebraic companion residual is

$$\begin{aligned} R^{\mathrm{cand}}_{\Theta_3,M} = {} & 2E_{\nu_{02}}^M - 2E_{\nu_{20}}^M + 3E_{\nu_{03}}^M + 2E_{\nu_{12}^{(1)}}^M - E_{\nu_{12}^{(2)}}^M \\ & + \mathsf{E}_{\theta_3,M} - 2E_{\nu_{21}^{(2)}}^M - 3E_{\nu_{30}}^M. \end{aligned}$$

This expression is the exact residual of the attempted companion route. It is not known to be zero.

The induced source-face transport is the diagonal cutoff

$$v_{\nu;\nu'}^{M,N} = \begin{cases} 1, & \nu = \nu' \text{ and } d(\nu) \leq N, \\ 0, & \text{otherwise}, \end{cases}$$

where $d(\nu_{02}) = d(\nu_{20}) = 2$ and the other six faces have degree 3. It satisfies the cocycle identity $v^{M,P} = v^{M,N} v^{N,P}$. For $N = 2$, however, the transported residual is

$$\pi_{M,2} R^{\mathrm{cand}}_{\Theta_3,M} = 2E_{\nu_{02}}^2 - 2E_{\nu_{20}}^2.$$

Thus even a fixed-window cancellation at $M \geq 3$ would not give a theorem-grade compatible tower unless the lower-window identity $E^2_{\nu_{02}} = E^2_{\nu_{20}}$ is proved in the chosen row basis, or the lower-window row presentation is changed and its v-matrix is recomputed.

Consequently the companion route is a construction/obstruction theorem with the following exact missing marked Costello datum:

$$\Big(d_{\mathrm{CE}}\zeta_M,\ \epsilon^{\mathrm{CE}}_{\Theta_3,\nu},\ K^M_{\Theta_3}(\zeta_{M,\nu}),\ \mu_\nu,\ v^{M,N}_{\nu;\nu'},\ C^{(2)}_{\nu,\nu'}\Big)_{\nu,\nu'}.$$

Here μ_ν is the marked output and marker-transport entry, and $C^{(2)}_{\nu,\nu'}$ is the codimension-two source-face incidence/weight table: the two orders of each pair of boundary operations, the output graph, the orientation/Koszul coefficient, the transported marker, and the resulting renormalized row coordinate in the declared degree-one basis. Until this table is supplied and proves both $R^{\mathrm{cand}}_{\Theta_3,M} = 0$ and its transported lower-window identities, the current theorem surface remains the one-row obstruction of Proposition 3.182. No companion cancellation and no primitive counterterm is claimed.

Proof. The Poisson rule $[H_{a,b}, H_{r,s}] = (as - br)H_{a+r-1,b+s-1}$ gives the displayed source-algebra faces from the differential of the two u-factors against the current label. In degree ≤ 3, constants are removed by the Hamiltonian quotient and the two $\rho_{1,1}$-faces have opposite coefficients, so the eight listed terms remain. Applying the row convention $R^{\mathrm{row}}_{\mathrm{CE}(\Theta_3,\nu),M} = \epsilon^{\mathrm{CE}}_{\Theta_3,\nu}E^M_\nu$ gives the signed residual formula, with the ν_3-summand equal to $\mathsf{E}_{\theta_3,M}$.

The diagonal transport is just finite-window projection on the Hamiltonian degree of the c-label. Such diagonal cutoffs compose, proving the cocycle identity. Projecting to $N = 2$ deletes every degree-three face and leaves exactly the two degree-two summands. Therefore the lower-window residual is $2E^2_{\nu_{02}} - 2E^2_{\nu_{20}}$.

The passage from this source-algebra table to a Costello cancellation theorem is precisely the missing marked incidence/weight computation. Source coefficients alone do not determine the renormalized graph weights, marker transports, codimension-two closure signs, or lower-window row coordinates. Those entries are the displayed missing datum. Without them, Definition 3.173 supplies no zero residual vector, and Proposition 3.180 still sees the current one-row obstruction. □

Theorem 3.185 *(Actual finite-window non-scalar decision datum).* Fix a finite-window graph/QME package $(I, w, M, \mathcal{G}_M)$, and fix the labelled CE cycle ζ_M on which the curvature is evaluated. Write K^M_Γ for the component of $\kappa^M_{\mathcal{G},w,I}$ attached to a graph Γ; for the zero-internal-edge component this is $\kappa^{\mathrm{coef},M}_{\hbar,w,I}$, and for a positive graph it is the renormalized weight $W^{R,M}_{\Gamma,L,w}$. If $|\Gamma|_\hbar$ denotes the $\hbar$-order assigned by the finite graph package, then the order-n graph curvature is

$$\begin{aligned}
\mathrm{Ob}^{\mathrm{graph}}_{\mathcal{G},M}[n](\zeta_M) &= [\hbar^n]\Big(d_M\kappa^M_{\mathcal{G},w,I}(\zeta_M) - \kappa^M_{\mathcal{G},w,I}(d_{\mathrm{CE}}\zeta_M) \\
&\qquad + \frac{1}{2}\sum_{(\zeta_M)}\{\kappa^M_{\mathcal{G},w,I}(\zeta'_M), \kappa^M_{\mathcal{G},w,I}(\zeta''_M)\}_{\hbar,M}\Big) \\
&= \sum_{|\Gamma|_\hbar=n}\varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{|\Gamma'|_\hbar=n}\varepsilon^{\mathrm{CE}}_{\Gamma'}K^M_{\Gamma'} \\
&\quad + \frac{1}{2}\sum_{|\Gamma_1|_\hbar+|\Gamma_2|_\hbar=n}\varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M}.
\end{aligned}$$

The signs are the orientation signs of the supplied finite graph differential and the CE coproduct. With lower counterterms $C_{<n,M}$, the cochain tested at order n is

$$R_{n,M} = \mathrm{Ob}^{\mathrm{graph}}_{\mathcal{G},M}[n] + [\hbar^n]\Big(d_M C_{<n,M} + \frac{1}{2}\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}\Big).$$

Its scalar gate is

$$\mathfrak{s}_{n,M} = \widehat{\sigma}_{\mathrm{sc},M}(R_{n,M}).$$

The first scalar-contact graph whose coefficient is actually computed in the present local model is the one-cross-contraction boundary graph with two Hamiltonian external legs and one open scalar loop. It has $\hbar$-order 1. For a full-equivariant marked boundary generator $X_{\Gamma,o,\mathfrak{m},M}$, the extracted order-one scalar gate is

$$\mathfrak{s}^{\mathrm{mk}}_{1,M}(X_{\Gamma,o,\mathfrak{m},M}) = \sum_{i:\, r(\beta_i)=1} (-1)^{i-1}\, \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m})\big)\, \omega(f_{\partial_{\beta_i}\Gamma}, g_{\partial_{\beta_i}\Gamma})\gamma_1.$$

Before the coefficient extraction this is the $\hbar^1$-term $\hbar\,\mathfrak{s}^{\mathrm{mk}}_{1,M}$. If $V = \mathbb{C}^{N|N}$ and the markers are full $\mathfrak{gl}(V)$-equivariant, then

$$\mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m})\big) = 0$$

for every elementary boundary term by Lemma E.29; hence

$$\mathfrak{s}^{\mathrm{mk}}_{1,M} = 0.$$

This is a scalar-gate vanishing, not a construction of a non-scalar primitive.

After the scalar gate vanishes, the first non-scalar order is the least n_0 for which the cochain $R_{n_0,M}$ is defined for the actual graph list and is not already killed by the lower-order choices. Its class is

$$[\mathfrak{o}^{\mathrm{fw}}_{n_0,M}] = [R_{n_0,M}] \in H^1\big(Q^\bullet_{\mathcal{G},M}(I), d_M\big).$$

It vanishes exactly when there exists $C_{n_0,M} \in Q^0_{\mathcal{G},M}(I)$ with $d_M C_{n_0,M} = -R_{n_0,M}$. In a compatible finite-window tower, after choosing any such windowwise primitives, the remaining compatibility obstruction is

$$\lambda^{\mathrm{fw}}_{n_0} = \big[\big([\pi_{M+1,M} C_{n_0,M+1} - C_{n_0,M}]\big)_M\big] \in \varprojlim{}^1_M\, H^0\big(Q^\bullet_{\mathcal{G},M}(I)\big).$$

Thus compatible finite-window counterterms exist precisely when all fixed-window classes $[\mathfrak{o}^{\mathrm{fw}}_{n_0,M}]$ vanish and $\lambda^{\mathrm{fw}}_{n_0} = 0$.

The data above are not recoverable from the abstract existence of $\mathcal{G}_M$. To decide n_0, the class $[\mathfrak{o}^{\mathrm{fw}}_{n_0,M}]$, or the Roos class $\lambda^{\mathrm{fw}}_{n_0}$, one must supply a valid finite-window coefficient array in the sense of Definition 3.173:

$$\mathsf{A}^{\mathrm{fw}}_{n,M}$$

whose rows specify graph type, edge labels, vertex labels, marker tensor, incidence coefficient, $\hbar$-order, coefficient degree, scalar-contact filtration, scalar-gate term, row curvature value, and truncation coefficients in M. Without this array the manuscript has only the formal obstruction symbol of Theorem 3.171; it has no theorem-grade computation of n_0, no proof that $[\mathfrak{o}^{\mathrm{fw}}_{n_0,M}] = 0$, and no defined $\lambda^{\mathrm{fw}}_{n_0}$.

Proof. The first displayed formula is the definition of the convolution curvature in Definition B.15, evaluated on ζ_M and then projected to the $\hbar^n$-coefficient. Expanding $-\kappa d_{\mathrm{CE}}$ and the CE coproduct bracket gives the three graph sums, with exactly the orientation signs supplied by the finite graph package. Adding the lower counterterm terms gives $R_{n,M}$, the residual of Theorem 3.171.

The order-one scalar-contact computation is Proposition E.21 applied to each first filtration-lowering marked boundary term of Definition E.26. The full-equivariant marker transport maps preserve the signed Brauer marker span. Lemma E.29 computes the cyclic supertrace of every such output marker as a power of $\operatorname{sdim} \mathbb{C}^{N|N} = 0$, and the empty marker word gives the same zero coefficient. Therefore the scalar gate vanishes on the full-equivariant marked habitat before any non-scalar cohomology class is formed.

Once the scalar gate is zero, $R_{n,M}$ lies in $\boldsymbol{Q}^{\bullet}_{\mathcal{G},M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},M}$ exactly as in the proof of Theorem 3.171. A degree-zero counterterm changes $R_{n,M}$ by a d_M-boundary, so the fixed-window solvability criterion is precisely the vanishing of its H^1-class. If primitives are chosen window by window, their truncation differences are closed degree-zero cochains; their classes form the standard Roos cocycle computing $\varprojlim^1_M H^0(\boldsymbol{Q}^{\bullet}_{\mathcal{G},M}(I))$. Vanishing of this class is exactly the condition that the primitives can be adjusted to a compatible family.

The final assertion is a dependency statement. The formulas require a finite graph list with order assignment, CE cycle, signs, coproduct decomposition, graph weights, lower counterterms, scalar projection, and truncation maps. Omitting any of these removes a term from the displayed cochain $R_{n,M}$, so neither its first nonzero order nor its cohomology class is defined. □

Remark 3.186 *(Scalar-gate boundary).* In the ordinary scalar-reduced $\mathfrak{gl}_N$ branch, the same two-Hamiltonian one-cross graph gives the order-one scalar class $N\,\omega(f,g)\gamma_1$, equivalently $N[\bar{c}]\gamma_1$ in cohomology after extracting the $\hbar^1$-coefficient. In the even block-diagonal marked habitat a single marker $T = \lambda_0 P_{\bar{0}} + \lambda_1 P_{\bar{1}}$ gives $N(\lambda_0 - \lambda_1)\omega(f,g)\gamma_1$. These vanish only under the corresponding central-character, balanced, or full-equivariant hypotheses; they are not non-scalar counterterms.

Remark 3.187 *(Relation to regulator independence).* Theorem 3.96 proves that, after the finite-window Mittag-Leffler hypotheses of Definition 3.90 are imposed, the Hadamard criterion is independent of the admissible exponential weight. This is weight-independence among regulated coefficient products. The unweighted product/direct-sum pair is separated by the endpoint obstruction of Theorem 3.109: the required coefficient Casimir has no strict product/direct-sum representative.

Remark 3.188 *(Reduced core and residual obstruction complex).* Theorem 2.70 consists exactly of the proved Hamiltonian, factorization, principal-part, and anomaly statements stated above. The reduced cotangent factor after de Rham-current contraction is proved by 3.237; the chain-level primitive projection 3.154 supplies only its primitive indecomposable P_0 shadow on the unreduced open complex. It does not construct the unreduced cotangent factorization-centre morphism or its centrality homotopies. The weighted mechanism closes the coefficient kernel and graphwise analytic estimates, but the all-order one-antifield quantum extension of the degree-zero Moyal/Weyl theorem requires vanishing of the mixed brane-defect QME obstruction class in Problem 3.87. The nilpotent truncation closes only the classical $\mathfrak{m}^3$ sector.

Remark 3.189 *(Stratified Weiss/Cech–Roos descent obstruction).* A global factorization-center morphism is not obtained by choosing Darboux coordinates at each point and gluing the resulting stalk maps. In the internal stratified lane the descent defect is the Cech cone

$$\Delta^{N,M}_{V,\mathcal{U}} = \operatorname{Cone}\Big(\mathcal{F}^{\mathrm{str}}_{\Omega,N,M}(V) \to \operatorname{Tot} \check{C}^{\bullet}(\mathcal{U}; \mathcal{F}^{\mathrm{str}}_{\Omega,N,M})\Big)$$

for collared stratified Weiss covers $\mathcal{U}$ of V, together with the Roos compatibility class

$$\mathfrak{w}^{\check{C}/R}_{V,\Omega,N,M}$$

under refinements, finite windows, interval inclusions, and admissible weight transports. Thus the internal descent entry is

$$\delta^{\check{C}/R}_{\mathrm{Weiss}} = \Big(\{H^\bullet(\Delta^{N,M}_{V,\mathcal{U}})\}_{V,\mathcal{U},N,M}, \mathfrak{w}^{\check{C}/R}_{V,\Omega,N,M}\Big).$$

Source comparison with Costello–Gwilliam style global descent is a separate external matched-conventions gate, not an internal proof of the local factorization-center morphism.

3.10 Operators in the open brane gauge theory and the large N limit

The local-operator calculation is performed in the reduced BV model with fields ϕ_1, ϕ_2 and the antifield $\psi = A^*$ for the Lagrange multiplier A. We work with degree-zero rational GL_N-invariant functions in characteristic zero. This is the algebraic invariant calculation used by the trace theorem, not the full Lie-algebra Chevalley–Eilenberg BRST cohomology of $\mathfrak{gl}_N$. The scalar gauge direction acts trivially and would leave scalar ghost classes in the unreduced Lie-CE complex; those stack-level stabilizer classes are outside the Hamiltonian trace calculation.

The gauge field has one algebraic job: it imposes the moment-map equation $[\phi_1, \phi_2] = 0$. In BV language this equation is imposed by the Koszul differential $Q\psi = [\phi_1, \phi_2]$. This is why the stable trace algebra starts noncommutative, as cyclic words in two matrices, and then collapses in cohomology to commutative monomials on the formal symplectic plane.

Construction 3.190 *(The reduced BV algebra at the insertion point).* At a fixed point of the brane line, introduce three matrix variables

$$\phi_1, \phi_2 \in \mathfrak{gl}_N, \qquad \psi \in \mathfrak{gl}_N[1].$$

The reduced local polynomial algebra is

$$\mathcal{R}_N = \mathbb{C}[(\phi_1)^i{}_j, (\phi_2)^i{}_j] \otimes \Lambda\big((\psi)^i{}_j\big).$$

Its reduced Koszul/BV differential is the derivation determined on generators by

$$Q(\phi_1)^i{}_j = 0, \qquad Q(\phi_2)^i{}_j = 0, \qquad Q(\psi)^i{}_j = ([\phi_1, \phi_2])^i{}_j.$$

This formula is the algebraic shadow of the A-equation of motion: varying

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to A gives $[\phi_1, \phi_2] = 0$, and the antifield $\psi = A^*$ records that equation as a Koszul differential.

LEMMA 3.191 *(De Rham jets).* Positive jets in the topological direction vanish in local operator cohomology. Equivalently, every cohomology class has a representative determined by the values of the fields and antifields at the insertion point.

Proof. For a coordinate t on the line, the de Rham part of the BRST operator is $d_{\mathbb{R}}$. On the jet algebra one has the contraction homotopy $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$. Hence a local expression containing a positive t-derivative is cohomologous to zero after applying the usual Euler homotopy on jet degree. This kills topological-direction jets and leaves the holomorphic z_i-derivatives used later on the closed side. □

Proposition 3.192 *(Stable trace invariants).* Let $F = \mathbb{C}\langle x_1, x_2\rangle$ and let $\mathrm{Cyc}(F) = F/[F,F]$ be the vector space of cyclic words. For each N, GL_N acts on $\mathfrak{gl}_N^2$ by simultaneous conjugation. The degreewise stable invariant algebra

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} := \bigoplus_{d\geq 0} \varprojlim_N \big(\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}\big)_d$$

with respect to block-restriction maps in each word-degree is the free commutative algebra generated by the nonempty cyclic words:

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} \cong \mathbf{S}\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}}\big),$$

where a cyclic word w maps to the trace function $(\phi_1, \phi_2) \mapsto \mathrm{Tr}\, w(\phi_1, \phi_2)$. More precisely, in total positive word-degree d this map is an isomorphism for every $N \geq d$, after omitting the scalar empty trace as a primitive generator.

Proof. Procesi–Razmyslov invariant theory [17, 18] gives two inputs. First, $\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}$ is generated by trace functions of words in the two generic matrices. Second, all polynomial relations among these trace functions are consequences of the fundamental trace identities of matrix size N; in particular every homogeneous trace relation has total word-degree at least $N+1$. Fix a total word-degree d. For every $N \geq d$ the degree-d relations are exactly cyclicity of the trace. Passing degree by degree to the stable inverse limit removes all finite-N trace identities. The surviving identification is $\mathrm{Tr}(uv) = \mathrm{Tr}(vu)$. This is exactly the symmetric algebra on cyclic words. □

Lemma 3.193 *(Stable marked trace invariants).* In the stable invariant theory used below, the primitive part linear or quadratic in an odd marked matrix ψ is obtained from the corresponding even three-matrix trace invariant by polarization and skew-symmetrisation in the marked letters. The only stable single-trace relation is graded cyclicity: moving one ψ through the trace has sign $+1$, while moving one ψ past another ψ gives a minus sign.

Proof. Apply Proposition 3.192 to three even generic matrices and take the homogeneous summand linear, respectively quadratic, in the third matrix. The Procesi–Razmyslov trace identities have no stable relations in the fixed total word-degree once N is large enough. Declaring the marked matrix odd is the usual super-polarization: the linear part is unchanged, and the quadratic part is projected to the alternating component under exchange of the two marked occurrences. The trace relation therefore becomes exactly graded cyclicity. These are the signs used in the finite complex for one- and two-ψ words. □

Remark 3.194. The empty trace is the finite-rank scalar $\mathrm{Tr}_N(1) = N$, which changes with N and fails to define a stable primitive generator. The stable symmetric algebra still has its ordinary unit; what is omitted is the empty trace as a generator of the connected primitive part. The class $\mathrm{Tr}(\psi)$ is a genuine stable odd primitive class. It is excluded from the Hamiltonian comparison because constants have been removed from the closed Hamiltonian algebra.

Varying

$$S = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to A gives the moment-map equation $[\phi_1, \phi_2] = 0$. In the reduced BV complex,

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0$$

on invariant trace functions.

Construction 3.195 *(Boundary evaluation of a closed Hamiltonian).* Let $f = \sum_{k,l} c_{k,l} z_1^k z_2^l \in \mathbb{C}[z_1, z_2]$. The brane replaces the two normal coordinates z_1, z_2 by the two adjoint scalar fields ϕ_1, ϕ_2. Choosing the ordered lift with all ϕ_1's to the left of all ϕ_2's gives the invariant function

$$\widetilde{\partial}_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l}\, \mathrm{Tr}(\phi_1^k \phi_2^l) \in \mathcal{R}_N^{GL_N}.$$

Let

$$H^0_{\mathrm{Ham}}(\mathcal{R}_N^{GL_N}, Q) := H^0(\mathcal{R}_N^{GL_N}, Q)/\mathbb{C} \cdot \mathrm{Tr}(1)$$

as a vector space, and as a Lie algebra for the Poisson bracket. This is not an associative quotient of the operator algebra by its unit. Constants give the scalar $c_{0,0}N$ and therefore vanish in this Hamiltonian quotient. Thus the construction descends to a linear map

$$\partial_{\mathrm{bb},N} : \bar{A} \to H^0_{\mathrm{Ham}}(\mathcal{R}_N^{GL_N}, Q), \qquad [f] \longmapsto \overline{\widetilde{\partial}_{\mathrm{bb},N}(f)}.$$

Let $\mathsf{W}_{k,l}$ be the set of words in two letters containing k copies of the first letter and l copies of the second. The first descendant is obtained by inserting one Koszul variable and summing over all ordered lifts: For $k + l > 0$ the descendant part of the Hamiltonian comparison is

$$\partial^{(1)}_{\mathrm{bb},N}(e^{\mathrm{PBW}}_{k,l}) = \Psi_{k,l}, \qquad \Psi_{k,l} := \sum_{w \in \mathsf{W}_{k,l}} \mathrm{Tr}\big(\psi\, w(\phi_1, \phi_2)\big).$$

Here $e^{\mathrm{PBW}}_{k,l}$ is an abstract special-fibre PBW label in bidegree (k, l), not the Taylor-dual coadjoint functional $\delta_{k,l}$ and not the Matlis principal part $\rho_{k,l}$. The same formula at $(k, l) = (0, 0)$ gives the decoupled scalar antifield $\mathrm{Tr}(\psi)$, which is excluded from the Hamiltonian pairing because constants have been removed from $\bar{A}$.

LEMMA 3.196 *(First descendants are cycles).* For every $k, l \geq 0$, the symmetrised trace $\Psi_{k,l}$ satisfies $Q\Psi_{k,l} = 0$.

Proof. Write $x = \phi_1$ and $y = \phi_2$. Since x and y are even and $Q\psi = [x, y]$, one has

$$Q\Psi_{k,l} = \sum_{w \in \mathsf{W}_{k,l}} \mathrm{Tr}\big((xy - yx)w(x, y)\big).$$

Group the resulting traces by cyclic word of multidegree $(k + 1, l + 1)$. The coefficient of a cyclic word in the positive sum is the number of cyclic adjacent occurrences of the block xy; the coefficient in the negative sum is the number of cyclic adjacent occurrences of the block yx. On a circle the number of transitions from an x-run to a y-run equals the number of transitions from a y-run to an x-run. The two grouped sums are therefore equal, so $Q\Psi_{k,l} = 0$. □

Proposition 3.197 *(Stable primitive one-antifield Koszul homology).* Fix $k, l \geq 0$. The middle homology of the primitive single-trace complex with one displayed ψ and k, l displayed x, y letters is one-dimensional, generated by the class of $\Psi_{k,l}$. The nonconstant part relevant to the Hamiltonian comparison is the direct sum over $k + l > 0$.

Proof. Write $x = \phi_1$ and $y = \phi_2$. Let $\mathsf{W}_{k,l}$ be the set of ordinary words with k letters x and l letters y, and let $\mathsf{N}_{a,b}$ be the set of cyclic words with a letters x and b letters y. By Lemma 3.193, primitive cyclic words are taken modulo the graded cyclic rule

$$\mathrm{Tr}(ab) = (-1)^{|a||b|}\mathrm{Tr}(ba), \qquad |x| = |y| = 0, \quad |\psi| \equiv 1 \pmod 2.$$

Graded cyclicity gives a unique representative $\mathrm{Tr}(\psi w)$ for every one-ψ cyclic word. Thus the relevant one-ψ chain group is

$$C_1(k, l) = \mathbb{C}\{\mathsf{W}_{k,l}\}.$$

The zero-ψ target is

$$C_0(k, l) = \mathbb{C}\{\mathsf{N}_{k+1,l+1}\},$$

and the differential is the incidence map

$$\partial_1(w) = [xyw]_{\mathrm{cyc}} - [yxw]_{\mathrm{cyc}}.$$

The two-ψ source is the alternating quotient generated by traces $\mathrm{Tr}(\psi u \psi v)$ with u and v ordinary words whose total bidegree is $(k-1, l-1)$:

$$\mathrm{Tr}(\psi u \psi v) = -\mathrm{Tr}(\psi v \psi u).$$

Hence diagonal pairs vanish, and if $k = 0$ or $l = 0$ this source is zero. Applying $Q\psi = xy - yx$ gives

$$\begin{aligned}\partial_2(u, v) &= Q\,\mathrm{Tr}(\psi u \psi v)\\ &= \mathrm{Tr}\big(\psi vxyu - \psi vyxu - \psi uxyv + \psi uyxv\big),\end{aligned}$$

so, after removing the common front ψ,

$$\partial_2(u, v) = vxyu - vyxu - uxyv + uyxv.$$

The displayed formula changes sign under $(u, v) \leftrightarrow (v, u)$, so it is well defined on the alternating quotient. A direct substitution shows $\partial_1\partial_2 = 0$.

If $k = 0$ or $l = 0$, then $C_1(k, l)$ is spanned by x^k or y^l, $\partial_1 = 0$, and $C_2(k, l) = 0$. Hence the middle homology is already the line spanned by $\Psi_{k,l}$. Assume from now on that $k, l > 0$.

We compute the first homology of this finite complex. Form a graph $\Gamma_{k,l}$ whose vertices are the necklaces in $\mathsf{N}_{k+1,l+1}$ and whose oriented edges are the words $w \in \mathsf{W}_{k,l}$, with

$$w : [yxw]_{\mathrm{cyc}} \longrightarrow [xyw]_{\mathrm{cyc}}.$$

Then ∂_1 is the cellular boundary of this graph. For each pair (u, v) as above attach the square whose oriented boundary is $vxyu - vyxu - uxyv + uyxv$. Denote the resulting two-complex by $X_{k,l}$.

Equivalently, $X_{k,l}$ is the two-skeleton of the chip-moving state complex $Y_{k,l}$. The full complex $Y_{k,l}$ has an r-cell for r mutually chosen chip moves on a cycle with $l + 1$ oriented edges and $k + 1$ indistinguishable

chips, modulo cyclic relabelling of the edge labels. A vertex is a cyclic distribution of the chips among the vertices of this cycle, hence a necklace in $\mathsf{N}_{k+1,l+1}$. A one-cell moves one chosen chip across one oriented edge; cutting the cycle at this moving edge records exactly a word $w \in \mathsf{W}_{k,l}$, and its two endpoints are $[yxw]_{\mathrm{cyc}}$ and $[xyw]_{\mathrm{cyc}}$. A two-cell records two chip moves performed independently, and its oriented boundary is precisely $vxyu - vyxu - uxyv + uyxv$. Therefore the cellular chain complex of $Y_{k,l}$ through degree two is exactly

$$C_2(k,l) \longrightarrow C_1(k,l) \longrightarrow C_0(k,l),$$

and the inclusion $X_{k,l} = Y_{k,l}^{(2)} \hookrightarrow Y_{k,l}$ induces an isomorphism on H_1, since cells of dimension at least three do not affect first homology.

The homology of $Y_{k,l}$ is the homology of the cycle. For completeness we recall the elementary circle-sum argument. Before quotienting by cyclic relabelling, send a chip configuration to the sum of the chip positions in $\mathbb{R}/(l+1)\mathbb{Z}$, and extend linearly over each cell. In the universal cover of the target circle, a fibre is described by weakly ordered lifted chip positions with fixed total sum; it is a convex polytope, subdivided by the same chip-crossing cells. Hence the circle-sum map is a cellular homotopy equivalence to S^1. The finite cyclic relabelling of the y-gaps is equivariant for this circle-sum map: it rotates the target circle, and every stabilizer acts affinely on a convex fibre, whose quotient is again contractible. Thus the quotient target is still a circle and the quotient complex $Y_{k,l}$ has the homology of S^1. Consequently

$$H_1(X_{k,l};\mathbb{C}) \cong H_1(S^1;\mathbb{C}) \cong \mathbb{C}.$$

Lemma 3.196 identifies this representative as a cycle in the original trace complex. The functional

$$\epsilon\big(\mathrm{Tr}(\psi w)\big) = 1, \qquad w \in \mathsf{W}_{k,l},$$

kills every displayed two-ψ boundary because each boundary has two positive and two negative summands, while

$$\epsilon(\Psi_{k,l}) = \binom{k+l}{k} \neq 0.$$

Thus the one-ψ homology in bidegree (k,l) is the one-dimensional line generated by $[\Psi_{k,l}]$. □

Remark 3.198. A finite exact homology check constructs the finite complex $C_2(k,l) \to C_1(k,l) \to C_0(k,l)$ over $\mathbb{Q}$, with the alternating two-ψ source above, and confirms $\dim\ker \partial_1/\mathrm{im}\,\partial_2 = 1$ for $0 \le k,l \le 5$. It also checks the open Hamiltonian descendant action of Proposition 3.223 (below) for exponents at most 3, and the closed coadjoint Taylor-dual formula together with the principal-part realization of Proposition 3.227 (below) for exponents at most 5. The finite exact check verifies the displayed finite models without adding assumptions.

COROLLARY 3.199 *(PBW special-fibre one-antifield pairing).* Let $\bar{A}_{\mathrm{desc}}[1]$ be the abstract shifted PBW label copy with basis $e_{k,l}^{\mathrm{PBW}}$ in nonconstant bidegree (k,l). The assignment

$$e_{k,l}^{\mathrm{PBW}} \longmapsto [\Psi_{k,l}]$$

extends to an isomorphism from $\bar{A}_{\mathrm{desc}}[1]$ to the nonconstant primitive one-ψ brane homology.

Proof. The monomials $z_1^k z_2^l$ with $k + l > 0$ form the polynomial basis of $\bar{A}$ and therefore index the PBW special-fibre label copy $\bar{A}_{\mathrm{desc}}$. By Proposition 3.197, the primitive one-ψ homology has one generator in each bidegree, namely $[\Psi_{k,l}]$. Discarding the scalar bidegree $(0, 0)$ and sending the abstract PBW label in every nonconstant bidegree to that generator gives a filtered vector-space isomorphism. This is the boundary pairing used in the PBW special fibre. It is not the continuous Taylor-dual/coadjoint module and not the Matlis principal-part realization. □

Lemma 3.200 *(Adjacent swaps).* In the invariant reduced BV cohomology,

$$\mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v)$$

is Q-exact for all ordinary words u, v in ϕ_1, ϕ_2.

Proof. By cyclicity of the trace,

$$\begin{aligned} Q\,\mathrm{Tr}(\psi vu) &= \mathrm{Tr}([\phi_1, \phi_2]vu) \\ &= \mathrm{Tr}(\phi_1\phi_2 vu - \phi_2\phi_1 vu) \\ &= \mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v). \end{aligned}$$

□

Theorem 3.201 *(Cut-dependent normal-ordering straightening).* Write $x = \phi_1$ and $y = \phi_2$. For every ordinary necklace $[w]$ in x, y, choose once and for all a linear representative $c([w])$. Suppose $c([w])$ has a occurrences of x and b occurrences of y, and choose a deterministic adjacent-swap path

$$c([w]) = w_0,\ w_1, \ldots,\ w_r = x^a y^b,$$

where each step has the form

$$w_s = A_s\, yx\, B_s, \qquad w_{s+1} = A_s\, xy\, B_s.$$

Define the one-ψ chain

$$H_c([w]) := -\sum_{s=0}^{r-1} \mathrm{Tr}(\psi B_s A_s).$$

Then

$$QH_c([w]) = \mathrm{Tr}(c([w])) - \mathrm{Tr}(x^a y^b).$$

Equivalently, if $N([w]) = z_1^a z_2^b$ and $i(z_1^a z_2^b) = \mathrm{Tr}(x^a y^b)$, then on this ordinary-word sector with the chosen cuts

$$QH_c = \mathrm{id} - iN.$$

The homotopy is not canonical as a cyclic construction. If a second cut c' is chosen, then $H_{c'}([w]) - H_c([w])$ is a closed one-ψ chain; its homology class is the corresponding scalar multiple, determined by the change of cut, of the primitive descendant $\Psi_{a-1,b-1}$ when $a, b > 0$. Thus this theorem is a cut-dependent straightening on the no-ψ sector, not a contraction of the full cyclic open complex.

Proof. For a single step $w_s = A_s yxB_s \to w_{s+1} = A_s xyB_s$, Lemma 3.200 gives

$$Q\,\mathrm{Tr}(\psi B_s A_s) = \mathrm{Tr}(A_s xyB_s) - \mathrm{Tr}(A_s yxB_s) = \mathrm{Tr}(w_{s+1}) - \mathrm{Tr}(w_s).$$

Summing over s and inserting the minus sign in the definition of $H_c([w])$ telescopes:

$$QH_c([w]) = -\sum_{s=0}^{r-1}\big(\mathrm{Tr}(w_{s+1}) - \mathrm{Tr}(w_s)\big) = \mathrm{Tr}(w_0) - \mathrm{Tr}(w_r).$$

Since $w_0 = c([w])$ and $w_r = x^a y^b$, this is the displayed identity. A deterministic sorting rule chooses one path for each chosen cut, so the formula extends linearly to the selected no-ψ necklace span.

If c' is a cyclic rotation of c, the two telescoping formulas have the same boundary because $\mathrm{Tr}(c([w])) = \mathrm{Tr}(c'([w]))$. Therefore $H_{c'}([w]) - H_c([w])$ is Q-closed. The one-ψ homology computation of Proposition 3.197 identifies its class with a scalar multiple of $\Psi_{a-1,b-1}$ determined by the change of cut; for example $[yxx] = [xxy]$, but the cuts yxx and xxy give homotopies differing by $-2[\psi x]$. This proves the announced non-canonicity and forbids interpreting H_c as a cyclic contraction. □

COROLLARY 3.202 *(Matrix evaluation is commutative in cohomology).* If two words in ϕ_1, ϕ_2 have the same numbers of ϕ_1 and ϕ_2, then their traces define the same class in the reduced BV cohomology. Consequently $\partial_{\mathrm{bb},N}(f)$ is determined by the commutative polynomial $f \in \mathbb{C}[z_1, z_2]$ and is independent of the chosen ordered lift.

Proof. Apply Theorem 3.201 to each word. Both traces are Q-homologous to the same normal-ordered trace $\mathrm{Tr}(\phi_1^a \phi_2^b)$, where a, b are their common numbers of letters. Hence they define the same reduced BV cohomology class. □

Definition 3.203 *(Unital Hamiltonian cyclic Koszul Lie model).* Let

$$E_{\mathrm{K}} = \mathbb{C}x \oplus \mathbb{C}y \oplus \mathbb{C}\psi, \qquad |x| = |y| = 0, \quad |\psi| = -1,$$

and let

$$\mathfrak{p}_{\mathrm{K}} = \mathrm{Cyc}\big(T(E_{\mathrm{K}})\big) = T(E_{\mathrm{K}})/[T(E_{\mathrm{K}}), T(E_{\mathrm{K}})]_{\mathrm{gr}}$$

be the graded cyclic words, including the scalar word $[1]$. Let $\bar{\mathfrak{p}}_{\mathrm{K}} := \mathfrak{p}_{\mathrm{K}}/\mathbb{C}[1]$ denote its scalar-reduced quotient. The differential is induced by

$$Qx = Qy = 0, \qquad Q\psi = xy - yx.$$

With cyclic derivatives ∂_x, ∂_y, define the constant necklace bracket by

$$\{F, G\}_{\mathrm{neck}} = \big[\partial_x F\, \partial_y G - \partial_y F\, \partial_x G\big]_{\mathrm{cyc}},$$

with the standard Koszul sign from cyclic rotation. Since x, y are even, the displayed formula has no additional signs on the ordinary Hamiltonian words.

THEOREM 3.204 *(Unital cyclic Koszul P_0 model).* The complex $(\mathfrak{p}_{\mathrm{K}}, Q, \{-,-\}_{\mathrm{neck}})$ is a dg Lie algebra. For every interval $I \subset \mathbb{R}$,

$$\mathfrak{p}_{\mathrm{K}}(I) := \Omega_c^\bullet(I) \otimes \mathfrak{p}_{\mathrm{K}}$$

with differential $d_{\mathbb{R}} + Q$ and bracket

$$[\alpha \otimes F, \beta \otimes G] = (-1)^{|F||\beta|}(\alpha \wedge \beta) \otimes \{F, G\}_{\mathrm{neck}}$$

is a compactly supported current dg Lie algebra. The graded-commutative algebra

$$\mathrm{Obs}^{\mathrm{cyc}}_{\mathrm{open}}(I) := \mathbf{S}\big(\mathfrak{p}_{\mathrm{K}}(I)\big)$$

with the biderivation extending this bracket is a P_0 factorization algebra on the brane line, with structure maps given by extension by zero and multiplication on disjoint intervals.

No cotangent CE/PV coordinate theorem for the full graded complex $\mathfrak{p}_{\mathrm{K}}(I)$ is asserted here. The CE/PV comparison used later is the separate degree-zero Hamiltonian identity of Theorem 2.44. Extending it to all ψ-words would require a graded coordinate theorem with the full signs and degree shifts supplied as data.

Proof. The necklace bracket is the standard constant-symplectic necklace bracket for the even pair x, y. Its graded antisymmetry and Jacobi identity are the usual cancellation of the cyclic sum of commutators in $T(E_{\mathrm{K}})$. The scalar word must be present: for example $\{[x], [y]\}_{\mathrm{neck}} = [1]$. The derivation Q replaces only uncontracted occurrences of ψ by $xy - yx$ and fixes the paired variables x, y. Expanding the bracket as the signed sum over cut pairs ($x \in F$, $y \in G$) and ($y \in F$, $x \in G$), Q never hits a contracted letter and carries the same super-cyclic cut sign as applying Q before bracketing; the terms where Q hits G acquire the usual factor $(-1)^{|F|}$. Hence

$$Q\{F, G\}_{\mathrm{neck}} = \{QF, G\}_{\mathrm{neck}} + (-1)^{|F|}\{F, QG\}_{\mathrm{neck}}.$$

Finally $Q^2 = 0$ is immediate from $Qx = Qy = 0$.

Tensoring with $\Omega^\bullet_c(I)$ gives the displayed current dg Lie algebra by the standard tensor-product sign rule; $d_{\mathbb{R}}$ is a derivation of $\wedge$, and Q is a derivation of the necklace bracket. For disjoint intervals $I_1, \dots, I_n \subset J$, extension by zero sends each current into J, and the factorization product is the product in the symmetric algebra. If two inputs have disjoint support, their current bracket has zero de Rham product, so the P_0-bracket is local for the factorization structure.

The final assertion is only the stated boundary on the claim: no all-degree cotangent-coordinate comparison is used in this theorem. □

Proposition 3.205 *(Normal-ordering quotient of the unital cyclic Koszul model).* Define

$$N : \mathfrak{p}_{\mathrm{K}} \longrightarrow \mathbb{C}[z_1, z_2]$$

on a cyclic word by

$$N([w]) = \begin{cases} z_1^{\#_x(w)} z_2^{\#_y(w)}, & w \text{ contains no } \psi, \\ 0, & w \text{ contains at least one } \psi. \end{cases}$$

Then N is a dg Lie morphism from $(\mathfrak{p}_{\mathrm{K}}, Q, \{-,-\}_{\mathrm{neck}})$ to the unreduced polynomial Hamiltonian Lie algebra $\mathbb{C}[z_1, z_2]$ with zero differential and Poisson bracket $\partial_{z_1} \wedge \partial_{z_2}$. Composing with $\mathbb{C}[z_1, z_2] \to \bar{A}$ gives the reduced Hamiltonian quotient $N_{\mathrm{red}} : \bar{\mathfrak{p}}_{\mathrm{K}} \to \bar{A}$. The section $i(z_1^a z_2^b) = [x^a y^b]$ and the cut-dependent homotopy H_c of Theorem 3.201 satisfy $QH_c = \mathrm{id} - iN$ on the chosen no-ψ necklace sector. No contraction of the full cyclic complex is asserted.

Proof. If a word contains ψ, every term of its necklace bracket with any word still contains ψ, because the bracket only differentiates in x and y. Hence both sides of the Lie identity vanish after N whenever one input contains ψ. For ordinary words F, G, cyclic derivative counting gives

$$N(\partial_x F) = \partial_{z_1} N(F), \qquad N(\partial_y F) = \partial_{z_2} N(F),$$

and similarly for G. Therefore

$$N\{F, G\}_{\text{neck}} = \partial_{z_1} N(F)\, \partial_{z_2} N(G) - \partial_{z_2} N(F)\, \partial_{z_1} N(G) = \{N(F), N(G)\}.$$

The chain-map property follows generatorwise and then by the derivation rule. If a word contains one ψ, each Q-summand replaces that ψ by $xy - yx$, and the two resulting ordinary commutative monomials agree after N. If a word contains two or more ψ's, every Q-summand still contains a ψ, hence maps to zero. On ordinary words $Q = 0$. Thus $NQ = 0$. The final displayed straightening identity is precisely Theorem 3.201 with the chosen cuts. □

LEMMA 3.206 *(No contraction onto the Hamiltonian quotient).* There is no degree -1 operator K on

$$\mathfrak{p}_{\mathrm{K}} = \mathrm{Cyc}\big(T(x, y, \psi)\big)$$

satisfying

$$QK + KQ = \mathrm{id} - iN$$

if N kills every ψ-containing word.

Proof. For each nonconstant bidegree (k, l), Proposition 3.197 gives a closed one-ψ primitive $\Psi_{k,l}$ whose class is nonzero. Since $N(\Psi_{k,l}) = 0$, the displayed identity would imply

$$\Psi_{k,l} = QK(\Psi_{k,l}) + KQ(\Psi_{k,l}) = QK(\Psi_{k,l}),$$

contradicting $[\Psi_{k,l}] \neq 0$. Hence any unreduced normal-ordering transfer must either retain the descendant module and higher coherence homotopies, or restrict to the cut-dependent no-ψ sector above. □

LEMMA 3.207 *(Gauge-paired necklace algebra does not have the naive Hamiltonian quotient).* Enlarge E_{K} to $E_{\mathrm{BV}} = \mathbb{C}x \oplus \mathbb{C}y \oplus \mathbb{C}a \oplus \mathbb{C}\psi$ with $|a| = 1$, $|\psi| = -1$, and add the constant pairing $\langle a, \psi \rangle = 1$ to the necklace bracket. The map that sends an x, y-word to its commutative monomial and kills every word containing a or ψ is not a Lie morphism.

Proof. With left cyclic derivatives and the displayed pairing convention,

$$\{[a], [\psi x]\}_{\text{neck}} = [x]$$

exactly. The naive projection sends the left-hand bracket to z_1, while it sends both $[a]$ and $[\psi x]$ individually to zero, so the bracket of their projected images is zero. Thus the full gauge-paired cyclic BV algebra requires an additional quotient, homotopy transfer, or explicit centrality data before it can map to the Hamiltonian formal-disk quotient. □

Theorem 3.208 *(Gauge-BRST cyclic algebra and the normal-ordering obstruction).* Let

$$E_{\mathrm{BRST}} = \mathbb{C}x \oplus \mathbb{C}y \oplus \mathbb{C}\gamma \oplus \mathbb{C}\psi, \qquad |x| = |y| = 0, \quad |\gamma| = 1, \quad |\psi| = -1,$$

and put

$$\mathfrak{p}_{\mathrm{BRST}} = \mathrm{Cyc}\big(T(E_{\mathrm{BRST}})\big).$$

Equip E_{BRST} with the even constant pairing

$$\langle x, y\rangle = 1, \qquad \langle y, x\rangle = -1, \qquad \langle \gamma, \psi\rangle = \langle \psi, \gamma\rangle = 1,$$

and the corresponding graded necklace bracket. The cyclic Hamiltonian

$$S = [\gamma(xy - yx)] + [\psi\gamma\gamma]$$

has degree 1 and satisfies $\{S, S\}_{\mathrm{neck}} = 0$. Hence $Q = \{S, -\}_{\mathrm{neck}}$ makes $\mathfrak{p}_{\mathrm{BRST}}$ a dg Lie algebra, with

$$Q\gamma = \gamma^2, \quad Qx = \gamma x - x\gamma, \quad Qy = \gamma y - y\gamma, \quad Q\psi = xy - yx + \gamma\psi + \psi\gamma.$$

The normal-ordering map

$$N_{\mathrm{BRST}} \colon \mathfrak{p}_{\mathrm{BRST}} \longrightarrow \mathbb{C}[z_1, z_2], \qquad [w] \longmapsto \begin{cases} z_1^{\#_x(w)} z_2^{\#_y(w)}, & w \text{ contains only } x, y, \\ 0, & w \text{ contains } \gamma \text{ or } \psi, \end{cases}$$

is a chain map, but it is not a Lie morphism and it is not part of a contraction $QK + KQ = \mathrm{id} - iN_{\mathrm{BRST}}$ on the full cyclic complex.

Proof. The constant-pairing necklace bracket has graded Jacobi by the standard two-cut cancellation in cyclic words. The displayed Hamiltonian gives the four formulas for Q by direct cyclic differentiation. A direct check of the generators gives $Q^2 = 0$: $Q\gamma^2 = 0$, $Q^2x = Q(\gamma x - x\gamma) = 0$, $Q^2y = 0$, and, for $\mu = xy - yx$, $Q\mu = \gamma\mu - \mu\gamma$, whence

$$Q^2\psi = Q\mu + Q(\gamma\psi) + Q(\psi\gamma) = (\gamma\mu - \mu\gamma) + (-\gamma\mu - \gamma\psi\gamma) + (\mu\gamma + \gamma\psi\gamma) = 0.$$

Therefore $Q = \{S, -\}$ is a square-zero derivation of the necklace bracket.

The chain-map statement for N_{BRST} follows as in Proposition 3.205: ordinary words are Q-closed after projection; a single ψ contributes $xy - yx$, whose two normal orderings agree, while the $\gamma\psi$ and $\psi\gamma$ terms are killed; every summand containing γ is killed. It is not a Lie morphism because the gauge-paired bracket gives $\{[\gamma], [\psi x]\}_{\mathrm{neck}} = [x]$, whose image is z_1, whereas both inputs have zero image. Finally $[\psi]$ is closed:

$$Q[\psi] = [xy - yx] + [\gamma\psi + \psi\gamma] = 0,$$

since $[xy] = [yx]$ and $[\psi\gamma] = -[\gamma\psi]$ in the cyclic quotient. No Q-image can equal $[\psi]$. With respect to cyclic word length, Q raises length by one: Qx, Qy, $Q\gamma$, and $Q\psi$ are quadratic words. Hence the length-one class $[\psi]$ cannot be a boundary. Equivalently, the derivation rule never produces a lone ψ-letter. Thus a full contraction would force $[\psi] = QK[\psi]$, contradiction. The extended BRST cyclic algebra is the correct unreduced primitive object; the Hamiltonian normal-ordering sector is only a quotient/chain-map sector and still requires separate centrality and factorization-center data. □

PROPOSITION 3.209 *(Stalk central multiplication).* Let

$$A_N^{\mathrm{pt}} = (\mathcal{R}_N^{GL_N}, Q)$$

be the invariant reduced open Koszul algebra at a brane insertion point. For $f \in \mathbb{C}[z_1, z_2]$, set

$$B_f = \widetilde{\partial}_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l} \mathrm{Tr}(\phi_1^k \phi_2^l).$$

Then $QB_f = 0$. Regard B_f as a Hochschild 0-cochain; its associated central operation is multiplication

$$L_{B_f} : A_N^{\mathrm{pt}} \to A_N^{\mathrm{pt}}, \qquad a \longmapsto B_f a.$$

This Hochschild 0-cochain is closed. The same statement holds for multiplication by a closed one-antifield class $\Psi_{k,l}$. After projection to the Hamiltonian quotient and stabilization in N, the degree-zero classes satisfy

$$\{B_f, B_g\}_{\mathrm{open}} = B_{\{f,g\}_{\bar{A}}}.$$

Proof. Since $Q\phi_1 = Q\phi_2 = 0$, every B_f is Q-closed. The algebra A_N^{pt} is graded-commutative, so the Hochschild differential of a 0-cochain b is

$$d_{\mathrm{Hoch}} b(a) = Qb\, a + ba - (-1)^{|b||a|} ab.$$

This vanishes for every closed graded-central element b. The classes $\Psi_{k,l}$ are closed by Lemma 3.196 and are central for the same graded-commutative reason. This is only a statement about multiplication by a closed polynomial descendant in the stalk algebra: $\Psi_{k,l}$ is not the principal-part cotangent boundary field $\rho_{k,l}$, and it does not realize the continuous Matlis coadjoint action of Theorem 3.230. Ordered-lift independence for B_f is cohomological and is exactly Corollary 3.202. The Poisson identity follows from Theorem 3.242 (below). □

Let $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The removed constant corresponds on the gauge side to the scalar $\mathrm{Tr}(1) = N$ and has zero Hamiltonian vector field.

PROPOSITION 3.210 *(Stable brane Hamiltonian algebra and connected quotient).* Put

$$V_H = \bar{A} \oplus \bar{A}_{\mathrm{desc}}[1].$$

In the stable large-N open BV model, the Hamiltonian descendant algebra generated by the nonempty primitive ghost-zero traces and the nonconstant primitive one-ψ Koszul homology is

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane},H,\mathrm{st}} \cong \mathbf{S}(V_H).$$

Its connected single-trace, or indecomposable, quotient is

$$Q\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane},H,\mathrm{st}} = \mathfrak{m}/\mathfrak{m}^2 \cong V_H.$$

The class of $z_1^k z_2^l$ with $k + l > 0$ corresponds to $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$. The corresponding descendant basis vector corresponds to the symmetrised trace with one ψ and k, l copies of ϕ_1, ϕ_2. All Lie comparisons on single traces use V_H after scalar reduction; CE/PV polyvector functions use the commutative algebra $\mathbf{S}(V_H)$.

Proof. Proposition 3.192 gives the free commutative algebra on cyclic words. Lemma 3.200 identifies any two cyclic words with the same numbers of ϕ_1 and ϕ_2, because adjacent transpositions generate all reorderings. Thus the connected ghost-zero cyclic words reduce to the monomials $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$ with $k + l > 0$. Proposition 3.197 computes the primitive one-ψ homology in each multidegree; the nonconstant generators are $\Psi_{k,l}$ with $k + l > 0$. Products of the connected ghost-zero primitive classes and these one-ψ primitive classes give the free graded-commutative multi-trace algebra $\mathbf{S}(V_H)$ in the Hamiltonian sector. Projection to indecomposables $\mathfrak{m}/\mathfrak{m}^2$ recovers the connected single-trace quotient V_H. The full primitive calculation proves that no stable primitive homology remains for ψ-degree $p \geq 2$; decomposable products of one-ψ classes remain only as products in the symmetric algebra. □

Definition 3.211 *(Three large-N operations).* Three different operations must not be conflated. They have different domains and are applied in the order displayed below.

(i) *Degreewise invariant stabilisation* $\mathrm{Prim}_{\mathrm{st}}$. At fixed word degree d, take $N \geq d$ so that Procesi–Razmyslov trace invariants have no extra finite-N trace identities. This is the operation used in Proposition 3.192.

(ii) *Connected single-trace extraction* $\mathrm{Conn}_{N\to\infty}$. After passing to the stable invariant algebra, project the symmetric algebra on cyclic-word generators to its primitive, or cumulant, summand. Products of two nonempty traces are decomposable and are killed by this projection.

(iii) *Hamiltonian scalar quotient* π_{Ham}. Remove the empty trace $\mathrm{Tr}(1) = N$ on the open side and the constant Hamiltonian 1 on the closed side. The removed line is not a primitive Hamiltonian generator; it is the scalar $U(1)$ line whose obstruction class is $[\bar{c}]$.

The reduced Hamiltonian sector used below is

$$\pi_{\mathrm{Ham}}\,\mathrm{Conn}_{N\to\infty}\,\mathrm{Prim}_{\mathrm{st}}\big(\mathcal{R}_N^{GL_N}, Q\big).$$

The formula is an ordered construction: first stabilize invariant theory degree by degree, then extract connected primitive trace classes, then quotient the scalar Hamiltonian line. It is not the unprojected $N \to \infty$ algebra and it is not invariant under permuting the three operations.

Remark 3.212. The word "sector" is literal. The calculation above presents the stable algebra generated by the primitive no-ψ Hamiltonians and the nonconstant primitive one-ψ homology generated by $\Psi_{k,l}$ for $k + l > 0$. The full invariant Koszul homology in higher ψ-count lies outside this sector, as does the decoupled scalar class $\mathrm{Tr}(\psi)$.

Corollary 3.213 *(Full primitive Koszul homology).* Let

$$A_\psi = \mathbb{C}\langle x, y, \psi\rangle, \qquad |x| = |y| = 0, \quad |\psi| \equiv 1 \pmod 2,$$

with differential $d\psi = xy - yx$ and $dx = dy = 0$. Put

$$\mathrm{Cyc}(A_\psi) = A_\psi/[A_\psi, A_\psi]_{\mathrm{gr}}, \qquad [uv] = (-1)^{|u||v|}[vu].$$

Give x, y, ψ weights

$$\mathrm{wt}(x) = (1, 0), \qquad \mathrm{wt}(y) = (0, 1), \qquad \mathrm{wt}(\psi) = (1, 1).$$

For fixed (R, S) define

$$K_{R,S}^p = \mathbb{C}\{[w]_{\mathrm{cyc}} : \#\psi(w) = p,\ \#x(w) = R - p,\ \#y(w) = S - p\},$$

with $0 \le p \le \min(R, S)$. The differential is

$$\partial[w] = \sum_{i:a_i=\psi} (-1)^{\#\{\psi \text{ before } i\}} [a_1 \cdots a_{i-1} x y\, a_{i+1} \cdots a_n \\ - a_1 \cdots a_{i-1} y x\, a_{i+1} \cdots a_n]_{\mathrm{cyc}}.$$

The primitive homology is

$$H_p(K_{R,S}) \cong \begin{cases} \mathbb{C} \cdot [x^R y^S], & p = 0, \\ \mathbb{C} \cdot [\Psi_{R-1,S-1}], & p = 1,\ R, S \ge 1, \\ 0, & p \ge 2. \end{cases}$$

Equivalently, the primitive stable Koszul homology is $\mathbb{C}[x, y] \oplus \mathbb{C}[x, y]\theta$ with $|\theta| = 1$ and $\mathrm{wt}(\theta) = (1, 1)$.

This is a universal stable primitive cyclic-word computation. The proof identifies the graded cyclic complex with the cellular chains of the chip-moving necklace state complex and uses the circle-sum map to compute the state complex as $H_\bullet(S^1; \mathbb{C})$; see Theorem 3.150 and Theorem D.12. The result does not identify the polynomial one-ψ span with Matlis principal parts or with the Moyal coadjoint module.

3.11 Operators in the closed Hamiltonian sector

The brane side has produced a stable algebra of trace classes generated by the primitive ghost-zero monomials $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$ and the primitive one-ψ descendants $\Psi_{k,l}$. The closed Hamiltonian sector must produce the same data as a Lie algebra of local modes whose Chevalley–Eilenberg cochains carry the residual BRST differential.

Let $\mathfrak{h} = \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ with the projected Poisson bracket, and let $\mathfrak{h}^\vee$ be its continuous Taylor dual. The cotangent field is a coadjoint module, so the local Hamiltonian Lie algebra is

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1].$$

For $f, g \in \mathfrak{h}$ and $\lambda, \mu \in \mathfrak{h}^\vee$, the brackets are

$$[f, g] = \{f, g\}, \qquad [f, \eta_\lambda] = \eta_{\mathrm{ad}_f^\vee \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0,$$

where $(\mathrm{ad}_f^\vee \lambda)(g) = -\lambda(\{f, g\})$. On polynomial monomials in the dense subalgebra $\bar{A}$ this reads

$$[z_1^k z_2^l, z_1^m z_2^n] = \overline{(kn - lm) z_1^{k+m-1} z_2^{l+n-1}},$$

where the bar means projection to $\bar{A}$; in particular the constant bracket of z_1 and z_2 is zero in the reduced Hamiltonian Lie algebra.

LEMMA 3.214 *(Shifted coadjoint signs).* Let η_λ denote the suspension of $\lambda \in \mathfrak{h}^\vee$ into $\mathfrak{h}^\vee[1]$. Since the Hamiltonians in $\mathfrak{h}$ are even,

$$[f, \eta_\lambda] = \eta_{\mathrm{ad}_f^\vee \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0.$$

Consequently the enveloping algebra relations involving two shifted coadjoint generators are odd-commutative:

$$\eta_\lambda \eta_\mu + \eta_\mu \eta_\lambda = 0.$$

Proof. The semidirect product bracket before shifting is the ordinary coadjoint action of the even Lie algebra $\mathfrak{h}$ on its continuous dual. Suspending the module raises the degree of every λ by one. The mixed-bracket sign is $+1$ because $|f| = 0$. The shifted module is abelian as a Lie ideal, so its bracket with itself is zero; in the associative enveloping algebra this zero Lie bracket gives $\eta_\lambda \eta_\mu - (-1)^{1\cdot 1}\eta_\mu \eta_\lambda = 0$. □

The kinematic BRST operator is

$$Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}.$$

The same jet contraction as in Lemma 3.191, together with the Dolbeault Poincare lemma in the formal local model, leaves precisely the holomorphic Taylor coefficients. Thus the linear local coordinates are the functionals

$$\partial^k_{z_1}\partial^l_{z_2}\alpha|_0, \qquad \partial^k_{z_1}\partial^l_{z_2}\beta|_0.$$

With the coefficient-dual convention fixed above, these are represented by $(k!l!)^{-1}\partial^k_{z_1}\partial^l_{z_2}|_0$. Since zero-form components of α and β have negative field ghost numbers, the corresponding linear functions have positive ghost numbers.

Proposition 3.215 *(Closed local operators are continuous CE cochains).* The classical local operators of the formal Hamiltonian cotangent BF sector are the continuous Chevalley-Eilenberg cochains

$$O^{\mathrm{cl}}_{\mathrm{grav}} \cong C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]),$$

with CE differential dual to the Lie brackets above.

Proof. Work at the formal insertion point. The jet algebra of the fields is the completed free graded-commutative algebra on the Taylor coefficients of α and β. The kinematic differential $Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ contracts all positive real and anti-holomorphic jets: in the real directions the homotopy is $[d_{\mathbb{R}^2}, \iota_{\partial_{x_i}}] = \partial_{x_i}$, and in the anti-holomorphic directions it is the formal Dolbeault Poincare homotopy. The Q_0-cohomology of the jet algebra is therefore generated by the holomorphic Taylor coefficients alone.

The remaining α-coefficients are Taylor Hamiltonians modulo constants, hence elements of $\mathfrak{h} = \widehat{\mathfrak{h}}$; degreewise polynomial coefficients form the dense subalgebra $\bar{A}$. The cotangent field β supplies the odd shifted continuous dual $\mathfrak{h}^\vee_{\mathrm{cont}}[1]$. Thus the linear space of residual local modes is $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$.

The nonlinear BRST differential is determined by the BV Hamiltonian

$$S = \int \left\langle \beta, D\alpha + \frac{1}{2}\{\alpha, \alpha\} \right\rangle.$$

The linear part is exactly Q_0, already killed in cohomology. The quadratic term sends two Hamiltonian α-modes f, g to their reduced Poisson bracket $\{f, g\}$, and the action on the cotangent variable is the coadjoint action displayed above. The quadratic term has zero component with two cotangent inputs, so $[\eta_\lambda, \eta_\mu] = 0$. Hence the residual BRST differential on functions is precisely the derivation dual to the Lie bracket on $\mathfrak{g}$, which is the Chevalley-Eilenberg differential. Continuity records the completion over polynomial degree. □

3.12 *Stratified CE/PV and admissible Koszul duality*

The closed local operators have been identified with continuous Chevalley–Eilenberg cochains of $\mathfrak{g}$. The brane Hamiltonian sector is a free graded-commutative algebra on $\bar{A}$ and $\bar{A}_{\mathrm{desc}}[1]$. To compare these as E_1-algebras one passes to the operation that exchanges commutative cochains and associative enveloping algebras: admissible Koszul duality. The universal calculation has staged input. The finite, coordinate,

and bracket-admissible CE/PV statements precede any kernel assertion; a kernel-admissible Maurer–Cartan tensor appears only when the diagonal converges. The enveloping/bar-cobar step is an admissible envelope criterion. The cochain layer is the completed cotangent CE/PV identity of Theorem 2.69, valid for the formal disk by finite coordinate tests. The enveloping layer is a bar-cobar criterion: it applies exactly when the chosen Tate or weighted coefficient category makes the CE coalgebra conilpotent and the filtered inverse limits exact. The strict product/direct-sum full-disk coefficient pair is therefore not discarded; it contributes the obstruction tuple whose vanishing or replacement is the input required for the enveloping upgrade.

Definition 3.216. For an augmented complete dg algebra B, its Koszul dual is

$$B^! = \mathrm{RHom}_B(\mathbb{C}, \mathbb{C})$$

[23]. We regard commutative Chevalley-Eilenberg cochains as E_1 algebras by restriction. All bar and cobar constructions below are taken in the category of complete filtered Tate vector spaces, with continuous maps. The completions are by symmetric or tensor length together with the coefficient topology coming from Taylor degree. For a Tate space V, the notation $V^\vee_{\mathrm{cont}}$ means the continuous dual with its dual Tate topology, $\widehat{\otimes}$ means the completed Tate tensor product, and

$$\widehat{\mathbf{S}}^n(V) = (V^{\widehat{\otimes} n})_{\mathbb{S}_n}.$$

Chevalley-Eilenberg chains below are direct sums in CE length with completed coefficients in each length; CE cochains, completed cobar algebras, and completed enveloping algebras are products in the corresponding length filtration.

LEMMA 3.217 *(Tate CE chains, completed PBW, and Koszul duality).* Work in the symmetric monoidal category **TateVec** of complete filtered Tate $\mathbb{C}$-vector spaces:

$$V = \varprojlim_{w\in\mathbb{N}} V_{\le w}, \qquad V_{\le w} \text{ finite-dimensional}, \qquad F^w V = \mathbf{Ker}\big(V \to V_{\le w-1}\big),$$

with $\widehat{\otimes}$ the completed tensor product $V\widehat{\otimes}W = \varprojlim_w (V_{\le w} \otimes W_{\le w})$ and $V^\vee_{\mathrm{cont}} = \varinjlim_w V^*_{\le w}$ the continuous dual. Admissible quasi-isomorphisms are the filtered ones: chain maps inducing a quasi-isomorphism on every quotient V/F^wV and on the associated graded.

Let $\mathfrak{g}$ be a Tate graded Lie algebra whose filtration is by closed dg Lie ideals, so that every $\mathfrak{g}_{\le w}$ used below is a finite-dimensional dg Lie quotient. Assume $\mathfrak{g}$ is lower-central Tate-pronilpotent in the sense that the descending Lie powers $\mathfrak{g}^{(\ge m)} = [\mathfrak{g}, \mathfrak{g}^{(\ge m-1)}]$ satisfy $\bigcap_m \big(\mathfrak{g}^{(\ge m)} + F^w\mathfrak{g}\big) = F^w\mathfrak{g}$ for every w. Then:

(i) For $V = \prod_d V_d$ with all V_d finite-dimensional and $W = \bigoplus_d W_d$ with all W_d finite-dimensional,

$$(V \oplus W)^\vee_{\mathrm{cont}} = \bigoplus_d V_d^\vee \oplus \prod_d W_d^\vee,$$

and the dualization $V \mapsto V^\vee_{\mathrm{cont}}$ is exact and continuous in the filtration F^w.

(ii) The continuous Chevalley–Eilenberg cochains

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \mathrm{Hom}_{\mathrm{cont}}\big(\widehat{\mathbf{S}}^c(\mathfrak{g}[1]), \mathbb{C}\big)$$

identify with $\widehat{\mathbf{S}}_{\mathrm{Tate}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big)$. The CE coalgebra $C^{\mathrm{CE}}_*(\mathfrak{g}) = \widehat{\mathbf{S}}^c(\mathfrak{g}[1])$ is conilpotent in CE length: every homogeneous CE chain has finite symmetric length and is killed by sufficiently high iterated reduced coproduct. Lower-central Tate-pronilpotence is a separate inverse-limit/bar-cobar convergence hypothesis, used below for the completed cobar and PBW comparison.

(iii) The continuous bar resolution of the augmentation $C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) \to \mathbb{C}$ has dual cobar algebra $\Omega C^{\mathrm{CE}}_{*}(\mathfrak{g})$, and the canonical Lie bar–cobar map

$$\Omega C^{\mathrm{CE}}_{*}(\mathfrak{g}) \longrightarrow \widehat{U}(\mathfrak{g})$$

is an admissible (filtered) quasi-isomorphism in **TateVec**.

(iv) The PBW filtration on $\widehat{U}(\mathfrak{g})$ by tensor length is complete and separated. Its associated graded is the completed symmetric algebra

$$\mathrm{gr}^{\bullet}_{F}\widehat{U}(\mathfrak{g}) \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{g}).$$

Proof. Item (1). A continuous functional on $V = \prod V_d$ depends on finitely many coordinates: the topology on $\mathbb{C}$ is discrete, so the kernel of any continuous map contains some $F^w V$. Therefore $V^{\vee}_{\mathrm{cont}} = \bigoplus_d V^*_d$. Conversely, a continuous functional on the discrete sum $W = \bigoplus W_d$ is given by an arbitrary product of degreewise functionals, hence $W^{\vee}_{\mathrm{cont}} = \prod_d W^*_d$. Direct sum of these gives the displayed formula. Exactness of dualization in **TateVec** is preservation of finite limits; continuity in the Tate filtration is direct.

Item (2). The completed CE coalgebra $\widehat{\mathbf{S}}^c(\mathfrak{g}[1]) = \bigoplus_n \widehat{\mathbf{S}}^n(\mathfrak{g}[1])$ has coaugmentation coideal $\widehat{\mathbf{S}}^{\geq 1}(\mathfrak{g}[1])$. An element in CE length n is killed by the $(n+1)$-fold reduced coproduct; this is the conilpotence used in the coalgebra. What is not automatic is the passage to completed cobar algebras and PBW enveloping algebras after the Tate inverse limit. That step uses the lower-central Tate-pronilpotence hypothesis: modulo every finite filtration step the descending Lie powers have empty intersection, so the completed bar-cobar spectral sequence is separated. The displayed equality identifies cochains as continuous functionals on the CE coalgebra, using item (1) windowwise.

Item (3). Filter the CE coalgebra by symmetric coalgebra length and by the supplied Tate quotient system. Project to a fixed bracket-compatible finite dg Lie quotient $\mathfrak{g}_{\leq w}$ and bound symmetric/tensor length to a fixed n. The resulting finite-dimensional diagram is the ordinary Lie bar–cobar comparison [24]: the bar resolution computes $\mathrm{RHom}_{C^{\bullet}_{\mathrm{CE}}(\mathfrak{g}_{\leq w}, \leq n)}(\mathbb{C}, \mathbb{C})$, and the cobar algebra maps to the universal enveloping algebra of the same finite Lie quotient. These finite comparisons are compatible with enlargement of the quotient and increase of the length cap because the transition maps are Lie-algebra quotient maps. Passing to the Tate limit and completing in tensor length gives the asserted filtered quasi-isomorphism in **TateVec**. The PBW filtration is, quotientwise, the standard PBW filtration on $U(\mathfrak{g}_{\leq w})$, hence preserved under inverse limits. Arbitrary Taylor-degree coordinate windows of the full formal Hamiltonian disk do not satisfy this hypothesis and are not used in this lemma.

Item (4). Tensor length is a separated, descending filtration on $\widehat{U}(\mathfrak{g})$ because the Lie relations $xy - (-1)^{|x||y|}yx - [x, y] = 0$ have leading tensor-length term $xy - (-1)^{|x||y|}yx$ of length 2 and lower-order term $-[x, y]$ of length 1; so the associated graded retains only the symmetric (length-fixed) part. Completeness is clear from the Tate construction. The associated graded is therefore the completed symmetric algebra $\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{g})$. □

Remark 3.218 *(Bar-cobar obstruction datum).* The lower-central Tate-pronilpotence hypothesis in item (2) is the exact bar-cobar criterion used here. It is satisfied by the nilpotent truncations and by the weighted admissible replacements used later. The full reduced formal Hamiltonian algebra with linear modes is already covered at cochain level by the pro coordinate-window/mixed-continuous CE/PV calculation; its enveloping/Koszul promotion is governed by the obstruction terms of Remark 3.27 and Theorem 2.69. The Beilinson–Feigin–Mazur explicit product/direct-sum Tate convention [22] is the special case $\mathfrak{g} = \left(\prod P_d\right) \oplus \left(\bigoplus Q_d\right)$ with the bracket bilinear in the two displayed pieces.

LEMMA 3.219 *(Linear Poisson Schouten differential).* Let $\mathfrak{h}$ be a pro-finite-dimensional Lie algebra in **TateVec**, and choose a mixed or weighted completion on which the linear Kirillov–Kostant–Souriau bracket is continuous. The bracket is $\{O_I, O_J\}_\partial = O_{[H_I,H_J]} = C_{IJ}^K\, O_K$ on a chosen homogeneous basis $\{H_I\}$ of $\mathfrak{h}$, where $O_I \in \mathfrak{h}^*$ is the linear coordinate. The associated Poisson bivector is the linear bivector

$$\pi_\partial = \tfrac{1}{2}\, C_{IJ}^K\, O_K\, \theta^I \wedge \theta^J, \qquad \theta^I = \partial/\partial O_I \in \mathrm{Der}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big).$$

Assume that this linear bivector and the Schouten contraction bracket are continuous on the chosen mixed or weighted completion. Then the Lichnerowicz/Schouten differential $d_\pi = [\pi_\partial, -]_S$ on

$$\mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big) = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\widehat{\otimes}\widehat{\Lambda}(\theta^I)$$

acts on generators by

$$d_\pi \theta^K = -\tfrac{1}{2}\, C_{IJ}^K\, \theta^I \theta^J, \qquad d_\pi O_K = C_{KJ}^L\, O_L\, \theta^J.$$

Proof. Direct calculation. The Schouten Leibniz rule with $[\theta^I, O_J]_S = \delta^I_J$, $[\theta^I, \theta^J]_S = 0$, $[O_I, O_J]_S = 0$ gives

$$\begin{aligned}
[\pi_\partial, O_K]_S &= \tfrac{1}{2} C_{IJ}^L\, O_L\, [\theta^I \theta^J, O_K]_S \\
&= \tfrac{1}{2} C_{IJ}^L\, O_L\, \big(\delta^I_K \theta^J - \delta^J_K \theta^I\big) \\
&= \tfrac{1}{2}\big(C_{KJ}^L - C_{JK}^L\big) O_L \theta^J = C_{KJ}^L O_L \theta^J,
\end{aligned}$$

using the antisymmetry $C_{IJ}^L = -C_{JI}^L$ in the lower indices. For $[\pi_\partial, \theta^K]_S$, the Schouten bracket with θ^K uses $[\theta^I \theta^J, \theta^K]_S = 0$ on the bivector factor and $[O_L, \theta^K]_S = -\delta^K_L$; the result is $-\tfrac{1}{2} C_{IJ}^K \theta^I \theta^J$. □

PROPOSITION 3.220 *(Admissible Tate Koszul-duality criterion).* Let $\mathfrak{g}_{\mathrm{adm}}$ be a Tate graded Lie algebra satisfying the lower-central Tate-pronilpotence hypothesis of Lemma 3.217; for the formal-local setting this means one of the nilpotent bracket-compatible finite-window replacements, the weighted admissible envelope where the hypothesis is explicitly imposed, or another stated admissible replacement. Put

$$B_{\mathrm{adm}} = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{adm,cont}}[-1]\big).$$

Then

$$B^!_{\mathrm{adm}} \simeq \widehat{U}(\mathfrak{g}_{\mathrm{adm}})$$

in the completed Tate category. For the strict full formal Hamiltonian algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$ with its linear and conformal-symplectic low-degree modes, the same conclusion is a criterion: it follows after supplying a coefficient replacement or relative model killing the obstruction tuple of Theorem 2.69.

Proof. This is exactly Lemma 3.217 applied to $\mathfrak{g}_{\mathrm{adm}}$. The lower-central Tate-pronilpotence condition is the load-bearing hypothesis: it ensures that the coefficient-completed CE coalgebra is conilpotent in every finite Tate window, that the continuous bar resolution is admissible, and that the completed Lie bar-cobar map

$$\Omega C_*^{\mathrm{CE}}(\mathfrak{g}_{\mathrm{adm}}) \longrightarrow \widehat{U}(\mathfrak{g}_{\mathrm{adm}})$$

is a filtered quasi-isomorphism.

For the full formal Hamiltonian algebra, the direct CE/PV generator calculation gives the cochain identity before any bar-cobar hypothesis is imposed. To turn that identity into an enveloping/Koszul-duality theorem one must supply an admissible replacement, relative semisimple model, or weighted coefficient category satisfying the displayed lower-central Tate-pronilpotence and convergence hypotheses. If this data is absent, the obstruction is exactly the Casimir, low-degree, semisimple, cone, and inverse-limit datum of Theorem 2.69. □

The completed PBW/order Rees enveloping algebra

$$U_\hbar(\mathfrak{g}) = \widehat{T}(\mathfrak{g})[[\hbar]] \Big/ \overline{\langle xy - (-1)^{|x||y|} yx - \hbar[x,y] \rangle},$$

with closure in tensor length and the Tate coefficient topology, makes the PBW special-fibre bookkeeping precise in every admissible replacement to which Proposition 3.220 applies. The Rees parameter $\hbar$ records word length in the enveloping algebra and is independent of Taylor degree. At $\hbar = 0$ the completed PBW special fibre is

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^\vee[1]\big).$$

After choosing the monomial-dual Taylor basis, the underlying label set of $\mathfrak{h}^\vee$ is identified with an abstract copy $\bar{A}_{\mathrm{desc}}$; no $\mathfrak{h}$-module structure is transported by this bookkeeping. The degreewise polynomial subalgebra $\mathbf{S}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1])$ is the stable classical brane Hamiltonian descendant algebra of Proposition 3.210: the degreewise polynomial part of the PBW associated graded gives the brane Hamiltonian descendant labels. For the full formal Hamiltonian algebra this is the PBW/Rees label level. A filtered enveloping/Koszul-duality theorem is obtained exactly when the admissible or relative replacement data of Proposition 3.220 are supplied. The interacting closed Hamiltonian bracket deforms the special fibre inside that admissible enveloping algebra; on the brane side the matching deformation at order one is the Hamiltonian trace bracket proved below.

3.12.1 First-order corrections, order hbar one

The PBW special-fibre label correspondence used in the reduced comparison reads

$$z_1^k z_2^l \leftrightarrow \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}.$$

The reduced Poisson bracket on the gravity side is:

$$\{z_1^k z_2^l,\, z_1^m z_2^n\}_{\bar{A}} = \overline{(kn - lm)\, z_1^{k+m-1} z_2^{l+n-1}}.$$

If an exponent in the displayed monomial is negative then the coefficient $kn - lm$ is already zero; if the displayed monomial is constant, its class in $\bar{A}$ is zero. The matching bracket on the open side is the canonical bracket induced by the first-order kinetic term $\int \mathrm{Tr}(\phi_1\, d\phi_2)$, with the sign convention $\iota_{X_f}\omega = -df$ fixed in the AKSZ remark above:

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j, \qquad \{(\phi_1)^i{}_j, (\phi_1)^k{}_l\} = 0 = \{(\phi_2)^i{}_j, (\phi_2)^k{}_l\}.$$

Lemma 3.221 *(Hamiltonian trace bracket).* The canonical matrix Poisson bracket descends to the invariant reduced BV cohomology and then to the Hamiltonian Lie quotient by constant trace classes.

Proof. For $\epsilon \in \mathfrak{gl}_N$, the component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[\phi_1, \phi_2])$ of the moment map generates the infinitesimal conjugation action, with the sign fixed by the convention for the bracket. Hence $\{F, \mu_\epsilon\} = 0$ for every invariant function F. If I_μ is the ideal generated by all μ_ϵ, then $\{F, \mu_\epsilon K\} = \mu_\epsilon\{F, K\}$ lies again in I_μ. Therefore the bracket of invariant functions descends to $(\mathcal{R}_N/I_\mu)^{GL_N}$, equivalently to the degree-zero reduced BV cohomology. The central line $\mathbb{C} \cdot \mathrm{Tr}(1)$ is central for this Lie bracket, so the Hamiltonian vector-space quotient by constants is compatible with the bracket. No associative quotient by the unit is taken. □

PROPOSITION 3.222 *(First-order trace bracket).* In the stable connected Hamiltonian trace cohomology,

$$\{\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}\} = \overline{(kn - lm)\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}.$$

Proof. Write $F_{k,l} = \mathrm{Tr}(\phi_1^k \phi_2^l)$. The matrix Poisson bracket induced by

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j$$

is

$$\{F, G\} = \mathrm{Tr}\left(\frac{\partial F}{\partial \phi_1}\frac{\partial G}{\partial \phi_2} - \frac{\partial F}{\partial \phi_2}\frac{\partial G}{\partial \phi_1}\right),$$

where the derivatives are cyclic matrix derivatives. Varying one occurrence at a time gives

$$\frac{\partial F_{k,l}}{\partial \phi_1} = \sum_{a=0}^{k-1} \phi_1^{k-1-a}\phi_2^l\phi_1^a,$$
$$\frac{\partial F_{k,l}}{\partial \phi_2} = \sum_{b=0}^{l-1} \phi_2^{l-1-b}\phi_1^k\phi_2^b.$$

Therefore the first summand in the bracket is a sum of kn traces, each obtained by removing one ϕ_1 from the first word and one ϕ_2 from the second word. The second summand is a sum of lm traces obtained by removing one ϕ_2 from the first word and one ϕ_1 from the second. Every trace appearing in either nonzero sum has multidegree $(k + m - 1, l + n - 1)$. Lemma 3.200 identifies all cyclic words with this multidegree in reduced Koszul cohomology. The coefficient is therefore $kn - lm$, and the common class is $\overline{\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}$. When the common word has multidegree $(0, 0)$, this class is the scalar trace and is zero in the Hamiltonian quotient by constants. □

PROPOSITION 3.223 *(Open action on primitive descendants).* Let

$$\widetilde{\Psi}_{a,b} = \binom{a+b}{a}^{-1}\Psi_{a,b}$$

be the averaged primitive one-ψ class. In the stable primitive one-ψ homology,

$$\{\overline{\mathrm{Tr}(\phi_1^p\phi_2^q)}, \widetilde{\Psi}_{a,b}\} = (pb - qa)\,\widetilde{\Psi}_{a+p-1,b+q-1},$$

with the convention that a term with a negative exponent is zero. Thus the unreduced polynomial descendant classes carry the adjoint polynomial Hamiltonian action. The Hamiltonian descendant sector is the quotient by the scalar line $\mathbb{C}\widetilde{\Psi}_{0,0}$, so the displayed action is projected further by killing the scalar target.

Proof. Put $x = \phi_1$ and $y = \phi_2$. The cyclic derivatives of $F_{p,q} = \mathrm{Tr}(x^p y^q)$ are

$$\frac{\partial F_{p,q}}{\partial x} = \sum_{i=0}^{p-1} x^{p-1-i} y^q x^i, \qquad \frac{\partial F_{p,q}}{\partial y} = \sum_{j=0}^{q-1} y^{q-1-j} x^p y^j.$$

For a one-ψ cyclic word $\mathrm{Tr}(\psi w)$ of bidegree (a, b), differentiating with respect to y gives one summand for each of the b occurrences of y in w, and differentiating with respect to x gives one summand for each of the a occurrences of x. Multiplying by the displayed cyclic derivatives of $F_{p,q}$ therefore produces pb positive one-ψ words and qa negative one-ψ words for each $w \in \mathsf{W}_{a,b}$, all of total bidegree $(a + p - 1, b + q - 1)$.

The bracket is a cycle because $F_{p,q}$ is invariant, hence $\{F_{p,q}, \mu_\epsilon\} = 0$ for every moment-map component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[x, y])$. Equivalently, the Hamiltonian vector field of $F_{p,q}$ preserves the Koszul differential $Q\psi = [x, y]$ and acts on Koszul homology. By Proposition 3.197, the target primitive one-ψ homology in that bidegree is one-dimensional. The functional $\epsilon(\mathrm{Tr}(\psi w)) = 1$ detects its coefficient. Since $\epsilon(\widetilde{\Psi}_{a,b}) = 1$, the averaged input gives

$$\epsilon\big(\{F_{p,q}, \widetilde{\Psi}_{a,b}\}\big) = pb - qa.$$

Also $\epsilon(\widetilde{\Psi}_{a+p-1,b+q-1}) = 1$. Hence the displayed formula follows. If the target bidegree is $(0, 0)$, the same formula is still the unreduced primitive one-ψ action; the Hamiltonian descendant quotient sends that scalar class to zero. □

Corollary 3.224 *(Descendant action and coadjoint action: structural separation).* Under the PBW special-fibre pairing of Corollary 3.199, the open Hamiltonian descendant action on the averaged polynomial $\widetilde{\Psi}_{a,b}$-basis is the symmetric PBW action

$$F_{p,q} : \widetilde{\Psi}_{a,b} \longmapsto (pb - qa)\, \widetilde{\Psi}_{a+p-1,b+q-1},$$

while the closed coadjoint action on the principal-part basis $\rho_{a,b} \leftrightarrow \delta_{a,b} \in \mathfrak{h}^\vee_{\mathrm{cont}}$ is the negative-transpose coadjoint action

$$\mathrm{ad}^\vee_{z_1^p z_2^q}\, \delta_{a,b} = -(pb - qa + p - q)\, \delta_{a-p+1,b-q+1},$$

with negative-index targets zero. The two actions agree only on the trivial Hamiltonian $p = q = 0$. The deformation quantisation of the cotangent factor sees the second action; the special-fibre PBW symmetric algebra sees the first. The structural reason is the PBW shadow versus coadjoint-module separation of Theorem 3.232 below: the $\widetilde{\Psi}_{a,b}$ and the $\rho_{a,b}$ have the same index set but different $\mathfrak{h}$-module structures.

Proof. Proposition 3.223 gives the symmetric PBW formula on $\widetilde{\Psi}_{a,b}$. The coadjoint action is dual to the Poisson bracket: for $f = z_1^p z_2^q$ and the Taylor-dual basis vector $\delta_{a,b}$,

$$(\mathrm{ad}^\vee_f\, \delta_{a,b})(z_1^m z_2^n) = -\delta_{a,b}\big((pn - qm) z_1^{p+m-1} z_2^{q+n-1}\big),$$

giving the displayed coadjoint formula. For example, z_1 acts on open descendants by $\widetilde{\Psi}_{a,b} \mapsto b\, \widetilde{\Psi}_{a,b-1}$, whereas it acts coadjointly by $\delta_{a,b} \mapsto -(b + 1)\, \delta_{a,b+1}$. The scalar edge $\{\, \mathrm{Tr}(\phi_1), \widetilde{\Psi}_{0,1} \,\} = \Psi_{0,0}$ is removed with the scalar descendant. The two formulas have opposite index shift directions and differ by $-(pb - qa + p - q)\rho_{a-p+1,b-q+1} - (pb - qa)\rho_{a+p-1,b+q-1}$ after any bidegree-only vector-space relabelling

between the two alphabets, which is non-zero unless both targets vanish identically; this happens exactly on the trivial $p = q = 0$ action. The PBW special-fibre product identification is Corollary 3.199; the coadjoint half is Theorem 3.230; the Laurent/distributional principal-part realization is Proposition 3.227 and the separation Theorem 3.232. □

Three notions of "local" on the brane line $\mathbb{R}$ coexist; the coadjoint identification on the principal-part side requires that they remain distinct.

LEMMA 3.225 *(Three notions of locality).* On the brane line $\mathbb{R}$ with formal holomorphic transverse coordinates z_1, z_2 at the brane point $p \in \mathbb{C}^2$, the following three notions of locality are independent.

(i) *Support locality*:distributional support on the brane: a section $T \otimes m$ of $\mathcal{D}'_c(I) \widehat{\otimes} M$ has support locality whenever $\operatorname{supp} T \subset I \subset \mathbb{R}$.

(ii) *Topological locality*:locally constant in the $\mathbb{R}$-direction up to Q-exact translations: every section is identified with its translate by a parameter on $\mathbb{R}$ modulo the open BV differential Q_{open}.

(iii) *Formal holomorphic locality*:polynomial, or Taylor-series, dependence on the transverse coordinates z_1, z_2.

The polynomial trace observables $O_f(a) = \int_{\mathbb{R}} a(t)\, \overline{\operatorname{Tr} f(\phi_1(t), \phi_2(t))}\, dt$ have all three. The shifted cotangent currents $\Theta_\rho(B)$ of Construction 3.236 have (1) and (2) but not (3): they are momentum modes attached to the brane and live in the $\mathbb{C}$-linear residue annihilator of scalar Hamiltonians inside the top local-cohomology principal-part module, $\operatorname{Ann}_{\text{res}}(\mathbb{C} \cdot 1)$.

Proof. Independence is a category-theoretic statement: a polynomial in z_1, z_2 with point support is constant in $\mathbb{R}$, so (1) does not imply (3); a delta-current at one point of $\mathbb{R}$ has compact support but only trivial Q-exact translation, so (1) does not imply (2). The cotangent currents $\Theta_\rho(B)$ are smooth or distributional sections of the line $\mathbb{R}$ (hence (1)) and translation-invariant up to Q_{open}-exact cocycles (hence (2)); their transverse content $\rho \in \mathcal{P}$ is a top-local-cohomology principal part, hence a residue-dual distributional mode rather than a Taylor-polynomial mode (hence not (3)). □

Remark 3.226. In canonical quantization of a constrained system, the configuration variables are locally polynomial in spatial coordinates and the conjugate momenta are dual distributions. This is exactly the separation Lemma 3.225 records: configuration observables on a brane have formal holomorphic locality (3), and momentum observables have only (1) and (2). The principal-part sector is the algebraic image of this separation in the local Hamiltonian BF model.

PROPOSITION 3.227 *(Principal-part realization of the coadjoint module).* Let

$$\mathcal{P} = \bigoplus_{\substack{a,b \geq 0 \\ a+b>0}} \mathbb{C}\, \rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

with residue pairing

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \operatorname{Res}_{z_1=z_2=0} z_1^{m-a-1} z_2^{n-b-1}\, dz_1 dz_2 = \delta_{a,m}\delta_{b,n}.$$

Let π_{pp} denote projection of Laurent two-forms to the span of the principal parts $\rho_{a,b}$. The action

$$f \cdot \rho = \pi_{\text{pp}}(\{f, \rho\})$$

identifies $\mathcal{P}$ with the continuous coadjoint module $\mathfrak{h}^\vee_{\mathrm{cont}}$. On monomials,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with the convention that a term with a negative index is zero.

Proof. The residue pairing identifies $\rho_{a,b}$ with the Taylor-dual functional $\delta_{a,b}$. Hamiltonian vector fields preserve the holomorphic volume form $dz_1 dz_2$, so integration by parts for the residue pairing gives

$$\langle \{f, \rho\}, g\rangle = -\langle \rho, \{f, g\}\rangle.$$

Hence the Laurent principal-part action realizes the coadjoint action. Directly, for $f = z_1^p z_2^q$ and $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$,

$$\begin{aligned}\{f, z_1^{-a-1} z_2^{-b-1}\} &= p z_1^{p-1} z_2^q\,(-(b+1)) z_1^{-a-1} z_2^{-b-2} \\ &\quad - q z_1^p z_2^{q-1}\,(-(a+1)) z_1^{-a-2} z_2^{-b-1} \\ &= -(pb - qa + p - q) z_1^{p-a-2} z_2^{q-b-2}.\end{aligned}$$

Since $z_1^{p-a-2} z_2^{q-b-2}\, dz_1 dz_2 = \rho_{a-p+1,b-q+1}$ when both indices are nonnegative, principal-part projection gives the displayed formula. □

Two rigidity statements pin the principal-part identification after the residue scalar is fixed: among continuous total-degree-zero pairings the residue pairing is unique up to a single nonzero scalar, and among continuous total-degree-zero $\mathfrak{h}$-module isomorphisms the coadjoint identification $\mathcal{P} \cong \mathfrak{h}^\vee_{\mathrm{cont}}$ is unique modulo that scalar.

Theorem 3.228 *(Residue pairing, normalized up to scalar).* Up to a nonzero scalar, the residue pairing

$$\langle\,\cdot\,,\,\cdot\,\rangle : \mathcal{P} \times (\mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1) \longrightarrow \mathbb{C}, \qquad \langle \rho_{a,b}, [z_1^m z_2^n]\rangle = \delta_{a,m}\delta_{b,n},$$

is the unique non-degenerate continuous $\mathfrak{h}$-equivariant pairing graded of total degree zero with the property that every element of $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$ annihilates the constant line.

Proof. Existence. The residue pairing is non-degenerate by the displayed Kronecker relation; it is continuous because $\mathcal{P}$ is graded discrete and $\mathbb{C}[[z_1, z_2]]$ carries the Taylor filtration; equivariance is integration by parts for the Poisson bracket, since Hamiltonian vector fields preserve the holomorphic two-form $dz_1 dz_2$:

$$\langle \{f, \rho\}, g\rangle = \mathrm{Res}(\{f, \rho\} g) = -\,\mathrm{Res}(\rho\,\{f, g\}) = -\langle \rho, \{f, g\}\rangle.$$

Uniqueness. Let $\beta : \mathcal{P} \times (\mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1) \to \mathbb{C}$ be a second pairing satisfying the same conditions, and write $\beta_{a,b;m,n} = \beta(\rho_{a,b}, z_1^m z_2^n)$. Since β is graded of total degree zero, $\beta_{a,b;m,n} = 0$ unless $a + b = m + n$.

Diagonal weight constraint. The Hamiltonian $z_1 z_2$ acts diagonally on monomials by $\{z_1 z_2, z_1^m z_2^n\} = (n - m) z_1^m z_2^n$. Direct Poisson computation on principal parts uses the Hamiltonian vector field $X_{z_1 z_2} = z_2\,\partial_{z_2} - z_1\,\partial_{z_1}$ (with $X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}$), giving $X_{z_1 z_2}(z_1^{-a-1} z_2^{-b-1}) = (a - b)\, z_1^{-a-1} z_2^{-b-1}$, so $\{z_1 z_2, \rho_{a,b}\} = (a - b)\rho_{a,b} = -(b - a)\rho_{a,b}$. Equivariance $\beta(\{z_1 z_2, \rho_{a,b}\}, z_1^m z_2^n) = -\beta(\rho_{a,b}, \{z_1 z_2, z_1^m z_2^n\})$ gives $-(b - a)\beta_{a,b;m,n} = -(n - m)\beta_{a,b;m,n}$, hence $\beta_{a,b;m,n} = 0$ unless $a - b = m - n$. Combined with the weight constraint $a + b = m + n$, this

gives $\beta_{a,b;m,n} = 0$ unless $(a,b) = (m,n)$. So β is diagonal in the residue basis: $\beta_{a,b;m,n} = c_{a,b}\,\delta_{a,m}\delta_{b,n}$ for some scalars $c_{a,b} \in \mathbb{C}$.

Coefficients are constant. The Hamiltonian generator $f = z_1$ has $X_{z_1} = \partial_{z_2}$, so $\{z_1, z_1^m z_2^n\} = n\, z_1^m z_2^{n-1}$ and a direct Poisson computation gives $\{z_1, \rho_{a,b}\} = -(b+1)\rho_{a,b+1}$. Equivariance reads $\beta(\{z_1, \rho_{a,b}\}, z_1^m z_2^n) = -\beta(\rho_{a,b}, \{z_1, z_1^m z_2^n\})$, i.e., $-(b+1)\beta_{a,b+1;m,n} = -n\,\beta_{a,b;m,n-1}$. Both sides vanish identically unless the diagonality condition is met on each pair. The right-hand side is non-zero only if $(a,b) = (m, n-1)$, i.e., $m = a, n = b+1$. The left-hand side is non-zero only if $(a, b+1) = (m,n)$, i.e., $m = a, n = b+1$. Both indices coincide, and we obtain $(b+1)\,c_{a,b+1} = (b+1)\,c_{a,b}$, hence $c_{a,b+1} = c_{a,b}$ when both coefficients occur in the reduced index set; that is, for $a + b > 0$. The case $a = b = 0$ is not a coefficient of $\mathcal{P}$. Symmetrically from $f = z_2$ (using $X_{z_2} = -\partial_{z_1}$, so $\{z_2, z_1^m z_2^n\} = -m\, z_1^{m-1} z_2^n$ and $\{z_2, \rho_{a,b}\} = (a+1)\rho_{a+1,b}$) we get $c_{a+1,b} = c_{a,b}$ when both coefficients occur in the reduced index set. Connecting these chains through $c_{1,1}$, all $c_{a,b}$ for $a + b \geq 1$ equal a single constant $c \in \mathbb{C}$. Non-degeneracy forces $c \neq 0$.

Therefore $\beta = c\,\langle\cdot,\cdot\rangle$ is the residue pairing scaled by c. The condition on constants holds because $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$; equivalently $\rho_{0,0} \notin \mathcal{P}$. □

Remark 3.229. Theorem 3.228 is better understood as Matlis-local-cohomology rigidity, not as an abstract Schur lemma. Let $R = \mathbb{C}[[z_1, z_2]]$ and $\mathfrak{m} = (z_1, z_2)$. The top local cohomology module, in the Čech-Laurent residue model of Grothendieck–Hartshorne local cohomology and Kunz–Cox–Dickenstein power-series residues [20, 21],

$$H^2_{\mathfrak{m}}(R)\, dz_1 dz_2 = \bigoplus_{a,b\geq 0} \mathbb{C}\, z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$$

is, after the trivialization $\omega_R = R\, dz_1 dz_2$, the injective hull $E_R(\mathbb{C})$ supplied by Matlis duality [19]. The reduced principal-part module $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$ is not an R-submodule and is not the literal R-module Matlis dual of $R/\mathbb{C}\cdot 1$; the scalar line is not an R-submodule. It is the $\mathbb{C}$-linear annihilator of the scalar Hamiltonian under the Matlis residue evaluation pairing. Hamiltonian vector fields preserve $dz_1 dz_2$, so this restricted pairing transports their action to the coadjoint action. The explicit coefficient proof above is the coordinate form of this local-duality statement.

THEOREM 3.230 *(Principal-part coadjoint isomorphism, Matlis form).* After fixing the residue scalar in Theorem 3.228, the residue pairing induces an isomorphism of $\mathfrak{h}$-modules

$$\Phi_{\mathrm{coad}} : \mathcal{P} \xrightarrow{\sim} \mathfrak{h}^\vee_{\mathrm{cont}}, \qquad \rho_{a,b} \longmapsto \delta_{a,b},$$

characterised uniquely (up to the residue scalar c of Theorem 3.228) by the property that for every $f \in \mathfrak{h}$ and every $g \in \mathfrak{h}$,

$$\langle \Phi^{-1}_{\mathrm{coad}}(\mathrm{ad}^\vee_f\, \delta), g\rangle = -\langle \Phi^{-1}_{\mathrm{coad}}(\delta), \{f, g\}\rangle.$$

In monomial form,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\, \rho_{a-p+1,b-q+1},$$

with the convention that any term with a negative index is zero. The isomorphism is continuous, of total degree zero, and the unique continuous total-degree-zero $\mathfrak{h}$-module isomorphism $\mathcal{P} \to \mathfrak{h}^\vee_{\mathrm{cont}}$ after the residue scalar is fixed; without that normalization it is unique modulo scalar.

Proof. Existence on monomials. Let $f = z_1^p z_2^q$ and $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2$. Since $\{z_1, z_2\} = 1$,

$$\{f, z_1^{-a-1} z_2^{-b-1}\} = p\, z_1^{p-1} z_2^q \cdot (-(b+1)) z_1^{-a-1} z_2^{-b-2} - q\, z_1^p z_2^{q-1} \cdot (-(a+1)) z_1^{-a-2} z_2^{-b-1},$$

which simplifies to $-(pb - qa + p - q)\, z_1^{p-a-2} z_2^{q-b-2}$. Multiplying by $dz_1 dz_2$ identifies this with $\rho_{a-p+1,b-q+1}$ when both indices are ≥ 0; otherwise the product lies outside $\mathcal{P}$ and the principal-part projection π_{pp} of Proposition 3.227 gives zero. The integration by parts formula $\langle \{f, \rho\}, g\rangle = -\langle \rho, \{f, g\}\rangle$ identifies this with the coadjoint action.

Continuity. The dual filtration on $\mathfrak{h}^\vee_{\mathrm{cont}}$ has $F^N \mathfrak{h}^\vee_{\mathrm{cont}}$ equal to the functionals vanishing on $\mathfrak{m}^{N+1} \subset \mathfrak{h}$, where $\mathfrak{m}$ is the maximal ideal. The grading of $\mathcal{P}$ by total weight $a + b$ corresponds under Φ_{coad} to the dual filtration: $F^N = \bigoplus_{a+b\leq N} \mathbb{C}\, \delta_{a,b}$ is the Φ_{coad}-image of $\bigoplus_{a+b\leq N} \mathbb{C}\, \rho_{a,b}$.

Verification on three test cases. With the convention $\{z_1, z_2\} = 1$, the Hamiltonian vector field is $X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}$.

(i) $(p, q, a, b) = (1, 0, 0, 1)$: the formula gives $z_1 \cdot \rho_{0,1} = -(1 \cdot 1 - 0 \cdot 0 + 1 - 0)\rho_{0-1+1,\,1-0+1} = -2\rho_{0,2}$. Direct: $X_{z_1} = \partial_{z_2}$, so $X_{z_1}(z_1^{-1} z_2^{-2}) = -2z_1^{-1} z_2^{-3}$, and multiplying by $dz_1 dz_2$ gives $-2\rho_{0,2}$. Match.

(ii) $(p, q, a, b) = (0, 1, 1, 0)$: the formula gives $z_2 \cdot \rho_{1,0} = -(0 \cdot 0 - 1 \cdot 1 + 0 - 1)\rho_{1-0+1,\,0-1+1} = 2\rho_{2,0}$. Direct: $X_{z_2} = -\partial_{z_1}$, so $X_{z_2}(z_1^{-2} z_2^{-1}) = 2z_1^{-3} z_2^{-1}$, multiplied by $dz_1 dz_2$ gives $2\rho_{2,0}$. Match.

(iii) $(p, q, a, b) = (2, 1, 1, 1)$: the formula gives $z_1^2 z_2 \cdot \rho_{1,1} = -(2 \cdot 1 - 1 \cdot 1 + 2 - 1)\rho_{0,1} = -2\rho_{0,1}$. Direct: $X_{z_1^2 z_2} = 2z_1 z_2\, \partial_{z_2} - z_1^2\, \partial_{z_1}$; $X_{z_1^2 z_2}(z_1^{-2} z_2^{-2}) = -4z_1^{-1} z_2^{-2} + 2z_1^{-1} z_2^{-2} = -2z_1^{-1} z_2^{-2}$, multiplied by $dz_1 dz_2$ gives $-2\rho_{0,1}$. Match.

These three cases are confirmed as part of the 1225-case exact check, which verifies the formula for $0 \leq p, q, a, b \leq 5$, $p + q > 0$, $a + b > 0$.

Uniqueness. Suppose $\Phi' : \mathcal{P} \to \mathfrak{h}^\vee_{\mathrm{cont}}$ is another continuous $\mathfrak{h}$-equivariant isomorphism of total degree zero. Define

$$\beta_{\Phi'}(\rho, g) = \Phi'(\rho)(g).$$

This is a non-degenerate continuous total-degree-zero pairing $\mathcal{P} \times \mathfrak{h} \to \mathbb{C}$, it kills the scalar line by definition of $\mathfrak{h}$, and $\mathfrak{h}$-equivariance of Φ' gives the coadjoint equivariance condition of Theorem 3.228. Therefore $\beta_{\Phi'} = c'\langle \cdot, \cdot \rangle$ for a unique $c' \in \mathbb{C}^\times$, and hence $\Phi' = c'\Phi_{\mathrm{coad}}$. The scalar is exactly the residue scalar of Theorem 3.228. □

Remark 3.231 *(Sign convention).* The sign $-(pb - qa + p - q)$ on the principal-part side and the sign $(pb - qa)$ on the averaged polynomial $\Psi_{a,b}$ side of Proposition 3.223 are both correct in their own conventions; they differ because the $\Psi_{a,b}$ are special-fibre PBW representatives and the $\rho_{a,b}$ are deformation-compatible coadjoint representatives. See Theorem 3.232 below for the precise separation.

Theorem 3.232 *(PBW shadow versus coadjoint deformation module).* Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ be the semidirect Lie algebra of Proposition 3.13. The polynomial one-ψ basis $\Psi_{a,b} \in \bar{A}_{\mathrm{desc}}$ and the principal-part basis $\rho_{a,b} \in \mathcal{P} \cong \mathfrak{h}^\vee_{\mathrm{cont}}$ are related as follows.

(i) *Special-fibre PBW basis.* The $\widetilde{\Psi}_{a,b}$ form the primitive one-ψ PBW label basis in the special-fibre symmetric algebra $\mathbf{S}(\mathfrak{h} \oplus \mathfrak{h}^\vee[1])\big|_{\hbar=0}$, and the open Hamiltonian Lie action is the *symmetric PBW action* on this generator span:

$$F_{p,q} : \widetilde{\Psi}_{a,b} \longmapsto (pb - qa)\, \widetilde{\Psi}_{a+p-1,b+q-1}.$$

The shift $(a + p - 1, b + q - 1)$ is determined by multiplication by the Hamiltonian monomial and by the scalar Hamiltonian quotient.

(ii) *Deformation-level coadjoint basis.* The $\rho_{a,b}$ form a basis of the deformation-compatible continuous coadjoint module $\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P}$ of Theorem 3.230, and the $\mathfrak{h}$-action is the *principal-part coadjoint action* on this basis:

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\, \rho_{a-p+1,b-q+1}.$$

The displayed shift is the negative-transpose residue action. Linear Hamiltonians raise pole order, while higher monomials have the mixed direction prescribed by the formula.

(iii) *No module interpolation in the present theorem.* The bidegree relabelling between the PBW special-fibre basis $\widetilde{\Psi}_{a,b}$ and the coadjoint basis $\rho_{a,b} \leftrightarrow \delta_{a,b}$ is only an isomorphism of graded vector spaces and label sets. It is *not* an isomorphism of $\mathfrak{h}$-modules: pulling the coadjoint $\mathfrak{h}$-action across this relabelling does not equal the symmetric PBW action displayed in (1), by exactly the difference $-(pb - qa + p - q)\rho_{a-p+1,b-q+1}$ versus $(pb - qa)\rho_{a+p-1,b+q-1}$. The formal Moyal/Weyl Rees deformation controls the Weyl algebra and its continuous dual $\mathfrak{h}^\vee_{\hbar,\mathrm{cont}}$; it does not identify the polynomial one-ψ descendant module with $\mathcal{P}$. The Fourier-polarized Rees label bridge is constructed in Theorem A.10; the same-action and unreduced quantum-observable realizations remain separate problems.

Thus the two formulas are not two bases of a single $\mathfrak{h}$-module in the formal-local category. They are the PBW special-fibre shadow and the continuous coadjoint cotangent module, respectively. The structural separation recorded in Corollary 3.224 follows.

Proof. (1) is Proposition 3.223, with the rephrasing in the PBW special-fibre basis. (2) is Theorem 3.230. For (3), the symmetric PBW action on the nonconstant $\widetilde{\Psi}_{a,b}$ labels is the PBW special-fibre action after the Hamiltonian scalar quotient. The coadjoint action on $\rho_{a,b}$ is the negative-transpose principal-part action. These two module behaviours are incompatible by the displayed formulas, already for the linear Hamiltonians z_1, z_2; hence any bidegree-only relabelling between $\widetilde{\Psi}_{a,b}$ and $\delta_{a,b}$ cannot be an $\mathfrak{h}$-module map. The Rees-deformation $A_\hbar = (\mathbb{C}[[z_1, z_2]][[\hbar]], *)$ has the property that the formal Moyal product $f * g = fg + \frac{\hbar}{2}\{f, g\} + O(\hbar^2)$ acts on its dual by the coadjoint formula at *all* orders of $\hbar$: the dual representation of the Moyal Lie algebra $(\mathfrak{h}_\hbar, [-,-]_{\mathfrak{h}_\hbar})$ on $\mathfrak{h}^\vee_\hbar$ is the residue pairing extended $\mathbb{C}[[\hbar]]$-linearly. Specialising to $\hbar = 0$ recovers the Poisson coadjoint action of the completed Hamiltonian algebra $\mathfrak{h}$ on $\mathfrak{h}^\vee = \mathcal{P}$, and its restriction to the dense polynomial subalgebra $\bar{A} \subset \mathfrak{h}$. The PBW filtration is the Rees filtration, and the special-fibre $\hbar = 0$ symmetric algebra is the associated graded; the symmetric PBW action on $\widetilde{\Psi}_{a,b}$ is exactly the $\hbar = 0$ *associated graded* image of the classical Hamiltonian vector-field action on the polynomial PBW descendant labels, not the adjoint or coadjoint action of the Lie algebra. It therefore cannot identify the descendant module with the continuous coadjoint module. This proves the asserted separation and identifies the Rees/Fourier bridge as additional Fourier-polarized structure rather than a same-action module identification. □

PROPOSITION 3.233 *(Polynomial realization obstruction for principal parts).* In the polynomial open BV/Koszul matrix model studied here, no family of ordinary polynomial boundary local cohomology classes

$$\Theta_{a,b}, \qquad a, b \geq 0, \quad a + b > 0,$$

can realize the principal-part coadjoint module of Proposition 3.227.

Proof. The linear Hamiltonians

$$O_{1,0} = \overline{\mathrm{Tr}(\phi_1)}, \qquad O_{0,1} = \overline{\mathrm{Tr}(\phi_2)}$$

act on polynomial boundary observables by the constant Hamiltonian vector fields ∂_{ϕ_2} and $-\partial_{\phi_1}$, respectively. On every ordinary polynomial observable these operators are locally nilpotent: after enough iterations the polynomial degree in the differentiated variable is exhausted.

On the principal-part coadjoint module, however, Proposition 3.227 gives

$$O_{1,0} \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad O_{0,1} \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Neither operator is locally nilpotent on any nonzero principal-part generator. Local nilpotence is preserved under isomorphism of modules. Therefore the desired $\Theta_{a,b}$ cannot be ordinary polynomial boundary observables in the current open BV/Koszul model. □

The categorical refinement records the same local-nilpotence obstruction at the level of polynomial sub-bimodules. It does not assert a classification of all $\bar{A}$-module morphisms out of $\mathcal{P}$.

Theorem 3.234 *(Categorical polynomial-realization obstruction for principal parts).* Let $\mathsf{P}_{\mathrm{pol}}$ denote the abelian category of $\mathbb{Z}$-graded bimodules $M = \bigoplus_{n\geq 0} M_n$ over the polynomial Hamiltonian Lie algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$, on which:

(i) every element acts as a Hamiltonian vector field;

(ii) every linear Hamiltonian acts locally nilpotently: for every $m \in M$ and every $i \in \{1,2\}$, there exists N with $(z_i\cdot)^N m = 0$.

This category models polynomial boundary observables of the open BV theory with respect to the polynomial sub-bimodule structure preserved by all morphisms of the boundary algebra into itself. Then $\mathcal{P} \notin \mathrm{Ob}(\mathsf{P}_{\mathrm{pol}})$. More generally, no injective $\bar{A}$-module map $\mathcal{P} \hookrightarrow M$ exists with $M \in \mathrm{Ob}(\mathsf{P}_{\mathrm{pol}})$. In particular $\mathcal{P}$ cannot be realized as a polynomial boundary submodule with the same Hamiltonian action.

Proof. By Theorem 3.230, $z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}$ and $z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}$ on every generator. Iteration gives $(z_1\cdot)^N \rho_{a,b} = (-1)^N (b+1)(b+2)\cdots(b+N)\,\rho_{a,b+N} \neq 0$ for every N, so z_1 is not locally nilpotent on any nonzero element of $\mathcal{P}$. The same argument applies to z_2.

Hence $\mathcal{P}$ itself violates condition (2) and is not an object of $\mathsf{P}_{\mathrm{pol}}$. If $\Phi : \mathcal{P} \hookrightarrow M$ were an injective $\bar{A}$-module map into an object of $\mathsf{P}_{\mathrm{pol}}$, then $m = \Phi(\rho_{a,b}) \neq 0$ for some generator. Equivariance gives

$$z_1^N \cdot m = \Phi(z_1^N \cdot \rho_{a,b}) = (-1)^N (b+1)\cdots(b+N)\Phi(\rho_{a,b+N}),$$

which is nonzero for every N by injectivity. This contradicts local nilpotence of z_1 on M. The same argument applies with z_2. □

Remark 3.235 *(Polynomial obstruction scope).* The categorical obstruction applies to every polynomial sub-bimodule of every boundary observable algebra with locally nilpotent z_i-action, not only to specific classes $\Theta_{a,b}$. The distributional/Laurent principal-part construction of Construction 3.236 is therefore required for this realization.

Construction 3.236 *(Stalkwise boundary local-dual principal-part sector).* Let M_∂^{pp} be the reduced continuous dual of the stable connected Hamiltonian trace sector, with basis $\Theta_{a,b}$ dual to

$$O_{m,n} = \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}, \qquad m+n > 0,$$

by

$$\langle \Theta_{a,b}, O_{m,n} \rangle = \delta_{a,m}\delta_{b,n}.$$

Equivalently,

$$M_\partial^{\mathrm{pp}} \cong \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) \subset H^2_{(z_1,z_2)}\big(\mathbb{C}[z_1,z_2]\big)\, dz_1 dz_2 \quad \text{after completing at } (z_1,z_2),$$

the continuous dual functionals vanishing on constants, with basis $\rho_{a,b}$ for $a+b>0$. Explicitly,

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \qquad a+b>0,$$

with

$$\Theta_{a,b} \longleftrightarrow \rho_{a,b}.$$

This is a $\mathbb{C}$-linear residue annihilator stable for the Hamiltonian coadjoint action, not an R-submodule and not a quotient by the non-invariant line $\mathbb{C}\,\rho_{0,0}$. Define the Hamiltonian action by negative transpose of the proved boundary Hamiltonian bracket:

$$\langle \{O_f,\Theta\}, O_g\rangle = -\langle \Theta, \{O_f, O_g\}_{\mathrm{open}}\rangle.$$

Then this stalkwise local-dual sector realizes the principal-part coadjoint action:

$$\{O_{p,q}, \Theta_{a,b}\}_{\mathrm{open}} = -(pb - qa + p - q)\Theta_{a-p+1,b-q+1},$$

with negative-index targets zero. Declare the odd local-dual ideal abelian:

$$\{\Theta_{a,b}, \Theta_{c,d}\}_{\mathrm{open}} = 0.$$

Equivalently, the assignment

$$z_1^p z_2^q \longmapsto O_{p,q}, \qquad \rho_{a,b}[1] \longmapsto \Theta_{a,b}$$

realizes stalkwise the Lie algebra $\bar{A} \ltimes \mathcal{P}[1]$.

THEOREM 3.237 *(Support-local principal-part current P_0 prefactorization datum).* After the de Rham contraction to the reduced local model on the brane line, the local-dual sector of Construction 3.236 extends to a support-local P_0 prefactorization datum on $\mathbb{R}$. In this theorem the de Rham differential has already been contracted; the Hamiltonian smearing uses degree-zero test functions, and the cotangent summand uses compactly supported distributions. Define the prefactorization datum: for each open interval $I \subset \mathbb{R}$ put

$$\mathfrak{J}_\partial^{\mathrm{pp}}(I) = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1], \qquad \mathfrak{J}_\partial^{\mathrm{Ham}}(I) = \Omega_c^0(I)\widehat{\otimes}\bar{A},$$

and

$$A_\partial^{\mathrm{Ham}}(I) = \widehat{\mathbf{S}}\big(\mathfrak{J}_\partial^{\mathrm{Ham}}(I)\big), \qquad A_\partial^{\mathrm{pp}}(I) = A_\partial^{\mathrm{Ham}}(I)\widehat{\otimes}\widehat{\mathbf{S}}\big(\mathfrak{J}_\partial^{\mathrm{pp}}(I)\big).$$

In this theorem the open factor is this reduced connected Hamiltonian current algebra $A_\partial^{\mathrm{Ham}}(I)$. The displayed brackets do not define a bracket of principal-part current generators with arbitrary reduced or unreduced open BV observables; that extension is the unreduced lift problem of Remark 3.241. We also write

$$\mathrm{Obs}^{\mathrm{cl,pp}}_{\partial,N}(I) = A_\partial^{\mathrm{pp}}(I),$$

where $\widehat{\mathbf{S}}$ is the degreewise completed symmetric algebra in the chosen current category: each finite symmetric degree is formed before completion, and the P_0 biderivation is extended continuously from generators. No exactness theorem for an unrestricted completed symmetric-algebra functor is used. The datum is equipped with extension by zero for inclusions of intervals and with the following structures:

(i) the differential Q_{open} on the open factor and zero on $\mathfrak{J}_\partial^{\mathrm{pp}}(I)$;

(ii) the factorization product on disjoint intervals $I_1 \sqcup I_2 \subset J$ given by extension by zero followed by graded-commutative multiplication;

(iii) the cosheaf structure on $I \mapsto \Omega_c^0(I)$ and $I \mapsto \mathcal{D}_c'(I)$ given by extension by zero;

(iv) the smearing maps

$$B_f(a) = \int_I a(t) B_f(t)\, dt, \qquad O_f(a) = a \otimes f, \qquad \Theta_\rho(B) = B \otimes \rho[1],$$

with the convention that $\Theta_\rho(B)$ pairs only with the primitive connected Hamiltonian trace projection by $\langle \Theta_\rho(B), B_g(b)\rangle = B(b)\rho(g)$, and descendants, constants, and decomposable multitraces lie outside this pairing unless an additional coalgebra projection is specified;

(v) the P_0 Poisson bracket

$$\begin{aligned} \{B_f(a), B_g(b)\} &= B_{\{f,g\}}(ab), \\ \{B_f(a), \Theta_\rho(B)\} &= \Theta_{f\cdot\rho}(aB), \\ \{\Theta_\rho(B), \Theta_\sigma(C)\} &= 0, \end{aligned}$$

where $a, b \in \Omega_c^0$, $B, C \in \mathcal{D}_c'$, and aB is the multiplication of a current by a smooth degree-zero form;

(vi) the smeared current morphism

$$\begin{aligned} \widetilde{\partial}_{\mathrm{bb},I} \colon \Omega_c^0(I) \otimes \bar{A} \oplus \mathcal{D}_c'(I) \otimes \mathcal{P}[1] &\longrightarrow A_\partial^{\mathrm{pp}}(I), \\ a \otimes f \longmapsto B_f(a), \qquad B \otimes \rho[1] &\longmapsto \Theta_\rho(B). \end{aligned}$$

This degreewise datum defines a support-local P_0 prefactorization datum on $\mathbb{R}$. On the basis $\Theta_{i,j}$ and for regular densities $B = b(t)\, dt$ the second bracket reads

$$\{O_{p,q}(a), \Theta_{i,j}(b)\} = -(pj - qi + p - q)\, \Theta_{i-p+1, j-q+1}(ab),$$

again with negative-index targets zero. The smeared map $\widetilde{\partial}_{\mathrm{bb},I}$ is the interval-level morphism into the reduced connected Hamiltonian P_0 current sector; there is no unsmeared canonical map on an interval, because no distinguished compactly supported unit or density has been chosen.

Proof. Cosheaf axioms. Both Ω_c^0 and $\mathcal{D}_c'$ are cosheaves on $\mathbb{R}$ for the extension-by-zero structure. Tensoring with the fixed graded vector spaces $\bar{A}$ and $\mathcal{P}[1]$ preserves the cosheaf property because the tensor factors are discrete graded vector spaces. Taking the degreewise completed symmetric algebra gives the prefactorization products used here: on disjoint intervals the structure map is extension by zero followed by the graded-commutative product, and the P_0 bracket is the continuous biderivation generated by the displayed current brackets. The odd generators Θ_ρ therefore generate the completed graded-symmetric, equivalently exterior, principal-part factor.

Disjoint-support locality. For disjoint $I_1, I_2 \subset J$, the multiplication map

$$\mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}(I_1) \otimes \mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}(I_2) \longrightarrow \mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}(J)$$

is the graded-commutative product after extending by zero. This is the standard prefactorization product. Because the bracket $\{O_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB)$ involves multiplication of a smooth function by a

distribution and is supported on $\operatorname{supp} a \cap \operatorname{supp} B$, disjoint I_1, I_2 have disjoint such supports and the corresponding bracket vanishes; the factorization product is therefore Poisson-commutative on disjoint intervals, as required for the P_0 structure.

Q-compatibility. $Q_{\mathrm{open}} B_f = 0$ on Hamiltonian trace classes by Proposition 3.209. By declaration $Q_{\mathrm{open}} \Theta_\rho = 0$ on the Θ_ρ-factor. Therefore the differential is zero on the Θ_ρ-factor and acts trivially on the bracket structure displayed above. This is consistent with the reduced model after de Rham-current contraction.

P_0 identities. The first bracket $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$ is Theorem 3.242 smeared by test functions, supplied by Corollary 3.243. The second bracket $\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB)$ is the negative transpose under the residue pairing of Theorem 3.228 applied to the first bracket; in coordinates with $\rho = \rho_{a',b'}$ and $f = z_1^p z_2^q$ this reads

$$\{O_{p,q}(a), \Theta_{a',b'}(B)\} = -(pb' - qa' + p - q)\, \Theta_{a'-p+1,b'-q+1}(aB),$$

exactly the principal-part coadjoint formula of Theorem 3.230. The third bracket $\{\Theta_\rho(B), \Theta_\sigma(C)\} = 0$ is by declaration of the abelian odd ideal.

Jacobi identity. The semidirect Lie algebra $\bar{A} \ltimes \mathcal{P}[1]$ has Jacobi identity $[\mathrm{ad}_f^\vee, \mathrm{ad}_g^\vee] = \mathrm{ad}_{\{f,g\}}^\vee$ for $f, g \in \bar{A}$ on the continuous dual. Smearing by test functions and distributions preserves Jacobi because the $C^\infty(\mathbb{R})$-module action on distributions is associative. Therefore the P_0 bracket on the smeared sector also satisfies Jacobi.

Bulk-boundary morphism. The map $\widetilde{\partial}_{\mathrm{bb},I}$ sends $a \otimes f$ to the smeared polynomial trace observable $B_f(a)$ by Theorem 3.242, and sends $B \otimes \rho[1]$, with $\mathcal{P}[1] \cong \mathfrak{h}_{\mathrm{cont}}^\vee[1]$ via Theorem 3.230, to the $\Theta_\rho(B)$-sector. The brackets just verified are exactly the smeared semidirect brackets on the degree-zero line-current version of $\bar{A} \ltimes \mathcal{P}[1]$. Therefore $\widetilde{\partial}_{\mathrm{bb},I}$ is a morphism into the reduced connected Hamiltonian P_0 current sector. □

LEMMA 3.238 *(Distribution-locality criterion for principal-part currents).* Let $a \in C_c^\infty(I)$, $B \in \mathcal{D}_c'(I)$, and $\rho, \sigma \in \mathcal{P}$. The product aB used in Theorem 3.237 is the standard multiplication of a distribution by a smooth function,

$$(aB)(\varphi) = B(a\varphi), \qquad \varphi \in C_c^\infty(I).$$

With this convention the reduced principal-part brackets are well-defined by

$$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\} = 0.$$

No product of two distributions is formed. This is only support-locality in the reduced line-current model; it is not a Costello-local functional statement and imposes no wavefront-set or RG-admissibility condition.

Proof. Multiplication $B \mapsto aB$ is a continuous endomorphism of $\mathcal{D}_c'(I)$, because $a\varphi \in C_c^\infty(I)$ for every test function φ. The first displayed bracket therefore lands again in $\mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P}[1]$. Its support is contained in $\operatorname{supp}(a) \cap \operatorname{supp}(B)$, so it is local on the line. The second bracket is zero by the abelian odd-ideal definition of the reduced principal-part sector; in particular no term BC appears. Jacobi with two Hamiltonian entries and one principal-part entry follows from associativity $a(bB) = (ab)B$ and from the coadjoint identity $[f\cdot, g\cdot]\rho = \{f, g\} \cdot \rho$. Jacobi with two principal-part entries is immediate from the zero bracket. □

Remark 3.239 *(Reduced principal-part current scope).* Theorem 3.237 is a reduced support-local P_0 prefactorization datum after de Rham-current contraction, with $\Omega^0_c(I)$ in the Hamiltonian current factor and $\mathcal{D}'_c(I)$ in the cotangent factor. It is not a locally constant current algebra on the raw distributional target: if $I \subsetneq J$, a delta-current supported in $J \setminus I$ is a nonzero degree-zero class in $\mathcal{D}'_c(J)$ not hit by extension from $\mathcal{D}'_c(I)$. It is not an unreduced BV/factorization algebra. In particular, the equation $Q\Theta_\rho = 0$ is part of the added reduced local-dual factor; it is not derived here from an unreduced de Rham current complex, a chain-level primitive projection, or Hochschild centrality homotopies against arbitrary open observables.

Theorem 3.240 *(Reduced prefactorization realization of the boundary local dual).* After de Rham contraction in the reduced setting, the assignment

$$\rho_{a,b}[1] \longmapsto \Theta_{a,b}$$

is a chain-level realization of the boundary local dual within the support-local P_0 prefactorization datum $\mathrm{Obs}^{\mathrm{cl,pp}}_{\partial,N}$ of Theorem 3.237. The intervalwise smeared current morphism $\widetilde{\partial}_{\mathrm{bb},I}$ extends, in this reduced principal-part target, the Hamiltonian bulk-to-boundary Lie algebra map of Theorem 3.242 to the closed-coadjoint sector $\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$.

Proof. Theorem 3.237 supplies the support-local P_0 prefactorization datum and verifies that $\widetilde{\partial}_{\mathrm{bb},I}$ is a smeared current Lie algebra morphism for each interval I. The identification $\rho_{a,b}[1] \leftrightarrow \Theta_{a,b}$ is by the basis of the local-dual factor, consistent with Construction 3.236. The closed source side $\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ is the semidirect product computed in Proposition 3.13, and its image in the support-local P_0 prefactorization datum is the displayed Lie subalgebra $\bar{A} \oplus \mathcal{P}[1]$. □

Remark 3.241 *(Unreduced lift obstruction datum).* Theorem 3.240 resolves the boundary local-dual prefactorization realization at the reduced level after de Rham-current contraction, with the cotangent current differential already contracted. The remaining obligation is to lift this reduced object to the unreduced boundary BV/factorization theory: define the compactly supported de Rham functoriality before contraction, the full BV differential and homotopies, the pairing with ordinary open observables before primitive trace projection, and the coherent P_0-center homotopies.

The candidate unreduced enlargement has the form

$$\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \widehat{\otimes} \widehat{\mathbf{S}}\big(\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I)\big),$$

where

$$\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I) = C^{\mathrm{dR}}_{c,\mathrm{cur}}(I) \widehat{\otimes} \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1)[1].$$

Here $C^{\mathrm{dR}}_{c,\mathrm{cur}}(I)$ is the compactly supported de Rham current cosheaf on the line, with differential dual to $d_{\mathbb{R}}$, and $\mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1)$ is the principal-part residue annihilator of constants. One must also construct a chain-level primitive, or cumulant, projection

$$\pi_{\mathrm{prim}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \longrightarrow \Omega^\bullet_c(I) \widehat{\otimes} \bar{A}$$

compatible with Q_{open}, constants, decomposable multi-traces, and factorization products. The required homotopies are the centrality homotopies for Θ_ρ against arbitrary unreduced open observables and the P_0 bracket homotopies

$$\{B_f, \Theta_\rho\} \simeq \Theta_{f \cdot \rho}, \qquad \{\Theta_\rho, \Theta_\sigma\} \simeq 0$$

before primitive projection. Only after this enlargement does the full cotangent factorization-center lift extend the reduced realization $\rho_{a,b}[1] \mapsto \Theta_{a,b}$ to a chain-level morphism on the unreduced sector. The desired unreduced lift would be a quantum-level homotopical lift, beyond the classical reduced theorem above; it sits in the residual Problem 3.273.

THEOREM 3.242 *(Classical bulk-to-boundary Hamiltonian map).* In the stable connected Hamiltonian trace sector of the formal local mixed holomorphic-topological theory, the boundary evaluation maps $\partial_{\mathrm{bb},N}$ of Construction 3.195 stabilize to a linear map

$$\partial_{\mathrm{bb}} : \bar{A} \longrightarrow H^{\circ}_{\mathrm{Ham}}(O^{\mathrm{cl}}_{\mathrm{brane}}), \qquad f \longmapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

This map is a morphism of Lie algebras from the closed Hamiltonian Poisson algebra $\bar{A}$ to the connected open Hamiltonian trace Poisson algebra:

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \partial_{\mathrm{bb}}\big(\{f, g\}_{\bar{A}}\big).$$

This is the degree-zero Hamiltonian bracket component of the reduced CE/PV central-operation map to the brane observables. The full Poisson factorization-center morphism is not constructed in this theorem.

Proof. The map is well-defined on $\bar{A}$ by Corollary 3.202; constants map to the scalar N and vanish in the Hamiltonian quotient by constants. Lemma 3.221 gives the open Hamiltonian trace Poisson bracket. For $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, Proposition 3.222 gives

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \overline{(kn - lm)\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}.$$

The closed Poisson bracket is

$$\{f, g\}_{\bar{A}} = \overline{(kn - lm) z_1^{k+m-1} z_2^{l+n-1}},$$

and applying ∂_{bb} gives the same expression. Bilinearity proves the identity for all polynomials. □

COROLLARY 3.243 *(Classical local bulk-boundary coupling with explicit U(1) center anomaly).* Let $a, b \in \Omega^{\circ}_c(\mathbb{R})$ be scalar compactly supported test functions on the brane line, and for $f \in \bar{A}$ define the smeared boundary Hamiltonian

$$I_\partial(a, f) = B_f(a) = \int_{\mathbb{R}} a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt \in A^{\mathrm{Ham}}_\partial(\mathbb{R}).$$

Then the following two strengthenings of the bracket identity hold jointly.

(i) Strict P_0 identity on disjoint supports. In the P_0 bracket of $A^{\mathrm{Ham}}_\partial$,

$$\{I_\partial(a, f), I_\partial(b, g)\}_{A^{\mathrm{Ham}}_\partial} = I_\partial(ab, \{f, g\}_{\bar{A}}),$$

and the bracket vanishes whenever the supports of a and b are disjoint. The locality input is the canonical equal-time bracket $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta^i_l \delta^k_j \delta(t - s)$ of 3.21; smearing against $a(t)b(s)$ produces the pointwise product ab after integration of the delta function. This is the level-one component of 3.22.

(ii) Explicit U(1) center anomaly constant. Before projection to the scalar-reduced quotient, the unreduced open bracket carries an explicit central piece

$$\{I_\partial(a,f), I_\partial(b,g)\}_{\text{open}} = I_\partial(ab, \{f,g\}_{\bar{A}}) + N \cdot \omega(f,g) \int_{\mathbb{R}} a(t)b(t)\, dt,$$

where ω is the cocycle of 2.4 and the prefactor N is the rank of the brane stack and is the value of the U(1) center character $\mathrm{Tr} : \mathbb{C} \cdot I \to \mathbb{C}, I \mapsto N$. After projection to $H^0_{\text{red}} = H^0/\mathbb{C} \cdot \mathrm{Tr}(1)$ the second term vanishes, and one recovers

$$\overline{\{I_\partial(a,f), I_\partial(b,g)\}_{\text{open}}} = \overline{I_\partial(ab, \{f,g\}_{\bar{A}})}.$$

Statements (i) and (ii) are compatible: (ii) is the unreduced refinement, (i) is the level-one factorization-algebra restatement on the scalar-reduced sector. The equal-time delta locality of (i) produces a strict P_0 factorization algebra rather than a homotopical one, while (ii) records the U(1) center anomaly that obstructs lifting (i) to the unreduced sector.

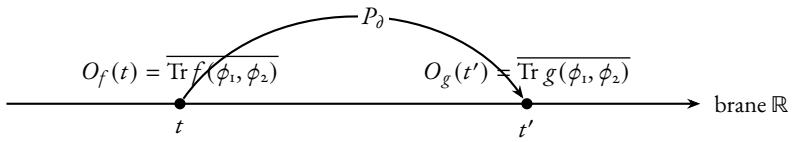
Proof. Proof of (i). By 3.18, the P_0 bracket on A^{Ham}_∂ is the biderivation with $\{a \otimes f, b \otimes g\}_{A_\partial} = ab \otimes \{f,g\}_{\bar{A}}$ on length-one generators, so $\{B_f(a), B_g(b)\}_{A^{\text{Ham}}_\partial} = B_{\{f,g\}}(ab) = I_\partial(ab, \{f,g\}_{\bar{A}})$. 3.20 then gives vanishing on disjoint supports because $ab \equiv 0$ pointwise. The same identity is the smeared content of the equal-time delta-function bracket of 3.21: tracing $f(\phi_1, \phi_2)(t)$ and $g(\phi_1, \phi_2)(s)$, applying the Leibniz rule, and integrating against $a(t)b(s)\delta(t-s)$ produces the pointwise product ab after the delta-function integration.

Proof of (ii). The unreduced bracket of two integrals is, by locality of the canonical bracket and 3.242,

$$\begin{aligned}\{I_\partial(a,f), I_\partial(b,g)\} &= \iint_{\mathbb{R}^2} a(t)b(t')\, \{\mathrm{Tr}\, f(\phi(t)), \mathrm{Tr}\, g(\phi(t'))\}\, dt\, dt' \\ &= \int_{\mathbb{R}} a(t)b(t)\, \mathrm{Tr}\big(\{f,g\}_{\bar{A}}\big)\, dt + \int_{\mathbb{R}} a(t)b(t)\, \omega(f,g)\, N\, dt,\end{aligned}$$

where the splitting is by the central decomposition of 2.6: the central scalar $\omega(f,g) \cdot N$ is removed by the projection to H^0_{red}, and the remaining piece is $I_\partial(ab, \{f,g\}_{\bar{A}})$ exactly. The constants are explicit: the prefactor of the central piece is $N \cdot \omega(f,g)$, and the smeared coefficient is $\int ab\, dt$. The classical-to-quantum reduction of 2.8 identifies $\omega(f,g) \cdot N$ with the leading-order Capelli shift $\hbar N$ on the linear generators $f = z_1$, $g = z_2$. Compact support of a, b kills boundary terms. □

$$P_\partial = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$$



$$|\mathrm{Aut}(\Gamma_1)| = 1, \quad \tfrac{1}{|\mathrm{Aut}(\Gamma_1)|} = 1, \quad \text{one edge}$$

$$\text{ordered: } \hbar P_\partial/2; \quad \text{commutator: } \hbar P_\partial$$

$$\text{output:} \quad \hbar(kn - lm)\, \overline{\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})} \quad \text{on } f = z_1^k z_2^l,\ g = z_1^m z_2^n$$

FIGURE 0.1: Notation for the reduced open-line order-$\hbar$ raw commutator graph Γ_1, not a Costello/Feynman-integral proof. Two ordered boundary insertions sit on the brane line $\mathbb{R}$ at distinct times t, t'; the unique propagator P_∂ contracts one z_1-leg on one side against one z_2-leg on the other (and the antisymmetric partner). The automorphism group is trivial. In the Weyl ordered product the ordered one-edge weight is $\hbar P_\partial/2$; after subtracting the reverse order the commutator weight is $\hbar P_\partial$. The bivector contraction on monomials reproduces the cocycle coefficient $(kn - lm)$ of Proposition 3.222, the order-$\hbar$ Moyal coefficient of Proposition 3.268, and the U(1) centre cocycle $\omega(f, g)$ of Lemma 2.4 realised at multidegree $(k + m, l + n) = (1, 1)$.

The graph Γ_1 of Figure 0.1 records the single canonical contraction producing the coefficient $(kn - lm)$ in Proposition 3.222; it is not a separate Feynman-integral proof.

3.12.2 FORMAL DEFORMATION-QUANTIZATION TARGET

The degree-zero Hamiltonian sector has a Weyl-normalized formal deformation-quantization target. The formal holomorphic disk has constant Poisson tensor, so after choosing the translation-invariant Weyl convention its deformation is the Weyl/Moyal algebra. This fixes the algebraic coefficients one would have to recover from a quantum closed-open BV construction. It is not, by itself, a construction of that quantum BV theory. The degree-zero Hamiltonian bracket part of the classical theorem is the proposed special fibre of this formal deformation; the cotangent/descendant part remains the separate problem stated below. Put

$$A_\hbar = \big(\mathbb{C}[[z_1, z_2]][[\hbar]], *\big),$$

where $*$ is the Weyl/Moyal product of the next subsection. The symbol $\bar{A}_\hbar$ denotes the central quotient of the underlying Moyal Lie algebra by $\mathbb{C}[[\hbar]]$; it is not used as an associative quotient. We write

$$[f, g]_{\mathrm{raw}} = f * g - g * f, \qquad \{f, g\}_\hbar = \hbar^{-1}[f, g]_{\mathrm{raw}}.$$

The degree-zero quantum Hamiltonian Lie algebra is

$$\mathfrak{h}_\hbar = \bar{A}_\hbar, \qquad [f, g]_{\mathfrak{h}_\hbar} = \{f, g\}_\hbar.$$

The specialization at $\hbar = 0$ is the Poisson Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ used above.

On the brane side, the open action

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2)$$

makes the matrix entries of ϕ_1 and ϕ_2 conjugate variables. The canonical algebraic quantization of the unconstrained first-order matrix phase space is the completed Weyl algebra $\mathcal{W}_N$ generated by those entries with

$$[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar\, \delta^i_l \delta^k_j, \qquad [(\phi_a)^i{}_j, (\phi_a)^k{}_l] = 0 \quad (a = 1, 2).$$

The gauge field imposes the conjugation moment-map constraint. With the commutator convention above, the normal-ordered quantum comoment is

$$\widehat{\mu}_\epsilon = -\mathrm{Tr}\big([\epsilon, \phi_1]\phi_2\big), \qquad \epsilon \in \mathfrak{gl}_N.$$

This formula is not cyclically reordered: for $\epsilon = 1$ it vanishes, as the scalar subgroup acts trivially. We use the convention

$$\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a$$

for the induced infinitesimal conjugation action on the Weyl algebra; changing to the opposite commutator convention changes the displayed comoment by the corresponding sign. Since the scalar comoment is zero, the effective constraint algebra is $\mathfrak{pgl}_N$, represented by $\mathfrak{sl}_N$ after choosing the trace pairing. The quantum brane local algebra used here is the reduced BRST, or degree-zero normalizer, quantum Hamiltonian reduction

$$\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N} = C^\bullet_{\mathrm{BRST}}\big(\mathfrak{sl}_N, \mathcal{W}_N, \widehat{\mu}\big),$$

whose degree-zero cohomology is the normalizer quotient

$$H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N} = \mathrm{N}_{\mathcal{W}_N}\big(\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)\big)/\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N).$$

The constant Hamiltonian trace is projected out in degree zero as before. The classical special fibre is the reduced BV algebra of Construction 3.190.

The quantum boundary map is Weyl-ordered trace evaluation:

$$\partial_{\mathrm{bb},\hbar,N}(f) = \overline{\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)}, \qquad f \in \bar{A}_\hbar,$$

where Op_{W} denotes total symmetrisation before passing to the trace. At $\hbar = 0$ this recovers Construction 3.195. After dividing the commutator by $\hbar$ and reducing modulo $\hbar$, the raw commutator identity recovers Theorem 3.242.

Proposition 3.244 *(Reduced BRST descent of Weyl-ordered Hamiltonian traces).* For every $f \in \bar{A}_\hbar$, the Weyl-ordered trace

$$F_{f,N} = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$$

is invariant under the quantum comoment:

$$[\widehat{\mu}_\epsilon, F_{f,N}] = 0, \qquad \epsilon \in \mathfrak{gl}_N.$$

Hence $F_{f,N}$ lies in the normalizer of the quantum moment-map ideal and defines a degree-zero class in $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ after removing the scalar Hamiltonian.

Proof. Total Weyl symmetrisation is equivariant for linear changes of the matrix variables, and simultaneous conjugation acts linearly on (ϕ_1, ϕ_2). Therefore $\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$ is invariant under infinitesimal conjugation. By the defining property of the quantum comoment,

$$\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a,$$

this invariance is exactly $[\widehat{\mu}_\epsilon, F_{f,N}] = 0$. If $I = \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$ is the left quantum moment-map ideal, then for every generator $\widehat{\mu}_\epsilon$, $\widehat{\mu}_\epsilon F_{f,N} = F_{f,N}\widehat{\mu}_\epsilon \in I$. Thus $F_{f,N}$ normalizes I and descends to the normalizer quotient describing $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$. □

PROPOSITION 3.245 *(Reduced BRST descent of arbitrary Weyl-ordered Hamiltonian products).* For every finite collection $f_1, \ldots, f_r \in \bar{\mathcal{A}}_\hbar$, the ordered product

$$F = \prod_{i=1}^{r} F_{f_i,N}, \qquad F_{f_i,N} = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f_i)(\phi_1, \phi_2),$$

is invariant under the quantum comoment:

$$[\widehat{\mu}_\epsilon, F] = 0, \qquad \epsilon \in \mathfrak{gl}_N.$$

Equivalently, every monomial in the Weyl-ordered single-trace generators normalizes the quantum moment-map ideal and defines a degree-zero class in $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ after removing the empty trace.

Proof. By the convention $\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a$, the operator $\hbar^{-1}\,\mathrm{ad}\,\widehat{\mu}_\epsilon$ acts on the associative algebra $\mathcal{W}_N$ by infinitesimal GL_N-conjugation, and is therefore a derivation of the associative product. By Proposition 3.244, each individual factor $F_{f_i,N}$ is annihilated by this derivation. Derivations of an associative algebra preserve products: the Leibniz expansion of $\hbar^{-1}\,\mathrm{ad}\,\widehat{\mu}_\epsilon$ on the ordered product $\prod_{i=1}^{r} F_{f_i,N}$ is a sum of summands each containing a factor $\hbar^{-1}[\widehat{\mu}_\epsilon, F_{f_i,N}] = 0$. Hence $[\widehat{\mu}_\epsilon, F] = 0$, and as in the proof of Proposition 3.244, $\widehat{\mu}_\epsilon F = F\widehat{\mu}_\epsilon \in \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)$, so F normalizes the left moment-map ideal and descends. □

Remark 3.246. The argument uses only that $\hbar^{-1}\,\mathrm{ad}\,\widehat{\mu}_\epsilon$ is a derivation. No identity on the Moyal commutator of symbols is invoked: the Capelli shift obstructs the bracket identity between traces, not their joint reduced BRST descent.

Remark 3.247 *(Normal-ordering obstruction).* Proposition 3.244 is not the quantum Moyal theorem. Put $X = \phi_1$ and $Y = \phi_2$. With the normal ordering above,

$$\widehat{\mu}_\epsilon = -\mathrm{Tr}([\epsilon, X]Y) = \mathrm{Tr}\big(\epsilon(YX - XY + \hbar N I)\big).$$

For $\epsilon = I$ this comoment is identically zero; equivalently the displayed matrix has trace zero inside the Weyl algebra. Thus the nontrivial quantum moment equations are represented in the quantum Hamiltonian reduction by the traceless matrix congruence

$$YX - XY + \hbar N I \equiv 0 \pmod{\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)},$$

not by the classical commuting-pair equation. Consequently, for a covariant matrix word A,

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\,\mathrm{Tr}(A) \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N).$$

This Capelli-type shift can feed lower-degree traces into the connected sector. A stable quantum theorem must prove that the chosen connected extraction removes or renormalizes this shift, or else change the sector or normalization.

Remark 3.248 *(Capelli-renormalized trace generators).* Let

$$T_{a,b} = \mathrm{Tr}(X^a Y^b)$$

denote the normal-ordered trace. The Weyl-ordered trace

$$J_N(z_1^a z_2^b) = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(z_1^a z_2^b)(X, Y)$$

is triangular over the normal-ordered traces after imposing the quantum moment relation:

$$J_N(z_1^a z_2^b) \equiv \sum_{r=0}^{\min(a,b)} \left(-\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} T_{a-r,b-r}.$$

For example,

$$J_N(z_1 z_2) = T_{1,1} - \frac{\hbar N^2}{2}, \qquad J_N(z_1^2 z_2) = T_{2,1} - \hbar N\, T_{1,0},$$

and

$$J_N(z_1 z_2^2) = T_{1,2} - \hbar N\, T_{0,1}, \qquad J_N(z_1^2 z_2^2) = T_{2,2} - 2\hbar N\, T_{1,1} + \frac{\hbar^2 N^3}{2}.$$

The naive normal-ordered basis therefore has the wrong commutator even in low degree; a direct calculation gives

$$\hbar^{-1}[T_{2,1}, T_{0,2}] = 4T_{1,2} - 2\hbar N\, T_{0,1} \mod \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N),$$

whereas the scalar Moyal bracket would give only $4T_{1,2}$. Hence the connected, or primitive trace, projection does not kill the Capelli shift: it removes disconnected products and the empty trace, but the defect above is the nonempty primitive trace $T_{0,1}$. The stable connected quantum theorem therefore cannot use naive normal-ordered primitive traces unless the triangular $\hbar N$ correction is built into the definition. The natural finite-N generators are the Capelli-renormalized Weyl traces $J_N(f)$, equivalently

$$J_N(f) = T_N\left(\exp\left(-\frac{\hbar N}{2}\partial_{z_1}\partial_{z_2}\right) f\right)$$

in the normal-ordered trace notation $T_N(z_1^a z_2^b) = T_{a,b}$.

Lemma 3.249 *(Capelli-renormalized stable connected trace).* Let $\mathrm{Tr}_N^{\mathrm{ren}}(f) = J_N(f)$ for $f \in \mathbb{C}[z_1, z_2][[\hbar]]$. Write $I_N = \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$ for the left moment-map ideal. Then:

(i) the map $\mathrm{Tr}_N^{\mathrm{ren}} : \bar{A}_\hbar \longrightarrow \mathcal{W}_N / I_N$ is well defined modulo the empty (constant) trace;

(ii) the inverse triangular system reads

$$T_{a,b} = \sum_{r=0}^{\min(a,b)} \left(\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} J_N(z_1^{a-r} z_2^{b-r}),$$

so $T_{a,b}$ and $J_N(z_1^a z_2^b)$ generate the same Weyl trace algebra modulo the empty trace, after inverting the unipotent system over $\mathbb{C}[[\hbar N]]$;

(iii) the Capelli contact term of Remark 3.247, $\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\, \mathrm{Tr}(A) \mod I_N$, reproduces, on J_N generators, exactly the $r = 1$ tail of the triangular formula. Consequently the displayed shift is *absorbed into the change of basis* between $T_{a,b}$ and $J_N(z_1^a z_2^b)$ and is not an additional term in the J_N-commutator;

(iv) the leading symbol of $J_N(f)$ is $\operatorname{Tr} f(X,Y)$ in associated graded order, so $\operatorname{Tr}_N^{\mathrm{ren}}$ stabilises in the large-N limit and its stable connected projection coincides with the leading primitive trace.

Proof. The triangular formula of Remark 3.248 is total Weyl symmetrisation of $z_1^a z_2^b$ inside the matrix Weyl algebra, in which one normal-ordered swap of $z_1 z_2$ commutes through $\hbar N$. This is the local bivariate Weyl-normal-ordering formula proved by the contraction count in Lemma F.7; the classical Capelli source context is [26, 27]. No primary source is being invoked for this exact bivariate triangular trace identity. A finite exact check verifies it as a round-trip $J_N \to T \to J_N = \mathrm{id}$ on exponents $a, b \le 10$. The system is unipotent over $\mathbb{C}[[\hbar N]]$ with $r = 0$ diagonal, hence invertible there; the sign in the inverse is $(+\hbar N/2)^r$, again verified by the same finite computation.

For (iii), expand

$$\operatorname{Tr}(AXY) - \operatorname{Tr}(AYX) = \operatorname{Tr}(A[X,Y]) = \hbar N \operatorname{Tr}(A) - \operatorname{Tr}\big(A(YX - XY + \hbar N I)\big)$$

using the Weyl moment relation $YX - XY + \hbar N I = 0 \bmod I_N$ from Remark 3.247; the subtracted term lies in I_N and the displayed image is $\hbar N \operatorname{Tr}(A)$. On J_N generators this is the $r = 1$ tail term of the triangular expansion, by direct comparison with Remark 3.248. Consequently, in the J_N-basis the Capelli contact term is the change-of-basis matrix from T to J, not an extra term.

For (iv), the shift $T_{a-r,b-r}$ has total degree $a + b - 2r < a + b$ for $r \ge 1$, so the $r = 0$ term is the unique top component in the $\mathfrak{m}$-adic filtration on $\bar{A}_\hbar$; the stable connected projection of f is $J_N(f) \bmod I_N$ in the associated graded sense. □

COROLLARY 3.250 *(Renormalized stable connected trace map).* The map

$$\operatorname{Tr}^{\mathrm{ren}} \colon \bar{A}_\hbar \longrightarrow H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty}), \qquad f \longmapsto \big[\overline{J_N(f)}\big]_{N\to\infty}$$

is well defined and is a $\mathbb{C}[[\hbar]]$-linear injection of $\bar{A}_\hbar$ onto the stable Capelli-renormalized single-trace generators. The bar $\overline{(-)}$ denotes the connected single-trace cumulant projection: it removes the empty trace and disconnected products, and it is consistent with the J_N-basis at every finite N by Lemma 3.249(iii). The notation $\big[\overline{J_N(f)}\big]_{N\to\infty}$ means this renormalized J-basis transition, or equivalently the associated-graded stable class determined by the leading primitive trace. It is not a literal block-restriction inverse-limit element in the naive normal-ordered trace basis over plain $\mathbb{C}[[\hbar]]$.

THEOREM 3.251 *(Reduced Moyal commutator: radial input and obstruction gate).* Let $\mathcal{W}_N$ be the Weyl algebra of $T^* \operatorname{Mat}_N$ and let $\widehat{\mu}$ be the normal-ordered conjugation comoment above. For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, put

$$J_N(f) = \operatorname{Tr} \operatorname{Op}_{\mathrm{W}}(f)(X,Y).$$

Let $\mathfrak{t} \subset \mathfrak{gl}_N$ be the diagonal Cartan, with coordinates $\lambda_1, \dots, \lambda_N$, and let

$$\Delta = \prod_{i<j} (\lambda_i - \lambda_j).$$

Put

$$S_N(f) = \sum_{i=1}^{N} \operatorname{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i}).$$

The ordinary-trace comparison has the following exact status. Assume the radial-parts input package of Statement F.8: after Vandermonde conjugation, the Harish-Chandra map on invariants has the quantum

moment-map ideal as kernel and therefore injects the reduced invariant quotient into Weyl-invariant differential operators on the regular Cartan. Under this input the radial-part map satisfies

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1} S_N(f)\Delta.$$

Consequently, for every $N \geq 1$ and all f, g the defect

$$D_N(f,g) = \hbar^{-1}[J_N(f), J_N(g)] - J_N(\{f,g\}_\hbar)$$

lies in the quantum moment-map ideal $I_N = \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$ and therefore vanishes in the degree-zero quantum Hamiltonian reduction

$$\mathrm{N}_{\mathcal{W}_N}(I_N)/I_N.$$

After the connected single-trace projection $\overline{(-)}\colon \mathcal{W}_N/I_N \twoheadrightarrow \mathcal{W}_N/I_N/(\text{empty trace} \oplus \text{disconnected products})$, the descended commutator on the J_N-images is the finite-rank image of the scalar Moyal bracket:

$$\left[\partial_{\mathrm{bb},\hbar,N}(f), \partial_{\mathrm{bb},\hbar,N}(g)\right]_{\mathrm{conn,raw}} = \hbar\, \partial_{\mathrm{bb},\hbar,N}(\{f,g\}_\hbar).$$

The identification with the scalar Moyal algebra itself is read only after passing to the degreewise stable range and then to the large-N connected sector. Here the large-N limit is the renormalized associated-graded connected transition system of Corollary 3.250. Finite-N trace identities remain in each rank, and the scalar character $\hbar N$ has already been absorbed into the J_N-basis and then quotiented along the Hamiltonian scalar line. The identity stabilises in this large-N sense, defining $\partial_{\mathrm{bb},\hbar}\colon \mathfrak{h}_\hbar \longrightarrow H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})$ satisfying

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f,g\}_\hbar),$$

which is the bracket equation recorded in Remark 3.254.

Without the image-normalization clause of Statement F.8, the missing theorem is not an unspecified all-order input. It is the quantum shear primitive: for every a, b, N,

$$E_{a,b,N} = \sum_{j=0}^{b} \mathrm{Tr}(Y^j X^a Y^{b-j}) - J_N\Big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\Big) \in \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N).$$

Equivalently one must kill the free normal-diagram obstruction

$$\mathfrak{o}_{a,b} = [R^{\mathrm{free}}_{a,b}] \in \operatorname{coker} C_{a,b}$$

by functorial lifts $K_{a,b} \in \ker \partial$. In the dual-potential formulation this is the vanishing of

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b},$$

equivalently the decorated PBW Stokes condition $D^{\square}_{a,b} = C^+_{a,b}\partial_2$; failure is exactly a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. The complete edge families $\mathfrak{o}_{2,b}$ and $\mathfrak{o}_{a,2}$ vanish by Proposition F.27. The finite free certificates cover the displayed interior windows, including the $K_{4,4}$ lift and the non-edge total-degree checks recorded in Proposition F.28 and the exact normal-form script. For $a, b \geq 3$ in all degrees, the exact remaining problem is to construct the associated-graded two-colour necklace homotopy, or prove the dual-potential theorem: every signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ satisfies $\lambda(E^+_{a,b}) = \phi(T_{a,b})$. Failure is exactly such a row with nonzero $\lambda(E^+_{a,b}) - \phi(T_{a,b})$.

Proof. Identify $\mathcal{W}_N$ with $\mathcal{D}_\hbar(\mathfrak{gl}_N)$, where X is multiplication and $Y = -\hbar\partial_X^t$. The quantum Hamiltonian reduction for conjugation is the normalizer quotient by the left ideal generated by the traceless comoment. The scalar matrix has zero comoment, as in Remark 3.247; equivalently the degree-zero function calculation is the $\mathfrak{sl}_N$ reduction together with rational GL_N-invariants. Since GL_N is linearly reductive in characteristic zero, the functor of rational algebraic invariants is exact on this function-level calculation. This is not an assertion that the unreduced Lie-CE cohomology of $\mathfrak{gl}_N$ has no scalar ghost or stabilizer classes.

The Harish-Chandra radial-parts package, whose source lineage is Harish-Chandra, Wallach, and Levasseur–Stafford [8–11], is exactly the assertion that the Vandermonde-conjugated radial map on invariants has the moment-map ideal as kernel, injects this quantum Hamiltonian reduction, and has the generator image normalization recorded in Statement F.8. The nonlinear shear obstruction in Lemma F.13 explains why the mixed Weyl-trace image is kept as part of that input package, rather than derived from pure-power radial images. Therefore the displayed radial formula for $J_N(f)$ is valid only under the exact input package named there.

Finally,

$$\hbar^{-1}[S_N(f), S_N(g)] = S_N(\{f, g\}_\hbar),$$

because the ith summand is the ordinary one-particle Weyl algebra and distinct summands commute. Conjugation by Δ preserves commutators, so $\mathrm{Rad}_0(D_N(f, g)) = 0$. Injectivity of the Harish-Chandra reduction map gives

$$D_N(f, g) \in \left(\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)\right)^{GL_N},$$

hence $D_N(f, g)$ is zero in the degree-zero quantum Hamiltonian reduction. The Cartan-rank-2 case has been checked by exact arithmetic: the direct $N = 2$ radial restriction of $J_2(f)$ is checked on 80 operator/test-polynomial pairs, and the $N = 3$ finite-rank sweep checks 80 corresponding pairs. The same computation also checks

$$[S_2(f), S_2(g)] = S_2([f, g]_*)$$

in the normal-form convention $z_2 = -\hbar\partial_\lambda$ on the primary pairs, and on the higher-order pairs that exhibit nonzero P^3 and P^5 coefficients. These are finite-rank falsification tests; they do not close the all-N source theorem.

For the connected-projection statement, the projection $\overline{(-)}$ acts on the basis $T_{a,b} \mapsto \overline{T_{a,b}} \equiv \overline{J_N(z_1^a z_2^b)}$ mod (Capelli triangular). By Lemma 3.249(iii) the Capelli contact term is absorbed into the change of basis between $T_{a,b}$ and J_N, so on $\overline{J_N(f)}$-classes the descended commutator is the image of $\hbar J_N(\{f, g\}_\hbar)$ mod I_N under the projection, i.e. $\hbar\, \partial_{\mathrm{bb},\hbar,N}(\{f, g\}_\hbar)$.

Stability in N: after passage to the renormalized associated-graded J_N-basis transition maps, the radial identity is compatible with increasing N. Enlarging $\mathfrak{t}_N \subset \mathfrak{gl}_N$ to $\mathfrak{t}_{N+k}$ adjoins k mutually commuting one-particle Weyl summands, and the bracket $[S_N(f), S_N(g)] = S_N(\{f, g\}_\hbar)$ is preserved at every N. By Lemma 3.249(iv), the connected extraction depends only on the leading symbol filtration, which the Capelli triangular system preserves. Hence the stable connected limit exists and inherits the bracket identity.

The final obstruction statement is Lemma F.15 together with Definition F.21 and Statement F.22. The edge vanishing is Proposition F.27. The finite interior certificates are the free normal-diagram identities of Proposition F.20, Proposition F.28, and the reproducible exact-arithmetic harness `scripts/quantum_shear_trace_diagram_normal_form.py`. These data are certificates and edge theorems; they do not supply the functorial all-interior splitting. □

Corollary 3.252 *(Degree-zero quantum comparison and decorated radial obstruction).* Under the same radial-parts input package as Theorem 3.251, the map

$$\partial_{\mathrm{bb},\hbar}\colon \mathfrak{h}_\hbar = \bar{A}_\hbar \longrightarrow H^{\circ}_{\mathrm{Ham,conn}}(O^{\mathrm{q}}_{\mathrm{brane},\infty}), \qquad f \longmapsto \left[\overline{J_N(f)}\right]_{N\to\infty},$$

is a Lie-algebra map for $\{\,,\,\}_\hbar$ on the source and $\hbar^{-1}[-,-]_{\mathrm{conn,raw}}$ on the target. Its associated graded at $\hbar = 0$ is the degree-zero part of Theorem 2.70. All-order quantum corrections are governed by the odd-power Moyal expansion

$$\{f, g\}_\hbar = \sum_{s\geq 0} \frac{\hbar^{2s}}{2^{2s}(2s+1)!} P^{2s+1}(f, g),$$

so its second-order normalized coefficient is $\frac{1}{24}P^3(f, g)$. This term need not vanish.

Without the full radial image package, the formal Moyal coefficients and the Capelli triangular trace basis remain proved. The quantum shear primitive needed for the ordinary Weyl-trace comparison is proved on the edge strips by Proposition F.27 and in the finite certified interior windows. Its all-interior extension is exactly the dual-potential obstruction theorem

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b}$$

of Proposition F.25: it vanishes if and only if the decorated PBW Stokes identity $D^{\square}_{a,b} = C^+_{a,b}\partial_2$ holds, and it fails exactly by a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$.

Proof. This is Theorem 3.251, combined with the Moyal odd-symmetry of Proposition 3.268: P^r changes by $(-1)^r$ under interchange of f and g, so only odd r survive in the raw star commutator. Dividing by $\hbar$ shifts these odd raw powers to even powers in the normalized Hamiltonian bracket. Hence the normalized $\hbar^2$ coefficient is $P^3(f, g)/24$, and the normalized $\hbar^4$ coefficient is $P^5(f, g)/1920$. □

Remark 3.253 *(Explicit second- and fourth-order corrections).* For higher-order test cases that exercise nontrivial P^3 and P^5:

$$[z_1^3 z_2, z_1 z_2^3]_* = 8\hbar\, z_1^3 z_2^3 - 3\hbar^3\, z_1 z_2 + O(\hbar^5),$$
$$[z_1^4 z_2^2, z_1^2 z_2^4]_* = 12\hbar\, z_1^5 z_2^5 - 40\hbar^3\, z_1^3 z_2^3 + 6\hbar^5\, z_1 z_2 + O(\hbar^7).$$

After normalization by $\hbar^{-1}$, the first example gives

$$\{z_1^3 z_2, z_1 z_2^3\}_\hbar = 8 z_1^3 z_2^3 - 3\hbar^2 z_1 z_2 + O(\hbar^4).$$

The $\hbar^3$ raw coefficient is therefore the normalized $\hbar^2$ correction $\frac{1}{24}P^3(f, g)$, in agreement with Proposition 3.268. The non-affine bracket correction reduces, in this constant Poisson tensor case, to exactly the higher Moyal coefficients.

Remark 3.254 *(Degree-zero quantum Hamiltonian-sector status).* Under the radial image package, Corollary 3.252 identifies the stable connected single-trace quotient of $O^{\mathrm{q}}_{\mathrm{brane},N}$ and proves that $\partial_{\mathrm{bb},\hbar,N}$ stabilizes to a map

$$\partial_{\mathrm{bb},\hbar} : \mathfrak{h}_\hbar \longrightarrow H^{\circ}_{\mathrm{Ham,conn}}(O^{\mathrm{q}}_{\mathrm{brane},\infty})$$

satisfying

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f, g\}_\hbar).$$

Equivalently, the degree-zero statement under that input is that the connected quantum brane algebra is the Weyl/Moyal deformation of the degree-zero Hamiltonian sector. The associated graded of this proposed degree-zero quantum boundary comparison at $\hbar = 0$ is the degree-zero part of Theorem 2.70. The finite-N radial-parts prerequisite is Theorem 3.251; the reduced-BRST descent prerequisite for arbitrary Weyl-ordered products is Proposition 3.245; the absorption of the Capelli contact term into the change of basis between $T_{a,b}$ and $J_N(z_1^a z_2^b)$ is Lemma 3.249. Combining these inputs, the closure of the degree-zero comparison on the Hamiltonian sector is Corollary 3.252: the stable primitive generators are the Weyl/Capelli-renormalized classes $\overline{J_N(f)}$. The J_N-basis change absorbs the finite-N Capelli contact and its triangular $\hbar N$ lower-trace terms by Lemma 3.249(iii); the connected projection then removes only the empty trace and decomposable products. The remaining interior construction problem is sharper than the old all-N radial slogan. One must either prove the exact Weyl-trace image clause of Statement F.8, or construct the equivalent quantum shear primitives $E_{a,b,N} \in \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$. In the free indexed normal-diagram model this is the vanishing of $\mathfrak{o}_{a,b}$ together with functorial lifts $K_{a,b}$. In the sharper dual-potential form the obstruction is $\Omega^{\mathrm{rad}}_{a,b}$, equivalently decorated PBW Stokes for $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$; a failure is a signed $B^{*}_{a,b}$-row with nonzero $\lambda(E^{+}_{a,b}) - \phi(T_{a,b})$. The edge strips are proved; the interior all-bidegree theorem is the associated-graded two-colour necklace homotopy, or this equivalent dual-potential signed-row vanishing theorem.

Remark 3.255. The degree-zero Hamiltonian part has a Weyl-normalized algebraic target: the Weyl/Moyal algebra of the formal symplectic disk, the constant-Poisson case of deformation quantization [7]. Costello's BV/RG formalism supplies the general renormalized BV construction under its elliptic, gauge-fixing, renormalization, and obstruction-vanishing hypotheses. Costello–Li construct quantum BCOV theory and prove a specific flat open-closed BCOV/HCS quantization theorem [13, 14]. Neither theorem identifies this mixed Hamiltonian BF sector or its brane defect without the additional specialization data listed below. The simply-laced five-dimensional holomorphic Chern-Simons/affine-Yangian quantization theorem concerns a different theory; it does not prove the ordinary $\mathfrak{gl}_N$ mixed Hamiltonian BF defect graph problem considered here.

Remark 3.256. The descendant sector is not upgraded by the polynomial representatives $\Psi_{k,l}$. Corollary 3.224 shows that their open Hamiltonian action is the adjoint polynomial action, while the closed cotangent field carries the coadjoint action. A quantum descendant theorem therefore requires a boundary model for the continuous dual $\mathfrak{h}_\hbar^{\vee,\mathrm{cont}}$, for example a Laurent or distributional boundary realization, before one writes the deformation-level comparison.

Remark 3.257. The full cotangent deformation would replace $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$ by

$$\mathfrak{g}_\hbar = \mathfrak{h}_\hbar \ltimes \mathfrak{h}_\hbar^{\vee,\mathrm{cont}}[1]$$

and would ask for $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_\hbar)$ as the closed quantum observable algebra. This is a separate descendant problem, not part of Remark 3.254, because the polynomial boundary representatives do not realize the coadjoint deformation-level action.

3.12.3 Graph formalism and specialization datum

The graph problem is the analytic realization problem for the same coefficient target. The algebra fixes the coefficients; a closed-open BV theory must produce those coefficients as renormalized observable products, with propagator, counterterm, anomaly, and trace-normalization choices specified. Costello's formalism supplies the general construction. The Hamiltonian input to that construction is the open task below.

Statement 3.258 *(Costello perturbative BV construction).* For an elliptic BV theory with a gauge fixing, heat kernel K_t, propagator

$$P_{\epsilon,L} = \int_{\epsilon}^{L} Q^{\mathrm{GF}} K_t \, dt,$$

and a renormalization scheme, Costello constructs finite-scale effective interactions by the RG operator

$$W(P_{\epsilon,L}, I[\epsilon]) = \hbar \log\bigl(\exp(\hbar \partial_{P_{\epsilon,L}}) \exp(I[\epsilon]/\hbar)\bigr).$$

The renormalized interaction is obtained by adding local counterterms and taking the $\epsilon \to 0$ limit. It has the connected-graph expansion

$$I[L] = \sum_{\Gamma \text{ connected}} \frac{\hbar^{b_1(\Gamma)+\sum_v g_v}}{|\operatorname{Aut}\Gamma|} W_\Gamma(P_{\epsilon,L}; I_{g_v}[\epsilon] + I^{\mathrm{CT}}_{g_v}(\epsilon)),$$

satisfies RG flow and locality, and satisfies the quantum master equation when the obstruction classes vanish [12]. The resulting quantum observables form a factorization algebra [15, 16].

Statement 3.259 *(Quarantined source: Costello–Li flat open-closed BCOV theorem).* Costello–Li prove a perturbative quantization of open-closed BCOV theory on flat $\mathbb{C}^d$, for d odd, unique up to the equivalences in their renormalization scheme, with open gauge algebra $\mathfrak{gl}(N \mid N)$ and compatibility under $\mathfrak{gl}(N \mid N) \hookrightarrow \mathfrak{gl}(N+k \mid N+k)$. In that flat open-closed BCOV/HCS theory they compute the anomaly cancellation needed for the graph construction [14]. Ordinary $\mathfrak{gl}_N$ and the mixed $\mathbb{R}^2 \times \mathbb{C}^2$ Hamiltonian brane model are not covered by this theorem without an additional reduction or summand argument. This statement is recorded only as an exclusion check.

Remark 3.260. The two source statements supply the universal renormalized BV framework and one major flat BCOV/HCS instance, subject to their hypotheses. They do not by themselves prove the graph realization for the Hamiltonian sector of this paper and are not inputs to Theorem 2.70. Here the brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ is a defect in the mixed space, not a spacetime boundary unless a half-space model is separately chosen.

Lemma 3.261 *(Finite-window obstruction for the full Hamiltonian sector).* The polynomial Hamiltonian Lie algebra of the formal symplectic plane admits no naive finite-dimensional Lie truncation compatible with the full Hamiltonian sector. More precisely:

(i) finite degree windows are not Lie subalgebras;
(ii) projected degree-window brackets need not satisfy Jacobi;
(iii) $\mathfrak{m}$-adic jet quotients are not Lie quotients in the presence of linear Hamiltonians;
(iv) there is no directed exhaustion by finite-dimensional Lie subalgebras containing the full polynomial Hamiltonian algebra.

Proof. Let H_d be the span of homogeneous polynomials of degree d, and let $\mathfrak{m} = (z_1, z_2)$. The Poisson bracket satisfies $\{H_p, H_q\} \subset H_{p+q-2}$. Thus

$$\{z_1^N, z_2^N\} = N^2 z_1^{N-1} z_2^{N-1},$$

which has degree $2N - 2 > N$ for $N > 2$, so degree $\leq N$ windows are not Lie subalgebras.

If one instead projects the bracket back to a degree window, Jacobi can fail. For the bracket $\pi_{\leq 3}\{-,-\}$, take

$$f = z_2, \qquad g = z_2^3, \qquad h = z_1^2 z_2.$$

A direct calculation gives the projected Jacobiator

$$[[f,g]_3,h]_3 + [[g,h]_3,f]_3 + [[h,f]_3,g]_3 = 6z_2^3.$$

The $\mathfrak{m}$-adic quotients are not stable under translations generated by linear Hamiltonians:

$$\{z_1^{N+1}, z_2\} = (N+1)z_1^N \notin \mathfrak{m}^{N+1}.$$

Hence $\mathfrak{h}/\mathfrak{m}^{N+1}$ is not a Lie quotient for the full Hamiltonian algebra.

Finally, any finite-dimensional Lie subalgebra containing

$$a = z_1^2 z_2^2, \qquad b = z_1^3 z_2$$

contains all iterated brackets

$$(\mathrm{ad}_a)^n(b) = (-4)^n z_1^{n+3} z_2^{n+1}, \qquad n \geq 0,$$

an infinite linearly independent set. Thus no directed finite-dimensional Lie-subalgebra exhaustion can contain all polynomial Hamiltonians. □

Remark 3.262 *(Linear heat operator versus BV kernel).* On the linear field complex

$$\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big),$$

the flat gauge fixing

$$Q^{\mathrm{GF}} = d^*_{\mathbb{R}^2} + \bar\partial^*_{\mathbb{C}^2}, \qquad \Delta = [d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}, Q^{\mathrm{GF}}]$$

gives finite-scale heat operators

$$K_t = K_t^{\mathrm{base}}\widehat{\otimes}\mathrm{id}, \qquad P^{\mathrm{op}}_{\epsilon,L} = \int_\epsilon^L Q^{\mathrm{GF}} K_t^{\mathrm{base}}\, dt\widehat{\otimes}\mathrm{id}$$

satisfying

$$[d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}, P^{\mathrm{op}}_{\epsilon,L}] = K_\epsilon - K_L$$

as continuous operators on the completed coefficient spaces. This is only an operator-level homotopy. To obtain Costello's BV propagator one also needs a coefficient coevaluation kernel for the Tate pair $\widehat{\mathfrak{h}} \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$. In the product/direct-sum topology used here, the formal sum of homogeneous coefficient coevaluations is not a convergent element of the completed tensor product. Thus the analytic graph problem requires a genuine Tate or weighted coefficient extension of the heat-kernel calculus, or a declared restriction to a smaller sector.

LEMMA 3.263 *(Tate Casimir obstruction for the full Hamiltonian sector).* In the product/direct-sum Tate topology

$$\mathfrak{h} = \prod_{d\geq 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d\geq 1} H_d^\vee,$$

the coefficient coevaluation

$$C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_{i} e_{d,i}\otimes e^{i}_{d}$$

is not an element of $\mathfrak{h}\widehat{\otimes}\mathfrak{h}^{\vee}_{\mathrm{cont}}$, nor of $\mathfrak{h}^{\vee}_{\mathrm{cont}}\widehat{\otimes}\mathfrak{h}$. Consequently the full Hamiltonian sector has a BV pairing but no Costello coefficient kernel in this topology. The operator-level homotopy of Remark 3.262 therefore does not define the BV propagator, the BV Laplacian, or renormalized graph weights for the full sector.

Proof. For each d, choose a basis $\{e_{d,i}\}_i$ of H_d and the dual basis $\{e^i_d\}_i$ of $H^{\vee}_d$. An element of a completed tensor product with the direct-sum factor $\bigoplus_d H^{\vee}_d$ has finite support in that direct-sum direction in the Tate convention used here. The displayed formal Casimir has a nonzero component in every $H^{\vee}_d$. Hence it is not a convergent tensor. The same support obstruction holds with the tensor factors reversed. Costello's heat-kernel BV formalism requires the inverse pairing as an actual distributional kernel, not merely an abstract continuous operator on the field complex. The weighted Tate completion of Theorem 3.85 resolves this Casimir obstruction by changing the coefficient category; the unweighted diagnosis here remains the structural reason that change is required. □

Problem 3.264 *(Hamiltonian specialization datum for Costello perturbative BV).* Costello supplies the universal renormalized BV graph calculus. For the Hamiltonian sector here, the required theorem data begin with an elliptic closed-open parent theory, or a stated dimensional reduction of one, whose restriction is this Hamiltonian BF sector and whose obstruction class vanishes. The required data are:

(i) choose the mechanism: either prove that the Hamiltonian cotangent BF complex

$$\Omega^{\bullet}(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1]\oplus\widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}[2]\big)$$

is literally a closed sector of a Costello-type open-closed BV theory, or prove that it is obtained by a specified dimensional reduction. The chosen mechanism must determine the field topology, degree -1 BV pairing, differential, local interaction, bracket degree, and a homotopy retract closed under the propagator and RG flow. Since $\widehat{\mathfrak{h}}$ is a completed Tate coefficient system, the proof cannot use a naive finite-degree or $\mathfrak{m}$-adic truncation for the full sector, by Lemma 3.261. It must instead construct a genuine Tate or weighted-coefficient extension of Costello's heat-kernel calculus [12] compatible with the Poisson bracket and RG flow, or restrict to a smaller Lie sector such as the classical $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ truncation and explicitly declare that translations, the conformal $\mathfrak{sl}_2$ sector, and the all-order Moyal closure have been removed. By Lemma 3.263, a damping factor inside the direct-sum dual fixed above is not enough; the coefficient category itself must be replaced by one in which the Casimir converges, or the sector must be restricted;

(ii) specify the asymptotic conditions on $\mathbb{C}^2$ and the brane-defect condition on $L = \mathbb{R}_{\mathrm{brane}}\times\{0\}\times\{p\}$;

(iii) choose the induced finite-scale gauge fixing, heat-kernel propagator, and BV Laplacian in this sector. The infrared limit $L\to\infty$ may be used only after the zero modes and harmonic projection have been fixed;

(iv) compute the obstruction/anomaly class for this mixed Hamiltonian sector. Concretely, the obstruction should be represented as a local functional class in the chosen bulk-defect local-functional complex. If the defect cotangent sector is represented by principal-part currents, the target must either be replaced by a current-compatible complex

$$\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}_w;\mathfrak{J}^{\mathrm{pp}}_{\partial}),$$

with wavefront and counterterm rules for $\mathcal{D}'_c$-coefficients, or the theorem must restrict B to regular compactly supported densities. In the weighted Tate category this means a class

$$\mathrm{Ob}_{w,n} \in H^1_{\mathrm{adm}}\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0^w, -\}\big)$$

coming from the n-loop failure of the quantum master equation after counterterms. Costello–Li's flat $\mathfrak{gl}(N \mid N)$ anomaly cancellation supplies a source theorem and a model; transfer is not automatic. To execute it one must prove that constant-Hamiltonian projection and connected trace extraction kill the ordinary $\mathfrak{gl}_N$ anomaly, or else replace the open sector by the appropriate super or special-linear variant. The first defect-local obstruction is the Capelli contact term

$$\omega_{\mathrm{Cap}}(A) = N\,\overline{\mathrm{Tr}(A)},$$

represented by

$$\mathrm{Tr}(A\,XY) - \mathrm{Tr}(A\,YX) = \hbar N\,\mathrm{Tr}(A) \quad \mod\ \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N).$$

It survives the naive connected trace projection, since the right-hand side is a nonempty connected trace when A is nonconstant. The anomaly datum must therefore be a filtered local class $\mathrm{Ob}_{\mathrm{bd}}$ whose scalar-symbol projection is

$$\sigma_{\mathrm{sc}}(\mathrm{Ob}_{\mathrm{bd}}) = \hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]].$$

A closed-exchange repair would have to supply $W_{\text{closed exch}}$ and a counterterm C with scalar leading class

$$[w_{\mathrm{ext}}] = -\hbar N[\bar{c}] \qquad \text{in } H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]]$$

and with

$$\omega_{\mathrm{Cap}} + W_{\text{closed exch}} + (Q + \{I_0, -\})C = 0.$$

The proved local mechanisms here are the specified supertrace and central-character normalizations; any exhaustive classification is controlled by the displayed obstruction complex;

(v) construct the bulk-to-defect kernel as a curved/cochain-level map κ to the Poisson central-operation object of the brane observables. After counterterms it must satisfy

$$\mathrm{Curv}(\kappa) = d\kappa + \frac{1}{2}[\kappa,\kappa] = 0.$$

On degree-zero Hamiltonians it must send $f(z_1, z_2)$ to the Weyl-ordered boundary observable $\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$, and it must satisfy the Ward identity and the quantum moment-map constraint. Its cotangent component must either restrict the principal-part current B to regular defect densities, or prove that the chosen Costello bulk-defect observable category admits the required compactly supported distributions with wavefront, counterterm, finite-window, and RG compatibility;

(vi) define the relevant boundary ribbon graphs: cyclic order, boundary components, automorphism factors, propagator labels, and the relation between Costello's loop-counting $\hbar$ and the Weyl contraction parameter. Equivalently, one must prove that a single boundary cross-contraction is $P/2$ in the Weyl convention;

(vii) fix the one-propagator boundary graph normalization. For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the raw first boundary commutator must have coefficient

$$\hbar(kn - lm) z_1^{k+m-1} z_2^{l+n-1},$$

equivalently the normalized commutator must have coefficient

$$(kn - lm) z_1^{k+m-1} z_2^{l+n-1};$$

(viii) after the first normalization, compute the three-propagator boundary graph normalization and prove that the connected order-$\hbar^3$ commutator contributes exactly

$$\frac{\hbar^3}{24} P^3(f, g)$$

with $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$;

(ix) give the convention for extracting connected single-trace and multi-trace coefficients as cumulants or primitive trace classes, and prove that this extraction cancels, renormalizes, or otherwise accounts for the normal-ordering shift of Remark 3.247;

(x) construct the renormalization scheme and counterterm locality statement in this mixed bulk-defect sector. For every divergent graph whose $\epsilon \to 0$ limit is subtracted, choose compatible representatives

$$C^{w_0}_{\Gamma,w}(\epsilon) \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{w_0}_w)$$

with

$$\pi_{w_0' w_0} C^{w_0'}_{\Gamma,w} = C^{w_0}_{\Gamma,w}, \qquad R^{w_0}_{w,w'} C^{w_0}_{\Gamma,w} = C^{w_0}_{\Gamma,w'},$$

prove support on the relevant bulk diagonal or brane-collision stratum, and check that RG flow carries these representatives inside the same local-functional subcomplex;

(xi) extend the bulk-to-defect kernel to the cotangent boundary-current target, or record the obstruction. In the reduced setting the extension must land in the principal-part current object of Theorem 3.237; before contraction it must supply the unreduced data of Remark 3.241. On the cotangent summand it must realize the principal-part coadjoint action of Theorem 3.230, and at the Moyal level the $\mathfrak{h}^\vee_\hbar$-action induced by the residue pairing. Polynomial one-ψ representatives are not an allowed substitute.

The computations here settle the formal polynomial Hamiltonian algebra and the formal Moyal coefficients. The analytic graph-realization problem is the sector-identification, local counterterm, brane-defect kernel, principal-part-current compatibility, and first/third normalization problem listed above.

Theorem 3.265 *(Scalar QME obstruction and cancellation branches).* Let $A = \mathbb{C}[z_1, z_2]$, $\bar{A} = A/\mathbb{C} \cdot 1$, and

$$\omega(f, g) = [\{f, g\}]_0.$$

Write $\bar{c}$ for the induced Lie two-cocycle on $\bar{A}$. In the ordinary scalar-reduced $\mathfrak{gl}_N$ brane-defect model, the scalar-symbol part of the mixed QME obstruction is represented by

$$\mathrm{Ob}_{\mathrm{sc}}(\gamma_1; a, f; b, g) = \hbar N\, \omega(f, g) \int_{\mathbb{R}} a(t) b(t) \gamma_1(t)\, dt,$$

and its associated-graded CE class is

$$\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]].$$

For $N > 0$, this class is nonzero in the scalar-reduced source. Hence no internal scalar-reduced local counterterm kills its leading scalar symbol.

The cancellation mechanisms have distinct types.

(i) On the unreduced source A, the functional $\eta(f) = -[f]_0$ satisfies $d_{\mathrm{CE}}\eta = \omega$. Thus

$$\hbar N\omega + d_{\mathrm{CE}}(-\hbar N\eta) = 0$$

before quotienting constants.

(ii) Equivalently, after replacing the scalar-reduced source by the unreduced source with central character $\chi_N(f) = N[f]_0$, the scalar term is absorbed before it becomes a reduced obstruction class.

(iii) Equivalently in the scalar-replacement category, define the opposite scalar central extension

$$\bar{A}^{\mathrm{cl}}_{-\hbar N\bar{c}} = \bar{A} \oplus \mathbb{C}[[\hbar]]K_{\mathrm{cl}},$$

with

$$[(f, aK_{\mathrm{cl}}), (g, bK_{\mathrm{cl}})] = \big(\{f, g\}_{\bar{A}}, -\hbar N\,\bar{c}(f, g)K_{\mathrm{cl}}\big), \qquad [K_{\mathrm{cl}}, -] = 0.$$

If $k^\vee(K_{\mathrm{cl}}) = 1$ and $k^\vee(\bar{A}) = 0$, then for $J_{\mathrm{op}} = -k^\vee$

$$d_{\mathrm{CE}}J_{\mathrm{op}} = -\hbar N\,\bar{c}, \qquad \mathrm{Ob}_{\mathrm{sc}} + d_{\mathrm{CE}}J_{\mathrm{op}} = 0.$$

The cancellation is compatible with scalar finite-window projections retaining z_1, z_2. It is a source replacement; after forgetting K_{cl}, the class $\hbar N[\bar{c}]$ on $\bar{A}$ reappears.

(iv) In the balanced supertrace model $V = \mathbb{C}^{N|N}$, the scalar coefficient is $\mathrm{sdim}\,V = 0$; the scalar brane-defect QME representative vanishes in that model.

(v) The finite-window algebraic central-contact cone is a genuine changed source branch. After adjoining a defect-supported central closed source line $\mathbb{C}[[\hbar]]e_M$ with

$$\Xi^{\mathrm{alg,sc}}_M(e_M) = -\hbar N\,\bar{c}_M$$

on the scalar-contact line, the choice $W_M = e_M$ and $C_M = 0$ cancels the finite-window scalar row, and the corresponding β, μ, λ_C classes vanish in that enlarged algebraic cone.

(vi) In the original scalar-reduced ordinary $\mathfrak{gl}_N$ branch, a same-branch closed-exchange or brane-defect contribution can cancel the scalar symbol only if its leading scalar class obeys

$$[w_{\mathrm{ext}}] = -\hbar N[\bar{c}] \qquad \text{in } H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]].$$

Equivalently, a Costello-local closed-exchange solution must supply finite-window closed-source complexes $\mathcal{X}^\bullet_{\mathrm{sc},w,M}(I)$, cochain maps

$$\Xi^{\mathrm{Cost,sc}}_M : \mathcal{X}^\bullet_{\mathrm{sc},w,M}(I) \to \mathfrak{Q}^\bullet_{\mathrm{sc},w,M}(I),$$

a filtered scalar chain projection $\widehat{\sigma}_{\mathrm{sc},M}$, cocycles W_M, and counterterms C_M such that

$$\mathrm{Ob}_{\mathrm{sc},M} + \Xi^{\mathrm{Cost,sc}}_M(W_M) + d_M C_M = 0, \qquad H^1(\widehat{\sigma}_{\mathrm{sc},M}\Xi^{\mathrm{Cost,sc}}_M)[W_M] = -\hbar N[\bar{c}_M],$$

with compatible truncation and vanishing inverse-limit β, μ, λ_C classes.

The algebraic cone in (4) is not a Costello-local closed-exchange theorem until its formal generator is replaced by a closed source complex, finite-scale propagator, bulk-to-defect graph map, source-face Hom differential, local counterterm support, scalar chain projection, and compatible finite-window representatives. For current genuine Costello-local towers the ordinary scalar-reduced branch is blocked first by

$$o^{(1),\mathrm{sc}}_{\sigma,M} = \hbar N[\bar{c}_M] \neq 0,$$

and, after any externally supplied scalar gate, by the image obstruction

$$\beta^{\mathrm{Cost,sc}}_M = [\hbar N[\bar{c}_M]] \in \operatorname{coker} H^1(\widehat{\sigma}_{\mathrm{sc},M}\Xi^{\mathrm{Cost,sc}}_M).$$

No exhaustive classification of scalar QME cancellations is asserted. Any further mechanism must be an explicitly constructed null-homotopy in the filtered QME obstruction complex whose scalar-symbol class cancels $\hbar N[\bar{c}]$.

Proof. The representative and its nonzero associated-graded class are exactly Theorem E.12. Non-exactness in the reduced source is the last part of that theorem: a primitive on $\bar{A}$ would lift to a primitive on A vanishing on 1, but $\omega(z_1, z_2) = 1$ and $\{z_1, z_2\} = 1$.

The first branch is Lemma E.11, multiplied by $\hbar N$. The central-character branch is Theorem E.13; it is a source replacement before scalar reduction, not a counterterm inside the original reduced obstruction complex. The opposite central-extension branch is the same CE sign calculation written as an honest central extension of $\bar{A}$: with $(d_{\mathrm{CE}}\lambda)(x, y) = -\lambda([x, y])$, the central bracket $-\hbar N\bar{c}(f, g)K_{\mathrm{cl}}$ gives $d_{\mathrm{CE}}k^\vee = \hbar N\bar{c}$, hence $d_{\mathrm{CE}}(-k^\vee) = -\hbar N\bar{c}$. Jacobi is exactly the cocycle identity for $\bar{c}$, and finite-window compatibility follows because $\bar{c}$ is detected on the retained pair z_1, z_2. The balanced-supertrace branch is Theorem E.14. The algebraic central-contact cone is the finite-window source extension recorded in the scalar QME obligation lane; its differential is zero and its scalar image is declared to be $-\hbar N\bar{c}_M$, so the displayed cancellation is immediate in that enlarged source. The same declaration is exactly why it is not yet a Costello-local closed exchange. The necessary condition for any same-branch ordinary cancellation is Theorem E.47. The additional Costello-local image and scalar-chain obstructions are the filtered Hom conditions in Problem 3.271. These results prove the listed branches and no more. □

Remark 3.266 *(Matched-conventions boundary).* This local Hamiltonian BF sector supplies only the $d = 2$ formal-disk normalization for the Weyl/Moyal target. It does not supply a compact Calabi-Yau target, a CoHA, a quintic sector, an HKQ/CDGP/GV/OSV input, an Abel–Jacobi normal function, a Vol III Φ_d-datum, a (Σ_{d-1}, C)-specialization, an Igusa product lattice, a Borcherds Weyl chamber, or a BKM orientation character. Compact Calabi-Yau, MNOP/chiral-volume, κ_{ch}, κ_{BKM}, compact CY_3 BCOV anomaly, $K3 \times E$, CoHA, quintic, OSV/GV, Abel–Jacobi, Igusa, Borcherds, and BKM uses therefore require their own matched-conventions theorem. The Weyl algebra fixes the target coefficients; the elliptic BV source is supplied by the descent, QME, completion, and defect data.

Remark 3.267 *(Parabolic functoriality of the local model).* The completion along the brane divisor $z_1 = 0$ used here breaks the formal symplectomorphism group $\mathrm{Symp}_{\mathrm{form}}(\mathbb{C}^2, 0)$ to the parabolic stabiliser

$$P_{(z_1)} = \{\varphi \in \mathrm{Symp}_{\mathrm{form}}(\mathbb{C}^2, 0) : \varphi^*(z_1) \in (z_1)\}.$$

Any cross-volume transfer therefore supplies a parabolic-equivariant comparison rather than a full $\mathrm{Symp}_{\mathrm{form}}$-equivariant one.

3.12.4 Formal Weyl/Moyal target, all orders

We use the Weyl/Moyal Rees convention for the formal quantization of the constant Poisson tensor [7]:

$$f * g = m \circ \exp\left(\frac{\hbar}{2}(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})\right)(f \otimes g).$$

The factor $1/2$ is already determined by linear Hamiltonians in this Weyl convention: if $u = az_1 + bz_2$ and $v = cz_1 + dz_2$, then $[\widehat{u}, \widehat{v}] = \hbar(ad - bc)$, so the Baker-Campbell-Hausdorff formula for Weyl-ordered exponentials contributes $\exp(\hbar P(u, v)/2)$. Differentiating these exponentials gives the cross-contraction $P/2$ in the formal Weyl convention. Then the star commutator is odd in $\hbar$:

$$[f, g]_* = \hbar\{f, g\} + \frac{\hbar^3}{24} P^3(f, g) + O(\hbar^5),$$

where $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$. Thus the $\hbar^2$ coefficient of the raw star commutator vanishes in this quantization convention. The normalized quantum Hamiltonian bracket is instead

$$\{f, g\}_\hbar = \hbar^{-1}[f, g]_* = \sum_{s \geq 0} \frac{\hbar^{2s}}{2^{2s}(2s+1)!} P^{2s+1}(f, g).$$

Proposition 3.268 *(Moyal coefficients on monomials).* Let $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$. For every $r \geq 0$,

$$P^r(f, g) = \sum_{a=0}^{r} (-1)^a \binom{r}{a} (k)_{r-a} (l)_a (m)_a (n)_{r-a} z_1^{k+m-r} z_2^{l+n-r},$$

where $(s)_q = s(s-1)\cdots(s-q+1)$ and $(s)_0 = 1$. Terms with a negative displayed exponent are understood to be zero; equivalently the corresponding falling factorial already vanishes. Consequently

$$[f, g]_* = \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{2}{2^r r!} \hbar^r P^r(f, g),$$

so the second-order coefficient of the raw commutator vanishes. The first four odd raw coefficients are

$$\hbar P(f, g), \qquad \frac{\hbar^3}{24} P^3(f, g), \qquad \frac{\hbar^5}{1920} P^5(f, g), \qquad \frac{\hbar^7}{322560} P^7(f, g).$$

Proof. Since

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1},$$

the two summands commute as differential operators on the tensor product. The binomial theorem gives

$$P^r = \sum_{a=0}^{r} (-1)^a \binom{r}{a} \partial_{z_1}^{r-a} \partial_{z_2}^{a} \otimes \partial_{z_1}^{a} \partial_{z_2}^{r-a}.$$

Applying this to $z_1^k z_2^l \otimes z_1^m z_2^n$ gives the displayed falling factorial formula. Interchanging f and g changes $P^r(f, g)$ by the sign $(-1)^r$, so the star commutator keeps exactly odd r and contributes the factor $2/(2^r r!)$ in order r. □

The all-order Moyal expansion, the reduced commutator, and the Capelli-renormalized stable trace have an unconditional formal target and an exact decorated radial obstruction theorem. The formal target is the Weyl/Moyal algebra of the constant symplectic formal disk. The ordinary Weyl-trace realization is governed by the quantum shear primitive $E_{a,b,N}$, equivalently by the free normal-diagram classes $\mathfrak{o}_{a,b}$, and more sharply by the dual-potential class $\Omega^{\mathrm{rad}}_{a,b}$ or decorated PBW Stokes identity $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$. This is the radial-parts algebraic component of the quantum P_0-operation comparison. It is not a construction of the descendant sector and it is not the Costello graph realization.

Theorem 3.269 *(Degree-zero Moyal target and decorated radial obstruction).* Let $\mathfrak{h}_\hbar = \bar{A}_\hbar$ be the Moyal Hamiltonian Lie algebra, and let $\mathfrak{h}_{\hbar,u}$ be its shifted-cotangent coordinate copy, with generator u_f labelled by $f \in \mathfrak{h}_\hbar$. Let

$$H^{\circ}_{\mathrm{Ham,conn}}(O^{\mathrm{q}}_{\mathrm{brane},\infty})_{\mathrm{ren}}$$

denote the stable connected Hamiltonian trace sector generated by the Capelli-renormalized J-basis transition classes $\left[\overline{J_N(f)}\right]_{N\to\infty}$. These are associated-graded stable classes in the renormalized transition system, not naive normal-ordered block-restriction limits.

The following data are proved without any unqualified all-bidegree radial claim.

(i) The Weyl/Moyal coefficients of Proposition 3.268 give the all-order formal Hamiltonian source. The odd raw coefficients are $2\hbar^r P^r/(2^r r!)$, and the first four are $1, 1/24, 1/1920, 1/322560$.

(ii) The Capelli-renormalized J_N-basis of Lemma 3.249 gives the stable connected trace target as a $\mathbb{C}[[\hbar]]$-module, after removing the empty trace and decomposable products.

(iii) The ordinary Weyl-trace image clause is equivalent, after the homomorphism, injectivity, and pure-power parts of the radial package, to the moment-ideal primitive $E_{a,b,N} \in \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)$ of Lemma F.15. In the free normal-diagram model this is the vanishing of

$$\mathfrak{o}_{a,b} = [R^{\mathrm{free}}_{a,b}] \in \operatorname{coker} C_{a,b}$$

with functorial lifts $K_{a,b} \in \ker \partial$. Equivalently, in the dual-potential obstruction theorem,

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^{+}_{a,b})] \in \operatorname{coker} B_{a,b}$$

must vanish, or equivalently the decorated PBW Stokes identity $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$ must hold; a failure is a signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$ with $\lambda(E^{+}_{a,b}) - \phi(T_{a,b}) \neq 0$. Proposition F.27 proves this primitive for every edge bidegree $(2, b)$ and $(a, 2)$. Proposition F.20, Proposition F.28, and the current exact normal-form script supply finite interior certificates. For all $a, b \geq 3$, the missing theorem is the associated-graded two-colour necklace homotopy, equivalently the dual-potential signed-row theorem above.

If the radial image package of Statement F.8 is supplied, or if the equivalent quantum shear primitive package above is constructed together with the homomorphism, injectivity, and pure-power clauses, then

$$\Phi^{(0)}_{\hbar} : \mathfrak{h}_{\hbar,u} \xrightarrow{\sim} H^{\circ}_{\mathrm{Ham,conn}}(O^{\mathrm{q}}_{\mathrm{brane},\infty})_{\mathrm{ren}}, \qquad u_f \longmapsto \partial_{\mathrm{bb},\hbar}(f) = [\overline{J_N(f)}]_{N\to\infty}$$

is the u_f-coordinate component of the desired quantum P_0-operation: it is a $\mathbb{C}[[\hbar]]$-linear isomorphism onto the renormalized degree-zero Hamiltonian sector. It is a Lie-algebra map on these Hamiltonian coordinates,

$$\Phi^{(0)}_{\hbar}(u_{\{f,g\}_\hbar}) = \hbar^{-1}[\Phi^{(0)}_{\hbar}(u_f), \Phi^{(0)}_{\hbar}(u_g)]_{\mathrm{conn,raw}}.$$

Under this assumption, the associated graded at $\hbar = 0$ is the degree-zero Hamiltonian restriction of Φ in Theorem 2.70. The descendant ($\mathfrak{h}^\vee$, one-antifield Koszul) sector remains governed by Problem 3.60; see Problem 3.273 below for the pinned obstruction.

Proof. Item (1) is Proposition 3.268. Item (2) is Corollary 3.250 and Lemma 3.249. Item (3) is the chain Proposition F.14, Lemma F.15, Definition F.21, and Statement F.22, sharpened by Proposition F.25 to $\Omega^{\mathrm{rad}}_{a,b} = 0$, equivalently $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$, with the signed-row failure criterion. The edge theorem is supplied by Proposition F.27. The finite interior statements cited there are exact free normal-diagram certificates; by Lemma F.19 they are all-rank identities in their displayed windows.

Under the radial image package, the displayed $\mathbb{C}[[\hbar]]$-linear bijection follows from the stable Capelli-renormalized generators. Flatness over $\mathbb{C}[[\hbar]]$ follows from the unipotent Capelli triangular system over $\mathbb{C}[[\hbar N]]$ (Lemma 3.249(ii)); the associated graded is the classical trace map because the $r = 0$ triangular term is the leading symbol (Lemma 3.249(iv)). The Lie identity is precisely Theorem 3.251, passed to the connected stable limit. The cotangent cochain summand is governed by the separate Tate-coefficient obstruction datum. □

THEOREM 3.270 *(Quantum coefficient surface, componentwise).* Fix an exponential degree weight $w(d) = q^d$, $q > 1$, and work over $\mathbb{C}[[\hbar]]$. Write

$$\mathfrak{h}_{w,\hbar} = \mathfrak{h}_w[[\hbar]], \qquad \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}} = \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[[\hbar]].$$

The Moyal bracket on $\mathfrak{h}_{w,\hbar}$ is transported from $\bar{A}_\hbar$ by the diagonal weight rescaling of Definition 3.74. Let

$$\mathcal{K}^w_{\mathrm{BV}} = (C_{\mathfrak{h},w}, P^w_{\epsilon,L}, \mathcal{E}_w)$$

be the weighted coefficient-kernel datum of Theorem 3.85. For an interval $I \subset \mathbb{R}$, let $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$ be the weighted reduced Moyal principal-part current envelope

$$\mathfrak{h}^w_{\hbar,I} = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}_{w,\hbar}, \qquad \mathcal{P}^w_{\hbar,I} = \mathcal{D}'_c(I)\widehat{\otimes}\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}},$$

$$A^{\mathrm{pp},w}_{\partial,\hbar}(I) = \widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}(\mathfrak{h}^w_{\hbar,I})\widehat{\otimes}\widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}(\mathcal{P}^w_{\hbar,I}[1]).$$

Its dense strict Matlis subtarget is the reduced Moyal principal-part target $A^{\mathrm{pp}}_{\partial,\hbar}(I)$ of Construction 3.156, obtained by replacing $\mathfrak{h}_{w,\hbar}$ with $\bar{A}_\hbar$ and $\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}$ with $\mathcal{P}[[\hbar]]$. The Moyal coadjoint action on the weighted envelope is the finite-window continuous extension of the action in that construction; the assertion is restricted to the vertexwise polynomial-degree-bounded weighted sector. Let $T_{\mathrm{Ham},\hbar} = H^{\circ}_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})_{\mathrm{ren}}$ be the stable connected Hamiltonian trace target of Theorem 3.269. Its formal Moyal and Capelli pieces are proved there; the ordinary-trace Lie comparison is pinned to the dual-potential radial obstruction $\Omega^{\mathrm{rad}}_{a,b}$, equivalently the decorated PBW Stokes identity $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$, with all edge strips proved and all-interior closure requiring the signed-row vanishing theorem. In the balanced $\mathfrak{gl}(N \mid N)$ supertrace branch put o_{sc} for the zero scalar QME representative of Theorem E.14; on a balanced scalar-contact habitat this is the zero filtered scalar projection of Proposition E.16.

The componentwise quantum coefficient surface is the product prefactorization target

$$\mathfrak{S}_{\hbar,w}(I) = \mathcal{K}^w_{\mathrm{BV}} \times \{\mathrm{o}_{\mathrm{sc}}\} \times A^{\mathrm{pp},w}_{\partial,\hbar}(I) \times T_{\mathrm{Ham},\hbar},$$

where the first, second, and fourth factors are constant in I, and the principal-part current factor uses extension by zero and the factorization product of Theorem 3.237. This product target carries the following proved data.

(i) The weighted Casimir $C_{\mathfrak{h},w}$ and the finite-scale BV propagator $P^{w}_{\epsilon,L}$ converge in the weighted Tate coefficient category. Each fixed Costello graph with vertexwise polynomial-degree-bounded Hamiltonian labels remains in that category.

(ii) On every balanced scalar-contact habitat of Definition E.15, the balanced $\mathfrak{gl}(N \mid N)$ supertrace model has zero filtered scalar projection $\widehat{\sigma}_{\mathrm{bal,sc}} = 0$, zero chain-map defect, and vanishing scalar-projection obstruction classes $o^{(r)}_{\sigma,w,\mathrm{bal}}(I)$. In the ordinary scalar-reduced $\mathfrak{gl}_N$ branch the scalar class is instead $\hbar N[\bar{c}]$ and requires the alternative mechanisms of Theorem 3.265.

(iii) The reduced weighted quantum cotangent current brackets are

$$\{B_f(a), B_g(b)\}_\hbar = B_{\{f,g\}_\hbar}(ab), \qquad \{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$

$$\{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0.$$

Here $\mathrm{ad}^\vee_{f,\hbar}$ is the Moyal coadjoint action on $\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}$, whose restriction to $\mathcal{P}[[\hbar]]$ is the action of Construction 3.156. It is not the polynomial one-ψ PBW action. Thus the classical principal-part current package is transported to the quantum target only after replacing $f \cdot \rho$ by the full Moyal coadjoint action and after imposing weighted finite-window continuity.

(iv) Under the radial image package, equivalently after solving the quantum shear primitive package of Theorem 3.269, the degree-zero Hamiltonian generator map

$$\Phi^{(0)}_\hbar : \mathfrak{h}_{\hbar,u} \xrightarrow{\sim} T_{\mathrm{Ham},\hbar}$$

is the $\mathbb{C}[[\hbar]]$-linear Lie comparison of Theorem 3.269.

Consequently the weighted coefficient/BV kernel, the balanced scalar-contact zero projection, the reduced weighted Moyal principal-part current target, and the radial-gated degree-zero Moyal coefficient comparison form a genuine componentwise theorem surface. This theorem does not assert a full non-scalar Costello graph realization, an unreduced factorization centre morphism, or an all-interior radial theorem beyond the obstruction formulation above.

Proof. Item (1) is Theorem 3.85: its parts (a)–(c) give Casimir convergence, heat-kernel propagator extension, and graphwise weighted RG compatibility. Item (2) is Theorem E.14 together with Proposition E.16; the ordinary scalar alternatives and obstruction criterion are assembled in Theorem 3.265.

At $\hbar = 0$, the current brackets are the reduced principal-part brackets of Theorem 3.237 and Corollary B.8. The quantum target is not obtained by deforming the polynomial one-ψ labels. On the dense strict Matlis subtarget it is the semidirect Moyal current complex of Construction 3.156, whose action is by $(\mathrm{ad}^\vee_{f,\hbar}\rho)(g) = -\rho(\{f,g\}_\hbar)$. The extension from $\mathcal{P}[[\hbar]]$ to $\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}$ is windowwise continuous: the Moyal coadjoint formula is finite for each polynomial Hamiltonian, and condition (W2) in Definition 3.71 gives the required finite-window bounds for vertexwise polynomial-degree-bounded inputs. Proposition 3.155 and Theorem 3.157 show why this hypothesis is necessary: the PBW one-ψ label map does not intertwine the Moyal coadjoint action. Hence item (3) is a target-side reduced current construction with the correct quantum action, not an unproved transport from polynomial descendants.

Item (4) is Theorem 3.269: its formal Moyal, Capelli, and edge-strip pieces are proved, while the all-interior ordinary-trace comparison is exactly Proposition F.25, namely $\Omega^{\mathrm{rad}}_{a,b} = 0$, equivalently decorated PBW Stokes $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$, with signed $B^{*}_{a,b}$-rows as the failure certificates. The product prefactorization structure is componentwise: the weighted BV kernel, scalar zero representative, and stable trace target are constant in the interval, while the principal-part current factor is functorial by extension by zero

and has the disjoint-interval product of Theorem 3.237. These components are independent targets, so their product carries exactly the displayed surface and no additional Costello graph/QME assertion. □

Problem 3.271 *(Non-scalar quantum P_0-operation obstruction complex).* The finite-window theorem after Theorem 3.270 is the non-scalar Costello graph/QME obstruction criterion. It is the vanishing problem in the filtered local-functional complex

$$\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I) = \Big(O^{\mathrm{bd}}_{\mathrm{loc}}\big(\mathcal{E}_w|_I;\ A^{\mathrm{pp},w}_{\partial,\hbar}(I)\big), d_{\mathrm{QME}} = Q + \{I^w_0, -\}_\hbar\Big),$$

where the local functionals have compact support on the brane line, coefficients in the weighted field complex $\mathcal{E}_w$, and target currents in $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$. The scalar-symbol projection is only associated-graded data until it is lifted through the scalar-contact filtration. Thus the first obstruction is not the reduced non-scalar class, but the filtered lift of the scalar projection. Choose a complete d_{QME}-stable scalar-contact filtration $F^\bullet\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$. For any filtered linear lift s of the scalar-symbol map, the successive chain-map defects define classes

$$o^{(r)}_{\sigma,w}(I) \in H^1\Big(\mathrm{Hom}\big(\mathrm{gr}_F\,\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I), C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]]\big)_{-r}\Big), \qquad r > 0,$$

with the Hom differential of Lemma E.4. The first theorem obligation is the successive vanishing of these $o^{(r)}_{\sigma,w}(I)$, with compatible choices in the completed finite-window inverse limit. Only then is there a filtered chain projection

$$\widehat{\sigma}_{\mathrm{sc}}\colon \mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I) \longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]]$$

whose associated graded is the scalar-symbol projection of Definition E.3.

The balanced scalar-contact branch has a proved positive habitat: on $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I) \subset \mathfrak{Q}^{\bullet}_{w,\partial,\hbar,\mathrm{str}}(I)$, Proposition E.16 gives the zero filtered chain projection $\widehat{\sigma}_{\mathrm{bal,sc}} = 0$ and kills every $o^{(r)}_{\sigma,w,\mathrm{bal}}(I)$. Extending this result to the full graph/counterterm habitat is a separate theorem obligation: either prove that the relevant scalar-contact symbols still factor through sdim $\mathbb{C}^{N|N}$, or compute and kill the extension classes of Remark E.17.

After this lift has been constructed, the balanced supertrace branch requires a continuous bulk-to-defect kernel

$$\kappa_{\hbar,w,I}\colon C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}) \longrightarrow \mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$$

for the current source of Definition B.14. Its curvature

$$\mathrm{Ob}_{w,\partial,\hbar} = d_{\mathrm{QME}}\kappa_{\hbar,w,I} + \frac{1}{2}[\kappa_{\hbar,w,I}, \kappa_{\hbar,w,I}]_\hbar$$

must be a cocycle satisfying

$$\widehat{\sigma}_{\mathrm{sc}}(\mathrm{Ob}_{w,\partial,\hbar}) = 0$$

and the remaining reduced non-scalar class

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1\big(\ker\widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}}\big)$$

must vanish. At finite windows this is the tower $\mathcal{Q}^{\bullet}_{w,M}(I) = \ker\widehat{\sigma}_{\mathrm{sc},M}$ of Definition 3.97. The compatible counterterm recursion is exactly the criterion of Theorem 3.98: at each $\hbar$-order the inverse-limit H^1-class and the $\varprojlim^1 H^0$-primitive-compatibility class must vanish.

More explicitly, let n_0 be the first order where lower-order choices leave a residual

$$\mathfrak{v}^{\mathrm{ns}}_{n_0,M} \in Z^1(Q^\bullet_{w,M}(I)).$$

Counterterms alone close this order exactly on the vanishing locus of the obstruction vector

$$\mathfrak{O}^{\mathrm{ns}}_{n_0} = \big(([\mathfrak{v}^{\mathrm{ns}}_{n_0,M}])_M, \lambda_{n_0}\big)$$

of Theorem 3.98. A closed-exchange repair is additional cochain data: finite-window complexes $\mathcal{X}^\bullet_{w,M}(I)$, compatible cochain maps

$$\Xi_M : \mathcal{X}^\bullet_{w,M}(I) \longrightarrow Q^\bullet_{w,M}(I),$$

and cochains satisfying the branch equation

$$\mathfrak{v}^{\mathrm{ns}}_{n_0,M} + \Xi_M(W_{n_0,M}) + d_M C_{n_0,M} = 0, \qquad W_{n_0,M} \in Z^1(\mathcal{X}^\bullet_{w,M}(I)), \quad C_{n_0,M} \in Q^0_{w,M}(I).$$

For a fixed window this equation is solvable if and only if

$$[\mathfrak{v}^{\mathrm{ns}}_{n_0,M}] \in -\operatorname{im}\Big(H^1(\Xi_M) \colon H^1(\mathcal{X}^\bullet_{w,M}(I)) \to H^1(Q^\bullet_{w,M}(I))\Big).$$

In the inverse-limit tower the exact closed-exchange obstruction is the triple

$$\Big(\beta^{\mathrm{cl}}_{n_0},\ \mu^{\mathrm{cl}}_{n_0},\ \lambda^{\mathrm{cl}}_{n_0}\Big),$$

where

$$\beta^{\mathrm{cl}}_{n_0} \in \operatorname{coker}\left(\varprojlim_M H^1(\mathcal{X}^\bullet_{w,M}(I)) \xrightarrow{\varprojlim H^1(\Xi_M)} \varprojlim_M H^1(Q^\bullet_{w,M}(I))\right),$$

$$\mu^{\mathrm{cl}}_{n_0} \in \varprojlim{}^1_M H^0(\mathcal{X}^\bullet_{w,M}(I)), \qquad \lambda^{\mathrm{cl}}_{n_0} \in \varprojlim{}^1_M H^0(Q^\bullet_{w,M}(I)).$$

Vanishing of this triple is the exact typed criterion for a compatible closed-exchange kernel plus local counterterm to kill the first non-scalar residual. Constructing $\mathcal{X}^\bullet$, Ξ, and such a class $[W]$ remains open.

For a fixed binary generator pair x, y, the row form of the centrality obstruction is the same elementary finite-window matrix problem. Once the finite package supplies a scalar-zero basis ρ^M_i for the degree $|x| + |y|$ residual rows and candidate homotopy rows η^M_j of degree $|x| + |y| - 1$, write

$$R^{\mathrm{cent}}_{x,y,M} = \sum_i r^M_i \rho^M_i, \qquad d_M \eta^M_j = \sum_i A^M_{ij} \rho^M_i.$$

Then $H^M_{x,y} = \sum_j h^M_j \eta^M_j$ is a binary centrality homotopy in the supplied finite row span if and only if

$$A^M h^M = -r^M.$$

The compatible inverse-limit homotopy additionally requires

$$[P^{M,N} h^M - h^N_{\pi x, \pi y}] = 0 \quad \text{in} \quad H^{|x|+|y|-1}(Q^\bullet_{w,N}(I)).$$

Thus the coefficient-current zero row is not evidence for an enlarged Costello row; the enlarged row is the displayed matrix obstruction.

Equivalently, the theorem obligation is to construct compatible finite-window local counterterms $C^{w_\circ}_{\Gamma,w}(\epsilon)$ and the bulk-to-defect kernel $\kappa_{\hbar,w,I}$ such that

$$\mathrm{Curv}(\kappa_{\hbar,w,I} + C_{\hbar,w}) = d_{\mathrm{QME}}(\kappa_{\hbar,w,I} + C_{\hbar,w}) + \frac{1}{2}[\kappa_{\hbar,w,I} + C_{\hbar,w}, \kappa_{\hbar,w,I} + C_{\hbar,w}]_\hbar = 0,$$

with finite-window compatibility

$$\pi_{w'_\circ w_\circ} C^{w'_\circ}_{\Gamma,w} = C^{w_\circ}_{\Gamma,w'}, \qquad R^{w_\circ}_{w,w'} C^{w_\circ}_{\Gamma,w} = C^{w_\circ}_{\Gamma,w'}.$$

This is the precise obstruction complex behind the desired

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]) \longrightarrow Z^{P_{\circ,\hbar}}_{\mathrm{fact}}(\mathrm{Obs}^\hbar_{\mathrm{open}})|_{\mathfrak{S}_{\hbar,w}}.$$

The degree-zero radial comparison supplies only the Hamiltonian generator restriction and is itself pinned to the moment-ideal primitive of Lemma F.15. The weighted Tate theorem supplies the coefficient kernel and graphwise boundedness. The balanced scalar-contact theorem constructs the zero filtered scalar projection only on its balanced habitat. It does not prove that the full Costello graph/counterterm habitat lies there, construct the bulk-to-defect kernel, solve the closed-exchange branch, or kill the reduced non-scalar class.

PROPOSITION 3.272 *(Finite-window acceptance gate for the first θ_3 row).* In a larger finite-window graph package, let Θ_3 be a genuine marked source-face row of $\hbar$-order 3. The exposed one-row scalar-zero summand is

$$Q^0_{\theta_3,M} = 0, \qquad Q^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, \qquad d_M = 0,$$

with

$$\mathsf{E}_{\theta_3,M} = -K^M_{\Theta_3}(\zeta_{M,\nu_3}), \qquad \mathfrak{v}^{\mathrm{ns}}_{\theta_3,M} = \mathsf{E}_{\theta_3,M}, \qquad \widehat{\sigma}_{\mathrm{sc},M}(\mathsf{E}_{\theta_3,M}) = 0.$$

In this supplied subcomplex the class $[\mathsf{E}_{\theta_3,M}]$ is nonzero. It is killed in an enlarged finite-window package precisely by one of the following data, together with scalar-zero and finite-window transport compatibility.

First, a CE-source ancestor

$$\eta_{\theta_3,M}, \qquad d_{\mathrm{CE}}\eta_{\theta_3,M} = \zeta_{M,\nu_3},$$

whose induced degree-zero row $C^{\mathrm{CE}}_{\theta_3,M} = K^M_{\Theta_3}(\eta_{\theta_3,M})$ satisfies

$$d_M C^{\mathrm{CE}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}.$$

Second, a Costello local counterterm primitive $C^{\mathrm{ct}}_{\theta_3,M} \in Q^0_{w,M}(I)$ satisfying

$$d_M C^{\mathrm{ct}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}, \qquad \widehat{\sigma}_{\mathrm{sc},M}(C^{\mathrm{ct}}_{\theta_3,M}) = 0.$$

Third, a complete companion-face table $F_{\Theta_3,M}$ with signs $\epsilon^{\mathrm{CE}}_{\Theta_3,\nu}$ and rows $E^M_{\Theta_3,\nu}$ such that

$$\sum_{\nu \in F_{\Theta_3,M}} \epsilon^{\mathrm{CE}}_{\Theta_3,\nu} E^M_{\Theta_3,\nu} = 0, \qquad E^M_{\Theta_3,\nu_3} = \mathsf{E}_{\theta_3,M}.$$

In the first two cases the finite-window tower must also satisfy

$$[\pi_{M,N} C_{\theta_3,M} - C_{\theta_3,N}] = 0 \quad \text{in} \quad H^0(Q^\bullet_{w,N}(I)).$$

In the companion-face case the source-face transport matrices must satisfy

$$\pi_{M,N} E^M_{\Theta_3,\nu} = \sum_{\nu'} v^{M,N}_{\Theta_3,\nu;\Theta_3,\nu'} E^N_{\Theta_3,\nu'}, \qquad v^{M,P} = v^{N,P} v^{M,N},$$

and must preserve the displayed signed zero sum. A formal symbol C_{θ_3} without the differential $d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$, scalar-zero value, and tower compatibility is not a counterterm.

The currently tested finite source routes are obstructed in a sharper sense. In the ordinary pure two-u CE source envelope, in every finite Hamiltonian degree, the covector

$$q_{2u} = e^*_{((H_{0,1},H_{2,0}),c_{\rho_{2,1}})} - \frac{1}{2} e^*_{((H_{0,1},H_{0,1}),c_{\rho_{0,2}})}$$

satisfies

$$q_{2u} d_{\mathrm{CE}} = 0, \qquad q_{2u}(\zeta_{M,\nu_3}) = 1.$$

Hence no ordinary pure two-u degree enlargement supplies a CE ancestor for ζ_{M,ν_3}. This is an exact obstruction theorem for that source envelope only, not a universal no-go for marked-source, Hom-valued companion, or local-functional counterterm habitats. The tested eight-face companion row has coordinates

$$(2, -2, 3, 2, -1, 1, -2, -3)$$

on

$$E_{\nu_{02}}, E_{\nu_{20}}, E_{\nu_{03}}, E_{\nu^{(1)}_{12}}, E_{\nu^{(2)}_{12}}, E_{\theta_3}, E_{\nu^{(2)}_{21}}, E_{\nu_{30}},$$

and diagonal transport to the $N = 2$ window leaves $2E^2_{\nu_{02}} - 2E^2_{\nu_{20}}$. Moreover q_{2u} evaluates to zero on this eight-face row, so the row cannot serve as the missing marked source column. The normalized lower-window cokernel functional $\lambda^{(2)}_{\nu_{02}} = \frac{1}{2}(E^2_{\nu_{02}})^*$ evaluates to 1 on the transported residual. A repair must therefore supply either a scalar-zero column $A^M_{\theta_3,C} = -1$, a marked source generator outside the ordinary pure two-u algebra with q_{2u}-value 1, or a companion relation whose lower-window transport kills this residual by a derived non-diagonal transport, lower primitive, or relation $E^2_{\nu_{02}} = E^2_{\nu_{20}}$. The canonical lower source-coordinate candidate is

$$\zeta^0_2 = u_{a,H_{1,0}} u_{b,H_{0,1}}, \qquad D^2_{02,20} = K^2_{\Theta_3}(\zeta^0_2),$$

with source differential

$$d_{\mathrm{CE}} \zeta^0_2 = 2\zeta_{\nu_{02}} - 2\zeta_{\nu_{20}}.$$

This proves source exactness of the transported lower row. It proves local-functional exactness only after the Hom-valued Bianchi defect

$$\mathfrak{b}^2_{02,20} = d_{\mathrm{ns},2} K^2_{\Theta_3}(\zeta^0_2) - K^2_{\Theta_3}(d_{\mathrm{CE}} \zeta^0_2)$$

is killed. Indeed

$$d_{\mathrm{ns},2} D^2_{02,20} = -2E^2_{\nu_{02}} + 2E^2_{\nu_{20}} + \mathfrak{b}^2_{02,20}.$$

In the Bianchi-exposed lower basis

$$(E^2_{\nu_{02}}, E^2_{\nu_{20}}, \mathfrak{b}^2_{02,20})$$

the column of $D^2_{02,20}$ and the residual are

$$A^2_D = (-2, 2, 1)^T, \qquad r_2 = (2, -2, 0)^T.$$

The equation $A^2_D c = -r_2$ has no solution in this free presentation. The cokernel

$$\widetilde{\lambda}_{02,\mathfrak{b}} = \frac{1}{2}(E^2_{\nu_{02}})^* + (\mathfrak{b}^2_{02,20})^*$$

satisfies

$$\widetilde{\lambda}_{02,\mathfrak{b}} A^2_D = 0, \qquad \widetilde{\lambda}_{02,\mathfrak{b}}(r_2) = 1.$$

Thus the lower source primitive is a genuine local-functional primitive only in the quotient where $\mathfrak{b}^2_{02,20} = 0$, or after a scalar-zero local counterterm $B^2_{02,20}$ with $d_{\mathrm{ns},2} B^2_{02,20} = -\mathfrak{b}^2_{02,20}$ is supplied, with locality and transport compatibility.

In the current Bianchi-exposed finite-row habitat no such $B^2_{02,20}$ exists. The only degree-zero lower column is $D^2_{02,20}$, so the desired Bianchi-counterterm target is the vector

$$-\mathfrak{b}^2_{02,20} = (0, 0, -1)^T.$$

It is not in the image of A^2_D, since

$$\widetilde{\lambda}_{02,\mathfrak{b}} A^2_D = 0, \qquad \widetilde{\lambda}_{02,\mathfrak{b}}((0, 0, -1)^T) = -1.$$

The minimal controlled enlargement must therefore add a genuine scalar-zero local-functional column

$$A^2_B = (0, 0, -1)^T, \qquad d_{\mathrm{ns},2} B^2_{02,20} = -\mathfrak{b}^2_{02,20},$$

with regular-density or wavefront-admissible locality and compatible finite-window transport. In a tower this requires

$$\Delta^1_{M,N} := -\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N = d_{\mathrm{ns},N}(\pi_{M,N} B^M - B^N)$$

together with vanishing of the H^1 Bianchi transport class on the right-hand side and then of the resulting secondary $\varprojlim^1 H^0$-class after any supplied transport correction.

Proof. The current one-row package has no degree-zero term and has one degree-one basis vector. Hence $H^1(\boldsymbol{Q}^\bullet_{\theta_3,M}) \cong \mathbb{C}\mathsf{E}_{\theta_3,M}$. The covector

$$\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$$

satisfies $\ell_{\theta_3} d_M = 0$ and evaluates nontrivially on the residual, so it is a left-cokernel certificate for the nonzero fixed-window H^1-class.

After adjoining any finite collection of degree-zero candidate rows, the fixed-window question is the elementary linear system

$$A^M c^M = -r^M,$$

where A^M is the matrix of d_M from candidate rows to residual rows and r^M is the coordinate vector of $\mathfrak{o}^{\mathrm{ns}}_{\theta_3,M}$. A CE ancestor or local counterterm with differential coefficient -1 is exactly a column of A^M equal to $-\mathsf{E}_{\theta_3,M}$, so $c^M = 1$ solves the equation. If the candidate is named but this differential entry is absent, the old covector ℓ_{θ_3} still annihilates the image of A^M and still evaluates to 1 on r^M; the obstruction survives.

The lower source-coordinate candidate illustrates the distinction between source exactness and local-functional exactness. The source calculation gives $d_{\mathrm{CE}}\zeta_2^0 = 2\zeta_{\nu_{02}} - 2\zeta_{\nu_{20}}$, but $K^2_{\Theta_3}$ is not a chain map until its Hom-valued Bianchi defect is killed. In the Bianchi-exposed basis the matrix equation is $A^2_D c = -r_2$, with $A^2_D = (-2, 2, 1)^T$ and $r_2 = (2, -2, 0)^T$. The first two coordinates force $c = 1$, while the Bianchi coordinate forces $c = 0$. Equivalently $\widetilde{\lambda}_{02,\mathfrak{b}} A^2_D = 0$ and $\widetilde{\lambda}_{02,\mathfrak{b}}(r_2) = 1$. Hence the current checkout contains an exact obstruction certificate, not a completed local-functional counterterm.

The same cokernel proves the no-$B^2_{02,20}$ statement in the current habitat. The target vector $(0, 0, -1)^T$ would require a scalar c with $c(-2, 2, 1) = (0, 0, -1)$; the first two coordinates force $c = 0$, while the third forces $c = -1$. Equivalently $\widetilde{\lambda}_{02,\mathfrak{b}}$ annihilates every current boundary column and evaluates to -1 on the target. A formal new column $A^2_B = (0, 0, -1)^T$ closes this fixed matrix equation, but it is theorem-grade only after the local functional, scalar-zero value, locality or wavefront habitat, and finite-window transport data are constructed.

A companion-face table is not a primitive. It changes the residual row by replacing the one-face CE source expression with the full signed CE boundary. It is accepted exactly when the signed sum is zero and the truncation matrices preserve that zero sum in the inverse system. The displayed Roos condition is precisely the obstruction to choosing the fixed-window primitives compatibly. These are necessary and sufficient finite-window conditions; they do not by themselves construct the analytic Costello graph package or any all-graph QME solution. □

Problem 3.273 *(Tate-coefficient lift of the descendant cochain isomorphism).* This is the quantum specialization of 3.60. The classical Tate obstruction is closed by the weighted Tate completion of Theorem 3.85 (parts (a)–(d)); the all-order quantum lift on the descendant sector requires the hypothesis of 3.87. The weighted-completion and nilpotent-truncation mechanisms are recorded in 3.8. Lemma 3.263 remains the diagnosis of the obstruction in the unweighted Tate pair; the resolution requires a category-level change of coefficient module, supplied by the weighted completion.

Remark 3.274. After the chain Proposition 3.244–Proposition 3.245–Lemma 3.249–Theorem 3.251–Corollary 3.252–Proposition 3.275–Theorem 3.269, under the radial image package of Statement F.8 or the equivalent quantum shear primitive package, the formal Moyal coefficient algebra, the Capelli renormalization, the finite-N radial-parts identity, and the reduced open-line boundary product form a self-contained algebraic construction. Two independent obstruction data lie outside that construction: the Tate completion needed for the descendant cochain summand, and the analytic Costello specialization of Problem 3.264. The classical analogue, Problem 3.60, is already pinned to the Tate obstruction at $\hbar = 0$.

Proposition 3.275 *(Open-line Weyl boundary product from midpoint kernel).* For Weyl-ordered Hamiltonian boundary observables placed on the open line $\mathbb{R}$ in the reduced first-order open model $S_0 = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2)$, with the zero-mode-removed midpoint propagator

$$G(t,s) = \tfrac{1}{2}\,\mathrm{sgn}(t-s), \qquad \langle \phi_1(t)\phi_2(s)\rangle = \hbar G(t,s), \qquad \langle \phi_2(t)\phi_1(s)\rangle = -\hbar G(t,s),$$

the boundary product on $\bar{A}_\hbar$ is

$$F \star G = m \circ \exp\left(\frac{\hbar}{2} P_\partial\right)(F \otimes G), \qquad P_\partial = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

Equivalently, every odd raw commutator coefficient is $\frac{2}{2^r r!}\hbar^r P^r(f,g)$. In particular the first and third coefficients are $\hbar P(f,g)$ and $\frac{\hbar^3}{24} P^3(f,g)$. This reduced open-line computation uses no Costello specialization. With these conventions, the Weyl normalization is determined by:

(i) the matrix Weyl algebra commutator $[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar\delta^i_l\delta^k_j$;
(ii) the zero-mode-removed midpoint propagator $G(\pm,\mp) = \pm\frac{1}{2}$;
(iii) the Baker–Campbell–Hausdorff determination of the cross-contraction weight $P/2$ on linear Hamiltonians;
(iv) the linear-Hamiltonian commutator $[\widehat{u},\widehat{v}] = \hbar(ad - bc)$ for $u = az_1 + bz_2$, $v = cz_1 + dz_2$.

Proof. Items (i)–(iv) form a normalization system. By (i)–(ii) the midpoint propagator at $t \neq s$ has weight $\pm\hbar/2$. By (iv) the linear-Hamiltonian Weyl commutator is $\hbar(ad - bc)$, so the cross-contraction weight on linear Hamiltonians is exactly $1/2$, giving $P/2$. Hence by (iii) the BCH formula on the Heisenberg algebra of linear Hamiltonians fixes the cross-contraction kernel on the boundary as $P_\partial = P$ with weight $\hbar/2$.

The all-order boundary product on $\bar{A}_\hbar$ is then determined by: (a) the local Weyl/BCH computation for the constant Poisson tensor, with Kontsevich deformation quantization [7] serving only as general context here; (b) Proposition 3.268 together with item (iii) above; (c) the antisymmetrisation argument of Theorem 3.276.

The "no additional factor of N, sign, or scalar moment-map term" clause is the content of Lemma 3.249(iii): the Capelli contact term is absorbed into the $T \leftrightarrow J_N$ change of basis, and the connected single-trace projection is the standard cumulant projection. Consequently all odd raw commutator coefficients are $\frac{2}{2^r r!}\hbar^r P^r(f,g)$, and for $r = 1, 3$ they are $\hbar P(f,g)$ and $\frac{\hbar^3}{24}P^3(f,g)$. The four primary cases (a)–(d) and a sweep of 14641 monomial pairs through order $\hbar^{11}$ are exact-arithmetic verification checks for this coefficient formula. They are not a substitute for the external finite-N radial-parts input. □

THEOREM 3.276 *(Reduced open-line Wick theorem).* In the reduced open first-order line model

$$S_0 = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2),$$

choose the zero-mode-removed midpoint propagator

$$G(t,s) = \frac{1}{2}\,\mathrm{sgn}(t-s), \qquad \langle\phi_1(t)\phi_2(s)\rangle = \hbar G(t,s), \quad \langle\phi_2(t)\phi_1(s)\rangle = -\hbar G(t,s).$$

For ordered insertions $t > s$, let $W_r^>(f,g)$ be the contribution of the graph with exactly r cross-contractions between the two Weyl-ordered boundary vertices. Let $W_r^<(f,g)$ be the reversed ordering contribution. Then

$$W_r^>(f,g) = \frac{1}{r!}\left(\frac{\hbar}{2}\right)^r P^r(f,g), \qquad W_r^<(f,g) = \frac{1}{r!}\left(-\frac{\hbar}{2}\right)^r P^r(f,g).$$

Consequently the reduced open-line commutator is

$$[f,g]_{\mathrm{Wick}} = \sum_{\substack{r\geq 1\\ r \text{ odd}}} \frac{2}{2^r r!}\hbar^r P^r(f,g),$$

and all even orders vanish. For monomials $f = z_1^k z_2^l$, $g = z_1^m z_2^n$,

$$P^r(f,g) = \sum_{a=0}^{r}(-1)^a\binom{r}{a}(k)_{r-a}(l)_a(m)_a(n)_{r-a}z_1^{k+m-r}z_2^{l+n-r}.$$

In particular the one-edge and three-edge commutator contributions are

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24}P^3(f,g).$$

Proof. The midpoint value $G(+,-) = 1/2$ assigns one copy of

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$$

to each cross-contraction, with weight $\hbar/2$. With Weyl vertices there are no self-contractions. The graph with r identical cross-edges has automorphism group S_r, so the ordered contribution is

$$W_r^{>}(f,g) = \frac{1}{r!}\left(\frac{\hbar}{2}\right)^r P^r(f,g).$$

Reversing the order changes the midpoint propagator from $1/2$ to $-1/2$, giving the displayed formula for $W_r^{<}$. Hence

$$W_r^{>}(f,g) - W_r^{<}(f,g) = \frac{1-(-1)^r}{2^r r!}\hbar^r P^r(f,g),$$

which is zero for even r and $2\hbar^r P^r(f,g)/(2^r r!)$ for odd r.

The monomial formula is the expansion of $P^r = (\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})^r$ by the binomial theorem: the summand with a copies of $-\partial_{z_2} \otimes \partial_{z_1}$ contributes the sign $(-1)^a$, the binomial coefficient $\binom{r}{a}$, and the four falling factorials shown above. The values at $r = 1$ and $r = 3$ give $\hbar P(f,g)$ and $\hbar^3 P^3(f,g)/24$. □

Remark 3.277. Theorem 3.276 is a computation in the reduced open-line Weyl model. It is not the Costello closed-open graph theorem. The latter still requires an elliptic parent or dimensional reduction, a brane-defect condition, finite-scale gauge fixing, the induced bulk-to-defect propagator, orientations and measures on configuration spaces, counterterms, anomaly cancellation, and compatibility with the quantum Hamiltonian reduction.

3.12.5 Third-order target for Costello's graph problem

For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the third-order Moyal coefficient from Proposition 3.268 is

$$P^3(f,g) = \sum_{a=0}^{3}(-1)^a\binom{3}{a}(k)_{3-a}(l)_a(m)_a(n)_{3-a}z_1^{k+m-3}z_2^{l+n-3},$$

where $(r)_s = r(r-1)\cdots(r-s+1)$ and $(r)_0 = 1$. Therefore

$$[f,g]_* = \hbar(kn-lm)z_1^{k+m-1}z_2^{l+n-1} + \frac{\hbar^3}{24}P^3(f,g) + O(\hbar^5).$$

The exact-arithmetic verification checks this first-order coefficient, the vanishing of the second-order coefficient in the raw commutator, the displayed third-order formula, and the star-commutator normalization on every monomial pair with exponents between 0 and 10 and commutator orders through 11.

This algebraic computation is an observable-product target, not an effective action computation. In Costello's formalism the connected effective action and the product/OPE of quantum observables are related but distinct. For two boundary observables F, G, the finite-scale free-field model has the schematic product

$$F \star_L G = m \circ \exp(\hbar\partial_{P_{\partial,L}})(F \otimes G),$$

where $P_{\partial,L}$ is the boundary or brane-defect propagator induced by the chosen kernel. The connected single-trace part is the cumulant, or primitive trace, extracted from this product. In this convention the order in $\hbar$ is the number of boundary cross-contractions; it is not the loop order of the connected effective interaction.

Thus there are two equivalent normalizations to keep separate. In Costello's observable-product notation one asks for the boundary contraction kernel

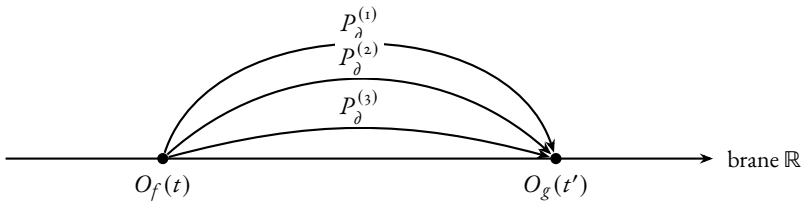
$$P_{\partial,L} \rightsquigarrow \frac{1}{2}P$$

on degree-zero Hamiltonians, so that

$$m \circ \exp(\hbar\partial_{P_{\partial,L}}) \rightsquigarrow m \circ \exp\left(\frac{\hbar}{2}P\right),$$

the Weyl/Moyal product used above. The ordered product has labeled inputs (f, g); its r identical cross-contractions contribute the denominator $r!$. No additional division by interchanging the two boundary vertices is made in the ordered product. The reverse order is the separate product $g * f$ whose subtraction forms the commutator.

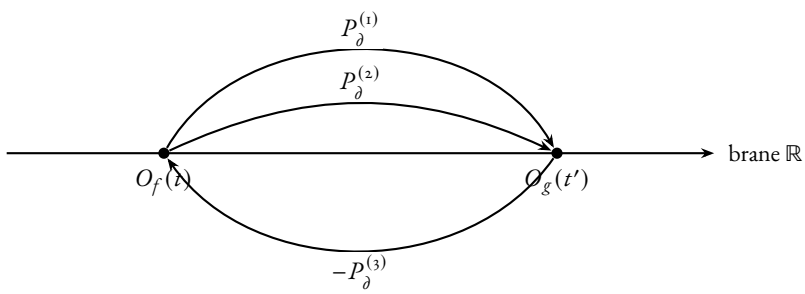
Γ_{3a}: three propagators, common orientation



$$\mathrm{Aut}(\Gamma_{3a}) = S_3, \quad |\mathrm{Aut}(\Gamma_{3a})| = 3! = 6, \quad \tfrac{1}{|\mathrm{Aut}(\Gamma_{3a})|} = \tfrac{1}{6}$$

orientations $a = 0, 3$: contributions $(k)_3(n)_3$ and $-(l)_3(m)_3$

Γ_{3b}: two propagators forward, one reversed



$$\mathrm{Aut}(\Gamma_{3b}) = S_2, \quad |\mathrm{Aut}(\Gamma_{3b})| = 2! = 2, \quad \tfrac{1}{|\mathrm{Aut}(\Gamma_{3b})|} = \tfrac{1}{2}$$

orientations $a = 1, 2$: contributions $-3(k)_2 lm(n)_2$ and $+3k(l)_2(m)_2 n$

Figure 0.2: The third-order graphs Γ_{3a} (top) and Γ_{3b} (bottom) for the order-$\hbar^3$ boundary contractions whose tensor target is $P^3(f, g)$. Each cross-contraction contributes one copy of the Poisson bivector P_∂; reversing the orientation of a propagator contributes a relative sign. The orientation multiplicities are the binomial coefficients $\binom{3}{a}$ of Proposition 3.268: Γ_{3a} realises $a \in \{0, 3\}$ with automorphism group S_3 permuting the three identical propagators (symmetry factor 1/6), and Γ_{3b} realises $a \in \{1, 2\}$ with automorphism group S_2 permuting the two same-orientation propagators (symmetry factor 1/2). These are two descriptions of the same count: either use the unexpanded aggregate P-edge count with automorphism S_3, or expand into orientation classes with residual automorphism $S_a \times S_{3-a}$; the equality is $\binom{3}{a}/3! = 1/(a!(3-a)!)$. The two counts must not be multiplied. Summing all four orientation classes, $P^3(f, g) = \sum_{a=0}^{3}(-1)^a \binom{3}{a}(k)_{3-a}(l)_a(m)_a(n)_{3-a}\, z_1^{k+m-3} z_2^{l+n-3}$. The third-order raw commutator weight after antisymmetrising is $\hbar^3/24$. The reduced open-line weights are computed in Theorem 3.276; the Costello counterterm and anomaly checks remain the analytic obligations of Problem 3.280.

The graphs Γ_{3a} and Γ_{3b} of Figure 0.2 record the displayed order-$\hbar^3$ boundary contraction types whose tensor target is $P^3(f, g)$. Each cross-contraction contributes one copy of the constant Poisson bivector P. Their reduced open-line weights and automorphism factors are computed above. Their realization as Costello brane-defect graphs still requires the sector specialization, propagator, counterterm, and anomaly checks in the problem below.

Theorem 3.278 *(Costello first/third normalization recognition criterion).* Assume the Hamiltonian specialization data of Problem 3.264 have been supplied for a Costello-type closed-open BV theory, and assume that the induced brane-defect observable product restricts on degree-zero Hamiltonians to the midpoint kernel $P/2$ of Theorem 3.276. Assume further that at order $\hbar^3$ every connected

local term not represented by the three-parallel-edge cross-contraction has been shown to vanish, to be counterterm-exact, or to land in the discarded scalar trace. Then the first- and third-order Costello boundary commutator coefficients are determined by the Weyl/Moyal coefficients:

(i) the connected one-propagator brane-defect ribbon graph carries coefficient $\hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$ for monomial inputs $f = z_1^k z_2^l$, $g = z_1^m z_2^n$;
(ii) the surviving three-propagator contribution carries coefficient $\frac{\hbar^3}{24}P^3(f,g)$;
(iii) any finite counterterm ambiguity that changes the first or third coefficient is fixed by matching the Moyal coefficients 1 and 1/24 at these orders. No all-order Costello uniqueness is asserted here.

Equivalently, the Costello graph theorem at these orders holds precisely when the specialization supplies the above data and the normalization defects

$$\mathfrak{n}_1 = P_{\partial,L}|_{\mathrm{Ham}^\circ} - \frac{1}{2}P,$$

$$\mathfrak{n}_3 = \left[\mathrm{ObsProd}^{(3)}_{\mathrm{Costello}}(f,g) - \frac{\hbar^3}{24}P^3(f,g)\right]_{\mathrm{Ham}^\circ,\mathrm{conn,red}}$$

vanish, together with the scalar-contact, reduced-kernel, and θ_3 graph-row obstructions. The obstruction tuple is

$$\mathfrak{O}^{1,3}_{\mathrm{Cost}} = (\mathfrak{o}_{\mathrm{Cas}}, \hbar N[\bar{c}] \text{ or } o^{(r)}_{\sigma,w}, [\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}], \mathfrak{O}_{\theta_3}, \mathfrak{n}_1, \mathfrak{n}_3).$$

Absent this datum, the theorem proved in Theorem 3.276 is the reduced open-line Wick normalization, not a Costello heat-kernel graph realization.

Proof. Under the stated hypotheses, the degree-zero boundary observable product is the reduced open-line Weyl product of Proposition 3.275. Thus each boundary cross-contraction contributes the operator P with weight $\hbar/2$. Here the S_3 count treats the three edges as copies of the aggregate bivector $P = A - B$. After expanding P^3, a mixed orientation class has residual automorphism $S_a \times S_{3-a}$ and binomial coefficient $\binom{3}{a}$. These are equivalent normalizations, since $\binom{3}{a}/3! = 1/(a!(3-a)!)$; one must not multiply the aggregate S_3 denominator by the residual orientation-class denominator again.

Item (i): one edge, no edge automorphisms, weight $\hbar P/2$ for the ordered product (Theorem 3.276); antisymmetrisation gives

$$\hbar P(f,g) = \hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$$

on monomials.

Item (ii): for three parallel edges between the two labelled boundary observables, the edge automorphism group is S_3. The ordered weight is

$$\hbar^3\frac{1}{3!}\left(\frac{1}{2}\right)^3 P^3(f,g) = \frac{\hbar^3}{48}P^3(f,g),$$

and antisymmetrisation doubles it to $\hbar^3 P^3(f,g)/24$. The hypotheses exclude the remaining connected local terms at the same order.

Item (iii): a finite local counterterm that changes the one- or three-edge boundary product changes the coefficient of P or P^3. The linear-Hamiltonian Heisenberg commutator fixes the coefficient of P to 1, and Proposition 3.268 fixes the coefficient of P^3 to 1/24 in the raw commutator. Thus matching these two coefficients removes precisely the counterterm freedom visible at these orders. The displayed defect tuple is a restatement of the hypotheses needed to make this reduced computation a Costello

graph theorem: the first defect fixes the brane-restricted propagator, the third-order defect fixes the connected reduced product, and the remaining entries are the scalar, reduced-kernel, and θ_3 graph-row gates developed in Problem 3.271. □

Remark 3.279. In the reduced open-line model the Weyl parameter $\hbar$ counts boundary cross-contractions. If a Costello specialization restricts to this free boundary kernel, then its observable-product expansion must be read with this edge-counting convention on degree-zero Hamiltonians. Without the specialization data of Problem 3.264, the statement is only the formal Weyl/Moyal normalization test above.

Problem 3.280 *(Costello first/third graph-normalization obstruction datum).* Work inside the observable product supplied by Statement 3.258, after solving the Hamiltonian specialization data of Problem 3.264. The first graph fixes the normalization; the third graph tests whether the Costello specialization really lands on the Weyl/Moyal product rather than on another deformation quantization convention. The analytic problem is reduced to the following checks:

(i) identify the connected one-propagator brane-defect ribbon graph whose tensor contraction is $P(f, g)$, and prove that its antisymmetrised boundary product gives the raw coefficient $\hbar P(f, g)$;

(ii) fix the sign, automorphism factor, boundary orientation, and trace normalization in this first-order graph so that the coefficient is $\hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$ on monomials;

(iii) identify the connected three-propagator brane-defect ribbon graphs whose tensor contraction is $P^3(f, g)$;

(iv) prove that every other connected order-$\hbar^3$ contribution in the Hamiltonian sector vanishes, is counterterm-exact, or lands in the discarded scalar trace;

(v) for every term declared counterterm-exact in the preceding item, exhibit compatible local counterterm representatives $C^{w_\circ}_{\Gamma,w}(\epsilon) \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{w_\circ}_w)$, prove Costello locality and RG compatibility, and verify that their restriction to degree-zero Hamiltonians does not alter the fixed one-edge coefficient except by the allowed Moyal normalization choice;

(vi) compute the graph weight and automorphism factor after the first-order normalization and show that the coefficient is $\frac{1}{24}$;

(vii) check that connected large-N extraction introduces no additional factor of N, sign, or scalar moment-map term;

(viii) verify that the same observable product, counterterm subtraction, and connected extraction commute with the principal-part current extension of Theorem 3.237 on the reduced target. If the claim is made before de Rham-current contraction, identify the additional unreduced homotopies of Remark 3.241. In particular the degree-zero Moyal product must act on the cotangent current summand through the principal-part coadjoint action, not through polynomial one-ψ descendants.

The target normalization is the pair of algebraic coefficients

$$\hbar P(f, g), \qquad \frac{\hbar^3}{24} P^3(f, g)$$

in the Weyl/Moyal commutator. The formal coefficients are proved above; Theorem 3.278 records the implication from these data once the Costello specialization supplies the checks. Items (i)–(viii) are precisely the analytic specialization and normalization data.

Remark 3.281. In characteristic zero, GL_N is linearly reductive on rational representations. Therefore taking GL_N-invariants is exact, and

$$H\big((\mathcal{R}_N)^{GL_N}, Q\big) \cong H(\mathcal{R}_N, Q)^{GL_N}.$$

This justifies the degree-zero rational invariant calculation used above. It does not identify that calculation with the full Lie-algebra Chevalley–Eilenberg BRST cohomology of $\mathfrak{gl}_N$. A stack-level presentation remembers stabilizers, including the trivially acting scalar $B\mathbb{G}_m$; those ghost and stabilizer classes are outside the Hamiltonian trace sector computed here. Linearly reductive exactness identifies derived algebraic invariants of functions with ordinary invariants in this setting.

4 Admissibility criteria and obstruction complexes

The coordinate CE/PV theorem and staged admissibility programme is governed by constructions, hypotheses, and obstruction complexes. Each item below is either a construction already used in the local theorem, an admissibility criterion for a frontier extension, or an obstruction complex whose vanishing is the exact condition for the next theorem.

External compact Calabi–Yau, CoHA, quintic, OSV/GV, Abel–Jacobi, MNOP, Igusa, BKM, and global BCOV material is not a core local obligation. It is a comparison surface only, and it cannot support the local QME, Ext, cyclic, Weyl/Moyal, radial-parts, or kernel theorems without a separate matched-conventions theorem and a vanishing obstruction vector.

(i) *Finite cotangent CE/PV data.* The finite model is attached to a finite-dimensional Lie algebra $\mathfrak{l}$, its ordinary dual $\mathfrak{l}^\vee$, and the cotangent Lie algebra

$$\mathfrak{g}_{\mathfrak{l}} = \mathfrak{l} \ltimes \mathfrak{l}^\vee[1].$$

The finite comparison is

$$\Phi_{\mathfrak{l}} \colon C^\bullet_{\mathrm{CE}}(\mathfrak{g}_{\mathfrak{l}}) \xrightarrow{\sim} \mathrm{PV}(\mathbf{S}(\mathfrak{l})), \qquad c \mapsto \theta, \quad u \mapsto O.$$

(ii) *Completed formal-disk CE/PV admissibility.* The comparison for $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ and $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ is accepted first as a pro coordinate-window CE/PV comparison. It becomes a completed bracketed statement only after a bracket-admissible realization $B \subset \mathrm{PV}_{\mathrm{coord}}(\mathcal{A}_{\partial,\mathrm{coord}})$ has a specified complete Hausdorff topology, continuous inclusion in the coordinate product, continuous multiplication, continuous d^{coord}_π, a continuous closed coordinate contraction bracket, and the transported continuous source structure on $\Phi^{-1}_{\mathrm{coord}}(B)$. Boundary Hamiltonian uses also require B to contain the relevant O_f's and $d^{\mathrm{coord}}_\pi O_f$'s. It becomes a kernel statement only after a kernel-admissible datum $(B, \mathcal{K}_B)$ is supplied, so that the diagonal tensor

$$\Theta_B = \sum_I H_I \otimes \theta^I + \sum_I \eta^I \otimes O_I$$

converges in $\mathcal{K}_B$ and satisfies

$$d\Theta_B + \frac{1}{2}[\Theta_B, \Theta_B] = 0.$$

The admissible formal-local interface datum is

$$\mathfrak{U}_{\mathfrak{h}} = (\Phi_{\mathrm{coord}}, \{\Phi^B_{P_0}\}_B, \{\Theta_B\}_B).$$

It is not a categorical universal object unless a category of admissible completions and a universal property are supplied. The coordinate obstruction terms are

$$\mathfrak{o}^{\mathrm{br}}_{M,N}(x,y) = \pi_{M,N}[x,y]_M - [\pi_{M,N}x, \pi_{M,N}y]_N,$$

$$\begin{aligned}\mathfrak{o}^{\mathrm{coad}}_{M,N}(x,\lambda) = \pi^{\vee}_{M,N}(x \cdot_M \lambda) \\ - (\pi_{M,N}x) \cdot_N (\pi^{\vee}_{M,N}\lambda),\end{aligned}$$

Their vanishing gives the pro coordinate-window comparison

$$\Phi_{\mathrm{coord}} \colon C^{\bullet}_{\mathrm{CE,coord}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}}).$$

On a bracket-admissible B the restricted map

$$\Phi^{B}_{P_0} \colon \Phi^{-1}_{\mathrm{coord}}(B) \xrightarrow{\sim} B$$

is the corresponding completed P_0-comparison. The Casimir transition term

$$\mathfrak{o}^{\mathrm{Cas}}_{M,N} = (\pi_{M,N} \otimes \pi^{\vee}_{M,N})C_M - C_N$$

diagnoses the failure of a strict global kernel, propagator, endpoint, ambient-product bracket, and bar-cobar/enveloping comparison in the unweighted product/direct-sum topology. It is not an additional hypothesis for each coordinate-window CE/PV identity.

(iii) *Admissible Tate bar-cobar/Quillen envelope.* For an admissible Tate replacement $\mathfrak{g}_{\mathrm{adm}}$, put

$$B_{\mathrm{adm}} = C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}).$$

The admissible Koszul duality theorem in the envelope is

$$B^{!}_{\mathrm{adm}} \simeq \widehat{U}(\mathfrak{g}_{\mathrm{adm}}),$$

together with the bar-cobar Quillen equivalence under lower-central Tate-pronilpotence, PBW completeness, and the completed symmetric associated graded. The acceptance datum lives in a chosen admissible finite-window model envelope, not in raw Tate vector spaces. Failure is the obstruction vector

$$O_{\mathrm{AdmTate}} = (O_{\mathrm{win}}, O_{\mathrm{top}}, O_{\mathrm{conilp}}, O_{\mathrm{MC}}, O_{\mathrm{cobar}}, O_{\mathrm{PBW}}, O_{P_0}, O_{\mathrm{cyc}}),$$

where O_{win} contains the bracket, coadjoint, and Casimir transition defects; O_{top} contains Milnor/Roos $\varprojlim^1$ terms; O_{conilp} records CE-length and lower-central separatedness defects; O_{MC} is the twisting-cochain curvature; O_{cobar} is the associated-graded cone plus tower defect; O_{PBW} is the PBW separatedness/associated-graded defect; O_{P_0} is bracket-admissibility and transferred-model failure; and O_{cyc} is the cyclic trace compatibility obstruction.

(iv) *Finite-window dg LQT comparison.* The repaired finite-window comparison uses the quotient dg algebra $A_{\psi,K} = A_{\psi}/F^{>K}A_{\psi}$, with $F^{>K}$ the two-sided dg ideal generated by words of total Koszul weight $> K$ for

$$\mathrm{wt}(x) = (1,0), \qquad \mathrm{wt}(y) = (0,1), \qquad \mathrm{wt}(\psi) = (1,1).$$

On the stable single-cycle image of the trace-cycle map

$$\Theta_{N,K,L}\colon CC_{\mathrm{red},\le L}(A_{\psi,K})[1] \to C^{\mathrm{CE}}_{*,\le L}(\mathfrak{gl}_N(A_{\psi,K}))^{GL_N},$$

the finite-window map is $\lambda_{\mathrm{LQT},K,L} = \Theta^{-1}_{N,K,L} P_{1\mathrm{cyc}}$. The theorem-grade projection is the stable block-sum Hopf Eulerian idempotent

$$P_{1\mathrm{cyc}} = \log^*(\mathrm{id}) = \sum_{m=1}^{L} \frac{(-1)^{m-1}}{m} (\mathrm{id} - u\epsilon)^{*m}.$$

This closes only after the stable range $N \ge \max(K, L+2)$, or $N \ge W + 2$ under one bound, controls both Koszul weight and CE length. Under one bound W, the notations $A_{\psi,W}$, $\Theta_{N,W}$, and $\lambda_{\mathrm{LQT},W}$ are abbreviations for $K, L \le W$. This projection also requires the scalar-Koszul chain quotient to kill $\Theta_1(\psi) \mapsto \Theta_1(xy - yx)$. The unqualified primitive formula is obstructed by

$$O_{\mathrm{LQT},K,L} = (O_{\mathrm{win}}, O_{\mathrm{Prim}}, O_{\mathrm{stab}}, O_{\mathrm{scalar}}, O_{\mathrm{CC}}, O_{\mathrm{Roos}}).$$

The raw same-rank operation that keeps one-cycle tensors and deletes fixed-rank multicycles is not a chain map: $d_{\mathrm{CE}}(\Theta_1(a) \wedge \Theta_1(b))$ has a single-cycle commutator component, while deletion before the differential gives zero. The finite-rank bridge must use separated-block stabilization: with $B(K, L) = \max(K, L+2)$ and $M = L\,B(K, L)$, $\beta_{M,K,L}$ is the normalized signed average over injections into the L labeled diagonal blocks, $\alpha_{M,K,L}$ forgets block labels, $\alpha_{M,K,L}\beta_{M,K,L} = \mathrm{id}$, and the strict separated-block idempotent is

$$P^{\mathrm{sep}}_{M,K,L} = \beta_{M,K,L} P_{1\mathrm{cyc}} \alpha_{M,K,L}.$$

The same-rank bridge is obstructed by

$$\Gamma_N = d_{\mathrm{same}} F_N - F_N d_{\oplus}, \qquad \Gamma_N(\Theta^N_1(a) \wedge \Theta^N_1(b)) = \pm\Theta^N_1(ab - (-1)^{|a||b|} ba).$$

(v) *Reduced CE/PV central-operation generator map.* The reduced formal central-operation map is determined on CE/PV generators by

$$c^I \mapsto \theta^I, \qquad u_I \mapsto O_I.$$

The boundary Hamiltonian observable is sourced by the shifted cotangent coordinate u_I. This type check is part of the theorem: c^I gives the vector-field operation, while u_I gives the Hamiltonian boundary coordinate.

(vi) *Constructed chiral algebra interface.* A constructed curve chiral algebra can enter the local mixed HT manuscript only through an interface

$$\mathfrak{I}_{\mathrm{ch}} = (\mathcal{R}_{\mathbb{C}\times\mathbb{R}}, \mathcal{F}_{\mathrm{red}}, V_{\mathrm{red}}, Q_{\mathrm{BRST}}, \mathbf{1}, T, Y, \mathrm{wt}, \zeta_\hbar, H_{\mathrm{anom}}),$$

not as the native $\mathbb{C}^2$ holomorphic E_2 object. The reduction $\mathcal{R}_{\mathbb{C}\times\mathbb{R}}$ must retain the z_2-mode or full two-index principal-part coefficient system; otherwise the Hamiltonian bracket, Moyal bivector, scalar cocycle, and second matrix brane coordinate are destroyed. The exact admission obstruction is

$$\mathrm{Ob}_{\mathfrak{I}_{\mathrm{ch}}} = (\mathrm{Ob}_{\mathrm{red}}, \mathrm{Ob}_{\mathrm{vert}}, \mathrm{Ob}_{\mathrm{Zhu}}, \mathrm{Ob}_{\mathrm{nat}}),$$

whose entries measure the retained-mode pushforward and anomaly transport, the finite Laurent/OPE and BRST vertex package, the zero-mode/Zhu multiplicativity, Capelli shift, and completed stable Hochschild/HKR/CE-PV comparison, and the comparison back to $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$.

(vii) *Matlis/local-cohomology cotangent source.* The continuous cotangent module is

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) \subset H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2,$$

with basis

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \qquad a + b > 0.$$

The coadjoint action is

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}.$$

(viii) *BMK one-pair pro-Matlis retract and native all-window obstruction.* The positive all-window datum presently constructed is one-pair and analytic:

$$\bar\partial H^{\circ}_{\chi} + H^{\circ}_{\chi}\bar\partial = \mathrm{id} - I_{\mathrm{an}} P_{\mathrm{an}}, \qquad P^{\mathrm{an}}_{\Pi} = \tau P_{\mathrm{an}}, \qquad \pi_N P^{\mathrm{an}}_{\Pi} = p_N.$$

It is not a native support-local factorization-current transfer. The strict native all-window current theorem is exactly the vanishing, or constructive replacement in a changed habitat, of

$$\mathrm{Ob}^{\Pi}_{\mathrm{BM}} = (\mathrm{ob}_{\oplus}, \mathrm{ob}_{I_\Pi}, \mathrm{ob}_{\mathfrak{h},\Pi}, \mathrm{ob}_{\mathrm{collar},\Pi}, \mathrm{ob}_{3,\Pi}, \mathrm{ob}_{\mathrm{unif},\Pi}).$$

Here $\mathrm{ob}_{\oplus}$ is the failure of all moments to land in the direct-sum Matlis module; ob_{I_Π} is the nonexistence of a compact-current section of the pro-product tower; and $\mathrm{ob}_{\mathfrak{h},\Pi}$ is the failure of the full formal Hamiltonian action on that target, detected by the divergent $\rho_{0,1}$-coefficient for $f = \sum_{p\geq 2} z_1^p$ acting on $\lambda = \sum_{p\geq 2} \rho_{p-1,0}$. The first, second, third, and sixth entries are nonzero in the native current/product topology; the collar and ternary entries are exact primitive construction targets.

(ix) *Primitive higher-ψ Hamiltonian extraction.* The full primitive Koszul complex for $Q\psi = [x, y]$ extracts the Hamiltonian sectors: nonconstant H_0 classes, primitive one-ψ classes $\Psi_{a,b}$, and no primitive $p \geq 2$ homology. The one-ψ action is

$$\{\overline{\mathrm{Tr}(\phi_1^p \phi_2^q)}, \widetilde{\Psi}_{a,b}\} = (pb - qa)\widetilde{\Psi}_{a+p-1,b+q-1},$$

with negative targets zero and scalar targets killed in the Hamiltonian quotient.

(x) *Principal-part versus polynomial-descendant criterion.* The polynomial one-ψ and principal-part modules have opposite linear-Hamiltonian behavior:

$$\{O_{1,0}, \widetilde{\Psi}_{a,b}\} = b\,\widetilde{\Psi}_{a,b-1}, \qquad \{O_{0,1}, \widetilde{\Psi}_{a,b}\} = -a\,\widetilde{\Psi}_{a-1,b},$$

while

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

This local-nilpotence obstruction is the necessary condition for any proposed polynomial realization of $\mathcal{P}$: it must change the target, the action, or the category.

(xi) *$\mathcal{D}_\hbar$ Fourier–Rees bridge.* With

$$\mathcal{D}_\hbar = \mathbb{C}[[\hbar]]\langle z_1, z_2, \partial_1, \partial_2\rangle \\ /([\partial_i, z_j] = \hbar\delta_{ij}),$$

the Fourier delta module $\Delta^{\mathsf{F}}_{\hbar} = \mathcal{D}_{\hbar}/(z_1, z_2)$ gives

$$z_1^a z_2^b \mapsto \partial_1^a \partial_2^b e_0 \mapsto (-1)^{a+b} \hbar^{a+b} a! b! \rho_{a,b}.$$

Before $\hbar$ is inverted, the image is the Rees sublattice $\bigoplus \hbar^{a+b}\mathbb{C}[[\hbar]]\rho_{a,b}$. The same-action obstruction forbids a flat Rees family that identifies polynomial one-ψ descendants with $\mathcal{P}$ as modules for the original Hamiltonian action.

(xii) *One-antifield Moyal target.* The reduced distributional principal-part target has bracket

$$\{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0.$$

A strict quantum descendant lift of the classical label shadow is excluded by the leading-term calculation

$$z_1 \cdot_0 \widetilde{\Psi}_{1,0} = 0, \qquad z_1 \cdot \rho_{1,0} = -\rho_{1,1}.$$

(xiii) *Weighted finite-window regulator independence.* A regulator-admissible sector is a strict Mittag-Leffler family on finite windows with graphwise bounded Hamiltonian and cotangent legs. For admissible weights w, w', diagonal transports $R^{w_0}_{w,w'}$ identify the local-functional complexes and carry the obstruction class

$$\mathrm{Ob}_w \in H^1_{\mathrm{adm}}(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0^w, -\})$$

to $\mathrm{Ob}_{w'}$. Thus QME vanishing is invariant under admissible finite-window weight.

(xiv) *Endpoint admissibility.* An endpoint for $w_q(d) = q^d$ is accepted by conditions (E1)–(E5): strict Mittag-Leffler quotients, a coefficient kernel projecting to every finite Casimir, transport of differential, bracket, coadjoint action, propagator, interaction, counterterms, and the generator map $c \mapsto \theta$, $u \mapsto O$, admissible filtered cohomology, and RG preservation. The endpoint obstruction tuple is

$$\mathfrak{o}_{\mathrm{BV,end}} = (\mathfrak{o}_{\mathrm{Cas}}, \mathfrak{o}_{\mathrm{def}}, \mathfrak{o}_{\mathrm{RG}}, \mathfrak{o}_{\mathrm{QME}}).$$

The strict product/direct-sum pair fails the criterion because no tensor K projects to all finite Casimirs $C^M_{\mathfrak{h}}$.

(xv) *Mixed brane-defect QME complex.* The local-functional obstruction lives in

$$H^1\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\}\big).$$

Its scalar component is computed by

$$\{I_\partial(a,f), I_\partial(b,g)\}_{\mathrm{open}} = I_\partial(ab, \{f,g\}_{\bar{A}}) + N\,\omega(f,g)\int_{\mathbb{R}} a(t)b(t)\,dt,$$

or, in Weyl normalization,

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\ \mathrm{Tr}(A) \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N).$$

The brane stack is still ordinary $\mathfrak{gl}_N$, but the quantum moment-map ideal is traceless after the scalar comoment vanishes. With the central ghost γ_1, the representative is

$$\mathrm{Ob}_{\mathrm{sc}} = \hbar N\, \omega(f,g) \int_{\mathbb{R}} a(t)b(t)\gamma_1(t)\, dt.$$

A scalar-reduced ordinary $\mathfrak{gl}_N$ lift requires W_{ext} with leading class

$$[w_{\mathrm{ext}}] = -\hbar N\,[\bar{c}] \qquad \text{in } H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]].$$

Alternatively, the unreduced central-character source and the opposite central extension $\bar{A}^{\mathrm{cl}}_{-\hbar N\bar{c}}$ absorb the scalar symbol before it becomes a reduced-source obstruction, while a Casimir-balanced supertrace model makes the scalar representative zero. In the opposite branch $J_{\mathrm{op}} = -k^\vee$ satisfies $d_{\mathrm{CE}} J_{\mathrm{op}} = -\hbar N\bar{c}$; forgetting the central line restores the class $\hbar N[\bar{c}]$ on $\bar{A}$. The finite-window algebraic repair obtained by adjoining a defect-supported central closed source line is a legitimate changed branch: with generator e_M and

$$\Xi^{\mathrm{alg,sc}}_M(e_M) = -\hbar N\, \bar{c}_M$$

on the scalar-contact line, one may take $W_M = e_M$ and $C_M = 0$, so that $\beta^{\mathrm{sc}}_M = \beta^{\mathrm{sc}}_{\mathrm{cl}} = \mu^{\mathrm{sc}}_{\mathrm{cl}} = \lambda^{\mathrm{sc}}_C = 0$ in that enlarged algebraic cone. This is not yet a Costello-local closed-exchange theorem. Without a closed field, bulk-to-defect propagator, locality/support rule, scalar-contact chain projection on the original ordinary complex, and compatible finite-window/Roos representatives, the original scalar-reduced branch remains blocked first by

$$o^{(1),\mathrm{sc}}_{\sigma,M} = \hbar N[\bar{c}_M] \neq 0,$$

and then, for any supplied genuine closed-exchange tower, by

$$\beta^{\mathrm{sc}}_M, \quad \beta^{\mathrm{sc}}_{\mathrm{cl}}, \quad \mu^{\mathrm{sc}}_{\mathrm{cl}}, \quad \lambda^{\mathrm{sc}}_C.$$

(xvi) *Componentwise quantum coefficient surface and non-scalar obstruction complex.* The componentwise theorem datum is

$$\mathfrak{S}_{\hbar,w}(I) = \mathcal{K}^w_{\mathrm{BV}} \times \{0_{\mathrm{sc}}\} \times A^{\mathrm{pp},w}_{\partial,\hbar}(I) \times T_{\mathrm{Ham},\hbar}.$$

Here $\mathcal{K}^w_{\mathrm{BV}}$ is the weighted coefficient-kernel datum, 0_{sc} is the balanced supertrace scalar QME representative, $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$ is the reduced weighted Moyal principal-part current target, and $T_{\mathrm{Ham},\hbar}$ is the degree-zero Hamiltonian trace target governed by the radial/Weyl dual-potential decorated-Stokes gate. The present theorem includes the algebraic Moyal coefficients, the Capelli scalar-contact relation, the all-edge PBW telescoping theorem for $(2,b)$ and $(a,2)$, and finite normal-form certificates for the balanced diagonal through $(6,6)$ and all non-edge bidegrees of total degree at most 12. The remaining all-interior theorem is not an external slogan: it is the vanishing of the dual-potential obstruction

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b},$$

equivalently the decorated PBW Stokes condition $D^{\square}_{a,b} = C^+_{a,b}\partial_2$ for every bidegree. Failure is exactly a signed row $(\phi,-\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. The reduced Moyal current brackets are

$$\{B_f(a), B_g(b)\}_\hbar = B_{\{f,g\}_\hbar}(ab), \qquad \{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$

$$\{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0,$$

with the full Moyal coadjoint action on the weighted continuous coefficient module. This is a proved componentwise target surface, not a Costello graph/QME construction.

The next theorem obligation is the constructive non-scalar Costello/QME problem. For each finite window one must first supply a codimension-two closed marked Costello list $\mathcal{G}_M^{\mathrm{mk}}$, with every field-differential, BV-edge, collision/contact, counterterm, and CE-source face carrying its output graph, coefficient, marker transport, incidence sign, and lower-window truncation. Missing any one of these entries means that the marked graph differential and the Hom-valued Bianchi identity have not been constructed.

The target complex is the completed local-functional complex

$$\mathfrak{Q}^\bullet_{\mathcal{G},M}(I) = \Big(O^{\mathrm{bd}}_{\mathrm{loc},\mathcal{G},M}\big(\mathcal{E}^M_w|_I;\ A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)\big), d_M = Q + \{I^w_{\mathrm{o},M}, -\}_{\hbar,M}\Big),$$

complete for the $\hbar$-adic, scalar-contact, and finite-window filtrations. The Hom kernel complex is

$$\mathbf{K}^\bullet_{\mathcal{G},M}(I) = \mathrm{Hom}_{\mathrm{cont}}\Big(C^\bullet_{\mathrm{lab}}(\mathcal{G}_M), \mathfrak{Q}^\bullet_{\mathcal{G},M}(I)\Big),$$

with

$$d_{\mathbf{K},M}T = d_M T - (-1)^{|T|} T\, d_{\mathrm{CE}}, \qquad \mathrm{Curv}_{\mathbf{K},M}(\kappa) = d_{\mathbf{K},M}\kappa + \frac{1}{2}[\kappa,\kappa]_{\mathbf{K},M}.$$

The source-face term $-\kappa d_{\mathrm{CE}}$ is therefore part of the QME row calculation, not a later convention.

The bulk-to-defect kernel is a continuous current-valued cochain candidate

$$\kappa_{\hbar,w,I}\colon C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}) \longrightarrow \mathfrak{Q}^\bullet_{\mathcal{G},M}(I),$$

where the current source is either restricted to regular compactly supported densities, placed in the sign-conic finite-scaling habitats $\mathcal{D}'_{c,\Lambda_\pm,\mathrm{fs}}(I)$ for ordinary collision-conormal graph kernels, or equipped with the graphwise wavefront/counterterm relation $\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,\mathcal{M})$. Its zero-edge values are fixed by the coefficient theorem,

$$u_{a(t)dt\otimes f} \mapsto \iota_I(B^{w,M}_f(a)), \qquad c_{B,\rho} \mapsto \iota_I(\Theta^{w,M}_\rho(B)),$$

while the positive-edge values must be actual renormalized graph weights

$$K^M_\Gamma = W^{R,M}_{\Gamma,L,w}(B^\Gamma_\bullet).$$

The exact current-kernel obstruction is the curvature class in $H^1(\ker\widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$; Costello's general BV/RG formalism supplies the habitat, not these local Hamiltonian current weights.

Before this class is defined, choose a complete d_{QME}-stable scalar-contact filtration and a filtered chain projection

$$\widehat{\sigma}_{\mathrm{sc},M}\colon \mathfrak{Q}^\bullet_{\mathcal{G},M}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]]$$

whose associated graded is the scalar-symbol projection. In the balanced supertrace branch the scalar-lift tower $o^{(r)}_{\sigma,w}(I)$ of Lemma E.4 must vanish before the reduced non-scalar complex is formed. Then $\mathcal{Q}^\bullet_{w,M}(I) = \ker\widehat{\sigma}_{\mathrm{sc},M}$ is the finite-window non-scalar complex, and the remaining class

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker\widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}})$$

must be killed.

At order n, after lower counterterms $C_{<n,M}$ have been chosen, the mandatory marked row types are

$$R^{\mathrm{row}}_{d\Gamma,M} = \varepsilon_\Gamma d_M K^M_\Gamma, \qquad R^{\mathrm{row}}_{\mathrm{CE}(\Gamma,\nu),M} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu}),$$

$$R^{\mathrm{row}}_{b(\Gamma_1,\Gamma_2),M} = \frac{1}{2}\varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M},$$

together with the homogeneous lower-counterterm contribution

$$[\hbar^n]\Big(d_M C_{<n,M} + \frac{1}{2}\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}\Big).$$

Their scalar row sum must vanish before non-scalar cohomology is tested. If it vanishes, the residual $R^{\mathrm{ns}}_{n,M} \in \mathcal{Q}^1_{w,M}(I)$ represents $\mathfrak{o}^{\mathrm{ns}}_{n,M}$.

Choose row coordinates ρ^M_i and degree-zero primitive candidates η^M_j:

$$R^{\mathrm{ns}}_{n,M} = \sum_i r^M_i \rho^M_i, \qquad d_{\mathrm{ns},M}\eta^M_j = \sum_i A^M_{ij}\rho^M_i.$$

A fixed-window counterterm exists exactly when the row equation

$$A^M c = -r^M$$

is solvable. Failure is certified by a left-cokernel functional

$$\ell A^M = 0, \qquad \ell(r^M) \neq 0.$$

Windowwise solutions form an inverse-limit theorem only after row and primitive transports satisfy

$$T^{M,N} r^M = r^N, \qquad T^{M,N} A^M = A^N P^{M,N},$$

and the Roos primitive class

$$[P^{M,N} c_M - c_N] \in H^0(\mathcal{Q}^\bullet_{w,N}(I))$$

vanishes. For larger graph packages the transition matrices $u^{M,N}$ for seed rows, $v^{M,N}$ for CE-source faces, $q^{M,N}$ for bracket rows, and $P^{M,N}$ for primitives are construction data. The order-n obstruction vector is

$$\mathfrak{O}^{\mathrm{ns}}_n = \big(([\mathfrak{o}^{\mathrm{ns}}_{n,M}])_M, \lambda_n\big),$$

as in Theorem 3.98. Thus changing admissible weight or passing to a finite window is not itself a cancellation mechanism.

A closed-exchange repair is extra cochain data. It consists of a strict Mittag–Leffler tower of closed-exchange local-functional complexes

$$\mathcal{X}^\bullet_{w,M}(I)$$

and compatible cochain maps

$$\Xi_M : \mathcal{X}^\bullet_{w,M}(I) \longrightarrow \mathcal{Q}^\bullet_{w,M}(I), \qquad \pi_{M,N}\Xi_M = \Xi_N \pi_{M,N}, \qquad R^M_{w,w'}\Xi_M = \Xi'_M R^M_{w,w'}.$$

The complex $\mathcal{X}^{\bullet}_{w,M}(I)$ must encode the closed propagator, bulk-to-defect restriction, wavefront/locality rule, and Costello counterterm support. At the first non-scalar order the branch equation is

$$\mathfrak{v}^{\mathrm{ns}}_{n_0,M} + \Xi_M(W_{n_0,M}) + d_M C_{n_0,M} = 0, \qquad W_{n_0,M} \in Z^1(\mathcal{X}^{\bullet}_{w,M}(I)), \quad C_{n_0,M} \in \mathcal{Q}^0_{w,M}(I).$$

The finite-window branch is theorem-grade exactly in the following sense. For a fixed M, such $W_{n_0,M}$ and $C_{n_0,M}$ exist if and only if

$$[\mathfrak{v}^{\mathrm{ns}}_{n_0,M}] \in -\operatorname{im}\Big(H^1(\Xi_M)\colon H^1(\mathcal{X}^{\bullet}_{w,M}(I)) \to H^1(\mathcal{Q}^{\bullet}_{w,M}(I))\Big).$$

In the compatible finite-window inverse limit this becomes the closed-exchange branch class

$$\beta^{\mathrm{cl}}_{n_0} = \big([\mathfrak{v}^{\mathrm{ns}}_{n_0,M}]\big)_M \quad \text{in} \quad \operatorname{coker}\Bigg(\varprojlim_M H^1(\mathcal{X}^{\bullet}_{w,M}(I)) \xrightarrow{\varprojlim H^1(\Xi_M)} \varprojlim_M H^1(\mathcal{Q}^{\bullet}_{w,M}(I))\Bigg),$$

together with a closed-exchange representative-compatibility class

$$\mu^{\mathrm{cl}}_{n_0} \in \varprojlim\nolimits^1_M H^0(\mathcal{X}^{\bullet}_{w,M}(I))$$

and the usual counterterm primitive-compatibility class

$$\lambda^{\mathrm{cl}}_{n_0} \in \varprojlim\nolimits^1_M H^0(\mathcal{Q}^{\bullet}_{w,M}(I)).$$

Vanishing of these three classes is equivalent to a compatible closed-exchange kernel plus local counterterm killing the first non-scalar residual. The proof is the elementary cohomology calculation: the displayed equation implies $[\mathfrak{v}^{\mathrm{ns}}_{n_0,M}] = -H^1(\Xi_M)[W_{n_0,M}]$; conversely that equality supplies a windowwise closed-exchange class and a primitive $C_{n_0,M}$. The inverse-limit obstruction to choosing the closed-exchange cocycles compatibly is $\mu^{\mathrm{cl}}_{n_0}$, and the obstruction to choosing the counterterm primitives compatibly is $\lambda^{\mathrm{cl}}_{n_0}$. A counterterm-only branch is the special case $\mathcal{X}^{\bullet}_{w,M}(I) = 0$. What remains open is not a scalar QME objection, but the construction of $\mathcal{X}^{\bullet}$, Ξ, and a class $[W]$ mapping to $-[\mathfrak{v}^{\mathrm{ns}}_{n_0}]$.

The first certified larger-row test is the θ_3 row. In the one-row finite-window package currently supplied by the graph-array calculation, the non-scalar scalar-zero subcomplex is

$$\mathcal{Q}^0_{\theta_3,M} = 0, \qquad \mathcal{Q}^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, \qquad d_M = 0,$$

with residual

$$\mathfrak{v}^{\mathrm{ns}}_{\theta_3,M} = \mathsf{E}_{\theta_3,M}, \qquad \ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1.$$

Thus the present one-row datum proves $[\mathsf{E}_{\theta_3,M}] \neq 0$ in the supplied subcomplex. Supremum repair does not weaken the non-scalar QME theorem; it identifies the exact additional construction that would make the stronger theorem true. The row is killed precisely by one of the following three pieces of added mathematics, with finite-window and weight compatibilities:

$$\text{source ancestor:} \quad d_{\mathrm{CE}}\eta_{\theta_3,M} = \zeta_{M,\nu_3}, \qquad C^{\mathrm{CE}}_{\theta_3,M} = K^M_{\Theta_3}(\eta_{\theta_3,M}),$$

together with the Hom-valued Bianchi identity giving $d_M C^{\mathrm{CE}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$; or

$$\text{local counterterm:} \quad C^{\mathrm{ct}}_{\theta_3,M} \in \mathcal{Q}^0_{w,M}(I), \qquad d_M C^{\mathrm{ct}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M},$$

with $\widehat{\sigma}_{\mathrm{sc},M}(C^{\mathrm{ct}}_{\theta_3,M}) = 0$; or a complete companion-face table whose differential changes the row residual from $\mathsf{E}_{\theta_3,M}$ to zero. In all three cases the tower obstruction

$$[\pi_{M,N} C_{\theta_3,M} - C_{\theta_3,N}] \in H^0(\mathcal{Q}^{\bullet}_{w,N}(I))$$

must vanish in the Roos $\varprojlim^1$-class. A name C_{θ_3} without its local-functional formula, CE ancestor, or companion-face differential is not a counterterm. The currently tested finite routes are obstructed. The source obstruction is sharper than the degree-≤ 3 test. In the ordinary pure two-u CE source envelope, in every finite Hamiltonian degree, the covector

$$q_{2u} = e^*_{((H_{0,1},H_{2,0}),c_{\rho_{2,1}})} - \frac{1}{2} e^*_{((H_{0,1},H_{0,1}),c_{\rho_{0,2}})}$$

satisfies

$$q_{2u} d_{\mathrm{CE}} = 0, \qquad q_{2u}(\zeta_{M,\nu_3}) = 1.$$

Hence no ordinary pure two-u degree enlargement supplies a CE ancestor for ζ_{M,ν_3}. This is an exact obstruction theorem for that source envelope only; it is not a universal no-go for scalar-zero local counterterms, marked source extensions, or Hom-valued companion rows. The tested eight-face companion expression has coordinate row

$$(2, -2, 3, 2, -1, 1, -2, -3)$$

on

$$E_{\nu_{02}}, E_{\nu_{20}}, E_{\nu_{03}}, E_{\nu_{12}^{(1)}}, E_{\nu_{12}^{(2)}}, E_{\theta_3}, E_{\nu_{21}^{(2)}}, E_{\nu_{30}},$$

and diagonal transport to $N = 2$ leaves $2E^2_{\nu_{02}} - 2E^2_{\nu_{20}}$. The same eight-face row has q_{2u}-value zero, so it cannot be reused as the missing marked source column. The lower-window functional $\lambda^{(2)}_{\nu_{02}} = \frac{1}{2}(E^2_{\nu_{02}})^*$ detects the transported residual. A repair must therefore supply exact data not present in the tested package: a counterterm column $A^M_{\theta_3,C} = -1$ with $d_M C^{\mathrm{ct}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$, scalar-zero value, and zero Roos class; or a marked source generator $\eta^{\mathrm{mark}}_{\theta_3,M}$ outside the ordinary pure two-u algebra, with a boundary column of q_{2u}-value 1, codimension-two closure, Hom-valued Bianchi cancellation, scalar-zero value, and compatible transport; or a marked companion relation $R^{\mathrm{cand}}_{\Theta_3,M} = 0$ together with lower-window transport $v^{M,N} R^{\mathrm{cand}}_{\Theta_3,M} = 0$, including the $N = 2$ relation killing $2E^2_{\nu_{02}} - 2E^2_{\nu_{20}}$ by a derived non-diagonal transport, lower primitive, or relation $E^2_{\nu_{02}} = E^2_{\nu_{20}}$. The canonical source-coordinate lower primitive candidate is

$$\zeta^0_2 = u_{a,H_{1,0}} u_{b,H_{0,1}}, \qquad D^2_{02,20} = K^2_{\Theta_3}(\zeta^0_2), \qquad d_{\mathrm{CE}}\zeta^0_2 = 2\zeta_{\nu_{02}} - 2\zeta_{\nu_{20}}.$$

This is source exactness. In the local-functional complex

$$d_{\mathrm{ns},2} D^2_{02,20} = -2E^2_{\nu_{02}} + 2E^2_{\nu_{20}} + \mathfrak{b}^2_{02,20},$$

where

$$\mathfrak{b}^2_{02,20} = d_{\mathrm{ns},2} K^2_{\Theta_3}(\zeta^0_2) - K^2_{\Theta_3}(d_{\mathrm{CE}}\zeta^0_2).$$

In the Bianchi-exposed lower basis $(E^2_{\nu_{02}}, E^2_{\nu_{20}}, \mathfrak{b}^2_{02,20})$, the column and residual are

$$A^2_D = (-2, 2, 1)^T, \qquad r_2 = (2, -2, 0)^T.$$

The cokernel

$$\widetilde{\lambda}_{02,\mathfrak{b}} = \frac{1}{2}(E^2_{\nu_{02}})^* + (\mathfrak{b}^2_{02,20})^*$$

kills A^2_D and evaluates to 1 on r_2. The source primitive becomes a genuine local-functional primitive only after $\mathfrak{b}^2_{02,20}$ is killed, or after adding a scalar-zero local counterterm $B^2_{02,20}$ with $d_{\mathrm{ns},2}B^2_{02,20} = -\mathfrak{b}^2_{02,20}$, locality or wavefront admissibility, and compatible transport. The current finite-row habitat contains no such $B^2_{02,20}$: the desired target $(0,0,-1)^T$ is not in the image of A^2_D, because

$$\widetilde{\lambda}_{02,\mathfrak{b}} A^2_D = 0, \qquad \widetilde{\lambda}_{02,\mathfrak{b}}((0,0,-1)^T) = -1.$$

The next construction target is therefore a genuine scalar-zero local-functional column $A^2_B = (0,0,-1)^T$, not a relabelling of the source primitive. In a tower the named Bianchi transport cocycle is

$$\Delta^1_{M,N} = -\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N.$$

A compatible family B^M would kill it by

$$\Delta^1_{M,N} = d_{\mathrm{ns},N}(\pi_{M,N}B^M - B^N).$$

Thus the tower must first kill the H^1 class of $\Delta^1_{M,N}$; after a choice of primitive, it must also kill the resulting secondary $\varprojlim^1 H^0$ class. The tested finite routes do not close the row.

(xvii) *Unreduced factorization-current lift.* The unreduced cotangent lift is a datum: distributional principal-part boundary fields, product centrality homotopies, P_0-bracket homotopies, a coefficient kernel, a brane-defect propagator, and a QME solution. Polynomial boundary representatives are ruled out by the local-nilpotence theorem; the accepted target is the principal-part current sector; the centrality homotopies must be supplied as additional unreduced lift data. In convolution form an unreduced coupling

$$\kappa\colon C^\bullet_{\mathrm{CE}}(\mathfrak{g}) \longrightarrow Z^{P_0}_{\mathrm{fact}}(\mathcal{A}_{\mathrm{BV}})$$

must first satisfy

$$\mathrm{Curv}(\kappa) = d\kappa + \frac{1}{2}[\kappa,\kappa] = 0,$$

and then pass the descent and QME tests. A chain-level normal-ordering lift is separate data: either a cut-dependent no-ψ straightening as in Theorem 3.201, or a genuinely larger target retaining the descendant module together with maps N, i, H satisfying the relevant deformation of $QH + HQ = \mathrm{id} - iN$. Only in such a larger target could one attempt to form a correction. After matching the bracket convention of Construction 3.236, the defect

$$\{O_f, \Theta_\rho\} - \Theta_{f\cdot\rho}$$

would have to be Q-exact; no fixed formula for that homotopy is asserted before the target and signs are constructed. Until these maps are constructed in the chosen target, the principal-part cotangent lift is an obligation rather than a theorem. Moreover, the explicit normal-ordering straightening now proved in the body is only the cut-dependent no-ψ calculation. A full cyclic contraction with N killing all one-ψ descendants is impossible by Lemma 3.206; any genuine unreduced lift must retain those descendant classes or replace the target by a larger homotopy-coherent object.

(xviii) *Stratified brane prefactorization and module datum.* Let

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}.$$

For a finite normal/Hamiltonian window (N, M), let $\mathsf{X}_{N,M}$ be the set of nonzero moving characters occurring in that window, containing the relevant characters

$$n\epsilon_s + a\epsilon_1 + b\epsilon_2, \qquad n\epsilon_s - a\epsilon_1 - b\epsilon_2.$$

The coefficient ring for the equivariant obligation is

$$R^{N,M}_{\Omega} = \mathbb{C}[\epsilon_s, \epsilon_1, \epsilon_2, \hbar_\omega][\chi^{-1} \mid \chi \in \mathsf{X}_{N,M}].$$

On the self-dual branch one first imposes $\epsilon_1 + \epsilon_2 = 0$, then inverts only the surviving nonzero characters; resonant summands stay in $\mathrm{pr}_{\chi=0}$. A field-valued theorem may replace this by a declared K_Ω, but not by an undeclared Laurent field in the three weights. The brane theorem datum is not an ordinary bulk factorization algebra with a boundary algebra added afterwards. The stronger datum is a stratified prefactorization algebra on the pair (X, L),

$$\mathcal{F}^{\mathrm{str}}_{\Omega,N} \colon \mathrm{Disk}^{\mathrm{str}}_{X,L} \longrightarrow \mathrm{Ch}_{R^{N,M}_{\Omega}[[\hbar_{\mathrm{QME}}]]},$$

where $\mathrm{Disk}^{\mathrm{str}}_{X,L}$ contains bulk mixed holomorphic-topological disks in $X \setminus L$, brane intervals in L, and collars meeting the lower stratum. Its missing values are part of the data:

$$\mathcal{F}^{\mathrm{str}}_{\Omega,N}(U \subset X \setminus L) = C^{\bullet}_{\mathrm{CE,cont}}\big(\mathfrak{g}^{\mathrm{mix}}_{\Omega,\hbar_\omega}(U)\big),$$

$$\mathcal{F}^{\mathrm{str}}_{\Omega,N}(I \subset L) = A^{q}_{N,\Omega}(I) \quad \text{or} \quad A^{\mathrm{pp},w}_{\partial,\Omega,\hbar_{\mathrm{W}}}(I),$$

together with a collar module

$$\mathcal{M}_{\Omega,N}(U; I)$$

carrying the bulk-to-brane structure maps. The finite-N brane algebra branch is the equivariant quantum Hamiltonian reduction

$$A^{q}_{N,\Omega} = \mathrm{Weyl}_{\hbar_{\mathrm{W}}}(\mathrm{Mat}^2_N)//_{\hbar_{\mathrm{W}}} GL_N, \qquad YX - XY + \hbar_{\mathrm{W}} NI = 0,$$

with $w(X) = \epsilon_1$, $w(Y) = \epsilon_2$, and $w(\hbar_{\mathrm{W}}) = \epsilon_1 + \epsilon_2$. The stable principal-part branch is the weighted current target already recorded above. Choosing one branch, or constructing a comparison between them, is not optional notation; it is part of the stratified object. The three formal parameters remain separate:

$$\hbar_{\mathrm{QME}}, \qquad \hbar_{\mathrm{W}}, \qquad \hbar_\omega.$$

Here $\hbar_{\mathrm{QME}}$ counts Costello/BV loop order, $\hbar_{\mathrm{W}}$ deforms the Weyl/Moyal brane algebra, and $\hbar_\omega$ scalarizes the equivariant Hamiltonian bracket. A Weyl branch may impose $\hbar_{\mathrm{W}} = \hbar_\omega$; no branch identifies either parameter with $\hbar_{\mathrm{QME}}$ without a separate QME obstruction calculation.
The required structure maps are the extension-by-zero products for mutually disjoint stratified disks, the collar restriction maps from bulk insertions to brane modules, and descent for stratified Weiss covers. The first obstruction cochains are

$$\delta_{\mathrm{pref}}\mu_{\mathrm{str}} = d\mu_{\mathrm{str}} - \mu_{\mathrm{str}}d, \qquad \delta_{\mathrm{assoc}}\mu_{\mathrm{str}} = \mu_{\mathrm{str}}(\mu_{\mathrm{str}} \otimes 1) - \mu_{\mathrm{str}}(1 \otimes \mu_{\mathrm{str}}),$$

together with the descent defect

$$\check{C}^p(\mathcal{U};\mathcal{F}^q) = \prod_{i_0<\cdots<i_p} \mathcal{F}^q(U_{i_0}\cap\cdots\cap U_{i_p}), \qquad \mathbb{D} = \check{\delta} + (-1)^p d_{\mathcal{F}},$$

and

$$\delta_{\mathrm{Weiss}}(V,\mathcal{U}) = \mathrm{Cone}\Big(\mathcal{F}^{\mathrm{str}}_{\Omega,N}(V) \to \mathrm{Tot}\,\check{C}^\bullet(\mathcal{U};\mathcal{F}^{\mathrm{str}}_{\Omega,N})\Big).$$

A brane restriction of a closed insertion x must also come with product centrality and $P_{0,\hbar_{\mathrm{QME}}}$-bracket homotopies against arbitrary unreduced open observables a:

$$d_{\mathrm{QME}} H^{\mathrm{prod}}_{x,a} = \rho(x)a - (-1)^{|x||a|} a\rho(x),$$

$$d_{\mathrm{QME}} H^{P_0}_{x,a} = \{\rho(x),a\}_{P_{0,\hbar_{\mathrm{QME}}}} - \rho_{\mathrm{mod}}(x;a),$$

where ρ_{mod} is the closed-induced brane module operation to be constructed, with compatibility under interval inclusions, disjoint products, finite-window projections, and weight transports. These homotopies are not supplied by the reduced current brackets $\{B_f(a), B_g(b)\}_{\hbar_{\mathrm{W}}}$ and $\{B_f(a), \Theta_\rho(B)\}_{\hbar_{\mathrm{W}}}$.

At a finite marked window the centrality obstruction is the Hom-curvature row, for $x, y \in \{u_{a(t)dt\otimes f}, c_{B,\rho}\}$,

$$\begin{aligned} R^{\mathrm{cent}}_{x,y,M} = &\sum_{\Gamma\in\mathsf{A}^d_{x,y,M}} \varepsilon_\Gamma d_M K^M_\Gamma - \sum_{\Gamma\in\mathsf{A}^{\mathrm{CE}}_{x,y,M}} \varepsilon^{\mathrm{CE}}_\Gamma K^M_\Gamma(\zeta_{\Gamma,x,y}) \\ &+ \frac{1}{2}\sum_{(\Gamma_1,\Gamma_2)\in\mathsf{A}^{\mathrm{br}}_{x,y,M}} \varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar_{\mathrm{QME}},M} + R^{\mathrm{ct}}_{x,y,M}. \end{aligned}$$

Only after $\widehat{\sigma}_{\mathrm{sc},M}(R^{\mathrm{cent}}_{x,y,M}) = 0$ does this row define a non-scalar class

$$\mathfrak{o}^{\mathrm{cent}}_{x,y,M} \in H^{|x|+|y|}(Q^\bullet_{w,M}(I)).$$

A product or P_0-centrality homotopy is then an explicit primitive row

$$H^M_{x,y} \in Q^{|x|+|y|-1}_{w,M}(I), \qquad R^{\mathrm{cent}}_{x,y,M} + d_M H^M_{x,y} = 0,$$

compatible with interval inclusions, finite-window projections, and admissible-weight transports. The corresponding $\varprojlim^1 H^{|x|+|y|-1}$-class is part of the centrality obstruction, not a consequence of scalar-shadow vanishing.

In the normal Ω-branch the same row is formed over $R^{N,M}_\Omega[[\hbar_{\mathrm{QME}}]]$ with $d_{M,\Omega}$ and the Ω-equivariant bracket. The primitive must be basic:

$$R^{\mathrm{cent}}_{x,y,\Omega,M} + d_{M,\Omega} H^\Omega_{x,y,M} = 0, \qquad L_{V_\Omega} H^\Omega_{x,y,M} = 0.$$

If a primitive is chosen before passing to T_Ω-invariants, the secondary class

$$[L_{V_\Omega} H_{x,y,\Omega,M}] \in H^{|x|+|y|}(Q^\bullet_{w,\Omega,M}(I), d_{\mathrm{ns},\Omega,M})$$

must vanish before the homotopy is a Q_Ω-homotopy.

The brane-defect Costello QME condition is the same curvature equation as the non-scalar lift, now required as a natural transformation of the stratified prefactorization object:

$$\mathrm{Curv}(K_{\Omega,N}) = d_{\mathrm{QME}} K_{\Omega,N} + \frac{1}{2}[K_{\Omega,N}, K_{\Omega,N}]_{\hbar_{\mathrm{QME}}} = 0.$$

The Omega coefficient ring, normal weights, and notation gate supply only the habitat for this equation. They do not prove the brane-defect QME. Its obstruction vector is

$$\mathrm{Ob}^{\mathrm{int}}_{\mathrm{str},\Omega,N} = (\delta_{\mathrm{pref}}, \delta_{\mathrm{assoc}}, \delta^{\check{C}/R}_{\mathrm{Weiss}}, \mathrm{ob}_{\mathrm{collar}}, \mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}, \mathrm{ob}^{P_0}_{\mathrm{cent}}, \mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}, \mathrm{ob}_{\Omega-\mathrm{basic}}, \mathrm{ob}_{\mathrm{tr-top}}).$$

where

$$\delta^{\check{C}/R}_{\mathrm{Weiss}} = \Big(\{H^{\bullet}(\Delta^{N,M}_{V,\mathcal{U}})\}_{V,\mathcal{U},N,M}, \mathfrak{w}^{\check{C}/R}_{V,\Omega,N,M}\Big).$$

This is the internal vector. The source-primary class $\mathrm{ob}^{\mathrm{src}}_{\mathrm{CFG}}$ is external. A Costello, Costello–Francis–Gwilliam, or stratified-homology analogy enters the stable core only after a source fixture identifies the source theorem, verifies its hypotheses, and supplies a matched-conventions morphism to the local mixed HT pair (X, L). Without that fixture it is motivation, not theorem support.

(xix) *Trace-state functional and physical large-N surface.* A protected brane trace is not produced by the algebraic stable trace theorem alone. The stronger theorem requires a continuous state

$$\omega_{N,\Omega} \colon \int_{L\subset X} \mathcal{F}^{\mathrm{str}}_{\Omega,N} \longrightarrow R^{N,M}_{\Omega}[[\hbar_{\mathrm{QME}}]]$$

on the stratified factorization homology of the chosen brane module. The finite-N input is a state on the quantum Hamiltonian reduction $A^{q}_{N,\Omega}$; the stable branch must specify the map from $A^{\mathrm{pp},w}_{\partial,\Omega,\hbar_{\mathrm{W}}}$ to the same state target. The trace datum is

$$\mathfrak{T}_{\Omega,N} = (\omega_{N,\Omega}, \nu_{\Omega,N}, \mathcal{T}_{\Omega}, \kappa_{\Omega,N}, C_{\Omega,N}, H^{\mathrm{prod}}, H^{P_0}),$$

where $\nu_{\Omega,N}$ fixes trace, N, the parameters $\hbar_{\mathrm{QME}}, \hbar_{\mathrm{W}}, \hbar_{\omega}$, and the residue/Euler normalization branch; $\mathcal{T}_{\Omega}$ is the topology in which $N \to \infty$ asymptotics are asserted; $\kappa_{\Omega,N}$ and $C_{\Omega,N}$ are the brane-defect kernel and counterterms; and H^{prod}, H^{P_0} are the centrality homotopies above. The normal Euler denominator is

$$e_{T_{\Omega}}(N_L X)^{-1} = (\epsilon_s \epsilon_1 \epsilon_2)^{-1}.$$

The normalization must decide among

$$\nu^{\mathrm{res}}_{\Omega} = 1, \qquad \nu^{\mathrm{Eu}}_{\Omega} = \sigma_s(\epsilon_s \epsilon_1 \epsilon_2)^{-1},$$

or an explicitly named Euler-rescaled residue convention

$$\nu^{\mathrm{Eres}}_{\Omega} = \sigma_s(\epsilon_s \epsilon_1 \epsilon_2)^{-1}.$$

Thus the Euler factor is built into $\omega_{N,\Omega}$, into $\nu^{\mathrm{Eres}}_{\Omega}$, or left external. Plain residue normalization never contains the Euler denominator silently.

The Ward identities are part of the theorem, not a consequence of the word trace:

$$\omega_{N,\Omega}(d_{\mathrm{QME}}F) = 0, \qquad \omega_{N,\Omega}(Q_\Omega F) = 0,$$

$$\omega_{N,\Omega}(\widehat{\mu}(\xi)F) = 0, \qquad \omega_{N,\Omega}(\mathrm{Curv}(K_{\Omega,N})\, F) = 0,$$

whenever the expressions are defined in the chosen brane-defect local-functional complex. These identities require the scalar anomaly branch, the non-scalar QME class, and the centrality homotopies to have been solved or absorbed by an explicitly named source replacement. They are not consequences of $R_\Omega^{N,M}$ localization.
Connected large-N cumulants are then defined by

$$\langle F_1, \dots, F_k\rangle^c_{N,\Omega} = [t_1 \cdots t_k]\ \log \omega_{N,\Omega}\left(\exp\left(\sum_i t_i F_i\right)\right),$$

after finite-N trace identities and the empty-trace scalar have been treated in the normalization $\nu_{\Omega,N}$. A physical large-N theorem is the asymptotic statement, in $\mathcal{T}_\Omega$,

$$\langle F_1, \dots, F_k\rangle^c_{N,\Omega} \sim \sum_{g\geq 0} N^{2-2g-k} F^\Omega_{g;k}(F_1, \dots, F_k),$$

or the same formula with a different explicitly chosen $\hbar_{\mathrm{QME}}, \hbar_{\mathrm{W}}, \hbar_\omega, N$-scaling. The algebraic degreewise stable trace theorem supplies candidate primitive generators only; it does not choose $\omega_{N,\Omega}, \nu_{\Omega,N}, \mathcal{T}_\Omega$, or the Ward identities.
The trace obstruction vector is

$$\mathrm{Ob}_{\mathrm{tr},\Omega} = (\mathrm{ob}_{\mathrm{state}}, \mathrm{ob}_{\mathrm{norm}}, \mathrm{ob}_{\mathrm{Ward}}, \mathrm{ob}_{\mathrm{cum}}, \mathrm{ob}_{\mathrm{conv}}, \mathrm{ob}_{\mathrm{QME}}, \mathrm{ob}_{\mathrm{src}}).$$

Vanishing means that the state exists, the normalization is fixed, the QME and moment-map Ward identities hold, connected cumulants are defined in the chosen topology, the large-N expansion converges asymptotically there, and all external factorization-homology or Costello analogies have been audited against primary sources and matched to this local brane model.

(xx) *Radial/Weyl finite certificates and the dual-potential decorated-Stokes gate.* The ordinary Weyl-trace part of the degree-zero Hamiltonian trace target is no longer a bare external input. The formal Moyal coefficients and the Capelli scalar-contact relation are proved algebraically, and the quantum shear primitive has a finite obstruction complex. In bidegree (a, b) let

$$R^{\mathrm{free}}_{a,b} = \mathcal{N}_{\mathrm{free}}(E_{a,b,N}) - \mathcal{N}_{\mathrm{free}}\left(\sum_W c^{\mathrm{cl}}_W\ \mathrm{Tr}(WM)\right)$$

be the free normal-form residual after the classical cyclic lift. The theorem-grade obstruction is the dual-potential class

$$B_{a,b}(W) = (\partial W, C^+_{a,b}(W)), \qquad \eta_{a,b} = (T_{a,b}, E^+_{a,b}),$$

$$\Omega^{\mathrm{rad}}_{a,b} = [\eta_{a,b}] \in \mathrm{coker}\, B_{a,b}.$$

The bidegree closes exactly when $\Omega^{\mathrm{rad}}_{a,b} = 0$. After fixing a classical lift, the free normal-diagram model is the subordinate map

$$C_{a,b}\colon\ \ker \partial \longrightarrow \mathsf{Diag}_{a,b},$$

and a displayed certificate is an explicit $K_{a,b} \in \ker \partial$ with $C_{a,b}(K_{a,b}) = R^{\text{free}}_{a,b}$. This is a normal-diagram presentation of the same obstruction, not the final theorem surface.
Proposition F.27 proves zero obstruction for the edge families $(2, b)$ and $(b, 2)$ for all b. The finite certificates additionally give zero obstruction for the balanced diagonal through $(6, 6)$, for all non-edge bidegrees of total degree at most 12, and for the total-degree 13 interior cases

$$(3, 10), (10, 3), (4, 9), (9, 4), (5, 8), (8, 5), (6, 7), (7, 6).$$

The total-degree 14 certificates also close

$$(3, 11), (11, 3), (4, 10), (10, 4), (5, 9), (9, 5).$$

The first undecided total-degree 14 systems are

$$(6, 8), (8, 6), (7, 7).$$

In the current basis these are timeout/frontier computations: the ordinary cyclic boundary equation closes, and no signed obstruction row is known. The balanced cases $(5, 5)$ and $(6, 6)$ require nonzero corrections with 16 and 54 terms respectively; a nonzero correction is not a nonzero obstruction.
The all-bidegree theorem is therefore $\Omega^{\text{rad}}_{a,b} = 0$ for every bidegree. The ordinary square-cell necklace cycle mechanism supplies $\ker \partial = \operatorname{im} \partial_2$ in the relevant quotient, so this is equivalently the decorated PBW Stokes equation

$$D^{\square}_{a,b} H_{a,b} = R^{\text{free}}_{a,b}, \qquad D^{\square}_{a,b} = C^{+}_{a,b} \partial_2.$$

A functorial splitting of the family $C_{a,b}$, or an associated-graded Hodge homotopy on the decorated two-colour necklace incidence graph with edges

$$[WXY] \longrightarrow [WYX].$$

is one construction route to this Stokes identity. A finite table, even a long one, is evidence for the mechanism but not the theorem. Conversely, an obstruction row for the free model may be written as a decorated harmonic projection

$$\Pi^{\text{harm}}_{a,b} R^{\text{free}}_{a,b},$$

or, in the older free-complex shadow, a functional $\lambda_{a,b}$ with $\lambda_{a,b} C_{a,b} = 0$ and $\lambda_{a,b}(R^{\text{free}}_{a,b}) \neq 0$; in the theorem-grade dual-potential system, failure is exactly a signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$ with

$$\lambda(E^{+}_{a,b}) - \phi(T_{a,b}) \neq 0.$$

Supremum repair asks first for the decorated Stokes identity, or equivalently the dual-potential identity $\lambda(E^{+}_{a,b}) = \phi(T_{a,b})$ for every such row; only a violating row proves that the stronger radial/Weyl theorem fails. The ordinary square-cell necklace cycle mechanism is now proved; the missing theorem is the decorated PBW Stokes identity for $D^{\square}_{a,b} = C^{+}_{a,b} \partial_2$.

(xxi) *Moyal and graph normalization.* The Weyl/Moyal convention is

$$f * g = m \circ \exp\Big(\frac{\hbar}{2}(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})\Big)(f \otimes g).$$

The normalized bracket is

$$\{f, g\}_\hbar = \sum_{s \geq 0} \frac{\hbar^{2s}}{2^{2s}(2s+1)!} P^{2s+1}(f, g).$$

The first and third raw commutator weights are 1 and 1/24; any Costello realization must reproduce these coefficients after its sign, orientation, automorphism, and counterterm conventions are fixed. These coefficients are necessary tests for a Costello realization, not sufficient data for analytic BV/QME existence.

(xxii) *Global and cross-volume descent.* A target theorem supplies a matched-conventions datum

$$\begin{aligned}\mathfrak{C}_{\mathrm{tar}} = (&d, Y, \eta, (\Sigma_{d-1}, C), L, \tau, \lambda, \mathfrak{f}, \Lambda,\\ &\hbar_{\mathrm{tar}}, \mathcal{O}_{\mathrm{tar}}, \chi_{\mathrm{cl}}, \chi_{\mathrm{op}}, \chi_{\mathrm{anom}},\\ &\mathcal{K}_{\mathrm{obs}}).\end{aligned}$$

Its admissibility vector is

$$\begin{aligned}\mathrm{Ob}_{\mathfrak{C}_{\mathrm{tar}}} = (&\mathrm{per}_{\mathrm{tar}}, \mathrm{ob}_{\mathrm{WR}}, \mathrm{ob}_{\mathrm{anom}}, \mathrm{ob}_{\mathrm{QME}},\\ &\mathrm{ob}_{\mathrm{conv}}, \mathrm{ob}_{\mathrm{hyp}}, \mathrm{ob}_{6r}, \mathrm{ob}_{\mathrm{MD}}).\end{aligned}$$

Here $\mathrm{ob}_{\mathrm{conv}}$ records convergence of the transported obstruction tower, $\mathrm{ob}_{\mathrm{hyp}}$ records hyperdescent and the absence of hidden $\varprojlim^1$ defects, and ob_{6r} records any asserted six-relation factorization constraint. A CoHA, compact BCOV, quintic, OSV/GV, Abel–Jacobi, MNOP/chiral-volume, Vol III, Igusa/BKM, or sister-volume theorem may use the formal-local theorem only after the target supplies this vanishing obstruction vector, chosen null-homotopies, and comparison morphisms identifying its formal restriction with the local Hamiltonian BF/Moyal normalization. The transfer theorem is target data, not a consequence of the formal disk. Missing source fixtures in those external lanes are not core local proof obligations and do not supply theorem support for the local QME, Ext, cyclic, Weyl/Moyal, radial-parts, or kernel theorem data.

5 Matched-conventions firewall

5.1 *Protected twisted-M motivation and local reduction*

Perturbative twisted M-theory in the sense of Costello [25] enters through a protected Hamiltonian realization datum at a brane point. The mathematical assertion is a realization criterion: vanishing of the period, closed-field comparison, scalar-anomaly, and brane-defect QME obstructions constructs the admissible Hamiltonian sector to which the formal local theorem of §2.15 applies.

Definition 5.1 *(Protected Hamiltonian sector at a brane point).* Let (X, ω) be a connected holomorphic-symplectic surface, let $p \in X$, and let $\mathcal{T}_{\mathrm{tw}M}(\mathbb{R} \times X; \mathbb{R} \times p)$ denote perturbative twisted-M-theory data

on $\mathbb{R} \times X$ in the presence of a stack of N one-dimensional topological branes supported on $\mathbb{R} \times p$. A *protected Hamiltonian sector at* $\widehat{p}$ is a restricted datum

$$\mathcal{T}_{\mathrm{tw}M}\big(\mathbb{R} \times X; \mathbb{R} \times p\big)\big|_{\mathrm{Ham},\widehat{p}} = \big(\mathcal{E}_{\mathrm{open,prot}}, \mathcal{E}_{\mathrm{cl,prot}}, \partial^{\mathrm{prot}}_{\mathrm{bb}}\big)$$

consisting of the protected open BV theory on the brane line, the protected closed BV theory coming from formal Hamiltonian operators at the symplectic disk $\widehat{X}_p$, and the protected bulk-to-boundary evaluation on the brane algebra.

The realization problem is to construct this triple from the ambient BV fields and to prove the vanishing of the obstruction datum below.

Definition 5.2 *(Admissible protected Hamiltonian restriction).* A protected Hamiltonian sector is *admissible for the local theorem* if it comes with the following data.

(i) *Formal Darboux Hamiltonian reduction.* The completed closed Hamiltonian algebra at p is identified, up to formal symplectomorphism, with $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, and the point brane Ext algebra is identified with $\mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p) \cong \Lambda^* T_p X$, with cyclic trace induced by ω_p.

(ii) *Brane-defect locality.* The protected operator restriction to $\mathbb{R} \times p$ factors through the weighted Tate field complex and is compatible with the interval-wise factorization map $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$, the chain-level primitive projection, and the weighted Hadamard parametrix used in the local theorem.

(iii) *Scalar anomaly compatibility.* The centre-of-mass scalar anomaly of the protected brane stack maps to the class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ used in the local scalar-anomaly comparison.

Definition 5.3 *(Protected twist obstruction datum).* Fix a candidate protected twisted-M background near $\mathbb{R} \times p$ and a candidate comparison with the Hamiltonian formal-disk model. The comparison carries four obstruction classes.

First, every holomorphic symplectic vector field v determines the closed one-form

$$\alpha_v = -\iota_v \omega.$$

The holomorphic de Rham exact sequence

$$0 \longrightarrow \mathbb{C}_X \longrightarrow \mathcal{O}_X \xrightarrow{d} \Omega^1_{X,\mathrm{cl}} \longrightarrow 0$$

gives the period obstruction

$$\mathrm{per}(v) = \delta(\alpha_v) \in H^1(X, \mathbb{C}_X).$$

This is the obstruction to choosing a single global holomorphic Hamiltonian primitive for v.

Second, a chosen local comparison of closed protected fields with the Hamiltonian BF fields gives a cone

$$C_{\mathrm{cl},p} = \mathrm{Cone}\Big(\mathcal{E}^!_{\mathrm{cl,prot},c}(\mathbb{R} \times \widehat{X}_p) \xrightarrow{\chi_{\mathrm{cl}}} \Omega^\bullet_c(\mathbb{R}) \widehat{\otimes} (\widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1])^!\Big).$$

The comparison is a field-level equivalence only after this cone is acyclic in the protected summand.

Third, the scalar anomaly mismatch is

$$\mathrm{ob}_{\mathrm{scal}} = \chi_{\mathrm{anom}}\big([\mathrm{anom}_{\mathrm{prot}}]\big) - [\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}),$$

where χ_{anom} is the anomaly map induced by the comparison and $[\bar{c}]$ is the scalar class used in the local theorem.

Fourth, after choosing the weighted coefficient model used by the local theorem, the brane-defect quantum obstruction is the class

$$\mathrm{ob}_{\mathrm{QME}} = \Big[QI_{\mathrm{prot}} + \frac{1}{2}\{I_{\mathrm{prot}}, I_{\mathrm{prot}}\} + \hbar\Delta I_{\mathrm{prot}} \Big] \in H^1\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_{\mathrm{prot},w}), Q + \{I_{0,\mathrm{prot}}, -\}\big).$$

Its vanishing is the precise statement that the protected brane-defect interaction satisfies the QME in the same normalization as Problem 3.87.

Definition 5.4 *(Casimir-balanced equivariant QME datum).* A Casimir-balanced equivariant brane-defect datum consists of:

(i) a matrix Lie superalgebra $\mathfrak{s}$ acting on the Chan-Paton factor, together with an even nondegenerate invariant form $B_{\mathfrak{s}}$ used to build the BV heat-kernel contraction;

(ii) a trace character $\mathrm{tr}_{\mathfrak{s}}$ on the matrix factor. Let $C_{B_{\mathfrak{s}}} = \sum_\alpha e_\alpha \otimes e^\alpha$ be the Casimir for $B_{\mathfrak{s}}$, put $K_{\mathfrak{s}} = m(C_{B_{\mathfrak{s}}})$, and define the scalar contraction coefficient $\kappa_{\mathfrak{s}}$ on the scalar trace line by

$$\mathrm{tr}_{\mathfrak{s}}(AK_{\mathfrak{s}}) = \kappa_{\mathfrak{s}}\,\mathrm{tr}_{\mathfrak{s}}(A).$$

The datum is Casimir-balanced when $\kappa_{\mathfrak{s}} = 0$;

(iii) the weighted brane-defect local-functional complex

$$\mathfrak{Q}_{\mathfrak{s}} = \Big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{\mathfrak{s}}_{\mathrm{bd},w}), d_{\mathrm{QME}} = Q + \{I_{0,\mathfrak{s}}, -\}\Big);$$

(iv) a d_{QME}-stable scalar filtration and a filtered chain projection

$$\widehat{\sigma}_{\mathrm{sc}} \colon \mathfrak{Q}_{\mathfrak{s}} \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{\mathbf{1}})[1]$$

whose associated graded is the scalar-symbol map σ_{sc};

(v) a $P_{(z_1)}$-equivariant coefficient topology on the formal symplectic disk, where

$$P_{(z_1)} = \{\varphi \in \mathrm{Symp}_{\mathrm{form}}(\widehat{\mathbb{C}^2}_0) : \varphi^*(z_1) \in (z_1)\}.$$

The scalar associated graded of $\mathfrak{Q}_{\mathfrak{s}}$ is the Chevalley–Eilenberg complex of the reduced Hamiltonian algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ with coefficients in the central boundary ghost line $\mathbb{C}\gamma_{\mathbf{1}}$.

THEOREM 5.5 *(Theorem F double-prime: Casimir-balanced equivariant scalar QME criterion).* Let $(\mathfrak{s}, B_{\mathfrak{s}}, \mathrm{tr}_{\mathfrak{s}})$ be a Casimir-balanced equivariant datum in the sense of Definition 5.4. For $a, b \in \Omega^0_c(\mathbb{R})$ and $f, g \in \mathbb{C}[z_1, z_2]$, the scalar part of the mixed brane-defect QME obstruction is represented in $\mathfrak{Q}_{\mathfrak{s}}$ by

$$\mathrm{Ob}^{\mathfrak{s}}_{\mathrm{sc}}(\gamma_{\mathbf{1}}; a, f; b, g) = \hbar\,\kappa_{\mathfrak{s}}\,\omega(f, g)\int_{\mathbb{R}} a(t)b(t)\gamma_{\mathbf{1}}(t)\,dt.$$

Hence the scalar obstruction vanishes identically on the Casimir-balanced locus $\kappa_{\mathfrak{s}} = 0$. The vanishing is $P_{(z_1)}$-equivariant. Equivalently, the associated graded scalar class is

$$[\mathrm{Ob}^{\mathfrak{s}}_{\mathrm{sc}}] = \hbar\,\kappa_{\mathfrak{s}}\,[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}\gamma_{\mathbf{1}}),$$

and the theorem applies exactly on the Casimir-balanced locus $\kappa_{\mathfrak{s}} = 0$. For the standard $\mathfrak{gl}(N|N)$ supertrace pairing, $K_{\mathfrak{s}} = \mathrm{sdim}(V)I = 0$, so $\kappa_{\mathfrak{s}} = 0$. Thus the scalar channel is an accepted conditional quantum input precisely on this locus: its acceptance component is the class $\hbar\,\kappa_{\mathfrak{s}}[\bar{c}]$.

If, in addition, the reduced non-scalar summand

$$\mathfrak{Q}_{\mathfrak{s}}^{\mathrm{nsc}} = \mathrm{Cone}\big(\widehat{\sigma}_{\mathrm{sc}} \colon \mathfrak{Q}_{\mathfrak{s}} \longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{\mathbf{1}})[1]\big)[-1]$$

has zero degree-one cohomology after the chosen weighted regulator and interval-factorization null-homotopies are imposed, then the complete degree-one brane-defect QME obstruction class in $H^1(\mathfrak{Q}_{\mathfrak{s}})$ vanishes.

Proof. The scalar contraction in the BV bracket is obtained by closing the matrix index in the single boundary loop. With the invariant form $B_{\mathfrak{s}}$ this contraction is the multiplication $K_{\mathfrak{s}} = m(C_{B_{\mathfrak{s}}})$ of the Casimir tensor, and its scalar trace-line coefficient is $\kappa_{\mathfrak{s}}$. The target part of the same local bracket is the constant Taylor coefficient cocycle

$$\omega(f, g) = [\{f, g\}]_0$$

of Lemma 2.4, and the brane-line locality gives the compactly supported factor $\int_{\mathbb{R}} a(t)b(t)\gamma_{\mathbf{1}}(t)\, dt$. This proves the displayed representative.

Passing to the scalar associated graded keeps only the Hamiltonian cocycle $[\bar{c}]$ and the matrix coefficient $\kappa_{\mathfrak{s}}$, giving the displayed cohomology class. Casimir-balance sets this coefficient to zero, so the degree-one representative is the zero cochain, not merely an exact cochain. The $P_{(z_1)}$-equivariance follows because $\kappa_{\mathfrak{s}}$ is matrix-factor data and ω is the symplectic Poisson cocycle on the target factor. The completion at the brane axis is functorial only under $P_{(z_1)}$, which is exactly the equivariance group in the hypothesis.

The last assertion uses the long exact sequence of cohomology for the displayed cone, which is defined only for the filtered chain projection $\widehat{\sigma}_{\mathrm{sc}}$ included in the datum. The scalar quotient class is zero by the first part, and the reduced degree-one cohomology is zero by hypothesis. Therefore the total degree-one obstruction class in $H^1(\mathfrak{Q}_{\mathfrak{s}})$ vanishes. □

Definition 5.6 *(Spin-Hecke queer coefficient input).* Let $N \geq 1$ and

$$\mathfrak{q}(N) = \left\{ \begin{pmatrix} A & B \\ B & A \end{pmatrix} : A, B \in \mathfrak{gl}_N \right\} \subset \mathfrak{gl}(N|N).$$

Put

$$J = \begin{pmatrix} 0 & I_N \\ -I_N & 0 \end{pmatrix}, \qquad P^{\mathfrak{q}} = \begin{pmatrix} I_N & 0 \\ 0 & -I_N \end{pmatrix},$$

and define the queer trace

$$\mathrm{otr}\begin{pmatrix} A & B \\ B & A \end{pmatrix} = \mathrm{Tr}(B) \in \Pi\mathbb{C}.$$

A spin-Hecke input for the queer brane is the Sergeev Hecke–Clifford action on $V^{\otimes n}$, $V = \mathbb{C}^{N|N}$, whose central-character comparison identifies

$$\mathrm{Tr}_{\mathfrak{gl}_N}(I_N) = \mathrm{otr}_{\mathfrak{q}(N)}(J) = N.$$

Theorem 5.7 *(Theorem G odd-trace: queer centre anomaly from the spin-Hecke input).* Assume the spin-Hecke input of Definition 5.6. For the queer-trace boundary evaluation $f \mapsto \mathrm{otr}\, f(\phi_1, \phi_2)$, the odd scalar QME representative is

$$\mathrm{Ob}^{\mathrm{otr}}_{\mathrm{sc}}(\gamma_{\mathbf{1}}; a, f; b, g) = \hbar N\, \omega(f, g) \int_{\mathbb{R}} a(t)b(t)\gamma_{\mathbf{1}}(t)\, dt \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{\mathfrak{q}}_{\mathrm{bd},w})_{\bar{1}}.$$

Its associated graded CE class is

$$\hbar N[\bar{c}]^{\mathrm{otr}} \in H^2_{\mathrm{Lie}}(\bar{A}, \Pi\mathbb{C})_{\bar{1}}.$$

This class is nonzero and is independent of the even class $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})_{\bar{0}}$. Since $N \geq 1$, the class does not vanish over $\mathbb{C}[[\hbar]]$ before imposing an odd scalar quotient or a target null-homotopy of the odd line $\Pi\mathbb{C} \cdot [\bar{c}]^{\mathrm{otr}}$. Hence the odd-trace channel is accepted by a conditional quantum target exactly after that quotient or null-homotopy is included in the target datum; otherwise its acceptance obstruction is the displayed nonzero odd class. The spin-Hecke input justifies the coefficient identity at the cohomology layer. It does not give a chain-level QME cancellation:

$$P^{\mathfrak{q}} J (P^{\mathfrak{q}})^{-1} = -J$$

is the obstruction to the unsigned parity-equivariant regulator used in the super-balanced theorem above.

Proof. The queer brane has two central coefficient lines: the even line generated by the identity and the odd line generated by J. The queer trace kills the even identity and evaluates the odd central element by

$$\mathrm{otr}(J) = \mathrm{Tr}(I_N) = N.$$

The spin-Hecke input asserts that this coefficient is preserved by the Sergeev central-character comparison; the Hamiltonian cocycle on the target remains the same $\omega(f, g) = [\{f, g\}]_0$. Therefore the scalar boundary loop gives exactly the displayed odd representative.

Passing to associated graded removes the brane-line smearing and keeps the Lie two-cocycle $\Pi\omega$, hence the class $\hbar N[\bar{c}]^{\mathrm{otr}}$. Nontriviality is the same calculation as Lemma 2.4: a primitive $\bar{\eta} : \bar{A} \to \Pi\mathbb{C}$ would lift with $\tilde{\eta}(1) = 0$, but then

$$\delta\tilde{\eta}(z_1, z_2) = -\tilde{\eta}(\{z_1, z_2\}) = -\tilde{\eta}(1) = 0$$

while $\omega(z_1, z_2) = 1$. The even and odd classes are independent because they lie in the two direct summands $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}) \oplus H^2_{\mathrm{Lie}}(\bar{A}, \Pi\mathbb{C})$. Since $N \geq 1$ in the datum and $\hbar$ is formal, the displayed odd class is not killed inside $H^2_{\mathrm{Lie}}(\bar{A}, \Pi\mathbb{C})[[\hbar]]$. Killing it requires an additional target quotient or null-homotopy; it is not a consequence of the spin-Hecke coefficient comparison.

Finally, $P^{\mathfrak{q}} J (P^{\mathfrak{q}})^{-1} = -J$ is an explicit matrix computation. Thus the queer channel has signed parity, not the unsigned parity condition used in the super-balanced cancellation. The algebraic spin-Hecke input gives the coefficient class, but not a null-homotopy of this odd obstruction. □

LEMMA 5.8 *(Primary obstruction test).* A protected twisted-M background can supply an admissible protected Hamiltonian restriction only if the period class, the comparison cone, the scalar mismatch, and the QME obstruction in Definition 5.3 vanish in the senses stated there. Conversely, if the period classes vanish for the Hamiltonian vector fields under consideration, the comparison cone is acyclic on the protected summand, the scalar mismatch is zero, and a solution of the displayed QME class has been chosen compatibly with interval factorization, then the candidate comparison supplies the data of Definition 5.2.

Proof. The period statement is the connecting morphism of $0 \to \mathbb{C}_X \to \mathcal{O}_X \xrightarrow{d} \Omega^1_{X,\mathrm{cl}} \to 0$: local Hamiltonian primitives h_i for α_v differ by constants on overlaps, and that Čech cocycle is exactly $\delta(\alpha_v)$. It vanishes precisely when the local primitives glue to a global Hamiltonian, modulo one global constant.

A comparison of protected closed fields with Hamiltonian BF fields is a quasi-isomorphism on the protected summand exactly when its mapping cone is acyclic. If the cone has cohomology, the unmatched class is a protected closed field invisible to the formal Hamiltonian model, or a Hamiltonian BF field not produced by the protected sector.

Scalar compatibility is an equality in $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$, since both the protected anomaly and the local scalar anomaly are central Lie algebra cocycles after the comparison. Finally, the displayed QME expression is the standard degree-one obstruction class for deforming the classical interaction to a quantum BV interaction in the fixed weighted coefficient category. The interval-factorization compatibility is precisely the brane-defect locality clause in the admissibility definition. □

Definition 5.9 *(Ambient-to-protected realization map).* An ambient-to-protected realization map is a morphism from the ambient twisted-M BV fields near $\mathbb{R} \times p$ to the triple

$$(\mathcal{E}_{\mathrm{open,prot}}, \mathcal{E}_{\mathrm{cl,prot}}, \partial^{\mathrm{prot}}_{\mathrm{bb}})$$

whose closed-field cone is $C_{\mathrm{cl},p}$, whose scalar anomaly map is χ_{anom}, and whose brane interaction has QME obstruction $\mathrm{ob}_{\mathrm{QME}}$.

Proposition 5.10 *(Admissible protected-sector reduction).* Let (X, ω) be a connected holomorphic-symplectic surface and let $p \in X$. Suppose that a perturbative twisted-M background with branes on $\mathbb{R} \times p$ supplies an admissible protected Hamiltonian sector at $\widehat{p}$. Then, after the Darboux coordinate and scalar-anomaly normalizations included in the admissibility data, that restricted sector is identified with the formal local reduced Hamiltonian CE/PV brane model of §2.15.

The open side is

$$\Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1],$$

the closed side is

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}), \qquad \mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1], \qquad \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1,$$

and the bulk-to-boundary evaluation is

$$f \longmapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

Hence the proved Hamiltonian, factorization, and scalar-anomaly statements of the local theorem apply to the protected sector. The all-order one-antifield quantum extension is still conditional on vanishing of the mixed brane-defect QME obstruction of Problem 3.87 for that sector.

Proof. The formal Darboux part of admissibility identifies the completed Hamiltonian algebra at the brane point with $\mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, after choosing formal Darboux coordinates. The point-brane Ext algebra and its cyclic trace give the open coefficient data of the Dirac probe model.

Brane-defect locality supplies the factorization-algebraic restriction to the brane line and matches it with the local factorization map, primitive projection, and weighted Hadamard reduction already used in the local theorem. These hypotheses put the protected sector in the same algebraic category as the local model. QME-flatness for the one-antifield quantum extension is the vanishing condition $\mathrm{ob}_{\mathrm{QME}} = 0$ in Theorem 5.11.

Scalar anomaly compatibility identifies the brane centre-of-mass anomaly with the class $[\bar{c}]$ appearing in the local anomaly comparison. The three admissibility inputs therefore supply exactly the data on which the local theorem acts. Applying that theorem gives the asserted protected-sector reduction. □

THEOREM 5.11 *(Protected twist-to-Hamiltonian comparison criterion).* Let (X, ω) be a connected holomorphic-symplectic surface and $p \in X$. Suppose a perturbative twisted-M background with branes on $\mathbb{R} \times p$ supplies a candidate protected sector together with the obstruction datum of Definition 5.3. Assume:

(i) the period classes $\mathrm{per}(v)$ vanish for the locally Hamiltonian symplectic vector fields used by the sector;

(ii) the closed-field comparison cone $C_{\mathrm{cl},p}$ is acyclic on the protected summand;

(iii) $\mathrm{ob}_{\mathrm{scal}} = 0$;

(iv) $\mathrm{ob}_{\mathrm{QME}} = 0$, with a chosen solution compatible with compactly supported interval factorization.

Then the candidate protected sector determines an admissible protected Hamiltonian restriction. Consequently the reduced brane-line Hamiltonian CE/PV, factorization-current, and scalar-anomaly statements of the formal local theorem apply to that protected sector. If, in addition, the QME solution includes the one-antifield weighted Tate lift of Problem 3.87, then the corresponding one-antifield quantum Hamiltonian comparison is valid for that protected sector and for that sector only.

Proof. By Lemma 5.8, the four hypotheses supply the Darboux Hamiltonian reduction, the field-level protected comparison, scalar anomaly compatibility, and brane-defect QME locality required in Definition 5.2. Applying Proposition 2.25 at the formal disk and then the admissible protected-sector reduction gives the local Hamiltonian BF model and its bulk-to-boundary evaluation. The last sentence is exactly the conditional weighted Tate/QME extension isolated in Problem 3.87, transported through the chosen protected-sector equivalence. □

LEMMA 5.12 *(Formal-restriction external acceptance criterion).* An admissible protected Hamiltonian restriction contributes precisely the formal Hamiltonian disk, the Dirac brane Ext algebra, the brane-line factorization restriction, and the scalar anomaly class $[\bar{c}]$. For an external compact BCOV, MNOP/chiral-volume, CY-to-chiral, Igusa, Borcherds, or BKM target T, this contribution remains only a conditional quantum input. The target theorem accepts it as matched-conventions global or cross-volume descent data exactly after T supplies its convention datum $\mathfrak{C}_T$, a morphism $\mathfrak{C}^{\mathrm{Ham}}_{\widehat{p}} \to \mathfrak{C}_T$, and null-homotopies for the obstruction vector

$$\mathrm{Ob}^{\mathrm{ext}}_T = \left(\mathrm{per}_T, \mathrm{ob}^{\mathrm{desc}}_T, \mathrm{ob}^{\mathrm{BCOV}}_T, \mathrm{ob}^{\mathrm{QME}}_T, \mathrm{ob}^{\mathrm{conv}}_T, \mathrm{ob}^{\mathrm{hyp}}_T, \mathrm{ob}^{\mathrm{VolIII}}_T, \mathrm{ob}^{\mathrm{Igusa/BKM}}_T, \mathrm{ob}^{6r/\mathrm{MD}}_T\right) \in H^1(\mathfrak{K}_T),$$

with a component omitted only when T does not assert the corresponding structure. A non-zero component is the exact obstruction to the corresponding external theorem.

Proof. The three admissibility clauses record Darboux Hamiltonians at $\widehat{X}_p$, the point-brane Ext algebra, brane-defect locality on $\mathbb{R} \times p$, and scalar anomaly compatibility. They contain no compact Calabi–Yau threefold, no negative-cyclic Calabi–Yau class, no Vol III specialization datum $\Phi_d = \mathrm{Sp}_{\Sigma_{d-1},C} \circ \Phi^{\mathrm{FA}}_d$, no compact worldsheet or S^3 framing, no BCOV propagator or holomorphic anomaly equation, no Borcherds input Jacobi form, no Weyl chamber, and no compact Hall orientation. Two ambient theories can have the same admissible formal restriction and different choices of all those global data. In the target convention complex $\mathfrak{K}_T$ these choices are independent coordinates of $\mathrm{Ob}^{\mathrm{ext}}_T$. Vanishing with specified null-homotopies is therefore precisely the acceptance datum for using the formal restriction in T; a non-zero coordinate is precisely the named target obstruction. □

Definition 5.13 *(Protected-sector matched-conventions contribution).* The protected-sector reduction contributes the following datum to any external matched-conventions theorem:

$$\mathfrak{C}^{\mathrm{Ham}}_{\widehat{p}} = \big(d = 2, \widehat{X}_p, \omega_p, \mathbb{R}_{\mathrm{brane}}, \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}, \mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p), [\bar{c}], \Phi^{\mathrm{Ham}}_{\mathrm{fact}}\big).$$

The remaining datum for a compact BCOV, CY-to-chiral, or Igusa/BKM theorem consists of the compact target, its Calabi–Yau or modular class, the worldsheet or specialization curve, the framing, the global propagator or Hall orientation, and the target obstruction complex.

Corollary 5.14 *(Formal uniformity after admissibility).* For every connected holomorphic-symplectic surface (X, ω), every brane point $p \in X$, and every protected twisted-M realization whose obstruction datum vanishes, the restricted sector is identified with the same formal Hamiltonian brane model

$$\big(\widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1], \Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1], [\bar{c}]\big)$$

after choosing a formal Darboux coordinate at p.

Proof. The period vanishing gives global Hamiltonian primitives for the vector fields used in the sector. The acyclic closed comparison cone identifies the protected closed fields with $\widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1]$. The Ext-algebra identification gives the Dirac brane probe, and scalar compatibility fixes the same class $[\bar{c}]$. These are exactly the inputs of Proposition 2.25. □

Remark 5.15 *(Dirac language inside the protected sector).* For an admissible protected Hamiltonian sector, the Dirac-thesis language of Remark 2.72 is a shorthand for the already identified formal local model. The brane stack probes the formal symplectic normal disk; Dirac reduction gives the open trace algebra; the closed Hamiltonian BF theory is the shifted cotangent theory of canonical transformations of that disk; and the closed observables identify with derived Poisson-central operations on the stable open Hamiltonian brane algebra only in the reduced local sense proved by Theorem 2.70. The scalar anomaly is the same class $[\bar{c}]$ on the closed and open sides. The corresponding ambient centrality statement is the matched-conventions interface of Corollary 5.16.

Corollary 5.16 *(Protected-sector matched-conventions acceptance criterion).* A theorem in a compact BCOV, CY-to-chiral, or Igusa/BKM target may use the protected-sector reduction only as a matched-conventions descent input: it must give a morphism $\mathfrak{C}^{\mathrm{Ham}}_{\widehat{p}} \to \mathfrak{C}_T$ and null-homotopies for every asserted component of $\mathrm{Ob}^{\mathrm{ext}}_T$.

Proof. Lemma 5.12 identifies the external acceptance class $\mathrm{Ob}^{\mathrm{ext}}_T \in H^1(\mathfrak{K}_T)$. A morphism of convention data and null-homotopies for this class supply exactly the compact, worldsheet, modular, orientation, and descent coordinates needed by the target theorem. Corollary 5.14 then supplies the $d = 2$ formal-disk normalization functorially in the chosen Darboux chart. □

5.2 *Cross-volume matched-conventions theorem*

The theorem is local on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$, with brane line $\mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{0\}$. It is the Hamiltonian BF theorem for the formal symplectic disk and its Dirac brane probe. Compact Calabi-Yau, CoHA, quintic, HKQ/CDGP/GV/OSV, Abel–Jacobi, MNOP/chiral-volume, Borcherds/BKM, and global twisted-M-theory consequences are not consequences of this local theorem. They can enter only through a supplied matched-conventions datum and the vanishing, with null-homotopies, of the obstruction vector in Theorem G.7.

The formal Weyl/Moyal calculation fixes the Capelli coefficients on the reduced degree-zero Hamiltonian sector. The Costello first/third graph-normalization criterion checks the same coefficients inside a putative brane-defect graph model. The all-order one-antifield quantum extension requires vanishing of the mixed brane-defect QME class of Problem 3.87.

Appendix G records the matched-conventions and obstruction-complex data required for any transfer to Vol III, Vol II, Vol I, the Igusa cusp-form program, a compact Calabi-Yau threefold, a $K3 \times E$ specialisation, CoHA, quintic, OSV/GV, Abel–Jacobi, or a BKM denominator identity.

Acknowledgements

This work belongs to the lineage of Bershadsky–Cecotti–Ooguri–Vafa 1993, Witten 1988, Costello 2011, and Beilinson–Drinfeld 2004. The local Hamiltonian BF analysis is a finite algebraic chain-level probe of the formal-disk corner of that program.

A Matlis Principal Parts and the Continuous Cotangent Module

This appendix centralizes the reduced Matlis calculation used in Theorem 2.59. Its stable core is the following: the m-adic continuous dual of the reduced Hamiltonian Lie algebra is the reduced scalar-annihilator inside the top local-cohomology module; the residue pairing gives the coadjoint action; and no ordinary polynomial descendant module realizes this coadjoint representation. The residue pairing is the universal dual module normalization for this lane. The Fourier–Rees construction below then gives the strongest flat positive bridge that exists: a genuine $\mathcal{D}_\hbar$-module bridge after Fourier conjugation of the polarization, together with an exact obstruction theorem for any bridge that tries to preserve the original Hamiltonian action. Over $\mathbb{C}[[\hbar]]$ its target is the Fourier delta Rees lattice inside local cohomology, not the full standard Čech lattice; the full principal-part lattice appears only after tensoring with $\mathbb{C}((\hbar))$. It does not identify $\Psi_{a,b}$ with $\rho_{a,b}$ as $\mathfrak{h}$-modules. The source conventions are Matlis duality [19], Grothendieck–Hartshorne local cohomology [20], and the power-series residue model of Kunz–Cox–Dickenstein, Chapter 5 [21].

Definition A.1 *(Topology and reduced Hamiltonians).* Put

$$R = \mathbb{C}[[z_1, z_2]], \qquad \mathfrak{m} = (z_1, z_2), \qquad \mathfrak{h} = R/\mathbb{C} \cdot 1.$$

The topology on R is the $\mathfrak{m}$-adic topology. The topology on $\mathfrak{h}$ is the quotient topology, equivalently the topology transported through the zero-constant section

$$\mathfrak{h} \simeq \mathfrak{m}, \qquad \overline{f} \longmapsto f - f(0).$$

A fundamental system of neighbourhoods of zero is

$$F^N\mathfrak{h} = \mathfrak{m}^N, \qquad N \geq 1,$$

via this splitting. The Lie bracket is the Poisson bracket

$$\{f, g\} = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g$$

followed by projection modulo constants. The adjoint maps are continuous for this topology: if $f \in \mathfrak{m}$, then $\{f, \mathfrak{m}^{N+1}\} \subset \mathfrak{m}^N$. The constant class acts by zero and is not part of $\mathfrak{h}$.

Lemma A.2 *(Continuous dual).* The continuous dual of $\mathfrak{h}$, with $\mathbb{C}$ discrete, is

$$\mathfrak{h}^\vee_{\mathrm{cont}} = \bigcup_{N\geq 1} (\mathfrak{h}/F^N\mathfrak{h})^\vee = \bigoplus_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\,\delta_{a,b},$$

where

$$\delta_{a,b}(z_1^m z_2^n) = \delta_{a,m}\delta_{b,n}.$$

Proof. A linear functional on the $\mathfrak{m}$-adic vector space $\mathfrak{h} \simeq \mathfrak{m}$ is continuous exactly when its kernel contains $F^N\mathfrak{h} = \mathfrak{m}^N$ for some N. Such a functional factors through the finite-dimensional quotient $\mathfrak{m}/\mathfrak{m}^N$, hence is a finite linear combination of Taylor coefficient functionals. Conversely every finite linear combination of the displayed $\delta_{a,b}$ kills all sufficiently high powers of $\mathfrak{m}$, hence is continuous. □

Definition A.3 *(Reduced Matlis principal parts).* The top local cohomology module of R is written in Čech Laurent form as

$$H^2_{\mathfrak{m}}(R)\,dz_1dz_2 = \bigoplus_{a,b\geq 0} \mathbb{C}\, z_1^{-a-1}z_2^{-b-1}\,dz_1dz_2.$$

Equivalently this is the Čech quotient

$$R[z_1^{-1}, z_2^{-1}]/(R[z_1^{-1}] + R[z_2^{-1}])$$

after multiplication by dz_1dz_2; the displayed direct sum is the standard representative basis in which both exponents are strictly negative. The basis vector $\rho_{0,0} = z_1^{-1}z_2^{-1}dz_1dz_2$ is not zero in $H^2_{\mathfrak{m}}(R)\,dz_1dz_2$; it is the functional which evaluates the constant coefficient. The reduced principal-part module is the $\mathbb{C}$-linear residue annihilator of constants,

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) = \bigoplus_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\,\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1}z_2^{-b-1}\,dz_1dz_2.$$

We give $\mathcal{P}$ the locally finite direct-sum topology

$$\mathcal{P} = \varinjlim_{N} \mathcal{P}_{\leq N}, \qquad \mathcal{P}_{\leq N} = \bigoplus_{\substack{a,b\geq 0\\ 0<a+b\leq N}} \mathbb{C}\,\rho_{a,b},$$

and give $\mathfrak{h}^\vee_{\mathrm{cont}}$ the corresponding inductive-limit topology through the basis $\delta_{a,b}$. Thus the residue identification below is a continuous map of the declared locally finite topological vector spaces, not a bare algebraic relabelling. The space $\mathcal{P}$ is a Hamiltonian-coadjoint module; it is not an R-submodule of $H^2_{\mathfrak{m}}(R)\,dz_1dz_2$, since multiplication by z_i can hit the excluded scalar-residue line. The residue pairing is

$$\langle \rho_{a,b}, z_1^m z_2^n\rangle = \mathrm{Res}_{z_1=z_2=0}\, z_1^{m-a-1}z_2^{n-b-1}\,dz_1dz_2 = \delta_{a,m}\delta_{b,n}.$$

Proposition A.4 *(Matlis realization of the continuous dual).* The residue pairing induces a continuous vector-space isomorphism

$$\mathcal{P} \xrightarrow{\sim} \mathfrak{h}^\vee_{\mathrm{cont}}, \qquad \rho_{a,b} \longmapsto \delta_{a,b}.$$

It is the scalar-reduced restriction of the Matlis evaluation pairing for the complete regular local ring R: after the trivialization $\omega_R = R\,dz_1dz_2$, $H^2_{\mathfrak{m}}(R)\,dz_1dz_2 \cong E_R(\mathbb{C})$. The subspace $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$ is the $\mathbb{C}$-linear scalar-annihilator dual to $\mathfrak{h} = R/\mathbb{C}\cdot 1$. The R-module structure on $E_R(\mathbb{C})$ does not restrict to $\mathcal{P}$; multiplication can hit the excluded scalar residue line. The Hamiltonian coadjoint action below is transported separately by the residue pairing and is the structure used by the brane cotangent sector.

Proof. The displayed residue formula identifies each basis vector $\rho_{a,b}$, $a+b>0$, with the continuous Taylor coefficient functional $\delta_{a,b}$. Lemma A.2 shows that these coefficient functionals form the whole continuous dual. The exclusion of $\rho_{0,0}$ is exactly the condition that the functional vanish on the scalar line $\mathbb{C}\cdot 1\subset R$; this is the continuous dual of the quotient $R/\mathbb{C}\cdot 1$. On finite stages the map sends $\mathcal{P}_{\le N}$ isomorphically to the functionals vanishing on $\mathfrak{m}^{N+1}$, so both the map and its inverse are continuous for the declared locally finite topologies. □

PROPOSITION A.5 *(Residue coadjoint action).* Under the isomorphism of Proposition A.4, the coadjoint action of $f\in\mathfrak{h}$ is the principal-part residue action

$$f\cdot\rho=\pi_{\mathrm{pp}}\{f,\rho\},$$

where π_{pp} projects Laurent two-forms to the span of the principal parts. On monomials,

$$z_1^p z_2^q\cdot\rho_{a,b}=-(pb-qa+p-q)\rho_{a-p+1,b-q+1},$$

with the convention that any term with a negative index is zero. If the displayed target is $\rho_{0,0}$, then the coefficient is zero; hence the scalar residue line is not produced from $\mathcal{P}$.

Proof. Let $X_f=(\partial_{z_1}f)\partial_{z_2}-(\partial_{z_2}f)\partial_{z_1}$. Hamiltonian vector fields preserve the volume form dz_1dz_2. For a Laurent coefficient ϕ and $g\in R$, the residue of a total z_i-derivative is zero, hence

$$0=\operatorname{Res}\big(L_{X_f}(\phi g\,dz_1dz_2)\big)=\operatorname{Res}\big((\{f,\phi\}g+\phi\{f,g\})\,dz_1dz_2\big).$$

Therefore the residue pairing satisfies integration by parts,

$$\langle\{f,\rho\},g\rangle=-\langle\rho,\{f,g\}\rangle,$$

which is exactly the coadjoint sign convention $(\mathrm{ad}_f^\vee\lambda)(g)=-\lambda(\{f,g\})$.

For $f=z_1^pz_2^q$ and $\rho_{a,b}=z_1^{-a-1}z_2^{-b-1}\,dz_1dz_2$,

$$\begin{aligned}\{f,z_1^{-a-1}z_2^{-b-1}\}&=pz_1^{p-1}z_2^q\,(-(b+1))z_1^{-a-1}z_2^{-b-2}\\&\quad-qz_1^pz_2^{q-1}\,(-(a+1))z_1^{-a-2}z_2^{-b-1}\\&=-(pb-qa+p-q)z_1^{p-a-2}z_2^{q-b-2}.\end{aligned}$$

Multiplication by dz_1dz_2 gives $\rho_{a-p+1,b-q+1}$ precisely when the two new indices are nonnegative; otherwise the principal-part projection gives zero. If $a-p+1=b-q+1=0$, then $a=p-1$ and $b=q-1$, and the coefficient $pb-qa+p-q=p(q-1)-q(p-1)+p-q$ is zero. Thus the action preserves the scalar-annihilator $\mathcal{P}$. □

LEMMA A.6 *(Continuity of the coadjoint action).* The formula of Proposition A.5 extends from polynomial Hamiltonians to all of $\mathfrak{h}=\mathbb{C}[[z_1,z_2]]/\mathbb{C}\cdot 1$. The bilinear action

$$\mathfrak{h}\times\mathcal{P}\longrightarrow\mathcal{P}$$

is continuous for the $\mathfrak{m}$-adic topology on $\mathfrak{h}$ and the locally finite direct-sum topology on $\mathcal{P}$.

Proof. Fix $\rho_{a,b}$. In the monomial formula a term $z_1^p z_2^q \cdot \rho_{a,b}$ can be nonzero only if $p \le a+1$ and $q \le b+1$. Thus an arbitrary formal Hamiltonian acts on $\rho_{a,b}$ through a finite jet, so the action is well-defined and continuous in the Hamiltonian variable. Conversely, if $a+b \le N$, every nonzero target has total index

$$(a-p+1)+(b-q+1) = a+b-(p+q)+2 \le N+1$$

because $p+q>0$ for a reduced Hamiltonian. Hence each fixed Hamiltonian sends $\mathcal{P}_{\le N}$ into $\mathcal{P}_{\le N+1}$. This is exactly continuity for the locally finite direct limit. □

Lemma A.7 *(Coadjoint centralizer).* Every $\mathfrak{h}$-module endomorphism of $\mathcal{P}$, with the coadjoint action above, is multiplication by one scalar.

Proof. The vector $\rho_{1,0}$ is cyclic. Indeed $z_1^2 \cdot \rho_{1,0} = -2\rho_{0,1}$; repeated action by the linear Hamiltonians z_1 and z_2 then reaches every $\rho_{a,b}$, $a+b>0$, with nonzero rising-factorial coefficient.

Let T be an $\mathfrak{h}$-module endomorphism. Since $z_1 z_2 \cdot \rho_{a,b} = (a-b)\rho_{a,b}$, the vector $T\rho_{1,0}$ lies in the eigenvalue-one subspace

$$\bigoplus_{r\ge 0} \mathbb{C}\,\rho_{r+1,r}.$$

It is a finite sum because $\mathcal{P}$ is a direct sum: $T\rho_{1,0} = \sum_{r=0}^{R} c_r \rho_{r+1,r}$. The Hamiltonian z_2^2 kills $\rho_{1,0}$, hence it kills $T\rho_{1,0}$. But for $r \ge 1$,

$$z_2^2 \cdot \rho_{r+1,r} = (2r+4)\rho_{r+2,r-1},$$

while the $r=0$ target has negative second index and is zero. The displayed nonzero targets are linearly independent, so $c_r = 0$ for all $r \ge 1$. Therefore $T\rho_{1,0} = c_0\rho_{1,0}$, and cyclicity gives $T = c_0\,\mathrm{id}_{\mathcal{P}}$. □

Theorem A.8 *(Residue pairing rigidity).* Up to multiplication by one scalar in $\mathbb{C}^\times$, the residue pairing is the unique perfect continuous pairing

$$\mathcal{P} \times \mathfrak{h} \longrightarrow \mathbb{C}$$

which is equivariant for the coadjoint action above. Here *perfect* means that the adjoint map $\mathcal{P} \to \mathfrak{h}^\vee_{\mathrm{cont}}$ is a topological vector-space isomorphism. In particular the same uniqueness holds in the narrower class of total-degree-zero pairings. Fixing the scalar by

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}$$

is therefore the universal dual-module normalization for the principal-part cotangent sector.

Proof. Let β be another perfect continuous equivariant pairing. It gives a topological vector-space isomorphism

$$\Phi_\beta : \mathcal{P} \longrightarrow \mathfrak{h}^\vee_{\mathrm{cont}}, \qquad \Phi_\beta(\rho)(g) = \beta(\rho, g).$$

Equivariance of β says

$$\beta(f\cdot\rho, g) = -\beta(\rho, \{f,g\}),$$

so Φ_β is an $\mathfrak{h}$-module map to the coadjoint module. Let Φ_{res} be the residue isomorphism of Proposition A.4. Then $T = \Phi_{\mathrm{res}}^{-1}\Phi_\beta$ is an $\mathfrak{h}$-module automorphism of $\mathcal{P}$. By Lemma A.7, $T = c\,\mathrm{id}_{\mathcal{P}}$. Since Φ_β is an isomorphism, $c \ne 0$, and $\beta = c\,\langle\,\cdot\,,\,\cdot\,\rangle_{\mathrm{res}}$. The displayed Kronecker normalization sets $c=1$. □

THEOREM A.9 *(Polynomial descendant obstruction for principal parts).* Let $\mathcal{M}$ be a polynomial boundary descendant module on which the linear Hamiltonians z_1, z_2 act locally nilpotently. Then $\mathcal{P}$ is not isomorphic to any nonzero submodule of $\mathcal{M}$ as an $\mathfrak{h}$-module. In particular the polynomial classes $\Psi_{a,b}$ cannot be identified with the principal parts $\rho_{a,b}$ by an ordinary $\mathfrak{h}$-module isomorphism.

Proof. In any ordinary polynomial boundary model, the linear Hamiltonians act as constant coefficient Hamiltonian vector fields; iterating either z_1 or z_2 eventually exhausts the polynomial degree. Thus these actions are locally nilpotent.

On $\mathcal{P}$, Proposition A.5 gives

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Hence

$$z_1^N \cdot \rho_{a,b} = (-1)^N (b+1)(b+2)\cdots(b+N)\rho_{a,b+N} \neq 0$$

for every N, and similarly for z_2. Local nilpotence is preserved by module isomorphism and by restriction to submodules. Therefore no polynomial descendant module with locally nilpotent linear Hamiltonians realizes $\mathcal{P}$.

For the manuscript's primitive PBW labels this obstruction is already visible at the boundary of the index quadrant. The special-fibre action has

$$z_1 \cdot \widetilde{\Psi}_{a,b} = b\,\widetilde{\Psi}_{a,b-1}, \qquad z_2 \cdot \widetilde{\Psi}_{a,b} = -a\,\widetilde{\Psi}_{a-1,b},$$

after the scalar descendant is removed. Thus z_1 kills $\widetilde{\Psi}_{a,0}$, $a > 0$, whereas $z_1 \cdot \rho_{a,0} = -\rho_{a,1} \neq 0$. No rescaling $\widetilde{\Psi}_{a,b} \mapsto \lambda_{a,b}\rho_{a,b}$ with all $\lambda_{a,b} \neq 0$ can intertwine the two actions. □

THEOREM A.10 *(Universal Fourier–Rees construction and exact obstruction).* Let

$$\mathcal{D}_\hbar = \mathbb{C}[[\hbar]]\langle z_1, z_2, \partial_1, \partial_2\rangle \big/ \big([\partial_i, z_j] = \hbar\delta_{ij}, [z_i, z_j] = [\partial_i, \partial_j] = 0\big)$$

be the Rees Weyl algebra. Let

$$\mathcal{O}_\hbar = \mathbb{C}[z_1, z_2][[\hbar]] \cong \mathcal{D}_\hbar/(\partial_1, \partial_2)$$

be the polynomial Rees module, and let

$$\Delta_\hbar^{\mathsf{F}} = \mathcal{D}_\hbar/(z_1, z_2)$$

be the Fourier delta Rees module, generated by a vector e_0 killed by z_1, z_2. These quotients are left cyclic $\mathcal{D}_\hbar$-modules, with the indicated left ideals generated by ∂_1, ∂_2 and by z_1, z_2. PBW normal form gives $\mathcal{O}_\hbar$ the free $\mathbb{C}[[\hbar]]$-basis $z_1^a z_2^b$ and gives $\Delta_\hbar^{\mathsf{F}}$ the free basis $\partial_1^a \partial_2^b e_0$; in particular both Rees modules are $\mathbb{C}[[\hbar]]$-flat. The Čech local-cohomology lattice

$$\Delta_\hbar^{\mathrm{std}} = H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2[[\hbar]]$$

carries the standard action in which z_i multiplies and $\partial_i = \hbar\partial_{z_i}$. Sending $e_0 \mapsto \rho_{0,0} = z_1^{-1}z_2^{-1}dz_1dz_2$ realizes $\Delta_\hbar^{\mathsf{F}}$ as the Rees sublattice

$$\iota_{\mathsf{F}}(\Delta_\hbar^{\mathsf{F}}) = \bigoplus_{a,b\geq 0} \mathbb{C}[[\hbar]]\,(-1)^{a+b}\hbar^{a+b}a!b!\,\rho_{a,b} \subset \Delta_\hbar^{\mathrm{std}}.$$

This inclusion becomes an isomorphism after tensoring with $\mathbb{C}((\hbar))$, but it is not an isomorphism of the two $\mathbb{C}[[\hbar]]$-lattices. The Rees Fourier automorphism

$$\mathsf{F}(z_i) = \partial_i, \qquad \mathsf{F}(\partial_i) = -z_i$$

identifies the Fourier twist $\mathsf{F}(\mathcal{O}_\hbar)$ with $\Delta_\hbar^{\mathsf{F}}$. This is the flat positive bridge: the action of $P \in \mathcal{D}_\hbar$ on the polynomial source is carried to the action of $\mathsf{F}(P)$ on the Fourier delta target. On basis labels in the Čech realization,

$$z_1^a z_2^b \longmapsto \partial_1^a \partial_2^b e_0 \overset{\iota_{\mathsf{F}}}{\longmapsto} \partial_1^a \partial_2^b \rho_{0,0} = (-1)^{a+b} \hbar^{a+b} a!b!\, \rho_{a,b}.$$

After inverting $\hbar$, this gives a filtered Fourier–Rees identification of label spaces between nonconstant polynomial labels and the reduced principal parts $\rho_{a,b}$, $a + b > 0$, but only after the Fourier twist of action just stated. Over the Rees lattice before inverting $\hbar$, the construction records the degree-$a + b$ shift by the explicit factor $\hbar^{a+b}$. The special fibre of the abstract Rees module keeps the labels $\partial_1^a \partial_2^b e_0$; the Čech inclusion into $\Delta_\hbar^{\mathrm{std}}$ sends the positive-degree labels into $\hbar\Delta_\hbar^{\mathrm{std}}$. Thus the comparison is an associated-graded label match, not an isomorphism of ordinary local-cohomology lattices.

The obstruction for the original Hamiltonian action is exact. The Hamiltonian action on polynomial descendants is by vector fields $X_f = \{f, -\}$ on $\mathcal{O}_\hbar((\hbar))$. In the Rees Weyl algebra the normalized linear operators are $\hbar X_{z_1} = \partial_2$ and $\hbar X_{z_2} = -\partial_1$. Their Fourier transforms are $\mathsf{F}(\hbar X_f)$, whereas the Matlis cotangent action on $\mathcal{P}((\hbar))$ is still $\hbar X_f$, acting on Laurent principal parts and projected to the reduced principal-part span. Already for the linear Hamiltonians,

$$\hbar X_{z_1} = \partial_2, \qquad \hbar X_{z_2} = -\partial_1,$$

but

$$\mathsf{F}(\hbar X_{z_1}) = -z_2, \qquad \mathsf{F}(\hbar X_{z_2}) = z_1.$$

The transformed linear operators are locally nilpotent on $\Delta_\hbar^{\mathsf{F}}$; the Rees-normalized coadjoint operators

$$\hbar X_{z_1} \rho_{a,b} = -(b+1)\hbar \rho_{a,b+1}, \qquad \hbar X_{z_2} \rho_{a,b} = (a+1)\hbar \rho_{a+1,b}$$

are not locally nilpotent on any nonzero reduced principal part after inverting $\hbar$. Dividing by $\hbar$ gives the same conclusion for the unnormalized Hamiltonian vector fields. Hence the Fourier–Rees construction cannot identify the polynomial descendant module with $\mathcal{P}$ as a module over the same reduced Hamiltonian Lie algebra. Thus any universal bridge must choose one of two precise alternatives: Fourier-conjugate the action as above, or keep the original Matlis coadjoint action and accept the same-action obstruction.

Proof. The displayed formulas for F preserve the Rees Weyl relations, since

$$[\mathsf{F}(\partial_i), \mathsf{F}(z_j)] = [-z_i, \partial_j] = \hbar\delta_{ij}.$$

The polynomial module $\mathcal{O}_\hbar$ is cyclic with generator 1 annihilated by ∂_1, ∂_2. After Fourier twist the cyclic generator is annihilated by z_1, z_2, so the twisted module is $\mathcal{D}_\hbar/(z_1, z_2) = \Delta_\hbar^{\mathsf{F}}$. In the Čech model, $z_i \rho_{0,0} = 0$, and applying $\partial_i = \hbar\partial_{z_i}$ to $\rho_{0,0}$ generates the Rees lattice displayed in the theorem. It generates the full standard local-cohomology lattice only after $\hbar$ is inverted. Therefore $\mathsf{F}(\mathcal{O}_\hbar) \cong \Delta_\hbar^{\mathsf{F}}$, with the Čech realization ι_{F}.

Since ∂_i acts as $\hbar\partial_{z_i}$ on Laurent representatives,

$$\partial_1^a \partial_2^b \rho_{0,0} = \hbar^{a+b} \partial_{z_1}^a \partial_{z_2}^b (z_1^{-1} z_2^{-1} dz_1 dz_2) = (-1)^{a+b} \hbar^{a+b} a!b! \rho_{a,b}.$$

Removing the scalar label $a = b = 0$ gives the reduced label comparison after the stated Rees normalization. Since the factors $a!b!$ and $(-1)^{a+b}$ are units over $\mathbb{C}$, the only genuine Rees-lattice shift is the power $\hbar^{a+b}$.

For the obstruction, compare the two linear actions. On polynomial descendants the Rees-normalized operators $\hbar X_{z_1} = \partial_2$ and $\hbar X_{z_2} = -\partial_1$ are locally nilpotent; after inverting $\hbar$, the same is true for X_{z_1} and X_{z_2}. Fourier sends the Rees-normalized operators to multiplication by $-z_2$ and z_1 on $\Delta^{\mathsf{F}}_{\hbar}$; multiplication by z_i lowers pole order and is again locally nilpotent. The coadjoint Matlis action is not the Fourier-transformed action. It keeps the same vector fields, now acting on Laurent principal parts:

$$\hbar X_{z_1}\rho_{a,b} = \hbar\partial_{z_2}\rho_{a,b} = -(b+1)\hbar\rho_{a,b+1}, \qquad \hbar X_{z_2}\rho_{a,b} = -\hbar\partial_{z_1}\rho_{a,b} = (a+1)\hbar\rho_{a+1,b}.$$

After tensoring with $\mathbb{C}((\hbar))$, no nonzero vector in $\mathcal{P}((\hbar))$ is killed by a high power of either linear Hamiltonian: for a finite nonzero linear combination, every power of X_{z_1} shifts each nonzero basis summand to a distinct $\rho_{a,b+N}$, with nonzero rising-factorial coefficient, and the same argument applies to X_{z_2}. Local nilpotence is invariant under module isomorphism. Thus any Fourier–Rees comparison that preserves the original Hamiltonian action is impossible; the actual positive construction changes the polarization and is a flat $\mathcal{D}_\hbar$-module bridge only for the Fourier-conjugated action stated above. □

COROLLARY A.11 *(Same-action obstruction criterion).* Let $\mathcal{M}_\hbar$ be a $\mathbb{C}[[\hbar]]$-flat family whose generic fibre carries the polynomial descendant action of the linear Hamiltonians z_1, z_2, so that X_{z_1} and X_{z_2} act locally nilpotently after tensoring with $\mathbb{C}((\hbar))$. Then $\mathcal{M}_\hbar((\hbar))$ is not isomorphic, as a module for the same Hamiltonian Lie algebra, to $\mathcal{P}((\hbar))$ with the Matlis coadjoint action. The same statement holds for the Rees-normalized operators $\hbar X_{z_i}$, since $\hbar$ is invertible on the generic fibre. Equivalently, local nilpotence of the two linear Hamiltonians on the generic fibre is a necessary condition for a flat same-action Rees bridge, and $\mathcal{P}((\hbar))$ violates that condition on every nonzero vector. Any same-action comparison must therefore either change the Hamiltonian action by a Fourier twist, enlarge the boundary target to a distributional principal-part sector, or state a different deformation problem.

Proof. This is the local-nilpotence obstruction of Theorem A.9 after extension of scalars to $\mathbb{C}((\hbar))$. The two linear Hamiltonians remain locally nilpotent on the polynomial descendant generic fibre and remain non-locally-nilpotent on every nonzero element of $\mathcal{P}((\hbar))$. Multiplication by the invertible scalar $\hbar$ cannot change local nilpotence. □

Remark A.12 *(What the Fourier–Rees construction proves).* The construction above justifies the shared index set (a, b) as Fourier–Rees associated-graded bookkeeping, with the factorial and $\hbar^{a+b}$ normalization displayed explicitly. It also records the lattice convention: the Fourier twist gives $\mathcal{D}_\hbar/(z_1, z_2)$, whose Čech realization is the Rees sublattice $\bigoplus \hbar^{a+b}\mathbb{C}[[\hbar]]\rho_{a,b}$, not the full $\bigoplus \mathbb{C}[[\hbar]]\rho_{a,b}$ before $\hbar$ is inverted. It does not weaken the no-polynomial theorem. The theorem says precisely where the apparent comparison breaks: Fourier sends the Hamiltonian vector fields to Fourier-transformed operators, while the Matlis cotangent module uses the original coadjoint vector fields on principal parts.

B FACTORIZATION CURRENT CONVENTIONS FOR THE CE/PV INTERFACE

This appendix fixes the algebraic source/target interface carried by the factorization-current theorem. The interface is used before the admissible bar-cobar/enveloping criterion: interval Hamiltonian currents form the CE source, distributional Matlis principal parts form the cotangent current target, and the CE/PV generator dictionary is

$$c_\lambda \longmapsto \theta_\lambda, \qquad u_{a(t)dt\otimes f} \longmapsto B_f(a).$$

The proved content is the reduced algebraic current model after de Rham-current contraction. The unreduced BV/QME current lift and the Costello brane-defect graph theorem are separate obstruction problems, recorded in Remark 3.241, Appendix E, and Problem 3.264. All completed symmetric algebras in this appendix are interpreted degreewise in the chosen current category; finite symmetric powers are formed before completion, and brackets extend continuously from the displayed length-one current formulas. The geometry throughout is the formal-local mixed HT model on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$. The interval I is an oriented topological current interval in the brane line, while the Hamiltonian labels live on the formal holomorphic symplectic disk with

$$\Omega_{\mathrm{loc}} = dz_1 \wedge dz_2, \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

No compact Calabi–Yau target, global BCOV moduli space, CoHA, quintic, OSV/GV, Abel–Jacobi, Igusa, or BKM datum is part of this appendix.

Definition B.1 *(Interval Hamiltonian current algebra).* For an open interval $I \subset \mathbb{R}$, set

$$\mathfrak{h}_I = \Omega^\bullet_c(I) \widehat{\otimes} \mathfrak{h},$$

with differential $d_I = d_{\mathrm{dR}} \otimes 1$. Here

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$$

is the Hamiltonian Lie algebra of the formal holomorphic symplectic disk $(\widehat{\mathbb{C}^2}_0, \Omega_{\mathrm{loc}})$, concentrated in degree 0. The graded bracket is

$$[\alpha \otimes f, \beta \otimes g] = \alpha \wedge \beta \otimes \{f, g\}, \qquad \alpha, \beta \in \Omega^\bullet_c(I),\ f, g \in \mathfrak{h}.$$

For degree-zero test functions $a, b \in \Omega^0_c(I)$,

$$[a \otimes f, b \otimes g] = ab \otimes \{f, g\}.$$

The cosheaf structure for inclusions of intervals is extension by zero on $\Omega^\bullet_c$, tensored with the identity on $\mathfrak{h}$. Polynomial formulas use the dense scalar-reduced subalgebra $\tilde{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1 \subset \mathfrak{h}$. For $j : I \hookrightarrow J$, extension by zero is smooth because every element of $\Omega^\bullet_c(I)$ has support compactly contained in I. On CE cochains the induced map is the dual coproduct/pullback; the covariant factorization structure belongs to the compact-support current target, not to a restriction map on raw CE cochains.

The notation $a(t)dt \otimes f$ is the oriented density label paired with distributional cotangent currents. It does not convert $a \in \Omega^0_c(I)$ into a degree-one Hamiltonian generator of $\mathfrak{h}_I$.

Definition B.2 *(Distributional Matlis current dual).* Let

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \cong \mathfrak{h}^\vee_{\mathrm{cont}}$$

be the reduced Matlis principal-part module of Section A. Its interval current form is

$$\mathcal{P}_I = \mathcal{D}'_c(I) \widehat{\otimes} \mathcal{P}.$$

This is the selected support-local current dual used after contraction, not an assertion that the full strong dual of $\Omega^\bullet_c(I) \widehat{\otimes} \mathfrak{h}$ has been imported. The pairing with degree-zero Hamiltonian currents is

$$\langle B \otimes \rho,\ a \otimes f \rangle = B(a)\, \mathrm{Res}(\rho f), \qquad B \in \mathcal{D}'_c(I),\ \rho \in \mathcal{P}.$$

The residue is the formal residue determined by $dz_1 \wedge dz_2$ on $\widehat{\mathbb{C}^2}_0$; it is not integration over a compact target. If $B = b(t)dt$ is a regular compactly supported density, then

$$B(a) = \int_I a(t)b(t)\,dt.$$

The shifted cotangent interval Lie algebra is

$$\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1],$$

where $\mathfrak{h}_I$ acts on $\mathcal{P}_I$ by the product of the $C_c^\infty(I)$-module operation $B \mapsto aB$ on compactly supported currents and the Matlis coadjoint action on $\mathcal{P}$. On degree-zero labels this action is

$$(a \otimes f) \cdot (B \otimes \rho) = (aB) \otimes (f \cdot \rho).$$

PROPOSITION B.3 *(Local unimodular input to the current kernels).* The current and kernel constructions in this appendix use the Calabi–Yau datum only in the following formal-local sense:

$$\left(\widehat{\mathbb{C}^2}_0, \Omega_{\mathrm{loc}} = dz_1 \wedge dz_2\right).$$

This datum supplies:

(i) the Poisson bracket on $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$;

(ii) the Hamiltonian vector-field convention

$$\iota_{X_f}\Omega_{\mathrm{loc}} = -df, \qquad X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1};$$

(iii) the divergence-free identity $\mathrm{div}_{\Omega_{\mathrm{loc}}} X_f = 0$, hence the residue integration-by-parts formula defining the Matlis coadjoint action;

(iv) the principal-part residue pairing

$$\mathrm{Res}_0\left(z_1^{-i-1} z_2^{-j-1} dz_1 dz_2\; z_1^m z_2^n\right) = \delta_{i,m}\delta_{j,n}.$$

At the reduced current-interface level the topological factor contributes the compactly supported de Rham current direction $I \subset \mathbb{R}_{\mathrm{brane}}$, the operation $B \mapsto aB$, and extension by zero. In the graph layer it also contributes the local heat-kernel variables on $\mathbb{R}^2_{\mathrm{top}}$. Finite-window and weighted kernels are completions of the formal Hamiltonian coefficient system. Compactness of a Calabi–Yau manifold is irrelevant to these constructions.

Proof. The bracket in Definition B.1 is the inverse bivector of $dz_1 \wedge dz_2$, and constants are quotiented because they give the zero Hamiltonian vector field. The displayed X_f satisfies $X_f(g) = \{f, g\}$, and

$$\mathrm{div}_{\Omega_{\mathrm{loc}}} X_f = -\partial_{z_1}\partial_{z_2} f + \partial_{z_2}\partial_{z_1} f = 0.$$

Thus Hamiltonian fields preserve the residue density. The transpose of this action under $\mathrm{Res}(\rho f)$ is exactly the Matlis coadjoint action used in Definition B.2 and Corollary B.8.

The interval variable is independent of the holomorphic disk. In the reduced current interface its operations are the de Rham differential, multiplication of a current by a smooth compactly supported test function, and extension by zero. In the graph layer the additional topological variables enter only through the local heat-kernel calculus on $\mathbb{R}^2_{\mathrm{top}}$. The finite-window constructions truncate only the Hamiltonian degree in the formal disk and then pass through the weighted Mittag–Leffler inverse limit. None of these operations uses compact Hodge theory, periods, worldsheet instantons, or global BCOV integration data. □

Lemma B.4 *(Multiplication of a current by a test function).* For $a \in C_c^\infty(I)$ and $B \in \mathcal{D}_c'(I)$, define

$$(aB)(\varphi) = B(a\varphi), \qquad \varphi \in C_c^\infty(I).$$

Then $aB \in \mathcal{D}_c'(I)$,

$$\operatorname{supp}(aB) \subset \operatorname{supp}(a) \cap \operatorname{supp}(B),$$

and $a(bB) = (ab)B$. If $B = b(t)dt$ is regular, then $aB = (ab)(t)dt$. No product of two distributions occurs in the reduced current model.

Proof. The function $a\varphi$ is a compactly supported smooth test function, so precomposition by $\varphi \mapsto a\varphi$ defines a compactly supported distribution. If φ is supported away from $\operatorname{supp}(a) \cap \operatorname{supp}(B)$, then $a\varphi$ is supported away from $\operatorname{supp}(B)$, hence $B(a\varphi) = 0$. Associativity follows from $a(b\varphi) = (ab)\varphi$. The regular-density formula is the displayed integral identity. □

Definition B.5 *(Current CE/PV coordinates).* Let $\lambda = B \otimes \rho \in \mathcal{P}_I$. The Hamiltonian CE coordinate dual to the Hamiltonian current through λ is written

$$c_\lambda, \qquad |c_\lambda| = 1.$$

Its PV/current target coordinate is

$$\theta_\lambda = \theta_{B\otimes\rho} := \Theta_\rho(B).$$

The label $a(t)dt \otimes f$ denotes the functional on $\mathcal{P}_I$

$$B \otimes \rho \longmapsto B(a)\operatorname{Res}(\rho f).$$

The CE coordinate dual to the shifted cotangent generator through this functional is

$$u_{a(t)dt\otimes f}, \qquad |u_{a(t)dt\otimes f}| = 0.$$

Its PV/current target coordinate is the smeared Hamiltonian boundary observable

$$O_{a,f} = B_f(a).$$

In the brane realization,

$$B_f(a) = \int_I a(t)\,\overline{\operatorname{Tr} f(\phi_1(t), \phi_2(t))}\,dt.$$

Definition B.6 *(Enlarged reduced defect-current target).* The reduced defect-current target attached to I is

$$A_\partial^{\mathrm{cur}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{P}_I[1]).$$

This is the same reduced principal-part target denoted $A_\partial^{\mathrm{pp}}(I)$ in the main text; the superscript cur records the current-interface role in this appendix. It enlarges the Hamiltonian target $A_\partial^{\mathrm{Ham}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$ by adjoining the odd principal-part current generators $\theta_\lambda = \Theta_\rho(B)$. The structure maps for inclusions of intervals are extension by zero on $\Omega_c^\bullet(I)$ and on $\mathcal{D}_c'(I)$. The factorization product on disjoint intervals is the completed graded-commutative product after extension by zero.

THEOREM B.7 *(Universal algebraic current interface).* For each interval I, the generator assignment

$$\Phi_I^{\mathrm{cur}}(c_\lambda) = \theta_\lambda, \qquad \Phi_I^{\mathrm{cur}}(u_{a(t)dt\otimes f}) = B_f(a)$$

extends uniquely to a continuous morphism of completed graded-commutative algebras from the interval CE cofactorization source $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_I)$ to the reduced defect-current central-operation target $\mathcal{A}^{\mathrm{cur}}_\partial(I)$. On length-one current generators the target open-current P_0 brackets are

$$\begin{aligned}\{B_f(a), B_g(b)\} &= B_{\{f,g\}}(ab),\\ \{B_f(a), \theta_{B\otimes\rho}\} &= \theta_{aB\otimes(f\cdot\rho)} = \Theta_{f\cdot\rho}(aB),\\ \{\theta_{B\otimes\rho}, \theta_{C\otimes\sigma}\} &= 0.\end{aligned}$$

The factorization product and the reduced current source-target P_0-equivariance defects in this reduced current target are strict:

$$H_{\mathrm{fact}} = 0, \qquad H_{P_0} = 0.$$

This is not a zero Poisson-centre statement for overlapping supports; the nonzero Hamiltonian current bracket is the moment-map bracket displayed above. Thus $u_{a(t)dt\otimes f} \mapsto B_f(a)$ and $c_\lambda \mapsto \theta_\lambda$ form the current-level source/target interface used before the admissible bar-cobar/enveloping criterion.

Proof. The CE algebra of $\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$ is the completed symmetric algebra on the continuous dual shifted by $[-1]$. The coordinate c_λ has degree 1, and the coordinate $u_{a(t)dt\otimes f}$ has degree 0. The target $\mathcal{A}^{\mathrm{cur}}_\partial(I)$ is the completed symmetric algebra on the corresponding current coordinates θ_λ and $B_f(a)$. Hence the displayed assignment extends uniquely and continuously to the completed graded-commutative algebras.

The CE differential and the target current brackets use the same semidirect Lie structure constants. The Hamiltonian-current bracket in Definition B.1 gives

$$[a\otimes f, b\otimes g] = ab\otimes\{f,g\},$$

and the boundary realization gives the first displayed bracket. The action of $a\otimes f$ on $B\otimes\rho$ is $(aB)\otimes(f\cdot\rho)$, giving the second displayed bracket by negative transpose under the product pairing

$$\langle B\otimes\rho,\ a\otimes f\rangle = B(a)\,\mathrm{Res}(\rho f).$$

The shifted cotangent current ideal is abelian, giving the third bracket. These three identities are exactly the current form of the CE/PV generator identities $c_\lambda \mapsto \theta_\lambda$ and $u \mapsto O$. Since the CE differential is a derivation and the target bracket is a biderivation, the generator identities determine the full completed current interface.

Factorization compatibility is strict in the cofactorization-source convention. Extension by zero on current Lie algebras induces the CE coproduct, and the target product is compatible with pointwise products ab and with the $C_c^\infty(I)$-module operation $B \mapsto aB$. If two intervals are disjoint, the products have empty support and the mixed brackets vanish. The target product is graded commutative, so multiplication by an image generator has zero Hochschild product-centrality defect.

For reduced current P_0-equivariance, let x be a length-one source generator and let Y be a homogeneous monomial in the CE source. The defect

$$D_x(Y) = \{\Phi_I^{\mathrm{cur}}(x), \Phi_I^{\mathrm{cur}}(Y)\} - \Phi_I^{\mathrm{cur}}(\mathrm{ad}_x Y)$$

is a derivation in Y. The three generator bracket identities give $D_x(Y) = 0$ for every length-one Y. The Leibniz rule gives $D_x(Y) = 0$ for every monomial, and continuity extends this to the completed algebra. Thus H_{fact} and H_{P_0} are zero equivariance-defect homotopies in the reduced current model. □

Corollary B.8 *(Principal-part current brackets in monomial coordinates).* For $f = z_1^p z_2^q$, $a \in C_c^\infty(I)$, $B \in \mathcal{D}_c'(I)$, and $\rho_{i,j} = z_1^{-i-1} z_2^{-j-1} dz_1 dz_2$, one has

$$\{B_{z_1^p z_2^q}(a), \Theta_{\rho_{i,j}}(B)\} = -(pj - qi + p - q)\Theta_{\rho_{i-p+1,j-q+1}}(aB),$$

with negative-index targets equal to zero. If $B = b(t)dt$ is regular, then the interval factor is $(ab)(t)dt$.

Proof. Proposition A.5 gives

$$z_1^p z_2^q \cdot \rho_{i,j} = -(pj - qi + p - q)\rho_{i-p+1,j-q+1},$$

with negative-index terms projected to zero in the principal-part module. Substituting this action into the second bracket of Theorem B.7 gives the displayed formula. The regular-current statement is Lemma B.4. □

Theorem B.9 *(BMK current data and finite-window obstruction vector).* Let $B_0 \Subset B_1 \Subset B_2 \Subset B \subset \mathbb{C}^2$ be nested polydisks around the collision point, and choose $\chi \in C_c^\infty(B_2)$ with $\chi = 1$ on B_1. Put

$$\mathcal{P}_{\le N} = \bigoplus_{\substack{a,b \ge 0 \\ 0 < a+b \le N}} \mathbb{C}\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2.$$

On $(B_\zeta \times B_z) \setminus \Delta$, set

$$\eta = \zeta - z, \qquad r^2 = |\eta_1|^2 + |\eta_2|^2, \qquad d\bar\eta_j = d\bar\zeta_j - d\bar z_j.$$

The Dolbeault homotopy kernel is the Bochner–Martinelli–Koppelman current

$$K_{\text{BM}}(\zeta, z) = \frac{1}{(2\pi i)^2} \frac{\overline{\eta_1}\, d\bar\eta_2 - \overline{\eta_2}\, d\bar\eta_1}{r^4} \wedge d\zeta_1 \wedge d\zeta_2.$$

The $d\bar z_j$-terms are part of the Koppelman current. The scalar $d\bar z = 0$ component is not enough for the Dolbeault homotopy.

Choose a radial cutoff $\vartheta_\epsilon(r)$, equal to 0 for $r \le \epsilon$ and to 1 for $r \ge 2\epsilon$, and put

$$K_{\chi,\epsilon} = \chi(\zeta)\chi(z)\vartheta_\epsilon(|\zeta - z|)K_{\text{BM}}(\zeta, z).$$

The current limit

$$H_\chi \alpha(z) = \lim_{\epsilon \to 0} \int_{\zeta \in B} K_{\chi,\epsilon}(\zeta, z) \wedge \alpha(\zeta)$$

exists on every fixed finite configuration stratum. For separated supports it satisfies

$$|\nabla^k K_{\text{BM}}(\zeta, z)| \le C_k |\zeta - z|^{-3-k}.$$

Let $i_N : \mathcal{P}_{\le N} \to \mathcal{D}_0'^{0,2}(B)$ be

$$i_N(\rho_{a,b}) = \Delta_{a,b} = \frac{(-1)^{a+b}}{a!\, b!} \partial_{z_1}^a \partial_{z_2}^b \delta_0\, d\bar z_1 \wedge d\bar z_2,$$

normalized by

$$\langle \Delta_{a,b}, f \rangle = \mathrm{Res}_0(\rho_{a,b} f).$$

Define the finite principal-part moment projection on top Dolbeault degree by

$$p_N(\alpha) = \sum_{0<a+b\le N} \left(\int_B z_1^a z_2^b \, \alpha^{0,2} \wedge dz_1 \wedge dz_2 \right) \rho_{a,b}.$$

The data above construct H_χ, p_N, and i_N, but not a native finite-window homotopy on the full compact-support/current complex. The desired identity

$$\bar\partial H_{\chi,N} + H_{\chi,N}\bar\partial = \mathrm{id} - i_N p_N + E_{\chi,N} \qquad (*_N^{\mathrm{BMK}})$$

is blocked by

$$\mathrm{Ob}_N^{\text{BMK-curr}} = \left(\mathrm{ob}_{\mathrm{diag}}, \mathrm{ob}_{\mathrm{an/fin},N}, \mathrm{ob}_{\mathfrak{h},N}, \mathrm{ob}_{\text{collar-fact},N} \right).$$

The first entry is the annular diagonal sign and scalar of $\bar\partial \vartheta_\epsilon \wedge K_{\mathrm{BM}}$ relative to the derivative-delta Matlis convention. With the displayed kernel and real outward boundary orientation the angular mass is -1. The second entry is the gap between the full analytic moment kernel $\ker(\mathcal{O}(B_1)^\vee \to \mathcal{P}_{\le N})$ and the smaller support-at-zero span generated by the scalar and high derivative-delta currents. The third entry is

$$\mathrm{ob}_{\mathfrak{h},N} = \pi_N(\mathfrak{h} \cdot \ker \pi_N),$$

which is nonzero because

$$z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1}.$$

Thus a strict finite support-local quotient killing high modes while retaining $\mathcal{P}_{\le N}$ is not compatible with the full Hamiltonian action. The fourth entry is the missing cosheaf/factorization compatibility of the collar relation with extension by zero, disjoint products, and the BMK homotopy.

A normalized finite-window identity is available only after changing habitat. If the full analytic BMK contraction is supplied and the relative complex is quotiented by representatives of $\ker(\mathcal{O}(B_1)^\vee \to \mathcal{P}_{\le N})$, then the normalized homotopy

$$(1 - I_{\mathrm{an}} P_{\mathrm{an}}) H_\chi (1 - I_{\mathrm{an}} P_{\mathrm{an}})$$

descends and gives

$$\bar\partial H_{\mathrm{BM},N} + H_{\mathrm{BM},N}\bar\partial = \mathrm{id} - i_N p_N.$$

This is a one-pair analytic quotient, not a native support-local factorization-current quotient.

Keeping the full Taylor tower gives the positive one-pair pro-Matlis retract. Put

$$\widehat{\mathcal{P}}_\Pi = \prod_{\substack{a,b\ge 0\\ a+b>0}} \mathbb{C}\rho_{a,b}, \qquad \mathcal{P}_{\mathrm{an}}(B_1) = \tau\big(\mathcal{O}(B_1)^\vee_{\mathrm{red}}\big) \subset \widehat{\mathcal{P}}_\Pi,$$

where

$$\tau(\lambda) = \big(\lambda(z_1^a z_2^b)\big)_{a+b>0}.$$

After the diagonal orientation is fixed and the collar quotient kills the cutoff error, the normalized analytic contraction $H_\chi^\circ = (1 - I_{\mathrm{an}} P_{\mathrm{an}}) H_\chi (1 - I_{\mathrm{an}} P_{\mathrm{an}})$ satisfies

$$\bar\partial H_\chi^\circ + H_\chi^\circ \bar\partial = \mathrm{id} - I_{\mathrm{an}} P_{\mathrm{an}}.$$

The pro-analytic moment map $P_{\Pi}^{\mathrm{an}} = \tau P_{\mathrm{an}}$ has compatible finite projections

$$\pi_N P_{\Pi}^{\mathrm{an}} = p_N, \qquad \pi_{M,N} p_M = p_N.$$

This is a one-pair analytic pro-Matlis statement, not a native support-local factorization-current transfer.

The binary coefficient operation

$$\ell_2^{\mathrm{BM},N}(x, y) = p_N[ix, iy]$$

is retained as the Hamiltonian/Matlis arity-two comparison, read in the output window $\mathcal{P}_{\leq N}$, not as a consequence of a native finite-window homotopy. For monomials,

$$\ell_2^{\mathrm{BM},N}(z_1^p z_2^q, z_1^r z_2^s) = (ps - qr)\ \mathrm{pr}_{\leq N}(z_1^{p+r-1} z_2^{q+s-1}) \quad (\mathrm{mod}\ \mathbb{C}),$$

and

$$\ell_2^{\mathrm{BM},N}(z_1^p z_2^q, \rho_{i,j}) = -(pj - qi + p - q)\ \mathrm{pr}_{\leq N}\, \rho_{i-p+1,j-q+1},$$

with negative-index targets equal to zero, while

$$\ell_2^{\mathrm{BM},N}(\rho, \sigma) = 0.$$

Thus the arity-two cotangent coefficient comparison is the Matlis coadjoint action of Proposition A.5.

Only after the finite-current gate $\mathrm{Ob}_N^{\mathrm{BMK\text{-}curr}}$ is killed, or replaced by a genuine native factorization-current habitat, does the strict all-window and arity-three extension become the following obstruction vector. Set

$$Q_{\oplus} = \widehat{\mathcal{P}}_{\Pi}/\mathcal{P}.$$

For $\alpha \in \Omega_c^{0,2}(B)$, put

$$\mathrm{mom}_{\infty}(\alpha) = \left(\int_B z_1^a z_2^b\, \alpha^{0,2} \wedge dz_1 \wedge dz_2\right)_{a+b>0} \in \widehat{\mathcal{P}}_{\Pi},$$

and

$$\mathrm{ob}_{\oplus}(\alpha) = [\mathrm{mom}_{\infty}(\alpha)] \in Q_{\oplus}.$$

This class is zero exactly when the all-window moment sequence lands in the direct-sum Matlis module $\mathcal{P}$.

The pro-product tower does not repair the native current problem by itself. There is no support-local compact-current section

$$i_{\Pi} : \widehat{\mathcal{P}}_{\Pi} \to \mathcal{D}_0'^{\,0,2}(B)$$

extending the finite derivative-delta sections i_N: a distribution supported at one point has finite order. This is $\mathrm{ob}_{I_{\Pi}}$. The full formal Hamiltonian algebra also fails to act on $\widehat{\mathcal{P}}_{\Pi}$. For $f = \sum_{p\geq 2} z_1^p$ and $\lambda = \sum_{p\geq 2} \rho_{p-1,0}$, the $\rho_{0,1}$-coefficient of $f \cdot \lambda$ would be $-\sum_{p\geq 2} p$. This undefined coefficient is $\mathrm{ob}_{\mathfrak{h},\Pi}$.

The collar part is the finite-window family

$$\mathrm{ob}_{\mathrm{collar},\Pi,N} = E_{\chi,N} \in \mathrm{Hom}\big(\Omega_c^{0,\bullet}(B), \mathcal{D}'^{\,0,\bullet}(B)\big)_{\mathrm{supp}\, d\chi}.$$

The first arity-three obstruction is

$$\mathrm{ob}_{3,\Pi,N}^{\chi}(x, y, z) = p_N[H_{\mathrm{BM},N}([ix, iy] - i\ell_2^{\mathrm{BM},N}(x, y)), iz] + \mathrm{cyc}_{x,y,z}\,.$$

It vanishes on separated-support strata by the displayed estimates. On the coefficient collision stratum its non-collar part is the Jacobiator of $\mathfrak{h} \ltimes \mathcal{P}[1]$, hence zero. The remaining part is supported on the small diagonal and on the cutoff collar.

Finally, let $C_{s,N}$ be the optimal constants for the chosen finite-window seminorm estimates of $H_{\chi,N}$, p_N, $E_{\chi,N}$, and $\mathrm{ob}^{\chi}_{3,\Pi,N}$. The uniformity obstruction is

$$\mathrm{ob}_{\mathrm{unif},\Pi} = \left(\left[(C_{s,N})_{N\geq 1}\right]\right)_s \in \prod_s \left(\prod_{N\geq 1} \mathbb{R}_{\geq 0}/\ell^{\infty}_N\right).$$

Thus the all-window obstruction vector is

$$\mathrm{Ob}^{\Pi}_{\mathrm{BM}} = \left(\mathrm{ob}_{\oplus}, \mathrm{ob}_{I_\Pi}, \mathrm{ob}_{\mathfrak{h},\Pi}, (\mathrm{ob}_{\mathrm{collar},\Pi,N})_N, (\mathrm{ob}^{\chi}_{3,\Pi,N})_N, \mathrm{ob}_{\mathrm{unif},\Pi}\right).$$

Any strict native all-window current E_2 transfer extending the displayed binary coefficient comparison forces this vector to vanish. The first, second, third, and sixth entries are nonzero in the native current topology; the collar and ternary entries are exact primitive construction targets. The proved theorem here is the BMK current data, the finite-window obstruction vector, the positive one-pair analytic pro-Matlis retract, and the arity-two Hamiltonian/Matlis comparison.

The principal-part face remains two-variable:

$$\frac{1}{(2\pi i)^2}\frac{d\zeta_1 \wedge d\zeta_2}{(\zeta_1 - z_1)(\zeta_2 - z_2)} = \sum_{a,b\geq 0} z_1^a z_2^b \rho_{a,b}(\zeta).$$

Killing z_2 kills the Hamiltonian bracket, and the $b = 0$ principal-part line is not stable since $z_1 \cdot \rho_{a,0} = -\rho_{a,1}$. No curve VOA, Zhu algebra, or one-variable transfer follows from this theorem.

Proof. The Bochner–Martinelli–Koppelman identity gives the Dolbeault homotopy on the punctured product $B_\zeta \times B_z \setminus \Delta$. Multiplication by the cutoff gives the exact derivative split

$$\bar\partial(\chi(\zeta)\chi(z)\vartheta_\epsilon K_{\mathrm{BM}}) = \chi\chi\,\vartheta_\epsilon\,\bar\partial K_{\mathrm{BM}} + \bar\partial(\chi\chi)\,\vartheta_\epsilon K_{\mathrm{BM}} + \chi\chi\,\bar\partial\vartheta_\epsilon \wedge K_{\mathrm{BM}}.$$

The middle term is collar-supported. The last term is the annular diagonal current, not a collar term; its sign and scalar are exactly $\mathrm{ob}_{\mathrm{diag}}$. Since K_{BM} has size r^{-3} in real dimension four, the cutoff current has a limit on every fixed finite configuration stratum. This proves the current data, not the finite-window identity on the full current complex.

The formula for i_N is fixed by integration by parts:

$$\left\langle \frac{(-1)^{a+b}}{a!\,b!}\partial^a_{z_1}\partial^b_{z_2}\delta_0,\, f\right\rangle = [z_1^a z_2^b]f = \mathrm{Res}_0(\rho_{a,b}f).$$

Hence $i_N p_N$ is the finite support-at-collision projector with the Matlis residue normalization, after the diagonal-current orientation is tied to that convention.

The native finite-window identity is obstructed already on closed top currents in the kernel of the finite moment map. If $T = \Delta_{N+1,0}$, then $p_N(T) = 0$ but $\langle T, z_1^{N+1}\rangle = 1$. A putative identity $T = \bar\partial S$ would give $\langle T, z_1^{N+1}\rangle = 0$ by Stokes. The scalar $\Delta_{0,0}$ is the same obstruction for the removed reduced principal-part line. More strongly, a strict support-local quotient must be an $\mathfrak{h}$-submodule quotient; the calculation $z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1}$ contradicts killing $\rho_{N+1,0}$ while retaining $\rho_{0,1}$.

The one-pair analytic quotient works for the reason stated in the theorem: after choosing the full analytic projector $I_{\mathrm{an}}P_{\mathrm{an}}$, the normalized homotopy kills representatives of $\ker \pi_N$, descends to the quotient, and identifies the analytic projector with $i_N p_N$. Keeping the Taylor tower instead gives the pro-analytic map $P^{\mathrm{an}}_{\Pi} = \tau P_{\mathrm{an}}$, and the equality $\pi_N P^{\mathrm{an}}_{\Pi} = p_N$ is just functoriality of Taylor moments under finite truncation. Its dependence on analytic representatives and on the chosen pair prevents it from being a native factorization quotient.

The arity-two coefficient comparison is the residue transpose. For Hamiltonian labels it is the Poisson bracket modulo constants, truncated only after the output coefficient is computed. For cotangent principal parts, Proposition A.5 gives

$$\mathrm{Res}_0((f \cdot \rho)g) = -\mathrm{Res}_0(\rho\,\{f, g\}),$$

which is the displayed coadjoint formula. The shifted cotangent ideal is abelian, so the (ρ, σ)-operation is zero.

The displayed ternary expression is only the next obstruction after the finite-current gate has been crossed. The separated-support estimate makes it vanish away from collisions. On the coefficient collision stratum the semidirect Lie algebra $\mathfrak{h} \ltimes \mathcal{P}[1]$ satisfies Jacobi. Therefore any surviving term is a collar, small-diagonal, or completed-window transfer residue, recorded as the finite-window representative $\mathrm{ob}^{\chi}_{3,\Pi,N}$ of the Π-tower class only after the finite-current obstruction has been removed.

A strict native all-window current transfer would have to replace the compatible family p_N by a single moment map landing in $\mathcal{P}$, construct a compact-current section of the pro-product tower, define the full formal Hamiltonian action on that tower, remove $E_{\chi,N}$ compatibly with extension by zero, kill the ternary collar residue by compatible primitives, and supply uniform estimates in the completed window topology. These six requirements are precisely the six entries of $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$. The all-window theorem target is a changed native habitat where these six witnesses exist, or the displayed obstruction theorem. It is not asserted from the finite-window calculation alone and does not bypass $\mathrm{Ob}^{\mathrm{BMK\text{-}curr}}_{N}$.

The Cauchy expansion is the Laurent expansion in the two holomorphic variables and records the same residue basis. The calculation $z_1 \cdot \rho_{a,0} = -\rho_{a,1}$, from Proposition A.5, rules out the $b = 0$ principal-part truncation as a native submodule. □

Theorem B.10 *(Normal Ω-weighted brane-current brackets).* Use the normal brane-preserving equivariant parameters of the local dictionary: t has weight 0, z_1, z_2 have weights ϵ_1, ϵ_2, and

$$\chi_\omega = \epsilon_1 + \epsilon_2, \qquad \mathrm{wt}_\Omega(\hbar_\omega) = \chi_\omega.$$

Put

$$B_{p,q}(a) := B_{z_1^p z_2^q}(a), \qquad \Theta_{r,s}(B) := \Theta_{\rho_{r,s}}(B),$$

with $p, q, r, s \geq 0$, $p + q > 0$, $r + s > 0$, and $\rho_{r,s} = z_1^{-r-1} z_2^{-s-1} dz_1 dz_2$. The compactly supported test function a and compactly supported current B live on the fixed brane line and have weight 0. Hence

$$\mathrm{wt}_\Omega\, B_{p,q}(a) = p\epsilon_1 + q\epsilon_2, \qquad \mathrm{wt}_\Omega\, \Theta_{r,s}(B) = -r\epsilon_1 - s\epsilon_2.$$

Off the self-dual locus the Hamiltonian current bracket is not a scalar bracket unless one either keeps the inverse symplectic-character line or adjoins $\hbar_\omega$. In the scalarized convention the length-one current brackets are

$$\begin{aligned}
\{B_{p,q}(a), B_{r,s}(b)\}_\Omega &= \hbar_\omega(ps - qr)\, B_{p+r-1,q+s-1}(ab),\\
\{B_{p,q}(a), \Theta_{r,s}(B)\}_\Omega &= -\hbar_\omega(ps - qr + p - q)\, \Theta_{r-p+1,s-q+1}(aB),\\
\{\Theta_{r,s}(B), \Theta_{u,v}(C)\}_\Omega &= 0.
\end{aligned}$$

Here $B_{\alpha,\beta} = 0$ if $\alpha < 0$, $\beta < 0$, or $(\alpha, \beta) = (0, 0)$, and $\Theta_{\alpha,\beta} = 0$ if $\alpha < 0$, $\beta < 0$, or $(\alpha, \beta) = (0, 0)$. The last convention only records the reduced scalar-annihilator target; when the coadjoint formula lands at $\rho_{0,0}$ its coefficient is already zero.

Equivalently, before adjoining $\hbar_\omega$, remove the displayed factor $\hbar_\omega$ and read the first two brackets as valued in the inverse symplectic-character line $\mathbb{L}_\omega^{-1}$, where $\mathrm{wt}_\Omega(\mathbb{L}_\omega) = \chi_\omega$. On the self-dual specialization $\epsilon_1 + \epsilon_2 = 0$, the line is trivial. After choosing the trivialization $\hbar_\omega = 1$, the formulas reduce to the non-equivariant brackets of Theorem B.7 and Corollary B.8. This specialization is a separate branch; it is not obtained by inverting χ_ω and then setting $\chi_\omega = 0$.

For an inclusion of intervals $j : I \hookrightarrow J$, extension by zero has weight 0 and satisfies

$$j_!B_{p,q,I}(a) = B_{p,q,J}(j_!a), \qquad j_!\Theta_{r,s,I}(B) = \Theta_{r,s,J}(j_!B),$$

together with

$$j_!\{x, y\}_{\Omega,I} = \{j_!x, j_!y\}_{\Omega,J}$$

for the displayed length-one generators. For disjoint intervals the mixed brackets vanish after extension to a common interval.

These statements are algebraic current identities. They hold for all $B, C \in \mathcal{D}'_c(I)$ because the only distributional operation is multiplication of a current by a smooth compactly supported function. They do not assert a normal-Ω Costello graph theorem for arbitrary distributional labels. A graph-level lift must still restrict cotangent labels to regular densities or to graphwise wavefront-admissible tuples as in Propositions B.17 and B.19, and must add the Q_Ω-Cartan package, normal-weight contraction, residue/Euler normalization choice, equivariant counterterms, and the scalar/non-scalar QME obstruction classes.

Proof. The weight assignments are forced by the normal torus action: $z_1^p z_2^q$ has character $p\epsilon_1 + q\epsilon_2$, while

$$z_1^{-r-1} z_2^{-s-1} dz_1 dz_2$$

has character $-(r+1)\epsilon_1 - (s+1)\epsilon_2 + \chi_\omega = -r\epsilon_1 - s\epsilon_2$. The brane-line variables and currents are in the fixed t-direction, so they add no equivariant character.

The Poisson bivector $\partial_{z_1} \wedge \partial_{z_2}$ has character $-\chi_\omega$. Thus the ordinary Hamiltonian bracket has weight $\mathrm{wt}(f) + \mathrm{wt}(g) - \chi_\omega$. Multiplying by $\hbar_\omega$ makes the bracket scalar and weight-preserving. On monomials this gives

$$[z_1^p z_2^q, z_1^r z_2^s]_\Omega = \hbar_\omega(ps - qr) z_1^{p+r-1} z_2^{q+s-1},$$

followed by the quotient by constants. Substitution in the first bracket of Theorem B.7 gives the displayed B-B formula.

For the cotangent current, the coadjoint action of Proposition A.5 gives

$$z_1^p z_2^q \cdot \rho_{r,s} = -(ps - qr + p - q)\rho_{r-p+1,s-q+1}.$$

Scalarizing the Hamiltonian action by the same $\hbar_\omega$ gives the B-Θ formula, and the shifted cotangent current ideal is abelian. Each right-hand side has total weight equal to the sum of the weights of the two inputs, so the formulas are T_Ω-homogeneous.

Extension by zero commutes with multiplication of compactly supported test functions and currents: $j_!(ab) = (j_!a)(j_!b)$ and $j_!(aB) = (j_!a)(j_!B)$. Hence it commutes with all three displayed brackets. If two intervals are disjoint inside a common interval, these products are zero, so the mixed brackets vanish.

Finally, no singular product of two compactly supported distributions is used in the algebraic current formulas. Costello graph weights are different: after pushing current labels through graph incidence maps one must multiply them by singular propagator boundary values and extend across collision diagonals. The wavefront and scaling-degree hypotheses in Propositions B.19 and B.20 are exactly the missing analytic conditions. The Ω-weights and the factor $\hbar_\omega$ do not remove those microlocal conditions. □

Corollary B.11 *(Smeared Hamiltonian source convention).* The smeared boundary Hamiltonian observable is sourced by the shifted-cotangent CE coordinate:

$$u_{a(t)dt\otimes f} \longmapsto B_f(a) = \int_I a(t)\,\overline{\operatorname{Tr} f(\phi_1(t),\phi_2(t))}\,dt.$$

The Hamiltonian CE coordinate c_λ maps to the PV/current vector θ_λ, not to $B_f(a)$. Consequently

$$\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$$

is the degree-zero moment-map shadow of the CE/PV identity $c_\lambda \mapsto \theta_\lambda,\ u \mapsto O$.

Proof. The coordinate $u_{a(t)dt\otimes f}$ is dual to the shifted cotangent generator through the pairing with $\mathcal{P}_I$. Under the CE/PV isomorphism it maps to the function coordinate $O_{a,f}$, realized on the brane by the displayed trace integral. The coordinate c_λ is dual to a Hamiltonian current and maps to the vector/current coordinate θ_λ. The bracket identity is the first bracket of Theorem B.7; equivalently, it is the equal-time Poisson bracket smeared against $a(t)b(s)\delta(t-s)$, which produces the pointwise product ab. □

Remark B.12 *(Orientation reversal).* The orientation convention is the ordinary convention for oriented one-forms and current one-forms. The labels $a(t)dt$ and regular currents $B = b(t)dt$ transform by pullback, $a(\psi(s))\psi'(s)\,ds$ and $b(\psi(s))\psi'(s)\,ds$. No additional sign is inserted from $u_{a(t)dt\otimes f}$, c_λ, θ_λ, or the principal-part label ρ. If the written one-form is held fixed while only the chosen orientation is reversed, the sign is exactly the sign of dt.

Remark B.13 *(Boundary of the algebraic current theorem).* Theorem B.7 proves the algebraic distributional-current interface. It does not identify $\Theta_\rho(B)$ with a finite polynomial BV observable, construct the primitive cumulant projection before reduction, solve the full de Rham-current BV differential, or prove the mixed brane-defect QME. It also does not supply Costello renormalized graph weights. Those statements require the unreduced current complex, scalar cancellation data, Tate/weighted coefficient kernel, and brane-defect propagator specified in Appendix E and Problem 3.264.

Definition B.14 *(Unreduced non-scalar current lift datum).* Fix an oriented interval $I \subset \mathbb{R}_{\mathrm{brane}} \subset \mathbb{R}^2_{\mathrm{top}}$, an admissible weight w on the Hamiltonian degree of $\widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$, and the Moyal parameter $\hbar$. A non-scalar Costello current lift of the reduced interface for the local mixed HT model consists of the following data.

(i) A source current complex

$$\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I} = \big(\Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}_{w,\hbar}\big) \ltimes \big(\mathcal{D}'_c(I)\widehat{\otimes}\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}\big)[1],$$

or, before scalar reduction, the corresponding central-character replacement of Definition E.8.

(ii) A target defect local-functional complex

$$\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I) = \Big(O^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}_w|_I;\ A^{\mathrm{pp},w}_{\partial,\hbar}(I)), d_{\mathrm{QME}} = Q + \{I^w_{\mathrm{o}}, -\}_\hbar\Big),$$

together with the finite-scale BV Laplacian and propagator whenever the Costello scale-L QME has not been absorbed into d_{QME}. Here $\mathcal{E}_w$ is the weighted local mixed HT field complex on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$, restricted to the brane-defect interval I; the coefficient algebra is the weighted Hamiltonian/Moyal algebra of the same formal holomorphic symplectic disk.

(iii) A continuous bulk-to-defect kernel, equivalently a Maurer–Cartan candidate in the completed convolution Lie algebra,

$$\kappa_{\hbar,w,I} : C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}) \longrightarrow \mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I),$$

whose generator values are representatives

$$u_{a(t)dt\otimes f} \longmapsto B^{\mathrm{BV}}_f(a), \qquad c_{B,\rho} \longmapsto \Theta^{\mathrm{BV}}_\rho(B).$$

(iv) A d_{QME}-stable scalar-contact filtration and a filtered chain projection

$$\widehat{\sigma}_{\mathrm{sc}} : \mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I) \longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]]$$

whose associated graded is the scalar-symbol map of Definition E.3.

(v) Product centrality homotopies and P_{o}-bracket homotopies against arbitrary unreduced open observables, not only against the freely adjoined current generators of the reduced target.

The reduced algebraic theorem supplies only the image of this datum after scalar reduction, de Rham-current contraction, and passage to the added principal-part current target.

Definition B.15 *(Bulk-to-defect convolution habitat).* With $I, w, \hbar$ as in Definition B.14, and with all Hamiltonian coefficients taken from the local formal holomorphic disk of Proposition B.3, put

$$\mathbf{K}^{\bullet}_{w,\partial,\hbar}(I) = \mathrm{Hom}_{\mathrm{cont}}\Big(C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}), \mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)\Big).$$

The topology is the complete filtration topology generated by the $\hbar$-adic filtration, the scalar-contact filtration, and the finite-window Mittag–Leffler filtration. Thus a sequence of kernels converges exactly when all of its finite-window, scalar-contact, and $\hbar$-adic projections converge. The differential and bracket are the completed convolution operations

$$d_{\mathbf{K}}T = d_{\mathrm{QME}}T - (-1)^{|T|}T\, d_{\mathrm{CE}}, \qquad [T_1, T_2]_{\mathbf{K}} = [-,-]_\hbar \circ (T_1 \widehat{\otimes} T_2) \circ \Delta_{\mathrm{CE}},$$

where Δ_{CE} is the completed CE coproduct and $[-,-]_\hbar$ is the target $P_{\mathrm{o},\hbar}$/BV bracket. For a degree-zero kernel candidate κ, its curvature is

$$\mathrm{Curv}_{\mathbf{K}}(\kappa) = d_{\mathbf{K}}\kappa + \frac{1}{2}[\kappa,\kappa]_{\mathbf{K}} \in \mathbf{K}^1_{w,\partial,\hbar}(I).$$

On generators, the term $-\kappa d_{\mathrm{CE}}$ is exactly the source-bracket subtraction appearing in the bracket-defect formulas below.

Proposition B.16 *(Bulk-to-defect coefficient-current kernel layer).* In the local mixed HT model on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$, there is a canonical degree-zero continuous coefficient-current kernel

$$\kappa^{\mathrm{coef}}_{\hbar,w,I} \in \mathbf{K}^{0}_{w,\partial,\hbar}(I)$$

whose generator values are

$$\kappa^{\mathrm{coef}}_{\hbar,w,I}(u_{a(t)dt\otimes f}) = \iota_I\big(B^w_f(a)\big), \qquad \kappa^{\mathrm{coef}}_{\hbar,w,I}(c_{B,\rho}) = \iota_I\big(\Theta^w_\rho(B)\big).$$

Here $a \in C^\infty_c(I)$, $f \in \mathfrak{h}_{w,\hbar}$, $B \in \mathcal{D}'_c(I)$, and $\rho \in \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}$. The map ι_I is the coefficient inclusion of $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$ as field-independent defect-local functionals in $\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$. The degrees are

$$|u_{a(t)dt\otimes f}| = |B^w_f(a)| = 0, \qquad |c_{B,\rho}| = |\Theta^w_\rho(B)| = 1.$$

The kernel has the following proved properties.

(i) For each finite Hamiltonian window M in the formal holomorphic Hamiltonian degree, its restriction $\kappa^{\mathrm{coef},M}_{\hbar,w,I}$ lands in

$$\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}\big(\mathcal{E}^M_w|_I;\ A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)\big)$$

and the finite-window maps satisfy

$$\pi_{M,N}\kappa^{\mathrm{coef},M}_{\hbar,w,I} = \kappa^{\mathrm{coef},N}_{\hbar,w,I}\pi^{\mathrm{CE}}_{M,N}, \qquad M \geq N.$$

For admissible weights w, w', the diagonal transport maps satisfy

$$R^M_{w,w'}\kappa^{\mathrm{coef},M}_{\hbar,w,I} = \kappa^{\mathrm{coef},M}_{\hbar,w',I}R^{M,\mathrm{CE}}_{w,w'}.$$

Hence the displayed finite-window kernels determine the unique strict Mittag–Leffler inverse-limit kernel $\kappa^{\mathrm{coef}}_{\hbar,w,I}$.

(ii) After setting $\hbar = 0$, applying scalar reduction and the de Rham-current contraction, the kernel is exactly Φ^{cur}_I of Theorem B.7.

(iii) Inside the freely adjoined coefficient-current subalgebra, the generator $P_{0,\hbar}$-defects vanish:

$$\{B^w_f(a), B^w_g(b)\}_\hbar = B^w_{\{f,g\}_\hbar}(ab),$$

$$\{B^w_f(a), \Theta^w_\rho(B)\}_\hbar = \Theta^w_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB), \qquad \{\Theta^w_\rho(B), \Theta^w_\sigma(C)\}_\hbar = 0.$$

(iv) Let $\kappa_{\hbar,w,I}$ be any Costello-local-functional lift of this coefficient kernel. If its curved Maurer–Cartan equation has been solved through order $\hbar^{n-1}$, then the order-n component of $\mathrm{Curv}_{\mathbf{K}}(\kappa_{\hbar,w,I})$ is d_{QME}-closed modulo the already solved lower-order terms. If the filtered scalar projection exists and its scalar image is zero, each fixed source-arity or graph component gives, by evaluation, a residual class in

$$H^1\big(\ker\widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}}\big).$$

Equivalently, the assembled curvature coefficient is a class in the corresponding Hom-valued obstruction complex; this is the class denoted $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]$ in Problem 3.271. Its compatible finite-window counterterm obstruction is precisely the inverse-limit H^1 class together with the $\varprojlim^1 H^0$ primitive-compatibility class of Theorem 3.98.

This proposition constructs only the local coefficient-current kernel layer. It does not construct a Costello graph kernel for arbitrary $B \in \mathcal{D}'_c(I)$. The missing analytic theorem is the following: for every finite window and every defect-current-labelled graph, the products of the finite-scale propagators with the pushed-forward compactly supported currents must satisfy Hörmander wavefront transversality, admit local extensions across the relevant small diagonals, and produce counterterms compatible with interval extension, finite-window truncation, and admissible weight transport. Without that theorem the graph-level lift is proved only after restricting the cotangent labels to regular compactly supported densities or to a wavefront-admissible class, or else remains an obstruction problem. No compact target or global BCOV theorem is invoked.

Proof. By Proposition B.3, the source Lie brackets and principal-part pairings are the formal-disk Hamiltonian brackets and residues tensored with interval currents. The continuous CE cochain algebra is the completed graded-symmetric algebra on the continuous dual of $\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}[1]$. The target coefficient algebra $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$ is the completed graded-symmetric algebra on the matching Hamiltonian-current generators $B^w_f(a)$ and cotangent-current generators $\Theta^w_\rho(B)$. The displayed assignment therefore extends uniquely to a continuous completed graded-commutative algebra map. Since the source and target generators have the displayed equal degrees, the resulting element of the convolution habitat has degree 0.

Windowwise, the same construction is finite-dimensional in the coefficient direction, with $D_{w,\leq M}$ and $A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)$ as in Definition 3.97. Truncation merely discards coefficient degrees $d > N$, while $R^M_{w,w'}$ rescales each surviving degree-d basis vector and its dual by inverse factors. The formulas for $B^w_f(a)$ and $\Theta^w_\rho(B)$ are degreewise identities, so they commute with truncation and with weight transport. The strict Mittag–Leffler universal property gives the inverse-limit kernel.

The reduction statement follows by forgetting the Moyal deformation, applying the scalar quotient, and contracting the de Rham current direction. The two generator formulas become $u_{a(t)dt\otimes f} \mapsto B_f(a)$ and $c_{B,\rho} \mapsto \Theta_\rho(B)$, which are precisely the generators of Theorem B.7.

The three displayed bracket identities are the weighted Moyal version of the semidirect-product calculation in Theorem B.7: the Hamiltonian-current bracket gives the pointwise product ab, the cotangent action is multiplication $B \mapsto aB$ together with the Moyal coadjoint action, and the shifted cotangent current ideal is abelian. By biderivation, the identities extend from generators to the completed coefficient-current subalgebra.

Finally, the convolution habitat is a filtered dg Lie algebra. The Bianchi identity

$$d_{\mathbf{K}}\,\mathrm{Curv}_{\mathbf{K}}(\kappa) + [\kappa, \mathrm{Curv}_{\mathbf{K}}(\kappa)]_{\mathbf{K}} = 0$$

implies, order by order in $\hbar$, that the first unsolved curvature coefficient is closed for the linearized differential. Under the chosen scalar projection, zero scalar image places every evaluated component in $\ker \widehat{\sigma}_{\mathrm{sc}}$, equivalently places the assembled coefficient in the Hom-valued kernel. Adding a degree-zero local counterterm changes the first unsolved coefficient by the corresponding d_{QME}-boundary, and the finite-window compatibility of such primitives is exactly the Milnor obstruction measured in Theorem 3.98. This proves the stated obstruction class and leaves only the analytic distributional graph-extension theorem named in the statement. □

PROPOSITION B.17 *(Regular-density Costello graph kernel layer).* Let

$$\mathcal{D}^{\mathrm{reg}}_c(I) = \{\, b(t)dt \mid b \in C^\infty_c(I) \,\} \subset \mathcal{D}'_c(I)$$

and let

$$\mathfrak{g}^{w,\mathrm{reg}}_{\hbar,I} = \big(\Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}_{w,\hbar}\big) \ltimes \big(\mathcal{D}^{\mathrm{reg}}_c(I)\widehat{\otimes}\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}\big)[1] \subset \mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}.$$

Put $\mathbf{K}^{\bullet,\mathrm{reg}}_{w,\partial,\hbar}(I)$ for the corresponding completed convolution Hom obtained from Definition B.15 by restricting the source to $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}^{w,\mathrm{reg}}_{\hbar,I})$. Assume that, on every finite Hamiltonian window M of the local formal holomorphic Hamiltonian sector, the target $\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$ is equipped with the regulator-admissible finite-scale Costello brane-defect graph calculus for $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$: the weighted propagator $P^{w,M}_{\epsilon,L}$, regularized BV Laplacian, local interaction $I^w_{\mathrm{o},M}$, interval extension by zero, truncation maps $\pi_{M,N}$, and admissible weight transports $R^M_{w,w'}$, satisfying the acceptance gate of Statement 3.93 and the graphwise Hadamard/Mittag–Leffler control of Theorem 3.166.

Then the restriction of $\kappa^{\mathrm{coef}}_{\hbar,w,I}$ to regular density labels has a graphwise Costello-local lift

$$\kappa^{\mathrm{graph,reg}}_{\hbar,w,I} \in \mathbf{K}^{\mathrm{o,reg}}_{w,\partial,\hbar}(I)$$

in the following precise sense. For every fixed connected Costello graph Γ, finite CE input, finite window M, and external cotangent labels $B_j = b_j(t)dt \in \mathcal{D}^{\mathrm{reg}}_c(I)$, the finite-scale graph contribution

$$W^{\mathrm{reg},M}_{\Gamma,\epsilon,L}(a_\bullet, b_\bullet; f_\bullet, \rho_\bullet) \in O^{\mathrm{bd}}_{\mathrm{loc}}\big(\mathcal{E}^M_w|_I;\ \mathcal{A}^{\mathrm{pp},w,M}_{\partial,\hbar}(I)\big)$$

is a well-defined smooth local-functional kernel for $0 < \epsilon \le L < \infty$. Its small-$\epsilon$ expansion admits Costello local counterterms $C^{\mathrm{reg},M}_{\Gamma,w}(\epsilon; b_\bullet)$, and the renormalized graph

$$W^{R,\mathrm{reg},M}_{\Gamma,L,w} = \lim_{\epsilon\to 0}\big(W^{\mathrm{reg},M}_{\Gamma,\epsilon,L} - C^{\mathrm{reg},M}_{\Gamma,w}(\epsilon; b_\bullet)\big)$$

is again a brane-defect local functional with regular density coefficient labels. These graph kernels are compatible with interval extension by zero, finite-window truncation, and admissible weight transport:

$$\pi_{M,N} W^{R,\mathrm{reg},M}_{\Gamma,L,w} = W^{R,\mathrm{reg},N}_{\Gamma,L,w}\pi^{\mathrm{CE}}_{M,N}, \qquad R^M_{w,w'} W^{R,\mathrm{reg},M}_{\Gamma,L,w} = W^{R,\mathrm{reg},M}_{\Gamma,L,w'} R^{M,\mathrm{CE}}_{w,w'}.$$

On the zero-internal-edge part, and after scalar reduction and de Rham-current contraction, the lifted kernel is Φ^{cur}_I.

Consequently, wherever a filtered scalar projection $\widehat{\sigma}_{\mathrm{sc}}$ has been constructed on the chosen regular-density graph/counterterm subhabitat, the curvature

$$\mathrm{Curv}_{\mathbf{K}}(\kappa^{\mathrm{graph,reg}}_{\hbar,w,I})$$

defines the same finite-window non-scalar obstruction system as in Theorem 3.98. Thus the regular-density theorem removes the analytic wavefront/counterterm gap attached to the cotangent labels, but it does not prove the filtered scalar projection outside its stated habitat and does not prove vanishing of

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker\widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}}).$$

No extension from $\mathcal{D}^{\mathrm{reg}}_c(I)$ to $\mathcal{D}'_c(I)$ is asserted. The exact remaining analytic theorem is: for every finite window M, every Costello graph Γ, and every compactly supported distributional label $B_j \in \mathcal{D}'_c(I)$, the product of the finite-scale brane-defect propagator kernels with the pushed-forward current labels must satisfy Hörmander wavefront transversality, admit local extensions across all graph small diagonals, and admit counterterms compatible with interval extension, $\pi_{M,N}$, and $R^M_{w,w'}$. This is a local brane-defect microlocal condition, not a compactness condition on the holomorphic target.

Proof. The regular subspace is stable under the current operations used in the source. Indeed, if $B = b(t)dt$ and $a \in C_c^\infty(I)$, then Lemma B.4 gives $aB = (ab)(t)dt$, again a compactly supported smooth density. Extension by zero along $I \hookrightarrow J$ is smooth because the support is compactly contained in I. Thus the regular labels define the stated source subcomplex and the restriction of the convolution habitat.

Fix a finite window M, a graph Γ, and $0 < \epsilon \leq L < \infty$. In this window the coefficient part of the propagator is finite rank, and the spacetime part of $P_{\epsilon,L}^{w,M}$ is the finite-scale heat-kernel propagator of Costello's elliptic BV calculus. It is smooth at positive scale. After restriction to the brane line, multiplication by the smooth compactly supported functions a_i and densities $b_j(t)dt$ produces a smooth compactly supported graph integrand on the relevant configuration space and edge-parameter domain. Hence no product of singular distributions is formed at finite scale, and the graph integral is a well-defined element of the displayed brane-defect local-functional target.

As $\epsilon \to 0$, the only singularities are the usual heat-kernel singularities along graph diagonals. Multiplication by smooth compact densities does not enlarge wavefront sets and does not change the Hadamard singular order. Therefore the local asymptotic expansion and local counterterm construction are exactly the windowwise Costello construction, tensored with the finite coefficient kernel and multiplied by smooth density coefficients. Theorem 3.166 supplies the graphwise Hadamard estimates and strict Mittag–Leffler passage in the weighted coefficient direction. The counterterms remain local on the brane line; their coefficients are differential expressions in smooth compactly supported labels, hence regular density labels.

The compatibility identities are degreewise. The maps $\pi_{M,N}$ discard coefficient degrees above N, while $R_{w,w'}^M$ rescales the surviving weighted Hamiltonian and cotangent bases. The regular density factors live in the interval variable and are untouched by these coefficient maps. The strict Mittag–Leffler compatibility of the finite-window propagators and counterterms in Statement 3.93 therefore gives the two displayed identities. Interval extension compatibility is the same locality statement: all labels and counterterms are extended by zero, and local graph weights are supported on the same brane diagonals after extension.

The zero-internal-edge graph is the generator assignment of Proposition B.16. After forgetting the Moyal deformation, applying scalar reduction, and contracting the de Rham-current direction, it is Theorem B.7.

The curvature assertion is formal in the filtered convolution dg Lie algebra. The Bianchi identity used in the proof of Proposition B.16 applies to the regular restricted habitat. Once a filtered scalar projection is available, the first unsolved curvature coefficient lands in $\ker \widehat{\sigma}_{\mathrm{sc}}$, and compatible counterterms change it by d_{QME}-boundaries. The finite-window obstruction is exactly the inverse-limit H^1 class and the $\varprojlim^1 H^0$ primitive-compatibility class of Theorem 3.98.

If B is an arbitrary compactly supported distribution, the preceding smooth-multiplier argument fails at the point where the pushed-forward brane current is multiplied with propagator boundary values and then extended across graph diagonals. The needed Hörmander transversality, extension, and compatible counterterm theorem is therefore exactly the residual theorem stated above. □

Definition B.18 *(Wavefront-admissible defect-current labels).* Fix a finite Hamiltonian window M, a connected Costello graph Γ, and a finite list of external cotangent-current labels $B_\bullet = (B_1, \ldots, B_r)$, $B_j \in \mathcal{D}'_c(I)$. Let $X = \mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ and $L_I = I \times \{0\} \times \{p\} \subset X$. If $V_{\mathrm{bulk}}(\Gamma)$ and $V_\partial(\Gamma)$ are the bulk and defect vertices, set

$$X_\Gamma^\circ(I) = X^{V_{\mathrm{bulk}}(\Gamma)} \times L_I^{V_\partial(\Gamma)} \times E_{\Gamma,M},$$

where $E_{\Gamma,M}$ is the finite-dimensional coefficient and edge parameter space in the window M. The final graph space $X_\Gamma(I)$ is the open complement of the graph collision diagonals in $X_\Gamma^\circ(I)$ before local extension across those diagonals. Choose an attachment map $\ell : \{1, \dots, r\} \to V_\partial(\Gamma)$ for the current labels and define the incidence map

$$e_\Gamma : X_\Gamma^\circ(I) \longrightarrow I^r, \qquad e_{\Gamma,j}(x) = t_{\ell(j)}.$$

Repeated labels at one defect vertex are therefore diagonal pullbacks by e_Γ. For $B^\boxtimes = B_1 \boxtimes \cdots \boxtimes B_r$, write $B_\Gamma = e_\Gamma^* B^\boxtimes$ only when this pullback is defined. Let $\mathrm{Diag}(\Gamma)$ be the finite set of graph collision diagonals on which the small-ϵ Hadamard coefficients may be singular.

The tuple $B_\bullet$ is called (Γ, M)-wavefront-admissible if the following four operations are defined by the usual Hörmander calculus [32, Ch. 8] and by the Costello finite-window graph package [12, Chs. 5–7].

(i) The incidence pullback is non-characteristic:

$$N_{e_\Gamma} \cap \mathrm{WF}(B^\boxtimes) = \emptyset, \qquad N_{e_\Gamma} = \{(e_\Gamma(x), \xi) \mid de_{\Gamma,x}^t \xi = 0\}.$$

Then

$$\mathrm{WF}(B_\Gamma) \subset e_\Gamma^\# \mathrm{WF}(B^\boxtimes) = \{(x, de_{\Gamma,x}^t \xi) \mid (e_\Gamma(x), \xi) \in \mathrm{WF}(B^\boxtimes)\}.$$

The resulting support is proper over the graph integration variables.

(ii) For every Hadamard coefficient $K_{\Gamma,\alpha}^M$ appearing in the small-ϵ expansion of the finite-window graph kernel,

$$\big(\mathrm{WF}(K_{\Gamma,\alpha}^M) + \mathrm{WF}(B_\Gamma)\big) \cap T^*_{X_\Gamma(I)} X_\Gamma(I) = \emptyset.$$

Equivalently, no covector in $\mathrm{WF}(B_\Gamma)$ is the negative of a graph-kernel covector. Thus the product $K_{\Gamma,\alpha}^M B_\Gamma$ is a canonical distribution.

(iii) For each collision diagonal $\Delta \in \mathrm{Diag}(\Gamma)$, the products $K_{\Gamma,\alpha}^M B_\Gamma$ have finite scaling degree along Δ. Their allowed extensions across Δ differ by distributions supported on $\Delta \cap \mathrm{supp}(B_\Gamma)$, hence by brane-defect local counterterms with distributional current coefficients.

(iv) The final graph pushforward has wavefront contained in the brane-defect local-functional wavefront class already used in Theorem 3.166, after tensoring with the finite coefficient kernel.

Denote the resulting multilinear relation by

$$\mathcal{D}'_{c,\mathrm{WF}}(I; \Gamma, M) \subset \big(\mathcal{D}'_c(I)\big)^r.$$

The universal wavefront-admissible relation is the intersection of these relations over the finite windows, graphs, and finite CE inputs under consideration. This is a relation on finite tuples. A linear subspace $\mathcal{V} \subset \mathcal{D}'_c(I)$ gives an honest restricted CE source only when every finite tuple from $\mathcal{V}$ lies in the corresponding relation.

A useful source of such subspaces is a diagonal-admissible conic set $\Lambda \subset T^*I \setminus 0$: for every $n \geq 1$,

$$\xi_1, \dots, \xi_n \in \Lambda \quad \Longrightarrow \quad \xi_1 + \cdots + \xi_n \neq 0.$$

Then

$$\mathcal{D}'_{c,\Lambda,\mathrm{fs}}(I) = \{\, B \in \mathcal{D}'_c(I) \mid \mathrm{WF}(B) \subset \Lambda,\ B \text{ has finite scaling degree along its singular support} \,\}$$

is the finite-scaling one-sided wavefront class. On an oriented interval, either open half-cone $\Lambda_+ = \{(t,\xi) \mid \xi > 0\}$ or $\Lambda_- = \{(t,\xi) \mid \xi < 0\}$ is diagonal-admissible. These two half-cones are maximal among sign-conic diagonal-admissible linear habitats for ordinary collision-conormal kernels: any conic subset of $T^*I \setminus 0$ containing both signs admits, after positive rescaling, a finite nonempty covector sum equal to zero, so it can cancel a collision conormal. Compactly supported one-sided conormal boundary values, for example cutoffs of $(t - t_0 - i0)^{-m}$ after choosing the Fourier sign convention, lie in one of these classes and are not regular densities.

PROPOSITION B.19 *(Wavefront-admissible distributional graph kernel layer).* Assume the same regulator-admissible finite-window Costello brane-defect graph calculus as in Proposition B.17. Let Γ be a fixed connected graph, $\mathcal{M}$ a finite Hamiltonian window, and $B_\bullet \in \mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,\mathcal{M})$. Then the graph contribution with external cotangent labels $B_\bullet$ has the following distributional extension.

(i) For every $0 < \epsilon \le L < \infty$, the finite-scale graph weight

$$W^{\mathrm{WF},\mathcal{M}}_{\Gamma,\epsilon,L}(a_\bullet, B_\bullet; f_\bullet, \rho_\bullet)$$

is a well-defined brane-defect local-functional kernel with compactly supported distributional current coefficients. At this positive scale the propagator is smooth, so the only distributional operation is multiplication of smooth kernels by the pushed-forward currents.

(ii) The small-ϵ expansion admits local counterterms

$$C^{\mathrm{WF},\mathcal{M}}_{\Gamma,w}(\epsilon; B_\bullet)$$

supported on

$$\bigcup_{\Delta \in \mathrm{Diag}(\Gamma)} \Delta \cap \mathrm{supp}(B_\Gamma).$$

The renormalized graph

$$W^{R,\mathrm{WF},\mathcal{M}}_{\Gamma,L,w} = \lim_{\epsilon \to 0}\big(W^{\mathrm{WF},\mathcal{M}}_{\Gamma,\epsilon,L} - C^{\mathrm{WF},\mathcal{M}}_{\Gamma,w}(\epsilon; B_\bullet)\big)$$

is a brane-defect local functional with coefficients in the distributional current target $\mathcal{A}^{\mathrm{pp},w,\mathcal{M}}_{\partial,\hbar}(I)$.

(iii) The construction is compatible with interval extension by zero, finite-window truncation, and admissible weight transport whenever the chosen tuple remains wavefront-admissible after applying the corresponding incidence maps:

$$\pi_{\mathcal{M},\mathcal{N}} W^{R,\mathrm{WF},\mathcal{M}}_{\Gamma,L,w} = W^{R,\mathrm{WF},\mathcal{N}}_{\Gamma,L,w}\, \pi^{\mathrm{CE}}_{\mathcal{M},\mathcal{N}}, \qquad R^{\mathcal{M}}_{w,w'} W^{R,\mathrm{WF},\mathcal{M}}_{\Gamma,L,w} = W^{R,\mathrm{WF},\mathcal{M}}_{\Gamma,L,w'} R^{\mathcal{M},\mathrm{CE}}_{w,w'}.$$

In particular, every finite tuple of labels in $\mathcal{D}'_{c,\Lambda,\mathrm{fs}}(I)$ for a diagonal-admissible cone Λ is wavefront-admissible for graph kernels whose singular wavefronts are conormal to ordinary graph collision diagonals. This gives a non-regular distributional extension of Proposition B.17.

This is the exact extension asserted here from the present Costello graph products: outside the relation $\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,\mathcal{M})$, at least one of the multiplication, extension, or graph-pushforward operations above is not defined by the cited wavefront calculus. No theorem for all of $\mathcal{D}'_c(I)$ follows. The obstruction and the admitted extension are entirely local on the brane interval and on the finite-window formal disk coefficient sector.

Proof. At positive scale the heat-kernel propagator $P^{w,\mathcal{M}}_{\epsilon,L}$ is smooth in the spacetime and defect variables. Multiplication of a compactly supported distribution by a smooth function is always defined, and the incidence maps are proper on the supports by Definition B.18. This proves the finite-scale graph weight.

As $\epsilon \to 0$, Theorem 3.166 gives graphwise Hadamard coefficients with wavefront contained in the union of conormals to graph collision diagonals. The second condition in Definition B.18 is exactly Hörmander's product criterion [32, Ch. 8], so $K^{\mathcal{M}}_{\Gamma,\alpha} B_\Gamma$ is a canonical distribution for every coefficient. The finite-scaling condition then gives the standard local extension across each collision diagonal. Its ambiguity is supported on the collision locus meeting the support of B_Γ; this is precisely a brane-defect local counterterm with distributional current coefficient. Subtracting these local counterterms gives the displayed renormalized graph.

The two compatibility identities are inherited degreewise from the finite-window Costello graph calculus. The maps $\pi_{\mathcal{M},N}$ and $R^{\mathcal{M}}_{w,w'}$ act on the weighted Hamiltonian coefficient factors and do not change the interval wavefront cone. Interval extension by zero is allowed exactly when the extended incidence tuple satisfies the same non-characteristic and proper-support conditions. Hence compatibility holds on the stated wavefront-admissible relation.

For the one-sided conic classes, let Λ be diagonal-admissible and let $B_j \in \mathcal{D}'_{c,\Lambda,\mathrm{fs}}(I)$. A conormal covector to an n-fold collision diagonal has components $(\eta_1, \dots, \eta_n)$ whose sum is zero on each collision block. If this covector were cancelled by covectors from the external labels, then a finite nonempty sum of elements of Λ would be zero, contradicting diagonal-admissibility. Thus the product criterion holds. The finite-scaling hypothesis gives the extension condition, and the remaining pushforward condition is the same proper-support condition as above.

Finally, the full compactly supported distribution space is too large. In the one-edge boundary graph take two external labels $B_1 = B_2 = \delta_{t_0}$. The singular boundary propagator has wavefront containing the conormal directions $(t_0, t_0; \tau, -\tau)$ to the diagonal $t = s$, while $\mathrm{WF}(\delta_{t_0} \boxtimes \delta_{t_0})$ contains the opposite directions $(t_0, t_0; -\tau, \tau)$. Hörmander's product criterion therefore fails. Equivalently, the graph asks for the value of the singular propagator on the diagonal against $\delta_{t_0} \boxtimes \delta_{t_0}$, which is not a canonical distributional operation without an additional renormalization rule. Thus arbitrary $\mathcal{D}'_c(I)$-labels are not supported by the current Costello graph products. □

Proposition B.20 *(Current-compatible D'_c theorem surface).* Assume the regulator-admissible finite-window Costello brane-defect graph package of Proposition B.17. The theorem surface compatible with compactly supported distributional cotangent labels is the following.

(i) For arbitrary $B \in \mathcal{D}'_c(I)$, the constructed object is the coefficient-current kernel

$$\kappa^{\mathrm{coef}}_{\hbar,w,I}(u_{a(t)dt\otimes f}) = \iota_I(B^w_f(a)), \qquad \kappa^{\mathrm{coef}}_{\hbar,w,I}(c_{B,\rho}) = \iota_I(\Theta^w_\rho(B)).$$

This is an algebraic current kernel. It is not a Costello graph-kernel theorem for all $B \in \mathcal{D}'_c(I)$.

(ii) For a fixed graph Γ, finite window $\mathcal{M}$, and admissible tuple $B_\bullet \in \mathcal{D}'_{c,\mathrm{WF}}(I; \Gamma, \mathcal{M})$, put

$$T^{\mathcal{M}}_{\Gamma,\alpha}(B_\bullet) = K^{\mathcal{M}}_{\Gamma,\alpha} B_\Gamma.$$

Its wavefront is bounded by the Hörmander product formula

$$\begin{aligned}\mathrm{WF}\Big(T^{\mathcal{M}}_{\Gamma,\alpha}(B_\bullet)\Big) \subset &\mathrm{WF}(K^{\mathcal{M}}_{\Gamma,\alpha}) \cup \mathrm{WF}(B_\Gamma) \\ &\cup \big(\mathrm{WF}(K^{\mathcal{M}}_{\Gamma,\alpha}) + \mathrm{WF}(B_\Gamma)\big),\end{aligned}$$

with the fibrewise sum taken over the same base point in $X_\Gamma(I)$. For the final graph projection $p_\Gamma : X_\Gamma(I) \to Y_\Gamma(I)$, proper on support, the pushed-forward graph has

$$\mathrm{WF}\big(p_{\Gamma,*}T^M_{\Gamma,\alpha}\big) \subset \{(y,\eta) \mid \exists x \in p_\Gamma^{-1}(y),\ (x, dp^t_{\Gamma,x}\eta) \in \mathrm{WF}(T^M_{\Gamma,\alpha})\}.$$

These are the only wavefront bounds used by the distributional graph layer.

(iii) Let $\Delta \in \mathrm{Diag}(\Gamma)$, let $c_\Delta = \operatorname{codim}\Delta$, and suppose

$$s_{\Gamma,\alpha,\Delta} = \mathrm{sd}_\Delta\Big(T^M_{\Gamma,\alpha}(B_\bullet)\Big) < \infty.$$

Set

$$N_{\Gamma,\alpha,\Delta} = \max\{-1,\ \lfloor s_{\Gamma,\alpha,\Delta} - c_\Delta \rfloor\}.$$

If $\widetilde{T}$ and $\widetilde{T}'$ are two local extensions of $T^M_{\Gamma,\alpha}(B_\bullet)$ across Δ, then locally in normal coordinates to Δ

$$\widetilde{T}' - \widetilde{T} = \sum_{|\beta| \le N_{\Gamma,\alpha,\Delta}} \partial_\nu^\beta \delta_\Delta \otimes S_{\Delta,\beta},$$

with no summand when $N_{\Gamma,\alpha,\Delta} = -1$. Here $S_{\Delta,\beta} \in \mathcal{D}'(\Delta)$ is supported in $\Delta \cap \mathrm{supp}(B_\Gamma)$ and carries the same finite-window Hamiltonian coefficient label. Thus the allowed counterterm ambiguity belongs to

$$\sum_{\Delta,\alpha} \sum_{|\beta| \le N_{\Gamma,\alpha,\Delta}} \partial_\nu^\beta \delta_\Delta\, \mathcal{D}'_{\Delta\cap\mathrm{supp}(B_\Gamma)}(\Delta) \widehat{\otimes} \mathcal{A}^{\mathrm{pp},w,M}_{\partial,\hbar}(I).$$

(iv) The finite-window counterterms are compatible precisely when the chosen extensions satisfy

$$j_* C^{\mathrm{WF},M}_{\Gamma,I}(\epsilon; B_\bullet) = C^{\mathrm{WF},M}_{\Gamma,J}(\epsilon; j_* B_\bullet),$$

for interval inclusions $j : I \hookrightarrow J$ for which the extended tuple remains admissible, and

$$\pi_{M,N} C^{\mathrm{WF},M}_{\Gamma,w}(\epsilon; B_\bullet) = C^{\mathrm{WF},N}_{\Gamma,w}(\epsilon; B_\bullet)\pi^{\mathrm{CE}}_{M,N},$$

$$R^M_{w,w'} C^{\mathrm{WF},M}_{\Gamma,w}(\epsilon; B_\bullet) = C^{\mathrm{WF},M}_{\Gamma,w'}(\epsilon; B_\bullet) R^{M,\mathrm{CE}}_{w,w'}.$$

Failure of the interval identity is a factorization-compatibility residual. Failure of the last two identities is exactly the finite-window primitive compatibility obstruction measured by Theorem 3.98 after passing to the relevant graph/counterterm habitat.

(v) There is no present theorem for all of $\mathcal{D}'_c(I)$. In particular, the coincident-current boundary graph with $B_1 = B_2 = \delta_{t_0}$ violates the product condition because conormal covectors $(\tau, -\tau)$ to $t = s$ can be cancelled by $\mathrm{WF}(\delta_{t_0} \boxtimes \delta_{t_0})$. Thus the arbitrary-$\mathcal{D}'_c$ graph theorem would require extra renormalization data not supplied by Costello's general finite-scale package or by the Costello–Gwilliam factorization vocabulary.

Hence the manuscript has a theorem-grade graph layer on $\mathcal{D}^{\mathrm{reg}}_c(I)$, on finite graphwise wavefront-admissible tuples, and on the sign-conic finite-scaling habitats $\mathcal{D}'_{c,\Lambda_\pm,\mathrm{fs}}(I)$ for ordinary collision-conormal graph kernels. The full distributional space $\mathcal{D}'_c(I)$ remains a named obstruction problem.

Proof. Item (1) is Proposition B.16: the assignment is degreewise algebraic in the compactly supported current B and uses only multiplication of a current by a smooth test function. No singular graph product is formed there.

Items (2) and (3) are the standard Hörmander product, pushforward, and finite-scaling extension calculus applied to Definition B.18. The product transversality condition makes $K^M_{\Gamma,\alpha} B_\Gamma$ canonical. Properness of the graph projection gives the displayed pushforward wavefront bound. Finite scaling degree along Δ gives extensions across Δ; the structure theorem for distributions supported on a submanifold gives the finite normal-derivative formula for the difference of two extensions. The support statement follows because all extensions agree off $\Delta \cap \operatorname{supp}(B_\Gamma)$.

Item (4) is the compatibility condition needed for the counterterms to define one interval-local element of the completed finite-window and weight-transport tower rather than unrelated graphwise choices. When the identities hold, the Bianchi argument of Proposition B.16 produces the Hom-valued obstruction class and the Milnor $\varprojlim^1 H^0$ primitive obstruction. Failure of interval extension compatibility is a failure of the factorization current structure itself; failure of finite-window or weight compatibility is precisely a primitive-compatibility obstruction.

Item (5) is the counterexample in the proof of Proposition B.19. It also matches the source audit: the local Costello sources support finite-scale BV/RG/QME graph calculus, local functionals, and local counterterms under their hypotheses, but they do not prove a current-valued arbitrary-$\mathcal{D}'_c$ brane-defect theorem. □

Proposition B.21 *(Obstruction maps for the non-scalar current lift).* The datum of Definition B.14 can realize the non-scalar Costello graph/QME theorem for the local mixed HT brane-defect model only if the following obstruction maps vanish, with the stated homotopies.

(i) The scalar projection of the curvature must vanish in the chosen scalar branch:

$$\widehat{\sigma}_{\mathrm{sc}}\big(\mathrm{Curv}_{\mathbf{K}}(\kappa_{\hbar,w,I})\big) = 0.$$

In the ordinary scalar-reduced $\mathfrak{gl}_N$ branch this condition is exactly the requirement that the Capelli contact class $\hbar N[\bar{c}]$ be cancelled by a closed-exchange term and a local counterterm. In the balanced supertrace branch it is the condition $\widehat{\sigma}_{\mathrm{sc}}(\mathrm{Ob}_{w,\partial,\hbar}) = 0$.

(ii) After scalar reduction and de Rham-current contraction, the kernel must recover the reduced current interface:

$$\overline{\kappa}_{\hbar,w,I}(u_{a(t)dt\otimes f}) = B_f(a), \qquad \overline{\kappa}_{\hbar,w,I}(c_{B,\rho}) = \Theta_\rho(B),$$

with brackets

$$\{B_f(a), B_g(b)\}_\hbar = B_{\{f,g\}_\hbar}(ab), \qquad \{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$

and $\{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0$. At $\hbar = 0$ this is Theorem B.7.

(iii) The unreduced bracket defects

$$D_{f,g}(a,b) = \{B^{\mathrm{BV}}_f(a), B^{\mathrm{BV}}_g(b)\}_\hbar - B^{\mathrm{BV}}_{\{f,g\}_\hbar}(ab),$$

$$D_{f,\rho}(a,B) = \{B^{\mathrm{BV}}_f(a), \Theta^{\mathrm{BV}}_\rho(B)\}_\hbar - \Theta^{\mathrm{BV}}_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$

and

$$D_{\rho,\sigma}(B,C) = \{\Theta^{\mathrm{BV}}_{\rho}(B), \Theta^{\mathrm{BV}}_{\sigma}(C)\}_{\hbar}$$

must be d_{QME}-exact by homotopies compatible with inclusions of intervals, disjoint factorization products, and the finite-window regulator maps.

(iv) After evaluation on each fixed finite CE input ξ, equivalently on each fixed graph/source-arity component, the residual non-scalar curvature class

$$\left[\mathrm{Curv}_{\mathbf{K}}(\kappa_{\hbar,w,I})(\xi)\right] \in H^1\big(\ker \widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}}\big)$$

must vanish after the compatible finite-window counterterms $C^{w_0}_{\Gamma,w}(\epsilon)$ have been inserted.

These conditions are necessary. They are also sufficient for the stated local graph/QME realization once the Costello finite-scale BV propagator, BV Laplacian, locality estimates, and counterterm system in the chosen coefficient category have been constructed. They do not assert a compact Calabi–Yau, CoHA, global BCOV, or cross-volume comparison theorem.

Proof. The source and target in Definition B.14 are the current versions of the weighted cotangent Hamiltonian source and the mixed brane-defect obstruction complex of Problem 3.271. A Costello graph realization is a solution of the QME in this local-functional complex, hence its curvature has zero scalar projection and zero residual class in $\ker \widehat{\sigma}_{\mathrm{sc}}$. This gives the first and fourth obstruction maps.

The second condition is forced by compatibility with the already proved reduced current theorem. After contraction and scalar reduction the only allowed target brackets are the Moyal principal-part current brackets of Theorem 3.270; at $\hbar = 0$ these reduce to the three brackets of Theorem B.7. Any different reduction would change the theorem already proved.

The three defects in the third condition are the generator-level failures of the source semidirect Lie brackets to match the target P_0-brackets. Since the CE differential and the target bracket are derivations, exactness of these generator defects, with homotopies compatible with factorization products and finite-window transition maps, is precisely the condition that the generator assignment extend to the desired homotopy P_0-central operation. The reduced algebraic target has zero defects by Theorem B.7; the unreduced polynomial BV target does not, by Theorem E.6. Therefore the missing theorem must construct these homotopies in a larger current-compatible Costello local-functional category. □

THEOREM B.22 *(Curved kernel centrality homotopy criterion).* Fix the local mixed HT brane-defect model on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$, a weighted current source $\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}$, and a Costello local-functional target $\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$ as in Definition B.14. Let κ_M be a finite-window component of a degree-zero continuous kernel in the convolution habitat $\mathbf{K}^{\bullet}_{w,\partial,\hbar}(I)$, and assume that the chosen scalar-contact projection is a chain map on the habitat under discussion.

For source generators

$$x \in \{u_{a(t)dt\otimes f}, c_{B,\rho}\}, \qquad y \in \{u_{b(t)dt\otimes g}, c_{C,\sigma}\},$$

define the binary closed-open curvature component by

$$\mathfrak{D}^{M}_{\kappa}(x,y) = \mathrm{Curv}_{\mathbf{K},M}(\kappa_M)(x \wedge y).$$

When no binary Taylor component has already been inserted, these are the three generator defects

$$D^{M}_{f,g}(a,b), \qquad D^{M}_{f,\rho}(a,B), \qquad D^{M}_{\rho,\sigma}(B,C)$$

displayed in Proposition B.21.

A compatible unreduced binary centrality homotopy for this kernel is exactly a family of cochains

$$H^M_{x,y} \in \mathcal{K}^{|x|+|y|-1}_{\mathrm{ns},M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},M} \cap \mathfrak{Q}^{|x|+|y|-1}_{w,M,\partial,\hbar}(I)$$

satisfying

$$\mathfrak{D}^M_\kappa(x,y) + d_M H^M_{x,y} = 0 \tag{14}$$

for every such pair, together with the compatibility identities

$$j_* H^M_{I;x,y} = H^M_{J;j_*x,j_*y},$$

for interval inclusions $j : I \hookrightarrow J$ for which the labels remain admissible, and

$$\pi_{M,N} H^M_{x,y} = H^N_{\pi x,\pi y}, \qquad R^M_{w,w'} H^M_{w;x,y} = H^M_{w';Rx,Ry}.$$

Equivalently, for every pair (x,y), the inverse-system cohomology class

$$\big([\mathfrak{D}^M_\kappa(x,y)]\big)_M \in \varprojlim_M H^{|x|+|y|}\Big(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_M\Big)$$

and the Milnor primitive-compatibility class represented by

$$\Big[\pi_{M,N} h^M_{x,y} - h^N_{\pi x,\pi y}\Big] \in H^{|x|+|y|-1}\Big(\mathcal{K}^\bullet_{\mathrm{ns},N}(I), d_N\Big)$$

must both vanish, where $h^M_{x,y}$ is any windowwise primitive solving $\mathfrak{D}^M_\kappa(x,y) + d_M h^M_{x,y} = 0$.

For the algebraic coefficient-current kernel $\kappa^{\mathrm{coef}}_{\hbar,w,I}$ of Proposition B.16, all $\mathfrak{D}^M_\kappa(x,y)$ vanish in the freely adjoined current target, and one may take $H^M_{x,y} = 0$. This is the strict homotopy of Theorem E.10. For a genuine unreduced Costello local-functional target, the same conclusion is not supplied by the coefficient-current formula. It requires the finite-window graph row representing $\mathfrak{D}^M_\kappa(x,y)$, a scalar-zero gate for that row, and a local counterterm or binary Taylor cochain $H^M_{x,y}$ satisfying equation (14) with the displayed interval, window, and weight compatibilities.

Proof. The arity-two part of the curved Maurer–Cartan equation in the convolution dg Lie algebra is precisely

$$\mathrm{Curv}_{\mathbf{K},M}(\kappa_M)(x \wedge y) = 0.$$

Expanding this identity before choosing a binary component gives the target bracket of the two generator images minus the image of the source semidirect bracket, with the source term supplied by $-\kappa d_{\mathrm{CE}}$. These are exactly the three defects $D_{f,g}$, $D_{f,\rho}$, and $D_{\rho,\sigma}$ of Proposition B.21.

A binary homotopy is, by definition, a degree $|x| + |y| - 1$ primitive for this curvature component in the scalar-zero kernel complex. This gives equation (14). Compatibility with extension by zero, finite-window truncation, and admissible weight transport is exactly the statement that these primitives assemble to a homotopy in the completed local factorization target rather than to unrelated windowwise choices. The inverse-limit obstruction and the Milnor class are the usual Roos obstruction for choosing compatible primitives, in the same form as Theorem 3.98.

In the algebraic coefficient-current target the three brackets are identities, hence the curvature components vanish and the zero primitive is compatible with every structural map. In a Costello local-functional target the terms are graph weights, source faces, collision/contact terms, and counterterms. Unless those rows and their primitive are supplied in the chosen finite-window package, the displayed equation is only an obstruction equation, not a constructed homotopy. □

C Sign conventions for the Hamiltonian comparison

This appendix fixes the signs used in the CE/PV comparison, the primitive one-ψ complex, the principal-part coadjoint action, and the Moyal deformation. The cohomological shift convention is

$$(V[1])^n = V^{n+1}.$$

The reduced matrix Koszul letter ψ is odd. All cyclic signs below use its parity; the absolute BV grading enters only through the global shift of the cotangent coordinate.

Definition C.1 *(Hamiltonian and cyclic conventions).* On $\mathbb{C}[z_1, z_2]$ the Poisson bracket is

$$\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g,$$

so

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad X_{z_1} = \partial_{z_2}, \quad X_{z_2} = -\partial_{z_1}.$$

In the graded cyclic quotient,

$$[uv]_{\mathrm{cyc}} = (-1)^{|u||v|}[vu]_{\mathrm{cyc}}, \qquad |x| = |y| = 0, \quad |\psi| \equiv 1 \pmod 2.$$

LEMMA C.2 *(CE/PV signs).* Let $\mathfrak{h}$ have basis e_I and $[e_I, e_J] = C_{IJ}^K e_K$. Let c^I be the CE ghost dual to e_I, and let u_K be the shifted cotangent coordinate paired with the basis vector e_K in the right-adjoint index convention. The Chevalley–Eilenberg differential is

$$d_{\mathrm{CE}} c^K = -\frac{1}{2} C_{IJ}^K c^I c^J, \qquad d_{\mathrm{CE}} u_K = C_{KJ}^L u_L c^J.$$

On the polyvector model with even coordinates O_K and odd coordinates θ^I,

$$[\theta^I, O_J] = \delta_J^I, \qquad d_\pi = [\pi, -],$$

and the signs are

$$d_\pi \theta^K = -\frac{1}{2} C_{IJ}^K \theta^I \theta^J, \qquad d_\pi O_K = C_{KJ}^L O_L \theta^J.$$

Therefore the assignment

$$c^I \longmapsto \theta^I, \qquad u_K \longmapsto O_K$$

intertwines d_{CE} and d_π.

Proof. The first CE formula is the standard dual of the bracket. The second is the right-adjoint action convention on the shifted cotangent coordinate:

$$e_J \cdot u_K = [u_K, e_J] = C_{KJ}^L u_L.$$

Applying d_{CE} once more gives terms proportional to

$$C_{KJ}^L C_{LA}^M u_M c^A c^J - \frac{1}{2} C_{KJ}^L C_{AB}^J u_L c^A c^B,$$

which vanish by the Jacobi identity for the right-adjoint representation.

For the PV model, take

$$\pi = \frac{1}{2} C_{IJ}^K O_K \theta^I \theta^J.$$

The Schouten rule with $[\theta^I, O_J] = \delta_J^I$ gives exactly the two displayed formulas. Hence $c^I \mapsto \theta^I$ and $u_K \mapsto O_K$ commute with the differentials. □

Lemma C.3 *(Primitive one-ψ boundary signs).* In the reduced matrix Koszul algebra $Q\psi = xy - yx$, $Qx = Qy = 0$. For ordinary words u, v,

$$Q\ \mathrm{Tr}(\psi u \psi v) = \mathrm{Tr}\big(\psi vxyu - \psi vyxu - \psi uxyv + \psi uyxv\big).$$

Hence the boundary in the one-ψ chain model is

$$\partial_2(u, v) = vxyu - vyxu - uxyv + uyxv.$$

Proof. The first displayed ψ contributes $+(xy - yx)$; the second displayed ψ contributes $-(xy - yx)$, because one odd letter precedes it. Move the even word created by the first replacement cyclically past the remaining ψ and the word v; no further sign is introduced. Removing the common front ψ gives the formula. □

Lemma C.4 *(Principal-part coadjoint signs).* Let

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \qquad a, b \geq 0.$$

The residue realization of the coadjoint action is

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with negative-index targets equal to zero. In particular,

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Proof. Hamiltonian vector fields preserve $dz_1 dz_2$, so the coadjoint sign is the residue integration-by-parts sign. Directly,

$$\begin{aligned}\{z_1^p z_2^q, z_1^{-a-1} z_2^{-b-1}\} &= p z_1^{p-1} z_2^q\,(-(b+1)) z_1^{-a-1} z_2^{-b-2} \\ &\quad - q z_1^p z_2^{q-1}\,(-(a+1)) z_1^{-a-2} z_2^{-b-1} \\ &= -(pb - qa + p - q) z_1^{p-a-2} z_2^{q-b-2}.\end{aligned}$$

Multiplication by $dz_1 dz_2$ identifies the last monomial with $\rho_{a-p+1,b-q+1}$ exactly when both indices are nonnegative. □

Lemma C.5 *(PBW descendant sign).* The polynomial primitive one-ψ descendants carry the PBW special-fibre action

$$z_1^p z_2^q : \widetilde{\Psi}_{a,b} \longmapsto (pb - qa)\widetilde{\Psi}_{a+p-1,b+q-1}.$$

This is a raising action and is not the principal-part coadjoint action of Lemma C.4.

Proof. The Hamiltonian vector field of $z_1^p z_2^q$ is

$$p z_1^{p-1} z_2^q \partial_{z_2} - q z_1^p z_2^{q-1} \partial_{z_1}.$$

Acting on the symmetrised trace with a x-letters and b y-letters, the first summand differentiates one of the b y-letters and contributes pb copies of the symmetrised target $\widetilde{\Psi}_{a+p-1,b+q-1}$. The second differentiates one of the a x-letters and contributes $-qa$ copies of the same symmetrised target. The total coefficient is $pb - qa$. □

LEMMA C.6 *(Current source convention).* For an interval $I \subset \mathbb{R}$, the Hamiltonian source complex is $\Omega_c^\bullet(I)\widehat{\otimes}\mathfrak{h}$. A degree-zero source density is written $a(t)dt \otimes f$, and the cotangent CE source coordinate maps to the boundary observable by

$$u_{a(t)dt\otimes f} \longmapsto B_f(a) = \int_I a(t)B_f(t)\,dt.$$

The companion ghost coordinate $c_{a\otimes f}$ maps to the PV coordinate $\theta_{a\otimes f}$. If $B \in \mathcal{D}_c'(I)$, the standard multiplication of a distribution by a smooth test function a is written aB, and

$$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\} = 0.$$

Proof. The de Rham factor supplies the compactly supported density on the brane line; the Hamiltonian current supplies f. Pairing this source with the boundary field gives the integral defining $B_f(a)$. The distributional cotangent factor is a module over $C_c^\infty(I)$, so aB is defined by $(aB)(\varphi) = B(a\varphi)$. The first bracket is the semidirect product bracket for $\mathfrak{h}_I \ltimes \mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P}[1]$; the second vanishes because the shifted principal-part summand is an abelian odd ideal. □

LEMMA C.7 *(Moyal coefficients).* With

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}, \qquad f * g = m\exp\left(\frac{\hbar}{2}P\right)(f \otimes g),$$

the normalized Moyal Lie bracket is

$$\{f, g\}_\hbar = \hbar^{-1}(f * g - g * f) = \sum_{s\geq 0} \frac{\hbar^{2s}}{2^{2s}(2s+1)!}P^{2s+1}(f, g).$$

The first quantum correction has coefficient $1/24$:

$$\{f, g\}_\hbar = \{f, g\} + \frac{\hbar^2}{24}P^3(f, g) + O(\hbar^4).$$

Proof. The even powers of P are symmetric under exchanging f and g, and the odd powers are antisymmetric. Thus

$$f * g - g * f = 2\sum_{s\geq 0} \frac{(\hbar/2)^{2s+1}}{(2s+1)!}P^{2s+1}(f, g),$$

and division by $\hbar$ gives the displayed formula. The $s = 1$ coefficient is $1/(2^2 \cdot 3!) = 1/24$. □

LEMMA C.8 *(Moyal coadjoint signs).* Let $(n)_r = n(n-1)\cdots(n-r+1)$, with $(n)_0 = 1$. For $f = z_1^p z_2^q$, the normalized Moyal coadjoint action on principal parts is

$$f \cdot_\hbar \rho_{a,b} = -\sum_{\substack{r\geq 1\\ r \text{ odd}}} \frac{\hbar^{r-1}}{2^{r-1}r!}C_r(p, q; a-p+r, b-q+r)\,\rho_{a-p+r,b-q+r},$$

where negative-index targets and the scalar target $\rho_{0,0}$ are set to zero, and

$$C_r(p, q; m, n) = \sum_{\alpha=0}^{r}(-1)^\alpha\binom{r}{\alpha}(p)_{r-\alpha}(q)_\alpha(m)_\alpha(n)_{r-\alpha}.$$

The $r = 1$ term is the Matlis coadjoint action

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}.$$

Proof. The r-fold bidifferential P^r applied to $z_1^p z_2^q \otimes z_1^m z_2^n$ has coefficient $C_r(p,q;m,n)$ and output $z_1^{p+m-r} z_2^{q+n-r}$. The normalized Moyal bracket keeps only odd r, with coefficient $\hbar^{r-1}/(2^{r-1}r!)$ by Lemma C.7. The coadjoint sign is the residue sign

$$(f \cdot_\hbar \rho)(g) = -\rho(\{f,g\}_\hbar).$$

To hit the coefficient functional $\rho_{a,b}$, the test monomial must have exponents $m = a - p + r$, $n = b - q + r$. Substitution gives the displayed formula. For $r = 1$, $C_1(p,q;a-p+1,b-q+1) = pb - qa + p - q$, giving the classical Matlis sign. □

Corollary C.9 *(Universal dual-module normalization).* The residue basis $\rho_{a,b}$ is normalized by

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n},$$

and the coadjoint convention is

$$(f \cdot \rho)(g) = -\rho(\{f,g\}), \qquad (f \cdot_\hbar \rho)(g) = -\rho(\{f,g\}_\hbar).$$

Therefore the linear Hamiltonians act by

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

This is the universal cotangent normalization used by the Matlis principal-part module. With it fixed, the averaged polynomial descendant formulas $z_1 \cdot \widetilde{\Psi}_{a,b} = b\widetilde{\Psi}_{a,b-1}$ and $z_2 \cdot \widetilde{\Psi}_{a,b} = -a\widetilde{\Psi}_{a-1,b}$ cannot be made into the same $\mathfrak{h}$-module action by changing signs, factorials, or the Moyal $1/24$ convention.

Proof. The first two displayed formulas are the residue pairing and the coadjoint sign convention used in Lemma C.8. Substituting $(p,q) = (1,0)$ and $(p,q) = (0,1)$ in the $r = 1$ term gives the two linear formulas. They raise pole order, whereas the polynomial descendant formulas lower polynomial degree. No scalar renormalization of basis vectors can turn a locally nilpotent polynomial-degree action into this non-locally-nilpotent principal-part action; the obstruction is structural, not a convention error. □

D Full Primitive Higher-ψ Koszul Homology

This appendix computes the universal primitive single-trace Koszul complex attached to the moment-map equation $[x,y] = 0$. Universal means that no choice of Matlis target, PBW model, finite N trace identity, or quantum deformation is used: after stable trace invariant theory has identified primitive traces with graded cyclic words, the answer is the cellular homology of a chip-moving state complex. Every finite-bidegree stable primitive descendant calculation over $\mathbb{C}$, with $d\psi = xy - yx$ and no trace relations beyond graded cyclicity, is obtained from this one. Passage to completed coefficient systems requires an exact coefficient functor, or an explicit condition excluding new $\varprojlim^1$ classes.

Definition D.1 *(Graded cyclic word complex).* Let

$$A_\psi = \mathbb{C}\langle x, y, \psi\rangle, \qquad |x| = |y| = 0, \quad |\psi| \equiv 1 \pmod 2,$$

and let d be the odd derivation determined by

$$d\psi = xy - yx, \qquad dx = dy = 0.$$

Put

$$\mathrm{Cyc}(A_\psi) = A_\psi/[A_\psi, A_\psi]_{\mathrm{gr}}, \qquad [uv] = (-1)^{|u||v|}[vu].$$

Give the letters weights

$$\mathrm{wt}(x) = (1,0), \qquad \mathrm{wt}(y) = (0,1), \qquad \mathrm{wt}(\psi) = (1,1).$$

For $R, S \geq 0$ define

$$K^p_{R,S} = \mathbb{C}\{[w]_{\mathrm{cyc}} : \#\psi(w) = p,\ \#x(w) = R - p,\ \#y(w) = S - p\},$$

with $K^p_{R,S} = 0$ unless $0 \leq p \leq \min(R, S)$. If $w = a_1 \cdots a_n$, the boundary is

$$\partial[w] = \sum_{i:a_i=\psi} (-1)^{\#\{j<i:a_j=\psi\}} [a_1 \cdots a_{i-1} x y\, a_{i+1} \cdots a_n - a_1 \cdots a_{i-1} y x\, a_{i+1} \cdots a_n]_{\mathrm{cyc}}.$$

Thus $\partial : K^p_{R,S} \to K^{p-1}_{R,S}$.

LEMMA D.2 *(Stable super-cyclic trace presentation).* Fix a total word length d. In the stable range $N \geq d$, the primitive GL_N-invariant single-trace functions in the variables x, y, ψ, homogeneous of total word length d, are exactly the graded cyclic words in $\mathrm{Cyc}(A_\psi)$. Equivalently, after connected single-trace extraction, the only stable relation among words in x, y, ψ is

$$\mathrm{Tr}(uv) = (-1)^{|u||v|}\,\mathrm{Tr}(vu).$$

Proof. First replace the odd letter ψ by p distinct even marked matrices $\eta_1, \ldots, \eta_p$, and take the summand multilinear in the η_i's and homogeneous of the prescribed x, y-degree. The Procesi–Razmyslov stable invariant theorem [17,18] identifies the primitive single-trace invariants of total word length d, for $N \geq d$, with ordinary cyclic words in the letters $x, y, \eta_1, \ldots, \eta_p$, modulo only cyclicity of the trace.

The symmetric group $\mathbb{S}_p$ permutes the marked letters. Passing from p even marked letters to one odd letter is the sign-isotypic projection for this action, followed by the identification $\eta_i \mapsto \psi$. Since the ground field has characteristic zero, this projection is exact. Interchanging two marked occurrences contributes the sign of the transposition, and rotating a block with a marked letters past a block with b marked letters contributes $(-1)^{ab}$. This is precisely the graded cyclic rule for a single odd letter ψ. No other stable trace relation remains in word length d. □

LEMMA D.3 *(Boundary signs and $\partial^2 = 0$).* The formula of Definition D.1 is well-defined on the graded cyclic quotient and satisfies $\partial^2 = 0$.

Proof. On the free algebra A_ψ, the displayed formula is the derivation rule for the odd derivation d: replacing the i-th occurrence of ψ crosses exactly the preceding odd letters and hence contributes $(-1)^{\#\{j<i:a_j=\psi\}}$. Since $d(x) = d(y) = 0$ and $d(\psi) = xy - yx$ is even and d-closed, one has $d^2 = 0$ on the generators, hence on all of A_ψ.

If u, v are homogeneous, then

$$d(uv - (-1)^{|u||v|}vu)$$

is a sum of graded commutators, using the odd Leibniz rule and the fact that d lowers the number of odd letters by one. Thus d preserves $[A_\psi, A_\psi]_{\mathrm{gr}}$, so it descends to $\mathrm{Cyc}(A_\psi)$. The descended operator is exactly ∂, and $\partial^2 = 0$ follows from $d^2 = 0$. □

Lemma D.4 *(Exact p-face formula).* Let $u_1, \dots, u_p$ be ordinary words in x, y. In $K^{p-1}_{R,S}$,

$$\partial[\psi u_1 \psi u_2 \cdots \psi u_p] = \sum_{j=1}^{p} (-1)^{j-1} [\psi u_1 \cdots \psi u_{j-1} x y\, u_j \psi u_{j+1} \cdots \psi u_p \\ - \psi u_1 \cdots \psi u_{j-1} y x\, u_j \psi u_{j+1} \cdots \psi u_p]_{\mathrm{cyc}},$$

where, for $j = 1$, the initial displayed $\psi u_1 \cdots \psi u_{j-1}$ is absent.

Proof. This is the boundary of Definition D.1 applied to the representative $\psi u_1 \psi u_2 \cdots \psi u_p$. The j-th displayed ψ has $j-1$ preceding odd letters, and replacing it by the even word $xy - yx$ produces no further Koszul sign. □

Lemma D.5 *(Two-ψ boundary sign).* For ordinary words u, v,

$$\partial[\psi u \psi v] = [\psi v x y u] - [\psi v y x u] - [\psi u x y v] + [\psi u y x v].$$

After removing the common front ψ from each term, this is the boundary

$$v x y u - v y x u - u x y v + u y x v.$$

Proof. Lemma D.4 with $p = 2$ gives

$$[(xy - yx) u \psi v] - [\psi u (xy - yx) v].$$

The first two terms are cyclically rotated to put the remaining ψ in front:

$$[(xy) u \psi v] = [\psi v x y u], \qquad [(yx) u \psi v] = [\psi v y x u].$$

The moved block is even, so no Koszul sign is produced. The second replacement already has the surviving ψ in front, and its overall sign is the sign -1 from the preceding odd letter. Expanding gives the displayed formula.

Interchanging u and v gives $u x y v - u y x v - v x y u + v y x u$, the negative of the displayed expression. Thus the formula is compatible with the alternating two-ψ quotient used below. □

Construction D.6 *(One-ψ three-term skeleton).* For $k, l \geq 0$, let $\mathsf{W}_{k,l}$ be the ordinary words with k letters x and l letters y, and let $\mathsf{N}_{a,b}$ be the necklaces with a letters x and b letters y. The one-ψ skeleton in bidegree (k, l) is the finite complex

$$C_2(k,l) \xrightarrow{\partial_2} C_1(k,l) \xrightarrow{\partial_1} C_0(k,l),$$

with

$$C_1(k,l) = \mathbb{C}\{\mathsf{W}_{k,l}\}, \qquad C_0(k,l) = \mathbb{C}\{\mathsf{N}_{k+1,l+1}\},$$

and

$$C_2(k,l) = \big(\mathbb{C}\{(u,v) : \deg u + \deg v = (k-1, l-1)\}\big)_{\mathrm{alt}},$$

where $(u,v) = -(v,u)$ and $(u,u) = 0$. If $k = 0$ or $l = 0$, then $C_2(k,l) = 0$. The boundaries are

$$\partial_1(w) = [xyw]_{\mathrm{cyc}} - [yxw]_{\mathrm{cyc}},$$

and

$$\partial_2(u,v) = vxyu - vyxu - uxyv + uyxv.$$

This is the truncation $K^2_{k+1,l+1} \to K^1_{k+1,l+1} \to K^0_{k+1,l+1}$ after putting the remaining ψ at the front of a one-ψ cyclic word. Equivalently, it is the cellular chain complex in dimensions 2, 1, 0 of the state complex constructed below: C_0 records chip configurations, C_1 records one moving chip, and C_2 records two independent moving chips with the orientation fixed by the ordered moving edges.

Proof. The identifications of C_1 and C_0 are exactly the graded cyclic normal forms in ψw and the ordinary necklaces. The source C_2 is the two-ψ summand of $K^2_{k+1,l+1}$: after cutting at one displayed ψ, a class is written $\psi u \psi v$, and moving the second ψ-block past the first contributes the sign -1. Lemma D.5 gives ∂_2, while the definition of ∂ gives ∂_1. □

Construction D.7 *(Labelled chip-moving state complex).* Assume $R, S > 0$. Let $\mathscr{C}_S$ be the oriented circle subdivided into S labelled oriented edges and S labelled vertices. Define $\widetilde{Y}_{R,S}$ as follows. A p-cell is a state with $R - p$ stationary indistinguishable chips on vertices and p moving chips in the interiors of p distinct edges. Several stationary chips may occupy the same vertex. No edge contains more than one moving chip.

Encode such a state by traversing $\mathscr{C}_S$. At a vertex write one x for each stationary chip sitting there. Then write y if the following edge is empty and ψ if the following edge contains a moving chip. This gives a word with $R - p$ letters x, $S - p$ letters y, and p letters ψ, with a distinguished starting edge.

The p-cell is oriented by the increasing order of the labelled edges carrying moving chips. Its j-th oriented coordinate has boundary

$$\text{terminal face } xy - \text{ initial face } yx.$$

Hence the cellular boundary in the labelled complex is

$$\sum_j (-1)^{j-1}(\text{replace the } j\text{-th moving edge by } xy - \text{replace it by } yx).$$

The cyclic group $\Gamma_S = \mathbb{Z}/S\mathbb{Z}$ rotates the edge labels and acts cellularly on $\widetilde{Y}_{R,S}$.

LEMMA D.8 *(Cellular chain identification with cyclic relabelling).* For $R, S > 0$, the cellular chain complex of $\widetilde{Y}_{R,S}$, after taking Γ_S-coinvariants, is naturally isomorphic to $K^\bullet_{R,S}$:

$$C_p(\widetilde{Y}_{R,S};\mathbb{C})_{\Gamma_S} \cong K^p_{R,S}.$$

Under this isomorphism the cellular boundary is the boundary ∂ of Definition D.1.

Proof. A labelled p-cell is equivalently a labelled cyclic sequence

$$x^{a_1} m_1 x^{a_2} m_2 \cdots x^{a_S} m_S, \qquad m_i \in \{y, \psi\},$$

with exactly p of the m_i's equal to ψ and $\sum_i a_i = R - p$. Every graded cyclic word in $K^p_{R,S}$ has a representative of this form: move the cut, if necessary, past even x-letters until it lies immediately before an edge marker y or ψ. This representative is unique after a starting edge has been chosen. The remaining ambiguity is rotation of the S edge markers, hence quotienting by Γ_S is exactly passage to graded cyclic words. No further relation is introduced: in the labelled complex the S edge markers are ordered, and in the coinvariant complex the only identifications are their rotations with the orientation character below.

It remains to check the sign. Suppose a rotation moves a block containing a occurrences of ψ past the complementary block containing b occurrences of ψ. On cellular orientations this rotation moves a oriented interval coordinates past b oriented interval coordinates, hence changes the orientation by $(-1)^{ab}$. This is exactly the graded cyclic sign $[uv] = (-1)^{|u||v|}[vu]$, since x and y are even and every ψ is odd.

With this orientation convention, the cellular boundary in Construction D.7 is

$$\sum_j (-1)^{j-1}([xy] - [yx])$$

in the j-th moving edge. The terminal face is the state in which the chip has crossed the edge and is read as xy; the initial face is read as yx. This is exactly Lemma D.4, term by term. □

Lemma D.9 *(Abel map for labelled states).* For $R, S > 0$, the labelled state complex $\widetilde{Y}_{R,S}$ has the homology of a circle:

$$H_q(\widetilde{Y}_{R,S}; \mathbb{C}) \cong \begin{cases} \mathbb{C}, & q = 0, \\ \mathbb{C}, & q = 1, \\ 0, & q \geq 2. \end{cases}$$

The cyclic rotation action of Γ_S is trivial on these two homology lines.

Proof. Work in the universal cover of $\mathscr{C}_S$, whose vertices are the integers and whose edges are the unit intervals. A lifted labelled cell is described by the positions of the R chips: stationary chips have integer positions, and a moving chip on the edge $[m, m+1]$ has coordinate $m + t$ with $0 < t < 1$. The closure of the cell permits $t = 0, 1$, which are exactly the two vertex states yx and xy in the boundary convention of Construction D.7. Define the Abel map

$$A : \widetilde{Y}_{R,S} \longrightarrow \mathbb{R}/S\mathbb{Z}$$

by summing the chip positions on the oriented circle, counted with multiplicity, and reducing modulo S. On every cell this map is affine.

Lift the target to $\mathbb{R}$ and fix a cyclic order of the R lifted chips. A lifted fibre is described by weakly ordered coordinates

$$t_1 \leq t_2 \leq \cdots \leq t_R \leq t_1 + S$$

with fixed sum. This is a convex polytope. Its faces are cut out by the equalities $t_i \in \mathbb{Z}$ and by the choices of those unit intervals which contain a moving chip; these faces are precisely the cells of the chip-moving subdivision meeting the fibre. Hence every lifted fibre is a contractible finite polyhedral complex. The map is a proper PL map of finite polyhedra with acyclic fibres. The Vietoris–Begle theorem, applied to this affine cellular map, gives a homology equivalence

$$H_\bullet(\widetilde{Y}_{R,S}; \mathbb{C}) \cong H_\bullet(S^1; \mathbb{C}).$$

A cyclic relabelling of the S edges adds a constant to every chip coordinate, so it rotates the target circle by an orientation-preserving rotation. Therefore Γ_S acts trivially on $H_0(S^1; \mathbb{C})$ and $H_1(S^1; \mathbb{C})$, and hence on the two homology lines of $\widetilde{Y}_{R,S}$. □

LEMMA D.10 *(Cyclic Abel quotient and stabilizer fibres).* Let $Y_{R,S}$ denote the cyclic relabelling quotient of $\widetilde{Y}_{R,S}$, with the orientation character encoded by the graded cyclic rule. Then the quotient chain complex $C_\bullet(\widetilde{Y}_{R,S};\mathbb{C})_{\Gamma_S}$ has the homology of S^1. Moreover, every stabilizer quotient of an Abel fibre is contractible.

Proof. By Lemma D.9, the labelled Abel map is a homology equivalence from $\widetilde{Y}_{R,S}$ to the circle. A cyclic relabelling by $r \in \Gamma_S$ adds r to each of the R chip positions, hence translates the Abel coordinate by Rr in $\mathbb{R}/S\mathbb{Z}$. The action on the target circle is by orientation-preserving rotations, so it is trivial on the two homology lines. Since $\mathbb{C}[\Gamma_S]$ is semisimple,

$$H_\bullet(C_\bullet(\widetilde{Y}_{R,S};\mathbb{C})_{\Gamma_S}) \cong H_\bullet(\widetilde{Y}_{R,S};\mathbb{C})_{\Gamma_S} \cong H_\bullet(S^1;\mathbb{C}).$$

Now fix an Abel value and a stabilizer subgroup $G \subset \Gamma_S$ of that value. In the universal cover, choose the lifted fibre before decomposing it into chip cells. It is a convex polytope cut out by weak ordering, the fixed-sum equation, and the chip-cell inequalities, and G acts affinely on it. Averaging a point over G gives a G-fixed point. Straight-line contraction to this fixed point is G-equivariant and respects the chip-cell subdivision. Hence the stabilizer quotient of the Abel fibre is contractible. □

LEMMA D.11 *(S^1 generator).* For $R, S > 0$, the cycle

$$\Psi_{R-1,S-1} = \sum_{w \in \mathsf{W}_{R-1,S-1}} [\psi w]_{\mathrm{cyc}}$$

represents the fundamental H_1-line of the cyclic chip-moving complex.

Proof. Its boundary is

$$\partial\Psi_{R-1,S-1} = \sum_{w \in \mathsf{W}_{R-1,S-1}} ([xyw]_{\mathrm{cyc}} - [yxw]_{\mathrm{cyc}}).$$

For a fixed necklace with R letters x and S letters y, the coefficient is the number of cyclic xy-transitions minus the number of cyclic yx-transitions. These two numbers are equal on a circle; hence $\Psi_{R-1,S-1}$ is closed.

Let $\epsilon : K^1_{R,S} \to \mathbb{C}$ send every one-ψ cyclic word to 1. Lemma D.5 shows that every two-ψ boundary has two positive and two negative terms, so it lies in $\ker \epsilon$. But

$$\epsilon(\Psi_{R-1,S-1}) = \binom{R+S-2}{R-1} \neq 0.$$

By Lemma D.10, the H_1-line is one-dimensional. Moreover, every one-cell in the displayed sum is oriented in the positive Abel direction: its terminal face has the moved chip one edge farther along the oriented circle. Thus the displayed cycle maps to $\binom{R+S-2}{R-1}$ times the oriented circle class and spans the fundamental H_1-line. □

THEOREM D.12 *(Full primitive higher-ψ Koszul homology).* For every $R, S \geq 0$,

$$H_p(K_{R,S}) \cong \begin{cases} \mathbb{C} \cdot [x^R y^S], & p = 0, \\ \mathbb{C} \cdot [\Psi_{R-1,S-1}], & p = 1,\ R, S \geq 1, \\ 0, & p \geq 2, \end{cases}$$

where

$$\Psi_{R-1,S-1} = \sum_{w \in \mathsf{W}_{R-1,S-1}} [\psi w]_{\mathrm{cyc}}.$$

Consequently the primitive stable Koszul homology is

$$\mathbb{C}[x, y] \oplus \mathbb{C}[x, y]\theta, \qquad |\theta| = 1, \quad \mathrm{wt}(\theta) = (1, 1),$$

with the scalar Hamiltonian removed later if one passes to the reduced Hamiltonian sector.

Proof. If $R = 0$ or $S = 0$, then $K^p_{R,S} = 0$ for $p > 0$, and $K^0_{R,S}$ is spanned by the unique cyclic word x^R or y^S (including the empty word when $R = S = 0$). This gives the stated homology.

Assume $R, S > 0$. By Lemma D.8, $K^\bullet_{R,S}$ is the Γ_S-coinvariant cellular chain complex of $\widetilde{Y}_{R,S}$. Since $\mathbb{C}[\Gamma_S]$ is semisimple, taking coinvariants is an exact functor on finite-dimensional $\mathbb{C}[\Gamma_S]$-modules. Therefore

$$H_p(K_{R,S}) \cong H_p(\widetilde{Y}_{R,S}; \mathbb{C})_{\Gamma_S}.$$

Lemma D.10 identifies this homology with $H_\bullet(S^1; \mathbb{C})$. Thus $H_p(K_{R,S})$ is one-dimensional for $p = 0, 1$ and vanishes for $p \geq 2$. In particular, the three-term skeleton $C_2 \to C_1 \to C_0$ already computes H_1: cells of dimension $p \geq 3$ have boundaries in C_{p-1} and cannot change $\ker(C_1 \to C_0)/\mathrm{im}(C_2 \to C_1)$.

The degree-zero line is generated by any necklace of multidegree (R, S). The functional sending every zero-ψ necklace to 1 kills every one-ψ boundary, so this line is nonzero; it is denoted $\mathbb{C} \cdot [x^R y^S]$.

Lemma D.11 identifies the degree-one generator as $\Psi_{R-1,S-1}$. □

Theorem D.13 *(Universal primitive descendant calculation).* Let $\mathsf{K}_{\mathrm{univ}}$ be the stable primitive cyclic Koszul complex generated by two even letters x, y and one odd letter ψ, with $d\psi = xy - yx$, decomposed by the weights of Definition D.1. Then

$$H_\bullet(\mathsf{K}_{\mathrm{univ}}) \cong \mathbb{C}[x, y] \oplus \mathbb{C}[x, y]\theta, \qquad |\theta| = 1, \quad \mathrm{wt}(\theta) = (1, 1),$$

with $x^a y^b \theta$ represented by the symmetrised primitive cycle $\Psi_{a,b}$. This is the universal primitive calculation in the following precise sense. If $E_\bullet$ is any stable primitive single-trace model whose generators are the images of x, y, ψ, whose differential sends ψ to $xy - yx$, and whose only stable primitive trace relation is graded cyclicity, then the induced map

$$\mathsf{K}_{\mathrm{univ}} \longrightarrow E_\bullet$$

factors on homology through the displayed two-line module. Hence there are no hidden universal primitive higher-ψ classes from which a further cotangent or descendant bridge can be built.

Proof. The first assertion is Theorem D.12 summed over all bidegrees (R, S), with $x^a y^b \theta = \Psi_{a,b}$ and $R = a + 1$, $S = b + 1$ in degree one. For the universal property, Lemma D.2 says that the stable primitive trace span is exactly the graded cyclic quotient of the free algebra on x, y, ψ. Therefore any stable primitive model satisfying the stated hypotheses receives a unique chain map from $\mathsf{K}_{\mathrm{univ}}$ after choosing the images of x, y, ψ. The homology of the source is the two-line module just computed. Since $H_p(K_{R,S}) = 0$ for $p \geq 2$ in every bidegree, every universal primitive class in the target comes from the degree-zero polynomial line or the degree-one Ψ-line. The statement is independent of the later choice of principal-part, PBW, or Moyal coefficient module. □

THEOREM D.14 *(Hamiltonian-sector extraction).* The full primitive homology decomposes as

$$\mathbb{C}\cdot[1]\oplus\bar{A}\oplus\mathbb{C}\cdot[\Psi_{0,0}]\oplus\bar{A}_{\mathrm{desc}}[1],$$

where

$$\bar{A}=\bigoplus_{R+S>0}\mathbb{C}\cdot[x^R y^S],\qquad \bar{A}_{\mathrm{desc}}[1]=\bigoplus_{a+b>0}\mathbb{C}\cdot[\Psi_{a,b}].$$

The Hamiltonian comparison uses only $\bar{A}\oplus\bar{A}_{\mathrm{desc}}[1]$. The complementary sectors are exactly the scalar unit $[1]$, the scalar odd primitive $\Psi_{0,0}$, and the vanishing higher-ψ primitive groups $H_p(K_{R,S})=0$ for $p\geq 2$.

Proof. The constant Hamiltonian has zero Hamiltonian vector field, so the reduced Hamiltonian algebra is $\bar{A}=\mathbb{C}[x,y]/\mathbb{C}\cdot 1$ degree by degree. This removes the degree-zero scalar class $[1]$. The odd scalar $\Psi_{0,0}$ is the one-ψ descendant of the same removed constant direction; it is a stable primitive trace class, but it has no partner in $\bar{A}_{\mathrm{desc}}[1]$. Theorem D.12 proves that no primitive classes remain in ψ-degree at least two. What remains is exactly the positive-degree Hamiltonian and one-ψ PBW label sector displayed above. This is a vector-space homology statement; it does not identify $\Psi_{a,b}$ with Matlis principal parts or with the Moyal coadjoint module. □

E UNREDUCED BV COTANGENT, SCALAR QME OBSTRUCTIONS, AND CANCELLATIONS

This appendix computes the scalar part of the mixed brane-defect QME, the algebraic current presentation available after passing to distributional principal-part currents, and the precise cancellations available before scalar reduction or after balanced superization. Every statement is formal-local on

$$\mathbb{R}^2_{\mathrm{top}}\times\widehat{\mathbb{C}^2}_0,\qquad \Omega_{\mathrm{loc}}=dz_1\wedge dz_2.$$

There is no compact Calabi–Yau target, global BCOV theorem, CoHA, quintic, OSV/GV, Abel–Jacobi, Igusa, Borcherds, or BKM input in this appendix. The calculation separates five formal-local facts: finite polynomial boundary observables cannot carry the principal-part module; the current principal-part model carries a strict algebraic P_0 presentation with zero homotopies inside the added current target; the ordinary scalar-reduced trace model has a nonzero central CE class; the unreduced central-character source absorbs the scalar symbol before scalar reduction; and the standard balanced supertrace model makes the scalar representative vanish.

Definition E.1 *(BV degrees and signs).* The open polynomial BV/Koszul model uses

$$|\phi_1|=|\phi_2|=0,\qquad |\psi=A^*|=-1,\qquad Q\psi=[\phi_1,\phi_2],\qquad |Q|=1.$$

A Hamiltonian generator H_f has degree 0. A shifted cotangent generator $\eta_\rho=\rho[1]$ is odd of BV degree -1. The CE coordinates satisfy

$$|c_f|=1,\qquad |u_\rho|=0.$$

This is the cotangent-coordinate grading of the CE/PV block; it is not the ordinary graded dual of the BV degree of $\rho[1]$. The CE/PV cotangent–Schouten bracket has degree -1. The reduced Hamiltonian boundary P_0 bracket has degree 0. The QME obstruction differential $Q+\{I_0,-\}$ has degree 1. The Poisson sign convention is the one determined by $\Omega_{\mathrm{loc}}=dz_1\wedge dz_2$, so $\{z_1,z_2\}=1$. The principal-part coadjoint action is

$$z_1^p z_2^q\cdot\rho_{a,b}=-(pb-qa+p-q)\rho_{a-p+1,b-q+1}.$$

Proposition E.2 *(Local geometry of scalar shadows).* All scalar-contact, finite-window, marked-graph, and supertrace calculations below use only the formal-local mixed holomorphic-topological geometry

$$I \subset \mathbb{R}_{\mathrm{brane}} \subset \mathbb{R}^2_{\mathrm{top}}, \qquad \widehat{\mathbb{C}^2}_0 = \operatorname{Spf} \mathbb{C}[[z_1, z_2]], \qquad \Omega_{\mathrm{loc}} = dz_1 \wedge dz_2.$$

The inverse Poisson bracket is

$$\{f, g\}_{\Omega_{\mathrm{loc}}} = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g.$$

With the convention $\iota_{X_f} \Omega_{\mathrm{loc}} = -df$, one has

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \operatorname{div}_{\Omega_{\mathrm{loc}}} X_f = 0.$$

Thus no divergence or volume-form anomaly is present in the scalar shadow. The only Hamiltonian scalar cocycle used in this appendix is

$$\omega(f, g) = [\{f, g\}_{\Omega_{\mathrm{loc}}}]_0, \qquad \omega(z_1, z_2) = 1.$$

The open index contraction then multiplies this local Hamiltonian scalar by the Chan–Paton trace coefficient:

$$\hbar N \, \omega(f, g)\gamma_{\mathbf{1}} \quad \text{in the ordinary } \mathfrak{gl}_N \text{ branch,}$$

$$\hbar \operatorname{sdim}(\mathbb{C}^{N|N}) \, \omega(f, g)\gamma_{\mathbf{1}} = 0 \quad \text{in the balanced supertrace branch,}$$

and, for a marked scalar loop,

$$\hbar \operatorname{Str}_{\mathrm{cyc}}(\mathfrak{m}) \, \omega(f, g)\gamma_{\mathbf{1}}.$$

A finite window M only restricts the polynomial Hamiltonian labels, graph labels, and counterterm table to a finite local coefficient package; it does not change Ω_{loc}, the Hamiltonian bracket, or the divergence-free condition.

Proof. The formula for X_f is obtained by solving $\iota_{X_f}(dz_1 \wedge dz_2) = -df$. Its divergence is

$$\partial_{z_1}(-\partial_{z_2} f) + \partial_{z_2}(\partial_{z_1} f) = 0$$

after writing $X_f = -(\partial_{z_2} f)\partial_{z_1} + (\partial_{z_1} f)\partial_{z_2}$. Therefore the local volume form is preserved and cannot contribute an additional scalar contact term. Taking the constant Taylor coefficient of the displayed Poisson bracket gives the cocycle ω, with $\omega(z_1, z_2) = 1$.

The remaining scalar factor is purely the open Chan–Paton index loop. The ordinary trace of the identity on $\mathbb{C}^N$ is N. The supertrace of the identity on $\mathbb{C}^{N|N}$ is $N - N = 0$. A marked loop replaces the identity by the cyclic product of its marker tensors and gives the cyclic supertrace shown above. Finite-window truncation is a restriction of labels and graph tables inside this same formal disk, so it preserves the local cocycle and the divergence computation. □

Definition E.3 *(Scalar-reduced brane-defect obstruction complex).* Let $\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}$ denote local functionals on the ordinary $\mathfrak{gl}_N$ brane-defect fields in the formal-local model $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_0$, with compact support on the brane line and with coefficients in the scalar-reduced Hamiltonian source $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The differential controlling a mixed brane-defect QME lift is

$$d_{\mathrm{QME}} = Q + \{I_0, -\}.$$

A scalar contact term is tested by adjoining the degree-one central ghost $\gamma_{\mathbf{1}}$ for the omitted constant Hamiltonian line. Its associated graded is the Chevalley–Eilenberg complex of $\bar{A}$ with trivial coefficients on this central line. Writing

$$\mathfrak{Q}^{\bullet}_{\mathrm{bd}} = (O^{\mathrm{bd}}_{\mathrm{loc}}, d_{\mathrm{QME}}),$$

the associated-graded scalar-symbol projection is

$$\sigma_{\mathrm{sc}}\colon\ \mathrm{gr}_{\mathrm{sc}}\,\mathfrak{Q}^{\bullet}_{\mathrm{bd}} \longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_{\mathbf{1}})[1].$$

A cone or long exact sequence is formed only after one has chosen a d_{QME}-stable scalar filtration and an actual filtered chain projection

$$\widehat{\sigma}_{\mathrm{sc}}\colon \mathfrak{Q}^{\bullet}_{\mathrm{bd}} \longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_{\mathbf{1}})[1]$$

whose associated graded is the displayed σ_{sc}. Without this extra projection, the assertion below is only an associated-graded scalar obstruction statement. Thus a degree-one QME representative with two Hamiltonian inputs maps to a Lie two-cocycle. Explicitly,

$$\sigma_{\mathrm{sc}}\Big(\kappa(f,g)\int_{\mathbb{R}} a(t)b(t)\gamma_{\mathbf{1}}(t)\,dt\Big) = \kappa(f,g)\gamma_{\mathbf{1}}.$$

LEMMA E.4 *(Filtered scalar projection obstruction).* Fix a complete decreasing scalar-contact filtration $F^{\bullet}\mathfrak{Q}^{\bullet}_{\mathrm{bd}}$ preserved by d_{QME}, and put the target $C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_{\mathbf{1}})[1]$ in filtration degree 0. Suppose the associated graded carries the scalar-symbol projection

$$\sigma_{\mathrm{sc}}\colon\ \mathrm{gr}_F\,\mathfrak{Q}^{\bullet}_{\mathrm{bd}} \longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_{\mathbf{1}})[1].$$

Choose any filtered linear lift $s\colon \mathfrak{Q}^{\bullet}_{\mathrm{bd}} \to C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_{\mathbf{1}})[1]$ of σ_{sc}. Its chain-defect

$$\delta_s = d_{\mathrm{CE}}s - s\,d_{\mathrm{QME}}$$

lowers the scalar-contact filtration. The first nonzero homogeneous piece defines an obstruction class

$$o^{(r)}_{\sigma} \in H^1\big(\mathrm{Hom}\big(\mathrm{gr}_F\,\mathfrak{Q}^{\bullet}_{\mathrm{bd}}, C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_{\mathbf{1}})[1]\big)_{-r}\big),$$

with the usual differential $D(\alpha) = d_{\mathrm{CE}}\alpha - (-1)^{|\alpha|}\alpha d_{\mathrm{gr}}$. A filtered chain projection $\widehat{\sigma}_{\mathrm{sc}}$ with associated graded σ_{sc} exists exactly when the obstruction classes $o^{(r)}_{\sigma}$ vanish successively for all $r > 0$, with compatible choices in the completed inverse limit.

Proof. Since the associated graded of s is already a chain map, the commutator $d_{\mathrm{CE}}s - sd_{\mathrm{QME}}$ strictly lowers the filtration. Its lowest nonzero component is closed for the Hom differential because $d^2_{\mathrm{CE}} = d^2_{\mathrm{QME}} = 0$. Replacing s by $s - \alpha$, where α lowers filtration by the same amount, changes this component by $D\alpha$. Thus the first defect can be removed exactly when its cohomology class vanishes. Iterating the same argument through the complete filtration gives a compatible sequence of corrections if and only if all successive classes vanish. □

PROPOSITION E.5 *(First scalar-lift obstruction in the ordinary branch).* Work in the ordinary scalar-reduced $\mathfrak{gl}_N$ weighted brane-defect local-functional complex $\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$, with $N > 0$. Let J_{sc} be the closed ideal of functionals whose associated-graded scalar-contact symbol vanishes, and put $F^r = J^r_{\mathrm{sc}}$,

completed $\hbar$-adically and in the finite-window inverse limit. For any filtered linear lift s of the scalar-symbol projection, the first chain-defect class has Hamiltonian two-input evaluation

$$\mathrm{ev}_{f,g}\big(o^{(1)}_{\sigma,w}\big) = \hbar N\, \omega(f,g)\gamma_{\mathbf{1}} \in H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_{\mathbf{1}})[[\hbar]].$$

In particular

$$\mathrm{ev}_{z_1,z_2}\big(o^{(1)}_{\sigma,w}\big) = \hbar N\, \gamma_{\mathbf{1}} \neq 0.$$

Hence the scalar-symbol projection has no filtered chain lift in the ordinary scalar-reduced branch unless one supplies additional data whose leading scalar class is $-\hbar N[\bar{c}]$, or replaces the source or open trace model before scalar reduction.

Proof. The filtration $F^\bullet$ is the naive scalar-contact filtration: its associated graded keeps the leading scalar contact term and discards higher scalar-contact corrections. The first test is the bracket of the two linear Hamiltonian boundary insertions. By Theorem E.12,

$$\{I_\partial(a,f), I_\partial(b,g)\}_{\mathrm{open},\hbar} - I_\partial(ab,\{f,g\}_{\bar{A}}) = \hbar N\, \omega(f,g)\int_I a(t)b(t)\,dt$$

after insertion of $\gamma_{\mathbf{1}}$. Applying the scalar-symbol lift gives the displayed representative of $d_{\mathrm{CE}}s - sd_{\mathrm{QME}}$. For $f = z_1$ and $g = z_2$, $\omega(z_1,z_2) = 1$.

If this first defect were a Hom coboundary, evaluation on the Hamiltonian two-input subcomplex would give a linear primitive $\bar{\eta} : \bar{A} \to \mathbb{C}\gamma_{\mathbf{1}}$ with $d_{\mathrm{CE}}\bar{\eta} = \hbar N\, \omega\gamma_{\mathbf{1}}$. Pulling $\bar{\eta}$ back to $A = \mathbb{C}[z_1,z_2]$ and forcing it to vanish on the constant line contradicts Lemma E.11, equivalently $\omega(z_1,z_2) = 1$ while $\{z_1,z_2\} = 1$ is zero in $\bar{A}$. Thus $o^{(1)}_{\sigma,w}$ is nonzero.

The obstruction is detected in the finite window containing z_1 and z_2. The admissible weight transports of Theorem 3.96 identify this finite-window class, so changing w does not affect its vanishing. The primitive exists on the unreduced source A, where $\eta(f) = -[f]_0$, and the central-character replacement uses exactly this primitive. In the balanced supertrace branch the displayed ordinary-trace evaluation is multiplied by $\mathrm{sdim}\,\mathbb{C}^{N|N} = 0$. The scalar-contact habitat and chain projection constructed in Proposition E.16 kill this obstruction on the balanced scalar-contact branch. Extension to all local functionals remains the separate non-scalar QME obstruction problem. □

Theorem E.6 *(Polynomial unreduced centrality no-go).* Let M_{pol} be any Q-stable subcomplex of the unreduced polynomial open BV observables in which every element has finite polynomial degree in ϕ_1, ϕ_2, ψ. There is no assignment

$$\rho_{a,b}[1] \longmapsto \Theta_{a,b} \in H^\bullet(M_{\mathrm{pol}}, Q)$$

and no system of centrality homotopies in M_{pol} whose induced Q-cohomology action realizes the principal-part module $\mathcal{P}$. Equivalently, the reduced map $\rho_{a,b}[1] \mapsto \Theta_{a,b}$ cannot be lifted to unreduced polynomial boundary representatives with homotopies satisfying

$$\{O_f, \Theta_\rho\} \simeq \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\} \simeq 0$$

for all polynomial Hamiltonians f.

Proof. The linear Hamiltonians act on polynomial open observables by the canonical scalar-translation derivations

$$O_{1,0} \mapsto \partial_{\phi_2}, \qquad O_{0,1} \mapsto -\partial_{\phi_1}.$$

These derivations commute with Q, since translating ϕ_1 or ϕ_2 by a scalar matrix leaves the moment-map equation $Q\psi = [\phi_1, \phi_2]$ unchanged. On every finite polynomial observable, repeated application of either derivation eventually vanishes. Hence their induced actions on $H^\bullet(\mathcal{M}_{\mathrm{pol}}, Q)$ are locally nilpotent.

On the principal-part module the same two Hamiltonians act by

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Iterating gives

$$z_1^r \cdot \rho_{a,b} = (-1)^r (b+1)\cdots(b+r)\rho_{a,b+r} \neq 0,$$

and similarly for z_2, for every $r \geq 1$. Thus no nonzero principal part is locally nilpotent for the linear Hamiltonian action.

If the stated homotopies existed in $\mathcal{M}_{\mathrm{pol}}$, then after passing to Q-cohomology the action of $O_{1,0}$ and $O_{0,1}$ on the classes $[\Theta_{a,b}]$ would be isomorphic to the displayed principal-part action. Local nilpotence is invariant under module isomorphism and under passage to a submodule. This contradicts the previous paragraph. Therefore the unreduced lift cannot live in the polynomial open BV/Koszul category. □

LEMMA E.7 *(Distributional locality of principal-part currents).* Let $I \subset \mathbb{R}$, $a \in C_c^\infty(I)$, and $B \in \mathcal{D}_c'(I)$. The product

$$(aB)(\varphi) = B(a\varphi), \qquad \varphi \in C_c^\infty(I),$$

is a compactly supported distribution with $\mathrm{supp}(aB) \subset \mathrm{supp}(a) \cap \mathrm{supp}(B)$. Consequently the reduced principal-part current brackets

$$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\} = 0$$

are distributionally local on the brane line. If the supports are disjoint, the brackets vanish.

Proof. Multiplication of a compactly supported distribution by a smooth function is the standard operation $B \mapsto aB$ defined by precomposition with multiplication by a. If φ is supported away from $\mathrm{supp}(a) \cap \mathrm{supp}(B)$, then $a\varphi$ is supported away from $\mathrm{supp}(B)$, hence $B(a\varphi) = 0$. This proves the support assertion. The first bracket is the negative transpose of the Hamiltonian action under the residue-current pairing, so its interval factor is exactly aB. The second bracket is zero because the principal-part current ideal is abelian. Disjoint-support vanishing follows from $aB = 0$. □

Definition E.8 *(Central-character principal-part current algebra).* Let $\mathcal{A} = \mathbb{C}[z_1, z_2]$, $\bar{\mathcal{A}} = \mathcal{A}/\mathbb{C}\cdot 1$, and let $\chi\colon \mathcal{A} \to \mathbb{C}$ be a linear central character on the constant Hamiltonian line. For an interval $I \subset \mathbb{R}$, write

$$\widetilde{\mathfrak{h}}_I = \Omega_c^\bullet(I)\widehat{\otimes}\mathcal{A}, \qquad \mathcal{P}_I = \mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P}.$$

The algebraic current target with central character is the completed graded-commutative algebra generated by symbols

$$B_f^\chi(a), \qquad \Theta_\rho(B), \qquad a \in \Omega_c^0(I),\ f \in \mathcal{A},\ B \in \mathcal{D}_c'(I),\ \rho \in \mathcal{P},$$

modulo

$$B^{\chi}_{f+c}(a) = B^{\chi}_{f}(a) + \chi(c \cdot 1) \int_I a(t)\, dt, \qquad \Theta_{\rho+\sigma}(B) = \Theta_{\rho}(B) + \Theta_{\sigma}(B),$$

and with P_0 brackets

$$\{B^{\chi}_{f}(a), B^{\chi}_{g}(b)\} = B^{\chi}_{\{f,g\}}(ab),$$
$$\{B^{\chi}_{f}(a), \Theta_{\rho}(B)\} = \Theta_{f\cdot\rho}(aB),$$
$$\{\Theta_{\rho}(B), \Theta_{\sigma}(C)\} = 0.$$

Constants act trivially on $\mathcal{P}$. The product is the ordinary graded-commutative factorization product after extension by zero on intervals. This is an algebraic current model; it is not the finite polynomial open BV/Koszul complex of Theorem E.6.

Construction E.9 *(Algebraic current presentation map).* Let

$$\widetilde{\mathfrak{g}}_I = \widetilde{\mathfrak{h}}_I \ltimes \mathcal{P}_I[1]$$

with bracket induced by the Hamiltonian bracket on A, the coadjoint action on $\mathcal{P}$, and the abelian bracket on $\mathcal{P}_I[1]$. In the CE algebra $C^{\bullet}_{\mathrm{CE,cont}}(\widetilde{\mathfrak{g}}_I)$, write

$$c_{B,\rho}$$

for the degree-one CE coordinate dual to Hamiltonian currents through the pairing

$$\langle B \otimes \rho, a(t)dt \otimes f\rangle = B(a)\rho(f),$$

and write

$$u_{a(t)dt\otimes f}$$

for the degree-zero CE coordinate dual to the shifted cotangent generator under the same pairing with $a(t)dt \otimes f \in \widetilde{\mathfrak{h}}_I$. Define

$$\Phi^{\chi}_{I}(c_{B,\rho}) = \Theta_{\rho}(B), \qquad \Phi^{\chi}_{I}(u_{a(t)dt\otimes f}) = B^{\chi}_{f}(a),$$

and extend continuously and multiplicatively.

Theorem E.10 *(Algebraic current presentation with zero homotopies).* The maps Φ^{χ}_{I} of Construction E.9 define a morphism of P_0 factorization algebras from the interval CE source of $\widetilde{\mathfrak{g}}_I$ to the algebraic current target of Definition E.8. The factorization centrality homotopies and the P_0-bracket compatibility homotopies in this algebraic model are strict:

$$H_{\mathrm{fact}} = 0, \qquad H_{P_0} = 0.$$

Thus the unreduced source with central character constructs the P_0 current presentation as far as the distributional algebraic current model permits. This is not constructed as a morphism into $Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}_{\mathrm{open}})$ for the genuine unreduced open BV factorization algebra. It does not contradict the polynomial no-go theorem: the image of $\Theta_{\rho}(B)$ lives in the added principal-part current sector, not in finite polynomial open observables.

Proof. The CE differential of $C^{\bullet}_{\mathrm{CE,cont}}(\widetilde{\mathfrak{g}}_I)$ is dual to exactly three brackets:

$$[\widetilde{\mathfrak{h}}_I, \widetilde{\mathfrak{h}}_I], \qquad \widetilde{\mathfrak{h}}_I \curvearrowright \mathcal{P}_I, \qquad [\mathcal{P}_I[1], \mathcal{P}_I[1]] = 0.$$

Under Φ^{χ}_I, these three structure maps become the three displayed P_0 brackets of Definition E.8. Hence the CE differential and the target Schouten differential agree on generators, while the degree-zero current P_0 brackets agree by the same source brackets. The biderivation property extends the equality to the completed symmetric algebra.

Factorization compatibility is also strict. Extension by zero sends a, B to their extensions as test functions and currents; the products ab and aB are computed after extension. If the supports are disjoint these products vanish, so mixed brackets vanish before any homotopy is introduced. Multiplication in the target is graded-commutative, hence multiplication by an image generator is a Hochschild central operation with zero product-centrality homotopy. The P_0-centrality defect on generators is the difference between the target bracket and the image of the corresponding source bracket; it is zero by the previous paragraph. Therefore H_{fact} and H_{P_0} may both be chosen to be zero.

The same statement holds against arbitrary monomials in the added algebraic current target, not only against length-one generators. Let $Y = Y_1 \cdots Y_r$ be a homogeneous source monomial and put $W = \Phi^{\chi}_I(Y)$, a monomial in the target generators $B^{\chi}_{\mathfrak{g}}(b)$ and $\Theta_\sigma(C)$. For a source generator x, the product-centrality defect is zero because the target product is graded-commutative:

$$\Phi^{\chi}_I(x)W - (-1)^{|\Phi^{\chi}_I(x)||W|} W\Phi^{\chi}_I(x) = 0.$$

Let δ_x denote the derivation of the source symmetric algebra induced by the corresponding bracket in $\widetilde{\mathfrak{g}}_I$. The bracket-compatibility defect

$$D_x(Y) = \{\Phi^{\chi}_I(x), \Phi^{\chi}_I(Y)\} - \Phi^{\chi}_I(\delta_x Y)$$

is a graded derivation in Y. It vanishes when $\Phi^{\chi}_I(Y)$ is $B^{\chi}_{\mathfrak{g}}(b)$ or $\Theta_\sigma(C)$ by the three displayed brackets. Hence the Leibniz rule gives $D_x(Y) = 0$ for every monomial and, by continuity, for every element of the completed algebra. Thus the centrality homotopies against arbitrary algebraic current-target monomials are strict zero homotopies in this model. Here "arbitrary" means arbitrary monomial in the freely adjoined algebraic current target. It does not mean arbitrary element of the genuine unreduced open BV observable factorization algebra, and it does not supply the unreduced $Z^{P_0}_{\mathrm{fact}}$-morphism. □

LEMMA E.11 *(Unreduced scalar primitive).* On the local unreduced Hamiltonian algebra $A = \mathbb{C}[z_1, z_2]$, the constant-coefficient cocycle

$$\omega(f, g) = [\{f, g\}_{\Omega_{\mathrm{loc}}}]_0$$

is the CE coboundary of the linear functional

$$\eta(f) = -[f]_0.$$

With the convention $(d_{\mathrm{CE}}\eta)(f, g) = -\eta(\{f, g\})$, one has $d_{\mathrm{CE}}\eta = \omega$. The primitive does not descend to $\bar{A} = A/\mathbb{C} \cdot 1$.

Proof. The identity is immediate:

$$(d_{\mathrm{CE}}\eta)(f, g) = -\eta(\{f, g\}_{\Omega_{\mathrm{loc}}}) = [\{f, g\}_{\Omega_{\mathrm{loc}}}]_0 = \omega(f, g).$$

If η descended to $\bar{A}$, then it would vanish on constants. But $\eta(1) = -1$. Equivalently $\omega(z_1, z_2) = 1$ while $\{z_1, z_2\} = 1$ is zero in the scalar-reduced Hamiltonian quotient. □

Theorem E.12 *(Local scalar contact QME obstruction).* Let

$$\omega(f,g) = [\{f,g\}_{\Omega_{\mathrm{loc}}}]_0$$

be the formal-local Hamiltonian scalar cocycle of Proposition E.2. On monomials,

$$\omega(z_1^k z_2^l, z_1^m z_2^n) = \begin{cases} kn - lm, & (k+m, l+n) = (1,1), \\ 0, & \text{otherwise.} \end{cases}$$

For compactly supported $a, b \in \Omega_c^0(\mathbb{R}_{\mathrm{brane}})$, the unreduced scalar boundary contact in the local mixed HT brane-defect model is

$$\{I_\partial(a,f), I_\partial(b,g)\}_{\mathrm{open}} = I_\partial(ab, \{f,g\}_{\bar{A}}) + N\,\omega(f,g)\int_{\mathbb{R}} a(t)b(t)\,dt.$$

In the quantum Weyl normalization it is equivalently represented by

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\ \mathrm{Tr}(A) \quad \mod\ \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N).$$

Equivalently,

$$\mathrm{Tr}(A[X,Y]) = \hbar N\ \mathrm{Tr}(A) \quad \mod\ \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N),$$

for the matrix Weyl commutator $[X,Y] = XY - YX$. After insertion of the central boundary ghost γ_1, $|\gamma_1| = 1$, the degree-one obstruction representative is

$$\mathrm{Ob}_{\mathrm{sc}}(\gamma_1; a, f; b, g) = \hbar N\,\omega(f,g)\int_{\mathbb{R}} a(t)b(t)\gamma_1(t)\,dt.$$

Its associated graded CE class is the formal scalar multiple $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]]$. This class is not exact in the scalar-reduced Hamiltonian source. Hence the ordinary scalar-reduced $\mathfrak{gl}_N$ brane-defect model has no internal scalar counterterm whose leading symbol cancels this obstruction.

Proof. The monomial formula for ω follows by applying

$$\{z_1^k z_2^l, z_1^m z_2^n\} = (kn - lm) z_1^{k+m-1} z_2^{l+n-1}$$

and then keeping only the constant Taylor coefficient. The first displayed formula is Corollary 3.243. The second is the quantum version of the same central term: the Weyl convention gives $[X^i{}_j, Y^k{}_l] = \hbar\delta^i_l\delta^k_j$, and the normal ordered moment relation is $Y_\hbar X_\hbar - X_\hbar Y_\hbar + \hbar N I \equiv 0$. Hence $XY - YX = \hbar N I$ in the quantum Hamiltonian reduction, and tracing against A gives the stated sign. The matrix-commutator form is the same identity written as $\mathrm{Tr}(A(XY - YX))$.

The scalar contact itself has degree 0. Multiplication by the central ghost γ_1 makes the QME obstruction representative degree 1, the degree of the obstruction complex. The CE associated graded of this representative is the scalar extension cocycle $\hbar N\omega$, hence the formal class $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]]$.

If it were exact in the scalar-reduced Hamiltonian source, the leading scalar term would give a linear primitive $\bar{\eta} : \bar{A} \to \mathbb{C}$ with $\partial\bar{\eta} = \omega$. Pulling $\bar{\eta}$ back to $\mathbb{C}[z_1, z_2]$ and setting the value on 1 to zero would contradict the computation

$$\omega(z_1, z_2) = 1, \qquad \{z_1, z_2\} = 1.$$

Equivalently, Lemma E.11 gives the unreduced scalar primitive used here. It lives on the unreduced polynomial Hamiltonian algebra and does not descend to $\bar{A}$. Thus the scalar-reduced source has no counterterm whose associated graded kills the class. □

THEOREM E.13 *(Central-character source replacement for the scalar QME symbol).* Let $A = \mathbb{C}[z_1, z_2]$ and let $\bar{f}$ denote the image of f in $\bar{A} = A/\mathbb{C} \cdot 1$ for the local Hamiltonian algebra of Proposition E.2. Define the ordinary trace central character

$$\chi_N(f) = N[f]_0$$

and the unreduced boundary generator

$$I_\partial^\chi(a, f) = I_\partial(a, \bar{f}) + \chi_N(f) \int_{\mathbb{R}} a(t)\, dt.$$

Then, for all $a, b \in C_c^\infty(\mathbb{R})$ and all $f, g \in A$,

$$\{I_\partial^\chi(a, f), I_\partial^\chi(b, g)\}_{\text{open}} = I_\partial^\chi(ab, \{f, g\}).$$

In the quantum Weyl normalization the scalar part is likewise absorbed by the same central character:

$$J_N^\chi(1) = N, \qquad \operatorname{Tr}(A[X, Y]) = J_N^\chi([z_1, z_2]_*) \operatorname{Tr}(A) = \hbar J_N^\chi(1) \operatorname{Tr}(A).$$

Thus the local scalar QME symbol is absorbed after replacing the scalar-reduced source by the unreduced source with central character χ_N. This is a source replacement before scalar reduction, not a null-homotopy inside the original scalar-reduced QME complex. Equivalently, if $\eta(f) = -[f]_0$, then

$$d_{\text{CE}}\eta = \omega, \qquad \chi_N(f) = N[f]_0 = -N\eta(f),$$

so

$$d_{\text{CE}}\chi_N = -N\omega, \qquad \hbar N\omega + d_{\text{CE}}(\hbar\chi_N) = 0$$

in the unreduced source.

Proof. Decompose the Poisson bracket into its scalar and reduced parts:

$$\{f, g\} = \overline{\{f, g\}} + \omega(f, g) \cdot 1.$$

By Theorem E.12,

$$\{I_\partial(a, \bar{f}), I_\partial(b, \bar{g})\}_{\text{open}} = I_\partial(ab, \overline{\{f, g\}}) + N\omega(f, g) \int_{\mathbb{R}} a(t)b(t)\, dt.$$

Since the central-character summand is scalar, it has zero open bracket with every boundary field. Therefore the right hand side is exactly

$$I_\partial(ab, \overline{\{f, g\}}) + \chi_N(\{f, g\}) \int_{\mathbb{R}} a(t)b(t)\, dt = I_\partial^\chi(ab, \{f, g\}).$$

The quantum formula is the same computation in the Weyl algebra: $[z_1, z_2]_* = \hbar$, $J_N^\chi(1) = N$, and the matrix moment relation gives $\operatorname{Tr}(A[X, Y]) = \hbar N \operatorname{Tr}(A)$. Hence the scalar Weyl commutator is the image of the unreduced Moyal scalar instead of an obstruction class.

This is the normal-ordering mechanism. The relation $YX - XY + \hbar NI \equiv 0$ says that an XY-to-YX interchange contributes the scalar $\hbar NI$. In the scalar-reduced source this scalar is invisible and therefore appears as Ob_{sc}. In the unreduced source with $J_N^\chi(1) = N$, the same scalar is $J_N^\chi([z_1, z_2]_*) = \hbar N$, so it is absorbed before forming the QME obstruction class. □

Theorem E.14 *(Local $\mathfrak{gl}(N|N)$ supertrace scalar QME cancellation).* Let $V = \mathbb{C}^{N|N}$, replace the ordinary trace by the supertrace, and use the canonical even Weyl relations on $\mathrm{End}(V)$ with the supertrace pairing in the same formal-local model of Proposition E.2. Put

$$\mathrm{sdim}\, V = \mathrm{Str}(\mathrm{id}_V) = 0.$$

The classical scalar contact term is

$$\{I_\partial^{\mathrm{str}}(a,f), I_\partial^{\mathrm{str}}(b,g)\}_{\mathrm{open}} = I_\partial^{\mathrm{str}}(ab, \{f,g\}_{\bar{A}}) + \mathrm{sdim}(V)\, \omega(f,g) \int_{\mathbb{R}} a(t)b(t)\, dt,$$

and hence has zero scalar component for $V = \mathbb{C}^{N|N}$. The quantum scalar commutator is

$$\mathrm{Str}(A[X,Y]) = \hbar\, \mathrm{sdim}(V)\, \mathrm{Str}(A) = 0 \mod \mathcal{W}_{\mathrm{End}(V)}\widehat{\mu}(\mathfrak{gl}(V)).$$

Consequently the scalar part of the local mixed brane-defect QME vanishes in the balanced $\mathfrak{gl}(N|N)$ supertrace model without adding a central-character counterterm.

Proof. The scalar term in the classical boundary bracket is the contraction of the identity endomorphism in the trace pairing. For an ordinary N-dimensional vector space this contraction is N. For a super vector space it is $\mathrm{Str}(\mathrm{id}_V) = \mathrm{sdim}\, V$. The same local calculation as in Theorem E.12 therefore gives the displayed classical formula. For $V = \mathbb{C}^{N|N}$, this coefficient is zero.

In coordinates, the super-Weyl contraction carries the Koszul sign of the supertrace pairing. Summing the diagonal contraction gives

$$YX - XY + \hbar\, \mathrm{sdim}(V) I \equiv 0 \pmod{\mathcal{W}_{\mathrm{End}(V)}\widehat{\mu}(\mathfrak{gl}(V))}.$$

Thus

$$\mathrm{Str}(A[X,Y]) = \hbar\, \mathrm{sdim}(V)\, \mathrm{Str}(A),$$

which is zero for $V = \mathbb{C}^{N|N}$. These are precisely the scalar terms entering the degree-one representative $\mathrm{Ob}_{\mathrm{sc}}$; hence that representative is zero in the balanced supertrace theory. □

Definition E.15 *(Balanced scalar-contact habitat).* Let $V = \mathbb{C}^{N|N}$. A balanced scalar-contact habitat over an interval $I \subset \mathbb{R}_{\mathrm{brane}}$ is a complete filtered subcomplex

$$\mathfrak{H}^\bullet_{\mathrm{bal,sc}}(I) \subset \mathfrak{Q}^\bullet_{w,\partial,\hbar,\mathrm{str}}(I)$$

of the local supertrace brane-defect local-functional complex, with differential d_{QME}, satisfying the following conditions. First, the filtration $F^\bullet$ is the closed scalar-contact filtration generated by identity-endomorphism contractions in $\mathrm{End}(V)$. Second, the associated-graded scalar-contact extractor factors through the superdimension:

$$\sigma_{\mathrm{bal,sc}} = \mathrm{sdim}(V)\, \tau_{\mathrm{sc}} \colon \mathrm{gr}_F\, \mathfrak{H}^\bullet_{\mathrm{bal,sc}}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]]$$

for a continuous $\mathbb{C}[[\hbar]]$-linear extractor τ_{sc}. Third, the same factorization holds after applying d_{QME}: no scalar-contact symbol is present in this habitat except those obtained by contracting a single identity endomorphism in the supertrace pairing.

PROPOSITION E.16 *(Balanced scalar-contact projection).* On every balanced scalar-contact habitat of Definition E.15 with $V = \mathbb{C}^{N|N}$, the scalar-symbol map is the zero associated-graded map, and it admits the filtered chain projection

$$\widehat{\sigma}_{\mathrm{bal,sc}} = 0 \colon \mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I) \longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{\mathbf{1}})[1][[\hbar]].$$

Its chain-map defect is

$$\delta_{\mathrm{bal}} = d_{\mathrm{CE}}\widehat{\sigma}_{\mathrm{bal,sc}} - \widehat{\sigma}_{\mathrm{bal,sc}} d_{\mathrm{QME}} = 0.$$

Hence all scalar-projection obstruction classes $o^{(r)}_{\sigma,w,\mathrm{bal}}(I)$ vanish on this habitat. In particular the first two-Hamiltonian-input defect is

$$\mathrm{ev}_{f,g}\big(o^{(1)}_{\sigma,w,\mathrm{bal}}(I)\big) = \hbar\ \mathrm{sdim}(V)\ \omega(f,g)\gamma_{\mathbf{1}} = 0.$$

Proof. By definition of the habitat, every scalar-contact symbol factors as the ordinary scalar contact multiplied by the contraction of the identity endomorphism in the supertrace pairing. That contraction is

$$\mathrm{Str}(\mathrm{id}_V) = \mathrm{sdim}(V) = N - N = 0.$$

Therefore $\sigma_{\mathrm{bal,sc}} = \mathrm{sdim}(V)\tau_{\mathrm{sc}} = 0$ on the associated graded. We choose the zero filtered lift to the stated scalar CE target. Since both composites $d_{\mathrm{CE}}\widehat{\sigma}_{\mathrm{bal,sc}}$ and $\widehat{\sigma}_{\mathrm{bal,sc}} d_{\mathrm{QME}}$ are zero, the chain-map defect is zero. Lemma E.4 then gives $o^{(r)}_{\sigma,w,\mathrm{bal}}(I) = 0$ for every $r > 0$.

Evaluating on two Hamiltonian boundary insertions gives the same local contraction as Theorem E.14: the ordinary coefficient N in $\hbar N\ \omega(f,g)\gamma_{\mathbf{1}}$ is replaced by $\mathrm{sdim}(V)$. Thus

$$\hbar\ \mathrm{sdim}(V)\ \omega(f,g)\gamma_{\mathbf{1}} = 0,$$

including the test $(f,g) = (z_1, z_2)$. □

Remark E.17 *(Extension obstruction outside the balanced habitat).* The preceding proposition is a positive scalar-contact projection theorem, not a full non-scalar Costello graph theorem. If one enlarges $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$ to a complete filtered subcomplex $\mathfrak{H}'$ carrying scalar symbols not known to factor through $\mathrm{sdim}(V)$, the obstruction is exactly the first nonzero homogeneous component of

$$d_{\mathrm{CE}}s - s d_{\mathrm{QME}} \in \mathrm{Hom}\big(\mathrm{gr}_F\, \mathfrak{H}', C^{\bullet}_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{\mathbf{1}})[1][[\hbar]]\big)$$

for a filtered scalar-symbol lift s. Its cohomology class is the $o^{(r)}_{\sigma}$-class of Lemma E.4. Thus an enlargement is admitted exactly when these classes vanish successively, with compatible finite-window choices.

Definition E.18 *(Balanced Costello graph/counterterm subhabitat).* Fix a Hamiltonian brane-defect specialization of the finite-scale Costello graph calculus and a finite-window tower for the formal-local mixed HT theory $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_0$. A Costello graph/counterterm enlargement

$$\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I) \subset \mathfrak{G}^{\bullet}_{\mathrm{bal,Cost}}(I) \subset \mathfrak{Q}^{\bullet}_{w,\partial,\hbar,\mathrm{str}}(I)$$

is a balanced graph subhabitat if it is a complete d_{QME}-stable filtered subcomplex generated, over $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$, by finite-window Costello graph contributions $G_{\Gamma,L,M}$ and local counterterm representatives $C_{\Gamma,L,M}$ such that every scalar-contact associated-graded term is an identity-loop supertrace contraction. Equivalently, for each homogeneous scalar-contact piece,

$$\sigma_{\mathrm{sc}}\big(\mathrm{gr}_F\, G_{\Gamma,L,M}\big) = \mathrm{Str}(\mathrm{id}_V)\ \tau_{\Gamma,L,M}, \qquad \sigma_{\mathrm{sc}}\big(\mathrm{gr}_F\, C_{\Gamma,L,M}\big) = \mathrm{Str}(\mathrm{id}_V)\ \tau^{C}_{\Gamma,L,M},$$

for continuous scalar extractors $\tau_{\Gamma,L,M}$ and $\tau^{C}_{\Gamma,L,M}$, and the same factorization holds after applying d_{QME} to these graph and counterterm generators. The factorization is required to commute with finite-window truncation and admissible weight transport.

Proposition E.19 *(Balanced graph/counterterm preservation).* Let $V = \mathbb{C}^{N|N}$. Every balanced graph subhabitat $\mathfrak{G}^{\bullet}_{\mathrm{bal,Cost}}(I)$ of Definition E.18 is a balanced scalar-contact habitat in the sense of Definition E.15. Hence the zero projection

$$\widehat{\sigma}_{\mathrm{bal,Cost}} = 0\colon \mathfrak{G}^{\bullet}_{\mathrm{bal,Cost}}(I) \longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]]$$

is a filtered chain map, and all scalar-projection obstruction classes vanish on this specified graph/counterterm subcomplex.

Proof. By Definition E.18, each new graph or counterterm scalar symbol is obtained by closing an identity endomorphism loop in the supertrace pairing. Its coefficient is therefore

$$\mathrm{Str}(\mathrm{id}_V) = \mathrm{sdim}(V) = N - N = 0.$$

Thus the associated-graded scalar extractor is zero on the graph and counterterm generators, and it is already zero on $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$ by Proposition E.16. The same factorization after applying d_{QME} says that the differential does not create a new scalar-contact symbol outside the identity-loop class. Therefore

$$d_{\mathrm{CE}}\widehat{\sigma}_{\mathrm{bal,Cost}} - \widehat{\sigma}_{\mathrm{bal,Cost}}d_{\mathrm{QME}} = 0.$$

The finite-window and weight-transport compatibility in the definition gives the same zero class in the completed inverse limit. Hence the scalar-contact projection obstruction tower of Lemma E.4 vanishes on $\mathfrak{G}^{\bullet}_{\mathrm{bal,Cost}}(I)$. □

Theorem E.20 *(Graph-extension scalar obstruction).* Let

$$\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I) \subset \mathfrak{G}^{\bullet}_{\mathrm{Cost}}(I) \subset \mathfrak{Q}^{\bullet}_{w,\partial,\hbar,\mathrm{str}}(I)$$

be any complete d_{QME}-stable filtered Costello graph/counterterm enlargement. Put

$$\overline{\mathfrak{G}}^{\bullet}_{\mathrm{Cost}}(I) = \mathfrak{G}^{\bullet}_{\mathrm{Cost}}(I)/\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I), \qquad S^{\bullet} = C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]].$$

Let

$$\bar{\sigma}_{\mathrm{Cost}}\colon \mathrm{gr}_F\,\overline{\mathfrak{G}}^{\bullet}_{\mathrm{Cost}}(I) \longrightarrow S^{\bullet}$$

be the associated-graded scalar-symbol extractor of the added graph/counterterm terms. The zeroth obstruction to extending the balanced scalar-contact projection to $\mathfrak{G}^{\bullet}_{\mathrm{Cost}}(I)$ is the actual associated-graded chain-map defect

$$\delta^{(0)}_{\mathrm{Cost},\sigma}(I) = d_{\mathrm{CE}}\bar{\sigma}_{\mathrm{Cost}} - \bar{\sigma}_{\mathrm{Cost}}\,d_{\mathrm{gr}} \in \mathrm{Hom}^{1}_{0}\big(\mathrm{gr}_F\,\overline{\mathfrak{G}}^{\bullet}_{\mathrm{Cost}}(I), S^{\bullet}\big).$$

If this defect vanishes, choose a filtered lift s of $\bar{\sigma}_{\mathrm{Cost}}$, extended by zero on $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$. If the first nonzero homogeneous component of

$$\delta_s = d_{\mathrm{CE}}s - sd_{\mathrm{QME}}$$

lowers the scalar-contact filtration by $r > 0$, then it defines

$$\mathfrak{o}^{(r)}_{\mathrm{Cost},\sigma}(I) = \left[\delta_s^{(-r)}\right] \in H^1\Big(\mathrm{Hom}_{-r}\big(\mathrm{gr}_F\, \overline{\mathfrak{G}}^{\bullet}_{\mathrm{Cost}}(I), S^{\bullet}\big)\Big).$$

These classes are independent of the chosen lift up to the usual Hom coboundary. A filtered chain projection extending the balanced projection exists exactly when

$$\delta^{(0)}_{\mathrm{Cost},\sigma}(I) = 0, \qquad \mathfrak{o}^{(r)}_{\mathrm{Cost},\sigma}(I) = 0 \quad (r > 0)$$

successively, with compatible finite-window representatives.

Explicitly, if $X_{\Gamma,M}$ is an added graph contribution or counterterm representative and, modulo F^{p-r+1},

$$d_{\mathrm{QME}} X_{\Gamma,M} = \sum_{\Gamma'} \varepsilon_{\Gamma,\Gamma'} X_{\Gamma',M}, \qquad X_{\Gamma,M} \in F^p,$$

with the signs $\varepsilon_{\Gamma,\Gamma'}$ determined by the graph orientation, BV bracket, and counterterm convention, then the obstruction cocycle is computed by

$$\mathfrak{o}^{(r)}_{\mathrm{Cost},\sigma}(X_{\Gamma,M}) = \left[d_{\mathrm{CE}} s(X_{\Gamma,M}) - \sum_{\Gamma'} \varepsilon_{\Gamma,\Gamma'} s(X_{\Gamma',M})\right]_{F^{p-r}/F^{p-r+1}}.$$

Thus a graph or counterterm term outside the identity-loop superdimension class is not admitted by the balanced theorem unless this displayed relative Hom class is killed by an explicit compatible counterterm correction.

Proof. The quotient by $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$ makes the statement relative to the already proved zero projection on the balanced scalar-contact habitat. If $d_{\mathrm{CE}}\bar{\sigma}_{\mathrm{Cost}} - \bar{\sigma}_{\mathrm{Cost}} d_{\mathrm{gr}}$ is nonzero, then the associated-graded scalar extractor is not a chain map; no filtered chain lift can have it as associated graded. This is the zeroth defect $\delta^{(0)}_{\mathrm{Cost},\sigma}(I)$.

Once the associated-graded defect is killed, the proof is the relative version of Lemma E.4. The commutator $d_{\mathrm{CE}} s - s d_{\mathrm{QME}}$ strictly lowers filtration. Its first nonzero homogeneous piece is closed for the Hom differential because both differentials square to zero and because the lower pieces have already been removed. Replacing s by $s - \alpha$, with α lowering filtration by the same amount and vanishing on $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$, changes this component by the Hom coboundary $D\alpha$. Therefore the class is independent of choices and vanishes exactly when the next correction to s exists. Completeness of the filtration and compatibility of the finite-window tower give the stated successive criterion.

The displayed graph formula is this same chain-defect evaluated on a homogeneous graph or counterterm generator, with the actual d_{QME}-boundary written as the signed sum of its graph and counterterm boundary terms. Hence it is computable from the supplied Costello graph differential, orientation signs, and scalar symbols. If all added terms satisfy Definition E.18, then $s = 0$ is the required lift for $V = \mathbb{C}^{N|N}$, and all the displayed classes vanish, recovering Proposition E.19. □

PROPOSITION E.21 *(First non-identity-loop scalar symbol).* Let $V = \mathbb{C}^{N|N}$. Fix a finite local Hamiltonian window M containing the polynomial Hamiltonian inputs under discussion, and let $T_\bullet = (T_1, \dots, T_m)$ be an ordered word of even $\mathrm{End}(V)$-labels carried by one open index loop. Write $M(T_\bullet) = T_1 \cdots T_m$, with the empty product equal to id_V. Let $\Gamma^{T_\bullet}_{1,M}$ be the connected local one-cross-contraction boundary graph with two Hamiltonian external legs (a, f) and (b, g), one Weyl edge, and

this marked scalar loop. Denote the degree-one scalar-contact representative obtained after inserting γ_1 by $X_{\Gamma^{T_\bullet}_{1,M}}(a, f; b, g)$.

Then $X_{\Gamma^{T_\bullet}_{1,M}}(a, f; b, g) \in F^1$, its leading term is not in F^2 unless the displayed coefficient vanishes, and

$$X_{\Gamma^{T_\bullet}_{1,M}}(a, f; b, g) = \hbar\, \mathrm{Str}\big(M(T_\bullet)\big)\, \omega(f, g) \int_I a(t)b(t)\gamma_1(t)\, dt \mod F^2.$$

Equivalently,

$$\sigma_{\mathrm{sc}}\Big(\mathrm{gr}_F\, X_{\Gamma^{T_\bullet}_{1,M}}(a, f; b, g)\Big) = \hbar\, \mathrm{Str}\big(M(T_\bullet)\big)\, \omega(f, g)\gamma_1.$$

For monomials,

$$\omega(z_1^k z_2^l, z_1^m z_2^n) = \begin{cases} kn - lm, & (k + m, l + n) = (1, 1), \\ 0, & \text{otherwise}, \end{cases}$$

so the first non-identity test is obtained by a single marker T:

$$\sigma_{\mathrm{sc}}\Big(\mathrm{gr}_F\, X_{\Gamma^{T}_{1,M}}(1, z_1; 1, z_2)\Big) = \hbar\, \mathrm{Str}(T)\gamma_1.$$

In particular, if P_+ is projection onto one even basis line, then

$$\sigma_{\mathrm{sc}}\Big(\mathrm{gr}_F\, X_{\Gamma^{P_+}_{1,M}}(1, z_1; 1, z_2)\Big) = \hbar\, \gamma_1 \neq 0.$$

Thus this is the first scalar symbol not killed by the identity-loop computation. Its associated CE class is

$$\hbar\, \mathrm{Str}\big(M(T_\bullet)\big)\, [\bar{c}]\, \gamma_1.$$

It is nonzero exactly when $\mathrm{Str}(M(T_\bullet)) \neq 0$. If every scalar-contact loop monodromy word in a complete d_{QME}-stable graph/counterterm subcomplex, and every such word appearing after d_{QME}, has supertrace zero, then the zero scalar projection is again a filtered chain map on that subcomplex.

Proof. The graph has one local scalar contact on the brane line, so it lies in F^1. There is no second scalar contact in the one-edge graph, hence its associated graded is read in F^1/F^2. The boundary Wick normalization fixed in Theorem 3.276 gives the one-edge antisymmetrized coefficient $\hbar P(f, g)$. Taking the scalar Taylor coefficient gives

$$[P(f, g)]_0 = \omega(f, g),$$

with positive sign for the ordered pair (z_1, z_2), since $\{z_1, z_2\} = 1$. Reversing the two external legs changes the sign through $\omega(g, f) = -\omega(f, g)$.

It remains only to compute the open index loop. In a homogeneous basis e_α of V, the closed marked loop contributes

$$\sum_\alpha (-1)^{|e_\alpha|} \big(M(T_\bullet)\big)^\alpha{}_\alpha = \mathrm{Str}\big(M(T_\bullet)\big).$$

For the empty word this is $\mathrm{Str}(\mathrm{id}_V) = \mathrm{sdim}\, V = 0$, which is exactly the identity-loop cancellation of Proposition E.19. For a non-identity marker T the same contraction gives $\mathrm{Str}(T)$. Projection onto one even basis line has supertrace 1, giving the displayed nonzero symbol.

The scalar contact has BV degree 0; multiplication by γ_1, of degree 1, puts the representative in the degree-one QME obstruction complex. The cocycle ω is the same scalar Hamiltonian cocycle as in

Theorem E.12. That theorem proves $[\bar{c}] \neq 0$ on the scalar-reduced Hamiltonian source, so multiplying by the scalar $\hbar\ \mathrm{Str}(M(T_\bullet))$ gives a nonzero class exactly when this supertrace is nonzero.

Finally suppose the stated marked-loop subcomplex has only supertraceless monodromy words, before and after applying d_{QME}. The formula above then gives zero associated scalar symbol on every graph and counterterm generator and also on every associated-graded boundary term. Hence $d_{\mathrm{CE}}0 - 0\, d_{\mathrm{QME}} = 0$, so the zero scalar projection is a filtered chain map on that larger, explicitly delimited subcomplex. □

Definition E.22 *(Supertraceless-monodromy prehabitat).* Let $V = \mathbb{C}^{N|N}$. For an ordered word $T_\bullet = (T_1, \dots, T_m)$ of even $\mathrm{End}(V)$-labels on a scalar open index loop, set

$$\mathrm{st}(T_\bullet) = \mathrm{Str}(T_1 \cdots T_m), \qquad \mathrm{st}(\varnothing) = \mathrm{Str}(\mathrm{id}_V) = 0.$$

The word is supertraceless if $\mathrm{st}(T_\bullet) = 0$. This condition is imposed on the ordered product $T_1 \cdots T_m$, not on the individual letters.

Fix a complete d_{QME}-stable finite-window graph/counterterm enlargement of the local balanced supertrace brane-defect complex. The supertraceless-monodromy prehabitat

$$\mathfrak{G}^{\bullet,\mathrm{pre}}_{\mathrm{st}=0}(I)$$

is the closed filtered span, over $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$, of homogeneous graph/counterterm representatives whose every scalar-contact associated-graded loop word is supertraceless. Let

$$\Pi_{\mathrm{st}\neq 0}$$

denote the quotient of the associated-graded scalar-contact monodromy module by the closed span of supertraceless loop words. The d_{QME}-preservation obstruction is the family of homogeneous quotient classes

$$\mathcal{B}_{\mathrm{st}=0}(X;q) = \Pi_{\mathrm{st}\neq 0}\big([d_{\mathrm{QME}}X]_{F^q/F^{q+1}}\big), \qquad X \in \mathfrak{G}^{\bullet,\mathrm{pre}}_{\mathrm{st}=0}(I),$$

over all filtration degrees q. We write $\mathcal{B}_{\mathrm{st}=0} = 0$ when every class in this family vanishes. If $\mathcal{B}_{\mathrm{st}=0} = 0$, the prehabitat is a complete d_{QME}-stable filtered subcomplex; in that case it is called the supertraceless-monodromy graph/counterterm subhabitat and is denoted $\mathfrak{G}^{\bullet}_{\mathrm{st}=0}(I)$.

THEOREM E.23 *(Supertraceless-monodromy scalar projection).* Assume the notation of Definition E.22. The prehabitat is preserved by d_{QME} if and only if

$$\mathcal{B}_{\mathrm{st}=0} = 0.$$

Under this condition the zero scalar projection

$$\widehat{\sigma}_{\mathrm{st}=0} = 0\colon \mathfrak{G}^{\bullet}_{\mathrm{st}=0}(I) \longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]]$$

is a filtered chain map. Hence all scalar-projection obstruction classes of Lemma E.4 vanish on the supertraceless-monodromy subhabitat.

More explicitly, let $X_{\Gamma,M} \in F^p$ be a homogeneous generator of the prehabitat and suppose that, modulo F^{p-r+1},

$$d_{\mathrm{QME}}X_{\Gamma,M} = \sum_{\Gamma',U_\bullet} \varepsilon_{\Gamma,\Gamma',U_\bullet} X_{\Gamma',U_\bullet,M},$$

where $U_\bullet$ runs over scalar-loop monodromy words appearing in the boundary terms. The first possible failure of preservation is the quotient class

$$\mathcal{B}^{(r)}_{\mathrm{st}=0}(X_{\Gamma,M}) = \left[\sum_{\Gamma',U_\bullet} \varepsilon_{\Gamma,\Gamma',U_\bullet} X_{\Gamma' U_\bullet,M}\right]_{\mathrm{st}\neq 0,\ F^{p-r}/F^{p-r+1}}.$$

Its scalar CE image on a two-Hamiltonian-input scalar loop is

$$\hbar \sum_{\Gamma',U_\bullet} \varepsilon_{\Gamma,\Gamma',U_\bullet} \operatorname{Str}\big(M(U_\bullet)\big)\, \omega(f_{\Gamma'}, g_{\Gamma'})\gamma_1.$$

In cohomology this becomes

$$\hbar \sum_{\Gamma',U_\bullet} \varepsilon_{\Gamma,\Gamma',U_\bullet} \operatorname{Str}\big(M(U_\bullet)\big)\, [\bar{c}]\, \gamma_1.$$

Thus a non-supertraceless boundary word is the exact obstruction to d_{QME}-preservation, and its scalar shadow is precisely the marked-loop class computed in Proposition E.21.

For the unmarked Hamiltonian boundary generators $I^{\mathrm{str}}_\partial(a,f)$ used in Theorem E.14, and for the identity-loop graph generators of Definition E.18, the obstruction $\mathcal{B}_{\mathrm{st}=0}$ vanishes. These generators carry only the empty monodromy word, and the displayed differential $Q + \{I_0, -\}$ contains no fixed noncentral $\mathrm{End}(V)$-marker. Therefore applying d_{QME} to this displayed Hamiltonian generator set cannot create a non-supertraceless marked word. This proves preservation for the Hamiltonian generators treated in this appendix; it does not assert preservation for a larger Costello specialization with additional marked idempotents, holonomies, or counterterm tensors.

Proof. By definition, $\Pi_{\mathrm{st}\neq 0}$ kills exactly the associated-graded scalar-contact monodromy terms whose loop words are supertraceless. Hence $\mathcal{B}_{\mathrm{st}=0} = 0$ says precisely that the d_{QME}-boundary of every generator of the prehabitat has only supertraceless loop words in every associated-graded filtration degree. Since the filtration is complete, this homogeneous condition is equivalent to preservation of the closed filtered span. This proves the preservation criterion.

On the preserved subhabitat every scalar-contact loop word $T_\bullet$ satisfies $\operatorname{Str}(M(T_\bullet)) = 0$. Proposition E.21 computes the associated-graded scalar symbol of such a loop as

$$\hbar\ \operatorname{Str}(M(T_\bullet))\, \omega(f,g)\gamma_1,$$

and this is zero. The same is true after applying d_{QME}, by the preservation criterion. Therefore both composites $d_{\mathrm{CE}}\widehat{\sigma}_{\mathrm{st}=0}$ and $\widehat{\sigma}_{\mathrm{st}=0}d_{\mathrm{QME}}$ are zero, so the zero projection is a filtered chain map. Lemma E.4 then kills the whole scalar-projection obstruction tower on this subhabitat.

The formula for $\mathcal{B}^{(r)}_{\mathrm{st}=0}(X_{\Gamma,M})$ is just the preceding quotient criterion evaluated on a homogeneous generator and then reduced modulo F^{p-r+1}. Applying the one-loop scalar computation of Proposition E.21 to each boundary term gives the displayed scalar CE representative. Passing to cohomology replaces ω by the nonzero class $[\bar{c}]$ from Theorem E.12.

For the unmarked Hamiltonian boundary generators, the open index loop contains no inserted label; its word is $\varnothing$, so its coefficient is $\operatorname{Str}(\mathrm{id}_V) = 0$. The differential $Q + \{I_0, -\}$ appearing in this appendix changes the BV fields and takes brackets with the unmarked classical interaction. It does not introduce an external even endomorphism marker on an open scalar loop. Thus the image of the unmarked Hamiltonian generator set again has only empty loop words. The identity-loop graph subhabitat has the same property by Definition E.18, which requires the same factorization after applying d_{QME}. This proves the

stated preservation result on the displayed generator set and leaves arbitrary marked graph/counterterm tensors governed by $\mathcal{B}_{\mathrm{st}=0}$. □

PROPOSITION E.24 *(Marked $\mathcal{B}_{\mathrm{st}=0}$ and gauge-equivariant markers).* Let $V = \mathbb{C}^{N|N}$, and impose full $\mathfrak{gl}(V)$-gauge equivariance on open Chan-Paton labels. A theorem-grade computation of $\mathcal{B}_{\mathrm{st}=0}$ for a marked Costello graph/counterterm enlargement requires, for every homogeneous generator $X_{\Gamma,\mathcal{M}} \in F^p$, an actual marked boundary expansion

$$d_{\mathrm{QME}} X_{\Gamma,\mathcal{M}} = \sum_{\Gamma',\lambda} \varepsilon_{\Gamma,\Gamma',\lambda}\, X_{\Gamma'\,\mathfrak{m}_{\Gamma,\Gamma',\lambda},\mathcal{M}} \qquad \mathrm{mod}\ F^{p-r+1},$$

where each

$$\mathfrak{m}_{\Gamma,\Gamma',\lambda} \in \left(\mathrm{End}(V)^{\otimes m_\lambda}\right)^{\mathfrak{gl}(V)}$$

is an even invariant marker tensor decorating the scalar open index loop. For $\mathfrak{m} = \sum_\alpha T_{\alpha,1} \otimes \cdots \otimes T_{\alpha,m}$, put

$$\mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}) = \sum_\alpha \mathrm{Str}(T_{\alpha,1} \cdots T_{\alpha,m}).$$

Then the first preservation obstruction is

$$\mathcal{B}^{(r)}_{\mathrm{st}=0}(X_{\Gamma,\mathcal{M}}) = \left[\sum_{\Gamma',\lambda} \varepsilon_{\Gamma,\Gamma',\lambda}\, X_{\Gamma'\,\mathfrak{m}_{\Gamma,\Gamma',\lambda},\mathcal{M}}\right]_{\mathrm{st}\neq 0,\ F^{p-r}/F^{p-r+1}},$$

and its two-Hamiltonian-input scalar shadow is

$$\hbar \sum_{\Gamma',\lambda} \varepsilon_{\Gamma,\Gamma',\lambda}\ \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}_{\Gamma,\Gamma',\lambda})\, \omega(f_{\Gamma'}, g_{\Gamma'})\gamma_{\mathbf{1}}.$$

In cohomology this becomes

$$\hbar \sum_{\Gamma',\lambda} \varepsilon_{\Gamma,\Gamma',\lambda}\ \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}_{\Gamma,\Gamma',\lambda})\, [\bar{c}]\, \gamma_{\mathbf{1}}.$$

On the displayed unmarked Hamiltonian generator set the actual differential is $Q + \{I_0, -\}$, with unmarked classical interaction; the balanced graph/counterterm subhabitat only allows identity-loop scalar contractions before and after d_{QME}. Hence

$$\mathcal{B}_{\mathrm{st}=0} = 0$$

on that displayed surface. For any larger marked Costello specialization not equipped with an oriented finite-window marked boundary complex as in Definition E.26, the missing datum blocking a theorem-grade scalar computation is exactly equation (E.24), including its signs and invariant marker tensors.

In particular, a single fixed marker $T \in \mathrm{End}(V)_{\bar{0}}$ can occur in a full $\mathfrak{gl}(V)$-equivariant marked differential only when $T = \lambda\, \mathrm{id}_V$. Its scalar shadow is then

$$\hbar\, \lambda\ \mathrm{Str}(\mathrm{id}_V)\, \omega(f, g)\gamma_{\mathbf{1}} = 0.$$

Thus the nonzero P_+ test of Proposition E.21 is a genuine obstruction to a non-equivariant or externally marked enlargement, but it is not an allowed single-marker term in the full $\mathfrak{gl}(N|N)$-equivariant Costello differential. If one imposes only the even block-diagonal group, or supplies external marker tensors, the admissible datum is the invariant tensor family above, and the first nonzero scalar shadow is detected exactly by the displayed cyclic-supertrace coefficient.

Proof. The formula for $\mathcal{B}^{(r)}_{\mathrm{st}=0}$ is Definition E.22 applied to the marked boundary expansion equation (E.24). The scalar-shadow formula is the linear extension of Proposition E.21: a decomposable marker word contributes the cyclic supertrace of the ordered product around the open index loop, and an invariant tensor is a finite linear combination of such words. Passing to cohomology replaces ω by $[\bar{c}]$ as in Theorem E.23.

It remains to record the equivariance constraint. A marked local functional with one fixed marker T transforms under $g \in GL(V)$ by replacing the marker with gTg^{-1}. Equivariance of the differential therefore forces $gTg^{-1} = T$, or infinitesimally $[A, T] = 0$ for every $A \in \mathfrak{gl}(V)$. Since T is even, write it in the parity decomposition as $T = T_{\bar{0}} \oplus T_{\bar{1}}$. Commutation with $\mathfrak{gl}(N) \oplus \mathfrak{gl}(N)$ gives $T_{\bar{0}} = \lambda_0 \,\mathrm{id}$ and $T_{\bar{1}} = \lambda_1 \,\mathrm{id}$. Commutation with the odd elementary endomorphisms forces $\lambda_0 = \lambda_1$. Hence $T = \lambda \,\mathrm{id}_V$. The cyclic supertrace is then $\lambda \,\mathrm{sdim}(V) = 0$ for $V = \mathbb{C}^{N|N}$.

The final assertion is bookkeeping, not an additional vanishing theorem. The only displayed d_{QME} in the current Hamiltonian generator calculation is $Q + \{I_0, -\}$ with no external endomorphism label, and Definition E.18 requires identity-loop factorization after applying d_{QME}. These terms have cyclic supertrace $\mathrm{Str}(\mathrm{id}_V) = 0$, so their $\mathcal{B}_{\mathrm{st}=0}$-class vanishes. Any further marked graph/counterterm term is governed by its chosen oriented marked boundary complex; without that data, equation (E.24) remains the residual. □

Definition E.25 *(Finite-window marked Costello list).* Fix a finite-window graph/QME package for the local mixed HT brane-defect theory, $(I, w, M, \mathcal{G}_M)$ in the sense of Definition 3.170. A finite-window marked Costello list

$$\mathcal{G}^{\mathrm{mk}}_M$$

is a finite set of connected brane-defect graphs and local counterterm graphs with the following labels and boundary table.

The admissible vertex labels are exactly the finite local labels visible in the chosen package:

(L1) external CE labels $u_{a(t)dt\otimes f}$ and $c_{B,\rho}$, with f, ρ in the Hamiltonian window M, and with $B \in \mathcal{D}^{\mathrm{reg}}_c(I)$ or $B_\bullet \in \mathcal{D}'_{c,\mathrm{WF}}(I; \Gamma, M)$;

(L2) interaction vertices from the finite-window local interaction $I^w_{0,M}$;

(L3) renormalized graph-weight vertices $W^{R,M}_{\Gamma,L,w}(B^\Gamma_\bullet)$ and local counterterm vertices $C^M_{\Gamma,w}(\epsilon; B^\Gamma_\bullet)$ supplied by the package;

(L4) the central scalar-contact ghost γ_1, only on the scalar-contact boundary vertices already used in Theorem E.12.

The admissible edge labels are the finite-scale local propagator $P^{w,M}_{\epsilon,L}$, the corresponding regularized BV contraction $\Delta^{w,M}_{\epsilon,L}$, and the boundary Weyl edge whose scalar part on two Hamiltonians is the formal-local scalar cocycle of Proposition E.2,

$$(f, g) \longmapsto \omega(f, g)\gamma_1.$$

The open index loops may carry marker tensors, defined below in Definition E.26.

For each $\Gamma \in \mathcal{G}^{\mathrm{mk}}_M$ one specifies a finite ordered set

$$\mathsf{B}_M(\Gamma) = \{\beta_1, \ldots, \beta_s\}$$

of elementary codimension-one boundary operations. If a differential, bracket, collision, or counterterm expansion is a finite sum, each summand is a separate β_i. Each operation has a coefficient $\varepsilon_{\beta_i} \in \mathbb{C}[[\hbar]]$, an

output graph $\partial_{\beta_i}\Gamma \in \mathcal{G}_M^{\mathrm{mk}}$, and a marker-transport map μ_{β_i}. The local-functional boundary operations are:

(B1) *field differential:* if a vertex has label ℓ_v and $Q\ell_v = \sum_\alpha q_\alpha \ell_{v,\alpha}$, then the summand (v, α) replaces ℓ_v by $\ell_{v,\alpha}$ and has $\varepsilon_{v,\alpha} = q_\alpha$, including the Koszul sign from moving Q to v;

(B2) *BV edge contraction:* if an oriented internal edge e joins vertices v_1, v_2 and

$$\{\ell_{v_1}, \ell_{v_2}\}_{\hbar,M} = \sum_\alpha q_{e,\alpha}\ell_{e,\alpha},$$

then the summand (e, α) contracts e, merges the two vertices with label $\ell_{e,\alpha}$, and has coefficient $q_{e,\alpha}$ multiplied by the ordinary BV orientation sign. The bracket with $I^w_{\mathrm{o},M}$ is this operation with one vertex labelled by $I^w_{\mathrm{o},M}$;

(B3) *boundary collision/contact:* if an ordered boundary collision stratum $S = (v_1, \dots, v_k)$ has local expansion

$$\mathrm{Coll}_S(\ell_{v_1}, \dots, \ell_{v_k}) = \sum_\alpha q_{S,\alpha}\ell_{S,\alpha},$$

then (S, α) collapses S to one boundary vertex labelled $\ell_{S,\alpha}$. For the two-Hamiltonian scalar contact this convention gives

$$\mathrm{Coll}^{\mathrm{sc}}((a, f), (b, g)) = \hbar\,\omega(f, g)\int_I a(t)b(t)\gamma_1(t)\,dt,$$

antisymmetric in (f, g);

(B4) *counterterm face:* if the small-diagonal extension of a graph subconfiguration has local counterterm expansion $C^M_{\Gamma,w}(\epsilon; B_\bullet) = \sum_\alpha q_{\Gamma,\alpha} C^M_{\Gamma,\alpha}$, then (Γ, α) replaces that subconfiguration by the local counterterm vertex $C^M_{\Gamma,\alpha}$;

For the Hom-valued graph kernel one adjoins one further operation: the source differential face $-\kappa d_{\mathrm{CE}}$, with the CE bracket coefficient included in ε_{β_i}. After evaluation on fixed CE cycles this extra face is absent.

The finite list is required to contain every output of (B1)–(B4) that appears in the chosen package, and also the source face when the Hom-valued kernel table is used. If an output graph, coefficient, marker transport, or counterterm vertex is not supplied in $\mathcal{G}_M^{\mathrm{mk}}$, the missing datum is exactly that omitted boundary entry; no scalar-shadow theorem is asserted for the larger specialization.

The orientation line used for signs is

$$\mathrm{or}_M(\Gamma) = \det \mathsf{B}_M(\Gamma) \otimes \det E_{\mathrm{int}}(\Gamma) \otimes \bigotimes_v \mathrm{or}(\ell_v).$$

The first determinant gives the incidence sign in the marked differential below. The edge, vertex, BV, and CE Koszul signs are included in the coefficients ε_{β_i}.

The list is *codimension-two closed* if, for every $\beta_i \neq \beta_j$, the two composites are again entries of the finite list and, after the canonical identification of output orientation lines,

$$\partial_{\beta_j|\beta_i}\partial_{\beta_i}\Gamma = \partial_{\beta_i|\beta_j}\partial_{\beta_j}\Gamma,$$

$$\varepsilon_{\beta_i}\varepsilon_{\beta_j|\beta_i} = \varepsilon_{\beta_j}\varepsilon_{\beta_i|\beta_j}, \qquad \mu_{\beta_j|\beta_i}\mu_{\beta_i} = \mu_{\beta_i|\beta_j}\mu_{\beta_j}.$$

Here $\beta_j|\beta_i$ denotes the surviving operation induced by β_j after applying β_i. Thus, from a seed graph Γ, the actual finite window consists of the finitely many faces obtained by deleting subsets of the finite set $\mathsf{B}_M(\Gamma)$:

$$\mathcal{G}_M^{\mathrm{mk}} = \{\partial_J\Gamma \,|\, \Gamma \in \mathcal{S}_M,\ J \subset \mathsf{B}_M(\Gamma)\}, \qquad \partial_J = \partial_{\beta_{j_k}} \cdots \partial_{\beta_{j_1}}.$$

Here $\mathcal{S}_M$ is the finite seed set of package graphs and counterterm graphs, and the codimension-two closure equations make ∂_J independent of the chosen order of J. Failure of this displayed equality is precisely failure to supply the finite graph list.

Definition E.26 *(Oriented full-equivariant marked boundary complex).* Let $V = \mathbb{C}^{N|N}$. For $m \geq 1$, let

$$\mathcal{M}_m^{\mathrm{full}}(V) \subset \left(\mathrm{End}(V)^{\otimes m}\right)^{\mathfrak{gl}(V)}$$

be the finite span of the signed Brauer permutation tensors $\mathfrak{C}_\sigma$, $\sigma \in S_m$. Here $\mathfrak{C}_\sigma$ is the string-diagram tensor obtained by writing $\mathrm{End}(V) = V \otimes V^\vee$, pairing the $V^\vee$ factor in the i-th copy with the V factor in the $\sigma(i)$-th copy, and using the Koszul signs of the symmetric monoidal category of super vector spaces. For $m = 0$ the marker is the empty word.

A full-equivariant marked graph/counterterm generator in a finite window M is a tuple

$$X_{\Gamma,o,\mathfrak{m},M},$$

where $\Gamma \in \mathcal{G}_M^{\mathrm{mk}}$ for a finite-window marked Costello list of Definition E.25, o is an orientation of the finite set $\mathsf{B}_M(\Gamma)$, and $\mathfrak{m} = (\mathfrak{m}_L)_L$ assigns to each marked scalar open index loop L a tensor $\mathfrak{m}_L \in \mathcal{M}_{m_L}^{\mathrm{full}}(V)$.

If

$$o = \beta_1 \wedge \cdots \wedge \beta_s$$

is the ordered orientation of $\mathsf{B}_M(\Gamma)$, put $o/\beta_i = \beta_1 \wedge \cdots \widehat{\beta_i} \cdots \wedge \beta_s$. The marker-transport map

$$\mu_{\beta_i} : \bigotimes_L \mathcal{M}_{m_L}^{\mathrm{full}}(V) \longrightarrow \bigotimes_{L'} \mathcal{M}_{m'_{L'}}^{\mathrm{full}}(V)$$

is obtained by the signed evaluation, coevaluation, braiding, loop splitting, and loop concatenation maps that define the open index contractions. Concretely, loop merging sends two cyclic words to their boundary-ordered concatenation with the Koszul braiding sign; loop splitting inserts the coevaluation tensor at the split points; deletion of a closed marked loop applies the cyclic supertrace. The signed marked differential is

$$d_{\mathrm{mk}} X_{\Gamma,o,\mathfrak{m},M} = \sum_{i=1}^{s} (-1)^{i-1} \varepsilon_{\beta_i}\, X_{\partial_{\beta_i}\Gamma,\, o/\beta_i,\, \mu_{\beta_i}(\mathfrak{m}),\, M}.$$

Its scalar-contact filtration is

$$p(\Gamma, \mathfrak{m}) = \#\{\text{scalar-contact open loops in } \Gamma\}, \qquad F^p = \overline{\mathrm{Span}}\{X_{\Gamma,o,\mathfrak{m},M} : p(\Gamma,\mathfrak{m}) \geq p\}.$$

For an elementary boundary operation set

$$r(\beta_i) = p(\Gamma, \mathfrak{m}) - p(\partial_{\beta_i}\Gamma, \mu_{\beta_i}(\mathfrak{m})).$$

The first filtration-lowering marked boundary is therefore

$$\left[d_{\mathrm{mk}} X_{\Gamma,o,\mathfrak{m},M}\right]_{F^{p-r}/F^{p-r+1}} = \sum_{i:\, r(\beta_i)=r} (-1)^{i-1} \varepsilon_{\beta_i}\, X_{\partial_{\beta_i}\Gamma,\, o/\beta_i,\, \mu_{\beta_i}(\mathfrak{m}),\, M}.$$

LEMMA E.27 *(Incidence signs).* If the finite marked Costello list is codimension-two closed in the sense of Definition E.25, then $d_{\mathrm{mk}}^2 = 0$.

Proof. This is the cellular boundary sign computation. The codimension-two output obtained from β_i and β_j, $i < j$, appears twice. The order β_i then β_j contributes

$$(-1)^{i-1}(-1)^{j-2}\varepsilon_{\beta_i}\varepsilon_{\beta_j|\beta_i},$$

because β_i has been removed before β_j is counted. The order β_j then β_i contributes

$$(-1)^{j-1}(-1)^{i-1}\varepsilon_{\beta_j}\varepsilon_{\beta_i|\beta_j}.$$

The incidence signs differ by a minus sign, while (E.25) identifies the coefficients, output graph, and marker tensor. Hence the two terms cancel. Summing over all unordered pairs gives $d_{\mathrm{mk}}^2 = 0$. □

PROPOSITION E.28 *(Finite-window marked habitat compatibility).* Let $(I, w, M, \mathcal{G}_M)$ be a finite-window graph/QME package for the local mixed HT brane-defect theory and let $\mathcal{G}_M^{\mathrm{mk}}$ be a codimension-two closed finite-window marked Costello list whose labels are the admissible local labels of Definition E.25. Let

$$\mathfrak{G}^{\bullet}_{\mathrm{mk,full},M}(I)$$

be the completed span of the full-equivariant marked generators $X_{\Gamma,o,\mathfrak{m},M}$. Then

$$\mathfrak{G}^{\bullet}_{\mathrm{mk,full},M}(I) \subset \mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I)$$

and d_{mk} is the restriction of the finite-window differential $d_M = Q + \{I^w_{\mathrm{o},M}, -\}_{\hbar,M}$ on the local-functional table. On the Hom-valued kernel table, adjoining the source face makes the same formula the convolution differential $d_{\mathbf{K}} = d_M - (-1)^{|\cdot|}(\cdot)d_{\mathrm{CE}}$ before evaluation. The graph kernel $\kappa^M_{\mathcal{G},w,I}$ and the local counterterms of the package land in this marked habitat exactly when their graph components and counterterm faces occur in $\mathcal{G}_M^{\mathrm{mk}}$. Truncation $\pi_{M,N}$ and admissible weight transport $R^M_{w,w'}$ preserve the marked habitat precisely when they send the finite boundary table, coefficients, and marker transports to the corresponding lower-window or transported table.

Proof. The vertex and edge labels are those of Definition 3.170: regular-density or wavefront-admissible graph labels, finite-window propagators, renormalized weights, and local counterterms. Hence the represented local functionals lie in $\mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I)$. The boundary table (B1)–(B4) is exactly the expansion of Q, the BV bracket with $I^w_{\mathrm{o},M}$, collision/contact faces, and Costello counterterm faces on these generators. Thus the signed sum (E.26) is d_M on the local functional span. In the convolution kernel before CE evaluation, the additional source face is the missing source differential term in $d_{\mathbf{K}} = d_M - (-1)^{|\cdot|}(\cdot)d_{\mathrm{CE}}$; for the degree-zero kernel this is $-\kappa d_{\mathrm{CE}}$.

The inclusion of the graph kernel and counterterms is tautological from the same finite label list. If a graph component or counterterm face is absent, the equality with d_M fails at that named face, which is the exact missing datum. The finite-window package requires compatibility of graph weights, counterterms, scalar projections, and kernels with $\pi_{M,N}$ and $R^M_{w,w'}$; applying this requirement to each entry of the finite boundary table gives the stated preservation criterion. □

LEMMA E.29 *(Cyclic supertrace of full-equivariant marker tensors).* Let $\tau_m = (1\,2\,\cdots\,m)$. For the signed Brauer permutation tensor $\mathfrak{C}_\sigma \in \mathcal{M}_m^{\mathrm{full}}(V)$,

$$\mathrm{Str}_{\mathrm{cyc}}(\mathfrak{C}_\sigma) = \big(\mathrm{sdim}\, V\big)^{c(\tau_m^{-1}\sigma)},$$

where $c(\pi)$ is the number of cycles of the permutation π. Hence, for $V = \mathbb{C}^{N|N}$,

$$\mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}) = 0 \qquad \text{for every } \mathfrak{m} \in \mathcal{M}_m^{\mathrm{full}}(V)$$

and also for the empty marker word.

Proof. Compose the string diagram for $\mathfrak{C}_\sigma$ with the cyclic supertrace functional $(T_1, \dots, T_m) \mapsto \mathrm{Str}(T_1 \cdots T_m)$. The cyclic multiplication identifies the output of the i-th endomorphism with the input of the $\tau_m(i)$-th endomorphism, while $\mathfrak{C}_\sigma$ identifies it with the input of the $\sigma(i)$-th endomorphism. The resulting closed diagram has $c(\tau_m^{-1}\sigma)$ closed supertraces. Each closed supertrace evaluates to $\mathrm{Str}(\mathrm{id}_V) = \mathrm{sdim}\, V$. The signs in $\mathfrak{C}_\sigma$ are exactly the Koszul signs needed for this diagrammatic evaluation, giving (E.29). Since $\mathrm{sdim}\, \mathbb{C}^{N|N} = 0$, every nonempty full-equivariant marker tensor has zero cyclic supertrace. The empty word gives the identity-loop coefficient $\mathrm{Str}(\mathrm{id}_V) = 0$. □

Theorem E.30 *(Full-equivariant marked scalar shadow vanishing).* Let $V = \mathbb{C}^{N|N}$. Let $\mathfrak{G}^\bullet_{\mathrm{mk,full}}(I)$ be a complete finite-window inverse-limit graph/counterterm habitat for the formal-local mixed HT brane-defect theory generated windowwise by codimension-two closed finite-window marked Costello lists of Definition E.25 and by the full-equivariant marked generators of Definition E.26, compatible with truncation and admissible weight transport as in Proposition E.28. This is a finite-list statement in every window; no summation over all graph topologies, compact target, or global BCOV datum is part of the hypothesis. Then the marked preservation obstruction has zero cyclic-supertrace shadow:

$$\mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m})\big) = 0 \qquad \text{for every elementary boundary term } \beta_i.$$

Consequently, for every homogeneous $X_{\Gamma,o,\mathfrak{m},M} \in F^p$, the first scalar CE shadow is

$$\hbar \sum_{i:\, r(\beta_i)=r} (-1)^{i-1} \varepsilon_{\beta_i}\ \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m})\big)\, \omega(f_{\partial_{\beta_i}\Gamma}, g_{\partial_{\beta_i}\Gamma})\gamma_1 = 0,$$

and its cohomology class on the $[\bar{c}]$-line is

$$\hbar \sum_{i:\, r(\beta_i)=r} (-1)^{i-1} \varepsilon_{\beta_i}\ \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m})\big)\, [\bar{c}]\, \gamma_1 = 0.$$

Thus the zero scalar projection is a filtered chain map on $\mathfrak{G}^\bullet_{\mathrm{mk,full}}(I)$, and all scalar projection obstruction classes vanish on this full-equivariant marked graph/counterterm habitat.

Proof. The marker transport maps μ_{β_i} are built from the evaluation, coevaluation, braiding, loop splitting, and loop concatenation maps of the $\mathfrak{gl}(V)$-module V. Therefore they send full-equivariant signed Brauer marker tensors to full-equivariant signed Brauer marker tensors. Lemma E.29 gives zero cyclic supertrace for every such output tensor when $V = \mathbb{C}^{N|N}$.

Formula (E.26) gives the actual signed first filtration-lowering boundary. Applying the one-loop scalar computation of Proposition E.21 to each boundary term gives exactly the displayed scalar CE representative. Every coefficient in that sum is zero by the preceding paragraph. Passing to cohomology replaces ω by the nonzero class $[\bar{c}]$, but it is multiplied by the same zero coefficient. Thus $\widehat{\sigma} = 0$ satisfies $d_{\mathrm{CE}}\widehat{\sigma} - \widehat{\sigma} d_{\mathrm{mk}} = 0$. Proposition E.28 identifies d_{mk} with the restricted finite-window QME differential on the chosen finite table. Hence this is the required filtered chain projection, and Lemma E.4 kills the scalar projection tower on the habitat. □

Definition E.31 *(Non-scalar kernel complex for a marked finite window).* Fix a finite-window graph/QME package $(I, w, M, \mathcal{G}_M)$, a codimension-two closed marked Costello list $\mathcal{G}_M^{\mathrm{mk}}$, and a scalar projection

$$\widehat{\sigma}_{\mathrm{sc},M} : \mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I) \longrightarrow S^{\bullet}_M, \qquad S^{\bullet}_M = C^{\bullet}_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]].$$

The non-scalar obstruction complex in that window is

$$\mathcal{K}^{\bullet}_{\mathrm{ns},M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},M}, \qquad d_{\mathrm{ns},M} = d_M|_{\mathcal{K}_{\mathrm{ns},M}}.$$

It is a complex precisely when $\widehat{\sigma}_{\mathrm{sc},M}$ is a filtered chain map. In the full-equivariant marked habitat of Theorem E.30 one has $\widehat{\sigma}_{\mathrm{sc},M} = 0$, hence

$$\mathcal{K}^{\bullet}_{\mathrm{ns},M}(I) = \mathfrak{G}^{\bullet}_{\mathrm{mk,full},M}(I).$$

Let $\mathsf{A}^{\mathrm{fw}}_{n,M}$ be a valid row array in the sense of Definition 3.173. If the scalar gate vanishes,

$$\sum_{\alpha \in \mathsf{A}^{\mathrm{fw}}_{n,M}} S_{\alpha,M} = 0,$$

the order-n kernel residual is the scalar-zero row sum

$$R^{\mathrm{ns}}_{n,M} = \sum_{\alpha \in \mathsf{A}^{\mathrm{fw}}_{n,M}} R^{\mathrm{row}}_{\alpha,M} \in \mathcal{K}^{1}_{\mathrm{ns},M}(I).$$

Its obstruction class is

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} = [R^{\mathrm{ns}}_{n,M}] \in H^1\Big(\mathcal{K}^{\bullet}_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}\Big).$$

A degree-zero counterterm $C_{n,M} \in \mathcal{K}^{0}_{\mathrm{ns},M}(I)$ exists at this order if and only if

$$d_M C_{n,M} = -R^{\mathrm{ns}}_{n,M}, \qquad \mathfrak{o}^{\mathrm{ns}}_{n,M} = 0.$$

Definition E.32 *(Native finite-window realization habitat).* A native Costello/QME realization habitat for the mixed HT brane-defect model on

$$\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$$

is the following finite-window datum. First, one fixes a graph/QME package $(I, w, M, \mathcal{G}_M)$ in the sense of Definition 3.170, a codimension-two closed marked list $\mathcal{G}_M^{\mathrm{mk}}$ in the sense of Definition E.25, and the completed local-functional complex

$$\mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I) = \Big(O^{\mathrm{bd}}_{\mathrm{loc},\mathcal{G},M}(\mathcal{E}^M_w|_I; A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)), d_M\Big), \qquad d_M = Q + \{I^w_{\mathrm{o},M}, -\}_{\hbar,M}.$$

The topology is the complete topology generated by the $\hbar$-adic, scalar-contact, and finite-window filtrations. The complex is part of the datum: every d_M-boundary, collision face, BV bracket row, and local counterterm face needed for closure must occur in the finite table.

Second, the source is the labelled finite CE object $C^{\bullet}_{\mathrm{lab}}(\mathcal{G}_M)$. The Hom-valued kernel complex is

$$\mathbf{K}^{\bullet}_{\mathcal{G},M}(I) = \mathrm{Hom}_{\mathrm{cont}}\Big(C^{\bullet}_{\mathrm{lab}}(\mathcal{G}_M), \mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I)\Big),$$

with

$$d_{\mathbf{K},M}T = d_M T - (-1)^{|T|} T\, d_{\mathrm{CE}}, \qquad \mathrm{Curv}_{\mathbf{K},M}(\kappa) = d_{\mathbf{K},M}\kappa + \frac{1}{2}[\kappa,\kappa]_{\mathbf{K},M}.$$

Thus the source-face term is not optional. For a degree-zero kernel it contributes $-\kappa d_{\mathrm{CE}}$.

Third, the kernel has a fixed zero-edge part:

$$\kappa^{\mathrm{coef},M}_{\hbar,w,I}(u_{a(t)dt\otimes f}) = \iota_I(B^{w,M}_f(a)), \qquad \kappa^{\mathrm{coef},M}_{\hbar,w,I}(c_{B,\rho}) = \iota_I(\Theta^{w,M}_\rho(B)).$$

Here $a \in C_c^\infty(I)$, B is a regular compactly supported density or a graphwise wavefront-admissible current for the chosen finite row, and f, ρ lie in the window M. Positive-edge components are exactly the renormalized graph weights $K^M_\Gamma = W^{R,M}_{\Gamma,L,w}(B^\Gamma_\bullet)$ supplied by the package. No graph theorem for arbitrary $B \in \mathcal{D}'_c(I)$ is included in this definition.

Fourth, one supplies a filtered chain scalar projection

$$\widehat{\sigma}_{\mathrm{sc},M} : \mathfrak{Q}^\bullet_{\mathcal{G},M}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]],$$

and puts

$$\mathcal{K}^\bullet_{\mathrm{ns},M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},M}.$$

The ambient QME complex is always denoted $\mathfrak{Q}^\bullet_{\mathcal{G},M}(I)$. If the windowed scalar-zero QME notation $\mathcal{Q}^\bullet_{w,M}(I)$ is used for the same kernel, then

$$\mathcal{Q}^\bullet_{w,M}(I) = \mathcal{K}^\bullet_{\mathrm{ns},M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},M}.$$

In this appendix $\mathcal{K}_{\mathrm{ns},M}$ is used for the non-scalar kernel, so that it is not confused with the ambient QME complex $\mathfrak{Q}_{\mathcal{G},M}$. If this projection is not a chain map, the non-scalar kernel complex is not formed.

Theorem E.33 *(Constructive non-scalar Costello/QME realization criterion).* Work in a native finite-window realization habitat of Definition E.32. Suppose that the QME has been solved through order $\hbar^{n-1}$ by counterterms $C_{<n,M}$. The order-n realization problem is decided by the following finite row construction.

For every valid row array $\mathsf{A}^{\mathrm{fw}}_{n,M}$ of Definition 3.173, the rows are precisely

$$R^{\mathrm{row}}_{d\Gamma,M} = \varepsilon_\Gamma\, d_M K^M_\Gamma,$$

$$R^{\mathrm{row}}_{\mathrm{CE}(\Gamma,\nu),M} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu}),$$

$$R^{\mathrm{row}}_{b(\Gamma_1,\Gamma_2),M} = \frac{1}{2}\varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M},$$

together with the homogeneous lower-counterterm rows in

$$[\hbar^n]\Big(d_M C_{<n,M} + \frac{1}{2}\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}\Big).$$

Their scalar gates are

$$S_{\alpha,M} = \widehat{\sigma}_{\mathrm{sc},M}(R^{\mathrm{row}}_{\alpha,M}),$$

and the scalar gate for the order is the single equation

$$\mathfrak{s}_{n,M} = \sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} S_{\alpha,M} = 0.$$

If (E.33) fails, the non-scalar problem has not started at order n. If it holds, then

$$R^{\mathrm{ns}}_{n,M} = \sum_{\alpha \in \mathsf{A}^{\mathrm{fw}}_{n,M}} R^{\mathrm{row}}_{\alpha,M} \in \mathcal{K}^{1}_{\mathrm{ns},M}(I)$$

is a cocycle whenever the finite row table is codimension-two closed and the lower orders have been solved. Its obstruction class is

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} = [R^{\mathrm{ns}}_{n,M}] \in H^{1}(\mathcal{K}^{\bullet}_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}).$$

Choose degree-one row coordinates $\rho^M_1, \dots, \rho^M_a$ and degree-zero counterterm candidates $\eta^M_1, \dots, \eta^M_b$ inside the same finite package, and write

$$R^{\mathrm{ns}}_{n,M} = \sum_i r^M_i \rho^M_i, \qquad d_{\mathrm{ns},M}\eta^M_j = \sum_i A^M_{ij}\rho^M_i.$$

A fixed-window counterterm in the supplied span exists exactly when

$$A^M c = -r^M$$

is solvable. If it is not solvable, any covector

$$\ell A^M = 0, \qquad \ell(r^M) \neq 0$$

is a checkable certificate that $\mathfrak{o}^{\mathrm{ns}}_{n,M} \neq 0$ in the displayed finite row subcomplex. A compatible finite-window tower also requires truncation matrices $T^{M,N}$ on the rows and $P^{M,N}$ on candidate primitives satisfying

$$T^{M,N} r^M = r^N, \qquad T^{M,N} A^M = A^N P^{M,N},$$

and the Roos class

$$[P^{M,N} c_M - c_N] \in H^{0}(\mathcal{K}^{\bullet}_{\mathrm{ns},N}(I))$$

must vanish.

Iterating these finite-window tests for all n, with graphwise regular-density or wavefront-admissible kernel data and with compatible counterterms, is equivalent to

$$\mathrm{Curv}_{\mathbf{K}}(\kappa_{\mathcal{G},w,I} + C_{\hbar,w}) = 0$$

in the completed native brane-defect kernel complex. This is the strongest realization statement supplied by the current manuscript. A larger Costello specialization must add the missing graph weight, source face, scalar gate, primitive candidate, or truncation matrix; none is supplied by analogy with compact BCOV theory.

Proof. The first three displayed row formulas are the homogeneous terms in

$$d_M \kappa^M_{\mathcal{G},w,I} - \kappa^M_{\mathcal{G},w,I} d_{\mathrm{CE}} + \frac{1}{2}[\kappa^M_{\mathcal{G},w,I}, \kappa^M_{\mathcal{G},w,I}]_{\mathbf{K},M}.$$

The minus sign in the source-face row is the term $-\kappa d_{\mathrm{CE}}$ in Definition E.32. The last displayed family is the order-n contribution of lower counterterms to the same curved Maurer–Cartan equation.

Applying the chain scalar projection gives the scalar gate. If the gate is nonzero, the residual is not in $\ker \widehat{\sigma}_{\mathrm{sc},M}$. If the gate is zero, the residual is in the kernel complex. Codimension-two closure is exactly the finite-row form of the Bianchi identity: the two ways of taking two elementary faces have opposite incidence signs and the same output graph, coefficient, and marker transport. Since lower orders are solved, this makes $R^{\mathrm{ns}}_{n,M}$ a $d_{\mathrm{ns},M}$-cocycle.

A counterterm $C_{n,M}$ changes the residual by $d_{\mathrm{ns},M} C_{n,M}$. Expanding $C_{n,M}$ in the supplied degree-zero row basis gives the linear system (E.33). Failure of solvability is equivalent to the residual vector being outside the image of A^M; a left-null covector with nonzero value on r^M is precisely the finite-dimensional cokernel certificate. The two truncation identities say that both residuals and boundaries commute with finite-window projection. After windowwise primitives are chosen, their failure to commute with truncation is the displayed Roos 1-cocycle in H^0. Thus the fixed-window H^1 class and the Roos primitive class are exactly the obstruction pair. Passing to all orders gives the completed curvature equation. □

Proposition E.34 *(Closed rows and the first supplied open source-face row).* In the full-equivariant branch $V = \mathbb{C}^{N|N}$, the one-cross two-Hamiltonian scalar-contact row α_{sc} with inputs (z_1, z_2) has

$$R^{\mathrm{row}}_{\alpha_{\mathrm{sc}},M} = 0, \qquad S_{\alpha_{\mathrm{sc}},M} = 0, \qquad C_{\alpha_{\mathrm{sc}},M} = 0.$$

The labelled scalar-zero row α_{II} with inputs (z_1, z_1) has

$$R^{\mathrm{row}}_{\alpha_{\mathrm{II}},M} = \mathrm{Str}_{\mathrm{cyc}}(\mu_{\beta_{\mathrm{II}}}(\mathfrak{m}))\, \omega(z_1, z_1) \int_I a(t) b(t) \gamma_{\mathbf{1}}(t)\, dt = 0,$$

$$S_{\alpha_{\mathrm{II}},M} = 0, \qquad C_{\alpha_{\mathrm{II}},M} = 0,$$

with truncation $t^{M,N}_{\alpha_{\mathrm{II}}\alpha_{\mathrm{II}}} = 1$ when the labelled zero row is retained and the lower window still contains z_1.

The first supplied nonzero scalar-zero source-face row in the present row machinery is the order-three symbolic row $\theta_3 = \mathrm{CE}(\Theta_3, \nu_3)$, with

$$R^{\mathrm{row}}_{\theta_3,M} = -K^M_{\Theta_3}(\zeta_{M,\nu_3}) =: \mathsf{E}_{\theta_3,M}, \qquad S_{\theta_3,M} = 0.$$

If the row-closure datum $d_{\mathrm{ns},M} \mathsf{E}_{\theta_3,M} = 0$ is supplied, it defines

$$\mathfrak{o}^{\mathrm{ns}}_{\theta_3,M} = [\mathsf{E}_{\theta_3,M}] \in H^1(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}).$$

In the finite row subcomplex currently supplied for this row,

$$\mathcal{K}^0_{\theta_3,M} = 0, \qquad \mathcal{K}^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, \qquad d_{\theta_3,M} = 0,$$

the covector

$$\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$$

certifies a nonzero finite-row obstruction. The exact missing healing datum is either a degree-zero local counterterm $C_{\theta_3,M}$ with

$$d_{\mathrm{ns},M} C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M},$$

compatible under finite-window truncation, or additional Hom-valued companion rows whose sum changes the residual vector. Without one of these data the order-three larger package is not a solved QME row.

Proof. The row α_{sc} is zero as a local functional by the cyclic supertrace computation of Lemma E.29; this is the closed row of Proposition 3.174. The row α_{II} is zero for the independent reason

$$\omega(z_1, z_1) = [\{z_1, z_1\}_{\Omega_{\mathrm{loc}}}]_0 = 0,$$

which is Proposition E.40. In both cases the zero counterterm is a primitive and the stated truncation identities send zero rows to zero rows.

The formula for θ_3 is the source-face row $-\kappa d_{\mathrm{CE}}$ from Theorem E.33, with the concrete row labels of Proposition 3.181. Its scalar gate is zero in the full-equivariant marked habitat. In the one-row finite complex, there is no degree-zero candidate primitive, so the boundary matrix is the 1×0 zero matrix and the residual vector is the basis vector $\mathsf{E}_{\theta_3,M}$. The displayed covector annihilates the image of the differential and not the residual. This proves nonvanishing only in the supplied finite row subcomplex. The displayed $C_{\theta_3,M}$ or companion-row datum is exactly what would enlarge the complex enough to heal the row. □

PROPOSITION E.35 *(Bianchi-exposed lower gate for θ_3).* In the lower window $N = 2$, let $E^2_{\nu_{02}}$ and $E^2_{\nu_{20}}$ be the two transported source-face rows attached to the θ_3 source differential. The canonical source-coordinate candidate is

$$\zeta_2^0 = u_{a,H_{1,0}} u_{b,H_{0,1}}, \qquad D^2_{02,20} = K^2_{\Theta_3}(\zeta_2^0),$$

with

$$d_{\mathrm{CE}}\zeta_2^0 = 2\zeta_{\nu_{02}} - 2\zeta_{\nu_{20}}.$$

The Hom-valued Bianchi defect is

$$\mathfrak{b}^2_{02,20} = d_{\mathrm{ns},2} K^2_{\Theta_3}(\zeta_2^0) - K^2_{\Theta_3}(d_{\mathrm{CE}}\zeta_2^0).$$

In the Bianchi-exposed basis

$$(E^2_{\nu_{02}}, E^2_{\nu_{20}}, \mathfrak{b}^2_{02,20}),$$

the current source-coordinate column and the lower transported residual are

$$A^2_D = (-2, 2, 1)^T, \qquad r_2 = (2, -2, 0)^T.$$

Hence

$$A^2_D c = -r_2$$

has no solution in the supplied lower local-functional row habitat. The normalized cokernel

$$\widetilde{\lambda}_{02,\mathfrak{b}} = \frac{1}{2}(E^2_{\nu_{02}})^* + (\mathfrak{b}^2_{02,20})^*$$

satisfies

$$\widetilde{\lambda}_{02,\mathfrak{b}} A^2_D = 0, \qquad \widetilde{\lambda}_{02,\mathfrak{b}}(r_2) = 1.$$

The missing Bianchi-counterterm target is

$$-\mathfrak{b}^2_{02,20} = (0, 0, -1)^T,$$

and it is not in the image of the current column A^2_D, since

$$\widetilde{\lambda}_{02,\mathfrak{b}}((0, 0, -1)^T) = -1.$$

A repair must therefore add a genuine scalar-zero local-functional column

$$A_B^2 = (0, 0, -1)^T, \qquad d_{\mathrm{ns},2} B_{02,20}^2 = -\mathfrak{b}_{02,20}^2, \qquad \widehat{\sigma}_{\mathrm{sc},2}(B_{02,20}^2) = 0,$$

with the required locality or wavefront admissibility. If such $B_{02,20}^2$ is supplied, then

$$d_{\mathrm{ns},2}(D_{02,20}^2 + B_{02,20}^2) = -2E_{\nu_{02}}^2 + 2E_{\nu_{20}}^2 = -r_2.$$

For a compatible finite-window tower, write $\mathfrak{b}^M$ for the corresponding Bianchi defect in window M. The transport obstruction is

$$\Delta_{M,N}^1 := -\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N.$$

A family B^M transports the fixed-window repair only when

$$\Delta_{M,N}^1 = d_{\mathrm{ns},N}(\pi_{M,N}B^M - B^N),$$

possibly after an explicitly supplied scalar-zero transport correction. Thus the H^1 class of $\Delta_{M,N}^1$ must vanish first, and the resulting primitive-compatibility class in

$$\varprojlim_N{}^1 H^0(\mathcal{K}_{\mathrm{ns},N}^\bullet(I))$$

must vanish next. Without these two gates, the lower θ_3 Bianchi repair is only a fixed-window matrix target, not a solved Costello/QME counterterm tower.

Proof. The source differential gives the two lower source-face rows with coefficients 2 and -2. Applying $K_{\Theta_3}^2$ is not a chain map until its Hom-valued Bianchi defect is killed, so

$$d_{\mathrm{ns},2} D_{02,20}^2 = -2E_{\nu_{02}}^2 + 2E_{\nu_{20}}^2 + \mathfrak{b}_{02,20}^2,$$

which is the column $A_D^2 = (-2, 2, 1)^T$ in the displayed basis. The residual transported from the lower two-face row is $r_2 = (2, -2, 0)^T$. If $A_D^2 c = -r_2 = (-2, 2, 0)^T$, the first two coordinates force $c = 1$, while the Bianchi coordinate forces $c = 0$. Thus the equation is inconsistent. The same conclusion is the cokernel calculation: $\widetilde{\lambda}_{02,\mathfrak{b}}$ annihilates A_D^2 and does not annihilate r_2.

The target $(0, 0, -1)^T$ is the boundary column required to remove only the Bianchi coordinate. Since the same cokernel evaluates to -1 on it, the current habitat does not contain such a boundary. Adding a scalar-zero local functional $B_{02,20}^2$ with the displayed boundary produces $A_D^2 + A_B^2 = (-2, 2, 0)^T = -r_2$, hence the lower residual is killed. This is only a formal finite-window repair until the local functional and its admissibility are supplied.

For $M \geq N$, applying $d_{\mathrm{ns},N}$ to $\pi_{M,N}B^M - B^N$ gives the difference between the transported Bianchi boundary and the N-window Bianchi boundary, namely $-\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N$. Therefore the transported family exists only after this degree-one class vanishes in $H^1(\mathcal{K}_{\mathrm{ns},N}^\bullet(I))$; after choosing primitives, their failure to be compatible is the usual Roos $\varprojlim^1 H^0$ obstruction. □

Definition E.36 *(Normal Ω-equivariant Costello kernel datum).* Let

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}_{z_1} \times \mathbb{C}_{z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\},$$

and let the normal torus be

$$T_\Omega = \mathbb{C}_{\epsilon_s}^* \times \mathbb{C}_{\epsilon_1}^* \times \mathbb{C}_{\epsilon_2}^*.$$

Its infinitesimal vector field is

$$V_\Omega = \epsilon_s s\partial_s + \epsilon_1 z_1\partial_{z_1} + \epsilon_2 z_2\partial_{z_2}, \qquad V_\Omega(t) = 0,$$

and the Cartan differential is

$$Q_\Omega = Q + \iota_{V_\Omega}, \qquad Q_\Omega^2 = L_{V_\Omega}.$$

All complexes below are taken in the T_Ω-invariant, or equivalently basic Cartan, subcomplex. Before this restriction the displayed operator is not a differential.

Put

$$\mathsf{X}_{N,M} \subset \{\, n\epsilon_s + a\epsilon_1 + b\epsilon_2,\ n\epsilon_s - a\epsilon_1 - b\epsilon_2 \,\}$$

for the finite set of nonzero moving normal characters occurring in the chosen N-brane, M-window package. The coefficient ring is

$$R_\Omega^{N,M} = \mathbb{C}[\epsilon_s, \epsilon_1, \epsilon_2, \hbar_\omega][\chi^{-1} \mid \chi \in \mathsf{X}_{N,M}], \qquad \mathrm{wt}(\hbar_\omega) = \epsilon_1 + \epsilon_2.$$

The self-dual branch is formed by base change to $\epsilon_1 + \epsilon_2 = 0$ before localization; only the surviving nonzero characters in the image of $\mathsf{X}_{N,M}$ are inverted. In particular $\epsilon_1 + \epsilon_2$ is never inverted and then specialized to zero. On that branch $\hbar_\omega$ has weight zero and may be retained as a scalar parameter or set to 1 after the specialization is explicit.

The three quantum parameters are distinct:

$$\hbar_{\mathrm{QME}}, \qquad \hbar_{\mathrm{W}}, \qquad \hbar_\omega.$$

The first counts Costello/BV graph order; the second deforms the boundary Weyl/Moyal algebra; the third scalarizes the equivariant Hamiltonian bracket. A Weyl branch may identify $\hbar_{\mathrm{W}} = \hbar_\omega$. No branch identifies $\hbar_{\mathrm{QME}}$ with either parameter unless a theorem states that extra specialization. The equivariant Hamiltonian bracket is the $R_\Omega^{N,M}$-linear bracket

$$\{f, g\}_\Omega = \hbar_\omega\big(\partial_{z_1} f\ \partial_{z_2} g - \partial_{z_2} f\ \partial_{z_1} g\big).$$

A normal Ω-equivariant finite-window Costello kernel datum over $(I, w, M, \mathcal{G}_M)$ is a finite-window marked Costello package as in Definitions 3.170 and E.25, together with the following additional data.

(Ω1) A T_Ω-locally finite weighted field complex $\mathcal{E}^M_{w,\Omega}|_I$, source CE complex, local interaction $I^w_{0,M,\Omega}$, and invariant local-functional target

$$\mathfrak{Q}^\bullet_{\mathcal{G},\Omega,M}(I) \subset O^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}^M_{w,\Omega}|_I; A^{\mathrm{pp},w,M}_{\partial,\Omega,\hbar_{\mathrm{W}}}(I)\widehat{\otimes} R_\Omega^{N,M}[[\hbar_{\mathrm{QME}}]])$$

with differential

$$d_{M,\Omega} = Q_\Omega + \{I^w_{0,M,\Omega}, -\}_{\hbar_{\mathrm{QME}},M,\Omega}$$

on the invariant subcomplex. The source CE differential $d_{\mathrm{CE},\Omega}$ is formed from Q_Ω and the bracket (E.36).

(Ω2) For every localized normal character $\chi \in \mathsf{X}_{N,M}$, or for every surviving nonzero image of such a character on the self-dual branch, a normal contraction h_χ satisfying

$$Q_\Omega h_\chi + h_\chi Q_\Omega = \mathrm{id} - \mathrm{pr}_{\chi=0}$$

on the corresponding χ-weight summand, after inverting χ. This is the normal-mode acyclicity datum. It is not a QME counterterm and does not decide any graph curvature class.

(Ω_3) An equivariant finite-scale gauge fixing and propagator

$$P^{w,M}_{\epsilon,L,\Omega} = \int_{\epsilon}^{L} (Q^{\mathrm{GF}}_{\Omega} \otimes 1) K^{w,M}_{t,\Omega}\, dt$$

such that

$$[Q_\Omega, P^{w,M}_{\epsilon,L,\Omega}] = K^{w,M}_{\epsilon,\Omega} - K^{w,M}_{L,\Omega}, \qquad [L_{V_\Omega}, P^{w,M}_{\epsilon,L,\Omega}] = 0,$$

and such that the coefficient Casimir, wavefront conditions, and finite-window weighted bounds required by the nonequivariant package remain valid over $R^{N,M}_{\Omega}[[\hbar_{\mathrm{QME}}]]$.

(Ω_4) A named normal-localization normalization ν_Ω. The residue branch has

$$\nu^{\mathrm{res}}_{\Omega} = 1.$$

The Euler branch has

$$\nu^{\mathrm{Eu}}_{\Omega} = \sigma_s(\epsilon_s\epsilon_1\epsilon_2)^{-1},$$

after the corresponding Euler denominators have been admitted in the chosen localized ring. Scalar-contact rows, graph weights, and trace normalizations are always read with the selected branch. The residue branch, Euler branch, and Euler-rescaled residue branch are not silently identified; a comparison map must state the real orientation sign σ_s and every denominator.

(Ω_5) Equivariant local counterterm representatives

$$C^{M}_{\Gamma,w,\Omega}(\epsilon; B^{\Gamma}_{\bullet}) \in \mathfrak{Q}^{\circ}_{\mathcal{G},\Omega,M}(I)$$

which are T_Ω-invariant, local on the relevant bulk diagonal or brane-collision stratum, compatible with finite-window truncation and admissible weight transport, and whose scalar-contact associated graded is computed using the same ν_Ω.

(Ω_6) An associated-graded scalar-symbol extractor candidate

$$\sigma_{\mathrm{sc},\Omega,M}\colon\ \mathrm{gr}_F\, \mathfrak{Q}^{\bullet}_{\mathcal{G},\Omega,M}(I) \longrightarrow S^{\bullet}_{\Omega,M}, \qquad S^{\bullet}_{\Omega,M} = C^{\bullet}_{\mathrm{Lie}}(\bar{\mathcal{A}}; \mathbb{C}\gamma_1)[1][[\hbar_{\mathrm{QME}}]]\widehat{\otimes} R^{N,M}_{\Omega},$$

which sends the two-Hamiltonian scalar contact to

$$\nu_\Omega\, \hbar_{\mathrm{QME}}\ \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m})\ [\{f, g\}_\Omega]_\circ\, \gamma_1.$$

A filtered chain lift $\widehat{\sigma}_{\mathrm{sc},\Omega,M}$ is not part of the datum until the obstruction test below has been passed.

(Ω_7) A degree-zero Hom-valued graph kernel

$$\kappa^{M}_{\mathcal{G},\Omega,w,I} \in \mathrm{Hom}_{\mathrm{cont}}\Big(C^{\bullet}_{\mathrm{CE},\Omega,\mathrm{lab}}(\mathcal{G}_M), \mathfrak{Q}^{\bullet}_{\mathcal{G},\Omega,M}(I)\Big)$$

whose zero-edge part is the coefficient-current kernel and whose positive graph components are the renormalized Ω-equivariant graph weights.

This datum gives a habitat in which the equivariant QME and centrality obstructions are defined. Localization supplies the coefficient ring, Q_Ω, normal contractions, and equivariant propagator data; it is not a QME proof and does not assert that any obstruction vanishes.

THEOREM E.37 *(Equivariant brane-defect QME obstruction criterion).* Fix a candidate normal Ω-equivariant finite-window package satisfying items (Ω1)–(Ω6) of Definition E.36. Let

$$\sigma_{\mathrm{sc},\Omega,M}\colon\ \mathrm{gr}_F\,\mathfrak{Q}^{\bullet}_{\mathcal{G},\Omega,M}(I)\longrightarrow S^{\bullet}_{\Omega,M}$$

be its associated-graded scalar extractor. The scalar projection is a theorem-grade chain projection exactly when the following relative Hom obstructions vanish. First,

$$\delta^{(0)}_{\Omega,\sigma,M}=d_{\mathrm{CE},\Omega}\sigma_{\mathrm{sc},\Omega,M}-\sigma_{\mathrm{sc},\Omega,M}d_{\mathrm{gr},\Omega}=0.$$

If this holds and $s_{\Omega,M}$ is any filtered lift of $\sigma_{\mathrm{sc},\Omega,M}$, the first nonzero homogeneous piece of

$$d_{\mathrm{CE},\Omega}s_{\Omega,M}-s_{\Omega,M}d_{M,\Omega}$$

lowering the scalar-contact filtration by $r>0$ gives

$$\mathfrak{o}^{(r)}_{\Omega,\sigma,M}\in H^1\Big(\mathrm{Hom}_{T_\Omega,-r}\big(\mathrm{gr}_F\,\mathfrak{Q}^{\bullet}_{\mathcal{G},\Omega,M}(I),S^{\bullet}_{\Omega,M}\big)\Big).$$

A filtered chain projection $\widehat{\sigma}_{\mathrm{sc},\Omega,M}$ exists if and only if $\delta^{(0)}_{\Omega,\sigma,M}=0$ and the classes $\mathfrak{o}^{(r)}_{\Omega,\sigma,M}$ vanish successively, with compatible finite-window and weight-transport choices.

Once this projection is supplied, put

$$\mathcal{K}^{\bullet}_{\mathrm{ns},\Omega,M}(I)=\ker\widehat{\sigma}_{\mathrm{sc},\Omega,M},\qquad d_{\mathrm{ns},\Omega,M}=d_{M,\Omega}|_{\mathcal{K}_{\mathrm{ns},\Omega,M}}.$$

Suppose the Hom-valued kernel of item (Ω7) is supplied, and counterterms have been chosen through order $\hbar^{n-1}_{\mathrm{QME}}$. The order-$n$ equivariant residual is

$$R^{\mathrm{ns}}_{n,\Omega,M}=[\hbar^{n}_{\mathrm{QME}}]\Big(\mathrm{Curv}_{\mathbf{K},\Omega,M}(\kappa^{M}_{\mathcal{G},\Omega,w,I})+d_{M,\Omega}C_{<n,\Omega,M}+\frac{1}{2}\{C_{<n,\Omega,M},C_{<n,\Omega,M}\}_{\hbar_{\mathrm{QME}},M,\Omega}\Big).$$

The scalar gate at this order is

$$\mathfrak{s}_{n,\Omega,M}=\widehat{\sigma}_{\mathrm{sc},\Omega,M}(R^{\mathrm{ns}}_{n,\Omega,M}).$$

Only when $\mathfrak{s}_{n,\Omega,M}=0$ and all lower orders have been solved does the residual define the non-scalar obstruction class

$$\mathfrak{o}^{\mathrm{ns}}_{n,\Omega,M}=[R^{\mathrm{ns}}_{n,\Omega,M}]\in H^1\Big(\mathcal{K}^{\bullet}_{\mathrm{ns},\Omega,M}(I),d_{\mathrm{ns},\Omega,M}\Big).$$

A compatible order-n counterterm exists precisely when

$$\big(\mathfrak{o}^{\mathrm{ns}}_{n,\Omega,M}\big)_M=0\quad\text{in}\quad\varprojlim_{M}H^1(\mathcal{K}^{\bullet}_{\mathrm{ns},\Omega,M}(I))$$

and the Milnor primitive-compatibility class

$$\lambda_{n,\Omega}\in\varprojlim{}^{1}_{M}\,H^0(\mathcal{K}^{\bullet}_{\mathrm{ns},\Omega,M}(I))$$

vanishes. Iterating this criterion for all n is equivalent to

$$\mathrm{Curv}_{\mathbf{K},\Omega}(\kappa_{\mathcal{G},\Omega,w,I}+C_{\Omega,w})=0$$

in the completed invariant kernel complex.

Therefore the normal Ω-background supplies only the habitat. A QME proof requires, in addition, the chain scalar projection, the non-scalar row classes displayed above, explicit local counterterms killing those classes, compatible primitive transport in the $\varprojlim^1 H^0$ sense, and the Q_Ω-centrality homotopies of Proposition E.45 for the source rows whose centrality is used.

Proof. The first paragraph is the equivariant version of Lemma E.4. On the T_Ω-invariant subcomplex $Q_\Omega^2 = L_{V_\Omega} = 0$, so the Hom differential is $D_\Omega(\alpha) = d_{\mathrm{CE},\Omega}\alpha - (-1)^{|\alpha|}\alpha d_{\mathrm{gr},\Omega}$. The associated-graded defect is the obstruction to even starting a filtered lift. After it vanishes, the first filtration-lowering chain defect is closed for D_Ω; changing the lift by a filtration-lowering cochain changes it by a Hom coboundary. Completeness gives the successive criterion and the compatibility condition in the inverse finite-window tower.

With a chain scalar projection fixed, its kernel is preserved by $d_{M,\Omega}$. The displayed residual is the order-n term of the curved Maurer–Cartan equation for the Hom-valued kernel plus lower counterterms. Applying the scalar projection gives the scalar gate. If the gate vanishes and the lower-order equations hold, the equivariant Bianchi identity in the invariant complex makes $R^{\mathrm{ns}}_{n,\Omega,M}$ a $d_{\mathrm{ns},\Omega,M}$-cocycle. Adding an order-n local counterterm changes this cocycle by a $d_{\mathrm{ns},\Omega,M}$-boundary, so the fixed-window obstruction is its H^1-class. Compatibility of primitives across the finite-window inverse system is exactly the usual $\varprojlim^1 H^0$ obstruction of Theorem 3.98. The final sentence follows because the normal contraction datum appears only in the construction of the equivariant propagator and localized coefficient complex; it does not identify the curvature residual with a boundary, does not produce scalar-projection chain data, and does not supply the centrality homotopies required for source rows. □

Theorem E.38 *(Minimal full-equivariant kernel vanishing).* Work in the local $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ full-equivariant marked branch with $V = \mathbb{C}^{N|N}$. Let the finite marked list be the minimal codimension-two closed list generated by the strict current-interface rows and by the one-cross two-Hamiltonian scalar-contact row of Proposition 3.174. Assume that no further $|\Gamma|_\hbar = 1$ graph row, no genuine $|\Gamma|_\hbar = 2$ seed graph, no order-2 CE-source face, and no order-2 graph counterterm face is added.

Then the first kernel residuals are

$$R^{\mathrm{ns}}_{1,M} = 0, \qquad R^{\mathrm{ns}}_{2,M} = 0,$$

and the compatible counterterms are

$$C_{1,M} = 0, \qquad C_{2,M} = 0.$$

Thus the minimal full-equivariant finite-window array has no genuinely non-scalar QME obstruction through the first codimension-two closure level. Its first possible non-scalar obstruction can only appear after a larger finite marked list supplies an additional scalar-zero row.

Proof. The order-0 current-interface rows have zero curvature by Theorem E.10 and by the finite-window compatibility recorded in Proposition E.28. The only order-1 row in the minimal list is the one-cross scalar-contact row. Its scalar coefficient is the cyclic supertrace of the transported full-equivariant marker tensor. By Lemma E.29, this coefficient is zero for $V = \mathbb{C}^{N|N}$, including the empty word. Hence the row is zero as a local functional in the full-equivariant branch, so $R^{\mathrm{ns}}_{1,M} = 0$ and $C_{1,M} = 0$.

Codimension-two closure adds only the two ordered composites of two first scalar-contact boundary faces. Their incidence signs are opposite and their output graphs, coefficients, and marker transports

are identified by Equation (E.25). This gives the pairwise cancellation of the marked differential. In the full-equivariant branch each scalar-loop factor is already zero by the same cyclic supertrace calculation. Since no genuine order-2 seed graph, CE-source face, or counterterm face is present, there is no remaining row in the kernel complex. Thus $R^{\mathrm{ns}}_{2,M} = 0$, and the zero counterterm is compatible under every finite-window truncation. □

PROPOSITION E.39 *(First missing non-scalar row).* Let $\mathcal{G}^{\mathrm{mk}}_M$ be any larger finite marked list in the same full-equivariant local branch, with scalar projection killed as in Theorem E.30. The first genuinely non-scalar obstruction not detected by the scalar projection is the least order n for which the row sum of Definition E.31 is nonzero in cohomology:

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} = \left[\sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} R^{\mathrm{row}}_{\alpha,M}\right] \in H^1\Big(\mathcal{K}^{\bullet}_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}\Big).$$

Since the minimal order-1 scalar-contact row vanishes, the first datum that can make $n = 1$ is an additional order-1 scalar-zero row

$$\alpha \in \mathsf{A}^{\mathrm{fw}}_{1,M} \setminus \{\alpha_{\mathrm{sc}}\}, \qquad S_{\alpha,M} = 0,$$

with its actual row value $R^{\mathrm{row}}_{\alpha,M}$. Equivalently, the missing row entry is one of the order-1 summands

$$\varepsilon_\Gamma d_M K^M_\Gamma, \qquad -\varepsilon^{\mathrm{CE}}_{\Gamma'} K^M_{\Gamma'}, \qquad \frac{1}{2}\varepsilon_{\Gamma_0,\Gamma_1}\{K^M_{\Gamma_0}, K^M_{\Gamma_1}\}_{\hbar,M},$$

together with its graph type, finite-window labels, full-equivariant marker tensor, incidence sign, scalar gate, and truncation coefficients $t^{M,N}_{\alpha\alpha'}$.

If the finite marked list explicitly has no such order-1 row, the next missing datum is the order-2 scalar-zero row sum

$$\begin{aligned} R^{\mathrm{ns}}_{2,M} = &\sum_{|\Gamma|_\hbar=2} \varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{|\Gamma'|_\hbar=2} \varepsilon^{\mathrm{CE}}_{\Gamma'} K^M_{\Gamma'} \\ &+ \frac{1}{2}\sum_{|\Gamma_1|_\hbar=|\Gamma_2|_\hbar=1} \varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M} + [\hbar^2] d_M C_{<2,M} \\ &+ \frac{1}{2}[\hbar^2]\{C_{<2,M}, C_{<2,M}\}_{\hbar,M}. \end{aligned}$$

Without these row values, signs, scalar gates, and truncation coefficients, no theorem-grade non-scalar obstruction class or counterterm is defined beyond the minimal vanishing theorem above.

Proof. The scalar projection is a chain map on the full-equivariant marked habitat by Theorem E.30, so its kernel is a d_M-stable complex. After the scalar gate vanishes, Theorem 3.171 puts the first unsolved residual in degree one of that kernel complex. Adding a degree-zero counterterm changes the residual by a d_M-boundary, and hence the displayed cohomology class is exactly the obstruction.

The minimal order-1 row is already zero by Theorem E.38. Thus an order-1 non-scalar class can only be supplied by an additional finite-window row outside α_{sc}. The three displayed forms are precisely the order-1 possibilities in the curvature expansion: target differential of an order-1 graph component, CE source face of such a component, or a convolution bracket whose orders add to 1. Each is a row only after the finite graph list supplies its labels, signs, marker transport, scalar gate, and truncation coefficients.

If there is no order-1 row, the next possible residual is the order-2 curvature expansion, including genuine order-2 graph components, CE-source faces, brackets of order-1 components, and lower-counterterm terms. These data are not determined by the scalar shadow theorem. They are exactly the row entries required by Definition 3.173. □

Proposition E.40 *(First scalar-zero order-one row outside the minimal array).* Work in the full-equivariant local branch of Theorem E.38. Let $M \geq 1$, and enlarge the minimal finite marked list by the labelled one-cross scalar-contact boundary operation α_{II} with ordered Hamiltonian inputs (a, z_1), (b, z_1), the boundary Weyl edge E_{Weyl}, the scalar-contact output vertex $\gamma_{\mathbf{I}}$, and the same full-equivariant marker habitat as before. Choose the orientation in which the displayed ordered pair has incidence coefficient +1.

Then α_{II} is a scalar-zero order-one row outside α_{sc}, with row data

$$|\alpha_{\mathrm{II}}|_{\hbar} = 1, \qquad d_{\max}(\alpha_{\mathrm{II}}) = 1, \qquad p_{\alpha_{\mathrm{II}}} = 1, \qquad r_{\alpha_{\mathrm{II}}} = 1, \qquad \iota_{\alpha_{\mathrm{II}},M} = +1.$$

Its row value and scalar gate are

$$R^{\mathrm{row}}_{\alpha_{\mathrm{II}},M} = \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_{\mathrm{II}}}(\mathfrak{m})\big)\omega(z_1, z_1) \int_I a(t)b(t)\gamma_{\mathbf{I}}(t)\, dt = 0,$$

$$S_{\alpha_{\mathrm{II}},M} = \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_{\mathrm{II}}}(\mathfrak{m})\big)\omega(z_1, z_1)\gamma_{\mathbf{I}} = 0.$$

Here the coefficient is the extracted $\hbar^1$-coefficient; the full scalar-contact representative has the additional factor $\hbar$. For $M \geq N \geq 1$, retain the same labelled row in window N and set

$$t^{M,N}_{\alpha_{\mathrm{II}}\alpha_{\mathrm{II}}} = 1, \qquad t^{M,N}_{\alpha_{\mathrm{II}}\alpha'} = 0 \quad (\alpha' \neq \alpha_{\mathrm{II}}).$$

These coefficients satisfy the truncation equations because $\pi_{M,N} R^{\mathrm{row}}_{\alpha_{\mathrm{II}},M} = 0$ and $\pi_{M,N} S_{\alpha_{\mathrm{II}},M} = 0$.

The associated kernel class is therefore

$$\mathfrak{o}^{\mathrm{ns}}_{1,M}(\alpha_{\mathrm{II}}) = [R^{\mathrm{row}}_{\alpha_{\mathrm{II}},M}] = 0 \in H^1\Big(\mathcal{K}^{\bullet}_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}\Big),$$

and the required counterterm contribution is $C_{1,M} = 0$.

Thus the finite closed graph-list axioms do not support an absence theorem for literal scalar-zero order-one rows beyond α_{sc}. The first explicit such row is harmless: it is zero before cohomology. If the row convention retains only nonzero summands, then α_{II} is omitted, and a nonzero order-one class still requires an additional supplied row $R^{\mathrm{row}}_{\alpha,M}$ of the form stated in Proposition E.39.

Proof. The scalar-contact boundary formula in Proposition 3.174 applies to every ordered pair of Hamiltonian labels in the finite window. For the pair (z_1, z_1),

$$\omega(z_1, z_1) = \big[\{z_1, z_1\}_{\Omega_{\mathrm{loc}}}\big]_0 = 0.$$

Multiplying by the transported full-equivariant cyclic supertrace cannot change this zero coefficient. Hence both the local functional row and its scalar projection vanish.

Since the row value is zero, $d_{\mathrm{ns},M}$-closedness is automatic and the cohomology class is the zero class. The displayed truncation coefficients send the labelled row to the same labelled row in lower windows containing z_1, and both sides of the row and scalar truncation identities are zero. The zero counterterm solves the row equation. The final assertion is logical: the finite-list axioms allow this added labelled zero row, so they cannot prove that no scalar-zero order-one row exists; they can only decide a nonzero order-one class after the actual nonzero order-one row values have been supplied. □

Remark E.41 *(Order two after no nonzero order-one row).* If zero rows such as α_{11} are suppressed and the finite marked list supplies no nonzero order-one row outside α_{sc}, then $C_{1,M} = 0$ and every order-two bracket row containing α_{sc} is zero by Proposition 3.175. The order-two scalar-zero residual reduces to

$$R^{\mathrm{ns}}_{2,M} = \sum_{|\Gamma|_\hbar=2} \varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{|\Gamma'|_\hbar=2} \varepsilon^{\mathrm{CE}}_{\Gamma'} K^M_{\Gamma'} + \frac{1}{2} \sum_{\substack{|\Gamma_1|_\hbar=|\Gamma_2|_\hbar=1 \\ \Gamma_i \neq \alpha_{\mathrm{sc}}}} \varepsilon_{\Gamma_1,\Gamma_2} \{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M}.$$

Under the stronger minimal hypotheses of Theorem E.38 the three sums are empty, and $R^{\mathrm{ns}}_{2,M} = 0$. In any larger finite package this displayed row sum is a cochain only after the genuine order-two graph weights, CE-source faces, bracket signs, scalar gates, and truncation coefficients have been supplied.

THEOREM E.42 *(Minimal full-equivariant all-order vanishing).* Work in the local $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$ full-equivariant marked branch with $V = \mathbb{C}^{N|N}$. Let $\mathcal{G}^{\mathrm{mk,o}}_M$ be the finite marked list generated by the strict current-interface rows, the two-Hamiltonian scalar-contact boundary row α_{sc}, and its subset-deletion closure in the sense of Definition E.25. Assume that no further positive-order seed graph, CE-source face, graph counterterm face, or nonzero scalar-zero row is added to the finite-window package.

Then for every $n \geq 1$

$$R^{\mathrm{ns}}_{n,M} = 0, \qquad \mathfrak{d}^{\mathrm{ns}}_{n,M} = 0, \qquad C_{n,M} = 0$$

If the row convention suppresses zero rows, then $\mathsf{A}^{\mathrm{fw,o}}_{n,M}$ is empty for every $n \geq 1$. If it retains labelled zero rows, every positive-order retained row has

$$R^{\mathrm{row}}_{\alpha,M} = 0, \qquad S_{\alpha,M} = 0,$$

and may be given the zero truncation row $t^{M,N}_{\alpha\alpha'} = 0$. In particular, the minimal full-equivariant order-three residual is zero.

Proof. Proposition 3.174 proves that the one-cross scalar-contact component satisfies $K^M_{\alpha_{\mathrm{sc}}} = 0$ as a local functional, not only after scalar projection. The proof is the cyclic-supertrace calculation of Lemma E.29: every transported full-equivariant marker on a scalar open loop has cyclic supertrace zero, including the empty marker word.

Every positive-order boundary row in $\mathcal{G}^{\mathrm{mk,o}}_M$ contains at least one such scalar-contact factor. Hence its local-functional value is zero. Bracket rows in the minimal package have at least one positive-order factor of this form, so bilinearity of the BV/convolution bracket gives zero. Lower counterterm rows vanish inductively because the chosen lower counterterms are zero. Genuine dK, CE-source, and graph-counterterm rows are absent by the minimality hypothesis. Therefore the order-n scalar-zero row sum is zero for all $n \geq 1$, its class in $H^1(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M})$ is zero, and the zero counterterm is a compatible primitive. Truncation sends zero rows to zero rows, giving the displayed truncation choice. □

PROPOSITION E.43 *(First order-three larger-package datum).* Keep the full-equivariant local branch, and suppose the minimal package has been closed through order two with $C_{1,M} = C_{2,M} = 0$ and no nonzero order-one or order-two row outside the scalar-contact closure. Then the finite graph-list

axioms do not by themselves supply a nonzero order-three obstruction. A nonzero order-three cochain exists only after the finite-window package supplies at least one of the following row data:

$$\begin{aligned} R^{\mathrm{ns}}_{3,M} = & \sum_{|\Gamma|_\hbar=3} \varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{|\Gamma'|_\hbar=3} \varepsilon^{\mathrm{CE}}_{\Gamma'} K^M_{\Gamma'} \\ & + \frac{1}{2} \sum_{\substack{|\Gamma_1|_\hbar+|\Gamma_2|_\hbar=3 \\ K^M_{\Gamma_1}\neq 0,\ K^M_{\Gamma_2}\neq 0}} \varepsilon_{\Gamma_1,\Gamma_2} \{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M} \\ & + [\hbar^3]\Big(d_M C_{<3,M} + \frac{1}{2}\{C_{<3,M}, C_{<3,M}\}_{\hbar,M}\Big). \end{aligned}$$

Under the preceding minimal hypotheses the bracket and lower-counterterm sums are empty. Thus the first missing order-three datum is a genuine $|\Gamma|_\hbar = 3$ graph-differential row

$$R^{\mathrm{row}}_{d\Gamma,M} = \varepsilon_\Gamma\, d_M K^M_\Gamma, \qquad S_{d\Gamma,M} = \varepsilon_\Gamma\, \widehat{\sigma}_{\mathrm{sc},M}(d_M K^M_\Gamma),$$

or an order-three CE-source row

$$R^{\mathrm{row}}_{\mathrm{CE}(\Gamma,\nu),M} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu}), \qquad S_{\mathrm{CE}(\Gamma,\nu),M} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} \widehat{\sigma}_{\mathrm{sc},M}\big(K^M_\Gamma(\zeta_{M,\nu})\big).$$

In a nonminimal package with supplied nonzero order-one and order-two components one must also supply each bracket row

$$R^{\mathrm{row}}_{b(\Gamma_1,\Gamma_2),M} = \frac{1}{2}\varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M}, \qquad S_{b,M} = \widehat{\sigma}_{\mathrm{sc},M}(R^{\mathrm{row}}_{b(\Gamma_1,\Gamma_2),M}),$$

for $|\Gamma_1|_\hbar + |\Gamma_2|_\hbar = 3$, together with its truncation coefficients $t^{M,N}_{bb'}$.

The scalar gate for order three is the single condition

$$\mathfrak{s}_{3,M} = \sum_{\alpha\in \mathsf{A}^{\mathrm{fw}}_{3,M}} S_{\alpha,M} = 0.$$

Only after this gate is supplied and vanishes is the non-scalar obstruction class defined:

$$\mathfrak{o}^{\mathrm{ns}}_{3,M} = [R^{\mathrm{ns}}_{3,M}] \in H^1\Big(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}\Big).$$

A fixed-window counterterm $C_{3,M}$ exists exactly when this class vanishes; a compatible tower also requires the corresponding $\varprojlim^1 H^0$-compatibility class to vanish.

Proof. The order-three expansion is Theorem 3.185 with $n = 3$, written in the row notation of Definition 3.173. Rows containing α_{sc} vanish by Proposition 3.175, and rows containing any retained labelled zero scalar-contact row vanish by the same bilinearity argument. Since $C_{1,M} = C_{2,M} = 0$, the lower-counterterm contribution is zero in the minimal package. Therefore no nonzero order-three cochain is produced unless the package supplies a genuine order-three graph row, an order-three CE-source row, or a bracket row between already supplied nonzero lower-order components. The displayed formulas are precisely the row values and scalar gates required by Definition 3.173. The final assertions are the kernel obstruction criterion of Definition E.31 and the finite-window compatibility criterion of Theorem 3.185. □

PROPOSITION E.44 *(Marked-row test for unreduced centrality homotopies).* Work in a finite-window marked Costello list for the local mixed HT brane-defect theory, with labels and source faces as in Definition E.25. Let $\kappa^{M}_{\mathcal{G},w,I}$ be the Hom-valued finite-window kernel carried by this list. For source generators $x, y \in \{u_{a(t)dt\otimes f}, c_{B,\rho}\}$, the marked row testing unreduced centrality is

$$R^{\mathrm{cent}}_{x,y,M} = \mathrm{Curv}_{\mathbf{K},M}(\kappa^{M}_{\mathcal{G},w,I})(x \wedge y).$$

In row notation this is the finite sum

$$\begin{aligned} R^{\mathrm{cent}}_{x,y,M} = &\sum_{\Gamma\in\mathsf{A}^{d}_{x,y,M}} \varepsilon_\Gamma\, d_M K^{M}_{\Gamma} - \sum_{\Gamma\in\mathsf{A}^{\mathrm{CE}}_{x,y,M}} \varepsilon^{\mathrm{CE}}_{\Gamma} K^{M}_{\Gamma}(\zeta_{\Gamma,x,y}) \\ &+ \frac{1}{2} \sum_{(\Gamma_1,\Gamma_2)\in\mathsf{A}^{\mathrm{br}}_{x,y,M}} \varepsilon_{\Gamma_1,\Gamma_2}\{K^{M}_{\Gamma_1}, K^{M}_{\Gamma_2}\}_{\hbar,M} + R^{\mathrm{ct}}_{x,y,M}. \end{aligned}$$

Here the four summands are respectively the target differential rows, the CE-source rows, the convolution bracket rows, and the supplied counterterm rows of the marked table. The scalar gate is

$$S^{\mathrm{cent}}_{x,y,M} = \widehat{\sigma}_{\mathrm{sc},M}(R^{\mathrm{cent}}_{x,y,M}).$$

Only after this gate is zero does the centrality row define a class

$$\mathfrak{o}^{\mathrm{cent}}_{x,y,M} = [R^{\mathrm{cent}}_{x,y,M}] \in H^{|x|+|y|}(\mathcal{K}^{\bullet}_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}).$$

A marked finite-window package supplies an actual unreduced centrality homotopy for the pair (x, y) exactly when it also supplies a degree $|x| + |y| - 1$ row $H^{M}_{x,y} \in \mathcal{K}^{|x|+|y|-1}_{\mathrm{ns},M}(I)$ with

$$R^{\mathrm{cent}}_{x,y,M} + d_M H^{M}_{x,y} = 0,$$

and these rows satisfy the truncation and weight identities of Theorem B.22. If such a row or compatibility identity is absent from the table, the missing datum is exactly the omitted centrality-homotopy row, not a consequence of scalar-shadow vanishing.

In the minimal full-equivariant branch of Theorem E.42, every positive-order row contains the zero scalar-contact factor $K^{M}_{\alpha_{\mathrm{sc}}} = 0$, and no additional positive-order seed graph, CE-source face, graph counterterm face, or nonzero scalar-zero row is present. Hence

$$R^{\mathrm{cent}}_{x,y,M} = 0, \qquad H^{M}_{x,y} = 0$$

is a valid homotopy inside that minimal finite marked habitat. This does not construct the centrality homotopy for an enlarged unreduced Costello graph/counterterm target; any larger package must supply the displayed row values, scalar gate, primitive $H^{M}_{x,y}$, and transition coefficients.

Proof. Proposition E.28 identifies the marked differential with the finite-window QME differential and, for Hom-valued kernels, adds the source face $-\kappa d_{\mathrm{CE}}$. Therefore the binary curvature on $x \wedge y$ is exactly the sum of the four row types displayed above: target differential, CE-source subtraction, convolution bracket, and counterterm contribution. Applying the scalar projection gives the scalar gate. If the gate vanishes, the row lies in the non-scalar kernel complex of Definition E.31; its cohomology class is the obstruction to finding a fixed-window primitive.

The equation $R^{\mathrm{cent}}_{x,y,M} + d_M H^M_{x,y} = 0$ is precisely the homotopy equation (14). The transition identities are necessary because the local kernel is the inverse limit over finite windows and admissible weights. Thus a row not present in the marked table cannot be inferred from the scalar projection; it is the missing graph, source-face, counterterm, or homotopy datum.

In the minimal full-equivariant package all positive-order scalar-contact rows are zero before cohomology by Theorem E.42. Since no other positive-order row is allowed by the minimal hypothesis, the centrality row is zero and the zero primitive is compatible with all truncation maps. The final assertion is the contrapositive of the row formula: once an enlarged package adds a nonzero graph, source, bracket, or counterterm row, the corresponding scalar gate and primitive must be supplied explicitly. □

Proposition E.45 *(Q_Ω-centrality homotopy criterion).* Let $(I, w, M, \mathcal{G}_M)$ carry a normal Ω-equivariant Costello kernel datum in the sense of Definition E.36, and assume the scalar projection has passed the Hom-obstruction test of Theorem E.37. For T_Ω-invariant source generators

$$x, y \in \{u_{a(t)dt\otimes f}, c_{B,\rho}\},$$

define the equivariant centrality row

$$R^{\mathrm{cent}}_{x,y,\Omega,M} = \mathrm{Curv}_{\mathbf{K},\Omega,M}(\kappa^M_{\mathcal{G},\Omega,w,I})(x \wedge y).$$

Its finite-row expansion is

$$R^{\mathrm{cent}}_{x,y,\Omega,M} = \sum_{\Gamma\in\mathsf{A}^d_{x,y,\Omega,M}} \varepsilon_\Gamma\, d_{M,\Omega} K^M_{\Gamma,\Omega} - \sum_{\Gamma\in\mathsf{A}^{\mathrm{CE}}_{x,y,\Omega,M}} \varepsilon^{\mathrm{CE}}_\Gamma K^M_{\Gamma,\Omega}(\zeta_{\Gamma,x,y,\Omega})$$
$$+ \frac{1}{2} \sum_{(\Gamma_1,\Gamma_2)\in\mathsf{A}^{\mathrm{br}}_{x,y,\Omega,M}} \varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1,\Omega}, K^M_{\Gamma_2,\Omega}\}_{\hbar_{\mathrm{QME}},M,\Omega} + R^{\mathrm{ct}}_{x,y,\Omega,M}.$$

The scalar gate is

$$S^{\mathrm{cent}}_{x,y,\Omega,M} = \widehat{\sigma}_{\mathrm{sc},\Omega,M}(R^{\mathrm{cent}}_{x,y,\Omega,M}).$$

Only after this gate vanishes does one obtain the obstruction class

$$\mathfrak{o}^{\mathrm{cent}}_{x,y,\Omega,M} = [R^{\mathrm{cent}}_{x,y,\Omega,M}] \in H^{|x|+|y|}(\mathcal{K}^\bullet_{\mathrm{ns},\Omega,M}(I), d_{\mathrm{ns},\Omega,M}).$$

A Q_Ω-centrality homotopy for (x, y) is exactly a T_Ω-invariant row

$$H^\Omega_{x,y,M} \in \mathcal{K}^{|x|+|y|-1}_{\mathrm{ns},\Omega,M}(I)$$

satisfying

$$R^{\mathrm{cent}}_{x,y,\Omega,M} + d_{M,\Omega} H^\Omega_{x,y,M} = 0, \qquad L_{V_\Omega} H^\Omega_{x,y,M} = 0,$$

together with the same interval, finite-window, and admissible-weight transition identities as in Theorem B.22. Equivalently, the inverse-system class

$$\left(\mathfrak{o}^{\mathrm{cent}}_{x,y,\Omega,M}\right)_M \in \varprojlim_M H^{|x|+|y|}(\mathcal{K}^\bullet_{\mathrm{ns},\Omega,M}(I))$$

and the corresponding $\varprojlim^1 H^{|x|+|y|-1}$-primitive compatibility class must both vanish.

If one works before passing to T_Ω-invariants, then $Q^2_\Omega = L_{V_\Omega}$ gives a secondary obstruction

$$\ell^{\mathrm{cent}}_{x,y,\Omega,M} = [L_{V_\Omega} H_{x,y,\Omega,M}] \in H^{|x|+|y|}(\mathcal{K}^\bullet_{\mathrm{ns},\Omega,M}(I), d_{\mathrm{ns},\Omega,M})$$

for any chosen primitive. This class must vanish, or the primitive must be replaced by a basic one, before the homotopy is a Q_Ω-homotopy.

Proof. The row expansion is the Hom-valued curvature computation of Proposition E.44 with d_M, the CE differential, the bracket, and the counterterm table replaced by their equivariant counterparts from Definition E.36. The scalar gate is necessary because only its vanishing places the row in $\mathcal{K}^{\bullet}_{\mathrm{ns},\Omega,M}$. Once there, the equivariant Bianchi identity makes the row closed, so its cohomology class is the obstruction to a fixed-window primitive.

The displayed equation is precisely the centrality homotopy equation in the invariant Q_Ω-complex. Since $Q_\Omega^2 = L_{V_\Omega}$, invariance of the primitive is part of the datum, not a consequence of scalar-shadow vanishing. The inverse-limit assertion is the same primitive-compatibility argument as in Theorem B.22. If a primitive is chosen outside the basic subcomplex, applying $d_{M,\Omega}$ a second time records $L_{V_\Omega} H$; its cohomology class is exactly the secondary obstruction displayed above. □

PROPOSITION E.46 *(Even-block marker obstruction).* Replace full $\mathfrak{gl}(V)$-equivariance by the smaller even block-diagonal habitat

$$\mathfrak{gl}(V_{\bar{0}}) \oplus \mathfrak{gl}(V_{\bar{1}}) \subset \mathfrak{gl}(V), \qquad V_{\bar{0}} \cong V_{\bar{1}} \cong \mathbb{C}^N.$$

Then a single fixed even marker is admissible exactly when

$$T = \lambda_0 P_{\bar{0}} + \lambda_1 P_{\bar{1}},$$

where $P_{\bar{0}}$ and $P_{\bar{1}}$ are the parity block projections. Its scalar shadow is

$$\mathrm{Str}(T) = N(\lambda_0 - \lambda_1).$$

For a two-Hamiltonian scalar contact the obstruction class is

$$\hbar N(\lambda_0 - \lambda_1)\, \omega(f, g)\gamma_{\mathbf{1}}, \qquad \hbar N(\lambda_0 - \lambda_1)[\bar{c}]\, \gamma_{\mathbf{1}} \text{ in cohomology.}$$

Since $[\bar{c}] \neq 0$ in the scalar-reduced source, this smaller habitat has zero $[\bar{c}]$-shadow for single markers if and only if $\lambda_0 = \lambda_1$, i.e. the marker is scalar and the habitat has collapsed back to the full-equivariant single-marker case. For higher marker tensors the strongest true positive statement is exactly the supertraceless-monodromy subhabitat of Theorem E.23; outside it the obstruction is the displayed $\mathrm{Str}_{\mathrm{cyc}}$-weighted $[\bar{c}]$-class.

Proof. The commutant of $\mathfrak{gl}(V_{\bar{0}}) \oplus \mathfrak{gl}(V_{\bar{1}})$ on $V_{\bar{0}} \oplus V_{\bar{1}}$ is $\mathbb{C}P_{\bar{0}} \oplus \mathbb{C}P_{\bar{1}}$, so the displayed form of T is necessary and sufficient. The supertrace is

$$\mathrm{Str}(\lambda_0 P_{\bar{0}} + \lambda_1 P_{\bar{1}}) = \lambda_0 \dim V_{\bar{0}} - \lambda_1 \dim V_{\bar{1}} = N(\lambda_0 - \lambda_1).$$

Proposition E.21 then gives the scalar CE representative and the cohomology class by multiplying ω, respectively $[\bar{c}]$, by this coefficient. The class $[\bar{c}]$ is nonzero by Theorem E.12; hence the class vanishes for all two-Hamiltonian tests exactly when $\lambda_0 = \lambda_1$. The higher tensor assertion is the same calculation with $\mathrm{Str}(T)$ replaced by $\mathrm{Str}_{\mathrm{cyc}}$ and with the preservation criterion supplied by Theorem E.23. □

THEOREM E.47 *(Ordinary scalar-reduced $\mathfrak{gl}_N$ obstruction criterion).* Work in the scalar-reduced source of Definition E.3 and keep the ordinary trace on the open $\mathfrak{gl}_N$ brane. Let W_{ext} be any additional formal-local closed-exchange or brane-defect graph contribution whose leading scalar associated graded is $w_{\mathrm{ext}} \in C^2_{\mathrm{Lie}}(\bar{A}; \mathbb{C})$. A cancellation

$$\mathrm{Ob}_{\mathrm{sc}} + W_{\mathrm{ext}} + d_{\mathrm{QME}} C = 0$$

with $C \in O^{\mathrm{bd}}_{\mathrm{loc}}$ can hold only if

$$[w_{\mathrm{ext}}] = -\hbar N[\bar{c}] \qquad \text{in } H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]].$$

In particular, with $W_{\mathrm{ext}} = 0$, or with w_{ext} exact, the ordinary scalar-reduced $\mathfrak{gl}_N$ brane-defect model has no scalar QME cancellation.

Proof. Taking associated graded with respect to the scalar-contact filtration sends the equation

$$\mathrm{Ob}_{\mathrm{sc}} + W_{\mathrm{ext}} + d_{\mathrm{QME}} C = 0$$

to

$$\hbar N \omega + w_{\mathrm{ext}} + \partial \bar{\eta} = 0 \qquad \text{in } C^2_{\mathrm{Lie}}(\bar{A};\mathbb{C}),$$

where $\bar{\eta}$ is the linear functional represented by the leading scalar part of C. Passing to cohomology gives the stated necessary condition. If $W_{\mathrm{ext}} = 0$, or if w_{ext} is exact, this condition would force $\hbar N[\bar{c}] = 0$ in $H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]]$. For $N > 0$ and formal $\hbar$, this contradicts Theorem E.12, which identifies $[\bar{c}] \neq 0$ on $\bar{A}$. Thus no internal local counterterm cancels the ordinary scalar-reduced obstruction. □

F Radial Parts and the Moyal Normalization

This appendix records the proof spine for Theorem 2.62. The stable core is the scalar-reduced Weyl/Moyal normalization, the Capelli change of basis, the stable connected trace bracket, and the reduced open-line Wick graph expansion with midpoint propagator $\frac{1}{2}\operatorname{sgn}(t-s)$. The finite-$N$ radial-parts comparison is isolated as the explicit input package stated below. Once that package is assumed, the radial argument is algebraic: Vandermonde conjugation removes the Dunkl–Calogero connection terms, injectivity detects the reduced quotient, and the stable connected cumulant projection extracts the single-trace Moyal bracket.

Definition F.1 *(Formal Weyl/Moyal convention).* Let

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

The formal Weyl product on $\mathbb{C}[[z_1, z_2]][[\hbar]]$ is

$$f * g = m \circ \exp\left(\frac{\hbar}{2} P\right)(f \otimes g),$$

and

$$[f, g]_{\mathrm{raw}} = f * g - g * f, \qquad \{f, g\}_\hbar = \hbar^{-1}[f, g]_{\mathrm{raw}}.$$

The reduced quantum Hamiltonian Lie algebra is the scalar quotient

$$\bar{A}_\hbar = \mathbb{C}[[z_1, z_2]][[\hbar]]/\mathbb{C}[[\hbar]]$$

as a Lie algebra. It is not used as an associative quotient.

Theorem F.2 *(Exact formal Weyl/Moyal coefficients).* For $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$,

$$P^r(f, g) = \sum_{a=0}^{r} (-1)^a \binom{r}{a} (k)_{r-a} (l)_a (m)_a (n)_{r-a} z_1^{k+m-r} z_2^{l+n-r}.$$

Consequently

$$[f,g]_{\mathrm{raw}} = \sum_{\substack{r\geq 1\\ r \text{ odd}}} \frac{2}{2^r r!}\hbar^r P^r(f,g),$$

equivalently, for $s \geq 0$, the raw coefficient of $\hbar^{2s+1}$ is

$$\frac{1}{2^{2s}(2s+1)!}\, P^{2s+1}(f,g),$$

and the normalized Hamiltonian bracket has coefficient

$$\frac{\hbar^{2s}}{2^{2s}(2s+1)!}\, P^{2s+1}(f,g).$$

The first and third raw commutator coefficients are

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24}P^3(f,g),$$

and all even raw commutator coefficients vanish. This is a theorem inside the formal Weyl algebra of the constant symplectic plane. It uses no finite-N radial-parts theorem and no Costello graph specialization.

Proof. The two summands in $P = \partial_{z_1}\otimes\partial_{z_2} - \partial_{z_2}\otimes\partial_{z_1}$ commute as differential operators on the tensor product. The binomial theorem gives

$$P^r = \sum_{a=0}^{r}(-1)^a\binom{r}{a}\partial_{z_1}^{r-a}\partial_{z_2}^{a}\otimes\partial_{z_1}^{a}\partial_{z_2}^{r-a}.$$

Applying this to the two monomials gives the falling-factorial formula. Interchanging f and g changes $P^r(f,g)$ by $(-1)^r$, so the commutator retains exactly the odd powers and contributes the coefficient $2/(2^r r!)$ in order r. □

COROLLARY F.3 *(All odd Moyal coefficients).* The Weyl/Moyal commutator has no even raw powers of $\hbar$. Its odd raw and normalized coefficients are

$$[f,g]_{\mathrm{raw}}^{(2s+1)} = \frac{\hbar^{2s+1}}{2^{2s}(2s+1)!}P^{2s+1}(f,g), \qquad \{f,g\}_{\hbar}^{(2s)} = \frac{\hbar^{2s}}{2^{2s}(2s+1)!}P^{2s+1}(f,g).$$

In particular the next two raw coefficients after the third-order term are

$$\frac{\hbar^5}{1920}P^5(f,g), \qquad \frac{\hbar^7}{322560}P^7(f,g).$$

Proof. Put $r = 2s+1$ in Theorem F.2. Then $2/(2^{2s+1}(2s+1)!) = 1/(2^{2s}(2s+1)!)$. Dividing by $\hbar$ gives the normalized coefficient. □

COROLLARY F.4 *(First and third formal graph weights).* In the ordered formal product $m \circ \exp(\hbar P/2)$, the one-cross-contraction term has weight $\hbar P(f,g)/2$, and the three-cross-contraction term has weight $\hbar^3 P^3(f,g)/48$. After antisymmetrising the ordered product, the commutator weights are

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24}P^3(f,g).$$

Proof. The ordered exponential contributes

$$\frac{1}{r!}\left(\frac{\hbar}{2}\right)^r P^r(f,g)$$

in order r. For $r = 1$ this is $\hbar P(f,g)/2$; for $r = 3$ it is $\hbar^3 P^3(f,g)/(3!\,2^3) = \hbar^3 P^3(f,g)/48$. Since $P(g,f) = -P(f,g)$ and $P^3(g,f) = -P^3(f,g)$, subtracting the reverse ordered product doubles these two ordered contributions. □

Example F.5 *(A higher-order coefficient check).* For

$$f = z_1^4 z_2^2, \qquad g = z_1^2 z_2^4,$$

the first nonzero odd contractions are

$$P(f,g) = 12 z_1^5 z_2^5, \qquad P^3(f,g) = -960 z_1^3 z_2^3, \qquad P^5(f,g) = 11520 z_1 z_2.$$

Hence

$$[f,g]_{\mathrm{raw}} = 12\hbar z_1^5 z_2^5 - 40\hbar^3 z_1^3 z_2^3 + 6\hbar^5 z_1 z_2.$$

The third coefficient is $-960/24 = -40$. The fifth coefficient is $11520/1920 = 6$. This example checks the numerical normalization beyond the first nontrivial term without using radial parts.

Proof. Apply the falling-factorial formula of Theorem F.2. For $r = 1$,

$$4 \cdot 4 - 2 \cdot 2 = 12.$$

For $r = 3$,

$$\begin{aligned} P^3(f,g) &= \big(24 \cdot 24 - 3 \cdot 12 \cdot 2 \cdot 2 \cdot 12 + 3 \cdot 4 \cdot 2 \cdot 2 \cdot 4\big) z_1^3 z_2^3 \\ &= -960 z_1^3 z_2^3. \end{aligned}$$

For $r = 5$, only the $a = 1$ and $a = 2$ summands survive, and they give

$$\big(-5 \cdot 24 \cdot 2 \cdot 2 \cdot 24 + 10 \cdot 24 \cdot 2 \cdot 2 \cdot 24\big) z_1 z_2 = 11520 z_1 z_2.$$

Substituting the odd Moyal coefficients 1, $1/24$, and $1/1920$ gives the displayed raw commutator. □

Definition F.6 *(Matrix Weyl algebra and Weyl traces).* Let $\mathcal{W}_N$ be the completed Weyl algebra generated by entries of two $N \times N$ matrices X, Y with

$$[X^i{}_j, Y^k{}_l] = \hbar\, \delta^i_l \delta^k_j, \qquad [X^i{}_j, X^k{}_l] = [Y^i{}_j, Y^k{}_l] = 0.$$

The normal-ordered quantum comoment for conjugation is

$$\widehat{\mu}_\epsilon = -\operatorname{Tr}([\epsilon, X]Y), \qquad \hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a.$$

For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, define the Weyl-ordered trace

$$J_N(f) = \operatorname{Tr} \operatorname{Op}_{\mathrm{W}}(f)(X,Y).$$

The degree-zero quantum Hamiltonian reduction uses the effective traceless comoment ideal

$$\mathrm{N}_{\mathcal{W}_N}(\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N))/\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N).$$

The scalar Hamiltonian line is not killed by this moment ideal itself; below it is removed by quotienting the empty trace before connected large-N extraction.

LEMMA F.7 *(Capelli–Weyl triangular normalization).* Let $T_{a,b} = \mathrm{Tr}(X^a Y^b)$ be the normal-ordered trace. In the quantum Hamiltonian reduction, the locally proved Weyl-normal-ordering formula is

$$J_N(z_1^a z_2^b) \equiv \sum_{r=0}^{\min(a,b)} \left(-\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} T_{a-r,b-r}.$$

The inverse triangular system is

$$T_{a,b} \equiv \sum_{r=0}^{\min(a,b)} \left(\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} J_N(z_1^{a-r} z_2^{b-r}).$$

Proof. Total Weyl symmetrisation is obtained from normal ordering by commuting Y's past X's. Each contraction contributes the Capelli contact term coming from

$$YX - XY + \hbar N I \equiv 0 \pmod{\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)}.$$

Choosing r contracted X-slots and r contracted Y-slots gives $\binom{a}{r}\binom{b}{r}$, pairing them gives $r!$, and Weyl ordering contributes 2^{-r}. The sign is negative because the displayed moment relation rewrites a normal-ordering swap as the lower triangular term $-\hbar N$. The diagonal term is $r = 0$, hence the system is unipotent over $\mathbb{C}[[\hbar N]]$; inversion changes $-\hbar N/2$ to $+\hbar N/2$.

The cited Capelli literature supplies the standard source for the Capelli contact phenomenon, and the displayed contraction count fixes the bivariate trace normalization used here. □

Statement F.8 *(Radial-parts input package).* The finite-N radial-parts step uses the following Harish-Chandra radial-parts theorem for invariant differential operators, in the form refined by Wallach and by Levasseur–Stafford [8–11]. After identifying $\mathcal{W}_N \simeq \mathcal{D}_\hbar(\mathfrak{gl}_N)$, this package uses the Harish-Chandra map on invariants whose kernel is the quantum moment-map ideal and hence injects the quantum Hamiltonian reduction by conjugation into Weyl-invariant differential operators on the regular Cartan. With $\lambda_1, \ldots, \lambda_N$ the Cartan coordinates and $\Delta = \prod_{i<j}(\lambda_i - \lambda_j)$, the radial-part map is conjugated by Δ.

The cited radial-parts literature sources the Harish-Chandra homomorphism, regular-Cartan landing, and kernel statement. The exact Weyl-trace image in item (3) is the matched normalization datum used in the formal-local comparison. It is not supplied by the cited radial-parts sources in this exact Weyl/Capelli convention. The finite exact checks below attack it in ranks 2 and 3, but they do not replace the all-N radial-parts input.

The input consists of the following three assertions.

(i) The Vandermonde-conjugated radial map is a homomorphism from the invariant quantum Hamiltonian reduction by conjugation to Weyl-invariant differential operators on the regular Cartan.

(ii) Its kernel is precisely the quantum moment-map ideal; hence it is injective on the reduced quotient.

(iii) The Weyl trace generators used here have image

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1}\left(\sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i})\right)\Delta.$$

The nonlinear-shear test in Lemma F.13 shows that pure z_1- and z_2-power images together with classical symbol-level shear compatibility do not determine item (3). A smaller source package would need a genuine quantum-corrected Egorov theorem in the Weyl algebra.

Definition F.9 *(Radially normalized Weyl trace class).* Assume only the homomorphism and injectivity clauses of Statement F.8. For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, a class $J_N^{\mathrm{rad}}(f)$ in the degree-zero quantum Hamiltonian reduction is called radially normalized if

$$\mathrm{Rad}_0(J_N^{\mathrm{rad}}(f)) = \Delta^{-1} S_N(f)\Delta.$$

Existence is a normalization hypothesis; uniqueness follows from radial injectivity. Statement F.8(3) is exactly the assertion that the ordinary Weyl trace $J_N(f)$ equals $J_N^{\mathrm{rad}}(f)$.

Definition F.10 *(Vandermonde gauge variables).* Set

$$\nabla_i^{\Delta} = \Delta^{-1}(-\hbar\partial_{\lambda_i})\Delta = -\hbar\partial_{\lambda_i} - \hbar \sum_{j\neq i} \frac{1}{\lambda_i - \lambda_j}.$$

The logarithmic summand is the type A Dunkl–Calogero correction in the Vandermonde gauge. For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, put

$$S_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i})$$

and

$$R_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, \nabla_i^{\Delta}) = \Delta^{-1} S_N(f)\Delta.$$

Lemma F.11 *(Dunkl–Calogero correction cancellation).* The Vandermonde-gauge variables satisfy

$$[\lambda_i, \lambda_j] = 0, \qquad [\nabla_i^{\Delta}, \nabla_j^{\Delta}] = 0, \qquad [\lambda_i, \nabla_j^{\Delta}] = \hbar\,\delta_{ij}.$$

Consequently

$$\hbar^{-1}[R_N(f), R_N(g)] = R_N(\{f, g\}_{\hbar})$$

for all f, g. In particular, all logarithmic Dunkl–Calogero terms introduced by expanding $\Delta^{-1} S_N(-)\Delta$ cancel in the commutator identity.

Proof. The operator ∇_i^{Δ} is conjugate to $-\hbar\partial_{\lambda_i}$ by multiplication by Δ. The three displayed commutation relations are therefore the ordinary Weyl relations for $(\lambda_i, -\hbar\partial_{\lambda_i})$, conjugated by Δ. The correction terms are a pure gauge connection: their curvature is zero because the unconjugated partial derivatives commute. Hence the i-th summand is again a one-particle Weyl representation of the formal Moyal algebra, and different summands commute. This gives the displayed commutator identity. No separate Calogero potential remains after the computation is kept in Vandermonde-conjugated form. □

Lemma F.12 *(Radial-map injectivity on the reduced quotient).* Under Statement F.8, if an invariant Weyl operator D in the normalizer of the quantum moment-map ideal satisfies

$$\mathrm{Rad}_0(D) = 0,$$

then D is zero in the degree-zero quantum Hamiltonian reduction.

Proof. Statement F.8 identifies the kernel of the Vandermonde-conjugated Harish-Chandra radial map with the quantum moment-map ideal. Passing to the normalizer quotient by that ideal gives an injective map into Weyl-invariant differential operators on the regular Cartan. Thus an element with zero radial image represents the zero class in the reduced quotient. □

LEMMA F.13 *(Nonlinear shear obstruction to a smaller radial package).* Pure-power radial images plus classical symbol-level compatibility with nonlinear shears do not determine the mixed Weyl-trace image normalization. In the one-particle Weyl algebra $[\lambda, y] = \hbar$, the coefficient of t in

$$(y + t\lambda^2)^4$$

is

$$4\lambda^2 y^3 - 12\hbar\lambda y^2 + 8\hbar^2 y,$$

whereas

$$4\,\mathrm{Op}_{\mathrm{W}}(z_1^2 z_2^3) = 4\lambda^2 y^3 - 12\hbar\lambda y^2 + 6\hbar^2 y.$$

The difference $2\hbar^2 y$ is non-scalar. Thus a proof of Statement F.8(3) cannot be reduced to classical nonlinear shear covariance without adding the missing quantum Egorov correction.

Proof. Expand to first order in t:

$$(y + t\lambda^2)^4 = y^4 + t\sum_{j=0}^{3} y^j \lambda^2 y^{3-j} + O(t^2).$$

Using $y\lambda = \lambda y - \hbar$, the sum normal-orders to $4\lambda^2 y^3 - 12\hbar\lambda y^2 + 8\hbar^2 y$. Weyl symmetrisation of $z_1^2 z_2^3$ gives $\lambda^2 y^3 - 3\hbar\lambda y^2 + \frac{3}{2}\hbar^2 y$, hence the displayed value after multiplication by 4. The discrepancy is a nonconstant one-particle operator, so it cannot be removed by the scalar Hamiltonian quotient. □

PROPOSITION F.14 *(Quantum shear congruence suffices for the all-N image).* Assume the homomorphism and injectivity clauses of Statement F.8, and assume the pure power images

$$\mathrm{Rad}_0(J_N(z_1^r)) = \Delta^{-1} S_N(z_1^r)\Delta, \qquad \mathrm{Rad}_0(J_N(z_2^r)) = \Delta^{-1} S_N(z_2^r)\Delta$$

for all $r \geq 0$. Suppose moreover that, for all $a, b \geq 0$, the quantum shear congruence

$$\hbar^{-1}\big[J_N(z_1^{a+1}/(a+1)), J_N(z_2^{b+1})\big] \equiv J_N\Big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\Big) \pmod{I_N}$$

holds in the degree-zero quantum Hamiltonian reduction, where I_N is the quantum moment-map ideal. Then the exact Weyl-trace image in Statement F.8(3) holds for every monomial $z_1^a z_2^b$, hence for every polynomial f.

Proof. The normalized Moyal bracket gives

$$\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar = (b+1) z_1^a z_2^b + \sum_{s\geq 1} \frac{\hbar^{2s}(a+1)_{2s+1}(b+1)_{2s+1}}{(a+1)2^{2s}(2s+1)!} z_1^{a-2s} z_2^{b-2s}.$$

Every summand after the leading term has lower total degree. The pure power cases start the induction. Assume the asserted radial image is known in all total degrees $< a + b$. Applying the radial homomorphism to the quantum shear congruence identifies the radial image of the left-hand commutator with the Moyal commutator of the already known pure-power radial images. Lemma F.29 computes this as the Vandermonde-conjugated Weyl quantization of the displayed Moyal bracket. By the induction hypothesis all lower-degree terms have the required images, so the leading coefficient $(b+1)$ determines the image of $J_N(z_1^a z_2^b)$. Radial injectivity then identifies the class in the reduced quotient. □

Lemma F.15 *(Moment-ideal primitive form of the quantum shear input).* In the notation of Proposition F.14, the quantum shear congruence is equivalent to the following explicit left-ideal membership problem. Put

$$E_{a,b,N} = \sum_{j=0}^{b} \mathrm{Tr}(Y^j X^a Y^{b-j}) - J_N\Big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\Big).$$

Then the congruence holds if and only if $E_{a,b,N} \in \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$. Equivalently, after choosing normal-ordered matrix generators $\widehat{\mu}^i{}_j$ for the traceless quantum moment-map ideal, one must construct coefficients $A^j{}_i(a,b,N) \in \mathcal{W}_N$ such that

$$E_{a,b,N} = \sum_{i,j} A^j{}_i(a,b,N)\widehat{\mu}^i{}_j.$$

Proof. Since the pure powers are already Weyl ordered, $J_N(z_1^{a+1}/(a+1)) = \mathrm{Tr}(X^{a+1})/(a+1)$ and $J_N(z_2^{b+1}) = \mathrm{Tr}(Y^{b+1})$. The Weyl relations give

$$\hbar^{-1}\big[\mathrm{Tr}(X^{a+1})/(a+1), Y\big] = X^a.$$

Applying the Leibniz rule to Y^{b+1} gives

$$\hbar^{-1}\big[J_N(z_1^{a+1}/(a+1)), J_N(z_2^{b+1})\big] = \sum_{j=0}^{b} \mathrm{Tr}(Y^j X^a Y^{b-j}).$$

Subtracting the J_N-image of the normalized Moyal bracket gives exactly $E_{a,b,N}$. Reduction modulo I_N is reduction modulo the left ideal $\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$, so the displayed congruence is precisely the asserted membership. The coefficient formula is the same membership written after a choice of generators for that left ideal. □

Statement F.16 *(Free trace-diagram quantum-lift obstruction).* Put $M = YX - XY + \hbar N I$, and interpret

$$\mathrm{Tr}(WM) = \sum_{i,j} W^j{}_i M^i{}_j$$

as a left-ideal expression. Let $\mathsf{C}_{a,b}$ be the rational vector space spanned by words W with $a-1$ letters X and $b-1$ letters Y, and let $\mathsf{Cyc}_{a,b}$ be the cyclic word space with a letters X and b letters Y. The leading term of the desired primitive is the cyclic boundary equation

$$\partial c = \sum_{j=0}^{b} [Y^j X^a Y^{b-j}] - \frac{b+1}{\binom{a+b}{a}} \sum_{U \in \mathrm{Sh}(a,b)} [U], \qquad \partial W = [WYX] - [WXY].$$

An all-N primitive is stronger. It requires equality after free indexed normal ordering, including every contraction term which splits a trace into multi-traces:

$$\mathcal{N}_{\mathrm{free}}\left(\sum_W c_W \,\mathrm{Tr}(WM)\right) = \mathcal{N}_{\mathrm{free}}(E_{a,b,N}).$$

The cyclic equation alone does not imply this quantum equation.

In bidegree $(a,b) = (4,4)$, the lexicographic pivot solution of the cyclic equation is

$$\begin{aligned} C^{\mathrm{cl}}_{4,4} &= \frac{31}{7}\operatorname{Tr}(XXXYYY\,M) - \frac{4}{7}\operatorname{Tr}(XXYXYY\,M) - \frac{4}{7}\operatorname{Tr}(XXYYXY\,M) \\ &\quad + \frac{27}{7}\operatorname{Tr}(XXYYYX\,M) + \frac{23}{7}\operatorname{Tr}(XYXXYY\,M) + \frac{1}{7}\operatorname{Tr}(XYXYXY\,M) \\ &\quad + \operatorname{Tr}(XYXYYX\,M) - \frac{2}{7}\operatorname{Tr}(XYYXXY\,M) + \frac{15}{7}\operatorname{Tr}(XYYXYX\,M). \end{aligned}$$

Its quantum residual is nonzero. In rank $N = 2$, exact normal ordering gives

$$\begin{aligned} R^{\mathrm{cl}}_{4,4,2} &:= E_{4,4,2} - C^{\mathrm{cl}}_{4,4} \\ &= \frac{43}{70}\hbar^2\big(4\operatorname{Tr}(XXYY) - 5\operatorname{Tr}(XYXY) + 2\operatorname{Tr}(XYYX) \\ &\qquad\qquad - 5\operatorname{Tr}(YXYX) + 4\operatorname{Tr}(YYXX)\big) \neq 0. \end{aligned}$$

After expanding the displayed traces into normal-ordered matrix entries, the coefficient of $\hbar^2 X^1{}_1 X^1{}_1 Y^1{}_2 Y^2{}_1$ is $43/7$.

The residual is killed by the classical-kernel correction

$$\begin{aligned} K_{4,4} = \frac{43}{14}\big(&\operatorname{Tr}(XXYXYY\,M) - \operatorname{Tr}(XYXXYY\,M) \\ &- \operatorname{Tr}(XYYXYX\,M) + \operatorname{Tr}(XYYYXX\,M)\big), \end{aligned}$$

whose cyclic boundary is zero. The free normal-form computation below proves that this correction cancels the $(4,4)$ residual before any finite-rank trace identity is imposed. The remaining theorem obligation is the decorated PBW Stokes identity: prove, for every bidegree, that

$$\Omega^{\mathrm{rad}}_{a,b} = 0 \quad\Longleftrightarrow\quad D^{\square}_{a,b} H_{a,b} = R^{\mathrm{free}}_{a,b}, \qquad D^{\square}_{a,b} = C^{+}_{a,b}\partial_2.$$

If this fails, the certificate is not a timed-out finite computation but a normalized signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$ with $\lambda(E^{+}_{a,b}) - \phi(T_{a,b}) = 1$.

Proof. The leading $\hbar^0$ component of $\operatorname{Tr}(WM)$ is $[WYX] - [WXY]$ in the cyclic trace quotient; comparing it with the leading term of $E_{a,b,N}$ gives the displayed cyclic equation. Normal ordering in the indexed Weyl algebra contracts every occurrence of a Y-entry moved past an X-entry. These contractions identify matrix indices and may split one cyclic trace into a product of cyclic traces. Hence equality in the cyclic quotient is only the associated graded condition.

The $(4,4)$ coefficients displayed above solve the cyclic equation in the stated pivot gauge. Direct exact normal ordering in $\mathcal{W}_2$ gives the displayed residual. Its nonvanishing follows from the indicated normal monomial coefficient. Applying ∂ to the four words in $K_{4,4}$ gives two copies of each resulting cyclic word with opposite signs, so $K_{4,4} \in \ker \partial$. The finite computation proves the obstruction to the uncorrected cyclic lift. The all-N status of $K_{4,4}$ is supplied by the free trace-diagram normal form below, not by finite-rank specialization. □

Definition F.17 *(Free indexed normal diagrams).* Let $u = L_1 \cdots L_m$ be a word in X, Y. Write the trace $\operatorname{Tr}(u)$ as the directed m-gon with vertices $v_0, \ldots, v_{m-1}$ and edge $e_r : v_r \to v_{r+1}$, labelled L_r, with indices read modulo m. A contraction matching Γ is a partial matching of pairs

$$p < q, \qquad L_p = Y, \qquad L_q = X.$$

For each pair $(p, q) \in \Gamma$, delete the two edges e_p, e_q and identify

$$v_q \sim v_{p+1}, \qquad v_p \sim v_{q+1}.$$

Let $e(u, \Gamma)$ be the number of isolated vertices left after these deletions and identifications. Let $\mathsf{D}(u, \Gamma)$ be the remaining directed labelled graph, with all X-edges placed before all Y-edges in the ambient commutative normal-ordered algebra of matrix entries. Disconnected components are retained; they are the split-trace terms. The empty graph is denoted 1.

Theorem F.18 *(Free indexed normal-ordering formula).* In the universal indexed matrix Weyl algebra with

$$Y^k{}_l X^i{}_j = X^i{}_j Y^k{}_l - \hbar\, \delta^i_l \delta^k_j,$$

the normal-ordered trace word is

$$\mathcal{N}_{\text{free}} \operatorname{Tr}(u) = \sum_{\Gamma} (-\hbar)^{|\Gamma|} N^{e(u,\Gamma)} \mathsf{D}(u, \Gamma),$$

where the sum runs over all contraction matchings Γ of Y-before-X inversions in u. Consequently

$$\begin{aligned} \mathcal{N}_{\text{free}} \operatorname{Tr}(WM) = \mathcal{N}_{\text{free}} \operatorname{Tr}(WYX) - \mathcal{N}_{\text{free}} \operatorname{Tr}(WXY) \\ + \hbar N\, \mathcal{N}_{\text{free}} \operatorname{Tr}(W), \end{aligned}$$

with $M = YX - XY + \hbar N I$, interpreted as the left-ideal expression $\operatorname{Tr}(WM) = \sum_{i,j} W^j{}_i M^i{}_j$.

Proof. Move the X-entries leftward through the Y-entries by the PBW relation displayed above. Choosing the commutator summand at a swap selects one pair Y_p, X_q, contributes the factor $-\hbar$, and imposes exactly the two Kronecker identifications $v_q \sim v_{p+1}$ and $v_p \sim v_{q+1}$. Choosing the ordered summand leaves both edges and continues the PBW ordering. Since a generator can be contracted only once, the possible choices are precisely the partial matchings of original inversions. After all X-entries have been moved left, the uncontracted entries form the normal diagram $\mathsf{D}(u, \Gamma)$. Each isolated index loop contributes a free summation over $1, \ldots, N$, hence N. This proves the formula for trace words. The formula for $\operatorname{Tr}(WM)$ is the defining expansion of M, with the scalar matrix term contributing $\hbar N \operatorname{Tr}(W)$. □

Lemma F.19 *(Stable injectivity for normal trace diagrams).* Fix d. Let $\mathsf{Diag}_{\leq d}$ be the rational span of free indexed normal diagrams with at most d labelled nonempty edges, with coefficients in $\mathbb{Q}[\hbar, N]$. The joint specialization map to $N \times N$ matrices for all ranks $N \geq d$ is injective. Thus a free normal-form equality in $\mathsf{Diag}_{\leq d}$ is an all-N matrix Weyl identity, and a nonzero free residual cannot be killed by a theorem which only checks finitely many ranks below the stable Procesi–Razmyslov range.

Proof. Polarize in the labelled edges. A linear relation among polarized closed directed trace diagrams of edge degree at most d is a trace identity for generic matrices of degree at most d. The Procesi–Razmyslov second fundamental theorem says that the trace identities are generated by the fundamental antisymmetrizer on $N + 1$ index lines. Hence no nonzero identity of degree at most d exists in ranks $N \geq d$. Since the coefficients are polynomials in the formal scalar N, vanishing in every rank $N \geq d$ forces each coefficient polynomial to vanish. Depolarizing gives the stated injectivity. □

PROPOSITION F.20 *(First free residual and its kernel correction).* Use the normal-diagram notation of Definition F.17; for instance $\mathsf{D}(X_{0\to1}Y_{1\to0})$ denotes the directed two-edge normal diagram with one X-edge and one Y-edge. Then

$$\begin{aligned}\mathcal{N}_{\mathrm{free}}(E_{4,4,N} - C^{\mathrm{cl}}_{4,4}) = \frac{43}{7}\hbar^2\Big(&\mathsf{D}(X_{0\to1}X_{1\to2}Y_{2\to3}Y_{3\to0})\\ &- \mathsf{D}(X_{0\to1}X_{2\to3}Y_{1\to2}Y_{3\to0})\\ &- \hbar\,\mathsf{D}(X_{0\to0} \mid Y_{0\to0}) + \hbar N\,\mathsf{D}(X_{0\to1}Y_{1\to0})\Big).\end{aligned}$$

Moreover

$$\mathcal{N}_{\mathrm{free}}(K_{4,4}) = \mathcal{N}_{\mathrm{free}}(E_{4,4,N} - C^{\mathrm{cl}}_{4,4}),$$

and therefore

$$\mathcal{N}_{\mathrm{free}}(E_{4,4,N} - C^{\mathrm{cl}}_{4,4} - K_{4,4}) = 0.$$

This is an all-N identity in the free indexed normal-diagram algebra. It is not inferred from ranks 2 and 3.

Proof. Apply Theorem F.18 to the nine trace words in $C^{\mathrm{cl}}_{4,4}$ and to the Weyl-trace expression defining $E_{4,4,N}$. The finite matching enumeration is rational and contains no finite-rank specialization. It leaves exactly the four displayed normal diagrams. Applying the same formula to the four trace words in $K_{4,4}$ gives the same four-diagram vector. The script

```
scripts/quantum_shear_trace_diagram_normal_form.py
```

is the reproducible enumeration: with `-case 4,4 -classical-only` it prints the four-term residual above, and with `-case 4,4` it verifies zero residual after adding $K_{4,4}$. □

Definition F.21 *(Kernel-correction residual and decorated necklace complex).* Let $\mathsf{W}_{a,b}$ be the rational span of words with $a-1$ letters X and $b-1$ letters Y, and let $\mathsf{V}_{a,b} = \mathsf{Cyc}_{a,b}$ be the rational span of cyclic words with a letters X and b letters Y. Define the two-colour necklace multigraph $G_{a,b}$ by taking $\mathsf{V}_{a,b}$ as its vertex span and by assigning to $W \in \mathsf{W}_{a,b}$ the oriented edge

$$e_W : [WXY] \longrightarrow [WYX].$$

Its incidence map is

$$\partial\colon \mathsf{W}_{a,b} \to \mathsf{V}_{a,b}, \qquad \partial W = [WYX] - [WXY],$$

and $Z_{a,b} := \ker\partial$ is the cycle space.

Let $\mathsf{Diag}_{a,b}$ denote the positive free indexed normal-diagram target in bidegree (a,b): the uncontracted $\hbar^0$ cyclic component is removed and all PBW contraction, split-trace, and Capelli $\hbar N$-terms are retained. Write π_+ for this projection. Define the decorated contraction cochain by

$$C^+_{a,b}(W) = \pi_+\mathcal{N}_{\mathrm{free}}\,\mathrm{Tr}(WM), \qquad M = YX - XY + \hbar NI.$$

The correction map is

$$C_{a,b} = A_{a,b} := C^+_{a,b}|_{Z_{a,b}}\colon Z_{a,b} \longrightarrow \mathsf{Diag}_{a,b}.$$

Put

$$T_{a,b} = \sum_{j=0}^{b} [Y^j X^a Y^{b-j}] - \frac{b+1}{\binom{a+b}{a}} \sum_{U \in \mathrm{Sh}(a,b)} [U].$$

For any solution $c^{\mathrm{cl}}_{a,b}$ of $\partial c = T_{a,b}$, define

$$E^+_{a,b} = \pi_+ \mathcal{N}_{\mathrm{free}}(E_{a,b,N})$$

and put

$$R^{\mathrm{free}}_{a,b} = E^+_{a,b} - C^+_{a,b}(c^{\mathrm{cl}}_{a,b})$$

and define

$$\mathfrak{o}_{a,b} = [R^{\mathrm{free}}_{a,b}] \in \operatorname{coker} \mathcal{C}_{a,b}.$$

This is the kernel-correction class after a choice of classical lift. The radial obstruction itself is the lift-independent class $\Omega^{\mathrm{rad}}_{a,b}$ and the signed-row criterion of Proposition F.25.

Statement F.22 *(Decorated kernel-correction Hodge criterion).* Give $Z_{a,b}$ and $\mathsf{Diag}_{a,b}$ the inner products for which the chosen word-cycle and free normal-diagram bases are orthonormal. Let $A = A_{a,b}$ and define

$$\Delta^{\mathrm{dec}}_{a,b} = AA^*, \qquad \mathsf{H}^{\mathrm{dec}}_{a,b} = \ker A^* = (\operatorname{im} A)^\perp.$$

Let

$$\Pi^{\mathrm{harm}}_{a,b} \colon \mathsf{Diag}_{a,b} \to \mathsf{H}^{\mathrm{dec}}_{a,b}$$

be the orthogonal projection. For every $a, b \geq 1$, the following conditions are equivalent.

(i) There exists $K_{a,b} \in Z_{a,b}$ such that

$$C^+_{a,b}(K_{a,b}) = R^{\mathrm{free}}_{a,b}.$$

(ii) The kernel-correction class $\mathfrak{o}_{a,b}$ vanishes.

(iii) The decorated harmonic projection vanishes:

$$\Pi^{\mathrm{harm}}_{a,b} R^{\mathrm{free}}_{a,b} = 0.$$

(iv) Every functional $\lambda \in \mathsf{Diag}^*_{a,b}$ with

$$\lambda C^+_{a,b}(K) = 0 \qquad (K \in Z_{a,b})$$

also satisfies $\lambda(R^{\mathrm{free}}_{a,b}) = 0$.

When these equivalent conditions hold, the canonical Hodge correction is

$$K_{a,b} = A^*_{a,b}(\Delta^{\mathrm{dec}}_{a,b})^{-1} R^{\mathrm{free}}_{a,b},$$

with the inverse taken on $\operatorname{im} A_{a,b}$. Thus the fixed-lift kernel-correction problem is equivalent to

$$\Pi^{\mathrm{harm}}_{a,b} R^{\mathrm{free}}_{a,b} = 0 \qquad \text{in the chosen bidegree.}$$

This is an intermediate Hodge form. The lift-independent exact obstruction theorem is the $\Omega^{\mathrm{rad}}_{a,b}$, $D^{\square}_{a,b}$, and normalized signed-row formulation in Proposition F.25. Proposition F.20 supplies the first nonzero lift; the statement above does not prove the all-bidegree vanishing.

Proof. If c and c' solve the cyclic equation, then $c - c' \in Z_{a,b}$. Hence $R(c') - R(c) = -A_{a,b}(c' - c)$, so the class in coker $A_{a,b}$, and its harmonic projection, are independent of the chosen cyclic lift.

The finite-dimensional Hodge decomposition for A gives

$$\mathsf{Diag}_{a,b} = \operatorname{im} A \oplus \ker A^*.$$

Therefore $R^{\text{free}}_{a,b} \in \operatorname{im} A$ if and only if its projection to $\ker A^*$ is zero. This proves the equivalence of the correction equation, the intermediate cokernel class, and the harmonic projection. The functional criterion is the annihilator form of the same statement: $R \in \operatorname{im} A$ if and only if every $\lambda \in (\operatorname{im} A)^\perp = \ker A^*$ vanishes on R. If $R \in \operatorname{im} A$, then $AA^*(AA^*)^{-1}R = R$ on $\operatorname{im} A$, so the displayed $K_{a,b}$ is a correction. Conversely, any correction places $R^{\text{free}}_{a,b}$ in $\operatorname{im} A$. □

Definition F.23 *(Square-cell PBW Stokes complex).* Let $\mathsf{S}_{a,b}$ be the rational span of square cells obtained from two commuting adjacent swaps in a cyclic word with a letters X and b letters Y. A based representative has the form

$$uXYvXYw,$$

and its oriented square is

$$uXYvXYw \to uYXvXYw \to uYXvYXw \to uXYvYXw \to uXYvXYw.$$

Each side is read as the corresponding edge $e_W : [WXY] \to [WYX]$ of the necklace graph. This defines

$$\partial_2 \colon \mathsf{S}_{a,b} \longrightarrow \mathsf{W}_{a,b}.$$

The decorated PBW Stokes map is

$$D^{\square}_{a,b} = C^{+}_{a,b}\partial_2 \colon \mathsf{S}_{a,b} \longrightarrow \mathsf{Diag}_{a,b}.$$

LEMMA F.24 *(Ordinary necklace Stokes theorem).* For $a, b \geq 1$,

$$\operatorname{im} \partial_2 = \ker \partial.$$

Proof. Work first with based words. A binary word with a letters X and b letters Y is the same as an order ideal in the rectangle poset $[a] \times [b]$, via its inversion set. An adjacent $XY \leftrightarrow YX$ move toggles one extremal box. The cubical complex of the distributive lattice of order ideals, with squares generated by commuting toggles, is contractible. Hence its rational one-cycles are generated by square boundaries. Passing to cyclic words is the quotient by a finite cellular rotation action; over $\mathbb{Q}$, transfer identifies the quotient rational homology with the invariant rational homology upstairs. Thus every necklace cycle is a rational sum of square boundaries. □

PROPOSITION F.25 *(Dual-potential form of the radial obstruction).* Define the finite-dimensional map

$$B_{a,b} \colon \mathsf{W}_{a,b} \longrightarrow \mathsf{V}_{a,b} \oplus \mathsf{Diag}_{a,b}, \qquad B_{a,b}(W) = (\partial W, C^{+}_{a,b}(W)),$$

and the target

$$\eta_{a,b} = (T_{a,b}, E^{+}_{a,b}).$$

Let

$$\Omega^{\text{rad}}_{a,b} = [\eta_{a,b}] \in \operatorname{coker} B_{a,b}.$$

For fixed (a, b), the following are equivalent.

(i) $\Omega^{\mathrm{rad}}_{a,b} = 0$.

(ii) For one, equivalently every, classical lift $c^{\mathrm{cl}}_{a,b}$ with $\partial c^{\mathrm{cl}}_{a,b} = T_{a,b}$, there is an $H_{a,b} \in \mathsf{S}_{a,b}$ such that

$$D^{\square}_{a,b} H_{a,b} = R^{\mathrm{free}}_{a,b}.$$

(iii) For one, equivalently every, classical lift $c^{\mathrm{cl}}_{a,b}$, the residual $R^{\mathrm{free}}_{a,b}$ lies in $\mathrm{im}(C^{+}_{a,b}|_{\ker \partial})$.

(iv) $\Pi^{\mathrm{harm}}_{a,b} R^{\mathrm{free}}_{a,b} = 0$.

(v) For every pair

$$(\phi, \lambda) \in \mathsf{V}^{*}_{a,b} \oplus \mathsf{Diag}^{*}_{a,b}$$

satisfying

$$\lambda C^{+}_{a,b}(W) = \phi(\partial W) \qquad (W \in \mathsf{W}_{a,b}),$$

one has

$$\lambda(E^{+}_{a,b}) = \phi(T_{a,b}).$$

Thus

$$\Omega^{\mathrm{rad}}_{a,b} = 0 \quad \Longleftrightarrow \quad D^{\square}_{a,b} H_{a,b} = R^{\mathrm{free}}_{a,b}$$

for one, equivalently every, classical lift. Failure in bidegree (a, b) is exactly a signed dual row

$$(\phi, -\lambda) \in \ker B^{*}_{a,b}$$

with

$$\lambda(E^{+}_{a,b}) - \phi(T_{a,b}) \neq 0.$$

Equivalently, after rescaling, it is a normalized row

$$\lambda \in \ker(D^{\square}_{a,b})^{*}, \qquad \lambda(R^{\mathrm{free}}_{a,b}) = 1.$$

Proof. The pair (ϕ, λ) in the statement corresponds to the signed dual row $(\phi, -\lambda)$ on $\mathsf{V}_{a,b} \oplus \mathsf{Diag}_{a,b}$. It annihilates $B_{a,b}(W)$ precisely when $\lambda C^{+}_{a,b}(W) = \phi(\partial W)$. Since the spaces are finite dimensional,

$$\eta_{a,b} \in \mathrm{im}\, B_{a,b} \quad \Longleftrightarrow \quad \ell(\eta_{a,b}) = 0 \text{ for every } \ell \in \ker B^{*}_{a,b}.$$

Evaluating $(\phi, -\lambda)$ on $\eta_{a,b}$ gives $\phi(T_{a,b}) - \lambda(E^{+}_{a,b})$, proving the equivalence of $\Omega^{\mathrm{rad}}_{a,b} = 0$ with the signed-row condition.

If $c^{\mathrm{cl}}_{a,b}$ is a classical lift, then

$$\eta_{a,b} - B_{a,b}(c^{\mathrm{cl}}_{a,b}) = (0, R^{\mathrm{free}}_{a,b}).$$

Adding $B_{a,b}(K)$ with $K \in \ker \partial$ changes only the decorated diagram component by $C^{+}_{a,b}(K)$. Hence $\eta_{a,b} \in \mathrm{im}\, B_{a,b}$ if and only if $R^{\mathrm{free}}_{a,b} \in \mathrm{im}(C^{+}_{a,b}|_{\ker \partial})$. By Lemma F.24, $\ker \partial = \mathrm{im}\, \partial_2$, so this image is exactly $\mathrm{im}\, D^{\square}_{a,b}$. The Hodge equivalence is the decomposition of Statement F.22.

Finally, finite-dimensional duality for $D^{\square}_{a,b}$ says that $R^{\mathrm{free}}_{a,b} \notin \mathrm{im}\, D^{\square}_{a,b}$ if and only if some $\lambda \in \ker(D^{\square}_{a,b})^{*}$ has $\lambda(R^{\mathrm{free}}_{a,b}) \neq 0$. Rescale the row to make the value 1. The equality $\lambda C^{+}_{a,b}(W) = \phi(\partial W)$ identifies this row with the signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$, and evaluating that signed row on $\eta_{a,b}$ gives $\phi(T_{a,b}) - \lambda(E^{+}_{a,b})$. $\square$

Remark F.26 *(Potential form and the failure of the bare graph Green operator).* A functional $\lambda \in \mathsf{Diag}^*_{a,b}$ satisfies $\lambda C^+_{a,b}|_{\ker \partial} = 0$ precisely when the edge cochain

$$W \longmapsto \lambda C^+_{a,b}(W)$$

has zero periods on the necklace graph $G_{a,b}$. The graph is connected for $a, b > 0$, since adjacent $XY \leftrightarrow YX$ moves connect all two-colour necklaces. Hence this zero-period condition is equivalent to a vertex potential $\phi_\lambda \in \mathsf{V}^*_{a,b}$, unique up to a constant, with

$$\lambda C^+_{a,b}(W) = \phi_\lambda([WYX]) - \phi_\lambda([WXY]) = \phi_\lambda(\partial W).$$

Because $T_{a,b}$ has total coefficient zero, $\phi_\lambda(T_{a,b})$ is independent of the additive constant. The radial obstruction vanishes, equivalently $\Omega^{\mathrm{rad}}_{a,b} = 0$, if and only if every such λ satisfies the potential identity

$$\lambda \pi_+ \mathcal{N}_{\mathrm{free}}(E_{a,b,N}) = \phi_\lambda(T_{a,b}).$$

By Lemma F.24, the same row condition is $\lambda D^{\square}_{a,b} = 0$. Indeed,

$$\lambda(R^{\mathrm{free}}_{a,b}) = \lambda E^+_{a,b} - \lambda C^+_{a,b}(c^{\mathrm{cl}}_{a,b}) = \lambda E^+_{a,b} - \phi_\lambda(T_{a,b}).$$

The ordinary graph Green operator for the incidence map ∂ is therefore insufficient. It constructs a classical cyclic lift $c^{\mathrm{cl}}_{a,b}$ of $T_{a,b}$, i.e. it solves only the uncontracted $\hbar^0$ incidence equation. The quantum correction equation is instead $A_{a,b}K_{a,b} = R^{\mathrm{free}}_{a,b}$, equivalently $D^{\square}_{a,b}H_{a,b} = R^{\mathrm{free}}_{a,b}$, inside the positive PBW diagram target, where contraction diagrams, split traces, and the Capelli term are visible. No canonical inverse from an arbitrary positive diagram residual to a necklace edge cochain has been constructed here. The correct finite-dimensional Green operator is the decorated square-cell one attached to $D^{\square}_{a,b}(D^{\square}_{a,b})^*$.

PROPOSITION F.27 *(Edge PBW telescoping primitive).* For every $b \geq 2$, set

$$C^{\mathrm{edge}}_{2,b} = \sum_{r=0}^{\lfloor (b-2)/2 \rfloor} (b-1-2r)\,\mathrm{Tr}(Y^r X Y^{b-1-r} M).$$

For every $a \geq 2$, set

$$C^{\mathrm{edge}}_{a,2} = \frac{3}{a+1} \sum_{r=0}^{\lfloor (a-2)/2 \rfloor} (a-1-2r)\,\mathrm{Tr}(X^{a-1-r} Y X^r M).$$

Then the free indexed normal-diagram identities

$$\mathcal{N}_{\mathrm{free}}(E_{2,b,N} - C^{\mathrm{edge}}_{2,b}) = 0, \qquad \mathcal{N}_{\mathrm{free}}(E_{a,2,N} - C^{\mathrm{edge}}_{a,2}) = 0$$

hold for all displayed edge bidegrees. Hence

$$\mathfrak{o}_{2,b} = 0, \qquad \mathfrak{o}_{a,2} = 0$$

with no separate kernel correction in this edge gauge. This is only the one-moving-letter edge strip; for $a, b \geq 3$ the remaining theorem is the decorated PBW Stokes problem $D^{\square}_{a,b}H_{a,b} = R^{\mathrm{free}}_{a,b}$, equivalently the normalized signed-row criterion of Proposition F.25.

Proof. We prove the $(2, b)$ strip. Put

$$\Theta_s^{(b)} = \mathsf{D}(XY^sXY^{b-s}), \qquad 0 \le s \le b,$$

and

$$\Phi_s^{(b)} = \mathsf{D}(XY^s) \mid \mathsf{D}(Y^{b-1-s}), \qquad 0 \le s \le b-1,$$

where the vertical bar denotes disjoint union and $\mathsf{D}(Y^0)$ is the isolated index loop, hence contributes the scalar factor N. For

$$W_r = Y^rXY^{b-1-r}, \qquad 0 \le r \le b-1,$$

Theorem F.18 gives the PBW column identity

$$\mathcal{N}_{\text{free}} \operatorname{Tr}(W_r\mathcal{M}) = \Theta_r^{(b)} - \Theta_{r+1}^{(b)} - \hbar\Phi_r^{(b)} + \hbar\Phi_{b-1-r}^{(b)}.$$

Indeed,

$$\operatorname{Tr}(W_r\mathcal{M}) = \operatorname{Tr}(W_rYX) - \operatorname{Tr}(W_rXY) + \hbar N \operatorname{Tr}(W_r).$$

PBW matchings in the first two trace words which do not use the adjacent crossing supplied by $\mathcal{M}$ cancel in pairs. The matchings using that adjacent crossing are cancelled, including their secondary contractions, by the scalar term $\hbar N \operatorname{Tr}(W_r)$. The uncontracted part leaves the boundary $\Theta_r^{(b)} - \Theta_{r+1}^{(b)}$. The two endpoint contraction families are the left and reflected right blocks, giving $-\hbar\Phi_r^{(b)}$ and $+\hbar\Phi_{b-1-r}^{(b)}$. No further term survives the pairing.

The target $E_{2,b,N}$ contains only the first and third Moyal terms:

$$J_N\Big(\{z_1^3/3, z_2^{b+1}\}_\hbar\Big) = (b+1)J_N(z_1^2z_2^b) + \frac{(b+1)b(b-1)}{12}\hbar^2 \operatorname{Tr}(Y^{b-2}).$$

Counting PBW matchings in the shear sum and in the Weyl average gives

$$\mathcal{N}_{\text{free}}(E_{2,b,N}) = \sum_{r=0}^{\lfloor (b-2)/2 \rfloor} (b-1-2r)\Big(\Theta_r^{(b)} - \Theta_{r+1}^{(b)} - \hbar\Phi_r^{(b)} + \hbar\Phi_{b-1-r}^{(b)}\Big).$$

In contraction degree 0, this is the discrete boundary coefficient of the two-X cyclic separation. In contraction degree 1, the coefficient of $\Phi_s^{(b)}$ is $-(b+1) + 2(s+1) = 2s - (b-1)$, the antisymmetric half-sum above. In contraction degree 2, the difference between the shear count and the first-Moyal Weyl count is exactly

$$\frac{(b+1)b(b-1)}{12}\mathcal{N}_{\text{free}} \operatorname{Tr}(Y^{b-2}),$$

which is cancelled by the third Moyal term in the displayed target. Combining this target expansion with the column identity proves $\mathcal{N}_{\text{free}}(E_{2,b,N} - C_{2,b}^{\text{edge}}) = 0$.

The $(a, 2)$ strip is the same one-moving-letter calculation with Y moving through an X-string. For $V_r = X^{a-1-r}YX^r$ one obtains

$$\mathcal{N}_{\text{free}} \operatorname{Tr}(V_r\mathcal{M}) = \widetilde{\Theta}_r^{(a)} - \widetilde{\Theta}_{r+1}^{(a)} - \hbar\widetilde{\Phi}_r^{(a)} + \hbar\widetilde{\Phi}_{a-1-r}^{(a)}.$$

Here

$$\widetilde{\Theta}_s^{(a)} = \mathsf{D}(X^{a-s}YX^sY), \qquad \widetilde{\Phi}_s^{(a)} = \mathsf{D}(X^{a-1-s}) \mid \mathsf{D}(X^sY),$$

with $\mathsf{D}(X^0)$ again interpreted as the isolated index loop. The first and third Moyal coefficients for $\{z_1^{a+1}/(a+1), z_2^3\}_\hbar$ are

$$3, \qquad \frac{a(a-1)}{4},$$

namely $3/(a+1)$ times the corresponding $(2, a)$ coefficients. The same count therefore gives the stated $C_{a,2}^{\mathrm{edge}}$ and proves the second identity. □

PROPOSITION F.28 *(Free obstruction certificates beyond* $(4,4)$*).* In the free indexed normal-diagram algebra, $\mathfrak{o}_{3,5} = \mathfrak{o}_{5,3} = 0$, and the lexicographic cyclic lifts already have zero quantum residual. Explicitly,

$$\begin{aligned} E_{3,5,N} = {} & \frac{36}{7}\operatorname{Tr}(XXYYYY\,M) + \frac{24}{7}\operatorname{Tr}(XYXYYY\,M) \\ & - \frac{6}{7}\operatorname{Tr}(XYYXYY\,M) - \frac{6}{7}\operatorname{Tr}(XYYYXY\,M) \\ & + \frac{30}{7}\operatorname{Tr}(XYYYYX\,M) + \frac{12}{7}\operatorname{Tr}(YXYXYY\,M), \end{aligned}$$

and

$$\begin{aligned} E_{5,3,N} = {} & \frac{24}{7}\operatorname{Tr}(XXXXYY\,M) - \frac{4}{7}\operatorname{Tr}(XXXYXY\,M) \\ & + \frac{20}{7}\operatorname{Tr}(XXXYYX\,M) + \frac{12}{7}\operatorname{Tr}(XXYXXY\,M) \\ & - \frac{8}{7}\operatorname{Tr}(XXYXYX\,M) + \frac{16}{7}\operatorname{Tr}(XXYYXX\,M). \end{aligned}$$

In total degree 9 the same free computation gives $\mathfrak{o}_{3,6} = \mathfrak{o}_{6,3} = 0$. The cyclic lifts already have zero free residual:

$$\begin{aligned} E_{3,6,N} = {} & \frac{25}{4}\operatorname{Tr}(XXYYYYY\,M) + \frac{19}{4}\operatorname{Tr}(XYXYYYY\,M) \\ & - \frac{3}{4}\operatorname{Tr}(XYYXYYY\,M) - \frac{3}{4}\operatorname{Tr}(XYYYXYY\,M) \\ & - \frac{3}{4}\operatorname{Tr}(XYYYYXY\,M) + \frac{11}{2}\operatorname{Tr}(XYYYYYX\,M) \\ & + \frac{13}{4}\operatorname{Tr}(YXYXYYY\,M) + \frac{1}{4}\operatorname{Tr}(YXYYXYY\,M) \\ & + \frac{7}{4}\operatorname{Tr}(YXYYYXY\,M), \end{aligned}$$

and

$$\begin{aligned} E_{6,3,N} = {} & \frac{25}{7}\operatorname{Tr}(XXXXXYY\,M) - \frac{3}{7}\operatorname{Tr}(XXXXYXY\,M) \\ & + \frac{22}{7}\operatorname{Tr}(XXXXYYX\,M) - \frac{3}{7}\operatorname{Tr}(XXXYXXY\,M) \\ & - \frac{6}{7}\operatorname{Tr}(XXXYXYX\,M) + \frac{19}{7}\operatorname{Tr}(XXXYYXX\,M) \\ & + \frac{16}{7}\operatorname{Tr}(XXYXXXY\,M) + \frac{1}{7}\operatorname{Tr}(XXYXXYX\,M) \\ & + \operatorname{Tr}(XXYXYXX\,M). \end{aligned}$$

Edge strips are not part of this finite-certificate assertion: Proposition F.27 proves $\mathfrak{o}_{2,b} = \mathfrak{o}_{a,2} = 0$ for all $a, b \geq 2$.

Proof. Apply Theorem F.18. For $(3,5)$ and $(5,3)$, the displayed coefficients solve the cyclic equation and the free normal-form residual has no terms. The total-degree-9 formulas for $(3,6)$ and $(6,3)$ are checked in the same way; their residual also has no terms, so $K_{3,6} = K_{6,3} = 0$ in this pivot gauge. The normal-form script reproduces these rational enumerations, and separately regression-checks the edge theorem, in the modes

```
-case 3,5 -case 5,3  -solve-free,
-case 3,6 -case 6,3  -solve-classical,
-case 3,6  -kernel-correction,    -case 6,3  -kernel-correction,
-case 2,7 -case 7,2  -kernel-correction.
```

The calculation is in the free diagram basis; finite-rank trace identities are not used. □

Lemma F.29 *(Cartan one-particle Moyal representation).* Put

$$S_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i}).$$

Then, for all $f, g \in \mathbb{C}[z_1, z_2][[\hbar]]$,

$$\hbar^{-1}[S_N(f), S_N(g)] = S_N(\{f, g\}_\hbar).$$

Proof. The i-th summand of $S_N(f)$ lies in the one-particle Weyl algebra generated by λ_i and $-\hbar\partial_{\lambda_i}$. Weyl quantization is an algebra isomorphism from the formal Moyal algebra of the constant Poisson plane to this one-particle Weyl algebra, so

$$\hbar^{-1}[\mathrm{Op}_{\mathrm{W}}(f)_i, \mathrm{Op}_{\mathrm{W}}(g)_i] = \mathrm{Op}_{\mathrm{W}}(\{f, g\}_\hbar)_i.$$

Summands with different i commute, because they use disjoint Cartan variables. Summing over i gives the displayed identity. □

Theorem F.30 *(Finite-N radial-parts identity from the input package).* Under the radial-parts input of Statement F.8, put

$$S_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i}).$$

Then

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1} S_N(f)\Delta,$$

and for all f, g

$$D_N(f, g) = \hbar^{-1}[J_N(f), J_N(g)] - J_N(\{f, g\}_\hbar)$$

lies in the quantum moment-map ideal. Equivalently, in the Vandermonde gauge,

$$\mathrm{Rad}_0(D_N(f, g)) = \hbar^{-1}[R_N(f), R_N(g)] - R_N(\{f, g\}_\hbar) = 0.$$

Thus $D_N(f, g)$ vanishes in degree-zero quantum Hamiltonian reduction.

Proof. The first displayed identity is the exact image normalization included in Statement F.8. The same radial-parts input makes the conjugated radial map injective on the reduced invariant quotient, so the commutator defect can be tested on the Cartan.

Lemma F.13 identifies the exact theorem target behind the full displayed formula: classical nonlinear shear covariance is not enough to recover mixed Weyl images in this convention. One must instead construct the all-N moment-ideal primitives, equivalently prove $\Omega^{\mathrm{rad}}_{a,b} = 0$ or the decorated PBW Stokes identity $D^{\square}_{a,b} H_{a,b} = R^{\mathrm{free}}_{a,b}$ in every bidegree. A genuine failure is a normalized signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$, not the absence of a finite computation.

Finally Lemma F.29 gives $\hbar^{-1}[S_N(f), S_N(g)] = S_N(\{f, g\}_\hbar)$. Equivalently, Lemma F.11 gives the same identity after Vandermonde conjugation:

$$\hbar^{-1}[R_N(f), R_N(g)] = R_N(\{f, g\}_\hbar).$$

Thus $\mathrm{Rad}_0(D_N(f, g)) = 0$. The radial-map injectivity of Lemma F.12 puts $D_N(f, g)$ in the quantum moment-map ideal. □

Definition F.31 *(Stable cumulant extraction).* Fix a total polynomial degree bound d and take $N \geq d$. Let

$$\kappa_{N,d}$$

denote the connected single-trace cumulant projection on the stable trace algebra in degrees $\leq d$: it kills the empty trace and all decomposable multi-trace products, and it sends the Capelli-renormalized generator

$$J_N(z_1^a z_2^b), \qquad 0 < a + b \leq d,$$

to its single-trace indecomposable class. The stable map $\kappa_{\infty,d}$ is the stable-tail identification of these projections in the Procesi–Razmyslov stable range, after passing through the renormalized associated-graded J_N-basis transition system.

LEMMA F.32 *(Connected-trace projection after Capelli triangularization).* Fix a polynomial degree bound d. In the stable range $N \geq d$, the Capelli triangular system

$$J_N(z_1^a z_2^b) \equiv \sum_{r=0}^{\min(a,b)} \left(-\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} T_{a-r,b-r}$$

identifies the span of nonempty Weyl traces $J_N(z_1^a z_2^b)$, $a + b \leq d$, with the span of nonempty normal-ordered traces $T_{a,b}$, modulo the empty trace and decomposable multi-traces. The cumulant projection

$$\kappa_{N,d}\big(J_N(z_1^a z_2^b)\big) \longmapsto z_1^a z_2^b$$

is therefore degreewise injective on the scalar-reduced connected single-trace quotient in the stable range.

Proof. The displayed Capelli formula is triangular for total degree: its $r = 0$ term is $T_{a,b}$, and every $r \geq 1$ term has total degree $a + b - 2r$. The diagonal coefficient is 1, hence the triangular matrix is unipotent and invertible over $\mathbb{C}[[\hbar N]]$. The only term which can become the empty trace is the term with $a - r = b - r = 0$, and the scalar-reduced connected quotient removes it. Decomposable products are removed by the connected projection. Stable Procesi–Razmyslov trace independence in degree d gives linear independence of the remaining connected trace generators for $N \geq d$. Thus the stated projection is injective degreewise in the stable range. □

Theorem F.33 *(Connected-trace bracket compatibility and stable cumulants).* Under the same radial-parts input as Theorem F.30, for each fixed polynomial degree the classes

$$\overline{J_N(f)}$$

have a stable connected large-N limit after removing the empty trace and disconnected products. This limit is the renormalized associated-graded connected transition system, not a naive inverse limit of finite-rank normal-ordered trace algebras over $\mathbb{C}[[\hbar]]$. In that limit,

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f, g\}_\hbar).$$

More explicitly, for $N \geq d$ and all f, g with all monomials in $f, g, \{f, g\}_\hbar$ of total degree $\leq d$,

$$\kappa_{N,d}\big(\hbar^{-1}[J_N(f), J_N(g)]\big) = \kappa_{N,d}\big(J_N(\{f, g\}_\hbar)\big),$$

and these identities are compatible under $N \mapsto N + 1$.

Proof. Lemma F.32 gives the degreewise stable connected single-trace projection and its injectivity after removing the empty trace and decomposable products. The finite-N identity of Theorem F.30 gives equality in the reduced quantum Hamiltonian quotient. Applying $\kappa_{N,d}$ gives the displayed connected-trace identity, since the cumulant projection kills only the empty trace and decomposable products and is injective on the Capelli-renormalized single-trace span by Lemma F.32. The identity is compatible with adjoining extra Cartan variables, because $S_{N+k}(f)$ is obtained from $S_N(f)$ by adding commuting one-particle summands. Therefore the bracket identity stabilizes degreewise in N. □

Definition F.34 *(Reduced open-line Wick graph data).* In the reduced first-order open-line model with action

$$S_0 = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2),$$

use Weyl-ordered boundary vertices $f(\phi_1, \phi_2)$, and choose the zero-mode-removed midpoint propagator

$$G(t, s) = \frac{1}{2}\,\mathrm{sgn}(t - s), \qquad \langle \phi_1(t)\phi_2(s)\rangle = \hbar G(t, s), \qquad \langle \phi_2(t)\phi_1(s)\rangle = -\hbar G(t, s).$$

Self-contractions at a Weyl vertex are set to zero. The only connected two-vertex graphs are r-fold cross-contractions between the two boundary vertices.

Theorem F.35 *(Reduced open-line Wick graph expansion).* With the graph data of Definition F.34, the ordered local product for two boundary Hamiltonians at $t > s$ is the formal Moyal product

$$F \star G = m \circ \exp\left(\frac{\hbar}{2} P\right)(F \otimes G).$$

More precisely, the graph with exactly r cross-edges contributes

$$W_r^{>}(f, g) = \frac{1}{r!}\left(\frac{\hbar}{2}\right)^r P^r(f, g),$$

while the reversed ordering $s > t$ contributes

$$W_r^{<}(f, g) = \frac{1}{r!}\left(-\frac{\hbar}{2}\right)^r P^r(f, g).$$

Hence the antisymmetrized factorization commutator is

$$[f,g]_{\text{graph}} = \sum_{r\geq 0} \frac{1-(-1)^r}{2^r r!}\hbar^r P^r(f,g) = \sum_{\substack{r\geq 1\\ r \text{ odd}}} \frac{2}{2^r r!}\hbar^r P^r(f,g).$$

The one-edge and three-edge commutator coefficients are exactly

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24}P^3(f,g).$$

All even commutator orders vanish.

Proof. For ordered insertions with $t > s$, the midpoint propagator gives $G(t,s) = 1/2$. A single cross-contraction therefore contributes

$$\frac{\hbar}{2}\big(\partial_{z_1}\otimes\partial_{z_2} - \partial_{z_2}\otimes\partial_{z_1}\big) = \frac{\hbar}{2}P.$$

Wick expansion of the free first-order model exponentiates the cross-contraction operator. With Weyl vertices there are no self-contractions; hence a connected two-vertex graph is determined by its number r of cross-edges. The exponential supplies the symmetry factor $1/r!$, and the ordered propagator supplies $(\hbar/2)^r$. This proves the formula for $W_r^{>}$. Reversing the order changes the propagator from $1/2$ to $-1/2$, giving $W_r^{<}$. Subtracting the two orderings gives the displayed commutator series. The coefficients in orders 1 and 3 are $2/(2\cdot 1!) = 1$ and $2/(2^3\cdot 3!) = 1/24$. □

COROLLARY F.36 *(All-order reduced Wick realization of Moyal).* The reduced open-line Wick graph expansion of Theorem F.35 realizes the Weyl/Moyal Lie bracket of Definition F.1 to all orders:

$$[f,g]_{\text{graph}} = [f,g]_{\text{raw}}, \qquad \hbar^{-1}[f,g]_{\text{graph}} = \{f,g\}_\hbar.$$

Proof. Compare the graph commutator series in Theorem F.35 with the formal Weyl commutator series in Theorem F.2. The coefficient of P^r is $2\hbar^r/(2^r r!)$ for odd r and 0 for even r in both series. □

Remark F.37 *(Finite exact arithmetic anchors).* The script `scripts/check_moyal_coefficients.py` is an exact rational-arithmetic falsification harness for the algebraic coefficients used above. It checks:

14641 monomial pairs $(z_1^k z_2^l, z_1^m z_2^n)$, $\quad 0 \leq k,l,m,n \leq 10$,

Moyal commutator orders $0 \leq r \leq 11$,

121 Capelli triangular round trips $J \to T \to J$ with $0 \leq a,b \leq 10$,

80 direct $N = 2$ radial restrictions and 80 direct $N = 3$ radial restrictions,

$[S_2(f), S_2(g)] = S_2([f,g]_*)$ on the primary and higher-order Cartan test pairs,

the pre-quotient leading scalar Moyal coefficient,

its $U(1)$-anomaly constant term, and

the open-line $r = 1, 3$ graph weights.

These checks verify the constants

$$1, \qquad \frac{1}{24}, \qquad \frac{1}{1920}, \qquad \frac{1}{322560}$$

in the tested Moyal orders, the Capelli signs $(-\hbar N/2)^r$ and $(+\hbar N/2)^r$, and the rank-two/rank-three shadows of the radial image formula. They do not prove the all-N Harish-Chandra radial-parts theorem or the analytic Costello brane-defect graph specialization.

The script `scripts/quantum_shear_trace_diagram_normal_form.py` is separate from the finite-rank harness. It enumerates the free indexed normal diagrams of Theorem F.18; finite matrices are used only in its optional PBW self-test. Its default run verifies the displayed low bidegree primitive candidates, the low edge-family regressions for Proposition F.27, and the free $K_{4,4}$-corrected residual. The command with `-case 4,4 -classical-only` prints the four-term residual of Proposition F.20. The `-solve-free` and `-kernel-correction` modes compute the intermediate kernel-correction class $\mathfrak{o}_{a,b}$ of Definition F.21; they give the additional finite certificates in Proposition F.28, including the total-degree-9 cases $(3,6)$ and $(6,3)$. The current exact obstruction theorem is the vanishing of $\Omega^{\mathrm{rad}}_{a,b}$, equivalently the decorated PBW Stokes equation $D^{\square}_{a,b}H_{a,b} = R^{\mathrm{free}}_{a,b}$, with normalized signed-row failure criterion. The total-degree 14 bidegrees $(6,8)$, $(8,6)$, and $(7,7)$ are timeout frontiers for the present exact enumeration, not known obstruction rows. The `-edge-family-max` mode checks any finite edge window against the closed PBW formula, but it is no longer the proof of the edge strip. These computations do not prove the all-bidegree decorated PBW Stokes identity.

Remark F.38 *(Scalar quotient and the $U(1)$ anomaly).* Constants are central in the Moyal Lie algebra and are killed in $\bar{A}_\hbar$. Before this scalar reduction the linear commutator carries the central term $[z_1, z_2]_* = \hbar$, and on the brane side the corresponding matrix trace is the Capelli shift $\hbar N$. The reduced theorem is the theorem after quotienting by this scalar line. The unreduced line is the $U(1)$ anomaly class of Theorem 2.5.

Proposition F.39 *(Moyal–Capelli normalization and realization boundary).* The following statements are the exact theorem surface used by the degree-zero Moyal sector.

(i) The formal Weyl algebra on the constant symplectic plane gives the raw commutator coefficients

$$[f,g]^{(2s+1)}_{\mathrm{raw}} = \frac{\hbar^{2s+1}}{2^{2s}(2s+1)!}P^{2s+1}(f,g), \qquad [f,g]^{(2s)}_{\mathrm{raw}} = 0.$$

In particular the third raw coefficient is $\hbar^3 P^3(f,g)/24$.

(ii) The finite-rank brane algebra is the degree-zero normalizer quotient of the matrix Weyl algebra by the traceless quantum moment ideal. With the normal-ordering convention of Definition F.6, the contact relation is

$$YX - XY + \hbar N I \equiv 0 \pmod{\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)}.$$

This scalar Capelli shift is absorbed by the triangular change of basis $T_{a,b} \leftrightarrow J_N(z_1^a z_2^b)$, and the remaining scalar Hamiltonian line is removed only after passing to $\bar{A}_\hbar$ and to the empty-trace quotient.

(iii) The equality

$$\hbar^{-1}[J_N(f), J_N(g)] = J_N(\{f,g\}_\hbar)$$

in degree-zero quantum Hamiltonian reduction is proved here relative to the radial-parts input of Statement F.8. Without that input, the exact algebraic obstruction theorem is

$$\Omega^{\mathrm{rad}}_{a,b} = 0 \quad \Longleftrightarrow \quad D^{\square}_{a,b}H_{a,b} = R^{\mathrm{free}}_{a,b},$$

with normalized signed-row failure criterion, as in Proposition F.25. The moment-ideal primitive $E_{a,b,N} = \sum A^j{}_i(a,b,N)\widehat{\mu}^i{}_j$ of Lemma F.15 and the class $\mathfrak{v}_{a,b}$ of Definition F.21 are equivalent intermediate forms after choosing a classical lift.

(iv) The quantum principal-part target is the continuous $\bar{A}_\hbar$-coadjoint module. It is defined by

$$(\mathrm{ad}^\vee_{f,\hbar}\,\rho)(g) = -\rho(\{f,g\}_\hbar).$$

For $f = z_1^p z_2^q$ and the principal-part basis $\rho_{a,b}$, this gives the finite formula

$$z_1^p z_2^q \cdot_\hbar \rho_{a,b} = -\sum_{\substack{r\geq 1\\ r \text{ odd}}} \frac{\hbar^{r-1}}{2^{r-1}r!}\, C_r(p,q;a-p+r,b-q+r)\,\rho_{a-p+r,b-q+r},$$

where

$$C_r(p,q;m,n) = \sum_{\alpha=0}^{r}(-1)^\alpha \binom{r}{\alpha}(p)_{r-\alpha}(q)_\alpha(m)_\alpha(n)_{r-\alpha},$$

and terms with negative indices or scalar target $(0,0)$ are zero. The $r = 1$ part is the classical Matlis coadjoint action. The $r \geq 3$ terms are genuine Moyal corrections and are not the polynomial one-ψ PBW action.

(v) The reduced open-line Wick graph computation realizes the same formal Moyal product with midpoint propagator. It does not prove an unreduced Costello graph/QME theorem. Such a theorem would require the brane-defect propagator, counterterms, anomaly cancellation, and d_{QME}-homotopies in the local BV complex; those data are outside the algebraic Weyl/Moyal target.

Proof. Item (1) is Theorem F.2. Example F.5 gives a concrete nontrivial check of the coefficients $1/24$ and $1/1920$.

Item (2) is the normal-ordered comoment calculation in Definition F.6 and Lemma F.7; the scalar quotient is exactly the quotient of Definition F.1 and Remark F.38.

Item (3) is Theorem F.30. The primitive formulation is Lemma F.15; the exact obstruction theorem is Proposition F.25, with Statement F.22 as its fixed-lift Hodge form. The edge strip is proved by Proposition F.27. The finite interior certificates in Proposition F.20 and Proposition F.28 do not supply the missing all-bidegree decorated PBW Stokes identity.

Item (4) is the residue-dual definition of the Moyal coadjoint action, expanded with the same P^r-coefficient formula as in item (1). Finiteness follows because the falling factorials vanish for $r > p+q$, and the scalar target is removed by the Hamiltonian scalar quotient. Its special fibre is Proposition 3.227; the quantum target is Construction 3.156.

Item (5) is Theorem F.35 together with Corollary F.36. The separation from the Costello graph/QME realization is part of the hypothesis boundary in Theorem 2.62. □

G Cross-volume matched conventions and obstruction complexes

The theorem is local on

$$\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}} = \mathbb{R}_{\mathrm{brane}} \times \mathbb{R}_{\mathrm{top,normal}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}},$$

with brane line $\mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{0\}$. Its closed sector is Hamiltonian BF theory on the formal holomorphic-symplectic disk together with the topological directions, and its open sector is the Dirac

brane probe on the brane line. The statement is at $d = \dim_{\mathbb{C}} Y = 2$ in the holomorphic factor and uses the brane line $\mathbb{R}_{\mathrm{brane}}$ as its one-dimensional factorization direction.

This appendix states the transfer problem in theorem form. The local result contributes a $d = 2$ formal-disk convention base for the external global or cross-volume descent problem. A global or cross-volume theorem using it is accepted exactly when it gives a target convention datum, comparison morphisms, and null-homotopies for the named obstruction classes.

Definition G.1 *(Local Hamiltonian convention base).* The local theorem supplies the convention base

$$\mathfrak{C}^{(2)}_{\mathrm{loc}} = \big(d = 2,\ \mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}},\ \mathbb{R}_{\mathrm{brane}},\ \widehat{\mathfrak{h}},\ \widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}[1],\ \mathbf{Ext}^*(\mathcal{O}_0, \mathcal{O}_0),\ [\bar{c}],\ \hbar_{\mathrm{Weyl}},\ \Phi^{\mathrm{Ham}}_{\mathrm{fact}}\big),$$

where $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$. The base includes the brane-line factorization convention, the scalar quotient, and the reduced degree-zero Capelli/Moyal normalization.

Lemma G.2 *(Formal-disk external descent criterion).* The formal-disk theorem contributes the formal stalk used by a global or cross-volume descent target $\mathfrak{T}$ exactly after the target supplies a matched-conventions datum $\mathfrak{C}_{\mathrm{tar}}$ and a null-homotopy of the acceptance class

$$\mathrm{Ob}_{\mathrm{UKD}}(\mathfrak{T}) = \Big(\mathrm{Ob}_{\mathfrak{C}_{\mathrm{tar}}}, \mathrm{Ob}_{\mathrm{glob}}, \mathrm{ob}_{6r}, \mathrm{ob}_{\mathrm{MD}}\Big),$$

with the entries omitted only when the corresponding structure is not asserted by $\mathfrak{T}$. The accepted morphism has formal stalk

$$\chi_{\mathrm{op}}^{-1} \Phi^{\mathrm{Ham}}_{\mathrm{fact}} \chi_{\mathrm{cl}}.$$

Proof. The local input consists of the formal Hamiltonian algebra $\mathbb{C}[[z_1, z_2]]/\mathbb{C}$, its continuous cotangent extension, the point brane Ext algebra, the brane-line factorization-current convention, the scalar quotient, and the Capelli/Weyl normalization. These data are invariant under the formal Darboux germ. Compact topology, a negative-cyclic Calabi–Yau class, a worldsheet or specialization curve, an S^3 framing, a global BV propagator, the holomorphic anomaly equation, Borcherds input data, a Weyl chamber, a compact Hall orientation, and a compact chiral-homology object therefore enter as components of $\mathfrak{C}_{\mathrm{tar}}$, not as consequences of the formal germ. The four displayed components record exactly the target convention, global descent, six-relation mixed-factorization, and mixed-Dunn acceptance tests. Vanishing with null-homotopies is the matched-conventions admissibility datum; a nonzero component is the corresponding obstruction class. □

Definition G.3 *(Matched-conventions theorem datum).* A matched-conventions theorem datum is a tuple

$$\mathfrak{C}_{\mathrm{tar}} = \big(d, Y, \eta, (\Sigma_{d-1}, C), L, \tau, \lambda, \mathfrak{f}, \Lambda, \hbar_{\mathrm{tar}}, \mathcal{O}_{\mathrm{tar}}, \chi_{\mathrm{cl}}, \chi_{\mathrm{op}}, \chi_{\mathrm{anom}}, \mathcal{K}_{\mathrm{obs}}, \mathfrak{K}_{\mathrm{KD}}, \Theta_{\mathrm{KD}}\big)$$

with the following meaning. Here $d = \dim_{\mathbb{C}} Y$; η is the negative-cyclic Calabi–Yau class, holomorphic volume form, or holomorphic-symplectic form used by the target theory; (Σ_{d-1}, C) is the worldsheet or factorization-homology specialization when a Φ_d-type theorem is asserted; L is the brane or boundary condition; τ is the trace pairing and scalar quotient; λ is the anomaly normalization; $\mathfrak{f}$ is the framing datum, including an S^3-framing for compact CY_3 statements; Λ is the coefficient topology; $\hbar_{\mathrm{tar}}$ is the deformation parameter; $\mathcal{O}_{\mathrm{tar}}$ is the QME or obstruction hypothesis; χ_{cl}, χ_{op}, and χ_{anom} are the closed-field, open-field, and anomaly comparison morphisms from the target formal restriction to $\mathfrak{C}^{(2)}_{\mathrm{loc}}$; and $\mathcal{K}_{\mathrm{obs}}$ is the obstruction complex in which descent, QME, and operadic coherence are solved; $\mathfrak{K}_{\mathrm{KD}}$ is the

filtered dg Lie algebra governing the target global or cross-volume descent Maurer–Cartan problem; and Θ_{KD} is the transported local reduced central-operation Maurer–Cartan element whose curvature is the target acceptance class.

Definition G.4 *(Compact Hamiltonian period complex).* Let (X, ω) be a compact connected holomorphic-symplectic surface. For a holomorphic symplectic vector field v, set

$$\alpha_v = -\iota_v \omega \in H^0(X, \Omega^1_{X,\mathrm{cl}}).$$

Choose a Stein cover $\mathfrak{U} = \{U_i\}$ and local holomorphic primitives $h_i \in \mathcal{O}_X(U_i)$ with $dh_i = \alpha_v|_{U_i}$. The differences $h_i - h_j \in \mathbb{C}$ define a Čech cocycle, and its class

$$\mathrm{per}_X(v) = [\, h_i - h_j \,] \in H^1(X, \mathbb{C}_X)$$

is the Hamiltonian period obstruction. Equivalently, it is the connecting image of α_v for the holomorphic de Rham exact sequence

$$0 \longrightarrow \mathbb{C}_X \longrightarrow \mathcal{O}_X \xrightarrow{d} \Omega^1_{X,\mathrm{cl}} \longrightarrow 0.$$

Definition G.5 *(Global Hamiltonian descent complex).* Let L be the brane locus in a proposed global target and suppose that formal Darboux charts have produced local Hamiltonian maps Φ_U on a Weiss/Ran cover of L. The descent target for these maps is the total complex

$$\mathfrak{G}_{\mathrm{WR}}(L) = \mathrm{Tot}\, C^{\bullet}_{\mathrm{WR}}\Big(\mathrm{Ran}(L); \underline{\mathrm{RHom}}\Big(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}, \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}})\Big)\Big).$$

The degree-zero Čech-Weiss cochain $\Phi_{\mathrm{loc}} = \{\Phi_U\}$ has first descent obstruction

$$\mathrm{ob}^{(1)}_{\mathrm{WR}}(\Phi_{\mathrm{loc}}) = [d_{\mathrm{WR}}\Phi_{\mathrm{loc}}] \in H^1(\mathfrak{G}_{\mathrm{WR}}(L)).$$

After choosing a filtration by number of factorization collisions and cotangent insertions, the higher coherence obstructions live in

$$\mathrm{ob}^{(r)}_{\mathrm{WR}}(\Phi_{\mathrm{loc}}) \in H^{r+1}\big(\mathrm{gr}^r\, \mathfrak{G}_{\mathrm{WR}}(L)\big), \qquad r \geq 1.$$

Vanishing of all these classes, together with chosen null-homotopies, is the exact assertion that the local Darboux maps descend to a global factorization-centre morphism.

THEOREM G.6 *(Formal local-to-global Hamiltonian descent theorem data).* Let (X, ω) be a compact connected holomorphic-symplectic surface and $L \subset X$ a brane locus for which the formal local Hamiltonian theorem is applied along Darboux charts. Fix the finite collection of locally Hamiltonian symplectic vector fields and local Darboux Hamiltonians used by the construction. Assume the target volume supplies the complete pronilpotent descent complex, hyperdescent for the relevant Weiss/Ran cosheaf, and the exact inverse-limit conditions excluding $\varprojlim^1$ defects in the obstruction tower. Under those supplied convergence and descent hypotheses, the following data are equivalent.

(i) A global Hamiltonian matched-conventions descent datum: a closed-open morphism

$$\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}})$$

whose Darboux restrictions are the local maps Φ_{loc}, together with its interval, Weiss/Ran, scalar, and connected-stable coherence homotopies.

(ii) Vanishing of the obstruction vector

$$\mathrm{Ob}_{\mathrm{glob}}(X, L, \Phi_{\mathrm{loc}}) = \Big(\{\mathrm{per}_X(v)\}_v,\ \mathrm{ob}_{\mathrm{scal}},\ \mathrm{ob}^{(1)}_{\mathrm{WR}}(\Phi_{\mathrm{loc}}),\ \{\mathrm{ob}^{(r)}_{\mathrm{WR}}(\Phi_{\mathrm{loc}})\}_{r\geq 1}\Big)$$

with null-homotopies compatible with disjoint unions, interval inclusions, cotangent boundary realization, P_0-centrality, and the stable connected quotient.

The morphism in (i) is the global accepted Koszul-duality surface

$$\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}})$$

and the entries in (ii) are:

(a) every locally Hamiltonian symplectic vector field used by the construction has period class

$$\mathrm{per}_X(v) = \delta(-\iota_v\omega) \in H^1(X, \mathbb{C}_X)$$

equal to zero;

(b) the first Weiss/Ran class $\mathrm{ob}^{(1)}_{\mathrm{WR}}$ vanishes;

(c) all higher coherence classes $\mathrm{ob}^{(r)}_{\mathrm{WR}}$, $r \geq 1$, vanish after the cotangent boundary realization and P_0-centrality homotopies have been fixed;

(d) the scalar anomaly class agrees with the local class $[\bar{c}]$.

When these conditions hold, the local Darboux maps glue, uniquely up to the corresponding contractible space of choices, to the displayed global Hamiltonian central-operation morphism. When one included component is nonzero, that component is the exact universal descent obstruction; when a component required by the asserted target structure is absent, the target datum is incomplete.

On a compact connected X the period condition is sharp:

$$\begin{aligned}\mathrm{per}_X(v) = 0 &\iff \alpha_v = 0\\ &\iff v = 0.\end{aligned}$$

Thus a nonzero global holomorphic symplectic vector field on a compact surface, for example a translation on an abelian surface, has nonzero $\mathrm{per}_X(v)$ and is rejected by the global Hamiltonian acceptance criterion.

Proof. The period calculation is Definition G.4. If $\mathrm{per}_X(v) = 0$, the local primitives h_i can be modified by constants and glued to a global holomorphic function h with $dh = -\iota_v\omega$. Since X is compact and connected, $H^0(X, \mathcal{O}_X) = \mathbb{C}$, hence $dh = 0$. Therefore $\iota_v\omega = 0$, and nondegeneracy of ω gives $v = 0$. The converse is immediate. This proves the sharp compact-surface obstruction: $\mathrm{per}_X(v)$ is the exact class preventing formal local Hamiltonians from becoming nonzero compact global Hamiltonians by descent alone.

Once local primitives and Darboux charts are fixed, the remaining problem is ordinary descent for morphisms of factorization objects. On double overlaps the failure of the local maps to agree is $d_{\mathrm{WR}}\Phi_{\mathrm{loc}}$, hence the first obstruction is the displayed class in $H^1(\mathfrak{G}_{\mathrm{WR}}(L))$. If it vanishes, a first homotopy can be chosen. The obstruction to making those homotopies coherent on higher overlaps and higher collision strata is computed by the associated graded pieces of the same filtered total complex, giving the classes $\mathrm{ob}^{(r)}_{\mathrm{WR}}$. This is the standard obstruction theory for a Maurer–Cartan element in a filtered dg Lie algebra of descent data. Scalar compatibility is separate because the local theorem has already quotiented by the distinguished scalar line. If all classes vanish and null-homotopies have been chosen, the Maurer–Cartan element is gauge equivalent to a global descent datum, which is exactly the displayed morphism. □

THEOREM G.7 *(Conditional matched-conventions transfer criterion).* Let $\mathfrak{C}_{\mathrm{tar}}$ be a matched-conventions datum and suppose that its comparison morphisms identify the formal restriction of the target closed and open theories with $\mathfrak{C}^{(2)}_{\mathrm{loc}}$:

$$\chi_{\mathrm{cl}}\colon \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed,tar}}\big|_{\widehat{p}} \xrightarrow{\simeq} \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}, \qquad \chi_{\mathrm{op}}\colon \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open,tar}}\big|_{\mathbb{R}_{\mathrm{brane}}} \xrightarrow{\simeq} \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},H}.$$

Assume that χ_{anom} sends the target scalar anomaly to $[\bar{c}]$, that the target period classes specified by $\mathcal{K}_{\mathrm{obs}}$ vanish for the locally Hamiltonian fields used by the comparison, that the Weiss/Ran obstructions in $\mathfrak{G}_{\mathrm{WR}}(L)$ vanish with chosen null-homotopies, and that the QME obstruction specified by O_{tar} vanishes. If $\mathfrak{C}_{\mathrm{tar}}$ also asserts a mixed topological-holomorphic operadic comparison, assume the map $\delta_{m,n}$ satisfies Theorem G.11. Equivalently, assume the target obstruction vector

$$\mathrm{Ob}_{\mathfrak{C}_{\mathrm{tar}}} = \Big(\mathrm{per}_{\mathrm{tar}},\ \mathrm{ob}_{\mathrm{WR}},\ \mathrm{ob}_{\mathrm{anom}},\ \mathrm{ob}_{\mathrm{QME}},\ \mathrm{ob}_{\mathrm{conv}},\ \mathrm{ob}_{\mathrm{hyp}},\ \mathrm{ob}_{\mathrm{VolIII}},\ \mathrm{ob}_{\mathrm{Igusa/BKM}},\ \mathrm{ob}_{6r},\ \mathrm{ob}_{\mathrm{MD}}\Big)$$

is zero, with $\mathrm{ob}_{\mathrm{VolIII}}$, $\mathrm{ob}_{\mathrm{Igusa/BKM}}$, ob_{6r}, and $\mathrm{ob}_{\mathrm{MD}}$ omitted only when the corresponding Vol III, Igusa/BKM, six-relation, or mixed topological-holomorphic operadic structure is not part of the target datum. The component $\mathrm{ob}_{\mathrm{conv}}$ records convergence of the transported Maurer–Cartan obstruction tower, and $\mathrm{ob}_{\mathrm{hyp}}$ records hyperdescent and the absence of hidden $\varprojlim^{1}$ defects. Let

$$\mathrm{Ob}_{\mathrm{UKD}}(\mathfrak{C}_{\mathrm{tar}}) = \Big(\mathrm{Ob}_{\mathfrak{C}_{\mathrm{tar}}}, [d\Theta_{\mathrm{KD}} + \tfrac{1}{2}[\Theta_{\mathrm{KD}}, \Theta_{\mathrm{KD}}]]\Big) \in H^{1}(\mathcal{K}_{\mathrm{obs}}) \oplus H^{1}(\mathfrak{K}_{\mathrm{KD}})$$

be the matched-conventions acceptance class of the transported local reduced Hamiltonian central-operation Maurer–Cartan element.

If $\mathrm{Ob}_{\mathrm{UKD}}(\mathfrak{C}_{\mathrm{tar}}) = 0$ and a null-homotopy of this class has been chosen, then the target descent datum, using the local formal-stalk component, determines an accepted target factorization-centre morphism

$$\Phi_{\mathfrak{C}_{\mathrm{tar}}}\colon \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed,tar}}\big|_{L} \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open,tar}})$$

whose formal restriction at p is

$$\chi_{\mathrm{op}}^{-1}\,\Phi^{\mathrm{Ham}}_{\mathrm{fact}}\,\chi_{\mathrm{cl}}.$$

If a specialization functor $\mathrm{Sp}_{\Sigma_{d-1},C}$ is part of $\mathfrak{C}_{\mathrm{tar}}$, the specialized morphism is $\mathrm{Sp}_{\Sigma_{d-1},C}(\Phi_{\mathfrak{C}_{\mathrm{tar}}})$. If the acceptance class is nonzero, that class is the obstruction to accepting the target theorem in the global or cross-volume descent problem.

Proof. The comparison morphisms transport the local reduced central-operation map to each Darboux chart of the target. Anomaly compatibility identifies the scalar quotient with the local class $[\bar{c}]$, so the transported bracket and connected-trace normalization agree with the Hamiltonian theorem. The vanishing of the target period classes gives Hamiltonian primitives for the local symplectic vector fields in every chart where the Hamiltonian model is used, and the vanishing of the Weiss/Ran classes in $\mathfrak{G}_{\mathrm{WR}}(L)$ glues the local maps with coherent factorization homotopies by Theorem G.6. The QME obstruction vanishing upgrades the classical comparison to the target quantum normalization specified by O_{tar}. When a mixed topological-holomorphic operad is present, the mixed-Dunn criterion gives the required equivalence of algebra categories before applying specialization. The displayed formula is the definition of the glued map on the formal stalk. The remaining Maurer–Cartan curvature is exactly $d\Theta_{\mathrm{KD}} + \frac{1}{2}[\Theta_{\mathrm{KD}}, \Theta_{\mathrm{KD}}]$; a chosen null-homotopy makes the transported local datum a global matched-conventions global descent datum. Descent gives the global morphism. □

Definition G.8 *(Functor of points of mixed six-relation factorization).* Let M be a real m-manifold and Y a complex n-fold. For an Artinian commutative dg algebra R, an R-point of $\mathrm{Fact}^{6r}_{\mathrm{mix}}(M,Y)$ is a datum

$$\mathcal{A}_R = (A_R, \mu_R, \ell_R, \bar\partial_R, \mathsf{W}_R, \mathsf{R}_R, \iota_R)$$

on product disks $D \times B \subset M \times Y$, where D is a real disk and B is a holomorphic polydisk: this is the product-disk site $\mathrm{Ran}_{\prod}(M,Y)$, generated by product disks and finite disjoint unions of such products. No cofinality of product-disk covers inside arbitrary Weiss covers of $M \times Y$, and no descent on the ordinary $\mathrm{Ran}(M \times Y)$, is asserted here. A target using arbitrary Weiss/Ran covers must supply $\mathrm{ob}_{\mathrm{WR}}$, $\mathrm{ob}_{\mathrm{hyp}}$, and compatible null-homotopies in the matched-conventions obstruction complex.

(i) $A_R(D \times B)$ is a nilcomplete R-linear cochain complex, functorial for inclusions of product disks;
(ii) μ_R gives factorization products for pairwise disjoint finite families of product disks;
(iii) ℓ_R gives local-constancy homotopies for inclusions of real disks in the M-direction;
(iv) $\bar\partial_R$ gives Dolbeault null-homotopies for anti-holomorphic translations in the Y-direction;
(v) W_R gives Weiss descent in the real direction;
(vi) R_R gives Ran descent in the holomorphic direction;
(vii) ι_R gives the interchange homotopies between real collision products and holomorphic factorization products.

These data are required to satisfy six relations. Put

$$\mathsf{Rel}^{6r}(\mathcal{A}_R) = \big(\mathrm{Assoc}(\mu_R),\ \mathrm{LC}(\ell_R),\ \bar\partial_R^2,\ d_{\mathrm{WR}}\mathsf{W}_R,\ d_{\mathrm{Ran}}\mathsf{R}_R,\ \mathrm{Int}(\iota_R)\big).$$

An R-point is precisely a seven-tuple with $\mathsf{Rel}^{6r}(\mathcal{A}_R) = 0$: associativity of μ_R, homotopy invariance for ℓ_R, Dolbeault closedness for $\bar\partial_R$, Weiss cocycle descent for W_R, Ran cocycle descent for R_R, and the mixed interchange identity for ι_R. A k-simplex of the derived functor of points is the same datum over $R \otimes \Omega^\bullet(\Delta^k)$. Base change in R defines the functor

$$\mathrm{Fact}^{6r}_{\mathrm{mix}}(M,Y)\colon \mathrm{dgArt}_{\mathbb{C}} \longrightarrow \mathrm{sSet}.$$

This functor is the mixed factorization surface used by the global or cross-volume descent problem. Its admissibility condition is the equation $\mathsf{Rel}^{6r} = 0$ together with coherent null-homotopies in square-zero families. For a partial lift $\widetilde{\mathcal{A}}_{R'}$ over a square-zero extension $R' \twoheadrightarrow R$ with kernel I, the six-relation obstruction is

$$\mathrm{ob}_{6r}(\widetilde{\mathcal{A}}_{R'}) = [\ \mathsf{Rel}^{6r}(\widetilde{\mathcal{A}}_{R'})\ ] \in H^2\Big(\mathrm{Der}^{\mathrm{fact,mix}}(\mathcal{A}_R) \otimes_R I\Big).$$

A mixed-Dunn comparison is a morphism of cofibrant operadic models

$$\delta_{m,n}\colon \mathbb{E}^{\mathrm{top}}_m \otimes \mathbb{E}^{\mathrm{hol}}_n \longrightarrow \mathbb{E}^{\mathrm{mix}}_{m,n}$$

whose induced algebra functor realizes this functor of points.

Proposition G.9 *(Recognition and tangent complex for mixed six-relation factorization).* Let $\mathcal{A}_R$ be an R-point of $\mathrm{Fact}^{6r}_{\mathrm{mix}}(M,Y)$. Then $\mathcal{A}_R$ determines a mixed factorization object on $\mathrm{Ran}_{\prod}(M,Y)$, locally constant in the M-directions and holomorphic in the Y-directions. Conversely, restriction to product disks sends every such mixed factorization object on this product-disk site to an R-point of $\mathrm{Fact}^{6r}_{\mathrm{mix}}(M,Y)$.

If $R' \twoheadrightarrow R$ is a square-zero extension with kernel I, the obstruction to lifting $\mathcal{A}_R$ to an R'-point is the six-relation class ob_{6r}, lying in

$$H^2\begin{pmatrix}\mathrm{Der}^{\mathrm{fact,mix}}(\mathcal{A}_R)\\ \otimes_R I\end{pmatrix},$$

and the space of lifts, when nonempty, is a torsor under

$$H^1\begin{pmatrix}\mathrm{Der}^{\mathrm{fact,mix}}(\mathcal{A}_R)\\ \otimes_R I\end{pmatrix}.$$

Proof. Product disks form the objects of the product-disk mixed Weiss/Ran site $\mathrm{Ran}_{\prod}(M, Y)$. The factorization products μ_R define the multiplication on disjoint configurations. The homotopies ℓ_R and $\bar{\delta}_R$ impose the two directional invariance conditions. The descent systems W_R and R_R glue the basis values to the mixed Ran object, and ι_R is exactly the compatibility between the two collision operations. Nilcompleteness over R identifies this glued object with the derived functor of points. Restricting a mixed factorization object to product disks recovers the seven pieces of data and the six relations.

For a square-zero extension $R' \to R$, deforming the seven structure maps while preserving the six relations is the Maurer–Cartan lifting problem in the dg Lie algebra of mixed factorization derivations $\mathrm{Der}^{\mathrm{fact,mix}}(\mathcal{A}_R) \otimes_R I$. The curvature of a partial lift is the degree-two component of Rel^{6r} for the lifted seven-tuple; its cohomology class is precisely ob_{6r} and is independent of choices. If this class vanishes, the remaining choices form the usual H^1-torsor. This proves the positive recognition criterion for the mixed surface: a target supplies a global or cross-volume descent factorization surface exactly by giving an R-point and killing the six-relation obstruction tower in square-zero thickenings. □

Definition G.10 *(Mixed-Dunn obstruction complex).* The deformation complex of the candidate operad map $\delta_{m,n}$ is

$$\mathfrak{D}_{\mathrm{MD}}(m,n) = \mathrm{Cone}\Big(C^\bullet_{\mathrm{Operad}}(\mathbb{E}^{\mathrm{mix}}_{m,n}, \mathbb{E}^{\mathrm{mix}}_{m,n}) \longrightarrow C^\bullet_{\mathrm{Operad}}(\mathbb{E}^{\mathrm{top}}_m \otimes \mathbb{E}^{\mathrm{hol}}_n, \mathbb{E}^{\mathrm{mix}}_{m,n})\Big)[-1].$$

After a cofibrant filtration of the source operad has been chosen, the mixed-Dunn curvature is

$$F_\delta = d\delta + \frac{1}{2}[\delta,\delta] \in Z^1(\mathfrak{D}_{\mathrm{MD}}(m,n)).$$

Its class

$$\mathrm{ob}_{\mathrm{MD}}(\delta_{m,n}) = [F_\delta] \in H^1(\mathfrak{D}_{\mathrm{MD}}(m,n))$$

measures the simultaneous failure of the topological E_m operations, the holomorphic E_n operations, and the mixed interchange law to assemble into a strict operadic comparison. Higher rectification obstructions lie in $H^{r+1}(\mathrm{gr}^r\,\mathfrak{D}_{\mathrm{MD}}(m,n))$.

THEOREM G.11 *(Mixed-Dunn matched-conventions criterion).* Let $\delta_{m,n}$ be a morphism between cofibrant operadic models as in Definition G.8, equipped with a complete exhaustive cofibrant filtration for which the obstruction tower below converges and has no hidden $\varprojlim^1$ defect. The mixed-Dunn surface is accepted by the global or cross-volume descent problem exactly under the following equivalent conditions.

(i) $\delta_{m,n}$ rectifies to a weak equivalence of operads and induces an equivalence

$$\mathrm{Alg}_{\mathbb{E}^{\mathrm{mix}}_{m,n}} \simeq \mathrm{Alg}_{\mathbb{E}^{\mathrm{top}}_m \otimes \mathbb{E}^{\mathrm{hol}}_n}$$

on the homotopy categories of algebras, with $\mathrm{Fact}^{6r}_{\mathrm{mix}}(M, Y)$ represented by the tensor product of the topological and holomorphic structures.

(ii) The aritywise homology map of $\delta_{m,n}$ is an isomorphism, the primary class $\mathrm{ob}_{\mathrm{MD}}(\delta_{m,n})$ vanishes, and every higher class in $H^{r+1}(\mathrm{gr}^r\, \mathfrak{D}_{\mathrm{MD}}(m, n))$ vanishes with chosen null-homotopies compatible with operadic composition.

When these classes do not vanish, their ordered tuple is the exact mixed-Dunn acceptance obstruction.

Proof. A weak equivalence between cofibrant operads induces a Quillen equivalence on their algebra categories [37, Thm. A.3.7.5]. This proves that (i) implies the aritywise homology isomorphism and that the curvature classes in $\mathfrak{D}_{\mathrm{MD}}(m, n)$ vanish.

Conversely, suppose (ii) holds. The complete exhaustive cofibrant filtration reduces rectification of $\delta_{m,n}$ to a convergent tower of lifting problems. The primary curvature F_δ is the obstruction at the first stage; the associated-graded obstruction groups record the successive failures of operadic composition, Beilinson–Drinfeld holomorphic compatibility, and the topological-holomorphic interchange law. By hypothesis each obstruction class is zero and a compatible null-homotopy has been chosen, so the tower has a rectified operad map. The aritywise homology isomorphism makes the rectified map a weak equivalence. Applying the cofibrant-operad Quillen-equivalence theorem gives (i), and Proposition G.9 identifies the resulting algebra objects with the displayed functor of points. □

Corollary G.12 *(Matched-conventions factorization).* Every target theorem that elects to use the local Hamiltonian BF/Moyal theorem through a matched-conventions datum $\mathfrak{C}_{\mathrm{tar}}$ with $\mathrm{Ob}_{\mathrm{UKD}}(\mathfrak{C}_{\mathrm{tar}}) = 0$ and chosen null-homotopies has, as its formal-stalk component, the transported morphism

$$\Phi_{\mathfrak{C}_{\mathrm{tar}}} = \chi_{\mathrm{op}}^{-1}\, \Phi^{\mathrm{Ham}}_{\mathrm{fact}}\, \chi_{\mathrm{cl}}$$

constructed in Theorem G.7. The remaining specialization, compactification, automorphic construction, and global descent maps are target data.

Proof. Theorem G.7 constructs the formal-stalk component obtained from the local theorem and the comparison morphisms $\chi_{\mathrm{cl}}, \chi_{\mathrm{op}}$. Any subsequent operation belongs to the target datum: specialization along (Σ_{d-1}, C), compact BV integration, or the Borcherds automorphic construction. Thus the local theorem enters the target proof only through the displayed formal-stalk component. □

Corollary G.13 *(External target acceptance criterion).* A compact BCOV, MNOP/chiral-volume, Vol III CY-to-chiral, Igusa/BKM, or other cross-volume theorem may use the local Hamiltonian BF/Moyal formal-stalk input only when supplied with a global or cross-volume descent datum $\mathfrak{C}_{\mathrm{tar}}$ with

$$\mathrm{Ob}_{\mathrm{UKD}}(\mathfrak{C}_{\mathrm{tar}}) = 0$$

and chosen null-homotopies for all components of the obstruction tower. The accepted theorem-data include the formal stalk

$$\chi_{\mathrm{op}}^{-1}\, \Phi^{\mathrm{Ham}}_{\mathrm{fact}}\, \chi_{\mathrm{cl}},$$

the global descent datum, the QME/anomaly null-homotopies, and any mixed-Dunn or six-relation factorization null-homotopies asserted by the target. If a component required by an asserted target structure is absent, or if any included component is nonzero, the exact residual is the corresponding component of $\mathrm{Ob}_{\mathrm{UKD}}(\mathfrak{C}_{\mathrm{tar}})$; no compact CY_3, global BCOV, MNOP/chiral-volume, Vol III, Igusa, or BKM conclusion is accepted without killing that class.

Proof. Theorem G.7 constructs the accepted target morphism after the target datum has been fixed and the matched-conventions acceptance class has been null-homotoped. The class contains the period, descent, anomaly, QME, six-relation, and mixed-Dunn components. Lemma G.2 identifies these components as target theorem data rather than consequences of the formal disk. Therefore a nonzero component is the named obstruction class, and a zero component with a null-homotopy is exactly the accepted theorem datum. □

Definition G.14 *(Named target convention data).* The cross-volume targets require the following matched-conventions data.

(i) *Vol III CY-to-chiral datum.*

$$\mathfrak{C}_{\mathrm{CY}\to\mathrm{ch}} = \big(d, Y, \eta_{\mathrm{CY}} \in HC_d^-(Y), (\Sigma_{d-1}, C), \Phi_d^{\mathrm{FA}}, \mathrm{Sp}_{\Sigma_{d-1},C}, \kappa_{\mathrm{cat}}, \tau, \mathfrak{f}, O_{\mathrm{QME}}, \mathfrak{K}_{\mathrm{KD}}, \mathrm{Ob}_{\mathrm{UKD}}\big).$$

The comparison map from $\mathfrak{C}_{\mathrm{loc}}^{(2)}$ is the formal $d = 2$ Darboux normalization at the chosen brane point. The Vol III convention is the two-stage factorization

$$\Phi_d = \mathrm{Sp}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}}.$$

For a compact CY_3 target the total-space κ_{cat} is the Hodge supertrace $\sum_q (-1)^q h^{0,q}(Y)$; in particular $\kappa_{\mathrm{cat}}(K3 \times E) = 0$. This component is distinct from any Heisenberg, BKM, or fibre normalization in Vol III. Any BKM weight comparison is therefore a separate component of $\mathrm{Ob}_{\mathrm{UKD}}$, not a consequence of the unqualified total-space supertrace.

(ii) *Compact BCOV datum.*

$$\mathfrak{C}_{\mathrm{BCOV}} = \big(Y^3, \Omega_Y, \mathrm{PV}^{*,*}(Y)[[t]], Q_{\mathrm{BCOV}}, P_{\mathrm{prop}}, L, \mathfrak{f}_{S^3}, I_{\mathrm{ct}}, O_{\mathrm{HAE}}, \mathfrak{K}_{\mathrm{KD}}, \mathrm{Ob}_{\mathrm{UKD}}\big).$$

Here P_{prop} is the global gauge-fixed propagator, I_{ct} is the counterterm scheme, and O_{HAE} is the holomorphic-anomaly/QME obstruction complex.

(iii) *Igusa/BKM datum.*

$$\mathfrak{C}_{\mathrm{BKM}} = \big(\varphi_{\mathrm{Jac}}, \Lambda_{\mathrm{Borch}}, W, \rho, \mathrm{ch}_{\mathrm{Hall}}, \ell_{\mathrm{or}}, \pi_{\mathrm{prim}}, \kappa_{\mathrm{BKM}}, \tau_{\mathrm{DT/PT/OP}}, \mathrm{Ob}_{\mathrm{UKD}}\big).$$

The entries are the Jacobi or modular input, Borcherds lattice, Weyl chamber, character, Hall or chiral source, orientation line, primitive recognition map, and scalar DT/PT/OP normalization. In the Δ_5 row the matched constants are

$$D_5 = 64^{-1}\Delta_5, \qquad \mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z),$$

$$\kappa_{\mathrm{BKM}}(\Phi_1) = c_1(0)/2 = 10/2 = 5, \qquad Z_{\mathrm{OP}}^X = -4096\,\Delta_5^{-2}.$$

The orientation comparison $\epsilon_o = \nu_{\Delta_5}$, compact Hall constants, and primitive BKM recognition are target obstructions; they are not supplied by the local $d = 2$ Hamiltonian theorem.

(iv) *Global Hamiltonian descent datum.*

$$\mathfrak{C}_{\mathrm{glob}} = \big(L, \mathcal{H}_X = \mathcal{O}_X/\mathbb{C}_X, \mathrm{per}(v) = \delta(-\iota_v\omega), \mathfrak{G}_{\mathrm{WR}}(L), \{\mathrm{ob}^{(r)}_{\mathrm{WR}}\}_{r\geq 1}, h_{\mathrm{cent}}, \mathrm{Ob}_{\mathrm{glob}}, \Theta_{\mathrm{KD}}\big).$$

The vanishing condition is the one in Theorem G.6; its output is the global matched-conventions descent surface.

(v) *Mixed-Dunn datum.*

$$\mathfrak{C}_{\mathrm{MD}} = \big(\mathbb{E}^{\mathrm{top}}_m, \mathbb{E}^{\mathrm{hol}}_n, \mathbb{E}^{\mathrm{mix}}_{m,n}, \mathrm{Fact}^{6r}_{\mathrm{mix}}, \mathsf{Rel}^{6r}, \delta_{m,n}, \mathrm{ob}_{6r}, \mathfrak{D}_{\mathrm{MD}}(m,n), \mathrm{ob}_{\mathrm{MD}}\big).$$

The vanishing condition is the simultaneous six-relation and rectification condition in Proposition G.9 and Theorem G.11.

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